



US012416034B2

(12) **United States Patent**
Houghton-Larsen et al.

(10) **Patent No.:** **US 12,416,034 B2**

(45) **Date of Patent:** ***Sep. 16, 2025**

(54) **RECOMBINANT PRODUCTION OF STEVIOL GLYCOSIDES**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 51 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/132,027**

(22) Filed: **Apr. 7, 2023**

(65) **Prior Publication Data**

US 2023/0407360 A1 Dec. 21, 2023

Related U.S. Application Data

(60) Continuation of application No. 17/471,615, filed on Sep. 10, 2021, now abandoned, which is a continuation of application No. 16/535,373, filed on Aug. 8, 2019, now abandoned, which is a continuation of application No. 15/439,660, filed on Feb. 22, 2017, now Pat. No. 10,435,730, which is a division of application No. 14/237,540, filed as application No. PCT/US2012/050021 on Aug. 8, 2012, now Pat. No. 9,631,215.

(60) Provisional application No. 61/521,084, filed on Aug. 8, 2011, provisional application No. 61/521,203, filed on Aug. 8, 2011, provisional application No. 61/521,051, filed on Aug. 8, 2011, provisional application No. 61/523,487, filed on Aug. 15, 2011, provisional application No. 61/567,929, filed on Dec. 7, 2011, provisional application No. 61/603,639, filed on Feb. 27, 2012.

(51) **Int. Cl.**

C12P 19/56 (2006.01)
C12N 1/21 (2006.01)
C12N 9/02 (2006.01)
C12N 9/06 (2006.01)
C12N 9/10 (2006.01)
C12N 9/88 (2006.01)
C12N 9/90 (2006.01)
C12N 15/63 (2006.01)
C12N 15/81 (2006.01)
C12N 15/82 (2006.01)
C12P 19/44 (2006.01)

(52) **U.S. Cl.**

CPC **C12P 19/56** (2013.01); **C12N 9/0032** (2013.01); **C12N 9/0071** (2013.01); **C12N 9/0073** (2013.01); **C12N 9/1051** (2013.01); **C12N 9/1085** (2013.01); **C12N 9/88** (2013.01); **C12N 9/90** (2013.01); **C12N 15/63** (2013.01); **C12N 15/81** (2013.01); **C12N 15/8243** (2013.01); **C12P 19/44** (2013.01); **Y02P 20/52** (2015.11)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Recombinant microorganisms, plants, and plant cells are disclosed that have been engineered to express recombinant genes encoding UDP-glycosyltransferases (UGTs). Such microorganisms, plants, or plant cells can produce steviol glycosides, e.g., Rebaudioside A and/or Rebaudioside D, which can be used as natural sweeteners in food products and dietary supplements.

30 Claims, 44 Drawing Sheets

Specification includes a Sequence Listing.

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FIG. 1

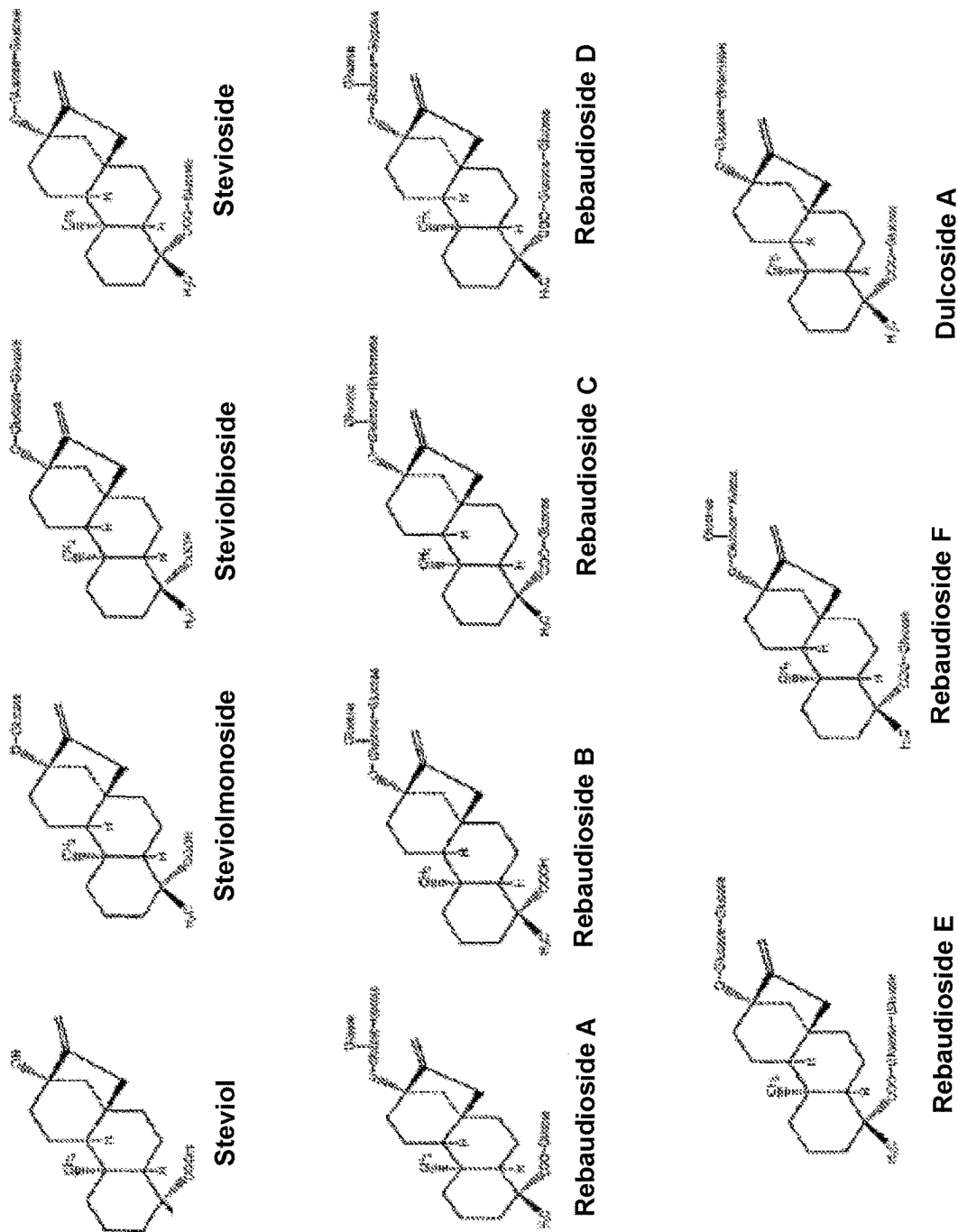


FIG. 2B

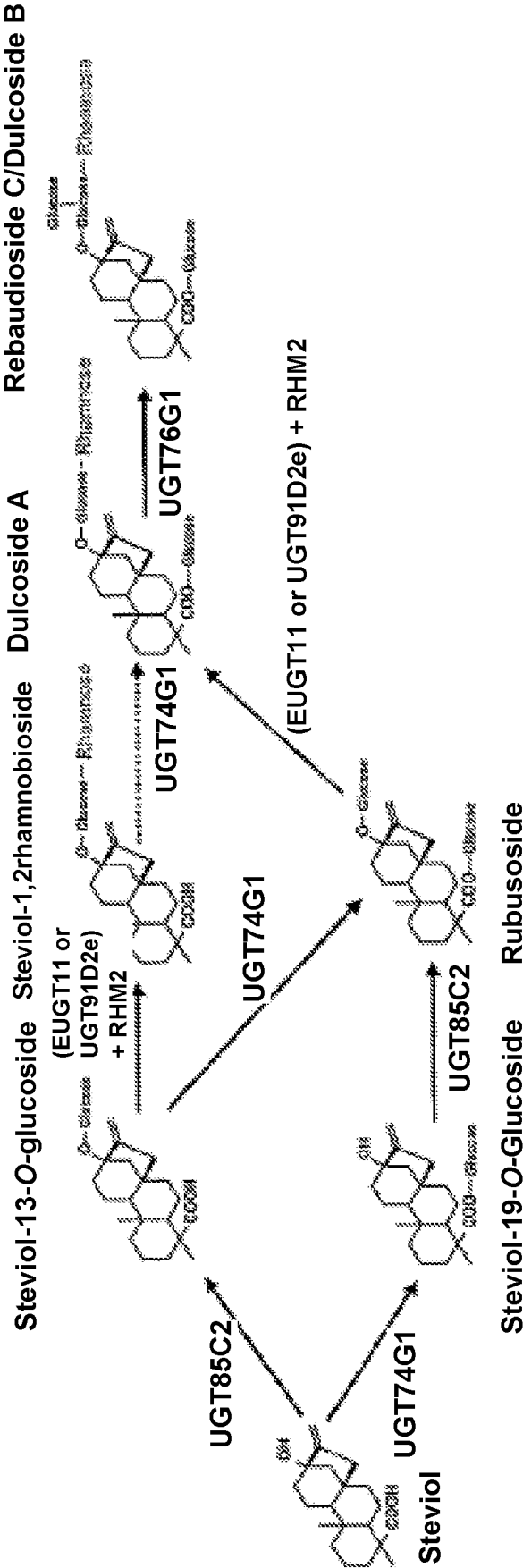


FIG. 2D

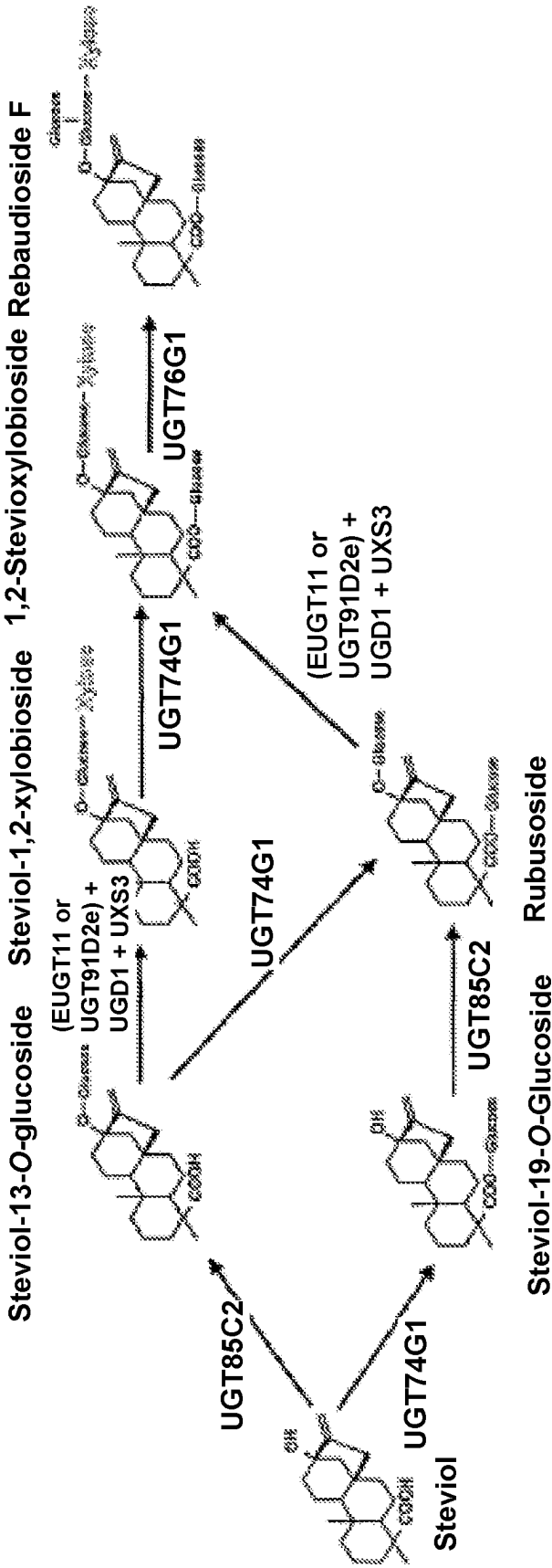


FIG. 3

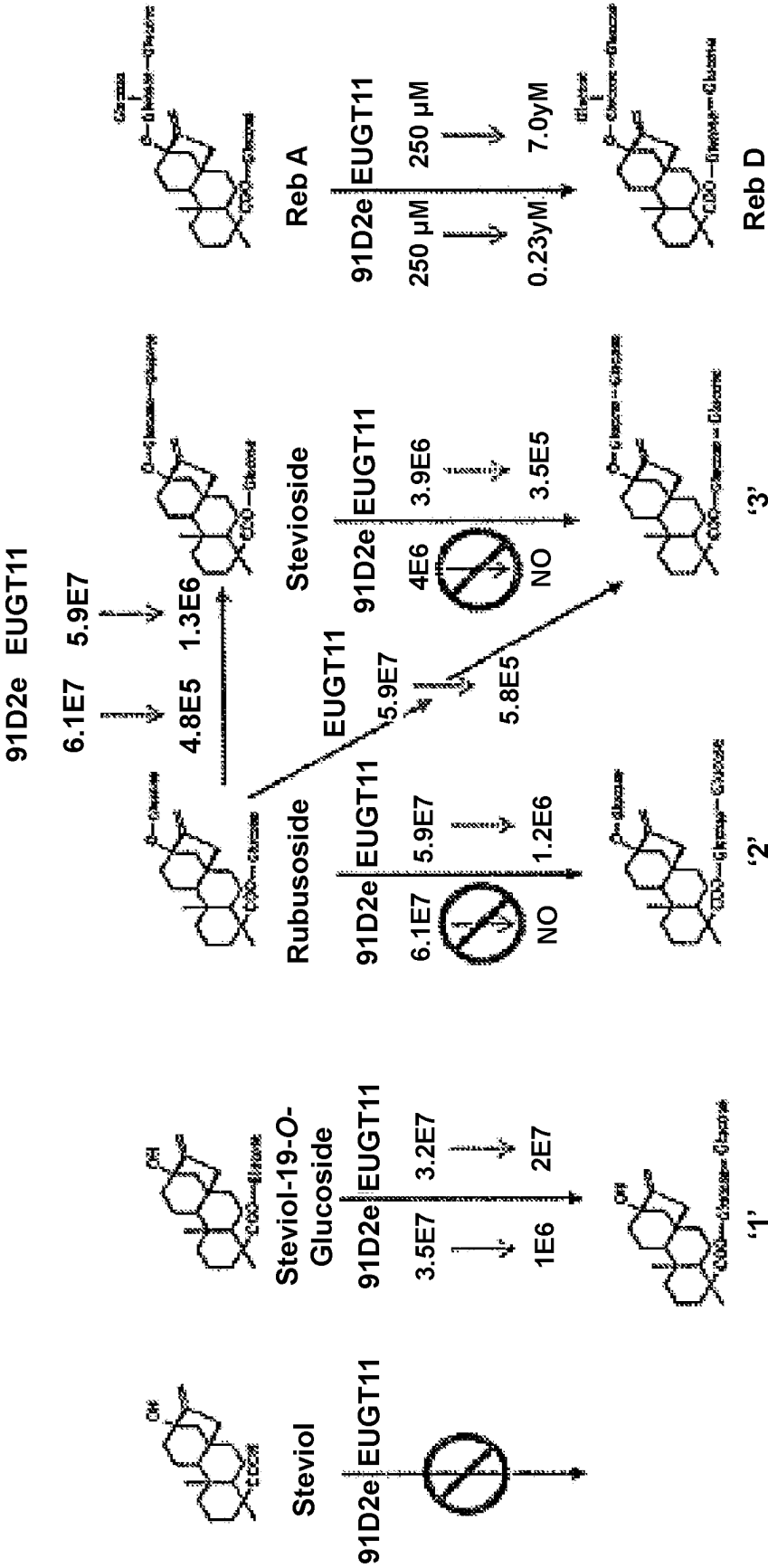


FIG. 4

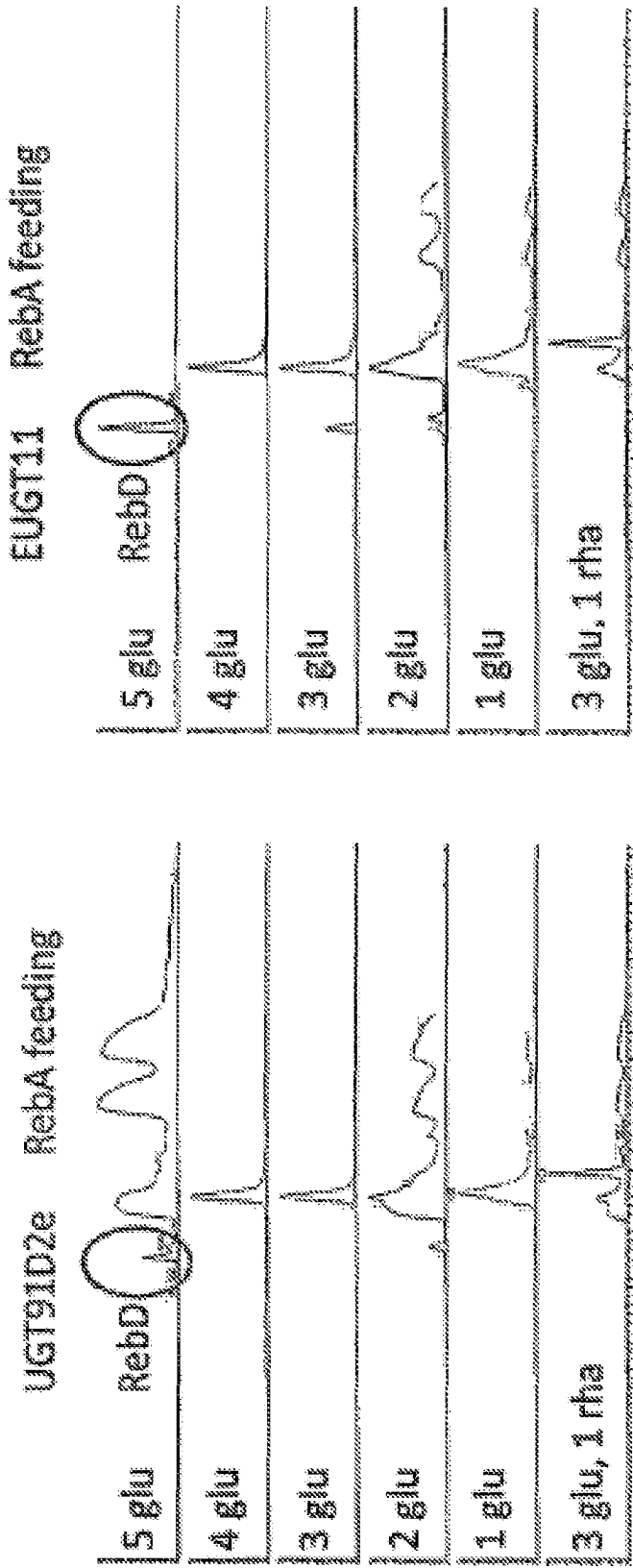
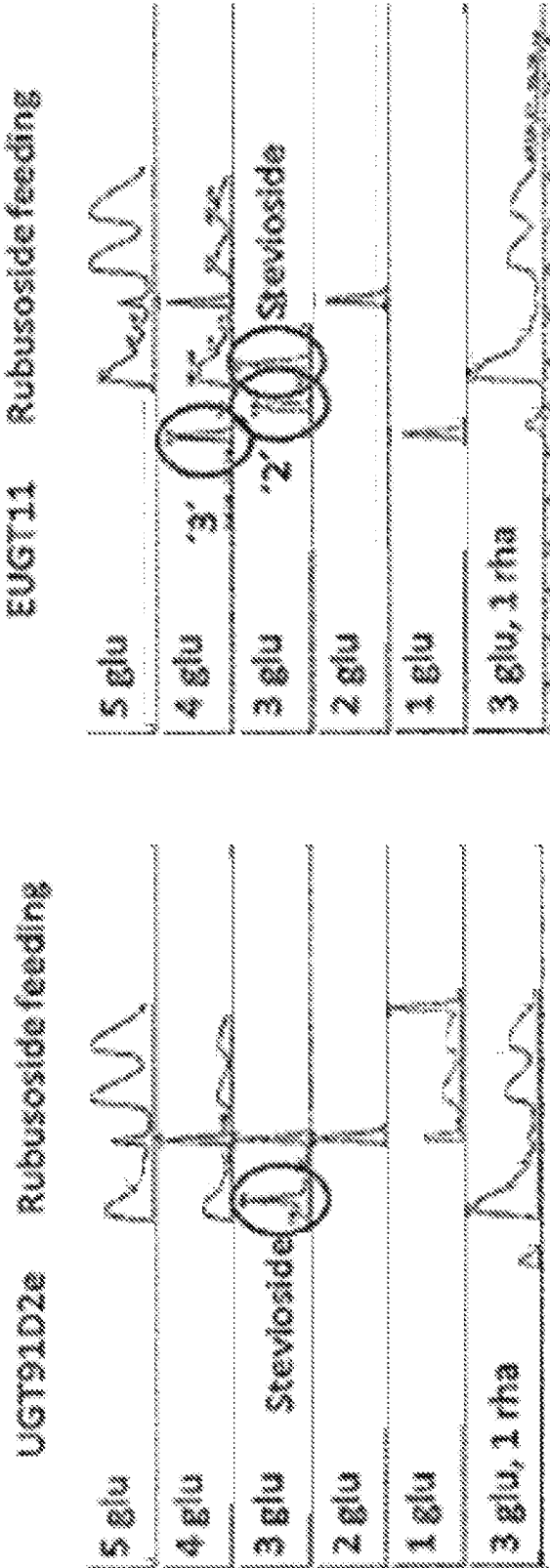


FIG. 5



[illegible]

FIG. 8

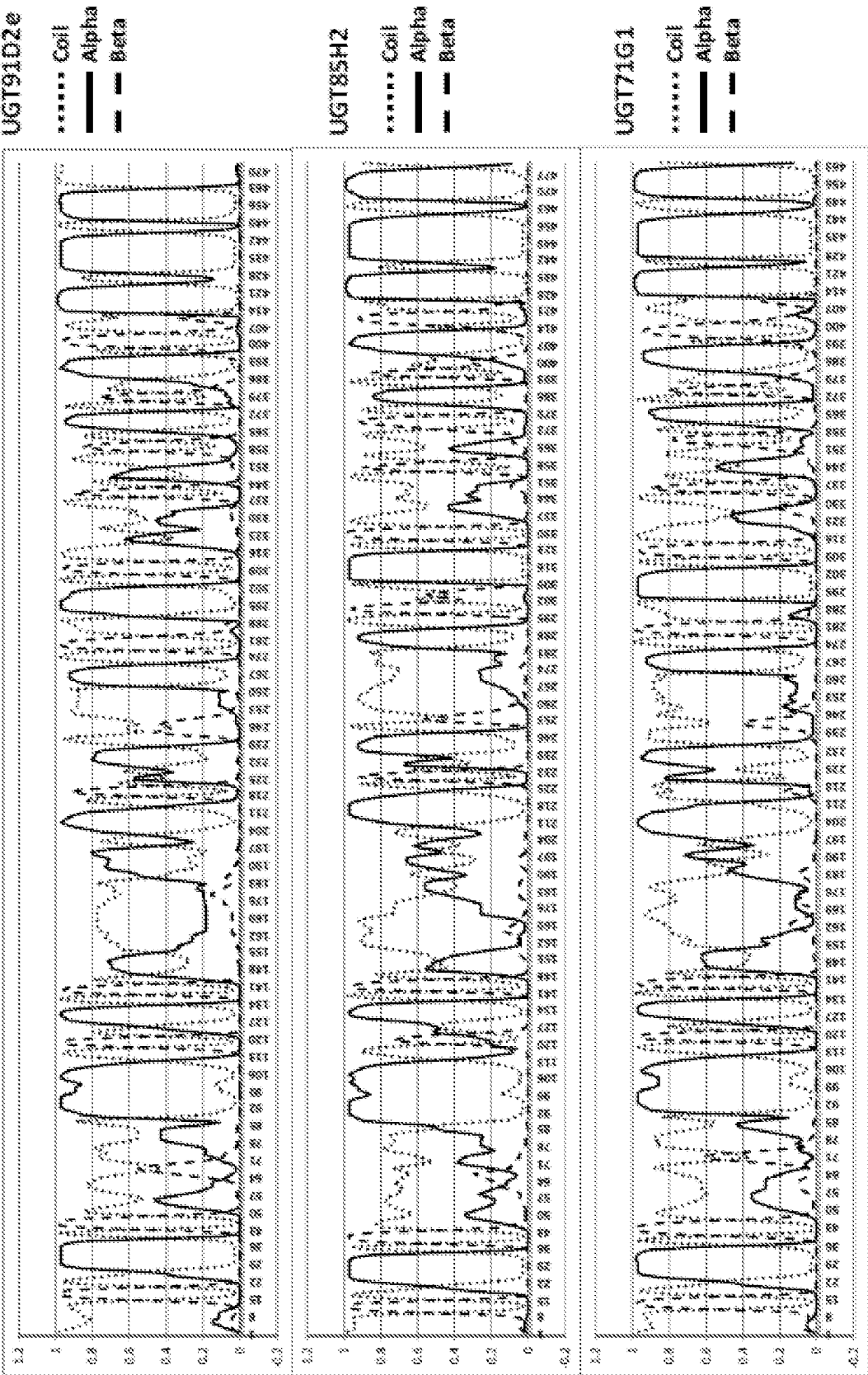


FIG. 9

UGT91D1	MATSDSIVDD	RKQLHVATFP	WLAFGH	LPF	LQLSKLIAEK	GHKV	SFLSTT	RNIQRLSSHI	60
UGT91D2e	MATSDSIVDD	RKQLHVATFP	WLAFGH	LPF	LQLSKLIAEK	GHKV	SFLSTT	RNIQRLSSHI	60
			***					*	120
UGT91D1	SPLINVVQELT	LPRVQELPED	AEATTDVHPE	DIQYL	KKAD	D	GLQPEVTRFL	EQHSPDWIYY	120
UGT91D2e	SPLINVVQELT	LPRVQELPED	AEATTDVHPE	DIQYL	KKAD	D	GLQPEVTRFL	EQHSPDWIYY	120
				***				*	160
UGT91D1	DTHYWLPSI	AASLGISRAY	FGV	TPWT	IA	Y	LAPS	SDAMI	180
UGT91D2e	DTHYWLPSI	AASLGISRAY	FSV	TPWA	IA	Y	MGPS	ADAMI	180
			***					*	220
UGT91D1	FPTKVCWRKH	DLARMEPYEA	PGISDGYRMG	MV	KGSDCLL		KCYHEFGTQ	WLPLETLHQ	240
UGT91D2e	FPTKVCWRKH	DLARLVPYKA	PGISDGYRMG	LV	KGSDCLL		SKCYHEFGTQ	WLPLETLHQ	240
			***					*	280
UGT91D1	VPVVPVGLLP	PEIPGDEKDE	TWVS	IKKWLD	GKQKGSVVYV		ALGSE	LVSD	300
UGT91D2e	VPVVPVGLLP	PEIPGDEKDE	TWVS	IKKWLD	GKQKGSVVYV		ALGSE	LVSD	300
							***		360
UGT91D1	ELSGLPFVWA	YRKPKGPAKS	DSVELPDGFV	ERTDRGLVW	TSWAPQLRIL		SHESVCGFLT	360	
UGT91D2e	ELSGLPFVWA	YRKPKGPAKS	DSVELPDGFV	ERTDRGLVW	TSWAPQLRIL		SHESVCGFLT	360	
			*						420
UGT91D1	HCGSGSIVEG	LMFGHPLIML	PIFQDQ	PLNA	RLLEDKQVGI		EIPRNEEDGC	LTKESVARSL	420
UGT91D2e	HCGSGSIVEG	LMFGHPLIML	PIFQDQ	PLNA	RLLEDKQVGI		EIPRNEEDGC	LTKESVARSL	420
			*						480
UGT91D1	RSVVVENEGE	IYKANARELS	KIYNDTKVEK	EYVSQFVDYL	EKNARAVAI	D	HES	474	
UGT91D2e	RSVVVEKEGE	IYKANARELS	KIYNDTKVEK	EYVSQFVDYL	EKNARAVAI	D	HES	474	

FIG. 10

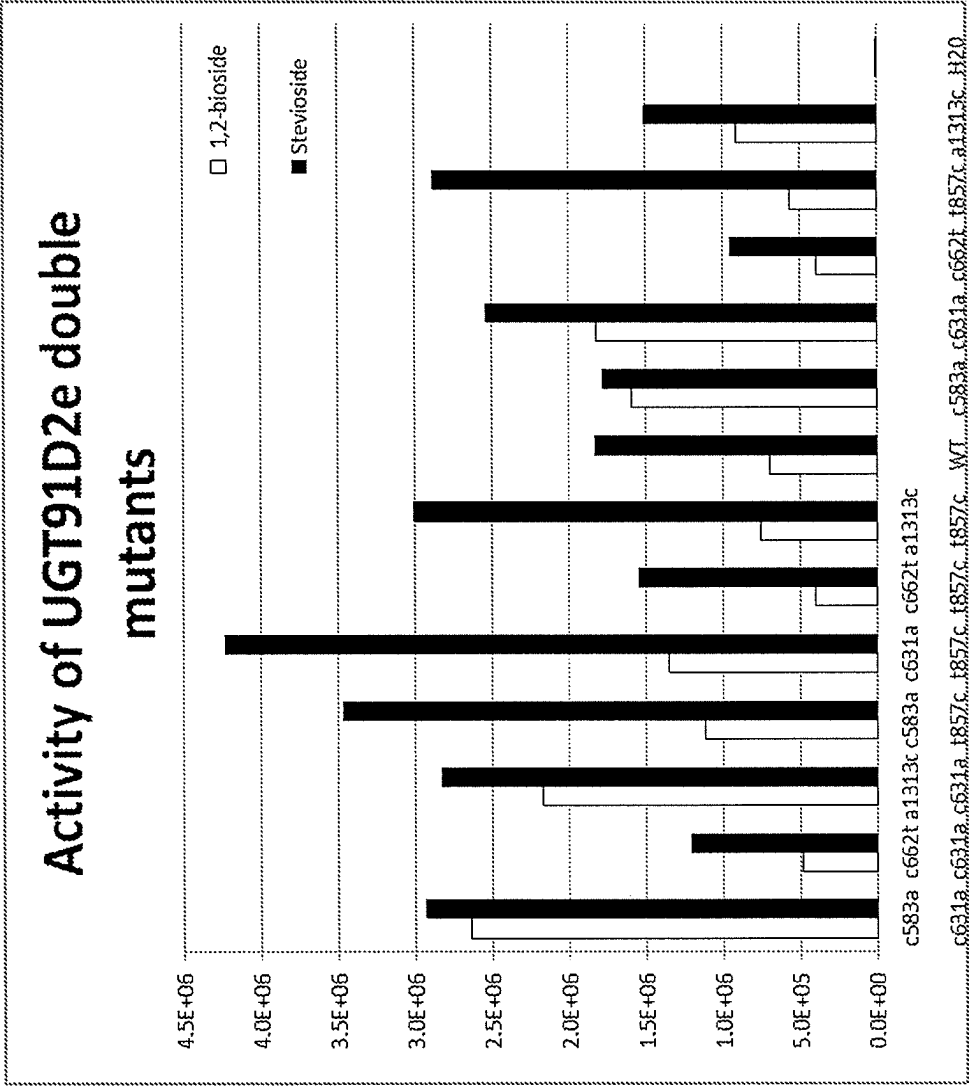


FIG. 11

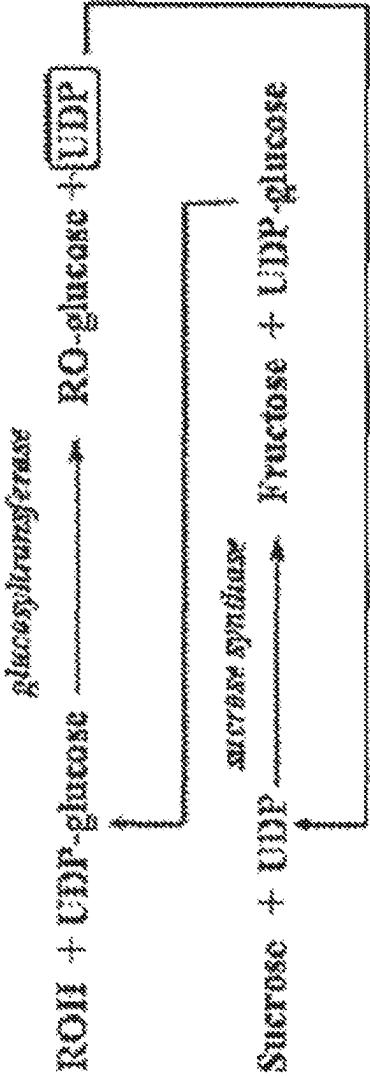


FIG. 12A

Stevia rebaudiana KAH nucleotide (SEQ ID NO: 163)

ATGGAGGCTTCATATCATATACATTTCATTCCTTCTAGCTTCGTACCTCTTCAACCACCAACTTCGTCTAATAATCCGCCAATCT
ACGCCGACGGTGTCCCATCCATCCCATATAATCGGTCACTCTACCTCTCAAAAACCACTCTATAGAACACTAGCCAAAATTTGCCG
CCAAATACGGCCCTATCCTCCAACTCCAACFAGGGFACCGCGTGTCTCGTAATCTCTCTCCCTTCGCGCCGCGAAGAAATGCTTCACC
AACACGACGTTATCTTCGCCAAACCGTCCGAAGACGCTATTCCGAAAAAATTGTAGGGGTACCAGCCTCGGGTCCCTGTCTGTAACGGGA
CCAGTGGGGCAACCTCCGCCCGCTTGCAATCCATCCGAGATTCTATCAGTCCACCGGCTCAACGAGTTTCACGACATACGTTGTTGACGAAA
ACCGCTTCTGATCCGTAAACATAAGATCCAGTCTCTCCGGTGAATCTGATAACGGTGTCTTACGCACTAACGTTAAATGTGATTATG
AGAATGATCTCCGGAAGAGAGGTAATTCGACTCGGGTGATCGGGAATTGGAGGAGGAAGGGAATTCGGGAGATACCTCGATGAGAC
ACTTTTGCTCGCGGGTGCTTCTAATGTTGGGATTAATCTCCGATTCTGAAATTGGTTGGGGTGAAGAGCTTGGAGAAAGAGCTAATCG
CATTCAGAAAAGAGAGAGATGATTTCTTTTCAGGGACTGATCGAGCAAGTTCGGAATCTAGAGGGCTAAAGTGGGAAAAGGAAGGAAG
ACGATGATCGAGTTGTTGTTATCCCTACAAGATCTGAACCTGAGTATTACACTGACGCCATGATCCGATCAATTTGTGCTGGGTTTATT
AGCAGCAGGGAGTGATACATCGGCTGGAACTATGGAATGGCGGATGTCTCTTTTGGTAAACCAACCCGCACTGTTAAAAAGGCACAGG
CTGAAATTGATCGAGTCAATCGGCAACAACCGTCTAATTGATGAGTCCGACATAGGGAATATACCTTACATTGGTTGCATCATAAACGAG
ACGCTCAGATTGTACCTCGGGCCCGTTGCTATTTCCTCATGAGTCAATCAGCGGACTGTGTTATCAGCGGTACACATCCCTCGTGG
GACGATGCTTATTGTCAACCAATGGCGGATACATCAATGACCCAAAGGTGTGGACGACCCCTGAGACATTCAAACCGGAAAGATTCCCAAG
GGCTTGAAGGGACACGAGACGGGTTTAAGCTGATGCCCTTTTGGGTCCGGAAGGAGGGGTTGTCCAGGGGAGGGATFGGCGATTCTGTTG
CTTGGGATGACTCTCGGGTCCGGTTATCCAAATGCTTTGATTGGGAACGAGTCCGTCACGAGATGGTTGATATGACCGAAGCTCTTGGGGT
CACGTTGCCCTAAAGCTGTACCAATTAGTCGCCCAAGTGCAGACCCCGGTTCCGAAATGACGGAATCTTACTCTCTGAGCTATGA

FIG. 12B

Stevia rebaudiana KAHel nucleotide (SEQ ID NO: 165)

ATGGAAGCCTCTTACCTATACATTTCTATTTTGGCTTTTACTGGCATCATACCTGTTCACCACTCAACTTAGAAGGAAGAGCGCTAATCT
ACCACCAACCCGTGTTTCCATCAATACCAATCATTCGACACTTATCTACTCAAAAAGCCCTCTTTATAGAACTTTAGCAAAAATTTGCCG
CTAAGTAGCGCAACCAATACTSCAATTACAACCTCGGCTACAGACCTGTTCCTCGTGATTTTCCTCACCATCAGCAGCAGAGAGTGTCTTACC
AATAACGATGTAACTTTCGCAAAATAGACCTAAGACATTTGTTGGCAAAATAGTGGTGGAAACATCCCTTGGCAGTTTATCTCTACGGCGA
TCAATGGCGTAATCTAAGGAGAGTAGCTTCTATCGAAATCCTATCAGTTTCATAGGTTGAACGAAATTTCAATGATATCAGAGTGGATGAGA
ACAGATTGTTAATTAGAAAACCTTAGAAGTTCACTTCTCTCTGTTACTCTTATAACAGTCTTTTATGCTCTAACATTTGAACGTCATTATG
AGAAATGATCTCTGGCAAAAGATATTTTCGACAGTGGGATAGAGAAATTTGGAGGAGGAAGGTAGAGATTTTCGAGAAATCTTTAGACGAAAC
GTTGCTTCTAGCCGGTGTCTAATGTTGGGCACTACTTACCAATATTTGAACCTGTTGGGAGTTAAAGTCTCTTGAAAGAAATTTGATCG
CTTTCAGAAAAGAGAGATGACATTTTCCAGGGTTTGATTTGAACAGGTTAGAAAATCTCGTGTGCTTAAAGTAGGCAAAAGTAGAAAA
ACGATGATCGAACTCTTATTTATCTTTGCAAGAGTCAGAACCTGAGTACTATACAGATGCTATGATAAGATCTTTTGTCTTAGGCTGTGCT
GGCTCCAGGTAGTGATACTTCAGCCGGCACTATGGAATGGGCCATGAGCTTACTGGTCAATCACCCACATGTATTTGAAGAAAGCTCAAG
CTGAAATCGATAGAGTTATCGGTAATAACAGATTGATTGACGAGTCAGACATTTGGAATAATCCCTTACATCGGGTGTATTATCAATGA
ACTCTAAGACTCTATCCAGCAGGGCCATTGTTGTTCCCATGAAAGTTCTGCCGACTGCGTTATTTCCGGTTACATATACCTAGAGG
TACAAATGTTAATCGTAACCAATGCGCGATTTCATCACCATCTTAAAGCTTGGGATGATCCTGAAACCTTTAAACCTGAAAGATTTCAG
GATTAGAAGCACTAGAGATGGTTTCAAACTTATGCCATTTCGGTCTCTGGGAGAGAGGATGTCCAGGTGAAGGTTTGGCAATAAGGCTG
TTAGGGATGACACTAGGCTCAGTGATCCAAATGTTTTCGATTGGGAGAGAGTAGAGATGAGAAGTGTGACATGACAGAGGTTTGGGTGT
CACACTTCCTAAGCCCGTTCCATTAGTTGCCAAATGTAAGCCACGTTCCGAAATGACATACTCTCTATCCGAACTTTAA

FIG. 12C

Stevia rebaudiana KAHel polypeptide (SEQ ID NO:164)

MEASYLYISILLLASYLEFTTQLRRKKSANLPPTVFPSIPTIGHLYLLKKPLRYRTLAKIAA
KYGPILQLQLGYRRVLVISSPSAAEECFNNNDVIEANRPKLTLEKIVGGTSLGSLSYGDQ
WRNLRVASIEILSVHRINEFHDIRVDENPLIRKLSSSSPVTLLITVFYALTINVMRM
ISGKRYFDSCDELEEGKRFREILDETLLACASNVDYLPILNLGVKSLEKKLLALQ
KKRDDFFOGLIEQVRNSRGAKVGNCRKTMIELLSLQSESEPEYYTDAMIRSFVLGLLAAG
SDTSAGTMEWAMSLLVNHPHVLKKAQAEIDRVIGNNRLIDESIGNIPYIGCIINETLRL
YPAGPLLPFHRESSADCVISGYNIPRGTMILVNQWAIHHDPKVWDDPETFKPERFQGLEGT
RDGFKLMPFGSGRRRCPCGGLAIRLLGMTLGSVIQCFDWERVGDENVDMTEGLGVTLPKA
VPLVANCKPRSEMTNLLSEL

FIG. 13A

S. cerevisiae CPR polypeptide encoded by NCP1 (SEQ ID NO: 166)
 MPEGIDNTDFTVLAVLLVYVKKNSIKELIMSDGDI TAVSSGNRDIAQVVTENNKNYLVLYASQTGTAEDYAKKFSKELVAKFNL
 NVMCADVENYDFESLNDVPVIVSIFISTYCEGDFPDGAVNFEDFICNAEAGALSNLRYNMFCLGNSTYEFFNGAAKAEKHL5AAGAIR
 LKGLCEADDCAGTTDEDYMAWKDSILEVLKDELHLDEQAEKFTSQOYTVLNEITDSMSLGEFSAHYLP5HQLNRRNADG1QLGPTDLSQ
 PYIAPIVKSRLEFSSNDRNCIHSEFDLSGSNIKYSTGDHLAVWPSNPLEKVEQFLSIFNLDRETIFDLKPLDPTVKVPFPTPTTIGAAI
 KHYLEITGPVSRQLFESSLIQFAPNADVKEKLTLLSKDNQFAVEITSKYFENIADALKYLSDCAKWDVTPMQFLVESVPQMTPRYYSISS
 SSLSEKQTVHVT5IVENFENPELFDAPPVVGVTNNLLRN1QLAQNNVNIAETNLPVHYDLNGCPKRLFANYKLPVHVRRSNERLPSNPST
 PVIMIGPGTGVAPFRGFIRERVAFLESQKKGNNVSLGKHILFYGSRNTDDFLYQDEWPEYAKKLDGSEFEMVVAHSRLPNTKKVYVQDK
 LKDYEDQVFEMINNGAFIYVCGDAKGMAGVSTALVGILSRGKSITTDEATELIKMLKTSGRYQEDVW

Arabidopsis thaliana CPR polypeptide encoded by ATR1 (SEQ ID NO: 148)

1 mtsalyaasl fklqlksimgt dslsddvvlv iattslalva gfvvllwkt tadrsgelkp
 61 lnipkslmak dedddldlgs gktrvsiffg tqtgtaegfa kalseeikar yekaavkvid
 121 lddyaaddddq yeeklkketl affcavatyg dgeptdnaarf ykwfteener diklqqalayg
 181 vfalgnrqye hfnkigivld eelckkgakr lievglgddd qsieddfnaw keslwseldk
 241 llkdeddkev atpytavipe yrvvthdprf ttqksmesnv angnttidih hpcrvdvavvq
 301 kelhthesdr scihlefdis rtqityetgd hvgvyaenhv eiveeagkl1 ghslldlvfsi
 361 hadkedgspl esavppfpq pctlgtglar yadllnpprk salvalaaya tepseaeclk
 421 hltspdgkde ysqwivasqr sllevmaafp sakpplgvff aaiaprlqpr ysyssssprl
 481 apsrivhvtsa lvygptptgr inhkvvcstwm knavpaeksh ecsgaplfir asnfklpsnp
 541 stpivmvpgg tglapfrgfl qermalkedg eelgssllff gcrnrqndfi yedelnfnvd
 601 qgviselima fsregaqkey vqhkmekaa qvwdlikeeg ylyvcgdakg nardvhrtlh
 661 tivqegegvs sseaeaiykk lqtegrylrd vw

A. thaliana CPR polypeptide encoded by ATR2 (SEQ ID NO: 168)

1 msssssssts midlmaalik gepvivsdpa nasayesvaa elssmlienr qfamivttsi
 61 avliqciavl vwrirsqsgns krveplkplv ikpreeeidd grkkvtlffg tqtgtaegfa

FIG. 13A (cont'd)

121 kalgeear yektrfkivd lddyaaddde yeekkkedv afflatygd geptdnaarf
161 ykwftegndr gewlknlkyy vfglgnrqye hfnkvakvvd dilveggagr lvqvglgddd
241 qcieddftaw realwpeidt ilreegdtav atpytaavle yrvsihdsed akfnditlan
301 gnqytfvdaq hpykanvavk relhtpesdr scihlefdia gqiltmklgd hvgtlcdnls
361 etvdealrll dmspdtyfsl haekedytpi ssslpppfpp cnlrtaltry aclesspkks
421 alvalaahas dpteaeerlkh laspagkdey shwvvesqrs lilevmaefps akpplgvffa
481 gvapriqprf ysisspskia etrihvtcal vyekmptgri hkqvcstwnk navpyeksek
541 lflgrplfvr qsnfkipsds kvplimlgpg tglapfryfl qerlaivesg velgpevlff
601 qcinrrmdfi yeeelqrive sgalaelsva fsregptkey vqhkmdkas diwnmisqga
661 ylyvcgdakg mardvhrslh tiaqeggsnd stkaegfvkn lqtsgrylrd vw

Stevia rebaudiana CPR7 (SEQ ID NO: 169)
MQSEVVEASTIDIMTAVLKDFVIDTANASDNGDSKMPALAMFEIRDLLILITTSVAVLVGCFVVLVWKRSSGKKSKELEPPKIIVP
KRRLQEVDEGKKKVTIFFCTQTGTAEQFAKALFEEAKARYEKAFFKVIDLDDYAADLDEYAEKIKKETYAFFFLATYGGDGEPTDNAAK
FYKWFTEGDEKGVWLQKLQYGVFGLGNRQYEHFNKIGIVVDDGLTECCAKRIVPVGLGDDQSIEDDFSANKELVWPELDLLRDEDDK
AAATFYTAAIPEYRVVFHDNPDADFSDHTQTNCHAVHDAQHPCRSNVAVKKELHTPESDRSCTHLEFDISHTCLSYETGDHVGVCENL
IEVVEEAGKLLGLSTDTYFSLHIDNEDGSEFLGCFSLQPFPPFCTLRKALTNYADELLSSPKKSTLLALAAHASDFTADRLRFLASREGK
DEYAEWVVANQRSLLLEVMEAFPSARPPLCGVFFAAVAPRLQPRYYSISSSPKMEPNRIHVTCALVYEKTPAGRIHKGICSTWMKNAVPLT
ESQDCSWAPIFVRTSNRFLPIDPKVPVIMIGPGTCLAPFRGLQERLALKESGTELGSSILFFGCRNRKVDYIYENELNNFVENGALSE
LDVAFSRDGETKEYVQHMKMTQKASEIWNMLSEGAYLYVCGDAKGMKDVHRTLHTIVQEQGLDSSKAELYVKNLQMSGRYL RDVW

Stevia rebaudiana CPR8 (SEQ ID NO: 170)
MQNSVKISPLDLVTALFSCKVLDTSNASESGESAMLP TIAMIMENRELLMILTTSVAVLIGCVVVLVWRRSSTRKSALEPPVIVPKR
VQEEEVDDGKKKVTVFFGTQTGTAEQFAKALVEEAKARYEKAFFKVIDLDDYAADDDEYEEKLKKESLAFFFLATYGGDGEPTDNAARFY
KWTEGDAKGEWLKLLQYGVFGLGNRQYEHFNKIAKVVDGLVEQGAKRIVPVGLGDDQCIEDDFTAWKELVWPELDQLLRDEDDTV
ATPYTAAVAEYRVVFHEKPDALSEDYSYTNGHAVHDAQHPCRSNVAVKKELHSPESDRSCTHLEFDISNTGLSYETGDHVGVCENLSE
VVNDAERLVCLPDDTYSSIHSDSDGSLGCASLPFPFPFCTLRKALTCTCYADVLSPPKKSALLALAAHATDPSEADRLKFLASPAKDE
YSQWIVASQRSLLLEVMEAFPSAKPSLGVFFASVAPRLQPRYYSISSSPKMAPDRIHVTCALVYEKTPAGRIHKGVCSTWMKNAVEMTES
QDCSWAPIYVRTSNRFLPSDPKVPVIMIGGTGLAPFRGFLQERLALKEAGTDGLSLIFFGCRNRKVDYIYENELNNFVETGALS ELI
VAFSEEGPTKEYVQHKNMSEKASDIWNLLSEGAYLYVCCDAKGMKDVHRTLHTIVQEQGLDSSKAELYVKNLQMSGRYL RDVW

FIG. 13B

>ATR1 codon optimized by DNA2.0. (SEQ ID NO: 171)
ATGACTTCTGCACCTTATGCCCTCCGATCTTTTCAACAACAAATGAAAAGTATCATCGGAACGGATTCTTTGTCCGATGATGTTGTATTAGT
TATTGCTACAACTTCTCTGCGACTGGTTGCTGCTTATTTGTGAAAAGACACCGGAGATCGTTCCGGCGAGCTAAAGC
CACTAATGATCCCTAAGTCTCTGATGGCGAAAGATGAGATGATCACTTAGATCTAGGTTCTGGA AAAACGAGAGTCTCTATCTTTCTTC
GGCACACAAACCGGAACAGCCGAAGGATTCGCTAAAGCACTTTCAGAAAGAGATCAAAAGCAAGATACGAAAAGGGCTGTAAAAGTAAT
CGATTTGGATGATTACGCTCCGATGATGACCAATATGAGGAAAAGTTGAAAAGCAATTCGCTTTCTTTTGTGTAGCCACGTAATG
GTGATGGTGAACCAACCGATAAACGGCGAAGATTCTCAAGTGGTTTACTGAAGAGAACGAAAGAGATATCAAGTTGCAGCAACTTGTGT
TACGGCGTTTTTGGCTTAGGTAACAGACAAATACGAGCACTTTTAAACAAGATAGGTATTGCTCTTAGATGAAGAGTTATGCAAAAAGGGTGC
GAAGAGATTGATTGAAGTCGGTTTAGGAGATGATCATCAATCTATCCAGGATGACTTTTAATGCATGGAAAGGAATCTTTTGTGGTCTGAAT
TAGATAAGTTACTTTAAGGACGAAGATGATAAATCCGTTGCCACTCCATACACAGCCGTCATTCAGAAATATAGAGTAGTTACTCATGAT
CCAAGATTCAACAACACAGAAATCAATGGAAAGTAATGTGGCTAATGGTAATACTAACCATCGATATTCATCATCCATGTAGAGTAGACGT
TGCAGTTCAAAAGGAATTGCACACTCATGAATCAGACAGATCTTGCATATACATCTTGAATTTGATATATCACCCTACTGGTATCACTTACG
AAACAGGTGATCACGTGGGTGTCTACGCTGAAAACCATGTTGAAAATTGTAGAGGAAGCTGGAAAGTTGTTGGGCCATAGTTTAGATCTT
GTTTTCTCAATTTCATGCCGATAAAGAGGATGGCTCAOCCACTAGAAAAGTGCAGTGCCTCCACCATTTCCAGGACCATGCACTAGGTAC
CGGTTTAGCTCGTTACGGGGATCTGTTAAAATCCTCCACGTAAATCAGCTCTAGTGGCTTGGCTCGTACGCCACAGAACCTTCTTGAGG
CAGAAAACCTGAACATCTAACCTTCAACCAGATGGTAAGGATGAATACCTCACAATGGATAGTAGCTAGTCAACCTTCTTTACTAGAAATT
ATGGCTGCTTTCCCATCCGCTAAACCTCCTTTGGGTGTTTTCTTCCCGGCAATAGCGGCTAGACTGCAACCAAGATACTATTCAAATTC
ATCCTCACCTAGACTGGCACCATCAAGAGTTCAATGTCAATCCGCTTTTAGTGTACGGTCCCACTCTACTGTGTAGATCCCATAAAGGGCG
TTTTGTTCAACAATGGATGAAAACCGGGTTCCAGCAGAGAAAGTCTCAGGAATGTTCTGGTGTCTCCAAATCTTTATCAGAGCTCTCCAACTTC
AAACTGCCCTTCCAAATCCTTCTACTCCTATTGTCTATGGTCCGTGGTACAGGTCTTGTCTCCATTCAGAGGTTTCTTACAAAGAGAGAAAT
GGCCTTAAAGGAGGATGGTGAAGCTTGGGATCTTTCTTTGTTGTTTTTCCGGCTGTAGAAAACAGACAAAATCGATTTTCATCTACCAAGATG
AACTGAATAAATTTGTAGATCAAGGAGTTATTTTCAGAGTTGATAATGGCTTTTCTTAGAGAAAGGTGTCTCAGAAAGGATACGTCCCAAC
AAAATGATGGAAAAGCCGACAAAGTTTGGGACTTAATCAAGAGGAAGGCTATCTATATGTCGTGTGATGCAAAAGGGTATGGCAAG
AGATGTTCAAGAACACTTCATATAGTCCAGGAACAGAGGCGTTAGTTCTCTCAAGCGGAAGCAATTTGTGAAAAGTTACAAA
CAGAGGGAAGATACTTGAGAGATGTGTGGTAA

FIG. 13B (cont'd)

>ATR2 codon optimized by Genscript (SEQ ID NO: 172)
ATGTCCTTCCTTCTCCAGTACCTCTATGATTTGATGGCTGCTTATTAAGGTGAACCCAGTTATCGTCTCGACCCAGC
AAATGCCCTCTGCTTATGAATCAGTTGCTCCAGAAATTCCTTCAATGTTGATCGAAACACAGACAATTCGCCATGATCGTAACATACATCAA
TCGCTGTGTTTTGATCGGTTGTATTGTCATGTTGGTATGGAGAAATCCGGTAGTGGTAAATTCCTAAAGAGTCGAACCTTTGAAACCCATTAA
GTAATTAAAGCCCAAGAGAAGAAATAGATGACCGGTAGAAAGAAAGTTACAAATATTTTTCCGTTACCCAACTGGTACAGCTGAAGCTTTT
TGCAAAAGCCCTTAGGTGAAGAAGCTAAGGCAAGATACGCAAAAGACTAGATTCGAAGATAGTCGAATTTGGATGACTATGCCGCTGATGACG
ATGAATACGAAGAAAAGTTGAAGAAAGAGATGTTGCATTTTTCTTTTGGCAACCTATGGTGACGGGTGAACCAACTGACAATGCAGCC
AGATTCACAAATGGTTTACAGAGGGTAATGATCGTGGTGAATGGTTGAAAACCTTAAAGTACGGTGTGTTTTCGGTTTGGGTAACAGACA
ATACGAACATTTCAACAAAGTTGCCAAAGGTTGTCCGCGATATTTTGGTCCGAACAAAGTGTCTCAAGATTAGTCCAAAGTAGGTTTGGGTG
ACGATGACCAATGATAGAAAGATGACCTTACTGCTCGAGAGAAGCTTTGTGCGCTGAATTTAGACACAATCTTGAGAGAGAAGAAAGGTGAC
ACCGCCGTTGCTACCCCATATACTGCTGCAGTATTAAGATACAGAGTTTCCATCCATGATAGTGAAGACGCCAAAGTTTAAATGATATCAC
TTTGGCCCAATGCTAAACGGTTATACAGTTTTCGATGCCACAACACCTTACAAAGCTAACGTTGCAGTCAAGAGAGAAATTACATACACCCAG
AATCCGACAGAAAGTTGTATAOACTTGGAAATTTGATATCGCTGGTTCGGTTTAAACCATGAAGTTGGGTGACCATGTAGGTGTTTTTATGC
GACAAATTTGTCGAAACCTGTTGATGAAGCAATTGAGATTGTTGGATATGTCCTGACACTTATTTTGTGACGCTTGAAAGAAAGAAAGA
TGGTACACCAATTTCCAGTTCTTTACCAACCTCCATTCCTCCATCTAACCTTAAGAAACAGCCCTTGACCACAGATACGCTTGCTTGTATCAT
CCCCTAAAAAGTCCGCTTGGTTGCTTTAGCCCTCATGCTAGTCACTCTAAGAGCAGAAAGATTGAAACACTTAGCATCTCCAGCC
GGTAAAGATGAATATTCAAAGTGGGTAGTTGAATCTCAAGATCAATGTTAGAAAGTTATGGCAGAAATTTTCCATCTGCCAAGCCTCCCAT
AGGTGCTCTTCTTTGCTGGTGTAGCACTAGATTGCCAACCAAGATTCTACTCAATCAGTTCTTTCACCTAAGATCCCTGAAACTAGAAATTC
ATGTTACATGTGCATTAGTCTACGAAAGAGATGCCAACCCGGTAGAAATTCACAAGGGGTGATGCTCTACTTGGATGAAAAAATGCTGTCTCT
TACGAAAAATCAGAAAAGTTGTTCTTAGGTAGACCAATCTCTCGTAAGACAAATCAAACTTCAAGTTGCCCTTCTGATTCAAAGGTTCCCAAT
AATCATGATAGGTCCTGGTACAGGTTTAGCCCATTCACAGGTTTCTTTGCCAAGAAAGATTGGCTTTAGTTGAATCTGGGTCTCCAAATTAG
GTCCTTCAGTTTGTTCCTTTGGTTGTAAGAACAGAGAAGAAATGGATTTCATCTATGAAGAAAGAAATTCGCAAGATTCGTCGAAATCTGGTCCA
TTGGCCGAATATCTGTAGCTTTTCAAGAGAGAGTCCCAACTAAGGAATACGTTCAACATAGATGATGATAAGCATCCGACATATG
GAACATGATCAGTCAAGGTGCTTATTTGCTACGTTTGGGTGACCGCAAGGGTATGCCAGAGATGTCCATAGATCTTTGCCACACAATTC
CTCAAGAACAAAGGTTCCATGTGATAGTACCAAAAGCTGAAGGTTTTTCGTAAAGAACTTACAAAATTCGGGTAGATACCTTGAGAGATGTCTGG
TGA

FIG. 13B (cont'd)

Stevia rebaudiana CER7 (SEQ ID NO: 173; cloned from *S. rebaudiana* cDNA)

ATGCAATCGGAATCCGTTGAAGCATCGACGATTGATGACTGCTGCTTTTGAAGGACACACAGTGAATACAGCGAAGCATCTGA
TAACCGAGACTCAAGATCGCCCGCGTTGGCGATGATGTTGGAATTCGTGATCTGTTGCTGATTTTGAAGTACGTCAGTTGCTGTTT
TGGTCGGATGTTTTCGTTGTTTGGTGTGGAAAGATCGTCCGGGAAGAAATTCGCGCAAGGAATTCGAGCCGCGGAAGATCGTTGTGCGG
AAGAGCGCGCTGGACGACGAGGCTTGATGATGTTGAAGAAAGGTTACGATTTTCTTCGGAACACAAACTGGAACGGCTGAAGGTTTCGC
TAAGGCACCTTTTCGAAAGACGGAAGCGCGATATGAAGAGCGAGCGTTTAAAGTGATTTGGAATGATTTGCTGCTGATTTGGATG
AGTATGCAGAGAAGCTGAAGAGGAACATATGCTTCTTCTTGGCTACATATGGAGATGGTGAGCCAACTGATATAATGCTGCCAAA
TTTTATATAATGGTTTACTGAGGAGACGAGAAAGCGGTTGGCTTCAAAAACCTTCAATATGGAGTATTTGGTCTTGGCAACACACAATA
TGAACATTTCAACAAAGATTGGAATAGTGGTTGATGATGCTCTCACCGCAGCAGGGTGCAAAACCCATTTGTTCCGTTGGTCTTTGGAGACG
ACGATCAATCAATTGAAGACGATTTTTCGGCATGGAAAGAGTTAGTGTGGCCGAAATGGAAATCTATTTGCTTGGCGATGAAGATGACAAA
GCTGCTGCAACTCCTTACACAGCTGCAATCCCTGAATACCGCGTCGTATTTCAAGCAAAACCCGATGCGTTTCTGAAGATCATACTCA
AACCAATGGTCACTGCTGTTCAATGCTCAACATCCATGCAATCCAAATGTGGCTGTTAAAAAAGAGCTTCACTACTCCTGAATCCGATC
GTTCAATGCACACATCTTGAAATTTTGACATTTCTCACACTGGATTATCTTATGAAACTGGGGATCATGTTGGTGTATATACTGTGAAAACCTA
ATTGAAGTAGTGAAGAGCTGGGAAATTTGTTAGGATTATCAACAGATACTTATTTCTCGTTACAPATTGATACCGAAGATGGTTCAAC
ACTTGGTGGACCTTCATTACAACTCCCTTTTCTCTCTTGTACTTTAAGAAAAGCAATGACTAATTAATGCAGATCTGTTAAGCTCTCCCCA
AAAAGTCAACTTTGCTTGCCTAGCTGCTCATGCTTCCGATCCCACTGAAGCTGATCGTTTAAGATTTTCTTGCATCTCGCGGAGGGCAAG
GATGAATATGCTGAATGGGTTGTTCGCAAAACCAAGAACTCTTCTTGAAGTCAATGGAAGCTTTCCCGTCAAGCTAGACCCGCCACTTGGTGT
TTTCTPTGCAGCGGTTGCACCGGPTTACAGCCTCGTTACTACTC PATTTCTTCTCTCCCAAGATGGAACCAACAGGATTCATGTTA
CTTGGCGGTTGGTTATGA AAAA ACTCCCGCAGGTCGTATCCAAAGGAATCTGCTCAACCTGGATGAAGAACGCTGTACCTTTGACC
GAAAGTCAAGATTGCCAGTTGGGCACCGATTTTGTGTTAGAACATCAAACTTCAGACTTCCAAATTGACCCGAAAGTCCCGGTTATCATGAT
TGCTCCGGAACCGGTTGCTCCATTTAGGGGTTTTCCTCAAGAAAGATTTGGCTCTTAAAGAAATCCGGAACCGAACTCGGGTCAATCTA
TTTTATTCTTCGGTTGTAGAAAACCCCAAGTSGATPACATATATGAGAAATGAATCAACAACITTTGTTGAAAATGCTGGCTTTCTGAG
CTTGATGTTGCTTTCCTCCCGATGCGCCGACGAAAGAAATACGTGCCAACAATAAATGACCCAAAAGGCTTCTGAATAATGGAATATGCT
TTCTGAGGAGGACATATTTATATGATGTGCTGATGCTAAAGGCAATGAGATGTACACCGTACACTTCACACACCATTTGTGCAAGAAC
AGGGAAGTTTGGACTCGTCTAAAGCGGAGTTGTATGTGAAGAAATCTACAAATGTCAAGGAATACCTCGGTGATGTTTGGTAA

FIG. 13B (cont'ed)

Stevia rebaudiana CERS (SEQ ID NO: 174; CDS cloned from *S. rebaudiana* cDNA)

ATGCAATCTAACTCCGTGAAGATTTCCCGCTTGATCTGGTAACCTGCTAACTCGCTGTTTAGCGGCAAGGTTTTCGACACATCGAAACGATCCGGA
ATCGGAGAAATCTGCTATGCTGCGCACTATAGCGATGATTAATGAGAAATCGTCAGCTGTTGATGATGATCTACAAACGTCGGTGTCTGTAT
TGAATCGGATCGGTTGTCGTTTGGTGTGGCGGAGATCCCTACGAAAGAAATCGGCGTTGGAGCCACCGGTGATTGTGGTTCCGAAAGAGA
GTGCAAGAGAGGAAAGTTGATGATGGTAAGAAAGAAAGTTACGGTTTTCCTTCGGCACCCCAAACCTGGAACACGCTGAAGGCTTCGCTAAGGC
ACTTGTTCAGGAAGCTAAAGCTCGATATGAAAGGCTGCTTTTAAAGTAATTGATTTGGATGATTAATGCTGCTGATGACGATGAGTATG
AGGAGAAACTAAAGAAAGAAATCTTTTGGCTTTTTTCTTTTGGCTACGTATGGAGATGGTGAAGCCAAACAGATAATGCTGCCAGATTTTAT
AAATGGTTTACTGAGGGAGATGCGAAAGGAGAAATGGCTTAATAAGCTTCAATATGAGTATTTTGGTTTGGTAAACAGACAAATATGAACA
TTTTAACAAAGATCGCAAAAGTGGTTGATGATGGTCTTTGTAGAACAGGGTGCAAAGCGTCTTGTTCCTGTTGGACTTGGAGATGATGATC
AATGTAATTGAAGATGACTTCACCGCATGGAAAGAGTTAGTATGGCCGAGTTGGATCAATTAATCTTCTGATGAGGATGACACAACCTGTT
GCTACTCCATACACAGCTGCTGTTGCCAGAAATATCCCGTTGTTTTTCATGAAAAACAGACCGCTTTCTGAAGATTATAGTTATACAAA
TGGCCATGCTGTTCATGATGCTCAACATCCATGCAGATCCAACTGGCTGTCAAAAAGGAACCTTCAATAGTCTGAATCTGACCCGGTCTT
GCACTCATCTTTGAATTTGACATCTCGAAACACCGGACTATCATATGAATAACTGGGGACCATGTTGGAGTTTACTGTGAAAAACCTTGAGTGAA
GTTGTGAATGATGCTGAAGATTAGTAGGATTAACCAACAGACACTTACTCCTCCATCCACACTGATAGTGAAGACGGTCCGCACTTGG
CGAGCCCTCATTTCCCGCTCCTTTCCCGCCATGCCACTTTAAGGAAGCATTGACGCTGTTATGCTGATGTTTTCAGTTCTCCCAAGAAAT
CGGCTTTGCTTGCATAGCTGCTCATGTCACCGCATCCAGTGAAGCTGATAGATTGAAATTTCTTTGCATCCCGCCGGAAGGATGAA
TATTCCTCAATGGATAGTTGCAAGCCAAAGAAAGTCTCCTTGAAGTCAATGGAAGCATTCCTCGTCAGCTPAAGCCCTTCACTTGGTGTTCCTT
TGCAATCTGTTGCCCGCTTACAACCAAGATACTACTCTATTTCTTCCTCACCCAGATGGCACCGGATAGGATTCATGTTACATGTG
CATTAGTCTATGAGAAACACCTGCAAGCCGCTATCCACAAAGGAGTTTGTTCAACTTGGATGAAGAACCGAGTGCCTATGACCCGAGAT
CAAGATTGCAGTTGGGCCCAATATACGTCCGAACATCCAAATTTTCAGACTACCATCTGACCCCTAAGGTCCCGGTTATCATGATTGGACC
TGGCACCTGGTTTGGCTCCTTTTAGAGGTTTTCCTTCAAGAGCGGTTAGCTTTAAGGAAGCCGGAACCTGACCTCGGTTTATCCATTTTAT
TCCTTCGGATGTAGGAATCCCAAGTGGATTTTCATATATATCAAAACCGAGCTTAAACAACTTTTGTGGAGACTGGTGTCTCTTTCTGAGCTTATT
GTTGCTTTCTCCCGTGAAGCCCGGACTAAGGAATATGTGCCAACACAAAGATGAGTGAAGGCTTCGGATATCTGGAACCTTGTCTCTGA
AGGAGCATATTTATACGTATGTGGTGTATGCCAAAGGATGCCAAAGATGTACATCGAAACCTCCACAAATTTGTGCAAGAACAGGAT
CTCTTGACTCGTCAAGGCAGAACTCTAAGTGAAGAAATCTACAAATGTGAGGAAGATACCTCCGTGACGTTTGGTAA

FIG. 14A

Zea mays CDPS nucleotide (SEQ ID NO: 157)

<u>atggttttgt</u>	<u>cttcttcttg</u>	<u>tactacagta</u>	<u>ccacacttat</u>	<u>cttcattagc</u>	<u>tgctcgtgcaa</u>	60
<u>cttggtcctt</u>	<u>ggagcagtag</u>	<u>gattaaaaag</u>	<u>aaaaccgata</u>	<u>ctgttgcaagt</u>	<u>accagccgct</u>	120
<u>gcaggaaggt</u>	<u>ggagaagggc</u>	<u>cttggctaga</u>	gcacagcaca	catcagaatc	cgcagctgtc	180
gcaaagggca	gcagtttgac	ccctatagtg	agaactgacg	ctgagtcaag	gagaacaaga	240
tggccaaccg	atgacgatga	cgccgaacct	ttagtggatg	agatcagggc	aatgcttact	300
tccatgtctg	atggtgacat	ttccgtgagc	gcatacgata	cagcctgggt	cggattgggt	360
ccaagattag	acggcgtga	aggtcctcaa	tttccagcag	ctgtgagatg	gataagaaat	420
aaccagttgc	ctgacggaag	ttggggcgat	gccgcattat	tctctgccta	tgacaggcct	480
atcaataccc	ttgcctgcgt	tgtaactttg	acaagggtgg	ccctagaacc	agagatgaga	540
ggtagaggac	tatctttttt	gggtaggaac	atgtggaaat	tagcaactga	agatgaagag	600
tcaatgccta	ttggcttcga	attagcattt	ccatctttga	tagagcttgc	taagagccta	660
ggtgtccatg	acttccctta	tgatcaccag	gccctacaag	gaatctactc	ttcaagagag	720
atcaaaatga	agaggattcc	aaaagaagtg	atgcataccg	ttccaacatc	aatattgcac	780
agtttgagg	gtatgcctgg	cctagattgg	gctaaactac	ttaaactaca	gagcagcgac	840
ggaagttttt	tgttctcacc	agctgccact	gcatatgctt	taatgaatac	cggagatgac	900
aggtgtttta	gctacatcga	tagaacagta	aagaaattca	acggcggcgt	ccctaattgt	960
tatccagtgg	atctatttga	acatatattg	gccgttgata	gacttgaaag	attaggaatc	1020
tccaggtact	tccaaaagga	gatcgaacaa	tgcatggatt	atgtaaacag	gcattggact	1080
gaggacggta	tttgttgggc	aaggaaactc	gatgtcaaag	aggtggacga	cacagctatg	1140
gccttttagac	ttcttaggtt	gcacggctac	agcgtcagtc	ctgatgtgtt	taaaaacttc	1200
gaaaaggacg	gtgaattttt	cgcatttgtc	ggacagtcta	atcaagctgt	taccggtagt	1260
tacaacttaa	acagagcaag	ccagatatcc	ttcccaggcg	aggatgtgct	tcatagagct	1320
ggtgccttct	catatgagtt	cttgaggaga	aaagaagcag	agggagcttt	gagggacaag	1380
tggatcattt	ctaaagatct	acctggtgaa	gttgtgtata	ctttggattt	tccatggtac	1440
ggcaacttac	ctagagtcga	ggccagagac	tacctagagc	aatacggagg	tggatgagac	1500
gtttggattg	gcaagacatt	gtataggatg	ccacttgtaa	acaatgatgt	atatttggaa	1560
ttggcaagaa	tggatttcaa	ccactgccag	gctttgcata	agtttagagt	gcaaggacta	1620
aaaagatgg	atactgaaaa	taggttgatg	gactttgggt	tcgccaaga	agatgccctt	1680
agagcttatt	ttcttgacg	cgcactctgt	tacgagcctt	gtagagctgc	cgagaggcct	1740
gcatgggcta	gagccgcaat	actagctaac	gccgtgagca	cccacttaag	aaatagccca	1800
tcattcagag	aaagggttag	gcattctctt	aggtgtagac	ctagtgaaga	gacagatggc	1860
tcctggttta	actcctcaag	tggctctgat	gcagtttttag	taaaggctgt	cttaagactt	1920
actgattcat	tagccaggga	agcacagcca	atccatggag	gtgaccacaga	agatattata	1980
cacaagttgt	taagatctgc	ttgggcccag	tgggttaggg	aaaaggcaga	cgctgccgat	2040
agcgtgtgca	atggtagttc	tgcaagtagaa	caagagggat	caagaatggg	ccatgataaa	2100
cagacctgtc	tattattggc	tagaatgatc	gaaatttctg	ccggtagggc	agctggtgaa	2160
gcagccagtg	aggacggcga	tagaagaata	attcaattaa	caggctccat	ctgcgacagt	2220
cttaagcaaa	aaatgctagt	ttcacaggac	cctgaaaaaa	atgaagagat	gatgtctcac	2280
gtggatgacg	aattgaagtt	gaggattaga	gagttcgttc	aatatttgct	tagactaggt	2340
gaaaaaaaga	ctggatctag	cgaaaaccagg	caaacatttt	taagtatagt	gaaatcatgt	2400
tactatgctg	ctcattgccc	acctcatgtc	gttgatagac	acattagtag	agtgattttc	2460
gagccagtaa	gtgccgcaaa	gtaaccgcgg				2490

FIG. 14B

Zea mays CDPS polypeptide (SEQ ID NO: 158)

Met	Val	Leu	Ser	Ser	Ser	Cys	Thr	Thr	Val	Pro	His	Leu	Ser	Ser	Leu	Ala	Val
Val	Gln	Leu	Gly	Pro	Trp	Ser	Ser	Arg	Ile	Lys	Lys	Lys	Thr	Asp	Thr	Val	Ala
Val	Pro	Ala	Ala	Ala	Gly	Arg	Trp	Arg	Arg	Ala	Leu	Ala	Arg	Ala	Gln	His	Thr
Ser	Glu	Ser	Ala	Ala	Val	Ala	Lys	Gly	Ser	Ser	Leu	Thr	Pro	Ile	Val	Arg	Thr
Asp	Ala	Glu	Ser	Arg	Arg	Thr	Arg	Trp	Pro	Thr	Asp	Asp	Asp	Asp	Ala	Glu	Pro
Leu	Val	Asp	Glu	Ile	Arg	Ala	Met	Leu	Thr	Ser	Met	Ser	Asp	Gly	Asp	Ile	Ser
Val	Ser	Ala	Tyr	Asp	Thr	Ala	Trp	Val	Gly	Leu	Val	Pro	Arg	Leu	Asp	Gly	Gly
Glu	Gly	Pro	Gln	Phe	Pro	Ala	Ala	Val	Arg	Trp	Ile	Arg	Asn	Asn	Gln	Leu	Pro
Asp	Gly	Ser	Trp	Gly	Asp	Ala	Ala	Leu	Phe	Ser	Ala	Tyr	Asp	Arg	Leu	Ile	Asn
Thr	Leu	Ala	Cys	Val	Val	Thr	Leu	Thr	Arg	Trp	Ser	Leu	Glu	Pro	Glu	Met	Arg
Gly	Arg	Gly	Leu	Ser	Phe	Leu	Gly	Arg	Asn	Met	Trp	Lys	Leu	Ala	Thr	Glu	Asp
Glu	Glu	Ser	Met	Pro	Ile	Gly	Phe	Glu	Leu	Ala	Phe	Pro	Ser	Leu	Ile	Glu	Leu
Ala	Lys	Ser	Leu	Gly	Val	His	Asp	Phe	Pro	Tyr	Asp	His	Gln	Ala	Leu	Gln	Gly
Ile	Tyr	Ser	Ser	Arg	Glu	Ile	Lys	Met	Lys	Arg	Ile	Pro	Lys	Glu	Val	Met	His
Thr	Val	Pro	Thr	Ser	Ile	Leu	His	Ser	Leu	Glu	Gly	Met	Pro	Gly	Leu	Asp	Trp
Ala	Lys	Leu	Leu	Lys	Leu	Gln	Ser	Ser	Asp	Gly	Ser	Phe	Leu	Phe	Ser	Pro	Ala
Ala	Thr	Ala	Tyr	Ala	Leu	Met	Asn	Thr	Gly	Asp	Asp	Arg	Cys	Phe	Ser	Tyr	Ile
Asp	Arg	Thr	Val	Lys	Lys	Phe	Asn	Gly	Gly	Val	Pro	Asn	Val	Tyr	Pro	Val	Asp
Leu	Phe	Glu	His	Ile	Trp	Ala	Val	Asp	Arg	Leu	Glu	Arg	Leu	Gly	Ile	Ser	Arg
Tyr	Phe	Gln	Lys	Glu	Ile	Glu	Gln	Cys	Met	Asp	Tyr	Val	Asn	Arg	His	Trp	Thr
Glu	Asp	Gly	Ile	Cys	Trp	Ala	Arg	Asn	Ser	Asp	Val	Lys	Glu	Val	Asp	Asp	Thr
Ala	Met	Ala	Phe	Arg	Leu	Leu	Arg	Leu	His	Gly	Tyr	Ser	Val	Ser	Pro	Asp	Val
Phe	Lys	Asn	Phe	Glu	Lys	Asp	Gly	Glu	Phe	Phe	Ala	Phe	Val	Gly	Gln	Ser	Asn
Gln	Ala	Val	Thr	Gly	Met	Tyr	Asn	Leu	Asn	Arg	Ala	Ser	Gln	Ile	Ser	Phe	Pro
Gly	Glu	Asp	Val	Leu	His	Arg	Ala	Gly	Ala	Phe	Ser	Tyr	Glu	Phe	Leu	Arg	Arg
Lys	Glu	Ala	Glu	Gly	Ala	Leu	Arg	Asp	Lys	Trp	Ile	Ile	Ser	Lys	Asp	Leu	Pro
Gly	Glu	Val	Val	Tyr	Thr	Leu	Asp	Phe	Pro	Trp	Tyr	Gly	Asn	Leu	Pro	Arg	Val
Glu	Ala	Arg	Asp	Tyr	Leu	Glu	Gln	Tyr	Gly	Gly	Gly	Asp	Asp	Val	Trp	Ile	Gly
Lys	Thr	Leu	Tyr	Arg	Met	Pro	Leu	Val	Asn	Asn	Asp	Val	Tyr	Leu	Glu	Leu	Ala
Arg	Met	Asp	Phe	Asn	His	Cys	Gln	Ala	Leu	His	Gln	Leu	Glu	Trp	Gln	Gly	Leu
Lys	Arg	Trp	Tyr	Thr	Glu	Asn	Arg	Leu	Met	Asp	Phe	Gly	Val	Ala	Gln	Glu	Asp
Ala	Leu	Arg	Ala	Tyr	Phe	Leu	Ala	Ala	Ala	Ser	Val	Tyr	Glu	Pro	Cys	Arg	Ala
Ala	Glu	Arg	Leu	Ala	Trp	Ala	Arg	Ala	Ala	Ile	Leu	Ala	Asn	Ala	Val	Ser	Thr
His	Leu	Arg	Asn	Ser	Pro	Ser	Phe	Arg	Glu	Arg	Leu	Glu	His	Ser	Leu	Arg	Cys
Arg	Pro	Ser	Glu	Glu	Thr	Asp	Gly	Ser	Trp	Phe	Asn	Ser	Ser	Ser	Gly	Ser	Asp
Ala	Val	Leu	Val	Lys	Ala	Val	Leu	Arg	Leu	Thr	Asp	Ser	Leu	Ala	Arg	Glu	Ala
Gln	Pro	Ile	His	Gly	Gly	Asp	Pro	Glu	Asp	Ile	Ile	His	Lys	Leu	Leu	Arg	Ser
Ala	Trp	Ala	Glu	Trp	Val	Arg	Glu	Lys	Ala	Asp	Ala	Ala	Asp	Ser	Val	Cys	Asn
Gly	Ser	Ser	Ala	Val	Glu	Gln	Glu	Gly	Ser	Arg	Met	Val	His	Asp	Lys	Gln	Thr
Cys	Leu	Leu	Leu	Ala	Arg	Met	Ile	Glu	Ile	Ser	Ala	Gly	Arg	Ala	Ala	Gly	Glu
Ala	Ala	Ser	Glu	Asp	Gly	Asp	Arg	Arg	Ile	Ile	Gln	Leu	Thr	Gly	Ser	Ile	Cys
Asp	Ser	Leu	Lys	Gln	Lys	Met	Leu	Val	Ser	Gln	Asp	Pro	Glu	Lys	Asn	Glu	Glu
Met	Met	Ser	His	Val	Asp	Asp	Glu	Leu	Lys	Leu	Arg	Ile	Arg	Glu	Phe	Val	Gln
Tyr	Leu	Leu	Arg	Leu	Gly	Glu	Lys	Lys	Thr	Gly	Ser	Ser	Glu	Thr	Arg	Gln	Thr
Phe	Leu	Ser	Ile	Val	Lys	Ser	Cys	Tyr	Tyr	Ala	Ala	His	Cys	Pro	Pro	His	Val
Val	Asp	Arg	His	Ile	Ser	Arg	Val	Ile	Phe	Glu	Pro	Val	Ser	Ala	Ala	Lys	

Codon-optimized nucleotide sequence for CDPS-KS (SEQ ID NO:161)

AATGCTCGTGAATAATTGAAAAATGGTACCCCAAAGCACCTCAAGACATGGAAATGATTTTGTCTCTGCTGTAAAGATTTACTAGATCGGACG
 TTTTCAAAAAGTCAATATCTCTACTACGATATATGCTCAACTTCATGTCAGGTTTATGATACAGCTTGCGGTTGCAATGATTCGAAAAACAA
 GAGATATATGTA AAAACAGTGGTGTATTCAGAAATGTTTCATTTACCTCTTAAAAACAAGCCGAGATGGCTCATGGGGTTCATATGGCTT
 ACAACAACAGACAGCGGGTATCCCTAGATACAGCCCTAGCTGTCTGGCAATTATGTGCCACGCACAAAGACCTTTACAAATATTTGGATGT
 AATCTCCAGATGAAAATGGGCTTGAGAAATAGAACACAGGTGCATATCTTGAAACGTCAATPACAGTTTGGAAATGATGTGGAGGACACCCA
 AACATATTTGGGCTCGAGTTTATCATACACGCTTACCTTCCATGCTAGAAAAGGAAATTAGATGTTCCATCTTTTGAATTTCCATGTATAG
 TCCCATCTTAGAGAGAAATCCACGGGGAGAAAATTAGTGCAATTTCCACCTGGAAACAAGTTTACGGCAAGCCATCATTTGTTGCACTCAT
 TGGGAAGCAATTTCTCGGTAAAGCTAGATTTTGTATCGCAATATCATCATCACCTATACCGGCAATATGATGCGCATCTCCATCTTCAACGCGCTG
 CTTATCTTATTTGGGGCTACAAAATGGAATGACGAAGCCGAAGATTACTCTAAGACATGTATATGCTAATGGTGACAGCATGCGAATGGAA
 TGGTATTTCTAGGTACATTTCCAACTACTCATTTTCGAATGTAGCTGGATATATAGCAACGTGTGTTAAAGTTGGCTTTTACTTTGAAGCATAAT
 TTGACGGGATGGCTTAAGAGGTTTATCAACCATCTTACTTTGAGCGCTTCTGTGATGAGAAATGGTGTCATAGGCTTTGGCCCTTAGAACAAG
 CAGATGTAGATACACAGCCAAAGCTCTATTTGGCTTTGTCAATTTGGTAAACCCAGCCAGTGTCACTGATATCATGATTAAGTCTTTTGGAG
 AAGACCAATTTTACCACCTTTTGGTTTCAGAAAAGAGATCCATCATTTGACTTCCAACTCCACCGTCCCTTTTATCTTTACTTAAACAATC
 TAACTGTGTCTCAATACCATCTCAAATCTCAAAAACAACATTATTCACCTTGATAGTGGTGGGGTTCGGATCATGTGTGTCAAGAGACA
 AAAATGGAAATTTGAGTCACCTATATACCAACTATGTTGTGGTTGAAGCCCTCCTACATGAAGTGTCTCCATCTCATTTGACGGTGGTGAATGTCT
 AGTCTGTGTTGATGAATCTTTAAGTGTAGATTGGTCTTAGCATCTTTCAAGCGGTACTTAGAAATAATCTCCACCCAAAGACAACGACGG
 CTTCTTGGAGAGGATACAGAGAACAGACGCTGTTACGCCAATATTTGGCTTTAGTTCAAGCGAGACAATGTATGCTTTTTCACCTCACATGTTTG
 ACAGACTGCCAATCATGTGTGTATCGAGGTTTCTCATGGTTGAAAATCTTGGCTCTTTTTCATCTCAAGACCTTGACTTGGACCTCTA AAAACA
 CTTATGAAGTGGGTTCTGTAGCTGAAGCATATAAACTAGCTGCTTTTAGCAATCTGCTTCCCTGGAGGTTCCCTGCTGCCAACCATTTGGACATA
 TTTCTGTCAAGTCTCGCCGTTCCATCAAGTGATCTTTGAAAAATACATGAGATTGGTGAGAAAAAATGCGGTTATTTCTCTCCACTGGCATGAGT
 TGGGGTCTAATGGCTCTCATCATCGAATCTTCATTTTTCGTACCATTTACTGCAAGGCAAAAGAGTTTGAATAATATACCTCTAGAGATAATATC
 AAAGGTGCACGAGATAAGTACTTGTCTATATATCCCATTCACATGGGCTCGGATGCCAATAATAGGTCTAGAACTTTTGGCAAGTAAACAGATG
 GCTATAAGATATGATGTAACCTTTTCATTTACTCGGCTATCAAAACCCAGACGATACATGGAAGCTGTAGCTGGGCCCCAGTGTTTTGGGGAATCTTT
 CCTTGTGTACATCAAAACAAATTGATAAGGTGATTTGATATAACATGGGTAACTTTGGAGAGGCCAAATGGAAACAGTACACACAGTGGTAAATGGA
 CATCAGCAAGAAATCTCTAAATATAGGTCAAGTCCAGGACACCTTTGACTGTGTTTCACAAAATTCAGTCTTGAATTCACAAAGACGCTCCTTAA
 CTTCTAGCTCATCTCATCAAGATATCTTTGACAGAGAGAGGTTTAGAACATTTCAATGCGCGCTCATATAACACAAAATCCGAAGATAACTCAAGCAT
 TCCAGTAAAGCAAGCCCTATCCGATGGGTTTCTCTCCCTCCGAAACAATCTTACTTTTCAATGGGTGAACCTCAACTGGTGGCTCACATGTCCCT
 TTGCGCCCTATTCAATTTCCCTCATGTCTGCAAAATTTGTTGCAAGGGTAAAGACGCAATTTCCAGCGGAACGCAAAAGTA
 CTTAAATCTCTCTGTATTAGACATGCCAACAAACATGTGTAGAAATGTATAACCACTTTGGCTCTATTGCCACAGACAAACGCTCAGAGAA
 ATGTTTAATAGTATTCATTTTCTCTGAGTTTACTCTCTGTAAACGGAACCTTCTCAAAACCTAGATGAAGGAACGGAAGACTTCTCAAAATC
 GCAACTTACGAACAAGGCTATTGGATACAGCACTAGAGGCTTTGGAAAGACAGACAGATGATGTCGCCGACACAGACTGGATCTTAA
 AGATATACGAAAAGTTGAAAATCTGTTAAGTTATTCTGTGATGTTACGGCACTTATACGATCAGCTCTCATGATTCAAAGATTTTGTCTATCTCT
 CTATGAAGTAA

FIG. 15B

Gibberella fujikuroi CDPS-KS polypeptide (SEQ ID NO:162)
MPGKIENGTPKDLKTGNDDEVSAKSLLDRAFNKSHHSYGYGLCSTSCQVYDPAWVAMIPKTR
DNVKQWLFFECFHYLLKTQAADGSWGSLLPTQTAGILDTASAVLALLCHAQEPQLILDVS
PDEMGLRIEHGVTSLKRQLAVWNVDEDTNHIGVEFIIPALLSMLEKELDVPSFEFPCRSI
LERMHGEKLGHFDELEQVYCKPSSLLHSLEAFLGKLDFDRLSHHLYHGSMMASSPSSTAAYL
IGATKWDDDEAEDYLIRHVMRNGAGHGNGGIGSTFPTTHFECSWIIATLLKVGFTLKQIDGD
GLRGLSTIILEALRDENGVI GFAPRTADVDDTAKALLA SLVNQPVSPD IMIKVFEKNDH
FYTFCGERDPSLTGSLNHLVLLSLLKQSNLSQYHPQILKTTLFTCRWWWGSDHCVDKNLS
HLYPTMLLVEAFTEVLHLIDGGE LSSLFDESFKCKIGLSIFQAVLRIILDTQDNDGSWRGY
REQTCYAILALVQARHVCFFTHMVDRLQSCVDRGFSWLKSCSEHSQDLTWTSKTAYEVGF
VAEAYKLAALQASASLEVPAATIGHSVTS AVPPSSDLEKYMRLVRKKTALFSPLDWGLMASI
IESSFFVPLLQAQRFVEIYPRDNIKVDKYL SIIPFTWVGONNRSRTFASNRLYDMMYL
SLLGYQTDEYMEAVAGPVFGDVSL LHQTIDKVIDNTMGNLARANGTVHSGNGCHQESPNI
GQVEDTLTRFTNSVLN HKDVLN SSSSDQDTERREFFTFMHAHITQIEDNSRFSKQASSDA
FSSPEQSYFQWVNSTGCSHVACAYSFAFSNCLMSANLLQCKDAFPSTQKYLIS SVMRHA
TNMCRMVNDPGSIARDNAERNVNSIHFPEFTLCNGTSQNLDERNERLLKIATYEQGYLDR
ALEALERQSRDDAGDRAGSKDMRKLKIVKLFCDVVTCLYDQLYVINKDLSSSMK

FIG. 16

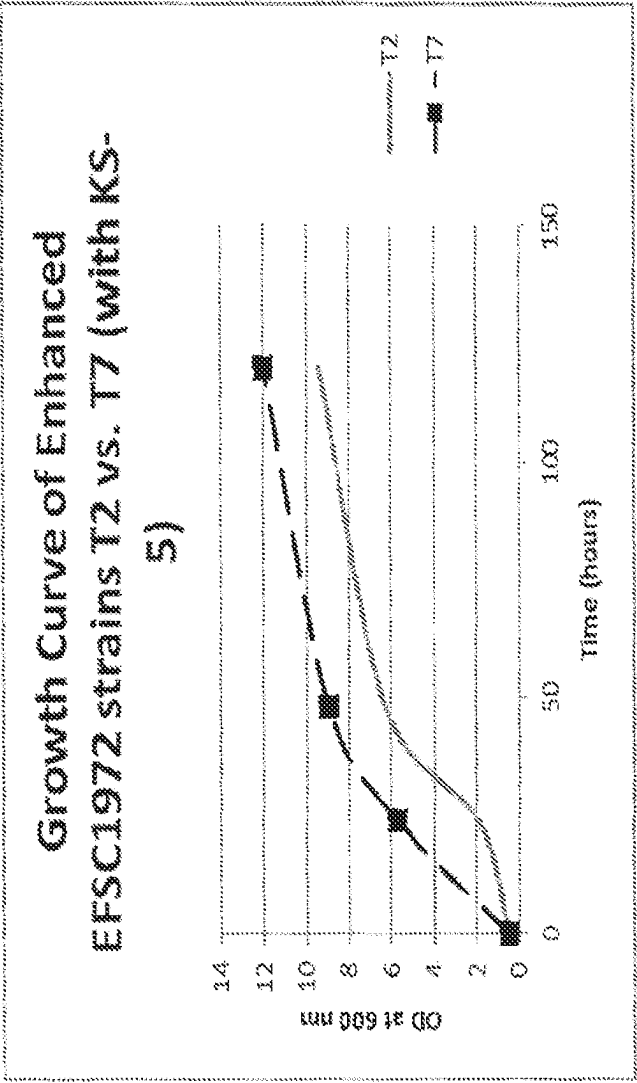


FIG. 17A

Native *A. thaliana* SUS1 nucleotide (SEQ ID NO: 175)

ATGCCAAACCGTGAACGTATGATAAACCGCGCTCCACAGCCAAACGTTGAGCGTTTGAACGAAACCGCTTGTTCGTGAGAGAAACGAAGTCCCT
TGCCCTTGCTTTCCAGGGTTGAAGCCAAAGGTAAAGGTATTTTACAAACAACCAAGATCATTTGCTGAATTCGAAGCTTTTGCCTGAACAAA
CCCCGAAGAAACTTCAAGGTGCTCTTTCTTGACCTTCTCAAAATCCCACTCAGGAAGCAAATTTGTGTGCCCAACCATGGGTGCTCTAGCT
GTGAGGCCAAGGCGTGGTGTGGGAATACCTTACGAGTCAATCTCCATGCTCTTGTGCTGAAGAACTCCAAACCTGCTGAGTTTCTTCA
TTTCAAGGAAGAACTCGTTGATGGAGTTAAGAAATGGTAATTTCACTCTTGAGCTTGATTTGAGGCCATTCAATGCGTCTATCCCTCGTC
CAACACTCCACAAATACATTTGGAAATGGTGTGACTTCTCTTAACCGTCAATTTATCGGCTAAGCTCTTCCATGACAAGGAGAGTTTGCTT
CCATTGCTTAAGTTCCCTTCGCTTTCACAGCCACAGGSCAAAGAACCTGATGTTGAGCGAGAGATTTCAGAACTCAACACTCTGCAACA
CACTTGAGGAAGCAGAGAGTATCTACAGAGGCTTAAGTCCGAAACACTGTATGAAGAGTTTGAGGCCAAAGTTTGAGGAGATTTGGTC
TTGAGAGGGGAATGGGAGACAAATGCAGAGCGGTGCTCCCTTGACATGATACCTCTTCTTTTGGACCTTCTTGAGGCCCTGATCCTTGCAC
CTTGAGACTTTTCTTGGAAGAGTACCAATGGTGTTCAAAGTTGTGATCCTCTCTCCACATGGTTACTTTTGCTCAGGACAAATGTTCTTGG
TTACCCCTGACACTGGTGGACAGGTTGTTTAOATTTCTTGATCAAGTTGCTGCTCTGAGATAGAGATGCTTCAACGTATTAAGCAACAAG
GACTCAACATTAACCAAGGATTTCTATTCTAACTCGACTTCTACCTGATGCGGTAGGAACCTACATGCGGTGAACGCTCTCGAGAGAGTT
TATGATTTCTGAGTACTGTGATATTTCTTCTGTGCTCCCTTCAGAAACAGAGAGAGGCTATTTGTTCCGAAATGGATCTCAAGGTTCCGAAGTCTG
GCCATATCTAGAGACTTACACCGAGGATGCTGCGGTTCAGCTATCGAAAGAAATTTGAATGCGAAAGCTTGACCTTATCATTTGTGTAACACTACA
GTGATGGAAATCTTTGTGCTTCTTTATTTGGCTCACAAACTTGGGTCTCACTCAGTGTACCAATTTGCTCATGCTCTTGAGAAACAAAGTAC
CCGGATTTCTGATATCTACTGGAAGAGCTTTGACGACAAGTACCATTTCATGCCCAGTTCACTGCGGATATTTTCCGAATGAACCCACAC
TGATTTTCATCATCATCAGTACTTTCCAGAAATTTGCTGGAGCAAGCAAGAAACTGTTGGCAGTATGAAGCCACACAGCTTTTACTCTTC
CCGGATTGTATCCAGTTGTTACCGGATTGATGTGTTGATCCCAAGTTCAACATTTGCTCTCTCTGCTGCTGATATGAGCATCTACTTC
CCTTACACAGAGGAGCGGTAGATTGACTAACTTCCACTCTGAGATCCAGGAGCTCCTCTACAGCGATGTTGAGAACAAAGAGCACTT
ATGTGTGCTCAAGGACAAGAAAGCCGATTCCTTCCAAATGGCTPAGGCTTGATCGTGTCAAGAACTTGTCAAGTCTTGTGAGTGGT
ACGGGAAGAACACCCGCTTCCGTGAGCTAGCTAACTTGGTTGTGTTGGAGGAGACAGGAGGAAAGAGTCAAAAGCACATGAAGAGAAA
GCAGAGATGAAGAAATGTATGATCTCATTTGAGGAATACAAAGCTAAACCGTCACTTCAGGTGGATCTCCTCTCAGATCGACCGGGTAAG
GAACGGTGAGCTGTACCGGTACATCTGTGACACCAAGGGTGTCTTTGTCCAACTGCTATTAATGAAGCCCTTTTGGTTAACTGTTGTGG
AGGCTATGACTTTGTGGTTTACCGACTTTTCGCCACTTGCAAAGGTGGTCCAGCTGAGATCATTTGTGCACGGTAAATCGGGTTTCCACATT
GAACCTTACCATGGTGATCAGGCTGCTGATACCTCTTGTGCTGATTTCTTCACCAAGTGAAGGAGATCCCATCTCACTGGGATGAGATCTC
AAAAGGAGGCTTCAGAGGAATTCAGGAGAAATACACTTGGCAAAATCTATTTCACAGAGGCTCTTTGACATTGCACTGCTGTGTATGGAATCT
GGAAGCATGTCTCGAAACCTTGACCGGCTTGAGGCTGCGCGTTAACCTTGAATGTTCTATGCAATTGAAGTATCGCCCATTTGGCTCAGGCT
GTTCTCTTGCACAAGATTTCA

FIG. 17B

S. rebaudiana SUS nucleotide (SEQ ID NO: 176) with the mutation that changes S11 to glutamate (E) in bold, lowercase letters

ATCCCGGAACGGTACTCTCACTCCGTTTCACgaaCTTCGTGAGCGTCTCGAATTCAACTCTCGCAACTCATCGTAATGAAATCCCTCTTGTT
TCCTTCAAGGATTGAAGCCAIGGAAAGGAATATTGAAGCCCTCATCAAGTTATGACTGAATTTGAAGCCTATCTGCAAAAGAAAGATCAGA
GCAAACTCTCTGATGCTGCTTTTATGAAGTTCTTAAATGCACACAGGAAGCAATAGTGCAACTCCATGGGTTGCACCTCGCATCCGT
CTTCGACCCGGTCTTTGGGAATATGTTAGAGTCAATGTTTTCGTGGTTGAAGAAATTAAGTGTTCCTCCTCAATATCTTCACTTCAA
AGAAGAAATTCGTTAATGGAAACATCGAAATCCCAACTTCGTGTGGAACTGCAATTTTGAACCTTTTACCGCACTCCTTCCCTCGACCAACTT
TAACCAAGTCTATTGGTAATGGTGTTCAGTTTCTAAACAGACATTTATCTGCTAAATGTTTTCATGATAAGGATAGCATGCACCCCTCTTT
CTTGATTTTCTACGGACTCACCACTATAAGGGAAGACAAATGATGTTGAATGATAGAAATCCAAAACCTCAATGCTCFACAAATCGGTGTT
GGAAAGGCGTCAGAGTACTTATCAACACTCGAAGCGGCAACACCGTACTCTCAGTTTGAACATAAGTTTCAAGAAATCGGGTTGGAGA
GAGTTGGGGTGATAAAGCGGAGGTGTAATGGAGATGATCCACATGCTTCTAGACCTTCTAGAAAGCAACCGACGCTATGCACACTCGAG
AAGTTTCTCGGAAGAAATCCCAATGGTTTTCAATGTTGTCAATTCCTTTCGCTTCATGCGTACTTTCGCGCAAGAAATGTGTTGGGATATCC
CGACACTGGCGGTCAGGTTGTTTACATCTTGGATCAAGTTCCCGCTCTCGGAACGGGAGATGCTCAAAAGGATTAAAGGACCAAGGACTCG
ATATCATCTCTCGTATATTGATTTGTTACGAGGCTTCTCTGAACGGGTTGGACCCACATGCGGCCAACGTTTAGAGAAAGTCTTTGGGA
GCCGAACACTCGCATATTCTTCGGGTCCCGTTTAGAACCCGAAAGGATTTCTTCGTAAATGGATCTCTCGTTTTGAGGTGTGGCCCTTA
CATCGAGACTTTCACCGAGGATGTTGCTAAAGAAGTTACAGCAGAGTTGCAAGCAAAACCAAGATTGATCATTTGGAAACTATAGTGAAG
GAAATTTGGTTGCATCTTTGCTAGCTCACAAAGTTGGGTGTCACCTCAGTGTACCATTTGCTCATGCTTTGGAGAAACTAAATACCCGGAT
TCTGATATCTACTGGAAAGAACTTTGAGGAGAAATATCATTTCTCTTCGCAAGTTTACCGCTGATCTTATCGCTATGAACCATACCGACTT
CATCATCAACCACTTTCCAGAAATTCCTGGAAAGTAAGGACACCGGTTGGACAGTACGAGAGTCAATACCGCTTCAACAAATGCCCGGAT
TGTATCGGGTGGTTCACGGGATCGATGTTTTTGGACCCCAATTCAAATATTGTTTCAACCCGGCGCGATATGGGAATTTACTACTCTGAT
ACCGAGAAAGAAAGAGGCTCACCTCGCTTCACCCCTGAAATCGATGAACCTCTCTTTAGTTCCGTCGAAACCGAAGAACACTTATGTGT
GTTGAAGGATAAGAGTAAACCAATCTTGTTCACAAATGGCCGATTTGGATAATGTGAAGAAATTTAACCGGACTGGTTGAATGGTACGCTA
AAACGACCCGCTTCGTGAGCTCGTGAACCTCGTGGTCTCGGTGTCACCGAAGGAAAGAGTCGAAAGATCTTTGAAGAGCAAGCTCAG
ATGAAGAAGATGCACTTATCGAAACCTTACAAACTCAACGGTTCAGTTTCAGGTGGATATCCTCACAAATGAACCGCTGAGGAACGG
TGAGTTGATACCGGTTATTGCTGACACCGGAGGTCGGTTTATCCAGCCCTGCGTTTTACGAGGCGTTTGGGTTGACGGTTGTGGAGGCCA
TGACTTGTGCGCTGCCGACATTCGCGACACTTCATATGGTGGCCCGCTGAGATTATTGTTACCGGAAATTCGCGGTTCCTATTTGACCCG
TATCACGGTGACCAAGGTCACCGAGTTGCTGGTCAATTTCTTTGAGAAAACTAAACACAGACCCGGTTCATTTGGAGGCGCATTTTCCAAAGG
TGGTCTGCAACGTAATPCAGGAGAAATACACGTGGCAGATTTATTACAGATAGGTTGTTGACGCTTCGCCGAGTTTATGGATTCCTGGAAGC
ATGTGTGCAAGCTTCACAGCGCTCGAGATCCGTCGTTATCTTGAAATGTTTTACCGGCTCAAGTATCGCAAACTGGCTGAATCTGTCTCA
TTCCCTGTTGATGAGTGA

FIG. 17C

Coffea arabica sus1 nucleotide (SEQ ID NO: 177)

ATGCCCGAAGCGTCTCTGACAGCCTCCGGGAACGCCCTTGATGCTACTTTGGCTGCCACCGCAACGATGTTTTCGTGTT
TATGTCGAGCGCTTGAAACCCATGGGAAGCGATCCCTGAACCCCAACCAACTTTTGGCTGAGTTTGAAGAAATTAAACAAGGATGGTAAAC
AAAAAATTCATGATCATGCCCTTTGAAGAAGTCTCTGAAGTCCACACAGGAAGCAATTGTGTGGCCCCCTGGGTGACACTTGCTATTTCGT
CTCAGACCTGGTGTCTGGGAGTATGTTGAGTCAATGTCATGCACTCGTGTGTTGAGGAGTTAAACCGTGCACAGAPACCTGCAATTTCAA
GGAAGAACTCGTTGATGGGAAGCAAAAATGGGAATTTGTTTGGAACTGGACCTTGAACCAATTTACAGCCTCTTTTCCCAAGCCAACTC
TAACTAAGTACATAGCGGACGGAGTTGAGTTCCCTCAACAGGCACTCTCTGCCCCAAATGTTCCATGACAAGGAGAGCATGGCCCCCTCTC
CTTGAAATTTCTCCGTGTTCAACCAATACAAGGCAAGACAAATGATGCTTAACGACAGGATCAAGGACCTTAACACTCTCCAAGCAGTTCT
GAGGAAGGCAAGGAGTACCCTAACACAACACTCTCTGCAGATACACCAATACTCTGAAATTCGAGCACAAATTCCAAGAAATTTGSACTGGAGA
GAGGTGGGGTGATACTGCTGAGCGTGTCTTGGAAATGATCTGCATGCTTCTGGATCTTCTGGGGCTCCTGACTCGTGCACACTAGAG
AAATTCCTAGGGAGAAATCCCTATGGTATTCAATGTTGTATTCTTTCCCCCCCATGGATACTTTGCCCAAGGAAAACGTATTGGGTTATCC
TGATAACGGGTGGCCAGGTTGTTTACATAATTGGATCAAGTTCCCTGCTTGGAGCTGAGATGCTCAAGAGAGGATAAAGGAACAAGCACTTG
ATGTCGAAGCCACGGCAATTTCTAATTATAAATACTAGGCTGCTCTCTGATGCCCTGGAAACCACTTGTGTCAACGGCTTGAGAAAGTATATGGA
TCAGAGTACTCCCATATACTCAGAGTCCCTTCAGAACTGAGAAGGGAGTTGTTCCGCAATATGGATCTCTCGCTTTGAAGTTTGGCCCCA
CATGGAACAATTTACTGAGGATGTTGGAAAAGAAAGTCACTGCGAATAATACAGGCAAGCCAGATTTGGTTATTGGTAACTACAGTGAGG
GTAACCTTGTGCTCCTTGCTTGCCTCAACAAGTTAGGTGTGAACACAGTGTACCATTTGCTCATGCTTTGGAAAAAACAACCAAGTATCCTGAT
TCTGATATTTATTTGAGCAAAATTTGATGAAGAAAGTACCACCTTCTCATGTCTAGTTCAGTCCGGATCTTATCGCAATCAACCATACAGATTT
CAATTATCACTAGCACCTTTCCAGAAATAGCTTGAAGCAAGGACACTGTTGGGCAATATGCAAGCCATATGGCCCTTCACAAATGCCAGGAT
TATACAGAGTTGTACATGGCAATTGATGTTTTTGTGATCCAAAATTCACATTTGCTCACTGAGCTGATACAAACCTCTACTTCCACAC
ACAGAGAAGCAAAACAGATTGACATCTTTCCATCCTGAAAATTCAGGAGCTTGCTTTTCAGCGGATGTGGAGAAATGAGCAACACCTATGTCT
GCTAAAAGACAAAAAGAGCCCTATCTTATTCACCAATGGCAAGACTGGATCGCGTAAAGAAATTTGACAGGGCTTGTGAAATTTGTATGCTA
AGAACCCAAAACATAAGGGAATTGGTCAATCTTTGTCTGGTGGTGGAGACCCGAAGGAAGGAATCCAAAGATTTTGGAAAGAAACAAGCTGAG
ATGAAGAAAAATGTATTCAATTGATAGAAAACCTTACAACTTGAACGGGCAATTCAGATGGATTTCTTCTCAGATGAACAGGGTTAGAAAATGG
TGAACCTCTATCGGTACATTCGCTGACACCCAGGGGAGCAATTCGTGCAACCTGCAATTTATGAGGCAATTTGGGTTGACTGTGGTCGAGGCCA
TGACATGTGTTTTGCCAACAATTTGCAACCAACCATGTTGGTTCCTGCTGAGATCATATTATTCATGGGAAATCTGGTTTCCACATTTGATCCA
TACCATGGTCAGCAGGTCAGCTCCTTGCCAAATTTCTTTGAAAAGGTCCAAAGAACAGCCCTTCTTACTGGGACACCAATTTTCAGCCGG
TGGCTTGAAGCGTATCCAGGAAAAGTACACCTGGCAAAATTTACTCAGATCGGTTGCTGACGCTGGAGTTTATGGATTTCTGCAAAAT
GTGTTTTCCAAAGCTTGATCGCCAGGAGATCCGCCGTTATCTGTGGAATGTTTTATGCTCTCAAGTATCGCAAGTTGGCTGAAGCTGTTCCA
TTGGCTGTTCAATCAGTAA

FIG. 18

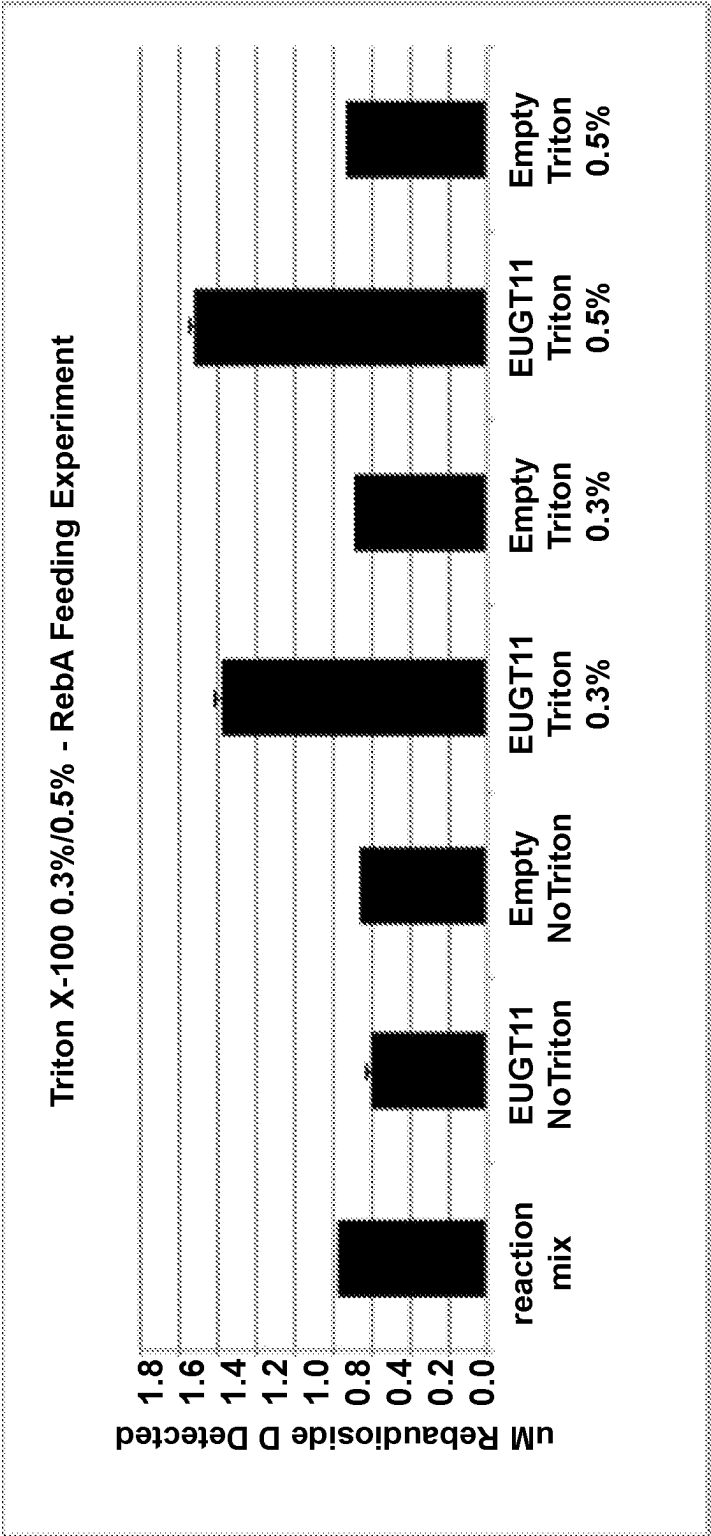


FIG. 19A

A. thaliana UDP-glycosyltransferase UGT72E2 polypeptide (SEQ ID NO: 178)
 MHITKPHAAMFSSPGMGHVIPVIELGKRLSANNGFHVTVFVLETDAAASAQSEKFNSTGVSDIVKLPSPDIYGLVDFPDHVVTKIGVINRA
 AVFALRSKIAAMHQPTALIVDLFGTDALCLAKEFNMLSYVFIPTNARELGVSIYYPNLDKNDIKEEHTVQRNPALAPGCEPVRFEEDTLD
 AYLVPDEPVYRDEVRHGLAYEKADGILVNTWEEMEPKSLKSLNPKLLGRVARVFVYPIGCLCRPIQSSETDHPVLDWINEQFNESVLY
 ISFGSGGCLSAKQLTELAWGLEQSQQRFFVVRPPFVSGCCSEYVSANGGTEENTPEYLLPEGFVSRTSDRGEVVPWSWAPQAEILSHRA
 VGGFLTHCGWSSTLESVVGCVPMIAWPLFAEQNMNAALLSDELGLAVRLDDPKEDISSWKIEALVRKVMTEKEGEAMRRKVNKLKRDSEAE
 MSLSIDGGGLAHESLCRVTKECQREFLERVVDLSRGA

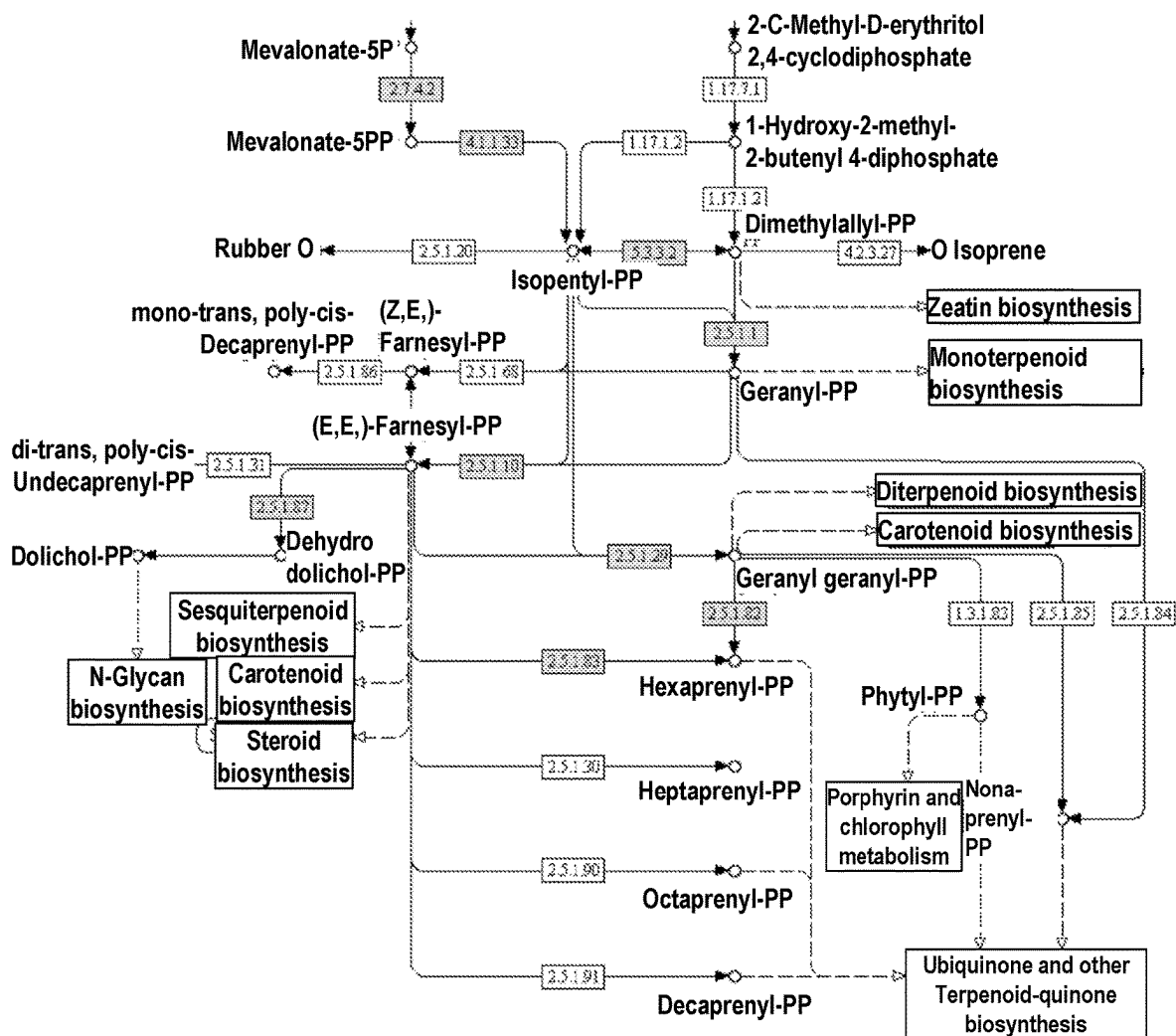
FIG. 19B

A. thaliana sucrose transporter SUC1 from (SEQ ID NO: 179)
 MGAYETEKPTKDAAALETQSPEDFDQFSPLRKIIISVASIAAGVQFGWALQLSLLTPYVQLLGIPHKWSSLIWLCCPVSGMIVQPIVGFH
 SDRCRSKFGRRRPFIATCAALVAVAVFLIGYAADFGYNMGDNLEENKVKVRAIGIFALGFWILDVANNTLQGPCRAFLADLAAGDAKTR
 VANAFSFFMAVGNVLGYAAGSYTNLHKMFPFTMTKACDIYCANLNTCFFLSITLLLVTVTSLWYVNDKQWSPFPRNADDDDEKTSVP
 LEGEILGAFKVKMKRPMWMLLIVTALNWIAWFFELLFPDIDWNGREVFGGSDSGNERSKKLYSLGVQSGANGLMFNSIVLGFMSLGVWEWIG
 RKLGCAKRLWGIWNEFIIAAGLAMTVLVTKEAEDHEKTAGDLAGPSASVKAGALSFAVLGIPLAITESTPFALASIFSSCSGAGQGLSL
 GVNLNAIVIPQMIVSLGGGPFDALEGGGNLPAFIVAAIAAISGVLAITVLPSPPDPAPKATTMGGFH

FIG. 19C

Coffea arabica sucrose synthase polypeptide (SEQ ID NO: 180)
 MAERVLT RVHS LRERLDATLAAHRNPVLLFMSRLETHGKGI LKPHQLLAETEEIN KDGKQKINHDAFEFV LKSTQEAIVLPFWVALAIR
 LRPGVWEYVFPVNVHALVVEELTVPEYLHFKKEBELVDGSKNGNFVLELDFEFTASEFPKPTLT KYIGDGV EFLNRRHLSAKMFHDKESMAPL
 LDFLRVHQYKGTMMINDRIKDINTLQAVLRKAAEYLTTLSDATFYSEFEHNFQEI GLERGWGDTAERVLEMICMLLDLLGAPDSCTLE
 KFLGRIPMVFNVVILSPHCYFAQENVLGYPDTGGQVVYILDQVPALEREMLNRIKEQGGLDVKPRILLITRLLPDAPGTTCCQRLEKVVY
 SEYSHILRVFFRTEKGVVRNWSRFEVWPYMETTEDVAKEVTAELQAKPDLVIGNYSEGNLVASLLAHKLGVTQCTIARALENTKYPD
 SDIYLSKFEDEKYHFSCQFTADLIAMNHTDFIITSTFQEIAGSKDITVGQYESHMAFTMPGLYRVVHGIDVPDPKFNIVSPGADTNLYFPH
 TEKEKRLTSFHPHEIEELLFSDVNEEHLV LKDKKKPILFTMARLDRVKNLTGLVELYAKNPKLRELVNLVVVGGPRRKESKDL EEQAE
 MKKMYSLIETYNLNGQFRWISSQMNVRVRNGELYRYIADTRGAFVQPAFYEAFLGTVVEAMTCGLPTFATNHGGPAEIIHGKSGFHI DP
 YHGEQVSELLANFFERCKKEPSYWDTISAGGLKRIQENYTWQIYSDRLLTLAGVYGFWKCVSKLDRQEIRRYLEMFYALKYRKLAEEAVP
 LAVDQ

FIG. 20



SEO ID NO: 185 (*Saccharomyces cerevisiae* Cycl promoter)

gcggttggttggtgatcaagccacgcgttaggcaatccctcgagcagatccgccaggcgtgtatatatagcgtggatggtccaggcaactt
tagtgtgaacacatacaggcatatatatgtgtcgacgcacatgatcatatggcatgcgtgtgtgtatgtatatataaaactctt
gttttctcttctctaaatattcttcttatcatattaggacctttgcagcatmaattactatacttctatagacacacacacacaa
atcacacacataaatttaata

SEO ID NO: 186 (*Saccharomyces cerevisiae* Kex2 promoter)

gggtcagcagcgtctgatgtagatacacagctatctcgacatgtttttatttttactatatacatataaaagaaatataaaaaatgataaacgtgtgtatattattattcatataatcaatgaggggtcattttctgaaacgcataaaacggttaattgaaaaataaagatagaaaaagaaaaacaaacaaaggaaaaggttagcatatttaattaaactgagctgatacttcaacagcatcgcgtgaagagaaacagttattgaaacgcataatttctcaaaaggcaaacaggtaactccatatttgcgtggaagtgtttttctctgttccatattctgttcataatctgttcataagcctgttttttttctctgtgcttaaacatcccggttttgtaaaaagagaaatctattccacatatttccattccatctactactaaggataaaactaatcccggtgtgttttttggccctgcacacabaattataaactactaaacccattatcaga

FIG. 22

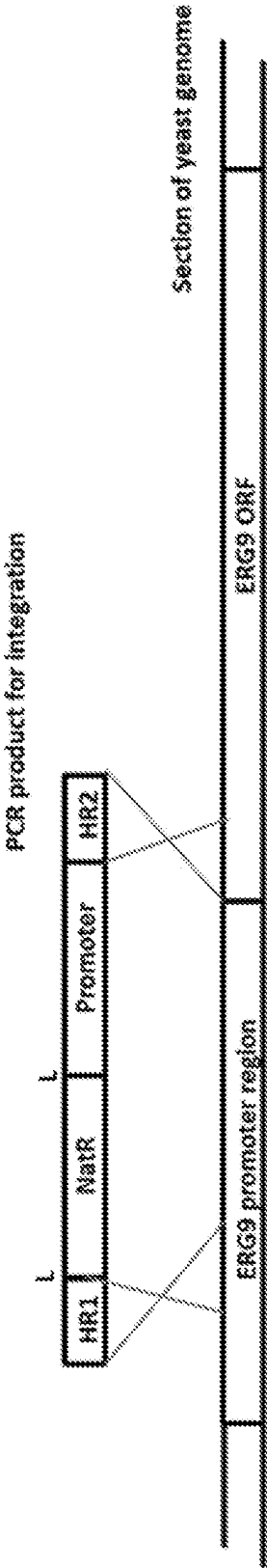


FIG. 23



#5-CACAAATACACACACTAAATTAATA-----atg	wt	100%
#4-CACAAATACACACACTAAATTAATATGAATTCGTTAACGAAATTatg		75%
#3-CACAAATACACACACTAAATTAATATGAATTCGTTAACGAAATCcatg		50%
#2-CACAAATACACACACTAAATTAATATGAATTCGTTAACGAAATCcatg		20%
#1-CACAAATACACACACTAAATTAATATGAATTCGTTAACGAAATTCcatg		5%

FIG. 24

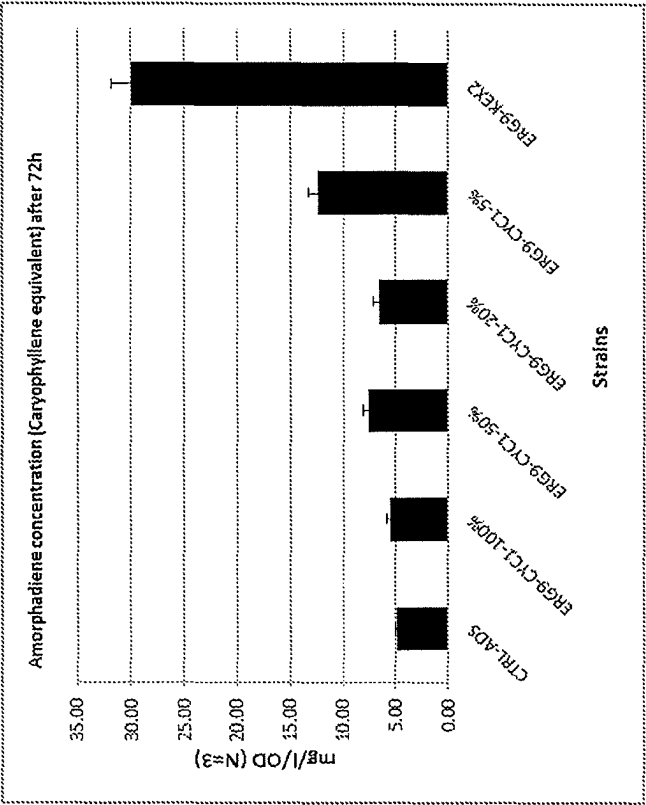


FIG. 25

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>sp|P29704|FDFT_YEAST Squalene synthase OS=Saccharomyces cerevisiae (strain ATCC 204508 / S288c) GN=ERG9 PE=1 SV=2 (SEQ ID NO: 192)
MCKLLQLALHPVEMKAALKIKFCRTPLFSYDQSTSPYLLHCFELLMLTSRFAAVIREL
HPELNCVTLFYLLRALDTIEDDMSIEHDLKIDLRHFHEKLLITKWSFDGNAPDVKDR
AVLTDFESILIEFHKLPKYQEVKEITEKMGNGMADYILDENYNLNGLOTVHDYDVYCH
YVAGLVGGLTRLIVIAKFANESLYSNEQLYESMGLFLQKTNIIIRDYNEDLVDGRSFWPK
EWSQYAPQLKDEMPKENEQLGLDCINHLVLNALSHVIDVLTYLAGIHEQSTQFCAIPQ
VMAIATPLALVFNNREVLHGNNVKIRKGTTCYLLKSRTLRGCVEIFDYLYLRDIKSKLAVQD
PNFLKLNIIQISKIEQFMEEMYQDKLPNVKPNETPIFLKVKERSRYDDELVFTQQEEYK
FNMVLSIILSVLLGFYIYTLHRA

>sp|P36596|FDFT_SCHPO Squalene synthase OS=Schizosaccharomyces pombe (strain ATCC 38366 / 972) GN=erg9 PE=1 SV=1 (SEQ ID NO: 193)
MSLANRIEIRCLCQYKLNWDLPSYGEDENVQONIRRCYQLLDMTSRSFAVVIKELFNGI
REAVMIFYLVLRGLDTEDDMTLPDKKLPILRDFYKTIEVEGWTFNESGPNEXDRQLLV
EFDVVIKEYLNLSGYNVISNITKEMGDMAYYASLAEKNDGFSVETIEDFNKYCHYVA
GLVGIGLSRLFAQSKLEPDLAHSQAISNSLGLFLQKVNIIIRDYREDEDDNRHFWPREIW
SKYTSSEFGDLCIPDNSEKALECLSDMTANALTHATDALVYLSQLKTQEIENFCAIPQVMA
IATLAAVFRNPDVFTNVKIRKQAVQIILHSVNLKNVCDLFLRYTRDIHYKNTPRKDPNF
LKISIECGKIEQVSESLFPPRFREMYEKAVVSKLSEQKKNGTQKAILNDEQKELYRKDL
QKLGISILEVEFIIILVCLAVIEFYVFNIRIHWSDFKELNLF

>sp|Q9Y753|FDFT_YARLI Squalene synthase OS=Yarrowia lipolytica (strain CLIB 122 / E 150)
GN=SQS1 PE=3 SV=1 (SEQ ID NO: 194)
MCKLIELLLHPSELSAAIHYKLWRQPLHPRDLSKESTELRRCYELLVDCSRSEAAVIREL
HPEVRDAMLFYLLRALDTIEDDMTILSRDIKIPILRDETKCMKTPGWKFTDSDENERDR
VVLQEFVVMTEFNKLPKYQEVYDITDRMGNGMADYVIDDDFNNGVDTIAAYDLYCH
HVAGIVCEGLTRITILAGEGTDVLHENPRLOESMGLFLQKVNIIIRDYREDIDVNRWFWR
EIWHKYAEEMRDEKPKYSKKALHCTSDLVANALGHATDCLDYLDNVTDPSPTFTTCAIPQ
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FIG. 25 (cont'd)

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VMAIATLQLYRNPOVFQKNVXLRKGTTSVLLEASNVSGVCCITETRYARARVYKXSDEND
PHYTRSVLCGKIEQHAALIKRQGPFAKTIAQLEGERKEMALSILIVCLAVITSNQGLMA
YIAYVSGFRWSPREIFDSKMFELRD

>sp|Q9HGZ6|FDFT_CANGA Squalene synthase OS=Candida glabrata (strain ATCC 2001 / CBS 138 /
JCM 3761 / NBRC 0622 / NRRL Y-65) GN=ERG9 PE=3 SV=2 (SEQ ID NO: 195)
MCKVLQDLALHPLELRAALKKFIROPLFSTNDTRATPQLERCYELLNLTSSFAAVIMEL
HPELRNVIMVFYLLRALDEVEDDMTIDPQLKGVLPREFDSKLDITWSDENMLKEKDR
VVLTEFPCLGEYHKLKPEYQVKIKRITGLMNGKADYILDENFNLCVQTVKDYDKYCH
YVAGLVGDGLTELIVLAGSGDDLYHGKNSFQLYESMGLFLQKTNILRDYAEIDLDDGRSF
WPKEIWSYATKLTDFRDEKNTQKGVDCINHLVINALTHVIDVLTYSSTHEQSSTQFCA
IPQVMAIATLAKVFNNPEVLKKNVKIRKGTTCDLILNSKTLKGCVDIFQYVLRDMKQRLP
VEDPNYLKFNIOVAKIEQFTEEMFQDNLPAQVEPREHMIYLVKQERLKWETQVIERVQEE
DYKFNMAISVFCVLLSYFFTK

>sp|Q752X9|FDFT_ASHGO Squalene synthase OS=Asbya gossypii (strain ATCC 10895 / CBS
109.51 / FGSC 9923 / NRRL Y-1036) GN=ERG9 PE=3 SV=1 (SEQ ID NO: 196)
MGKVVOQLFTHFLELKAALKFIREPLYPADDTQGSABLKRCYQLLOFTSPSPAIVIMEL
HPELRNAVMLFYLLRALDTVEDEMTISPKNVPLIREFDQKILDTWSTFDGNAKTEKDR
DVLVEFSTILAEFHKLKPEYQQVIADITHKMGNGMADYILDEKFNLSGLETIQDYDRYCH
YVAGLVGDGLTHLIMLAKFSSPGLYQSPDIYESMGLFLQKTNILRDYAEIDLDDGRSPWE
KEIMSHYADDLASFSPENATAGVYCINHLVINALGHVQHVLTLYLASUREQSSFOFCAIE
QVMAIATLALVFONERVLOTSVKIRKGTTCYLILKERTFQCCVEIFEYVLRDINRRLTVA
DPNYLKLNIETIAKLDXFEEMVQDKLPVGAKPQETETIYKKVPEERSAYDLEVLPREQEEEF
KENVLLSILFTVFGALYNYAK

>sp|O74165|FDFT_CYBJA Squalene synthase OS=Cyberlindnera jadinii CN=ERG9 PE=3 SV=1
(SEQ ID NO: 197)
MGALLQLALHPDELASIVQKLFRENENARNPATESABLRCYELLNLTSSFAAVIEEL
HPELRNVIMVFYLLRALDTVEVMSIENSVPVLRQFHEKLDTKDWTDCGNSPNEKDR
CVLVEFDRILQYHKLKPKQYQXVIRHETKMGNGMADYILENENFNSMGLTIEDYDLYCY

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FIG. 25 (cont'd)

YVACLVDCLTQLIYVLAKFONSELSEVVKQLFZSMGLFLOKNTNIIRDYEEHQVDQCRATWPK
 EEWKAYANELSDENKPEMQSGLWCISELVONALDHYVDLYLALVEEQTSNECAIPO
 VMAIATLELVFQNPQVLTQHVKIRKGTIVSLLESRLCCARIFSRYLKIRHKSHPSD
 PNILRIGITIGKIEQFLDGMYPHYVPKGITPQTTSIRTVVKRIQLDEMRKDIDEELK
 TRILLSLGVAVFQVVGVRIL

 >sp|P78589|DET CANAL Squalene synthase OS=Candida albicans GN=ERG9 PE=3 SV=1
 (SEQ ID NO: 198)
 MGRQLQLSAPTELKAVIQLFGERQPLHPGKRVDNDKELGRQYELLNLTSRSPAIVIELL
 HPELSDAVWIEFYLVIRALDTIEDDMFIKSSIKIPLDNEETTKINTKNTFDGYPNEKOR
 TVIVPEFDKILINVYHRLKPOYQDIKSIPTFMGNGMADYILDEEFNVYGVATVEDYNLYCH
 YVAGLVGEGILTNIPLVLANFGDKILTENNFAKADSMGLFLOKNTNIIRDYHEDLQDGRSFWP
 RETWQKYTEKLDQDFHKVKTPAKETAGVSCINELVNLALGVATCCLDYLSLVKDPSSFSFC
 AIFQMAVATILAEVYNNPKVILHGVVKIRKGTICCLLLESHLPGVVKLFEKYEIQVINHAS
 SVRDENVILKIGIKCGEIEQYCEMIYENKQALPPSRKSLPENKZTKIVASRESIDLVSQRR
 IEPGNFNCNVVLEFCIGALLSLIYFVLY

 >sp|P28604|ERG7 YEAST Lanosterol synthase OS=Saccharomyces cerevisiae (strain ATCC 204508
 / S288c) GN=ERG7 PE=1 SV=6 (SEQ ID NO: 199)
 MTEFYSTIGLPKTDPRLMRLRTDELGRESWEYLTFQDAANDEFTTFQWLLQDPKFPQ
 HPERKHSPPFSAPDACHNGASTFKLLOEPDSCITFCQYKGMENITCYVAVNYIAGLET
 PERERELIRYIVNTAHPVDGGWGLHSVDKSTVECTVLAIVLIRLLGLPKDRFVCAKARS
 TELRLCGAIGSEPHWCKIWLISALNLYKREGVNPAPPETALLPYSLPMHPGRWVHTRGVYI
 PVYLISLVKFCSEMTPLLEELRNEYTKPFDKINFSKNNNTVCGVDINYPHSTLINIANS
 LVVFYEKYLENRFTYLSKRVYDLIKTELONTDSCIAFVNCATCALVTLIEGVDSEA
 FQRLQYRFEDALFIGPQGMIMGTNGVQWDCAFALQYTVAGLAERPEFYNTIVSAYKK
 LCHAQDTECYPSYRDNRKGCWCFSTKIQQYTVADCTAATKATIMWNSPVFSEVHHM
 ISSERLEGIDVLLNLQNLGSEFYCGSFATYERIKAPLAMELTINPAEVEGNIMWEYPYVEC
 TDSSVGLTYTFHYFYDYRKEEIRTRIRIAIEFIKKSOLFQGSWYGSWGTCTFYAGMFALE
 ALHTVGTENSSSTVRKGCDFLVSKQMDGGWGESMKSELSHSYVDSERSLVVQTAMALI
 ALLFAEYPNKEVIDRGIDLLKNROEESGEWKTESVRGVTHNSCAIEYPSYRFLFEIKALGMYSRAIETHLL

FIG. 25 (cont'd)

```
>sp|P37268|FDFT_HUMAN Squalene synthase OS=Homo sapiens GN=FDFT1 PE=1 SV=1
(SEQ ID NO: 200)
MEFYKCLGHPEEFYNLVRFRIGGKRKVMFKMDQDSSLSSSLKTCYKYLNQTSRFAAVIQ
LDGEMRNAVCIFYLVLRALDTLEDMDTISVEKKVPLLNPHSFYLPDWRFMESKEKDRQ
VLEDFPTISLEFRNLAEKYQTVIADICRMGCGMAEFLDKHVTSEQEWDKCHYVAGLVG
IGLSRLFSASEFDPVGEDTERANSMGLFLQKTNIIIRDYLEDDQCGREFWFOEVMGRYV
KKLEDFAKPENIDLAVQCLNELITNALHPIPDVITYLSRLRNQSVNFCAIPQVMAIATL
AACYNNQVFKGAVKIRKQQAATLMDATNMPAVKAIYQYIEEYHAIIPNSDPSSSKTK
QITSTRTQNLNQCQLISRSYSPYYLSFVMLLAALSQYLTLSQVTEYVQVQGEH

>sp|P53798|FDFT_MOUSE Squalene synthase OS=Mus musculus GN=Fdft1 PE=2 SV=2
(SEQ ID NO: 201)
MEFYKCLGHPEEFYNLLRFRMGGRNFIPEKMDQDSSLSSSLKTCYKYLNQTSRFAAVIQ
LDGDIRHAICVFYLVLRALDTVEDDMSISVEKKIPLICNEHFFLYDPESWRFTESKEKDRQ
VLEDFPTISLEFRNLAEKYQTVIIDDICRMGCGMAEFDKDVTSKQDMDKCHYVAGLVG
IGLSRLFSASEFDPVGEDIECANSMGLFLQKTNIIIRDYLEDDQCGREFWFOEVMGRYV
KKLEDFAKPENVDVAVQCLNELITNTLQHIPDVLITYLSRLRNQSVNFCAIPQVMAIATL
AACYNNQVFKGVVKIRKQQAATLMDATNMPAVKAIYQYIEEYHAIIPNSDPSSSKTK
QVTSKIRTONLNCQLISRSYSPYYLSFIMLLAALSQYLTLSQVTEYVQVQREH

>sp|Q02769|FDFT_RAT Squalene synthase OS=Rattus norvegicus GN=Fdft1 PE=2 SV=1
(SEQ ID NO: 202)
MEFYKCLGHPEEFYNLLRFRMGGRNFIPEKMDRNSLSNLKTCYKYLDQTSRFAAVIQ
LDGDIRHAVCVFYLLIRAMDTVEDDMAISVEKKIPLIRNEHFFLYEPESWRFTESKEKHRV
VLEDFPTISLEFRNLAEKYQTVIADICRMGCGMAEFLNEDVTSKQDMDKCHYVAGLVG
IGLSRLFSASEFDPVGEDTECANSMGLFLQKTNIIIRDYLEDDQCGREFWFOEVMGRYV
KKLEDVFKPENVDVAVKCLNELITNALQHIPDVITYLSRLRNQSVNFCAIPQVMAIATL
AACYNNHQVFKGVVKIRKQQAATLMDATNMPAVKAIYQYIEEYHAIIPNSDPSSSKTK
QLLSNIRTSQNLNQCQLISRSYSPYYLSFIMLLAALSQYLTLSQVTEYVQVQREH
```

FIG. 25 (cont'ed)

Aspergillus nidulans GGPPS-2 polypeptide (SEQ ID NO: 203)

MTSDSHETPPHAIPPRISSNRMSGASTRDKAALMGNEFEKDWLSKGDKLQENTDLSKRHTR
NQSSLDCTKXKDGKWSQENEEVINGPYDYMLQHFGKDLRRQMINAFNVWLKVPSESLLAI
TKVVANLHTASILLIDVEDNSLLRRGIPVAHSIYGTAQTINSANYVYFLALQEVQKLSR
AAIDYVQELLNLRGQGMDFWRDTLTCPSEDEYLEMVGNKYGGLFRLAVKLMQAESST
GKDCVALNVVLGLVFIQICDDYLNLSDTTYTONKGLCEDLTECKFSFPIIHSIRSNPGNHQ
LINILRQRTKDEEVERYALQYMESTGSEKHTQDVVRQLRARALQLIEEISENSENGEQPEE
HNDGTMVRAILDKITESTIADENTTTTRDINGNCATR

Saccharomyces cerevisiae GGPPS-3 polypeptide (SEQ ID NO: 167)

MEAKIDELINNDEFVWSSQNESLISKPYNHILLKPKKNERNLNLVQINRVMNLPKDQLAIV
SQIVELLHNSSLIIDIEDNAPFLRRGQTTSHLIFGVPESTINTANMYFRAMQLVSQLTTR
EPLYGNLLTIFNEELINLRGCGGLDIYWRDFLPHIIPTOEMYLNMVMNKTGGLFRLTLRL
MEALSPSSHNCHSLVFFINLLGIITYQIRDDYLNKDFQMSSEKGFREDITEGKLSFPIVH
ALNFTKTKGQTEQHNEILRILLRTSDKDIKLKLIIQILEFDTNSLAYTKNFINQLVNMIK
NDNENKYLPLDASHSDTATNLHDELLYITDHLSEL

FIG. 26

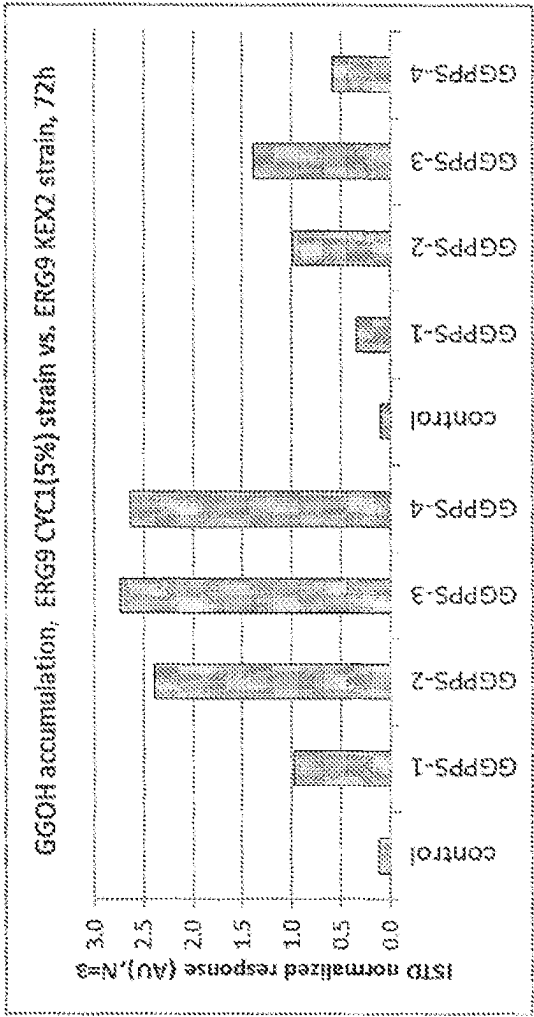
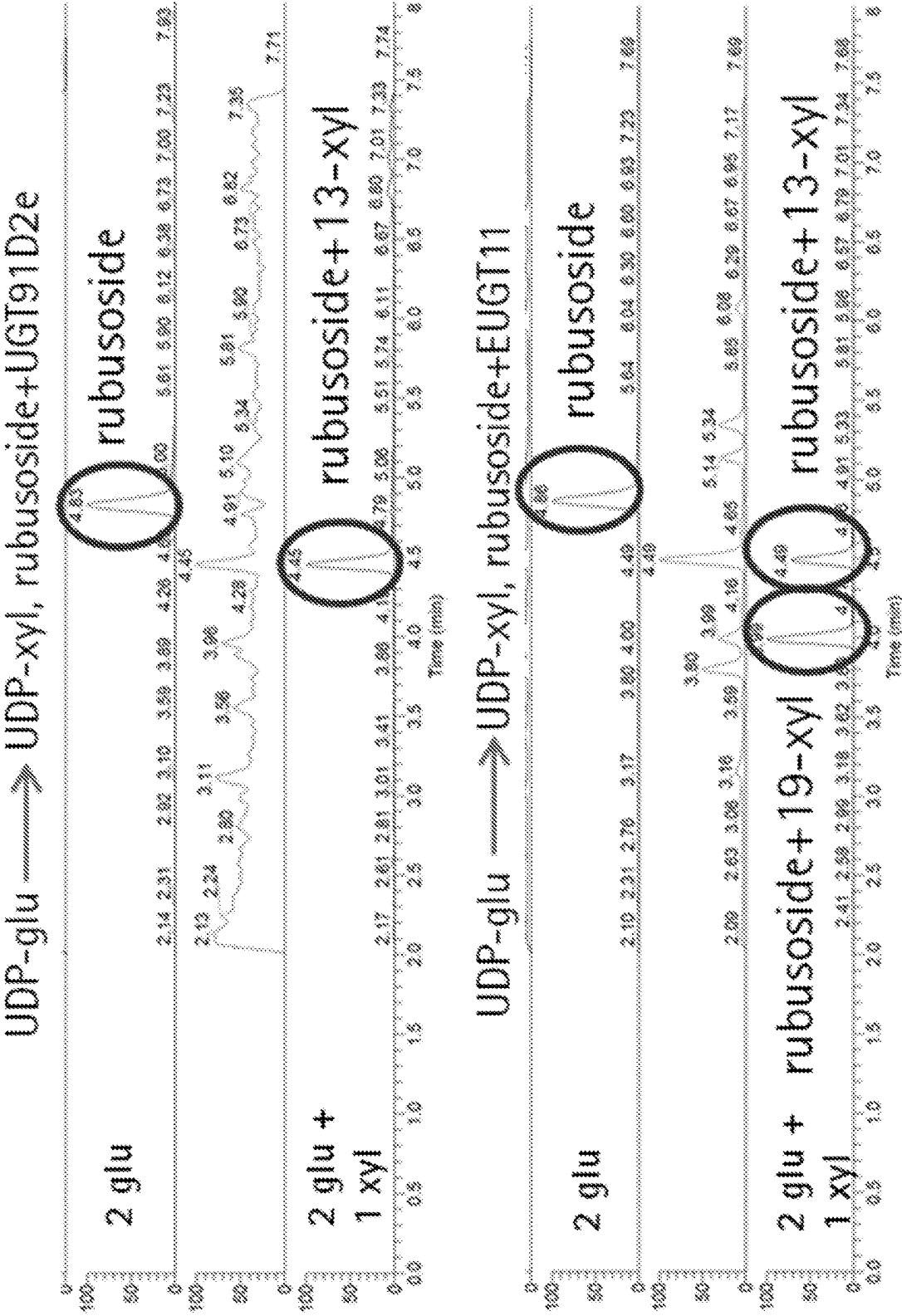


FIG. 27



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RECOMBINANT PRODUCTION OF STEVIOL GLYCOSIDES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/471,615, filed on Sep. 10, 2021, which is a continuation of Ser. No. 16/535,373, filed on Aug. 8, 2019, which is a continuation of Ser. No. 15/439,660, filed on Feb. 22, 2017, now U.S. Pat. No. 10,435,730, which is a divisional of Ser. No. 14/237,540, filed on May 15, 2014, now U.S. Pat. No. 9,631,215, which is a U.S. national phase of International Application No. PCT/US2012/050021, filed on Aug. 8, 2012, which claims priority to U.S. Application Ser. No. 61/521,084, filed Aug. 8, 2011; U.S. Application Ser. No. 61/521,203, filed Aug. 8, 2011; U.S. Application Ser. No. 61/521,051, filed Aug. 8, 2011; U.S. Application Ser. No. 61/523,487, filed Aug. 15, 2011; U.S. Application Ser. No. 61/567,929, filed Dec. 7, 2011; and U.S. Application Ser. No. 61/603,639, filed Feb. 27, 2012. The entire disclosure contents of these applications are herewith incorporated by reference in their entirety into the present application.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING PROVIDED ELECTRONICALLY

This application contains a Sequence Listing submitted as an electronic text file named "13-1234-US-CON3_SequenceListing_ST26.xml," having a size of 438, 583 bytes, and created on Jun. 8, 2023. The information contained in this electronic file is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

This disclosure relates to the recombinant production of steviol glycosides. In particular, this disclosure relates to the production of steviol glycosides such as rebaudioside D by recombinant hosts such as recombinant microorganisms, plants, or plant cells. This disclosure also provides compositions containing steviol glycosides. The disclosure also relates to tools and methods for producing terpenoids by modulating the biosynthesis of terpenoid precursors of the squalene pathway.

BACKGROUND

Sweeteners are well known as ingredients used most commonly in the food, beverage, or confectionary industries. The sweetener can either be incorporated into a final food product during production or for stand-alone use, when appropriately diluted, as a tabletop sweetener or an at-home replacement for sugars in baking. Sweeteners include natural sweeteners such as sucrose, high fructose corn syrup, molasses, maple syrup, and honey and artificial sweeteners such as aspartame, saccharine and sucralose. Stevia extract is a natural sweetener that can be isolated and extracted from a perennial shrub, *Stevia rebaudiana*. *Stevia* is commonly grown in South America and Asia for commercial production of stevia extract. Stevia extract, purified to various degrees, is used commercially as a high intensity sweetener in foods and in blends or alone as a tabletop sweetener.

Extracts of the Stevia plant contain rebaudiosides and other steviol glycosides that contribute to the sweet flavor, although the amount of each glycoside often varies among

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different production batches. Existing commercial products are predominantly rebaudioside A with lesser amounts of other glycosides such as rebaudioside C, D, and F. Stevia extracts may also contain contaminants such as plant-derived compounds that contribute to off-flavors. These off-flavors can be more or less problematic depending on the food system or application of choice. Potential contaminants include pigments, lipids, proteins, phenolics, saccharides, spathulenol and other sesquiterpenes, labdane diterpenes, monoterpenes, decanoic acid, 8,11,14-eicosatrienoic acid, 2-methyloctadecane, pentacosane, octacosane, tetracosane, octadecanol, stigmasterol, β -sitosterol, α - and β -amyrin, lupeol, β -amyrin acetate, pentacyclic triterpenes, centaureidin, quercetin, epi- α -cadinol, carophyllenes and derivatives, beta-pinene, beta-sitosterol, and gibberellin.

SUMMARY

Provided herein is a recombinant host, such as a micro-organism, plant, or plant cell, comprising one or more biosynthesis genes whose expression results in production of steviol glycosides such as rebaudioside A, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F, or dulcoside A. In particular, EUGT11, a uridine 5'-diphospho (UDP) glycosyl transferase described herein, can be used alone or in combination with one or more other UDP glycosyl transferases such as UGT74G1, UGT76G1, UGT85C2, and UGT91D2e, to allow the production and accumulation of rebaudioside D in recombinant hosts or using in vitro systems. As described herein, EUGT11 has a strong 1,2-19-O-glucose glycosylation activity, which is an important step for rebaudioside D production.

Typically, stevioside and rebaudioside A are the primary compounds in commercially-produced stevia extracts. Stevioside is reported to have a more bitter and less sweet taste than rebaudioside A. The composition of stevia extract can vary from lot to lot depending on the soil and climate in which the plants are grown. Depending upon the sourced plant, the climate conditions, and the extraction process, the amount of rebaudioside A in commercial preparations is reported to vary from 20 to 97% of the total steviol glycoside content. Other steviol glycosides are present in varying amounts in stevia extracts. For example, Rebaudioside B is typically present at less than 1-2%, whereas Rebaudioside C can be present at levels as high as 7-15%. Rebaudioside D is typically present in levels of 2% or less, and Rebaudioside F is typically present in compositions at 3.5% or less of the total steviol glycosides. The amount of the minor steviol glycosides affects the flavor profile of a Stevia extract. In addition, Rebaudioside D and other higher glycosylated steviol glycosides are thought to be higher quality sweeteners than Rebaudioside A. As such, the recombinant hosts and methods described herein are particularly useful for producing steviol glycoside compositions having an increased amount of Rebaudioside D for use, for example, as a non-caloric sweetener with functional and sensory properties superior to those of many high-potency sweeteners.

In one aspect, this document features a recombinant host that includes a recombinant gene encoding a polypeptide having at least 80% identity to the amino acid sequence set forth in SEQ ID NO:152.

This document also features a recombinant host that includes a recombinant gene encoding a polypeptide having the ability to transfer a second sugar moiety to the C-2' of a 19-O-glucose of rubusoside. This document also features a recombinant host that includes a recombinant gene encoding

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a polypeptide having the ability to transfer a second sugar moiety to the C-2' of a 19-O-glucose of stevioside.

In another aspect, this document features a recombinant host that includes a recombinant gene encoding a polypeptide having the ability to transfer a second sugar moiety to the C-2' of the 19-O-glucose of rubusoside and to the C-2' of the 13-O-glucose of rubusoside.

This document also features a recombinant host that includes a recombinant gene encoding a polypeptide having the ability to transfer a second sugar moiety to the C-2' of a 19-O-glucose of rebaudioside A to produce rebaudioside D, wherein the catalysis rate of the polypeptide is at least 20 times faster (e.g., 25 or 30 times faster) than a 91D2e polypeptide having the amino acid sequence set forth in SEQ ID NO: 5 when the reactions are performed under corresponding conditions.

In any of the recombinant hosts described herein, the polypeptide can have at least 85% sequence identity (e.g., 90%, 95%, 98%, or 99% sequence identity) to the amino acid sequence set forth in SEQ ID NO:152. The polypeptide can have the amino acid sequence set forth in SEQ ID NO: 152.

Any of the hosts described herein further can include a recombinant gene encoding a UGT85C polypeptide having at least 90% identity to the amino acid sequence set forth in SEQ ID NO:3. The UGT85C polypeptide can include one or more amino acid substitutions at residues 9, 10, 13, 15, 21, 27, 60, 65, 71, 87, 91, 220, 243, 270, 289, 298, 334, 336, 350, 368, 389, 394, 397, 418, 420, 440, 441, 444, and 471 of SEQ ID NO:3.

Any of the hosts described herein further can include a recombinant gene encoding a UGT76G polypeptide having at least 90% identity to the amino acid sequence set forth in SEQ ID NO:7. The UGT76G polypeptide can have one or more amino acid substitutions at residues 29, 74, 87, 91, 116, 123, 125, 126, 130, 145, 192, 193, 194, 196, 198, 199, 200, 203, 204, 205, 206, 207, 208, 266, 273, 274, 284, 285, 291, 330, 331, and 346 of SEQ ID NO:7.

Any of the hosts described herein further can include a gene (e.g., a recombinant gene) encoding a UGT74G1 polypeptide.

Any of the hosts described herein further can include a gene (e.g., a recombinant gene) encoding a functional UGT91D2 polypeptide. The UGT91D2 polypeptide can have at least 80% sequence identity to the amino acid sequence set forth in SEQ ID NO:5. The UGT91D2 polypeptide can have a mutation at position 206, 207, or 343 of SEQ ID NO:5. The UGT91D2 polypeptide also can have a mutation at positions 211 and 286 of SEQ ID NO:5 (e.g., L211M and V286A, referred to as UGT91D2e-b). The UGT91D2 polypeptide can have the amino acid sequence set forth in SEQ ID NOs: 5, 10, 12, 76, 78, or 95.

Any of the hosts described herein further can include one or more of

- (i) a gene encoding a geranylgeranyl diphosphate synthase;
- (ii) a gene encoding a bifunctional copalyl diphosphate synthase and kaurene synthase, or a gene encoding a copalyl diphosphate synthase and a gene encoding a kaurene synthase;
- (iii) a gene encoding a kaurene oxidase; and
- (iv) a gene encoding a steviol synthetase. Each of the genes of (i), (ii), (iii), and (iv) can be a recombinant gene.

Any of the hosts described herein further can include one or more of

- (v) a gene encoding a truncated HMG-CoA;
- (vi) a gene encoding a CPR;

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- (vii) a gene encoding a rhamnose synthetase;
- (viii) a gene encoding a UDP-glucose dehydrogenase; and
- (ix) a gene encoding a UDP-glucuronic acid decarboxylase. At least one of the genes of (i), (ii), (iii), (iv), (v), (vi), (vii), (viii), or (ix) can be a recombinant gene.

The geranylgeranyl diphosphate synthase can have greater than 90% sequence identity to one of the amino acid sequences set forth in SEQ ID NOs: 121-128. The copalyl diphosphate synthase can have greater than 90% sequence identity to the amino acid sequence set forth in SEQ ID NOs: 129-131. The kaurene synthase can have greater than 90% sequence identity to one of the amino acid sequences set forth in SEQ ID NOs: 132-135. The kaurene oxidase can have greater than 90% sequence identity to one of the amino acid sequences set forth in SEQ ID NOs: 138-141. The steviol synthetase can have greater than 90% sequence identity to one of the amino acid sequences set forth in SEQ ID NOs: 142-146.

Any of the recombinant hosts can produce at least one steviol glycoside when cultured under conditions in which each of the genes is expressed. The steviol glycoside can be selected from the group consisting of rubusoside, rebaudioside A, rebaudioside B, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F, dulcoside A, stevioside, steviol-19-O-Glucoside, steviol-13-O-glucoside, steviol-1, 2-bioside, steviol-1,3-bioside, 1,3-stevioside, as well as other rhamnosylated or xylosylated intermediates. The steviol glycoside (e.g., rebaudioside D) can accumulate to at least 1 mg/liter (e.g., at least 10 mg/liter, 20 mg/liter, 100 mg/liter, 200 mg/liter, 300 mg/liter, 400 mg/liter, 500 mg/liter, 600 mg/liter, or 700 mg/liter, or greater) of culture medium when cultured under said conditions.

This document also features a method of producing a steviol glycoside. The method includes growing any of the hosts described herein in a culture medium, under conditions in which the genes are expressed; and recovering the steviol glycoside produced by the host. The growing step can include inducing expression of one or more of the genes. The steviol glycoside can be a 13-O-1,2-diglycosylated and/or a 19-O-1,2-diglycosylated steviol glycoside (e.g., stevioside, steviol 1,2 bioside, rebaudioside D, or rebaudioside E). For example, the steviol glycoside can be rebaudioside D or rebaudioside E. Other examples of steviol glycosides can include rebaudioside A, rebaudioside B, rebaudioside C, rebaudioside F, and dulcoside A.

This document also features a recombinant host. The host includes (i) a gene encoding a UGT74G1; (ii) a gene encoding a UGT85C2; (iii) a gene encoding a UGT76G1; (iv) a gene encoding a glycosyltransferase having the ability to transfer a second sugar moiety to the C-2' of a 19-O-glucose of rubusoside or stevioside; and (v) optionally a gene encoding a UGT91D2e, wherein at least one of the genes is a recombinant gene. In some embodiments, each of the genes is a recombinant gene. The host can produce at least one steviol glycoside (e.g., rebaudioside D) when cultured under conditions in which each of the genes (e.g., recombinant genes) is expressed. The host further can include (a) a gene encoding a bifunctional copalyl diphosphate synthase and kaurene synthase, or a gene encoding a copalyl diphosphate synthase and a gene encoding a kaurene synthase; (b) a gene encoding a kaurene oxidase; (c) a gene encoding a steviol synthetase; (d) a gene encoding a geranylgeranyl diphosphate synthase.

This document also features a steviol glycoside composition produced by any of the hosts described herein. The

composition has reduced levels of stevia plant-derived contaminants relative to a stevia extract.

In another aspect, this document features a steviol glycoside composition produced by any of the hosts described herein. The composition has a steviol glycoside composition enriched for rebaudioside D relative to the steviol glycoside composition of a wild-type *Stevia* plant.

In yet another aspect, this document features a method of producing a steviol glycoside composition. The method includes growing a host described herein in a culture medium, under conditions in which each of the genes is expressed; and recovering the steviol glycoside composition produced by the host (e.g., a microorganism). The composition is enriched for rebaudioside A, rebaudioside B, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F or dulcoside A relative to the steviol glycoside composition of a wild-type *Stevia* plant. The steviol glycoside composition produced by the host (e.g., microorganism) can have a reduced level of stevia plant-derived contaminants relative to a stevia extract.

This document also features a method for transferring a second sugar moiety to the C-2' of a 19-O-glucose or the C-2' of a 13-O-glucose in a steviol glycoside. The method includes contacting the steviol glycoside with a EUGT11 polypeptide described herein or UGT91D2 polypeptide described herein (e.g., UGT91D2e-b) and a UDP-sugar under suitable reaction conditions for the transfer of the second sugar moiety to the steviol glycoside. The steviol glycoside can be rubusoside, wherein the second sugar moiety is glucose, and stevioside is produced upon transfer of the second glucose moiety. The steviol glycoside can be stevioside, wherein the second sugar moiety is glucose, and Rebaudioside E is produced upon transfer of the second glucose moiety. The steviol glycoside can be Rebaudioside A, and Rebaudioside D is produced upon transfer of the second glucose moiety.

In another embodiment of an improved downstream steviol glycoside pathway as disclosed herein, materials and methods are provided for the recombinant production of sucrose synthase, and to materials and methods for increasing production of UDP-glucose in a host, specifically for increasing the availability of UDP-glucose in vivo, with the purpose of promoting glycosylation reactions in the cells, and methods for reducing UDP concentrations in the cells are provided.

The document also provides a recombinant host comprising one or more exogenous nucleic acids encoding a sucrose transporter and a sucrose synthase, wherein expression of the one or more exogenous nucleic acids with a glucosyltransferase results in increased levels of UDP-glucose in the host. Optionally, the one or more exogenous nucleic acids comprise a SUS1 sequence. Optionally, the SUS1 sequence is from *Coffea arabica*, or encodes a functional homolog of the sucrose synthase encoded by the *Coffea arabica* SUS1 sequence, but equally an *Arabidopsis thaliana* or *Stevia rebaudiana* SUS may be used as described herein. In the recombinant host of the invention, the one or more exogenous nucleic acids may comprise a sequence encoding a polypeptide having the sequence set forth in SEQ ID NO:180, or an amino acid sequence at least 90 percent identical thereto, and optionally the one or more exogenous nucleic acids comprise a SUC1 sequence. In one embodiment, the SUC1 sequence is from *Arabidopsis thaliana*, or the SUC1 sequence encodes a functional homolog of the sucrose transporter encoded by the *Arabidopsis thaliana* SUC1 sequence. In the recombinant host, the one or more exogenous nucleic acids may comprise a sequence encoding

a polypeptide having the sequence set forth in SEQ ID NO:179, or an amino acid sequence at least 90 percent identical thereto. The recombinant host has reduced ability to degrade external sucrose, as compared to a corresponding host that lacks the one or more exogenous nucleic acids.

The recombinant host may be a microorganism, such as a Saccharomycete, for example *Saccharomyces cerevisiae*. Alternatively, the microorganism is *Escherichia coli*. In an alternative embodiment, the recombinant host is a plant or host cell.

The invention also provides a method for increasing the level of UDP-glucose and reducing the level of UDP in a cell, the method comprising expressing in the cell a recombinant sucrose synthase sequence and a recombinant sucrose transporter sequence, in a medium comprising sucrose, wherein the cell is deficient in sucrose degradation.

The invention additionally provides a method for promoting a glycosylation reaction in a cell, comprising expressing in the cell a recombinant sucrose synthase sequence and a recombinant sucrose transporter sequence, in a medium comprising sucrose, wherein the expressing results in a decreased level of UDP in the cell and an increased level of UDP-glucose in the cell, such that glycosylation in the cell is increased.

In either method for increasing the level of UDP-glucose or promoting glycosylation, the cell may produce vanillin glucoside, resulting in increased production of vanillin glucoside by the cell, or may produce steviol glucoside, resulting in increased production of steviol glucoside by the cell. Optionally, the SUS1 sequence is a *A. thaliana*, *S. rebaudiana*, or *Coffea arabica* SUS1 sequence (see e.g., FIG. 17, SEQ ID NOs. 175-177), or is a sequence that encodes a functional homolog of the sucrose synthase encoded by the *A. thaliana*, *S. rebaudiana*, or *Coffea arabica* SUS1 sequence. The recombinant sucrose synthase sequence optionally comprises a nucleic acid encoding a polypeptide having the sequence set forth in SEQ ID NO:180, or an amino acid sequence at least 90% identical thereto, wherein optionally the recombinant sucrose transporter sequence is a SUC1 sequence, or wherein optionally the SUC1 sequence is an *Arabidopsis thaliana* SUC1 sequence, or is a sequence that encodes a functional homolog of the sucrose transporter encoded by the *Arabidopsis thaliana* SUC1 sequence, or wherein optionally the recombinant sucrose transporter sequence comprises a nucleic acid encoding a polypeptide having the sequence set forth in SEQ ID NO:179, or an amino acid sequence at least 90% identical thereto. In either method, the host is a microorganism, for example a Saccharomycete, optionally such as *Saccharomyces cerevisiae*. Or the host may be *Escherichia coli*. Or the host may be a plant cell.

Also provided herein is a recombinant host, such as a microorganism, comprising one or more biosynthesis genes whose expression results in production of diterpenoids. Such genes include a gene encoding an ent-copalyl diphosphate synthase (CDPS) (EC 5.5.1.13), a gene encoding an ent-kaurene synthase, a gene encoding an ent-kaurene oxidase; or a gene encoding a steviol synthetase. At least one of the genes is a recombinant gene. The host can also be a plant cell. Expression of these gene(s) in a *Stevia* plant can result in increased steviol glycoside levels in the plant. In some embodiments the recombinant host further comprises a plurality of copies of a recombinant gene encoding a CDPS polypeptide (EC 5.5.1.13) lacking a chloroplast transit peptide sequence. The CDPS polypeptide can have at least 90%, 95%, 99%, or 100% identity to the truncated CDPS amino acid sequence set forth in FIG. 14. The host can further

comprise a plurality of copies of a recombinant gene encoding a KAH polypeptide, e.g., a KAH polypeptide that has at least 90%, 95%, 99%, or 100% identity to the KAH amino acid sequence set forth in FIG. 12. The host can further comprise one or more of: (i) a gene encoding a geranylgeranyl diphosphate synthase; (ii) a gene encoding an ent-kaurene oxidase; and (iii) a gene encoding an ent-kaurene synthase. The host can further comprise one or more of (iv) a gene encoding a truncated HMG-CoA; (v) a gene encoding a CPR; (vi) a gene encoding a rhamnose synthetase; (vii) a gene encoding a UDP-glucose dehydrogenase; and (viii) a gene encoding a UDP-glucuronic acid decarboxylase. Two or more exogenous CPRs can be present, for example. The expression of one or more of such genes can be inducible. At least one of genes (i), (ii), (iii), (iv), (v), (vi), (vii), or (viii) can be a recombinant gene, and in some cases each of the genes of (i), (ii), (iii), (iv), (v), (vi), (vii), and (viii) is a recombinant gene. The geranylgeranyl diphosphate synthase can have greater than 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:127; the kaurene oxidase can have greater than 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:138; a CPR can have greater than 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:168; a CPR can have greater than 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:170, and a kaurene synthase can have greater than 90% sequence identity to the amino acid sequence set forth in SEQ ID NO: 156.

In one aspect, this document features an isolated nucleic acid encoding a polypeptide having the amino acid sequence set forth in SEQ ID NO:5, wherein the polypeptide contains substitutions position 211 and 286 of SEQ ID NO:5. For example, the polypeptide can include a methionine at position 211 and an alanine at position 286.

In one aspect, this document features an isolated nucleic acid encoding a polypeptide having at least 80% identity (e.g., at least 85%, 90%, 95%, or 99% identity) to the amino acid sequence set forth in FIG. 12C (SEQ ID NO:164). The polypeptide can have the amino acid sequence set forth in FIG. 12C.

In another aspect, this document features a nucleic acid construct that included a regulatory region operably linked to a nucleic acid encoding a polypeptide having at least 80% identity (e.g., at least 85%, 90%, 95%, or 99% identity) to the amino acid sequence set forth in FIG. 12C (SEQ ID NO:164). The polypeptide can have the amino acid sequence set forth in FIG. 12C.

This document also features a recombinant host that includes a recombinant gene (e.g., a plurality of copies of a recombinant gene) encoding a KAH polypeptide having at least 80% identity (e.g., at least 85%, 90%, 95%, or 99% identity) to the amino acid sequence set forth in FIG. 12C. The polypeptide can have the amino acid sequence set forth in FIG. 12C. The host can be a microorganism such as a saccharomycete (e.g., *Saccharomyces cerevisiae*) or *Escherichia coli*. The host can be a plant or plant cell (e.g., a *Stevia*, *Physcomitrella*, or tobacco plant or plant cell). The *Stevia* plant or plant cell is a *Stevia rebaudiana* plant or plant cell. The recombinant host can produce steviol when cultured under conditions in which each of the genes is expressed. The recombinant host can further comprise a gene encoding a UGT74G1 polypeptide; a gene encoding a UGT85C2 polypeptide; a gene encoding a UGT76G1 polypeptide; a gene encoding a UGT91D2 polypeptide; and/or a gene encoding a EUGT11 polypeptide. Such a host can produce at least one steviol glycoside when cultured under conditions in which each of the genes is expressed. The

steviol glycoside can be steviol-13-O-glucoside, steviol-19-O-glucoside, rubusoside, rebaudioside A, rebaudioside B, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F, and/or dulcoside A. The recombinant host can further comprise one or more of: a gene encoding a deoxyxylulose 5-phosphate synthase (DXS); a gene encoding a D-1-deoxyxylulose 5-phosphate reductoisomerase (DXR); a gene encoding a 4-diphosphocytidyl-2-C-methyl-D-erythritol synthase (CMS); a gene encoding a 4-diphosphocytidyl-2-C-methyl-D-erythritol kinase (CMK); a gene encoding a 4-diphosphocytidyl-2-C-methyl-D-erythritol 2,4-cyclodiphosphate synthase (MCS); a gene encoding a 1-hydroxy-2-methyl-2(E)-butenyl 4-diphosphate synthase (HDS); and a gene encoding a 1-hydroxy-2-methyl-2(E)-butenyl 4-diphosphate reductase (HDR). The recombinant host can further comprise one or more of: a gene encoding an acetoacetyl-CoA thiolase; a gene encoding a truncated HMG-CoA reductase; a gene encoding a mevalonate kinase; a gene encoding a phosphomevalonate kinase; and a gene encoding a mevalonate pyrophosphate decarboxylase. In another aspect, this document features a recombinant host that further comprises a gene encoding an ent-kaurene synthase (EC 4.2.3.19) and/or a gene encoding a gibberellin 20-oxidase (EC 1.14.11.12). Such a host produces gibberellin GA3 when cultured under conditions in which each of the genes is expressed.

This document also features an isolated nucleic acid encoding a CPR polypeptide having at least 80% sequence identity (e.g., at least 85%, 90%, 95%, or 99% sequence identity) to the *S. rebaudiana* CPR amino acid sequence set forth in FIG. 13. In some embodiments, the polypeptide has the *S. rebaudiana* CPR amino acid sequence set forth in FIG. 13 (SEQ ID NOs: 169 and 170).

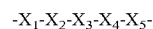
In any of the hosts described herein, expression of one or more of the genes can be inducible.

In any of the hosts described herein, one or more genes encoding endogenous phosphatases can be deleted or disrupted such that endogenous phosphatase activity is reduced. For example, the yeast gene DPP1 and/or LPP1 can be disrupted or deleted such that the degradation of farnesyl pyrophosphate (FPP) to farnesol is reduced and the degradation of geranylgeranylpyrophosphate (GGPP) to geranylgeraniol (GGOH) is reduced.

In another aspect, as described herein, ERG9 can be modified as defined below, resulting in the decreased production of squalene synthase (SQS) and an accumulation of terpenoid precursors. The precursors may or may not be secreted into the culture medium and can in turn be used as substrates to enzymes capable of metabolizing the terpenoid precursors into desired terpenoids.

Thus, in a main aspect the present invention relates to a cell comprising a nucleic acid sequence, said nucleic acid comprising

- i) a promoter sequence operably linked to
 - ii) a heterologous insert sequence operably linked to
 - iii) an open reading frame operably linked to
 - iv) a transcription termination signal,
- wherein the heterologous insert sequence has the general formula (I):



wherein X_2 comprises at least 4 consecutive nucleotides being complementary to, and forming a hairpin secondary structure element with at least 4 consecutive nucleotides of X_4 , and

wherein X_3 is optional and if present comprises unpaired nucleotides involved in forming a hairpin loop between X_2 and X_4 , and

wherein X_1 and X_5 individually and optionally comprises one or more nucleotides, and

wherein the open reading frame upon expression encodes a polypeptide sequence having at least 70% identity to a squalene synthase (EC 2.5.1.21) or a biologically active fragment thereof, said fragment having at least 70% sequence identity to said squalene synthase in a range of overlap of at least 100 amino acids.

The cell of the present invention is useful in enhancing yield of industrially interesting terpenoids. Accordingly, in another aspect the present invention relates to a method for producing a terpenoid compound synthesized through the squalene pathway, in a cell culture, said method comprising the steps of

- (a) providing the cell as defined herein above,
- (b) culturing the cell of (a).
- (c) recovering the terpenoid product compound.

By providing the cell comprising the genetically modified construct defined herein above, the accumulation of terpenoid precursors is enhanced (see e.g., FIG. 20).

Thus, in another aspect, the invention relates to a method for producing a terpenoid derived from a terpenoid precursor selected from the group consisting of Farnesyl-pyrophosphate (FPP), Isopentenyl-pyrophosphate (IPP), Dimethylallyl-pyrophosphate (DMAPP), Geranyl-pyrophosphate (GPP) and/or Geranylgeranyl-pyrophosphate (GGPP), said method comprising:

- (a) contacting said precursor with an enzyme of the squalene synthase pathway,
- (b) recovering the terpenoid product.

The present invention may operate by at least partly, sterically hindering binding of the ribosome to the RNA thus reducing the translation of squalene synthase.

Accordingly, in one aspect the present invention relates to a method for reducing the translation rate of a functional squalene synthase (EC 2.5.1.21) said method comprising:

- (a) providing the cell defined herein above,
- (b) culturing the cell of (a).

Similarly, the invention in another aspect relates to a method for decreasing turnover of farnesyl-pp to squalene, said method comprising:

- (a) providing the cell defined herein above,
- (b) culturing the cell of (a).

As depicted in FIG. 20, the knocking down of the ERG9 results in build-up of precursors to squalene synthase. Thus in one aspect, the present invention relates to a method for enhancing accumulation of a compound selected from the group consisting of Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate and Geranylgeranyl-pyrophosphate, said method comprising the steps of:

- (a) providing the cell defined herein above, and
- (b) culturing the cell of (a).

In one embodiment the invention relates to the production of Geranylgeranyl Pyrophosphate (GGPP) as well as other terpenoids, which can be prepared from Geranylgeranyl Pyrophosphate (GGPP).

In this embodiment of the invention the above described decrease of production of squalene synthase (SQS) may be combined with an increase in activity of Geranylgeranyl Pyrophosphate Synthase (GGPPS), which converts FPP to Geranylgeranyl Pyrophosphate (GGPP), leading to increased production of GGPP.

Thus, in one embodiment the invention relates to a microbial cell comprising a nucleic acid sequence, said nucleic acid comprising

- i) a promoter sequence operably linked to
- ii) a heterologous insert sequence operably linked to
- iii) an open reading frame operably linked to
- iv) a transcription termination signal,

wherein the heterologous insert sequence and the open reading frame are as defined herein above,

wherein said microbial cell furthermore comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell.

In addition, the document relates to a method for producing steviol or a steviol glycoside, wherein the method comprises use of any one of the above-mentioned microbial cells.

Any of the hosts described herein can be a microorganism (e.g., a *Saccharomycete* such as *Saccharomyces cerevisiae*, or *Escherichia coli*), or a plant or plant cell (e.g., a *Stevia* such as a *Stevia rebaudiana*, *Physcomitrella*, or tobacco plant or plant cell).

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the invention belongs. Although methods and materials similar or equivalent to those described herein can be used to practice the invention, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and are not intended to be limiting. Other features and advantages of the invention will be apparent from the following detailed description. Applicants reserve the right to alternatively claim any disclosed invention using the transitional phrase "comprising," "consisting essentially of," or "consisting of," according to standard practice in patent law.

DESCRIPTION OF DRAWINGS

FIG. 1 is the chemical structure of various steviol glycosides.

FIGS. 2A-D show representative pathways for the biosynthesis of steviol glycosides from steviol.

FIG. 3 is a schematic representation of 19-O-1,2-diglycosylation reactions by EUGT11 and UGT91D2e. The numbers are the average signal intensity for the substrates or the products of the reaction, from liquid chromatography-mass spectrometry (LC-MS) chromatograms. FIG. 4 contains LC-MS chromatograms showing the production of rebaudioside D (RebD) from Rebaudioside A (RebA) using in vitro transcribed and translated UGT91D2e (SEQ ID NO:5) (left panel) or EUGT11 (SEQ ID NO:152) (right panel). The LC-MS was set to detect certain masses corresponding to steviol+5 glucoses (such as RebD), steviol+4 glucoses (such as RebA) etc. Each 'lane' is scaled according to the highest peak.

FIG. 5 contains LC-MS chromatograms showing the conversion of rubusoside to stevioside and compounds '2' and '3' (RebE) by UGT91D2e (left panel) and EUGT11 (right panel).

FIG. 6 is an alignment of the amino acid sequence of EUGT11 (SEQ ID NO:152, top line) with the amino acid sequence of UGT91D2e (SEQ ID NO:5, bottom line).

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FIG. 7 contains the amino acid sequence of EUGT11 (SEQ ID NO:152), the nucleotide sequence (SEQ ID NO:153) encoding EUGT11, and the nucleotide sequence encoding EUGT11 that has been codon optimized for expression in yeast (SEQ ID NO: 154).

FIG. 8 is an alignment of the secondary structure predictions of UGT91D2e with UGT85H2 and UGT71G1. Secondary structure predictions were made by subjecting the amino acid sequences of the three UGTs to NetSurfP ver. 1.1—Protein Surface Accessibility and Secondary Structure Predictions, at the world wide web at cbs.dtu.dk/services/NetSurfP/. This predicted the presence and location of alpha helices, beta sheets and coils in the proteins. These were subsequently labeled as shown for UGT91D2e. For example, the first N-terminal beta-sheet was labeled Nβ1. The y-axis represents the certainty of the prediction, the higher the more confident and the x-axis represents amino acid position. Although the primary sequence identity between these UGTs is very low, the secondary structures show a very high degree of conservation.

FIG. 9 is an alignment of the amino acid sequences of UGT91D1 and UGT91D2e (SEQ ID NO: 5).

FIG. 10 is a bar graph of the activity of double amino acid substitution mutants of UGT91D2e. The filled bars represent stevioside production and the open bars represent 1,2-bioside production.

FIG. 11 is a schematic representation of UDP-glucose regeneration for the biosynthesis of steviol glycosides. SUS=sucrose synthase; Steviol=steviol or steviol glycoside substrate; UGT=UDP glycosyl transferase.

FIG. 12A is the nucleotide sequence encoding the *Stevia rebaudiana* KAH (SEQ ID NO:163), designated SrKAHe1 herein.

FIG. 12B is the nucleotide sequence encoding the *Stevia rebaudiana* KAHel that has been codon-optimized for expression in yeast (SEQ ID NO:165).

FIG. 12C is the amino acid sequence of the *Stevia rebaudiana* KAHel (SEQ ID NO:164).

FIG. 13A contains the amino acid sequences of CPR polypeptides from *S. cerevisiae* (encoded by NCP1 gene) (SEQ ID NO:166), *A. thaliana* (encoded by ATR1 and encoded by ATR2) ((SEQ ID NOs: 148 and 168), and *S. rebaudiana* (encoded by CPR7 and encoded by CPR8) (SEQ ID NOs: 169 and 170).

FIG. 13B contains ATR1 nucleotide sequence (Accession No. CAA23011) that has been codon optimized for expression in yeast (SEQ ID NO:171); ATR2 nucleotide sequence that has been codon optimized for expression in yeast (SEQ ID NO:172); the *Stevia rebaudiana* CPR7 nucleotide sequence (SEQ ID NO:173); and the *Stevia rebaudiana* CPR8 nucleotide sequence (SEQ ID NO:174).

FIG. 14A contains the nucleotide sequence (SEQ ID NO:157) encoding a CDPS polypeptide (SEQ ID NO:158) from *Zea mays*. The sequence that is in bold and underlined can be deleted to remove the sequence encoding the chloroplast transit sequence.

FIG. 14B contains the amino acid sequence of the CDPS polypeptide (SEQ ID NO:158) from *Zea mays*. The sequence that is in bold and underlined can be deleted to remove the chloroplast transit sequence.

FIG. 15A contains a codon-optimized nucleotide sequence (SEQ ID NO:161) encoding a bifunctional CDPS-KS polypeptide (SEQ ID NO:162) from *Gibberella fujikuroi*. FIG. 15B contains the amino acid sequence of the bifunctional CDPS-KS polypeptide (SEQ ID NO:162) from *Gibberella fujikuroi*.

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FIG. 16 is a graph of the growth of two strains of *S. cerevisiae*, enhanced EFSC1972 (designated T2) and enhanced EFSC1972 with further overexpression of the *Arabidopsis thaliana* kaurene synthase (KS-5) (designated T7, squares). Numbers on the y-axis are OD600 values of the cell culture, while numbers on the x-axis represent hours of growth in synthetic complete based medium at 30° C.

FIG. 17 contains the nucleic acid sequences encoding the *A. thaliana*, *S. rebaudiana* (from contig10573 selection_ ORF S11E, with the mutation that changes S11 to glutamate (E) in bold, lowercase letters), and coffee (*Coffea arabica*) sucrose synthases, SEQ ID NOs:175, 176, and 177, respectively.

FIG. 18 is a bar graph of rebD production in permeabilized *S. cerevisiae*, which had been transformed with EUGT11 or an empty plasmid ("Empty"). Cells were grown to exponential growth phase, washed in PBS buffer and subsequently treated with Triton X-100 (0.3% or 0.5% in PBS) 30° C., 30 min. After permeabilization cells were washed in PBS and resuspended in reaction mix containing 100 μM RebA and 300 μM UDP-glucose. Reactions proceeded for 20 h, 30° C.

FIG. 19A is the amino acid sequence of the *A. thaliana* UDP-glycosyltransferase UGT72E2 (SEQ ID NO:178).

FIG. 19B is the amino acid sequence of the sucrose transporter SUC1 from *A. thaliana* (SEQ ID NO:179).

FIG. 19C is the amino acid sequence of the sucrose synthase from coffee (SEQ ID NO:180).

FIG. 20 is a schematic of the isoprenoid pathway in yeast, showing the position of ERG9.

FIG. 21 contains the nucleotide sequence of the *Saccharomyces cerevisiae* Cycl promoter (SEQ ID NO:185) and *Saccharomyces cerevisiae* Kex2 promoter (SEQ ID NO:186).

FIG. 22 is a schematic of the PCR product containing two regions, HR1 and HR2, which are homologous to parts of the genome sequence within the ERG9 promoter or 5' end of the ERG9 open reading frame (ORF), respectively. Also, on the PCR product is an antibiotic marker, NatR, which can be embedded between two Lox sites (L) for subsequent excision with Cre recombinase. The PCR product further can include a promoter, such as either the wild type ScKex2, wild type ScCyc1, and the promoter further can include a heterologous insert such as a hairpin (SEQ ID NO: 181-184) at its 3'-end (See FIG. 23).

FIG. 23 is a schematic of promoter and ORF with a hairpin stemloop immediately upstream of the translation startsite (arrow) and an alignment of a portion of the wild-type *S. cerevisiae* Cycl promoter sequence and initial ATG of the ERG9 OPR without a heterologous insert (SEQ ID NO:187) and with four different heterologous inserts (SEQ ID NOs. 188-191). 75% refers to construct comprising the ScCyc1 promoter followed by SEQ ID NO: 184 (SEQ ID NO:191); 50% refers to construct comprising the ScCyc1 promoter followed by SEQ ID NO: 183 (SEQ ID NO:190); 20% refers to construct comprising the ScCyc1 promoter followed by SEQ ID NO: 182 (SEQ ID NO:189); 5% refers to construct comprising the ScCyc1 promoter followed by SEQ ID NO: 181 (SEQ ID NO:188).

FIG. 24 is a bar graph showing amorphaadiene produced in yeast strains with different promoter constructs inserted in front of the ERG9 gene of the host genome. CTRL-ADS refers to control strain with no modification; ERG9-CYC1-100% refers to construct comprising the ScCyc1 promoter and no insert; ERG9-CYC1-50% refers to construct comprising the ScCyc1 promoter followed by SEQ ID NO: 183 (SEQ ID NO:190); ERG9-CYC1-20% refers to construct

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comprising the ScCyc1 promoter followed by SEQ ID NO: 182 (SEQ ID NO:189); ERG9-CYC1-5% refers to construct comprising the ScCyc1 promoter followed by SEQ ID NO: 181 (SEQ ID NO:188); ERG9-KEX2-100% refers to construct comprising the ScKex2 promoter.

FIG. 25 contains the amino acid sequence of squalene synthase polypeptides from *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Yarrowia lipolytica*, *Candida glabrata*, *Ashbya gossypii*, *Cyberlindnera jadinii*, *Candida albicans*, *Saccharomyces cerevisiae*, *Homo sapiens*, *Mus musculus*, and *Rattus norvegicus* (SEQ ID NOs:192-202), and the amino acid sequence of a geranylgeranyl diphosphate synthase (GGPPS) from *Aspergillus nidulans* and *S. cerevisiae* (SEQ ID NOs. 203 and 167).

FIG. 26 is a bar graph of geranylgeraniol (GGOH) accumulation in the ERG9-CYC1-5% strain and ERG9-KEX2 strain after 72 hours.

FIG. 27 is a representative chromatograph showing the conversion of rubusoside to xylosylated intermediates for RebF production by UGT91D2e and EUGT11.

Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

This document is based on the discovery that recombinant hosts such as plant cells, plants, or microorganisms can be developed that express polypeptides useful for the biosynthesis of steviol glycosides such as rebaudioside A, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F, or dulcoside A. The recombinant hosts described herein are particularly useful for producing Rebaudioside D. Such hosts can express one or more Uridine 5'-diphospho (UDP) glycosyl transferases suitable for producing steviol glycosides. Expression of these biosynthetic polypeptides in various microbial chassis allows steviol glycosides to be produced in a consistent, reproducible manner from energy and carbon sources such as sugars, glycerol, CO₂, H₂, and sunlight. The proportion of each steviol glycoside produced by a recombinant host can be tailored by incorporating preselected biosynthetic enzymes into the hosts and expressing them at appropriate levels, to produce a sweetener composition with a consistent taste profile. Furthermore, the concentrations of steviol glycosides produced by recombinant hosts are expected to be higher than the levels of steviol glycosides produced in the *Stevia* plant, which improves the efficiency of the downstream purification. Such sweetener compositions contain little or no plant based contaminants, relative to the amount of contaminants present in *Stevia* extracts.

At least one of the genes is a recombinant gene, the particular recombinant gene(s) depending on the species or strain selected for use. Additional genes or biosynthetic modules can be included in order to increase steviol glycoside yield, improve efficiency with which energy and carbon sources are converted to steviol and its glycosides, and/or to enhance productivity from the cell culture or plant. Such

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additional biosynthetic modules include genes involved in the synthesis of the terpenoid precursors, isopentenyl diphosphate and dimethylallyl diphosphate. Additional biosynthetic modules include terpene synthase and terpene cyclase genes, such as genes encoding geranylgeranyl diphosphate synthase and copalyl diphosphate synthase; these genes may be endogenous genes or recombinant genes.

I. STEVIOL AND STEVIOL GLYCOSIDE BIOSYNTHESIS POLYPEPTIDES

A. Steviol Biosynthesis Polypeptides

Chemical structures for several of the compounds found in *Stevia* extracts are shown in FIG. 1, including the diterpene steviol and various steviol glycosides. CAS numbers are shown in Table A below. See also, *Steviol Glycosides Chemical and Technical Assessment 69th JECFA*, prepared by Harriet Wallin, Food Agric. Org. (2007).

TABLE A

COMPOUND	CAS #
Steviol	471-80-7
Rebaudioside A	58543-16-1
Steviolbioside	41093-60-1
Stevioside	57817-89-7
Rebaudioside B	58543-17-2
Rebaudioside C	63550-99-2
Rebaudioside D	63279-13-0
Rebaudioside E	63279-14-1
Rebaudioside F	438045-89-7
Rubusoside	63849-39-4
Dulcoside A	64432-06-0

It has been discovered that expression of certain genes in a host such as a microorganism confers the ability to synthesize steviol glycosides upon that host. As discussed in more detail below, one or more of such genes may be present naturally in a host. Typically, however, one or more of such genes are recombinant genes that have been transformed into a host that does not naturally possess them.

The biochemical pathway to produce steviol involves formation of geranylgeranyl diphosphate, cyclization to (-) copalyl diphosphate, followed by oxidation and hydroxylation to form steviol. Thus, conversion of geranylgeranyl diphosphate to steviol in a recombinant microorganism involves the expression of a gene encoding a kaurene synthase (KS), a gene encoding a kaurene oxidase (KO), and a gene encoding a steviol synthetase (KAH). Steviol synthetase also is known as kaurenoic acid 13-hydroxylase.

Suitable KS polypeptides are known. For example, suitable KS enzymes include those made by *Stevia rebaudiana*, *Zea mays*, *Populus trichocarpa*, and *Arabidopsis thaliana*. See, Table 1 and SEQ ID NOs: 132-135 and 156. Nucleotide sequences encoding these polypeptides are set forth in SEQ ID NOs: 40-47 and 155. The nucleotide sequences set forth in SEQ ID NOs:40-43 were modified for expression in yeast while the nucleotide sequences set forth in SEQ ID NOs: 44-47 are from the source organisms from which the KS polypeptides were identified.

TABLE 1

KS Clones						
Enzyme Source Organism	gi Number	Accession Number	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Stevia rebaudiana</i>	4959241	AAD34295	MM-12	2355	40	132

TABLE 1-continued

KS Clones						
Enzyme Source Organism	gi Number	Accession Number	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Stevia rebaudiana</i>	4959239	AAD34294	MM-13	2355	41	133
<i>Zea mays</i>	162458963	NP_001105097	MM-14	1773	42	134
<i>Populus trichocarpa</i>	224098838	XP_002311286	MM-15	2232	43	135
<i>Arabidopsis thaliana</i>	3056724	AF034774	EV-70	2358	155	156

Suitable KO polypeptides are known. For example, suitable KO enzymes include those made by *Stevia rebaudiana*, *Arabidopsis thaliana*, *Gibberella fujikori* and *Trametes versicolor*. See, Table 2 and SEQ ID NOs: 138-141. Nucleotide sequences encoding these polypeptides are set forth in SEQ ID NOs: 52-59. The nucleotide sequences set forth in SEQ ID NOs: 52-55 were modified for expression in yeast. The nucleotide sequences set forth in SEQ ID NOs: 56-59 are from the source organisms from which the KO polypeptides were identified.

TABLE 2

KO Clones						
Enzyme Source Organism	gi Number	Accession Number	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Stevia rebaudiana</i>	76446107	ABA42921	MM-18	1542	52	138
<i>Arabidopsis thaliana</i>	3342249	AAC39505	MM-19	1530	53	139
<i>Gibberella fujikori</i>	4127832	CAA76703	MM-20	1578	54	140
<i>Trametes versicolor</i>	14278967	BAB59027	MM-21	1500	55	141

Suitable KAH polypeptides are known. For example, suitable KAH enzymes include those made by *Stevia rebaudiana*, *Arabidopsis thaliana*, *Vitis vinifera* and *Medicago trunculata*. See, e.g., Table 3, SEQ ID NOs: 142-146; U.S. Patent Publication No. 2008-0271205; U.S. Patent Publication No. 2008-0064063 and Genbank Accession No. gi 189098312. The steviol synthetase from *Arabidopsis thali-*

ana is classified as a CYP714A2. Nucleotide sequences encoding these KAH enzymes are set forth in SEQ ID NOs: 60-69. The nucleotide sequences set forth in SEQ ID NOs: 60-64 were modified for expression in yeast while the nucleotide sequences from the source organisms from which the polypeptides were identified are set forth in SEQ ID NOs: 65-69.

TABLE 3

KAH Clones							
Enzyme Source Organism	gi Number	Accession Number	Plasmid Name	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Stevia rebaudiana</i>	—*	—	pMUS35	MM-22	1578	60	142
<i>Stevia rebaudiana</i>	189418962	ACD93722	pMUS36	MM-23	1431	61	143
<i>Arabidopsis thaliana</i>	15238644	NP_197872	pMUS37	MM-24	1578	62	144
<i>Vitis vinifera</i>	225458454	XP_002282091	pMUS38	MM-25	1590	63	145
<i>Medicago trunculata</i>	84514135	ABC59076	pMUS39	MM-26	1440	64	146

*= Sequence is shown in U.S. patent publication No. 2008-0064063.

In addition, a KAH polypeptide from *Stevia rebaudiana* that was identified herein is particularly useful in a recombinant host. The nucleotide sequence (SEQ ID NO:163) encoding the *S. rebaudiana* KAH (SrKAHe1) (SEQ ID NO:164) is set forth in FIG. 12A. A nucleotide sequence encoding the *S. rebaudiana* KAH that has been codon-optimized for expression in yeast (SEQ ID NO:165) is set forth in FIG. 12B. The amino acid sequence of the *S. rebaudiana* KAH is set forth in FIG. 12C. The *S. rebaudiana* KAH shows significantly higher steviol synthase activity as compared to the *Arabidopsis thaliana* *ent*-kaurenoic acid hydroxylase described by Yamaguchi et al. (U.S. Patent Publication No. 2008/0271205 A1) when expressed in *S. cerevisiae*. The *S. rebaudiana* KAH polypeptide set forth in FIG. 12C has less than 20% identity to the KAH from U.S. Patent Publication No. 2008/0271205, and less than 35% identity to the KAH from U.S. Patent Publication No. 2008/0064063.

In some embodiments, a recombinant microorganism contains a recombinant gene encoding a KO and/or a KAH polypeptide. Such microorganisms also typically contain a recombinant gene encoding a cytochrome P450 reductase (CPR) polypeptide, since certain combinations of KO and/or KAH polypeptides require expression of an exogenous CPR polypeptide. In particular, the activity of a KO and/or a KAH polypeptide of plant origin can be significantly increased by the inclusion of a recombinant gene encoding an exogenous CPR polypeptide. Suitable CPR polypeptides are known. For example, suitable CPR enzymes include those made by *Stevia rebaudiana* and *Arabidopsis thaliana*. See, e.g., Table 4 and SEQ ID NOs: 147 and 148. Nucleotide sequences encoding these polypeptides are set forth in SEQ ID NOs: 70, 71, 73, and 74. The nucleotide sequences set forth in SEQ ID NOs: 70-72 were modified for expression in yeast. The nucleotide sequences from the source organisms from which the polypeptides were identified are set forth in SEQ ID NOs:73-75.

TABLE 4

CPR Clones							
Enzyme Source Organism	gi Number	Accession Number	Plasmid Name	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Stevia rebaudiana</i>	93211213	AB88839	pMUS40	MM-27	2133	70	147
<i>Arabidopsis thaliana</i>	15233853	NP_194183	pMUS41	MM-28	2079	71	148
<i>Gibberella fujikuroi</i>	32562989	CAE09055	pMUS42	MM-29	2142	72	149

For example, the steviol synthase encoded by SrKAHe1 is activated by the *S. cerevisiae* CPR encoded by gene NCPI (YHR042W). Even better activation of the steviol synthase encoded by SrKAHe1 is observed when the *Arabidopsis thaliana* CPR encoded by the gene ATR2 or the *S. rebaudiana* CPR encoded by the gene CPR8 are co-expressed. FIG. 13A contains the amino acid sequence of the *S. cerevisiae*, *A. thaliana* (from ATR1 and ATR2 genes) and *S. rebaudiana* CPR polypeptides (from CPR7 and CPR8 genes) (SEQ ID NOs: 166-170). FIG. 13 B contains the nucleotide sequence encoding the *A. thaliana* and *S. rebaudiana* CPR polypeptides (SEQ ID NOs:171-174).

For example, the yeast gene DPP1 and/or the yeast gene LPP1 can be disrupted or deleted such that the degradation of farnesyl pyrophosphate (FPP) to farnesol is reduced and

the degradation of geranylgeranylpyrophosphate (GGPP) to geranylgeraniol (GGOH) is reduced. Alternatively, the promoter or enhancer elements of an endogenous gene encoding a phosphatase can be altered such that the expression of their encoded proteins is altered. Homologous recombination can be used to disrupt an endogenous gene. For example, a "gene replacement" vector can be constructed in such a way to include a selectable marker gene. The selectable marker gene can be operably linked, at both 5' and 3' end, to portions of the gene of sufficient length to mediate homologous recombination. The selectable marker can be one of any number of genes that complement host cell auxotrophy, provide antibiotic resistance, or result in a color change. Linearized DNA fragments of the gene replacement vector then are introduced into the cells using methods well known in the art (see below). Integration of the linear fragments into the genome and the disruption of the gene can be determined based on the selection marker and can be verified by, for example, Southern blot analysis. Subsequent to its use in selection, a selectable marker can be removed from the genome of the host cell by, e.g., Cre-loxP systems (see, e.g., Gossen et al. (2002) *Ann. Rev. Genetics* 36:153-173 and U.S. Application Publication No. 20060014264). Alternatively, a gene replacement vector can be constructed in such a way as to include a portion of the gene to be disrupted, where the portion is devoid of any endogenous gene promoter sequence and encodes none, or an inactive fragment of, the coding sequence of the gene. An "inactive fragment" is a fragment of the gene that encodes a protein having, e.g., less than about 10% (e.g., less than about 9%, less than about 8%, less than about 7%, less than about 6%, less than about 5%, less than about 4%, less than about 3%, less than about 2%, less than about 1%, or 0%) of the activity of the protein produced from the full-length coding sequence of the gene. Such a portion of the gene is inserted in a vector in such a way that no known promoter sequence is operably linked to the gene sequence, but that a stop codon and a

transcription termination sequence are operably linked to the portion of the gene sequence. This vector can be subsequently linearized in the portion of the gene sequence and transformed into a cell. By way of single homologous recombination, this linearized vector is then integrated in the endogenous counterpart of the gene.

Expression in a recombinant microorganism of these genes results in the conversion of geranylgeranyl diphosphate to steviol.

B. Steviol Glycoside Biosynthesis Polypeptides

A recombinant host described herein can convert steviol to a steviol glycoside. Such a host (e.g., microorganism) contains genes encoding one or more UDP Glycosyl Transferases, also known as UGTs. UGTs transfer a monosaccharide unit from an activated nucleotide sugar to an acceptor

moiety, in this case, an —OH or —COOH moiety on steviol or steviol derivative. UGTs have been classified into families and subfamilies based on sequence homology. Li et al. *J. Biol. Chem.* 276:4338-4343 (2001).

B. 1 Rubusoside Biosynthesis Polypeptides

The biosynthesis of rubusoside involves glycosylation of the 13-OH and the 19-COOH of steviol. See FIG. 2A. Conversion of steviol to rubusoside in a recombinant host such as a microorganism can be accomplished by the expression of gene(s) encoding UGTs 85C2 and 74G1, which transfer a glucose unit to the 13-OH or the 19-COOH, respectively, of steviol.

A suitable UGT85C2 functions as a uridine 5'-diphospho glucosyl:steviol 13-OH transferase, and a uridine 5'-diphospho glucosyl:steviol-19-O-glucoside 13-OH transferase. Functional UGT85C2 polypeptides also may catalyze glucosyl transferase reactions that utilize steviol glycoside substrates other than steviol and steviol-19-O-glucoside.

A suitable UGT74G1 polypeptide functions as a uridine 5'-diphospho glucosyl: steviol 19-COOH transferase and a uridine 5'-diphospho glucosyl: steviol-13-O-glucoside 19-COOH transferase. Functional UGT74G1 polypeptides also may catalyze glycosyl transferase reactions that utilize steviol glycoside substrates other than steviol and steviol-13-O-glucoside, or that transfer sugar moieties from donors other than uridine diphosphate glucose.

A recombinant microorganism expressing a functional UGT74G1 and a functional UGT85C2 can make rubusoside and both steviol monosides (i.e., steviol 13-O-monoglucoside and steviol 19-O-monoglucoside) when steviol is used as a feedstock in the medium. One or more of such genes may be present naturally in the host. Typically, however, such genes are recombinant genes that have been transformed into a host (e.g., microorganism) that does not naturally possess them.

As used herein, the term recombinant host is intended to refer to a host, the genome of which has been augmented by at least one incorporated DNA sequence. Such DNA sequences include but are not limited to genes that are not naturally present, DNA sequences that are not normally transcribed into RNA or translated into a protein ("expressed"), and other genes or DNA sequences which one desires to introduce into the non-recombinant host. It will be appreciated that typically the genome of a recombinant host described herein is augmented through the stable introduction of one or more recombinant genes. Generally, the introduced DNA is not originally resident in the host that is the recipient of the DNA, but it is within the scope of the invention to isolate a DNA segment from a given host, and to subsequently introduce one or more additional copies of that DNA into the same host, e.g., to enhance production of the product of a gene or alter the expression pattern of a gene. In some instances, the introduced DNA will modify or even replace an endogenous gene or DNA sequence by, e.g., homologous recombination or site-directed mutagenesis. Suitable recombinant hosts include microorganisms, plant cells, and plants.

The term "recombinant gene" refers to a gene or DNA sequence that is introduced into a recipient host, regardless of whether the same or a similar gene or DNA sequence may already be present in such a host. "Introduced," or "augmented" in this context, is known in the art to mean introduced or augmented by the hand of man. Thus, a recombinant gene may be a DNA sequence from another species, or may be a DNA sequence that originated from or is present in the same species, but has been incorporated into a host by recombinant methods to form a recombinant host.

It will be appreciated that a recombinant gene that is introduced into a host can be identical to a DNA sequence that is normally present in the host being transformed, and is introduced to provide one or more additional copies of the DNA to thereby permit overexpression or modified expression of the gene product of that DNA.

Suitable UGT74G1 and UGT85C2 polypeptides include those made by *Stevia rebaudiana*. Genes encoding functional UGT74G1 and UGT85C2 polypeptides from *Stevia* are reported in Richman, et al. *Plant J.* 41: 56-67 (2005). Amino acid sequences of *S. rebaudiana* UGT74G1 and UGT85C2 polypeptides are set forth in SEQ ID NOs: 1 and 3, respectively. Nucleotide sequences encoding UGT74G1 and UGT85C2 that have been optimized for expression in yeast are set forth in SEQ ID NOs: 2 and 4, respectively. DNA 2.0 codon-optimized sequence for UGTs 85C2, 91D2e, 74G1 and 76G1 are set forth in SEQ ID NOs: 82, 84, 83, and 85, respectively. See also the UGT85C2 and UGT74G1 variants described below in the "Functional Homolog" section. For example, an UGT85C2 polypeptide containing substitutions at positions 65, 71, 270, 289, and 389 can be used (e.g., A65S, E71Q, T270M, Q289H, and A389V).

In some embodiments, the recombinant host is a microorganism. The recombinant microorganism can be grown on media containing steviol in order to produce rubusoside. In other embodiments, however, the recombinant microorganism expresses one or more recombinant genes involved in steviol biosynthesis, e.g., a CDPS gene, a KS gene, a KO gene and/or a KAH gene. Suitable CDPS polypeptides are known. For example, suitable CDPS enzymes include those made by *Stevia rebaudiana*, *Streptomyces clavuligerus*, *Bradyrhizobium japonicum*, *Zea mays*, and *Arabidopsis*. See, e.g., Table 5 and SEQ ID NOs: 129-131, 158, and 160. Nucleotide sequences encoding these polypeptides are set forth in SEQ ID NOs: 34-39, 157, and 159. The nucleotide sequences set forth in SEQ ID NOs: 34-36 were modified for expression in yeast. The nucleotide sequences from the source organisms from which the polypeptides were identified are set forth in SEQ ID NOs: 37-39.

In some embodiments, CDPS polypeptides that lack a chloroplast transit peptide at the amino terminus of the unmodified polypeptide can be used. For example, the first 150 nucleotides from the 5' end of the *Zea mays* CDPS coding sequence shown in FIG. 14 (SEQ ID NO:157) can be removed. Doing so removes the amino terminal 50 residues of the amino acid sequence shown in FIG. 14 (SEQ ID NO:158), which encode a chloroplast transit peptide. The truncated CDPS gene can be fitted with a new ATG translation start site and operably linked to a promoter, typically a constitutive or highly expressing promoter. When a plurality of copies of the truncated coding sequence are introduced into a microorganism, expression of the CDPS polypeptide from the promoter results in an increased carbon flux towards ent-kaurene biosynthesis.

TABLE 5

CDPS Clones							
Enzyme Source Organism	gi Number	Accession Number	Plasmid Name	Construct Name	Length (nts)	SEQ ID: (DNA)	SEQ ID (protein)
<i>Stevia rebaudiana</i>	2642661	AAB87091	pMUS22	MM-9	2364	34	129
<i>Streptomyces clavuligerus</i>	197705855	EDY51667	pMUS23	MM-10	1584	35	130
<i>Bradyrhizobium japonicum</i>	529968	AAC28895.1	pMUS24	MM-11	1551	36	131
<i>Zea mays</i>	50082774	AY562490		EV65	2484	157	158
<i>Arabidopsis thaliana</i>	18412041	NM_116512		EV64	2409	159	160

CDPS-KS bifunctional proteins (SEQ ID NOs: 136 and 137) also can be used. Nucleotide sequences encoding the CDPS-KS bifunctional enzymes shown in Table 6 were modified for expression in yeast (see SEQ ID NOs: 48 and 49). The nucleotide sequences from the source organisms from which the polypeptides were originally identified are set forth in SEQ ID NOs: 50 and 51. A bifunctional enzyme from *Gibberella fujikuroi* (SEQ ID NO:162) also can be used. A nucleotide sequence encoding the *Gibberella fujikuroi* bifunctional CDPS-KS enzyme was modified for expression in yeast (see FIG. 15A, SEQ ID NO: 161).

TABLE 6

CDPS-KS Clones						
Enzyme Source Organism	gi Number	Accession Number	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Phomopsis amygdali</i>	186704306	BAG30962	MM-16	2952	48	136
<i>Physcomitrella patens</i>	146325986	BAF61135	MM-17	2646	49	137
<i>Gibberella fujikuroi</i>	62900107	Q9UVY5.1		2859	161	162

Thus, a microorganism containing a CDPS gene, a KS gene, a KO gene and a PG-7T KAH gene in addition to a UGT74G1 and a UGT85C2 gene is capable of producing both steviol monosides and rubusoside without the necessity for using steviol as a feedstock.

In some embodiments, the recombinant microorganism further expresses a recombinant gene encoding a geranylgeranyl diphosphate synthase (GGPPS). Suitable GGPPS polypeptides are known. For example, suitable GGPPS enzymes include those made by *Stevia rebaudiana*, *Gibberella fujikuroi*, *Mus musculus*, *Thalassiosira pseudonana*, *Streptomyces clavuligerus*, *Sulfolobus acidocaldarius*, *Synechococcus* sp. and *Arabidopsis thaliana*. See, Table 7 and SEQ ID NOs: 121-128. Nucleotide sequences encoding

these polypeptides are set forth in SEQ ID NOs:18-33. The nucleotide sequences set forth in SEQ ID NOs: 18-25 were modified for expression in yeast while the nucleotide sequences from the source organisms from which the polypeptides were identified are set forth in SEQ ID NOs: 26-33.

TABLE 7

GGPPS Clones							
Enzyme Source Organism	gi Number	Accession Number	Plasmid Name	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Stevia rebaudiana</i>	90289577	ABD92926	pMUS14	MM-1	1086	18	121
<i>Gibberella fujikuroi</i>	3549881	CAA75568	pMUS15	MM-2	1029	19	122
<i>Mus musculus</i>	47124116	AAH69913	pMUS16	MM-3	903	20	123
<i>Thalassiosira pseudonana</i>	223997332	XP_002288339	pMUS17	MM-4	1020	21	124
<i>Streptomyces clavuligerus</i>	254389342	ZP_05004570	pMUS18	MM-5	1068	22	125
<i>Sulfolobus acidocaldarius</i>	506371	BAA43200	pMUS19	MM-6	993	23	126
<i>Synechococcus</i> sp.	86553638	ABC98596	pMUS20	MM-7	894	24	127

TABLE 7-continued

GGPPS Clones							
Enzyme Source Organism	gi Number	Accession Number	Plasmid Name	Construct Name	Length (nts)	SEQ ID (DNA)	SEQ ID (protein)
<i>Arabidopsis thaliana</i>	15234534	NP_195399	pMUS21	MM-8	1113	25	128

In some embodiments, the recombinant microorganism further can express recombinant genes involved in diterpene biosynthesis or production of terpenoid precursors, e.g., genes in the methylerythritol 4-phosphate (MEP) pathway or genes in the mevalonate (MEV) pathway discussed below, have reduced phosphatase activity, and/or express a sucrose synthase (SUS) as discussed herein.

B. 2 Rebaudioside A. Rebaudioside D. And Rebaudioside E Biosynthesis Polypeptides

The biosynthesis of rebaudioside A involves glucosylation of the aglycone steviol. Specifically, rebaudioside A can be formed by glucosylation of the 13-OH of steviol which forms the 13-O-steviolmonoside, glucosylation of the C-2' of the 13-O-glucose of steviolmonoside which forms steviol-1,2-bioside, glucosylation of the C-19 carboxyl of steviol-1,2-bioside which forms stevioside, and glucosylation of the C-3' of the C-13-O-glucose of stevioside. The order in which each glucosylation reaction occurs can vary. See FIG. 2A.

The biosynthesis of rebaudioside E and/or rebaudioside D involves glucosylation of the aglycone steviol. Specifically, rebaudioside E can be formed by glucosylation of the 13-OH of steviol which forms steviol-13-O-glucoside, glucosylation of the C-2' of the 13-O-glucose of steviol-13-O-glucoside which forms the steviol-1,2-bioside, glucosylation of the C-19 carboxyl of the 1,2-bioside to form 1,2-stevioside, and glucosylation of the C-2' of the 19-O-glucose of the 1,2-stevioside to form rebaudioside E. Rebaudioside D can be formed by glucosylation of the C-3' of the C-13-O-glucose of rebaudioside E. The order in which each glycosylation reaction occurs can vary. For example, the glucosylation of the C-2' of the 19-O-glucose may be the last step in the pathway, wherein Rebaudioside A is an intermediate in the pathway. See FIG. 2C.

It has been discovered that conversion of steviol to rebaudioside A, rebaudioside D, and/or rebaudioside E in a recombinant host can be accomplished by expressing the following functional UGTs: EUGT11, 74G1, 85C2, and 76G1, and optionally 91D2. Thus, a recombinant microorganism expressing combinations of these four or five UGTs can make rebaudioside A and rebaudioside D when steviol is used as a feedstock. Typically, one or more of these genes are recombinant genes that have been transformed into a microorganism that does not naturally possess them. It has also been discovered that UGTs designated herein as SM12UGT can be substituted for UGT91D2.

In some embodiments, less than five (e.g., one, two, three, or four) UGTs are expressed in a host. For example, a recombinant microorganism expressing a functional EUGT11 can make rebaudioside D when rebaudioside A is used as a feedstock. A recombinant microorganism expressing two functional UGTs, EUGT11 and 76G1, and optionally a functional 91D2, can make rebaudioside D when rubusoside or 1,2-stevioside is used as a feedstock. As another alternative, a recombinant microorganism expressing three functional UGTs, EUGT11, 74G1, 76G1, and

optionally 91D2, can make rebaudioside D when fed the monoside, steviol-13-O-glucoside, in the medium. Similarly, conversion of steviol-19-O-glucoside to rebaudioside D in a recombinant microorganism can be accomplished by the expression of genes encoding UGTs EUGT11, 85C2, 76G1, and optionally 91D2, when fed steviol-19-O-glucoside. Typically, one or more of these genes are recombinant genes that have been transformed into a host that does not naturally possess them.

Suitable UGT74G1 and UGT85C2 polypeptides include those discussed above. A suitable UGT76G1 adds a glucose moiety to the C-3' of the C-13-O-glucose of the acceptor molecule, a steviol 1,2 glycoside. Thus, UGT76G1 functions, for example, as a uridine 5'-diphospho glucosyl: steviol 13-O-1,2 glucoside C-3' glucosyl transferase and a uridine 5'-diphospho glucosyl: steviol-19-O-glucose, 13-O-1,2 bioside C-3' glucosyl transferase. Functional UGT76G1 polypeptides may also catalyze glucosyl transferase reactions that utilize steviol glycoside substrates that contain sugars other than glucose, e.g., steviol rhamnosides and steviol xylosides. See, FIGS. 2A, 2B, 2C and 2D. Suitable UGT76G1 polypeptides include those made by *S. rebaudiana* and reported in Richman, et al. *Plant J.* 41: 56-67 (2005). The amino acid sequence of a *S. rebaudiana* UGT76G1 polypeptide is set forth in SEQ ID NO:7. The nucleotide sequence encoding the UGT76G1 polypeptide of SEQ ID NO:7 has been optimized for expression in yeast and is set forth in SEQ ID NO:8. See also the UGT76G1 variants set forth in the "Functional Homolog" section.

A suitable EUGT11 or UGT91D2 polypeptide functions as a uridine 5'-diphospho glucosyl: steviol-13-O-glucoside transferase (also referred to as a steviol-13-monoglucoside 1,2-glucosylase), transferring a glucose moiety to the C-2' of the 13-O-glucose of the acceptor molecule, steviol-13-O-glucoside.

A suitable EUGT11 or UGT91D2 polypeptide also functions as a uridine 5'-diphospho glucosyl: rubusoside transferase transferring a glucose moiety to the C-2' of the 13-O-glucose of the acceptor molecule, rubusoside, to produce stevioside. EUGT11 polypeptides also can transfer a glucose moiety to the C-2' of the 19-O-glucose of the acceptor molecule, rubusoside, to produce a 19-O-1,2-diglycosylated rubusoside (compound 2 in FIG. 3).

Functional EUGT11 or UGT91D2 polypeptides also can catalyze reactions that utilize steviol glycoside substrates other than steviol-13-O-glucoside and rubusoside. For example, a functional EUGT11 polypeptide may utilize stevioside as a substrate, transferring a glucose moiety to the C-2' of the 19-O-glucose residue to produce Rebaudioside E (see compound 3 in FIG. 3). Functional EUGT11 and UGT91D2 polypeptides may also utilize Rebaudioside A as a substrate, transferring a glucose moiety to the C-2' of the 19-O-glucose residue of Rebaudioside A to produce Rebaudioside D. As set forth in the Examples, EUGT11 can convert Rebaudioside A to Rebaudioside D at a rate that is at least 20 times faster (e.g., as least 25 times or at least 30 times faster) than the corresponding rate of UGT91D2e (SEQ ID NO: 5) when the reactions are performed under similar conditions, i.e., similar time, temperature, purity, and

substrate concentration. As such, EUGT11 produces greater amounts of RebD than UGT91D2e when incubated under similar conditions.

In addition, a functional EUGT11 exhibits significant C-2' 19-O-diglycosylation activity with rubusoside or stevioside as substrates, whereas UGT91D2e has no detectable diglycosylation activity with these substrates. Thus, a functional EUGT11 can be distinguished from UGT91D2e by the differences in steviol glycoside substrate-specificity. FIG. 3 provides a schematic overview of the 19-O-1,2 diglycosylation reactions that are performed by EUGT11 and UGT91D2e.

A functional EUGT11 or UGT91D2 polypeptide typically does not transfer a glucose moiety to steviol compounds having a 1,3-bound glucose at the C-13 position, i.e., transfer of a glucose moiety to steviol 1,3-bioside and 1,3-stevioside does not occur.

Functional EUGT11 and UGT91D2 polypeptides can transfer sugar moieties from donors other than uridine diphosphate glucose. For example, a functional EUGT11 or UGT91D2 polypeptide can act as a uridine 5'-diphospho D-xylosyl: steviol-13-O-glucoside transferase, transferring a xylose moiety to the C-2' of the 13-O-glucose of the acceptor molecule, steviol-13-O-glucoside. As another example, a functional EUGT11 or UGT91D2 polypeptide can act as a uridine 5'-diphospho L-rhamnosyl: steviol-13-O-glucoside transferase, transferring a rhamnose moiety to the C-2' of the 13-O-glucose of the acceptor molecule, steviol-13-O-glucoside.

Suitable EUGT11 polypeptides are described herein and can include the EUGT11 polypeptide from *Oryza sativa* (GenBank Accession No. AC133334). For example, an EUGT11 polypeptide can have an amino acid sequence with at least 70% sequence identity (e.g., at least 75, 80, 85, 90, 95, 96, 97, 98, or 99% sequence identity) to the amino acid sequence set forth in SEQ ID NO: 152 (see FIG. 7). The nucleotide sequence encoding the amino acid sequence of SEQ ID NO: 152 is set forth in SEQ ID NO: 153. SEQ ID NO: 154 is a nucleotide sequence encoding the polypeptide of SEQ ID NO: 152 that has been codon optimized for expression in yeast.

Suitable functional UGT91D2 polypeptides include those disclosed herein, e.g., the polypeptides designated UGT91D2e and UGT91D2m. The amino acid sequence of an exemplary UGT91D2e polypeptide from *Stevia rebaudiana* is set forth in SEQ ID NO: 5. SEQ ID NO: 6 is a nucleotide sequence encoding the polypeptide of SEQ ID NO: 5 that has been codon optimized for expression in yeast. The *S. rebaudiana* nucleotide sequence encoding the polypeptide of SEQ ID NO: 5 is set forth in SEQ ID NO: 9. The amino acid sequences of exemplary UGT91D2m polypeptides from *S. rebaudiana* are set forth in SEQ ID NOs: 10 and 12, and are encoded by the nucleic acid sequences set forth in SEQ ID NOs: 11 and 13, respectively. In addition, UGT91D2 variants containing a substitution at amino acid residues 206, 207, and 343 of SEQ ID NO: 5 can be used. For example, the amino acid sequence set forth in SEQ ID NO: 95 and having the following mutations with respect to wild-type UGT92D2e (SEQ ID NO: 5) G206R, Y207C, and W343R can be used. In addition, a UGT91D2 variant containing substitutions at amino acid residues 211 and 286 can be used. For example, a UGT91D2 variant can include a substitution of a methionine for leucine at position 211 and a substitution of an alanine for valine at position 286 of SEQ ID NO: 5 (UGT91D2e-b).

As indicated above, UGTs designated herein as SM12UGT can be substituted for UGT91D2. Suitable func-

tional SM12UGT polypeptides include those made by *Ipomoea purpurea* (Japanese morning glory) and described in Morita et al. *Plant J.* 42, 353-363 (2005). The amino acid sequence encoding the *I. purpurea* IP3GGT polypeptide is set forth in SEQ ID NO: 76. SEQ ID NO: 77 is a nucleotide sequence encoding the polypeptide of SEQ ID NO: 76 that has been codon optimized for expression in yeast. Another suitable SM12UGT polypeptide is a Bp94B1 polypeptide having an R25S mutation. See Osmani et al. *Plant Phys.* 148: 1295-1308 (2008) and Sawada et al. *J. Biol. Chem.* 280:899-906 (2005). The amino acid sequence of the *Bellis perennis* (red daisy) UGT94B1 polypeptide is set forth in SEQ ID NO: 78. SEQ ID NO: 79 is the nucleotide sequence encoding the polypeptide of SEQ ID NO: 78 that has been codon optimized for expression in yeast.

In some embodiments, the recombinant microorganism is grown on media containing steviol-13-O-glucoside or steviol-19-O-glucoside in order to produce rebaudioside A and/or rebaudioside D. In such embodiments, the microorganism contains and expresses genes encoding a functional EUGT11, a functional UGT74G1, a functional UGT85C2, a functional UGT76G1, and an optional functional UGT91D2, and is capable of accumulating rebaudioside A and rebaudioside D when steviol, one or both of the steviolmonosides, or rubusoside is used as feedstock.

In other embodiments, the recombinant microorganism is grown on media containing rubusoside in order to produce rebaudioside A and/or rebaudioside D. In such embodiments, the microorganism contains and expresses genes encoding a functional EUGT11, a functional UGT76G1, and an optional functional UGT91D2, and is capable of producing rebaudioside A and/or rebaudioside D when rubusoside is used as feedstock.

In other embodiments the recombinant microorganism expresses one or more genes involved in steviol biosynthesis, e.g., a CDPS gene, a KS gene, a KO gene and/or a KAH gene. Thus, for example, a microorganism containing a CDPS gene, a KS gene, a KO gene and a KAH gene, in addition to a EUGT11, a UGT74G1, a UGT85C2, a UGT76G1, and optionally a functional UGT91D2 (e.g., UGT91D2e), is capable of producing rebaudioside A, rebaudioside D, and/or rebaudioside E without the necessity for including steviol in the culture media.

In some embodiments, the recombinant host further contains and expresses a recombinant GGPPS gene in order to provide increased levels of the diterpene precursor geranylgeranyl diphosphate, for increased flux through the steviol biosynthetic pathway. In some embodiments, the recombinant host further contains a construct to silence the expression of non-steviol pathways consuming geranylgeranyl diphosphate, ent-Kaurenoic acid or farnesyl pyrophosphate, thereby providing increased flux through the steviol and steviol glycosides biosynthetic pathways. For example, flux to sterol production pathways such as ergosterol may be reduced by downregulation of the ERG9 gene. See, the ERG9 section below and Examples 24-25. In cells that produce gibberellins, gibberellin synthesis may be downregulated to increase flux of ent-kaurenoic acid to steviol. In carotenoid-producing organisms, flux to steviol may be increased by downregulation of one or more carotenoid biosynthetic genes. In some embodiments, the recombinant microorganism further can express recombinant genes involved in diterpene biosynthesis or production of terpenoid precursors, e.g., genes in the MEP or MEV pathways discussed below, have reduced phosphatase activity, and/or express a SUS as discussed herein.

One with skill in the art will recognize that by modulating relative expression levels of different UGT genes, a recombinant host can be tailored to specifically produce steviol glycoside products in a desired proportion. Transcriptional regulation of steviol biosynthesis genes and steviol glycoside biosynthesis genes can be achieved by a combination of transcriptional activation and repression using techniques known to those in the art. For in vitro reactions, one with skill in the art will recognize that addition of different levels of UGT enzymes in combination or under conditions which impact the relative activities of the different UGTs in combination will direct synthesis towards a desired proportion of each steviol glycoside. One with skill in the art will recognize that a higher proportion of rebaudioside D or E or more efficient conversion to rebaudioside D or E can be obtained with a diglycosylation enzyme that has a higher activity for the 19-O-glucoside reaction as compared to the 13-O-glucoside reaction (substrates rebaudioside A and stevioside).

In some embodiments, a recombinant host such as a microorganism produces rebaudioside D-enriched steviol glycoside compositions that have greater than at least 3% rebaudioside D by weight total steviol glycosides, e.g., at least 4% rebaudioside D at least 5% rebaudioside D, 10-20% rebaudioside D, 20-30% rebaudioside D, 30-40% rebaudioside D, 40-50% rebaudioside D, 50-60% rebaudioside D, 60-70% rebaudioside D, 70-80% rebaudioside D. In some embodiments, a recombinant host such as a microorganism produces steviol glycoside compositions that have at least 90% rebaudioside D, e.g., 90-99% rebaudioside D. Other steviol glycosides present may include those depicted in FIG. 2 C such as steviol monosides, steviol glucobiosides, rebaudioside A, rebaudioside E, and stevioside. In some embodiments, the rebaudioside D-enriched composition produced by the host (e.g., microorganism) can be further purified and the rebaudioside D or rebaudioside E so purified can then be mixed with other steviol glycosides, flavors, or sweeteners to obtain a desired flavor system or sweetening composition. For instance, a rebaudioside D-enriched composition produced by a recombinant host can be combined with a rebaudioside A, C, or F-enriched composition produced by a different recombinant host, with rebaudioside A, F, or C purified from a *Stevia* extract, or with rebaudioside A, F, or C produced in vitro.

In some embodiments, rebaudioside A, rebaudioside D, rebaudioside B, steviol monoglucosides, steviol-1,2-bioside, rubusoside, stevioside, or rebaudioside E can be produced using in vitro methods while supplying the appropriate UDP-sugar and/or a cell-free system for regeneration of UDP-sugars. See, for example, Jewett M C, et al. *Molecular Systems Biology*, Vol. 4, article 220 (2008); Masada S et al. *FEBS Letters*, Vol. 581, 2562-2566 (2007). In some embodiments, sucrose and a sucrose synthase may be provided in the reaction vessel in order to regenerate UDP-glucose from the UDP generated during glycosylation reactions. See FIG. 11. The sucrose synthase can be from any suitable organism. For example, a sucrose synthase coding sequence from *Arabidopsis thaliana*, *Stevia rebaudiana*, or *Coffea arabica* can be cloned into an expression plasmid under control of a suitable promoter, and expressed in a host such as a microorganism or a plant.

Conversions requiring multiple reactions may be carried out together, or stepwise. For example, rebaudioside D may be produced from rebaudioside A that is commercially available as an enriched extract or produced via biosynthesis, with the addition of stoichiometric or excess amounts of UDP-glucose and EUGT11. As an alternative, rebaudioside

D may be produced from steviol glycoside extracts that are enriched for stevioside and rebaudioside A, using EUGT11 and a suitable UGT76G1 enzyme. In some embodiments, phosphatases are used to remove secondary products and improve the reaction yields. UGTs and other enzymes for in vitro reactions may be provided in soluble forms or in immobilized forms.

In some embodiments, rebaudioside A, rebaudioside D, or rebaudioside E can be produced using whole cells that are fed raw materials that contain precursor molecules such as steviol and/or steviol glycosides, including mixtures of steviol glycosides derived from plant extracts. The raw materials may be fed during cell growth or after cell growth. The whole cells may be in suspension or immobilized. The whole cells may be entrapped in beads, for example calcium or sodium alginate beads. The whole cells may be linked to a hollow fiber tube reactor system. The whole cells may be concentrated and entrapped within a membrane reactor system. The whole cells may be in fermentation broth or in a reaction buffer. In some embodiments, a permeabilizing agent is utilized for efficient transfer of substrate into the cells. In some embodiments, the cells are permeabilized with a solvent such as toluene, or with a detergent such as Triton-X or Tween. In some embodiments, the cells are permeabilized with a surfactant, for example a cationic surfactant such as cetyltrimethylammonium bromide (CTAB). In some embodiments, the cells are permeabilized with periodic mechanical shock such as electroporation or a slight osmotic shock. The cells can contain one recombinant UGT or multiple recombinant UGTs. For example, the cells can contain UGT 76G1 and EUGT11 such that mixtures of stevioside and RebA are efficiently converted to RebD. In some embodiments, the whole cells are the host cells described in section III A. In some embodiments, the whole cells are a Gram-negative bacterium such as *E. coli*. In some embodiments, the whole cell is a Gram-positive bacterium such as *Bacillus*. In some embodiments, the whole cell is a fungal species such as *Aspergillus*, or a yeast such as *Saccharomyces*. In some embodiments, the term "whole cell biocatalysis" is used to refer to the process in which the whole cells are grown as described above (e.g., in a medium and optionally permeabilized) and a substrate such as rebA or stevioside is provided and converted to the end product using the enzymes from the cells. The cells may or may not be viable, and may or may not be growing during the bioconversion reactions. In contrast, in fermentation, the cells are cultured in a growth medium and fed a carbon and energy source such as glucose and the end product is produced with viable cells.

B. 3 Dulcoside A and Rebaudioside C Biosynthesis Polypeptides

The biosynthesis of rebaudioside C and/or dulcoside A involves glucosylation and rhamnosylation of the aglycone steviol. Specifically, dulcoside A can be formed by glucosylation of the 13-OH of steviol which forms steviol-13-O-glucoside, rhamnosylation of the C-2' of the 13-O-glucose of steviol-13-O-glucoside which forms the 1,2 rhamnobioside, and glucosylation of the C-19 carboxyl of the 1,2 rhamnobioside. Rebaudioside C can be formed by glucosylation of the C-3' of the C-13-O-glucose of dulcoside A. The order in which each glycosylation reaction occurs can vary. See FIG. 2B.

It has been discovered that conversion of steviol to dulcoside A in a recombinant host can be accomplished by the expression of gene(s) encoding the following functional UGTs: 85C2, EUGT11 and/or 91D2e, and 74G1. Thus, a recombinant microorganism expressing these three or four

UGTs and a rhamnose synthetase can make dulcoside A when fed steviol in the medium. Alternatively, a recombinant microorganism expressing two UGTs, EUGT11 and 74G1, and rhamnose synthetase can make dulcoside A when fed the monoside, steviol-13-O-glucoside or steviol-19-O-glucoside, in the medium. Similarly, conversion of steviol to rebaudioside C in a recombinant microorganism can be accomplished by the expression of gene(s) encoding UGTs 85C2, EUGT11, 74G1, 76G1, optionally 91D2e, and rhamnose synthetase when fed steviol, by the expression of genes encoding UGTs EUGT11 and/or 91D2e, 74G1, and 76G1, and rhamnose synthetase when fed steviol-13-O-glucoside, by the expression of genes encoding UGTs 85C2, EUGT11 and/or 91D2e, 76G1, and rhamnose synthetase when fed steviol-19-O-glucoside, or by the expression of genes encoding UGTs EUGT11 and/or 91D2e, 76G1, and rhamnose synthetase when fed rubusoside. Typically, one or more of these genes are recombinant genes that have been transformed into a microorganism that does not naturally possess them.

Suitable EUGT11, UGT91D2, UGT74G1, UGT76G1 and UGT85C2 polypeptides include the functional UGT polypeptides discussed herein. Rhamnose synthetase provides increased amounts of the UDP-rhamnose donor for rhamnosylation of the steviol compound acceptor. Suitable rhamnose synthetases include those made by *Arabidopsis thaliana*, such as the product of the *A. thaliana* RHM2 gene.

In some embodiments, a UGT79B3 polypeptide is substituted for a UGT91D2 polypeptide. Suitable UGT79B3 polypeptides include those made by *Arabidopsis thaliana*, which are capable of rhamnosylation of steviol 13-O-monoside in vitro. *A. thaliana* UGT79B3 can rhamnosylate glucosylated compounds to form 1,2-rhamnosides. The amino acid sequence of an *Arabidopsis thaliana* UGT79B3 is set forth in SEQ ID NO:150. The nucleotide sequence encoding the amino acid sequence of SEQ ID NO:150 is set forth in SEQ ID NO:151.

In some embodiments rebaudioside C can be produced using in vitro methods while supplying the appropriate UDP-sugar and/or a cell-free system for regeneration of UDP-sugars. See, for example, "An integrated cell-free metabolic platform for protein production and synthetic biology" by Jewett M C, Calhoun K A, Voloshin A, Wu J J and Swartz J R in *Molecular Systems Biology* 4, article 220 (2008); Masada S et al. *FEBS Letters*, Vol. 581, 2562-2566 (2007). In some embodiments, sucrose and a sucrose synthase may be provided in the reaction vessel in order to regenerate UDP-glucose from UDP during the glycosylation reactions. See FIG. 11. The sucrose synthase can be from any suitable organism. For example, a sucrose synthase coding sequence from *Arabidopsis thaliana*, *Stevia rebaudiana*, or *Coffea arabica* can be cloned into an expression plasmid under control of a suitable promoter, and expressed in a host (e.g., a microorganism or a plant). In some embodiments a RHM2 enzyme (Rhamnose synthase) may also be provided, with NADPH, to generate UDP-rhamnose from UDP-glucose.

Reactions may be carried out together, or stepwise. For instance, rebaudioside C may be produced from rubusoside with the addition of stoichiometric amounts of UDP-rhamnose and EUGT11, followed by addition of UGT76G1 and an excess or stoichiometric supply of UDP-glucose. In some embodiments, phosphatases are used to remove secondary products and improve the reaction yields. UGTs and other enzymes for in vitro reactions may be provided in soluble forms or immobilized forms. In some embodiments, rebaudioside C, Dulcoside A, or other steviol rhamnosides can be

produced using whole cells as discussed above. The cells can contain one recombinant UGT or multiple recombinant UGTs. For example, the cells can contain UGT 76G1 and EUGT11 such that mixtures of stevioside and RebA are efficiently converted to RebD. In some embodiments, the whole cells are the host cells described in section III A.

In other embodiments, the recombinant host expresses one or more genes involved in steviol biosynthesis, e.g., a CDPS gene, a KS gene, a KO gene and/or a KAH gene. Thus, for example, a microorganism containing a CDPS gene, a KS gene, a KO gene and a KAH gene, in addition to a UGT85C2, a UGT74G1, a EUGT11 gene, optionally a UGT91D2e gene, and a UGT76G1 gene, is capable of producing rebaudioside C without the necessity for including steviol in the culture media. In addition, the recombinant host typically expresses an endogenous or a recombinant gene encoding a rhamnose synthetase. Such a gene is useful in order to provide increased amounts of the UDP-rhamnose donor for rhamnosylation of the steviol compound acceptor. Suitable rhamnose synthetases include those made by *Arabidopsis thaliana*, such as the product of the *A. thaliana* RHM2 gene.

One with skill in the art will recognize that by modulating relative expression levels of different UGT genes as well as modulating the availability of UDP-rhamnose, a recombinant host can be tailored to specifically produce steviol and steviol glycoside products in a desired proportion. Transcriptional regulation of steviol biosynthesis genes, and steviol glycoside biosynthesis genes can be achieved by a combination of transcriptional activation and repression using techniques known to those in the art. For in vitro reactions, one with skill in the art will recognize that addition of different levels of UGT enzymes in combination or under conditions which impact the relative activities of the different UGTs in combination will direct synthesis towards a desired proportion of each steviol glycoside.

In some embodiments, the recombinant host further contains and expresses a recombinant GGPPS gene in order to provide increased levels of the diterpene precursor geranylgeranyl diphosphate, for increased flux through the rebaudioside A biosynthetic pathway. In some embodiments, the recombinant host further contains a construct to silence or reduce the expression of non-steviol pathways consuming geranylgeranyl diphosphate, ent-Kaurenoic acid or farnesyl pyrophosphate, thereby providing increased flux through the steviol and steviol glycosides biosynthetic pathways. For example, flux to sterol production pathways such as ergosterol may be reduced by downregulation of the ERG9 gene. See, the ERG9 section below and Examples 24-25. In cells that produce gibberellins, gibberellin synthesis may be downregulated to increase flux of ent-kaurenoic acid to steviol. In carotenoid-producing organisms, flux to steviol may be increased by downregulation of one or more carotenoid biosynthetic genes.

In some embodiments, the recombinant host further contains and expresses recombinant genes involved in diterpene biosynthesis or production of terpenoid precursors, e.g., genes in the MEP or MEV pathway, have reduced phosphatase activity, and/or express a SUS as discussed herein.

In some embodiments, a recombinant host such as a microorganism produces steviol glycoside compositions that have greater than at least 15% rebaudioside C of the total steviol glycosides, e.g., at least 20% rebaudioside C, 30-40% rebaudioside C, 40-50% rebaudioside C, 50-60% rebaudioside C, 60-70% rebaudioside C, 70-80% rebaudioside C, 80-90% rebaudioside C. In some embodiments, a recombinant host such as a microorganism produces steviol

glycoside compositions that have at least 90% rebaudioside C, e.g., 90-99% rebaudioside C. Other steviol glycosides present may include those depicted in FIGS. 2 A and B such as steviol monosides, steviol glucobiosides, steviol rhamnobilosides, rebaudioside A, and Dulcoside A. In some embodiments, the rebaudioside C-enriched composition produced by the host can be further purified and the rebaudioside C or Dulcoside A so purified may then be mixed with other steviol glycosides, flavors, or sweeteners to obtain a desired flavor system or sweetening composition. For instance, a rebaudioside C-enriched composition produced by a recombinant microorganism can be combined with a rebaudioside A, F, or D-enriched composition produced by a different recombinant microorganism, with rebaudioside A, F, or D purified from a *Stevia* extract, or with rebaudioside A, F, or D produced in vitro.

B. 4 Rebaudioside F Biosynthesis Polypeptides

The biosynthesis of rebaudioside F involves glucosylation and xylosylation of the aglycone steviol. Specifically, rebaudioside F can be formed by glucosylation of the 13-OH of steviol which forms steviol-13-O-glucoside, xylosylation of the C-2' of the 13-O-glucose of steviol-13-O-glucoside which forms steviol-1,2-xylobioside, glucosylation of the C-19 carboxyl of the 1,2-xylobioside to form 1,2-stevioxyloside, and glucosylation of the C-3' of the C-13-O-glucose of 1,2-stevioxyloside to form rebaudioside F. The order in which each glycosylation reaction occurs can vary. See FIG. 2D.

It has been discovered that conversion of steviol to rebaudioside F in a recombinant host can be accomplished by the expression of genes encoding the following functional UGTs: 85C2, EUGT11 and/or 91D2e, 74G1, and 76G1, along with endogenous or recombinantly expressed UDP-glucose dehydrogenase and UDP-glucuronic acid decarboxylase. Thus, a recombinant microorganism expressing these four or five UGTs along with endogenous or recombinant UDP-glucose dehydrogenase and UDP-glucuronic acid decarboxylase can make rebaudioside F when fed steviol in the medium. Alternatively, a recombinant microorganism expressing two functional UGTs, EUGT11 or 91D2e, and 76G1, can make rebaudioside F when fed rubusoside in the medium. As another alternative, a recombinant microorganism expressing a functional UGT 76G1 can make rebaudioside F when fed 1,2-steviorhamnoside. As another alternative, a recombinant microorganism expressing 74G1, EUGT11 and/or 91D2e, 76G1, and can make rebaudioside F when fed the monoside, steviol-13-O-glucoside, in the medium. Similarly, conversion of steviol-19-O-glucoside to rebaudioside F in a recombinant microorganism can be accomplished by the expression of genes encoding UGTs 85C2, EUGT11 and/or 91D2e, and 76G1, when fed steviol-19-O-glucoside. Typically, one or more of these genes are recombinant genes that have been transformed into a host that does not naturally possess them.

Suitable EUGT11, UGT91D2, UGT74G1, UGT76G1 and UGT85C2 polypeptides include the functional UGT polypeptides discussed herein. In some embodiments, a UGT79B3 polypeptide is substituted for a UGT91, as discussed above. UDP-glucose dehydrogenase and UDP-glucuronic acid decarboxylase provide increased amounts of the UDP-xylose donor for xylosylation of the steviol compound acceptor. Suitable UDP-glucose dehydrogenases and UDP-glucuronic acid decarboxylases include those made by *Arabidopsis thaliana* or *Cryptococcus neoformans*. For example, suitable UDP-glucose dehydrogenase and UDP-glucuronic acid decarboxylases polypeptides can be

encoded by the *A. thaliana* UGD1 gene and UXS3 gene, respectively. See, Oka and Jigami, *FEBS J.* 273:2645-2657 (2006).

In some embodiments rebaudioside F can be produced using in vitro methods while supplying the appropriate UDP-sugar and/or a cell-free system for regeneration of UDP-sugars. See, for example, Jewett M C, et al. *Molecular Systems Biology*, Vol. 4, article 220 (2008); Masada S et al. *FEBS Letters*, Vol. 581, 2562-2566 (2007). In some embodiments, sucrose and a sucrose synthase are provided in the reaction vessel in order to regenerate UDP-glucose from UDP during the glycosylation reactions. See FIG. 11. The sucrose synthase can be from any suitable organism. For example, a sucrose synthase coding sequence from *Arabidopsis thaliana*, *Stevia rebaudiana*, or *Coffea arabica* can be cloned into an expression plasmid under control of a suitable promoter, and expressed in a host, e.g., a microorganism or a plant. In some embodiments, UDP-xylose can be produced from UDP-glucose by supplying suitable enzymes, for example, the *Arabidopsis thaliana* UGD1 (UDP-glucose dehydrogenase) and UXS3 (UDP-glucuronic acid decarboxylase) enzymes along with NAD⁺ cofactor.

Reactions may be carried out together, or stepwise. For instance, rebaudioside F may be produced from rubusoside with the addition of stoichiometric amounts of UDP-xylose and EUGT11, followed by addition of UGT76G1 and an excess or stoichiometric supply of UDP-glucose. In some embodiments, phosphatases are used to remove secondary products and improve the reaction yields. UGTs and other enzymes for in vitro reactions may be provided in soluble forms or immobilized forms. In some embodiments, rebaudioside F or other steviol xylosides can be produced using whole cells as discussed above. For example, the cells may contain UGT 76G1 and EUGT11 such that mixtures of stevioside and RebA are efficiently converted to RebD. In some embodiments, the whole cells are the host cells described in section III A.

In other embodiments, the recombinant host expresses one or more genes involved in steviol biosynthesis, e.g., a CDPS gene, a KS gene, a KO gene and/or a KAH gene. Thus, for example, a microorganism containing a CDPS gene, a KS gene, a KO gene and a KAH gene, in addition to a EUGT11, UGT85C2, a UGT74G1, an optional UGT91D2 gene, and a UGT76G1 gene, is capable of producing rebaudioside F without the necessity for including steviol in the culture media. In addition, the recombinant host typically expresses an endogenous or a recombinant gene encoding a UDP-glucose dehydrogenase and a UDP-glucuronic acid decarboxylase. Such genes are useful in order to provide increased amounts of the UDP-xylose donor for xylosylation of the steviol compound acceptor. Suitable UDP-glucose dehydrogenases and UDP-glucuronic acid decarboxylases include those made by *Arabidopsis thaliana* or *Cryptococcus neoformans*. For example, suitable UDP-glucose dehydrogenase and UDP-glucuronic acid decarboxylases polypeptides can be encoded by the *A. thaliana* UGD1 gene and UXS3 gene, respectively. See, Oka and Jigami, *FEBS J.* 273:2645-2657 (2006).

One with skill in the art will recognize that by modulating relative expression levels of different UGT genes as well as modulating the availability of UDP-xylose, a recombinant microorganism can be tailored to specifically produce steviol and steviol glycoside products in a desired proportion. Transcriptional regulation of steviol biosynthesis genes can be achieved by a combination of transcriptional activation and repression using techniques known to those in the art. For in vitro reactions, one with skill in the art will recognize

that addition of different levels of UGT enzymes in combination or under conditions which impact the relative activities of the different UGTs in combination will direct synthesis towards a desired proportion of each steviol glycosides.

In some embodiments, the recombinant host further contains and expresses a recombinant GGPPS gene in order to provide increased levels of the diterpene precursor geranylgeranyl diphosphate, for increased flux through the steviol biosynthetic pathway. In some embodiments, the recombinant host further contains a construct to silence the expression of non-steviol pathways consuming geranylgeranyl diphosphate, ent-Kaurenoic acid or farnesyl pyrophosphate, thereby providing increased flux through the steviol and steviol glycosides biosynthetic pathways. For example, flux to sterol production pathways such as ergosterol may be reduced by downregulation of the ERG9 gene. See, the ERG9 section below and Examples 24-25. In cells that produce gibberellins, gibberellin synthesis may be down-regulated to increase flux of ent-kaurenoic acid to steviol. In carotenoid-producing organisms, flux to steviol may be increased by downregulation of one or more carotenoid biosynthetic genes. In some embodiments, the recombinant host further contains and expresses recombinant genes involved in diterpene biosynthesis, e.g., genes in the MEP pathway discussed below.

In some embodiments, a recombinant host such as a microorganism produces rebaudioside F-enriched steviol glycoside compositions that have greater than at least 4% rebaudioside F by weight total steviol glycosides, e.g., at least 5% rebaudioside F, at least 6% of rebaudioside F, 10-20% rebaudioside F, 20-30% rebaudioside F, 30-40% rebaudioside F, 40-50% rebaudioside F, 50-60% rebaudioside F, 60-70% rebaudioside F, 70-80% rebaudioside F. In some embodiments, a recombinant host such as a microorganism produces steviol glycoside compositions that have at least 90% rebaudioside F, e.g., 90-99% rebaudioside F. Other steviol glycosides present may include those depicted in FIGS. 2A and D such as steviol monosides, steviol glucobiosides, steviol xylobiosides, rebaudioside A, stevi-oxylsides, rubusoside and stevioside. In some embodiments, the rebaudioside F-enriched composition produced by the host can be mixed with other steviol glycosides, flavors, or sweeteners to obtain a desired flavor system or sweetening composition. For instance, a rebaudioside F-enriched composition produced by a recombinant microorganism can be combined with a rebaudioside A, C, or D-enriched composition produced by a different recombinant microorganism, with rebaudioside A, C, or D purified from a *Stevia* extract, or with rebaudioside A, C, or D produced in vitro.

C. Other Polypeptides

Genes for additional polypeptides whose expression facilitates more efficient or larger scale production of steviol or a steviol glycoside can also be introduced into a recombinant host. For example, a recombinant microorganism, plant, or plant cell can also contain one or more genes encoding a geranylgeranyl diphosphate synthase (GGPPS, also referred to as GGPPS). As another example, the recombinant host can contain one or more genes encoding a rhamnose synthetase, or one or more genes encoding a UDP-glucose dehydrogenase and/or a UDP-glucuronic acid decarboxylase. As another example, a recombinant host can also contain one or more genes encoding a cytochrome P450 reductase (CPR). Expression of a recombinant CPR facilitates the cycling of NADP⁺ to regenerate NADPH, which is utilized as a cofactor for terpenoid biosynthesis. Other methods can be used to regenerate NADHP levels as well.

In circumstances where NADPH becomes limiting; strains can be further modified to include exogenous transhydrogenase genes. See, e.g., Sauer et al., *J. Biol. Chem.* 279: 6613-6619 (2004). Other methods are known to those with skill in the art to reduce or otherwise modify the ratio of NADH/NADPH such that the desired cofactor level is increased.

As another example, the recombinant host can contain one or more genes encoding one or more enzymes in the MEP pathway or the mevalonate pathway. Such genes are useful because they can increase the flux of carbon into the diterpene biosynthesis pathway, producing geranylgeranyl diphosphate from isopentenyl diphosphate and dimethylallyl diphosphate generated by the pathway. The geranylgeranyl diphosphate so produced can be directed towards steviol and steviol glycoside biosynthesis due to expression of steviol biosynthesis polypeptides and steviol glycoside biosynthesis polypeptides.

As another example the recombinant host can contain one or more genes encoding a sucrose synthase, and additionally can contain sucrose uptake genes if desired. The sucrose synthase reaction can be used to increase the UDP-glucose pool in a fermentation host, or in a whole cell bioconversion process. This regenerates UDP-glucose from UDP produced during glycosylation and sucrose, allowing for efficient glycosylation. In some organisms, disruption of the endogenous invertase is advantageous to prevent degradation of sucrose. For example, the *S. cerevisiae* SUC2 invertase may be disrupted. The sucrose synthase (SUS) can be from any suitable organism. For example, a sucrose synthase coding sequence from, without limitation, *Arabidopsis thaliana*, *Stevia rebaudiana*, or *Coffea arabica* can be cloned into an expression plasmid under control of a suitable promoter, and expressed in a host (e.g., a microorganism or a plant). The sucrose synthase can be expressed in such a strain in combination with a sucrose transporter (e.g., the *A. thaliana* SUC1 transporter or a functional homolog thereof) and one or more UGTs (e.g., one or more of UGT85C2, UGT74G1, UGT76G1, and UGT91D2e, EUGT11 or functional homologs thereof). Culturing the host in a medium that contains sucrose can promote production of UDP-glucose, as well as one or more glucosides (e.g., steviol glycosides).

In addition, a recombinant host can have reduced phosphatase activity as discussed herein.

C. 1 MEP Biosynthesis Polypeptides

In some embodiments, a recombinant host contains one or more genes encoding enzymes involved in the methylerythritol 4-phosphate (MEP) pathway for isoprenoid biosynthesis. Enzymes in the MEP pathway include deoxyxylulose 5-phosphate synthase (DXS), D-1-deoxyxylulose 5-phosphate reductoisomerase (DXR), 4-diphosphocytidyl-2-C-methyl-D-erythritol synthase (CMS), 4-diphosphocytidyl-2-C-methyl-D-erythritol kinase (CMK), 4-diphosphocytidyl-2-C-methyl-D-erythritol 2,4-cyclodiphosphate synthase (MCS), 1-hydroxy-2-methyl-2(E)-butenyl 4-diphosphate synthase (HDS) and 1-hydroxy-2-methyl-2(E)-butenyl 4-diphosphate reductase (HDR). One or more DXS genes, DXR genes, CMS genes, CMK genes, MCS genes, HDS genes and/or HDR genes can be incorporated into a recombinant microorganism. See, Rodriguez-Concepcion and Boronat, *Plant Phys.* 130: 1079-1089 (2002).

Suitable genes encoding DXS, DXR, CMS, CMK, MCS, HDS and/or HDR polypeptides include those made by *E. coli*, *Arabidopsis thaliana* and *Synechococcus leopoliensis*. Nucleotide sequences encoding DXR polypeptides are described, for example, in U.S. Pat. No. 7,335,815.

C. 2 Mevalonate Biosynthesis Polypeptides

In some embodiments, a recombinant host contains one or more genes encoding enzymes involved in the mevalonate pathway for isoprenoid biosynthesis. Genes suitable for transformation into a host encode enzymes in the mevalonate pathway such as a truncated 3-hydroxy-3-methylglutaryl (HMG)-CoA reductase (tHMG), and/or a gene encoding a mevalonate kinase (MK), and/or a gene encoding a phosphomevalonate kinase (PMK), and/or a gene encoding a mevalonate pyrophosphate decarboxylase (MPPD). Thus, one or more HMG-CoA reductase genes, MK genes, PMK genes, and/or MPPD genes can be incorporated into a recombinant host such as a microorganism.

Suitable genes encoding mevalonate pathway polypeptides are known. For example, suitable polypeptides include those made by *E. coli*, *Paracoccus denitrificans*, *Saccharomyces cerevisiae*, *Arabidopsis thaliana*, *Kitasatospora griseola*, *Homo sapiens*, *Drosophila melanogaster*, *Gallus gallus*, *Streptomyces* sp. KO-3988, *Nicotiana attenuata*, *Kitasatospora griseola*, *Hevea brasiliensis*, *Enterococcus faecium* and *Haematococcus pluvialis*. See, e.g., Table 8 and U.S. Pat. Nos. 7,183,089, 5,460,949, and 5,306,862.

TABLE 8

Sources of HMG CoA Reductases and other Mevalonate Genes

Accession#	Organism	Enzyme	Size (nt)	Gene name	SEQ ID (codon optimized)	SEQ ID (protein)
XM_001467423	<i>Leishmania infantum</i>	Acetyl-CoA C-acetyltransferase	1323	MEV-4	103	104
YML075C	<i>Saccharomyces cerevisiae</i>	Truncated HMG (tHMG1)	1584	tHMG1	105	106
EU263989	<i>Ganoderma lucidum</i>	3-HMG-CoA reductase	3681	MEV-11	107	108
BC153262	<i>Bos taurus</i>	3-HMG-CoA reductase	2667	MEV-12	109	110
AAD47596	<i>Artemisia annua</i>	3-HMG-CoA reductase	1704	MEV-13	111	112
AAB62280	<i>Trypanosoma cruzi</i>	3-HMG-CoA reductase	1308	MEV-14	113	114
CAG41604	<i>Staph aureus</i>	3-HMG-CoA reductase	1281	MEV-15	115	116
DNA2.0 sequence	<i>Archaeoglobus fulgidus</i>	3-HMG-CoA reductase	1311	HMG reductase	117	118
DNA2.0 sequence	<i>Pseudomonas mevalonii</i>	3-HMG-CoA reductase	1287	HMG reductase	119	120

C.3 Sucrose Synthase Polypeptides

Sucrose synthase (SUS) can be used as a tool for generating UDP-sugar. SUS (EC 2.4.1.13) catalyzes the formation of UDP-glucose and fructose from sucrose and UDP (FIG. 11). UDP generated by the reaction of UGTs thus can be converted into UDP-glucose in the presence of sucrose. See, e.g., Chen et al. (2001) *J. Am. Chem. Soc.* 123:8866-8867; Shao et al. (2003) *Appl. Env. Microbiol.* 69:5238-5242; Masada et al. (2007) *FEBS Lett.* 581:2562-2566; and Son et al. (2009) *J. Microbiol. Biotechnol.* 19:709-712.

Sucrose synthases can be used to generate UDP-glucose and remove UDP, facilitating efficient glycosylation of compounds in various systems. For example, yeast deficient in the ability to utilize sucrose can be made to grow on sucrose by introducing a sucrose transporter and a SUS. For example, *Saccharomyces cerevisiae* does not have an efficient sucrose uptake system, and relies on extracellular SUC2 to utilize sucrose. The combination of disrupting the endogenous *S. cerevisiae* SUC2 invertase and expressing recombinant SUS resulted in a yeast strain that was able to metabolize intracellular but not extracellular sucrose (Ries-

meier et al. ((1992) *EMBO J.* 11:4705-4713). The strain was used to isolate sucrose transporters by transformation with a cDNA expression library and selection of transformants that had gained the ability to take up sucrose.

As described herein, the combined expression of recombinant sucrose synthase and a sucrose transporter in vivo can lead to increased UDP-glucose availability and removal of unwanted UDP. For example, functional expression of a recombinant sucrose synthase, a sucrose transporter, and a glycosyltransferase, in combination with knockout of the natural sucrose degradation system (SUC2 in the case of *S. cerevisiae*) can be used to generate a cell that is capable of producing increased amounts of glycosylated compounds such as steviol glycosides. This higher glycosylation capability is due to at least (a) a higher capacity for producing UDP-glucose in a more energy efficient manner, and (b) removal of UDP from growth medium, as UDP can inhibit glycosylation reactions.

The sucrose synthase can be from any suitable organism. For example, a sucrose synthase coding sequence from, without limitation, *Arabidopsis thaliana*, *Stevia rebaudiana*, or *Coffea arabica* (see, e.g., FIGS. 19A-19C, SEQ ID

NOs:178, 179, and 180) can be cloned into an expression plasmid under control of a suitable promoter, and expressed in a host (e.g., a microorganism or a plant). As described in the Examples herein, a SUS coding sequence may be expressed in a SUC2 (sucrose hydrolyzing enzyme) deficient *S. cerevisiae* strain, so as to avoid degradation of extracellular sucrose by the yeast. The sucrose synthase can be expressed in such a strain in combination with a sucrose transporter (e.g., the *A. thaliana* SUC1 transporter or a functional homolog thereof) and one or more UGTs (e.g., one or more of UGT85C2, UGT74G1, UGT76G1, EUGT11, and UGT91D2e, or functional homologs thereof). Culturing the host in a medium that contains sucrose can promote production of UDP-glucose, as well as one or more glucosides (e.g., steviol glucoside). It is to be noted that in some cases, a sucrose synthase and a sucrose transporter can be expressed along with a UGT in a host cell that also is recombinant for production of a particular compound (e.g., steviol).

C. 4 Modulation of ERG9 Activity

It is an object of the disclosure to produce terpenoids based on the concept of increasing the accumulation of

terpenoid precursors of the squalene pathway. Non-limiting examples of terpenoids include Hemiterpenoids, 1 isoprene unit (5 carbons); Monoterpenoids, 2 isoprene units (10 C); Sesquiterpenoids, 3 isoprene units (15 C); Diterpenoids, 4 isoprene units (20 C) (e.g. ginkgolides); Triterpenoids, 6 isoprene units (30 C); Tetraterpenoids, 8 isoprene units (40 C) (e.g. carotenoids); and polyterpenoid with a larger number of isoprene units.

Hemiterpenoids include isoprene, prenol and isovaleric acid. Monoterpenoids include Geranyl pyrophosphate, Eucalyptol, Limonene and Pinene. Sesquiterpenoids include Farnesyl pyrophosphate, Artemisinin and Bisabolol. Diterpenoids include Geranylgeranyl pyrophosphate, steviol, Retinol, Retinal, Phytol, Taxol, Forskolin and Aphidicolin. Triterpenoids include Squalene and Lanosterol. Tetraterpenoids include Lycopene and Carotene.

Terpenes are hydrocarbons resulting from the combination of several isoprene units. Terpenoids can be thought of as terpene derivatives. The term terpene is sometimes used broadly to include the terpenoids. Just like terpenes, the terpenoids can be classified according to the number of isoprene units used. The present invention is focused on terpenoids and in particular terpenoids derived through the squalene pathway from the precursors Farnesyl-pyrophosphate (FPP), Isopentenyl-pyrophosphate (IPP), Dimethylallyl-pyrophosphate (DMAPP), Geranyl-pyrophosphate (GPP) and/or Geranylgeranyl-pyrophosphate (GGPP).

By terpenoids is understood terpenoids of the Hemiterpenoid class such as but not limited to isoprene, prenol and isovaleric acid; terpenoids of the Monoterpenoid class such as but not limited to geranyl pyrophosphate, eucalyptol, limonene and pinene; terpenoids of the Sesquiterpenoids class such as but not limited to farnesyl pyrophosphate, artemisinin and bisabolol; terpenoids of the diterpenoid class such as but not limited to geranylgeranyl pyrophosphate, steviol, retinol, retinal, phytol, taxol, forskolin and aphidicolin; terpenoids of the Triterpenoid class such as but not limited to lanosterol; terpenoids of the Tetraterpenoid class such as but not limited to lycopene and carotene.

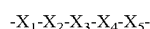
In one embodiment the invention relates to production of terpenoids, which are biosynthesized from Geranylgeranyl-pyrophosphate (GGPP). In particular such terpenoids may be steviol.

In one embodiment the invention relates to production of terpenoids, which are biosynthesized from Geranylgeranyl-pyrophosphate (GGPP). In particular such terpenoids may be steviol.

The Cell

The present invention relates to a cell, such as any of the hosts described in section III, modified to comprise the construct depicted in FIG. 22. Accordingly, in a main aspect, the present invention relates to a cell comprising a nucleic acid, said nucleic acid comprising

- i) a promoter sequence operably linked to
 - ii) a heterologous insert sequence operably linked to
 - iii) an open reading frame operably linked to
 - iv) a transcription termination signal,
- wherein the heterologous insert sequence has the general formula (I):



wherein X_2 comprises at least 4 consecutive nucleotides being complementary to, and forms a hairpin secondary structure element with at least 4 consecutive nucleotides of X_4 , and

wherein X_3 is optional and if present comprises nucleotides involved in forming a hairpin loop between X_2 and X_4 , and

wherein X_1 and X_5 individually and optionally comprise one or more nucleotides, and

wherein the open reading frame upon expression encodes a squalene synthase (EC 2.5.1.21), e.g., a polypeptide sequence having at least 70% identity to a squalene synthase (EC 2.5.1.21) or a biologically active fragment thereof, said fragment having at least 70% sequence identity to said squalene synthase in a range of overlap of at least 100 amino acids.

In addition to above mentioned nucleic acid comprising a heterologous insert sequence, the cell may also comprise one or more additional heterologous nucleic acid sequences (e.g., nucleic acids encoding any of the steviol and steviol glycoside biosynthesis polypeptides of section I). In one preferred embodiment the cell comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell.

Heterologous Insert Sequence

The heterologous insert sequence can adapt the secondary structure element of a hairpin with a hairpin loop. The hairpin part comprises sections X_2 and X_4 which are complementary and hybridize to one another. Sections X_2 and X_4 flank section X_3 , which comprises nucleotides that form a loop—the hairpin loop. The term complementary is understood by the person skilled in the art as meaning two sequences compared to each other, nucleotide by nucleotide counting from the 5' end to the 3' end, or vice versa.

The heterologous insert sequence is long enough to allow a hairpin to be completed, but short enough to allow limited translation of an ORF that is present in-frame and immediately 3' to the heterologous insert sequence. Thus, in one embodiment, the heterologous insert sequence comprises 10-50 nucleotides, preferably 10-30 nucleotides, more preferably 15-25 nucleotides, more preferably 17-22 nucleotides, more preferably 18-21 nucleotides, more preferably 18-20 nucleotides, more preferably 19 nucleotides.

X_2 and X_4 may individually consist of any suitable number of nucleotides, so long as a consecutive sequence of at least 4 nucleotides of X_2 is complementary to a consecutive sequence of at least 4 nucleotides of X_4 . In a preferred embodiment X_2 and X_4 consist of the same number of nucleotides.

X_2 may for example consist of in the range of 4 to 25, such as in the range of 4 to 20, for example of in the range of 4 to 15, such as in the range of 6 to 12, for example in the range of 8 to 12, such as in the range of 9 to 11 nucleotides.

X_4 may for example consist of in the range of 4 to 25, such as in the range of 4 to 20, for example of in the range of 4 to 15, such as in the range of 6 to 12, for example in the range of 8 to 12, such as in the range of 9 to 11 nucleotides.

In one preferred embodiment, X_2 consists of a nucleotide sequence, which is complementary to the nucleotide sequence of X_4 , i.e. it is preferred that all nucleotides of X_2 are complementary to the nucleotide sequence of X_4 .

In one preferred embodiment X_4 consists of a nucleotide sequence, which is complementary to the nucleotide sequence of X_2 , i.e. it is preferred that all nucleotides of X_4 are complementary to the nucleotide sequence of X_2 . Very preferably, X_2 and X_4 consists of the same number of nucleotides, wherein X_2 is complementary to X_4 over the entire length of X_2 and X_4 .

X_3 may be absent, i.e., X_3 may consist of zero nucleotides. It is also possible that X_3 consists of in the range of 1 to 5, such as in the range of 1 to 3 nucleotides.

X_1 may be absent, i.e., X_1 may consist of zero nucleotides. It is also possible that X_1 consists of in the range of 1 to 25, such as in the range of 1 to 20, for example in the range of 1 to 15, such as in the range of 1 to 10, for example in the range of 1 to 5, such as in the range of 1 to 3 nucleotides.

X_5 may be absent, i.e., X_5 may consist of zero nucleotides. It is also possible that X_5 may consist of in the range 1 to 5, such as in the range of 1 to 3 nucleotides.

The sequence may be any suitable sequence fulfilling the requirements defined herein above. Thus, the heterologous insert sequence may comprise a sequence selected from the group consisting of SEQ ID NO: 181, SEQ ID NO:182, SEQ ID NO:183, and SEQ ID NO:184. In a preferred embodiment the insert sequence is selected from the group consisting of SEQ ID NO: 181, SEQ ID NO:182, SEQ ID NO:183, and SEQ ID NO:184.

Squalene Synthase

Squalene synthase (SQS) is the first committed enzyme of the biosynthesis pathway that leads to the production of sterols. It catalyzes the synthesis of squalene from farnesyl pyrophosphate via the intermediate presqualene pyrophosphate. This enzyme is a critical branch point enzyme in the biosynthesis of terpenoids/isoprenoids and is thought to regulate the flux of isoprene intermediates through the sterol pathway. The enzyme is sometimes referred to as farnesyl-diphosphate farnesyltransferase (FDFT1).

The mechanism of SQS is to convert two units of farnesyl pyrophosphate into squalene.

SQS is considered to be an enzyme of eukaryotes or advanced organisms, although at least one prokaryote has been shown to possess a functionally similar enzyme.

In terms of structure and mechanics, squalene synthase most closely resembles phytoene synthase, which serves a similar role in many plants in the elaboration of phytoene, a precursor of many carotenoid compounds.

A high level of sequence identity indicates likelihood that the first sequence is derived from the second sequence. Amino acid sequence identity requires identical amino acid sequences between two aligned sequences. Thus, a candidate sequence sharing 70% amino acid identity with a reference sequence requires that, following alignment, 70% of the amino acids in the candidate sequence are identical to the corresponding amino acids in the reference sequence. Identity may be determined by aid of computer analysis, such as, without limitations, the ClustalW computer alignment program as described in section D. Using this program with its default settings, the mature (bioactive) part of a query and a reference polypeptide are aligned. The number of fully conserved residues are counted and divided by the length of the reference polypeptide. The ClustalW algorithm may similarly be used to align nucleotide sequences. Sequence identities may be calculated in a similar way as indicated for amino acid sequences.

In one important embodiment, the cell of the present invention comprises a nucleic acid sequence coding, as defined herein, upon expression for a squalene synthase wherein the squalene synthase is at least 75%, such as at least 76%, such as at least 77%, such as at least 78%, such as at least 79%, such as at least 80%, such as at least 81%, such as at least 82%, such as at least 83%, such as at least 84%, such as at least 85%, such as at least 86%, such as at least 87%, such as at least 88%, such as at least 89%, such as at least 90%, such as at least 91%, such as at least 92%, such as at least 93%, such as at least 94%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99%, such as at least 99.5%, such as at least 99.6%, such as at least 99.7%, such as at least

99.8%, such as at least 99.9%, such as 100% identical to a squalene synthase wherein the squalene synthase is selected from the group consisting of SEQ ID NO:192, SEQ ID NO:193, SEQ ID NO:194, SEQ ID NO:195, SEQ ID NO:196, SEQ ID NO:197, SEQ ID NO:198, SEQ ID NO:199, SEQ ID NO:200, SEQ ID NO:201, and SEQ ID NO:202.

Promoter

A promoter is a region of DNA that facilitates the transcription of a particular gene. Promoters are located near the genes they regulate, on the same strand and typically upstream (towards the 5' region of the sense strand). In order for the transcription to take place, the enzyme that synthesizes RNA, known as RNA polymerase, must attach to the DNA near a gene. Promoters contain specific DNA sequences and response elements which provide a secure initial binding site for RNA polymerase and for proteins called transcription factors that recruit RNA polymerase. These transcription factors have specific activator or repressor sequences of corresponding nucleotides that attach to specific promoters and regulate gene expressions.

In bacteria, the promoter is recognized by RNA polymerase and an associated sigma factor, which in turn are often brought to the promoter DNA by an activator protein binding to its own DNA binding site nearby. In eukaryotes the process is more complicated, and at least seven different factors are necessary for the binding of an RNA polymerase II to the promoter. Promoters represent critical elements that can work in concert with other regulatory regions (enhancers, silencers, boundary elements/insulators) to direct the level of transcription of a given gene.

As promoters are normally immediately adjacent to the open reading frame (ORF) in question, positions in the promoter are designated relative to the transcriptional start site, where transcription of RNA begins for a particular gene (i.e., positions upstream are negative numbers counting back from -1, for example -100 is a position 100 base pairs upstream).

Promoter Elements

Core promoter—the minimal portion of the promoter required to properly initiate transcription

Transcription Start Site (TSS)

Approximately -35 bp upstream and/or downstream of the start site

A binding site for RNA polymerase

RNA polymerase I: transcribes genes encoding ribosomal RNA

RNA polymerase II: transcribes genes encoding messenger RNA and certain small nuclear RNAs

RNA polymerase III: transcribes genes encoding tRNAs and other small RNAs

General transcription factor binding sites

Proximal promoter—the proximal sequence upstream of the gene that tends to contain primary regulatory elements

Approximately -250 bp upstream of the start site

Specific transcription factor binding sites

Distal promoter—the distal sequence upstream of the gene that may contain additional regulatory elements, often with a weaker influence than the proximal promoter

Anything further upstream (but not an enhancer or other regulatory region whose influence is positional/orientation independent)

Specific transcription factor binding sites

Prokaryotic Promoters

In prokaryotes, the promoter consists of two short sequences at -10 and -35 positions upstream from the transcription start site. Sigma factors not only help in enhancing RNAP binding to the promoter but also help RNAP target specific genes to transcribe.

The sequence at -10 is called the Pribnow box, or the -10 element, and usually consists of the six nucleotides TATAAT. The Pribnow box is essential to start transcription in prokaryotes.

The other sequence at -35 (the -35 element) usually consists of the seven nucleotides TTGACAT. Its presence allows a very high transcription rate.

Both of the above consensus sequences, while conserved on average, are not found intact in most promoters. On average only 3 of the 6 base pairs in each consensus sequence is found in any given promoter. No promoter has been identified to date that has intact consensus sequences at both the -10 and -35; artificial promoters with complete conservation of the -10/-35 hexamers has been found to promote RNA chain initiation at very high efficiencies.

Some promoters contain a UP element (consensus sequence 5'-AAAWWTWTTTNNNAAANN-3'; W=A or T; N=any base) centered at -50; the presence of the -35 element appears to be unimportant for transcription from the UP element-containing promoters.

Eukaryotic Promoters

Eukaryotic promoters are typically located upstream of the gene (ORF) and can have regulatory elements several kilobases (kb) away from the transcriptional start site. In eukaryotes, the transcriptional complex can cause the DNA to fold back on itself, which allows for placement of regulatory sequences far from the actual site of transcription. Many eukaryotic promoters, contain a TATA box (sequence TATAAA), which in turn binds a TATA binding protein which assists in the formation of the RNA polymerase transcriptional complex. The TATA box typically lies very close to the transcriptional start site (often within 50 bases).

The cell of the present invention comprises a nucleic acid sequence which comprises a promoter sequence. The promoter sequence is not limiting for the invention and can be any promoter suitable for the host cell of choice.

In one embodiment of the present invention the promoter is a constitutive or inducible promoter.

In a further embodiment of the invention, the promoter is selected from the group consisting of an endogenous promoter, PGK-1, GPD1, PGK1, ADH1, ADH2, PYK1, TPI1, PDC1, TEF1, TEF2, FBA1, GAL1-10, CUP1, MET2, MET14, MET25, CYC1, GAL1-S, GAL1-L, TEF1, ADH1, CAG, CMV, human UbiC, RSV, EF-1alpha, SV40, Mt1, Tet-On, Tet-Off, Mo-MLV-LTR, Mx1, progesterone, RU486 and Rapamycin-inducible promoter.

Post-Transcriptional Regulation

Post-transcriptional regulation is the control of gene expression at the RNA level, therefore between the transcription and the translation of the gene.

The first instance of regulation is at transcription (transcriptional regulation) where due to the chromatin arrangement and due to the activity of transcription factors, genes are differentially transcribed.

After being produced, the stability and distribution of the different transcripts is regulated (post-transcriptional regulation) by means of RNA binding protein (RBP) that control the various steps and rates of the transcripts: events such as alternative splicing, nuclear degradation (exosome), processing, nuclear export (three alternative pathways), sequestration in DCP2-bodies for storage or degradation, and

ultimately translation. These proteins achieve these events thanks to a RNA recognition motif (RRM) that binds a specific sequence or secondary structure of the transcripts, typically at the 5' and 3' UTR of the transcript.

Modulating the capping, splicing, addition of a Poly(A) tail, the sequence-specific nuclear export rates and in several contexts sequestration of the RNA transcript occurs in eukaryotes but not in prokaryotes. This modulation is a result of a protein or transcript which in turn is regulated and may have an affinity for certain sequences.

Capping

Capping changes the five prime end of the mRNA to a three prime end by 5'-5' linkage, which protects the mRNA from 5' exonuclease, which degrades foreign RNA. The cap also helps in ribosomal binding.

Splicing

Splicing removes the introns, noncoding regions that are transcribed into RNA, in order to make the mRNA able to create proteins. Cells do this by spliceosomes binding on either side of an intron, looping the intron into a circle and then cleaving it off. The two ends of the exons are then joined together.

Polyadenylation

Polyadenylation is the addition of a poly(A) tail to the 3' end, i.e. the poly(A) tail consists of multiple adenosine monophosphates. The poly-A sequence acts as a buffer to the 3' exonuclease and thus increases half-life of mRNA. In addition, a long poly(A) tail can increase translation. Thus the poly-A tail may be used to further modulate translation of the construct of the present invention, in order to arrive at the optimal translation rate.

In eukaryotes, polyadenylation is part of the process that produces mature messenger RNA (mRNA) for translation.

The poly(A) tail is also important for the nuclear export, translation, and stability of mRNA.

In one embodiment the nucleic acid sequence of the cell of the present invention, as defined herein above, further comprises a polyadenyl/polyadenylation sequence, preferably the 5' end of said polyadenyl 1/polyadenylation sequence is operably linked to the 3' end of the open reading frame, such as to the open reading frame encoding squalene synthase.

RNA Editing

RNA editing is a process which results in sequence variation in the RNA molecule, and is catalyzed by enzymes. These enzymes include the Adenosine Deaminase Acting on RNA (ADAR) enzymes, which convert specific adenosine residues to inosine in an mRNA molecule by hydrolytic deamination. Three ADAR enzymes have been cloned, ADAR1, ADAR2 and ADAR3, although only the first two subtypes have been shown to have RNA editing activity. Many mRNAs are vulnerable to the effects of RNA editing, including the glutamate receptor subunits GluR2, GluR3, GluR4, GluR5 and GluR6 (which are components of the AMPA and kainate receptors), the serotonin_{2C} receptor, the GABA-alpha3 receptor subunit, the tryptophan hydroxylase enzyme TPH2, the hepatitis delta virus and more than 16% of microRNAs. In addition to ADAR enzymes, CDAR enzymes exist and these convert cytosines in specific RNA molecules, to uracil. These enzymes are termed 'APOBEC' and have genetic loci at 22q13, a region close to the chromosomal deletion which occurs in velocardiofacial syndrome (22q11) and which is linked to psychosis. RNA editing is extensively studied in relation to infectious diseases, because the editing process alters viral function.

Post-Transcriptional Regulatory Elements

Use of a post-transcriptional regulatory elements (PRE) is often necessary to obtain vectors with sufficient performance for certain applications. Schambach et al in Gene Ther. (2006) 13(7):641-5 reports that introduction of a post-transcriptional regulatory element (PRE) of woodchuck hepatitis virus (WHV) into the 3' untranslated region of retroviral and lentiviral gene transfer vectors enhances both titer and transgene expression. The enhancing activity of the PRE depends on the precise configuration of its sequence and the context of the vector and cell into which it is introduced.

Thus use of a PRE such as a woodchuck hepatitis virus post-transcriptional regulatory elements (WPRE) may be useful in the preparation of the cell of the present invention when using a gene therapeutic approach.

Accordingly, in one embodiment the nucleic acid sequence of the cell defined herein further comprises a post-transcriptional regulatory element.

In a further embodiment, the post-transcriptional regulatory element is a Woodchuck hepatitis virus post-transcriptional regulatory element (WPRE).

Terminal Repeats

To insert genetic sequences into host DNA, viruses often use sequences of DNA that repeats up to thousands of times, so called repeats, or terminal repeats including long terminal repeats (LTR) and inverted terminal repeats (ITR), wherein said repeat sequences may be both 5' and 3' terminal repeats. ITRs aid in concatamer formation in the nucleus after the single-stranded vector DNA is converted by host cell DNA polymerase complexes into double-stranded DNA. ITR sequences may be derived from viral vectors, such as AAV, e.g. AAV2.

In one embodiment, the nucleic acid sequence or the vector of the cell defined herein comprises a 5' terminal repeat and a 3' terminal repeat.

In one embodiment said 5' and 3' terminal repeats are selected from Inverted Terminal Repeats [ITR] and Long Terminal Repeats [LTR].

In one embodiment of said 5' and 3' terminal repeats are AAV Inverted Terminal Repeats [ITR].

Geranylgeranyl Pyrophosphate Synthase

The microbial cells of the present invention may in preferred embodiments contain a heterologous nucleic acid sequence encoding Geranylgeranyl Pyrophosphate Synthase (GGPPS). See, e.g., Table 7. GGPPS is an enzyme, which catalyzes the chemical reaction that turns one farnesyl pyrophosphate (FPP) molecule into one Geranylgeranyl Pyrophosphate (GGPP) molecule. Genes encoding GGPPS may for example be found in organisms that contain the mevalonate pathway.

The GGPPS to be used with the present invention may be any useful enzyme, which is capable of catalysing conversion of a farnesyl pyrophosphate (FPP) molecule into a Geranylgeranyl Pyrophosphate (GGPP) molecule. In particular, the GGPPS to be used with the present invention may be any enzyme capable of catalysing the following reaction:

(2E,6E)-farnesyl diphosphate+isopentenyl diphosphate->diphosphate+geranylgeranyl diphosphate.

It is preferred that the GGPPS used with the present invention is an enzyme categorised under EC 2.5.1.29.

The GGPPS may be GGPPS from a variety of sources, such as from bacteria, fungi or mammals. The GGPPS may be any kind of GGPPS, for example GGPPS-1, GGPPS-2, GGPPS-3 or GGPPS-4. The GGPPS may be wild type GGPPS or a functional homologue thereof.

For example, the GGPPS may be GGPPS-1 of *S. acidocaldarius* (SEQ ID NO: 126), GGPPS-2 of *A. nidulans* (SEQ ID NO: 203), GGPPS-3 of *S. cerevisiae* (SEQ ID NO: 167) or GGPPS-4 of *M. musculus* (SEQ ID NO:123) or a functional homologue of any of the aforementioned.

The heterologous nucleic acid encoding said GGPPS may be any nucleic acid sequence encoding said GGPPS. Thus, in embodiments of the invention where GGPPS is a wild type protein, the nucleic acid sequence may for example be a wild type cDNA sequence encoding said protein. However, it is frequently the case that the heterologous nucleic acid is nucleic acid sequence encoding any particular GGPPS, where said nucleic acid has been codon optimised for the particular microbial cell. Thus, by way of example, if the microbial cell is *S. cerevisiae*, then the nucleic acid encoding GGPPS has preferably been codon optimised for optimal expression in *S. cerevisiae*.

Functional homologues of GGPPS are preferably protein having above-mentioned activity and sharing at least 70% amino acid identity with the sequence of a reference GGPPS. Methods for determining sequence identity are described herein above in the section "Squalene synthase" and in section D.

In one embodiment, the cell, such as the microbial of the present invention comprises a nucleic acid sequence coding a GGPPS or a functional homologue thereof, where said functional homologue is at least 75%, such as at least 76%, such as at least 77%, such as at least 78%, such as at least 79%, such as at least 80%, such as at least 81%, such as at least 82%, such as at least 83%, such as at least 84%, such as at least 85%, such as at least 86%, such as at least 87%, such as at least 88%, such as at least 89%, such as at least 90%, such as at least 91%, such as at least 92%, such as at least 93%, such as at least 94%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99%, such as at least 99.5%, such as at least 99.6%, such as at least 99.7%, such as at least 99.8%, such as at least 99.9%, such as 100% identical to a GGPPS selected from the group consisting of SEQ ID NO:123, SEQ ID NO:126, SEQ ID NO:167 and SEQ ID NO:203.

Said heterologous nucleic acid sequence encoding a GGPPS is in general operably linked to a nucleic acid sequence directing expression of GGPPS in the microbial cell. The nucleic acid sequence directing expression of GGPPS in the microbial cell may be a promoter sequence, and preferably said promoter sequence is selected according to the particular microbial cell. The promoter may for example be any of the promoters described herein above in the section "Promoter"

Vectors

A vector is a DNA molecule used as a vehicle to transfer foreign genetic material into another cell. The major types of vectors are plasmids, viruses, cosmids, and artificial chromosomes. Common to all engineered vectors is an origin of replication, a multicloning site, and a selectable marker.

The vector itself is generally a DNA sequence that consists of an insert (transgene) and a larger sequence that serves as the "backbone" of the vector. The purpose of a vector which transfers genetic information to another cell is typically to isolate, multiply, or express the insert in the target cell. Vectors called expression vectors (expression constructs) specifically are for the expression of the transgene in the target cell, and generally have a promoter sequence that drives expression of the transgene. Simpler vectors called transcription vectors are only capable of being transcribed but not translated: they can be replicated in a

target cell but not expressed, unlike expression vectors. Transcription vectors are used to amplify their insert.

Insertion of a vector into the target cell is usually called transformation for bacterial cells, transfection for eukaryotic cells, although insertion of a viral vector is often called transduction.

Plasmids

Plasmid vectors are double-stranded generally circular DNA sequences that are capable of automatically replicating in a host cell. Plasmid vectors minimally consist of an origin of replication that allows for semi-independent replication of the plasmid in the host and also the transgene insert. Modern plasmids generally have many more features, notably including a "multiple cloning site" which includes nucleotide overhangs for insertion of an insert, and multiple restriction enzyme consensus sites to either side of the insert. In the case of plasmids utilized as transcription vectors, incubating bacteria with plasmids generates hundreds or thousands of copies of the vector within the bacteria in hours, and the vectors can be extracted from the bacteria, and the multiple cloning site can be cut by restriction enzymes to excise the hundredfold or thousandfold amplified insert. These plasmid transcription vectors characteristically lack crucial sequences that code for polyadenylation sequences and translation termination sequences in translated mRNAs, making protein expression from transcription vectors impossible. plasmids may be conjugative/transmissible and non-conjugative:

conjugative: mediate DNA transfer through conjugation and therefore spread rapidly among the bacterial cells of a population; e.g., F plasmid, many R and some col plasmids.

nonconjugative—do not mediate DNA through conjugation, e.g., many R and col plasmids.

Viral Vectors

Viral vectors are generally genetically-engineered viruses carrying modified viral DNA or RNA that has been rendered noninfectious, but still contain viral promoters and also the transgene, thus allowing for translation of the transgene through a viral promoter. However, because viral vectors frequently are lacking infectious sequences, they require helper viruses or packaging lines for large-scale transfection. Viral vectors are often designed for permanent incorporation of the insert into the host genome, and thus leave distinct genetic markers in the host genome after incorporating the transgene. For example, retroviruses leave a characteristic retroviral integration pattern after insertion that is detectable and indicates that the viral vector has incorporated into the host genome.

In one embodiment the invention concerns a viral vector capable of transfecting a host cell, such as a cell that can be cultured, e.g. a yeast cell or any other suitable eukaryotic cell. The vector is then capable of transfecting said cell with a nucleic acid that includes the heterologous insert sequence as described herein.

The viral vector can be any suitable viral vector such as a viral vector selected from the group consisting of vectors derived from the Retroviridae family including lentivirus, HIV, SIV, FIV, EAV, CIV.

The viral vector may also be selected from the group consisting of alphavirus, adenovirus, adeno associated virus, baculovirus, HSV, coronavirus, Bovine papilloma virus, Mo-MLV and adeno associated virus.

In embodiments of the invention wherein the microbial cell comprises a heterologous nucleic acid encoding GGPPS, then said heterologous nucleic acid may be positioned on the vector also containing the nucleic acid encod-

ing squalene synthase, or the heterologous nucleic acid encoding GGPPS may be positioned on a different vector. Said heterologous nucleic acid encoding GGPPS may be contained in any of the vectors described herein above.

In embodiments of the invention wherein the microbial cell comprises a heterologous nucleic acid encoding HMCR, then said heterologous nucleic acid may be positioned on the vector also containing the nucleic acid encoding squalene synthase, or the heterologous nucleic acid encoding HMCR may be positioned on a different vector. Said heterologous nucleic acid encoding HMCR may be contained in any of the vectors described herein above. It is also contained within the invention that the heterologous nucleic acid encoding GGPPS and the heterologous nucleic acid encoding HMCR may be positioned on the same or on individual vectors.

Transcription

Transcription is a necessary component in all vectors: the premise of a vector is to multiply the insert (although expression vectors later also drive the translation of the multiplied insert). Thus, even stable expression is determined by stable transcription, which generally depends on promoters in the vector. However, expression vectors have a variety of expression patterns: constitutive (consistent expression) or inducible (expression only under certain conditions or chemicals). This expression is based on different promoter activities, not post-transcriptional activities. Thus, these two different types of expression vectors depend on different types of promoters.

Viral promoters are often used for constitutive expression in plasmids and in viral vectors because they normally reliably force constant transcription in many cell lines and types.

Inducible expression depends on promoters that respond to the induction conditions: for example, the murine mammary tumor virus promoter only initiates transcription after dexamethasone application and the *Drosophila* heat shock promoter only initiates after high temperatures. transcription is the synthesis of mRNA. Genetic information is copied from DNA to RNA

Expression

Expression vectors require sequences that encode for e.g. polyadenylation tail (see herein above): Creates a polyadenylation tail at the end of the transcribed pre-mRNA that protects the mRNA from exonucleases and ensures transcriptional and translational termination: stabilizes mRNA production.

Minimal UTR length: UTRs contain specific characteristics that may impede transcription or translation, and thus the shortest UTRs or none at all are encoded for in optimal expression vectors.

Kozak sequence: Vectors should encode for a Kozak sequence in the mRNA, which assembles the ribosome for translation of the mRNA.

Above conditions are necessary for expression vectors in eukaryotes but not in prokaryotes.

Modern vectors may encompass additional features besides the transgene insert and a backbone such as a promoter (discussed above), genetic markers to e.g. allow for confirmation that the vector has integrated with the host genomic DNA, antibiotic resistance genes for antibiotic selection, and affinity tags for purification.

In one embodiment the cell of the present invention comprises a nucleic acid sequence integrated in a vector such as an expression vector.

In one embodiment the vector is selected from the group consisting of plasmid vectors, cosmids, artificial chromosomes and viral vectors.

The plasmid vector should be able to be maintained and replicated in bacteria, fungi and yeast.

The present invention also concerns cells comprising plasmid and cosmid vectors as well as artificial chromosome vectors.

The important factor is that the vector is functional and that the vector comprises at least the nucleic acid sequence comprising the heterologous insert sequence as described herein.

In one embodiment the vector is functional in fungi and in mammalian cells.

In one embodiment the invention concerns a cell transformed or transduced with the vector as defined herein above.

Methods for Producing Terpenoids

As mentioned herein above, the cell of the present invention (e.g., recombinant host cells) is useful in enhancing yield of industrially relevant terpenoids.

The cell of the invention may therefore be used in various set-ups in order to increase accumulation of terpenoid precursors and thus to increase yield of terpenoid products resulting from enzymatic conversion of said (upstream) terpenoid precursors.

Accordingly, in one aspect the present invention relates to a method for producing a terpenoid compound synthesized through the squalene pathway, in a cell culture, said method comprising the steps of

- (a) providing the cell as defined herein above,
- (b) culturing the cell of (a).
- (c) recovering the terpenoid product compound.

By providing the cell comprising the genetically modified construct defined herein above, the accumulation of terpenoid precursors is enhanced (FIG. 20).

Thus, in another aspect, the invention relates to a method for producing a terpenoid derived from a terpenoid precursor selected from the group consisting of Farnesyl-pyrophosphate (FPP), Isopentenyl-pyrophosphate (IPP), Dimethylallyl-pyrophosphate (DMAPP), Geranyl-pyrophosphate (GPP) and/or Geranylgeranyl-pyrophosphate (GGPP), said method comprising:

- (a) contacting said precursor with an enzyme of the squalene synthase pathway,
- (b) recovering the terpenoid product.

In one embodiment the terpenoid (product) of the method of the present invention as defined herein above, is selected from the group consisting of hemiterpenoids, monoterpenes, sesquiterpenoids, diterpenoids, sesterpenes, triterpenoids, tetraterpenoids and polyterpenoids.

In a further embodiment the terpenoid is selected from the group consisting of farnesyl phosphate, farnesol, geranylgeranyl, geranylgeraniol, isoprene, prenyl, isovaleric acid, geranyl pyrophosphate, eucalyptol, limonene, pinene, farnesyl pyrophosphate, artemisinin, bisabolol, geranylgeranyl pyrophosphate, retinol, retinal, phytol, taxol, forskolin, aphidicolin, lanosterol, lycopene and carotene.

The terpenoid product can be used as starting point in an additional refining process. Thus, in one embodiment said method further comprises dephosphorylating the farnesyl phosphate to produce farnesol.

The enzyme or enzymes used in the process of preparing the target product terpenoid compound is preferably an enzyme "located downstream" of the terpenoid precursors Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate and Geranylgeranyl-pyrophosphate such as an enzyme located downstream of the terpenoid precursors Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophos-

phate, Geranyl-pyrophosphate and Geranylgeranyl-pyrophosphate depicted in the squalene pathway of FIG. 20. The enzyme used in the process of preparing the target product terpenoid, based on the accumulation of precursors achieved through the present invention, may thus be selected from the group consisting of Dimethylallyltransferase (EC 2.5.1.1), Isoprene synthase (EC 4.2.3.27) and Geranyltranstransferase (EC 2.5.1.10).

The present invention may operate by at least partly, sterically hindering binding of the ribosome to the RNA thus reducing the translation of squalene synthase. Accordingly, in one aspect the present invention relates to a method for reducing the translation rate of a functional squalene synthase (EC 2.5.1.21) said method comprising:

- (a) providing the cell defined herein above,
- (b) culturing the cell of (a).

Similarly, the invention in another aspect relates to a method for decreasing turnover of farnesyl-pp to squalene, said method comprising:

- (a) providing the cell defined herein above,
- (b) culturing the cell of (a).

As depicted in FIG. 20, the knocking down of the ERG9 results in build-up of precursors to squalene synthase. Thus in one aspect, the present invention relates to a method for enhancing accumulation of a compound selected from the group consisting of Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate and Geranylgeranyl-pyrophosphate, said method comprising the steps of:

- (a) providing the cell defined herein above, and
- (b) culturing the cell of (a).

In one embodiment the method of the invention as defined herein above further comprises recovering the Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate or Geranylgeranyl-pyrophosphate compound. The recovered compound may be used in further processes for producing the desired terpenoid product compound. The further process may take place in the same cell culture as the process performed and defined herein above, such as the accumulation of the terpenoid precursors by the cell of the present invention. Alternatively, the recovered precursors may be added to another cell culture, or a cell free system, to produce the desired products.

As the precursors are intermediates, however mainly stable intermediates, a certain endogenous production of terpenoid products may occur based on the terpenoid precursor substrates. Also, the cells of the invention may have additional genetic modifications such that they are capable of performing both the accumulation of the terpenoid precursors (construct of the cell of the invention) and whole or substantially the whole subsequent biosynthesis process to the desired terpenoid product.

Thus, in one embodiment the method of the invention further comprises recovering a compound synthesized through the squalene pathway, said compound being derived from said Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate and/or Geranylgeranyl-pyrophosphate.

Occasionally it may be advantageous to include a squalene synthase inhibitor when culturing the cell of the present invention. Chemical inhibition of squalene synthase, e.g. by lapaquistat, is known in the art and is under investigation e.g. as a method of lowering cholesterol levels in the prevention of cardiovascular disease. It has also been suggested that variants in this enzyme may be part of a genetic

association with hypercholesterolemia. Other squalene synthase inhibitors include Zaragozic acid and RPR 107393.

Thus, in one embodiment the culturing step of the method(s) defined herein above is performed in the presence of a squalene synthase inhibitor.

The cell of the invention may furthermore be genetically modified to further enhance production of certain key terpenoid precursors. In one embodiment the cell is additionally genetically modified to enhance activity of and/or overexpress one or more enzymes selected from the group consisting of Phosphomevalonate kinase (EC 2.7.4.2), Diphosphomevalonate decarboxylase (EC 4.1.1.33), 4-hydroxy-3-methylbut-2-en-1-yl diphosphate synthase (EC 1.17.7.1), 4-hydroxy-3-methylbut-2-enyl diphosphate reductase (EC 1.17.1.2), Isopentenyl-diphosphate Delta-isomerase 1 (EC 5.3.3.2), Short-chain Z-isoprenyl diphosphate synthase (EC 2.5.1.68), Dimethylallyltransferase (EC 2.5.1.1), Geranyltranstransferase (EC 2.5.1.10) and Geranylgeranyl pyrophosphate synthetase (EC 2.5.1.29).

As described herein above in one embodiment of the invention the microbial cell comprises both a nucleic acid encoding a squalene synthase as described herein above as well as a heterologous nucleic acid encoding a GGPPS. Such microbial cells are particularly useful for the preparation of GGPP as well as terpenoids, wherein GGPP is an intermediate in their biosynthesis.

Accordingly, in one aspect the invention relates to a method for preparing GGPP, wherein the method comprises the steps of

- a. providing a microbial cell comprising a nucleic acid sequence, said nucleic acid comprising
 - i) a promoter sequence operably linked to
 - ii) a heterologous insert sequence operably linked to
 - iii) an open reading frame operably linked to
 - iv) a transcription termination signal,
 wherein the heterologous insert sequence and the open reading frame are as defined herein above, wherein said microbial cell furthermore comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell;
- b. Cultivating the microbial cell of a.;
- c. Recovering the GGPP.

In another aspect the invention relates to a method for preparing a terpenoid of which GGPP is an intermediate in the biosynthesis pathway, wherein the method comprises the steps of

- a. providing a microbial cell, wherein said microbial cell comprises a nucleic acid sequence, said nucleic acid comprising
 - i) a promoter sequence operably linked to
 - ii) a heterologous insert sequence operably linked to
 - iii) an open reading frame operably linked to
 - iv) a transcription termination signal,
 wherein the heterologous insert sequence and the open reading frame are as defined herein above, wherein said microbial cell furthermore comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell;
- b. Cultivating the microbial cell of a.; and
- c. Recovering the terpenoid,

wherein said terpenoid may be any terpenoid described herein above in the section "Terpenoids" having GGPP as intermediate in its biosyntheses; and said microbial cell may be any of the microbial cells described herein above in the section "The cell"; and said promoter may be any

promoter, such as any of the promoters described herein above in the section "Promoter"; and said heterologous insert sequence may be any of the heterologous insert sequences described herein above in the section "Heterologous insert sequence"; and said open reading frame encodes a squalene synthase, which may be any of the squalene synthases described herein above in the section "Squalene synthase"; and said GGPPS may be any of the GGPPS described herein above in the section "Geranylgeranyl Pyrophosphate Synthase".

In this embodiment said microbial cell may also optionally contain one or more additional heterologous nucleic acids encoding one or more enzymes involved in the biosynthesis pathway of said terpenoid.

In one particular aspect the invention relates to a method for preparing steviol, wherein the method comprises the steps of

- a. providing a microbial cell, wherein said microbial cell comprises a nucleic acid sequence, said nucleic acid comprising
 - i) a promoter sequence operably linked to
 - ii) a heterologous insert sequence operably linked to
 - iii) an open reading frame operably linked to
 - iv) a transcription termination signal,
 wherein the heterologous insert sequence and the open reading frame are as defined herein above, wherein said microbial cell furthermore comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell;
- b. Cultivating the microbial cell of a.;
- c. Recovering steviol,

wherein said microbial cell may be any of the microbial cells described herein above in the section "The cell"; and said promoter may be any promoter, such as any of the promoters described herein above in the section "Promoter"; and said heterologous insert sequence may be any of the heterologous insert sequences described herein above in the section "Heterologous insert sequence"; and said open reading frame encodes a squalene synthase, which may be any of the squalene synthases described herein above in the section "Squalene synthase"; and said GGPPS may be any of the GGPPS described herein above in the section "Geranylgeranyl Pyrophosphate Synthase".

In this embodiment said microbial cell may also optionally contain one or more additional heterologous nucleic acids encoding one or more enzymes involved in the biosynthesis pathway of steviol.

In another aspect the invention relates to a method for preparing a terpenoid of which GGPP is an intermediate in the biosynthesis pathway, wherein the method comprises the steps of

- a. providing a microbial cell, wherein said microbial cell comprises a nucleic acid sequence, said nucleic acid comprising
 - i) a promoter sequence operably linked to
 - ii) a heterologous insert sequence operably linked to
 - iii) an open reading frame operably linked to
 - iv) a transcription termination signal,
 wherein the heterologous insert sequence and the open reading frame are as defined herein above, wherein said microbial cell furthermore comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell

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and wherein said microbial cell furthermore comprises a heterologous nucleic acid encoding HMCR operably linked to a nucleic acid sequence directing expression of HMCR in said cell;

b. Cultivating the microbial cell of a.;

c. Recovering the terpenoid,

wherein said terpenoid may be any terpenoid described herein above in the section "Terpenoids" having GGPP as intermediate in its biosyntheses; and said microbial cell may be any of the microbial cells described herein above in the section "The cell"; and said promoter may be any promoter, such as any of the promoters described herein above in the section "Promoter"; and said heterologous insert sequence may be any of the heterologous insert sequences described herein above in the section "Heterologous insert sequence"; and said open reading frame encodes a squalene synthase, which may be any of the squalene synthases described herein above in the section "Squalene synthase"; and said GGPPS may be any of the GGPPS described herein above in the section "Geranylgeranyl Pyrophosphate Synthase"; and said HMCR may be any of the HMCR described herein above in the section "HMCR".

In this embodiment said microbial cell may also optionally contain one or more additional heterologous nucleic acids encoding one or more enzymes involved in the biosynthesis pathway of said terpenoid.

In one particular aspect the invention relates to a method for preparing steviol, wherein the method comprises the steps of

a. providing a microbial cell, wherein said microbial cell comprises a nucleic acid sequence, said nucleic acid comprising

- i) a promoter sequence operably linked to
- ii) a heterologous insert sequence operably linked to
- iii) an open reading frame operably linked to
- iv) a transcription termination signal,

wherein the heterologous insert sequence and the open reading frame are as defined herein above,

wherein said microbial cell furthermore comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell;

b. Cultivating the microbial cell of a.;

c. Recovering steviol,

wherein said microbial cell may be any of the microbial cells described herein above in the section "The cell"; and said promoter may be any promoter, such as any of the promoters described herein above in the section "Promoter"; and said heterologous insert sequence may be any of the heterologous insert sequences described herein above in the section "Heterologous insert sequence"; and said open reading frame encodes a squalene synthase, which may be any of the squalene synthases described herein above in the section "Squalene synthase"; and said GGPPS may be any of the GGPPS described herein above in the section "Geranylgeranyl Pyrophosphate Synthase" and said HMCR may be any of the HMCR described herein above in the section "HMCR".

In this embodiment said microbial cell may also optionally contain one or more additional heterologous nucleic acids encoding one or more enzymes involved in the biosynthesis pathway of steviol.

In one embodiment the cell is additionally genetically modified to enhance activity of and/or overexpress one or more enzymes selected from the group consisting of acetoacetyl CoA thiolase, HMG-CoA reductase or the catalytic domain thereof, HMG-CoA synthase, mevalonate

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kinase, phosphomevalonate kinase, phosphomevalonate decarboxylase, isopentenyl pyrophosphate isomerase, farnesyl pyrophosphate synthase, D-1-deoxyxylulose 5-phosphate synthase, and 1-deoxy-D-xylulose 5-phosphate reductoisomerase and farnesyl pyrophosphate synthase.

In one embodiment of the method of the present invention, the cell comprises a mutation in the ERG9 open reading frame.

In another embodiment of the method of the present invention the cell comprises an ERG9[Delta]::HIS3 deletion/insertion allele.

In yet another embodiment the step of recovering the compound in the method of the present invention further comprises purification of said compound from the cell culture media.

D. Functional Homologs

Functional homologs of the polypeptides described above are also suitable for use in producing steviol or steviol glycosides in a recombinant host. A functional homolog is a polypeptide that has sequence similarity to a reference polypeptide, and that carries out one or more of the biochemical or physiological function(s) of the reference polypeptide. A functional homolog and the reference polypeptide may be natural occurring polypeptides, and the sequence similarity may be due to convergent or divergent evolutionary events. As such, functional homologs are sometimes designated in the literature as homologs, or orthologs, or paralogs. Variants of a naturally occurring functional homolog, such as polypeptides encoded by mutants of a wild type coding sequence, may themselves be functional homologs. Functional homologs can also be created via site-directed mutagenesis of the coding sequence for a polypeptide, or by combining domains from the coding sequences for different naturally-occurring polypeptides ("domain swapping"). Techniques for modifying genes encoding functional UGT polypeptides described herein are known and include, inter alia, directed evolution techniques, site-directed mutagenesis techniques and random mutagenesis techniques, and can be useful to increase specific activity of a polypeptide, alter substrate specificity, alter expression levels, alter subcellular location, or modify polypeptide:polypeptide interactions in a desired manner. Such modified polypeptides are considered functional homologs. The term "functional homolog" is sometimes applied to the nucleic acid that encodes a functionally homologous polypeptide.

Functional homologs can be identified by analysis of nucleotide and polypeptide sequence alignments. For example, performing a query on a database of nucleotide or polypeptide sequences can identify homologs of steviol or steviol glycoside biosynthesis polypeptides. Sequence analysis can involve BLAST, Reciprocal BLAST, or PSI-BLAST analysis of nonredundant databases using a GGPPS, a CDPS, a KS, a KO or a KAH amino acid sequence as the reference sequence. Amino acid sequence is, in some instances, deduced from the nucleotide sequence. Those polypeptides in the database that have greater than 40% sequence identity are candidates for further evaluation for suitability as a steviol or steviol glycoside biosynthesis polypeptide. Amino acid sequence similarity allows for conservative amino acid substitutions, such as substitution of one hydrophobic residue for another or substitution of one polar residue for another. If desired, manual inspection of such candidates can be carried out in order to narrow the number of candidates to be further evaluated. Manual inspection can be performed by selecting those candidates

that appear to have domains present in steviol biosynthesis polypeptides, e.g., conserved functional domains.

Conserved regions can be identified by locating a region within the primary amino acid sequence of a steviol or a steviol glycoside biosynthesis polypeptide that is a repeated sequence, forms some secondary structure (e.g., helices and beta sheets), establishes positively or negatively charged domains, or represents a protein motif or domain. See, e.g., the Pfam web site describing consensus sequences for a variety of protein motifs and domains on the World Wide Web at sanger.ac.uk/Software/Pfam/ and pfam.janelia.org/. The information included at the Pfam database is described in Sonnhammer et al., *Nucl. Acids Res.*, 26:320-322 (1998); Sonnhammer et al., *Proteins*, 28:405-420 (1997); and Bateman et al., *Nucl. Acids Res.*, 27:260-262 (1999). Conserved regions also can be determined by aligning sequences of the same or related polypeptides from closely related species. Closely related species preferably are from the same family. In some embodiments, alignment of sequences from two different species is adequate.

Typically, polypeptides that exhibit at least about 40% amino acid sequence identity are useful to identify conserved regions. Conserved regions of related polypeptides exhibit at least 45% amino acid sequence identity (e.g., at least 50%, at least 60%, at least 70%, at least 80%, or at least 90% amino acid sequence identity). In some embodiments, a conserved region exhibits at least 92%, 94%, 96%, 98%, or 99% amino acid sequence identity.

For example, polypeptides suitable for producing steviol glycosides in a recombinant host include functional homologs of EUGT11, UGT91D2e, UGT91D2m, UGT85C, and UGT76G. Such homologs have greater than 90% (e.g., at least 95% or 99%) sequence identity to the amino acid sequence of EUGT11 (SEQ ID NO: 152), UGT91D2e (SEQ ID NO:5), UGT91D2m (SEQ ID NO:10), UGT85C (SEQ ID NO:3), or UGT76G (SEQ ID NO:7). Variants of EUGT11, UGT91D2, UGT85C, and UGT76G polypeptides typically have 10 or fewer amino acid substitutions within the primary amino acid sequence, e.g., 7 or fewer amino acid substitutions, 5 or conservative amino acid substitutions, or between 1 and 5 substitutions. However, in some embodiments, variants of EUGT11, UGT91D2, UGT85C, and UGT76G polypeptides can have 10 or more amino acid substitutions (e.g., 10, 15, 20, 25, 30, 35, 10-20, 10-35, 20-30, or 25-35 amino acid substitutions). The substitutions may be conservative, or in some embodiments, non-conservative. Non-limiting examples of non-conservative changes in UGT91D2e polypeptides include glycine to arginine and tryptophan to arginine. Non-limiting examples of non-conservative substitutions in UGT76G polypeptides include valine to glutamic acid, glycine to glutamic acid, glutamine to alanine, and serine to proline. Non-limiting examples of changes to UGT85C polypeptides include histidine to aspartic acid, proline to serine, lysine to threonine, and threonine to arginine.

In some embodiments, a useful UGT91D2 homolog can have amino acid substitutions (e.g., conservative amino acid substitutions) in regions of the polypeptide that are outside of predicted loops, e.g., residues 20-26, 39-43, 88-95, 121-124, 142-158, 185-198, and 203-214 are predicted loops in the N-terminal domain and residues 381-386 are predicted loops in the C-terminal domain of SEQ ID NO:5. For example, a useful UGT91D2 homolog can include at least one amino acid substitution at residues 1-19, 27-38, 44-87, 96-120, 125-141, 159-184, 199-202, 215-380, or 387-473 of SEQ ID NO:5. In some embodiments, a UGT91D2 homolog can have an amino acid substitution at one or more residues

selected from the group consisting of residues 30, 93, 99, 122, 140, 142, 148, 153, 156, 195, 196, 199, 206, 207, 211, 221, 286, 343, 427, and 438 of SEQ ID NO:5. For example, a UGT91D2 functional homolog can have an amino acid substitution at one or more of residues 206, 207, and 343, such as an arginine at residue 206, a cysteine at residue 207, and an arginine at residue 343 of SEQ ID NO:5. See, SEQ ID NO:95. Other functional homologs of UGT91D2 can have one or more of the following: a tyrosine or phenylalanine at residue 30, a proline or glutamine at residue 93, a serine or valine at residue 99, a tyrosine or a phenylalanine at residue 122, a histidine or tyrosine at residue 140, a serine or cysteine at residue 142, an alanine or threonine at residue 148, a methionine at residue 152, an alanine at residue 153, an alanine or serine at residue 156, a glycine at residue 162, a leucine or methionine at residue 195, a glutamic acid at residue 196, a lysine or glutamic acid at residue 199, a leucine or methionine at residue 211, a leucine at residue 213, a serine or phenylalanine at residue 221, a valine or isoleucine at residue 253, a valine or alanine at residue 286, a lysine or asparagine at residue 427, an alanine at residue 438, and either an alanine or threonine at residue 462 of SEQ ID NO:5. In another embodiment, a UGT91D2 functional homolog contains a methionine at residue 211 and an alanine at residue 286.

In some embodiments, a useful UGT85C homolog can have one or more amino acid substitutions at residues 9, 10, 13, 15, 21, 27, 60, 65, 71, 87, 91, 220, 243, 270, 289, 298, 334, 336, 350, 368, 389, 394, 397, 418, 420, 440, 441, 444, and 471 of SEQ ID NO:3. Non-limiting examples of useful UGT85C homologs include polypeptides having substitutions (with respect to SEQ ID NO:3) at residue 65 (e.g., a serine at residue 65), at residue 65 in combination with residue 15 (a leucine at residue 15), 270 (e.g., a methionine, arginine, or alanine at residue 270), 418 (e.g., a valine at residue 418), 440 (e.g., an aspartic acid at residue at residue 440), or 441 (e.g., an asparagine at residue 441); residues 13 (e.g., a phenylalanine at residue 13), 15, 60 (e.g., an aspartic acid at residue 60), 270, 289 (e.g., a histidine at residue 289), and 418; substitutions at residues 13, 60, and 270; substitutions at residues 60 and 87 (e.g., a phenylalanine at residue 87); substitutions at residues 65, 71 (e.g., a glutamine at residue 71), 220 (e.g., a threonine at residue 220), 243 (e.g., a tryptophan at residue 243), and 270; substitutions at residues 65, 71, 220, 243, 270, and 441; substitutions at residues 65, 71, 220, 389 (e.g., a valine at residue 389), and 394 (e.g., a valine at residue 394); substitutions at residues 65, 71, 270, and 289; substitutions at residues 220, 243, 270, and 334 (e.g., a serine at residue 334); or substitutions at residues 270 and 289. The following amino acid mutations did not result in a loss of activity in 85C2 polypeptides: V13F, F15L, H60D, A65S, E71Q, I87F, K220T, R243W, T270M, T270R, Q289H, L334S, A389V, I394V, P397S, E418V, G440D, and H441N. Additional mutations that were seen in active clones include K9E, K10R, Q21H, M27V, L91P, Y298C, K350T, H368R, G420R, L431P, R444G, and M471T. In some embodiments, an UGT85C2 contains substitutions at positions 65 (e.g., a serine), 71 (a glutamine), 270 (a methionine), 289 (a histidine), and 389 (a valine).

The amino acid sequence of *Stevia rebaudiana* UGTs 74G1, 76G1 and 91D2e with N-terminal, in-frame fusions of the first 158 amino acids of human MDM2 protein, and *Stevia rebaudiana* UGT85C2 with an N-terminal in-frame fusion of 4 repeats of the synthetic PMI peptide (4 X TSFAEYWNLLSP, SEQ ID NO:86) are set forth in SEQ ID

NOs: 90, 88, 94, and 92, respectively; see SEQ ID NOs: 89, 92, 93, and 95 for the nucleotide sequences encoding the fusion proteins.

In some embodiments, a useful UGT76G homolog can have one or more amino acid substitutions at residues 29, 74, 87, 91, 116, 123, 125, 126, 130, 145, 192, 193, 194, 196, 198, 199, 200, 203, 204, 205, 206, 207, 208, 266, 273, 274, 284, 285, 291, 330, 331, and 346 of SEQ ID NO:7. Non-limiting examples of useful UGT76G homologs include polypeptides having substitutions (with respect to SEQ ID NO:7) at residues 74, 87, 91, 116, 123, 125, 126, 130, 145, 192, 193, 194, 196, 198, 199, 200, 203, 204, 205, 206, 207, 208, and 291; residues 74, 87, 91, 116, 123, 125, 126, 130, 145, 192, 193, 194, 196, 198, 199, 200, 203, 204, 205, 206, 207, 208, 266, 273, 274, 284, 285, and 291; or residues 74, 87, 91, 116, 123, 125, 126, 130, 145, 192, 193, 194, 196, 198, 199, 200, 203, 204, 205, 206, 207, 208, 266, 273, 274, 284, 285, 291, 330, 331, and 346. See, Table 9.

TABLE 9

Clone	Mutations
76G_G7	M29I, V74E, V87G, L91P, G116E, A123T, Q125A, I126L, T130A, V145M, C192S, S193A, F194Y, M196N, K198Q, K199I, Y200L, Y203I, F204L, E205G, N206K, I207M, T208I, P266Q, S273P, R274S, G284T, T285S, 287-3 bp deletion, L330V, G331A, L346I
76G_H12	M29I, V74E, V87G, L91P, G116E, A123T, Q125A, I126L, T130A, V145M, C192S, S193A, F194Y, M196N, K198Q, K199I, Y200L, Y203I, F204L, E205G, N206K, I207M, T208I, P266Q, S273P, R274S, G284T, T285S, 287-3 bp deletion
76G_C4	M29I, V74E, V87G, L91P, G116E, A123T, Q125A, I126L, T130A, V145M, C192S, S193A, F194Y, M196N, K198Q, K199I, Y200L, Y203I, F204L, E205G, N206K, I207M, T208I

Methods to modify the substrate specificity of, for example, EUGT11 or UGT91D2e, are known to those skilled in the art, and include without limitation site-directed/rational mutagenesis approaches, random directed evolution approaches and combinations in which random mutagenesis/saturation techniques are performed near the active site of the enzyme. For example see Sarah A. Osmani, et al. *Phytochemistry* 70 (2009) 325-347.

A candidate sequence typically has a length that is from 80 percent to 200 percent of the length of the reference sequence, e.g., 82, 85, 87, 89, 90, 93, 95, 97, 99, 100, 105, 110, 115, 120, 130, 140, 150, 160, 170, 180, 190, or 200 percent of the length of the reference sequence. A functional homolog polypeptide typically has a length that is from 95 percent to 105 percent of the length of the reference sequence, e.g., 90, 93, 95, 97, 99, 100, 105, 110, 115, or 120 percent of the length of the reference sequence, or any range between. A percent identity for any candidate nucleic acid or polypeptide relative to a reference nucleic acid or polypeptide can be determined as follows. A reference sequence (e.g., a nucleic acid sequence or an amino acid sequence) is aligned to one or more candidate sequences using the computer program ClustalW (version 1.83, default parameters), which allows alignments of nucleic acid or polypeptide sequences to be carried out across their entire length (global alignment). Chenna et al., *Nucleic Acids Res.*, 31(13):3497-500 (2003).

ClustalW calculates the best match between a reference and one or more candidate sequences, and aligns them so that identities, similarities and differences can be determined. Gaps of one or more residues can be inserted into a reference sequence, a candidate sequence, or both, to maxi-

mize sequence alignments. For fast pairwise alignment of nucleic acid sequences, the following default parameters are used: word size: 2; window size: 4; scoring method: percentage; number of top diagonals: 4; and gap penalty: 5. For multiple alignment of nucleic acid sequences, the following parameters are used: gap opening penalty: 10.0; gap extension penalty: 5.0; and weight transitions: yes. For fast pairwise alignment of protein sequences, the following parameters are used: word size: 1; window size: 5; scoring method: percentage; number of top diagonals: 5; gap penalty: 3. For multiple alignment of protein sequences, the following parameters are used: weight matrix: blosum; gap opening penalty: 10.0; gap extension penalty: 0.05; hydrophilic gaps: on; hydrophilic residues: Gly, Pro, Ser, Asn, Asp, Gln, Glu, Arg, and Lys; residue-specific gap penalties: on. The ClustalW output is a sequence alignment that reflects the relationship between sequences. ClustalW can be run, for example, at the Baylor College of Medicine Search Launcher site on the World Wide Web (searchlauncher.bcm.tmc.edu/multi-align/multi-align.html) and at the European Bioinformatics Institute site on the World Wide Web (ebi.ac.uk/clustalw).

To determine percent identity of a candidate nucleic acid or amino acid sequence to a reference sequence, the sequences are aligned using ClustalW, the number of identical matches in the alignment is divided by the length of the reference sequence, and the result is multiplied by 100. It is noted that the percent identity value can be rounded to the nearest tenth. For example, 78.11, 78.12, 78.13, and 78.14 are rounded down to 78.1, while 78.15, 78.16, 78.17, 78.18, and 78.19 are rounded up to 78.2.

It will be appreciated that functional UGTs can include additional amino acids that are not involved in glucosylation or other enzymatic activities carried out by the enzyme, and thus such a polypeptide can be longer than would otherwise be the case. For example, a EUGT11 polypeptide can include a purification tag (e.g., HIS tag or GST tag), a chloroplast transit peptide, a mitochondrial transit peptide, an amyloplast peptide, signal peptide, or a secretion tag added to the amino or carboxy terminus. In some embodiments, a EUGT11 polypeptide includes an amino acid sequence that functions as a reporter, e.g., a green fluorescent protein or yellow fluorescent protein.

II. STEVIOL AND STEVIOL GLYCOSIDE BIOSYNTHESIS NUCLEIC ACIDS

A recombinant gene encoding a polypeptide described herein comprises the coding sequence for that polypeptide, operably linked in sense orientation to one or more regulatory regions suitable for expressing the polypeptide. Because many microorganisms are capable of expressing multiple gene products from a polycistronic mRNA, multiple polypeptides can be expressed under the control of a single regulatory region for those microorganisms, if desired. A coding sequence and a regulatory region are considered to be operably linked when the regulatory region and coding sequence are positioned so that the regulatory region is effective for regulating transcription or translation of the sequence. Typically, the translation initiation site of the translational reading frame of the coding sequence is positioned between one and about fifty nucleotides downstream of the regulatory region for a monocistronic gene.

In many cases, the coding sequence for a polypeptide described herein is identified in a species other than the recombinant host, i.e., is a heterologous nucleic acid. Thus, if the recombinant host is a microorganism, the coding

sequence can be from other prokaryotic or eukaryotic microorganisms, from plants or from animals. In some case, however, the coding sequence is a sequence that is native to the host and is being reintroduced into that organism. A native sequence can often be distinguished from the naturally occurring sequence by the presence of non-natural sequences linked to the exogenous nucleic acid, e.g., non-native regulatory sequences flanking a native sequence in a recombinant nucleic acid construct. In addition, stably transformed exogenous nucleic acids typically are integrated at positions other than the position where the native sequence is found.

"Regulatory region" refers to a nucleic acid having nucleotide sequences that influence transcription or translation initiation and rate, and stability and/or mobility of a transcription or translation product. Regulatory regions include, without limitation, promoter sequences, enhancer sequences, response elements, protein recognition sites, inducible elements, protein binding sequences, 5' and 3' untranslated regions (UTRs), transcriptional start sites, termination sequences, polyadenylation sequences, introns, and combinations thereof. A regulatory region typically comprises at least a core (basal) promoter. A regulatory region also may include at least one control element, such as an enhancer sequence, an upstream element or an upstream activation region (UAR). A regulatory region is operably linked to a coding sequence by positioning the regulatory region and the coding sequence so that the regulatory region is effective for regulating transcription or translation of the sequence. For example, to operably link a coding sequence and a promoter sequence, the translation initiation site of the translational reading frame of the coding sequence is typically positioned between one and about fifty nucleotides downstream of the promoter. A regulatory region can, however, be positioned as much as about 5,000 nucleotides upstream of the translation initiation site, or about 2,000 nucleotides upstream of the transcription start site.

The choice of regulatory regions to be included depends upon several factors, including, but not limited to, efficiency, selectability, inducibility, desired expression level, and preferential expression during certain culture stages. It is a routine matter for one of skill in the art to modulate the expression of a coding sequence by appropriately selecting and positioning regulatory regions relative to the coding sequence. It will be understood that more than one regulatory region may be present, e.g., introns, enhancers, upstream activation regions, transcription terminators, and inducible elements.

One or more genes can be combined in a recombinant nucleic acid construct in "modules" useful for a discrete aspect of steviol and/or steviol glycoside production. Combining a plurality of genes in a module, particularly a polycistronic module, facilitates the use of the module in a variety of species. For example, a steviol biosynthesis gene cluster, or a UGT gene cluster, can be combined in a polycistronic module such that, after insertion of a suitable regulatory region, the module can be introduced into a wide variety of species. As another example, a UGT gene cluster can be combined such that each UGT coding sequence is operably linked to a separate regulatory region, to form a UGT module. Such a module can be used in those species for which monocistronic expression is necessary or desirable. In addition to genes useful for steviol or steviol glycoside production, a recombinant construct typically also contains an origin of replication, and one or more selectable markers for maintenance of the construct in appropriate species.

It will be appreciated that because of the degeneracy of the genetic code, a number of nucleic acids can encode a particular polypeptide; i.e., for many amino acids, there is more than one nucleotide triplet that serves as the codon for the amino acid. Thus, codons in the coding sequence for a given polypeptide can be modified such that optimal expression in a particular host is obtained, using appropriate codon bias tables for that host (e.g., microorganism). SEQ ID NOs:18-25, 34-36, 40-43, 48-49, 52-55, 60-64, 70-72, and 154 set forth nucleotide sequences encoding certain enzymes for steviol and steviol glycoside biosynthesis, modified for increased expression in yeast. As isolated nucleic acids, these modified sequences can exist as purified molecules and can be incorporated into a vector or a virus for use in constructing modules for recombinant nucleic acid constructs.

In some cases, it is desirable to inhibit one or more functions of an endogenous polypeptide in order to divert metabolic intermediates towards steviol or steviol glycoside biosynthesis. For example, it may be desirable to downregulate synthesis of sterols in a yeast strain in order to further increase steviol or steviol glycoside production, e.g., by downregulating squalene epoxidase. As another example, it may be desirable to inhibit degradative functions of certain endogenous gene products, e.g., glycohydrolases that remove glucose moieties from secondary metabolites or phosphatases as discussed herein. As another example, expression of membrane transporters involved in transport of steviol glycosides can be inhibited, such that secretion of glycosylated steviosides is inhibited. Such regulation can be beneficial in that secretion of steviol glycosides can be inhibited for a desired period of time during culture of the microorganism, thereby increasing the yield of glycoside product(s) at harvest. In such cases, a nucleic acid that inhibits expression of the polypeptide or gene product may be included in a recombinant construct that is transformed into the strain. Alternatively, mutagenesis can be used to generate mutants in genes for which it is desired to inhibit function.

III. HOSTS

A. Microorganisms

A number of prokaryotes and eukaryotes are suitable for use in constructing the recombinant microorganisms described herein, e.g., gram-negative bacteria, yeast and fungi. A species and strain selected for use as a steviol or steviol glycoside production strain is first analyzed to determine which production genes are endogenous to the strain and which genes are not present. Genes for which an endogenous counterpart is not present in the strain are assembled in one or more recombinant constructs, which are then transformed into the strain in order to supply the missing function(s).

Exemplary prokaryotic and eukaryotic species are described in more detail below. However, it will be appreciated that other species may be suitable. For example, suitable species may be in a genus selected from the group consisting of *Agaricus*, *Aspergillus*, *Bacillus*, *Candida*, *Corynebacterium*, *Escherichia*, *Fusarium*/*Gibberella*, *Claviceps*, *Laetiporus*, *Lentinus*, *Phaffia*, *Phanerochaete*, *Pichia*, *Physcomitrella*, *Rhodotorula*, *Saccharomyces*, *Schizosaccharomyces*, *Sphaceloma*, *Xanthophyllomyces* and *Yarrowia*. Exemplary species from such genera include *Lentinus tigrinus*, *Laetiporus sulphureus*, *Phanerochaete chrysosporium*, *Pichia pastoris*, *Physcomitrella patens*, *Rhodotorula glutinis* 32, *Rhodotorula mucilaginosa*,

Phaffia rhodozyma UBV-AX, *Xanthophyllomyces dendrorhous*, *Fusarium fujikuroi*/*Gibberella fujikuroi*, *Candida utilis* and *Yarrowia lipolytica*. In some embodiments, a microorganism can be an Ascomycete such as *Gibberella fujikuroi*, *Kluyveromyces lactis*, *Schizosaccharomyces pombe*, *Aspergillus niger*, or *Saccharomyces cerevisiae*. In some embodiments, a microorganism can be a prokaryote such as *Escherichia coli*, *Rhodobacter sphaeroides*, or *Rhodobacter capsulatus*. It will be appreciated that certain microorganisms can be used to screen and test genes of interest in a high throughput manner, while other microorganisms with desired productivity or growth characteristics can be used for large-scale production of steviol glycosides. *Saccharomyces cerevisiae*

Saccharomyces cerevisiae is a widely used chassis organism in synthetic biology, and can be used as the recombinant microorganism platform. There are libraries of mutants, plasmids, detailed computer models of metabolism and other information available for *S. cerevisiae*, allowing for rational design of various modules to enhance product yield. Methods are known for making recombinant microorganisms.

A steviol biosynthesis gene cluster can be expressed in yeast using any of a number of known promoters. Strains that overproduce terpenes are known and can be used to increase the amount of geranylgeranyl diphosphate available for steviol and steviol glycoside production. *Aspergillus* spp.

Aspergillus species such as *A. oryzae*, *A. niger* and *A. sojae* are widely used microorganisms in food production, and can also be used as the recombinant microorganism platform. Nucleotide sequences are available for genomes of *A. nidulans*, *A. fumigatus*, *A. oryzae*, *A. clavatus*, *A. flavus*, *A. niger*, and *A. terreus*, allowing rational design and modification of endogenous pathways to enhance flux and increase product yield. Metabolic models have been developed for *Aspergillus*, as well as transcriptomic studies and proteomics studies. *A. niger* is cultured for the industrial production of a number of food ingredients such as citric acid and gluconic acid, and thus species such as *A. niger* are generally suitable for the production of food ingredients such as steviol and steviol glycosides.

Escherichia coli

Escherichia coli, another widely used platform organism in synthetic biology, can also be used as the recombinant microorganism platform. Similar to *Saccharomyces*, there are libraries of mutants, plasmids, detailed computer models of metabolism and other information available for *E. coli*, allowing for rational design of various modules to enhance product yield. Methods similar to those described above for *Saccharomyces* can be used to make recombinant *E. coli* microorganisms.

Agaricus, *Gibberella*, and *Phanerochaete* spp.

Agaricus, *Gibberella*, and *Phanerochaete* spp. can be useful because they are known to produce large amounts of gibberellin in culture. Thus, the terpene precursors for producing large amounts of steviol and steviol glycosides are already produced by endogenous genes. Thus, modules containing recombinant genes for steviol or steviol glycoside biosynthesis polypeptides can be introduced into species from such genera without the necessity of introducing mevalonate or MEP pathway genes.

Arxula adeninivorans (*Blastobotrys adeninivorans*)

Arxula adeninivorans is a dimorphic yeast (it grows as a budding yeast like the baker's yeast up to a temperature of 42° C., above this threshold it grows in a filamentous form) with unusual biochemical characteristics. It can grow on a

wide range of substrates and can assimilate nitrate. It has successfully been applied to the generation of strains that can produce natural plastics or the development of a biosensor for estrogens in environmental samples.

Yarrowia lipolytica

Yarrowia lipolytica is a dimorphic yeast (see *Arxula adeninivorans*) that can grow on a wide range of substrates. It has a high potential for industrial applications but there are no recombinant products commercially available yet.

Rhodobacter spp.

Rhodobacter can be used as the recombinant microorganism platform. Similar to *E. coli*, there are libraries of mutants available as well as suitable plasmid vectors, allowing for rational design of various modules to enhance product yield. Isoprenoid pathways have been engineered in membranous bacterial species of *Rhodobacter* for increased production of carotenoid and CoQ10. See, U.S. Patent Publication Nos. 20050003474 and 20040078846. Methods similar to those described above for *E. coli* can be used to make recombinant *Rhodobacter* microorganisms.

Candida boidinii

Candida boidinii is a methylotrophic yeast (it can grow on methanol). Like other methylotrophic species such as *Hansenula polymorpha* and *Pichia pastoris*, it provides an excellent platform for the production of heterologous proteins. Yields in a multigram range of a secreted foreign protein have been reported. A computational method, IPRO, recently predicted mutations that experimentally switched the cofactor specificity of *Candida boidinii* xylose reductase from NADPH to NADH.

Hansenula polymorpha (*Pichia angusta*)

Hansenula polymorpha is another methylotrophic yeast (see *Candida boidinii*). It can furthermore grow on a wide range of other substrates; it is thermo-tolerant and can assimilate nitrate (see also *Kluyveromyces lactis*). It has been applied to the production of hepatitis B vaccines, insulin and interferon alpha-2a for the treatment of hepatitis C, furthermore to a range of technical enzymes.

Kluyveromyces lactis

Kluyveromyces lactis is a yeast regularly applied to the production of kefir. It can grow on several sugars, most importantly on lactose which is present in milk and whey. It has successfully been applied among others to the production of chymosin (an enzyme that is usually present in the stomach of calves) for the production of cheese. Production takes place in fermenters on a 40,000 L scale.

Pichia pastoris

Pichia pastoris is a methylotrophic yeast (see *Candida boidinii* and *Hansenula polymorpha*). It provides an efficient platform for the production of foreign proteins. Platform elements are available as a kit and it is worldwide used in academia for the production of proteins. Strains have been engineered that can produce complex human N-glycan (yeast glycans are similar but not identical to those found in humans).

Physcomitrella sp.

Physcomitrella mosses, when grown in suspension culture, have characteristics similar to yeast or other fungal cultures. This genera is becoming an important type of cell for production of plant secondary metabolites, which can be difficult to produce in other types of cells.

B. Plant Cells or Plants

In some embodiments, the nucleic acids and polypeptides described herein are introduced into plants or plant cells to increase overall steviol glycoside production or enrich for the production of specific steviol glycosides in proportion to others. Thus, a host can be a plant or a plant cell that includes

at least one recombinant gene described herein. A plant or plant cell can be transformed by having a recombinant gene integrated into its genome, i.e., can be stably transformed. Stably transformed cells typically retain the introduced nucleic acid with each cell division. A plant or plant cell can also be transiently transformed such that the recombinant gene is not integrated into its genome. Transiently transformed cells typically lose all or some portion of the introduced nucleic acid with each cell division such that the introduced nucleic acid cannot be detected in daughter cells after a sufficient number of cell divisions. Both transiently transformed and stably transformed transgenic plants and plant cells can be useful in the methods described herein.

Transgenic plant cells used in methods described herein can constitute part or all of a whole plant. Such plants can be grown in a manner suitable for the species under consideration, either in a growth chamber, a greenhouse, or in a field. Transgenic plants can be bred as desired for a particular purpose, e.g., to introduce a recombinant nucleic acid into other lines, to transfer a recombinant nucleic acid to other species, or for further selection of other desirable traits. Alternatively, transgenic plants can be propagated vegetatively for those species amenable to such techniques. As used herein, a transgenic plant also refers to progeny of an initial transgenic plant provided the progeny inherits the transgene. Seeds produced by a transgenic plant can be grown and then selfed (or outcrossed and selfed) to obtain seeds homozygous for the nucleic acid construct.

Transgenic plants can be grown in suspension culture, or tissue or organ culture. For the purposes of this invention, solid and/or liquid tissue culture techniques can be used. When using solid medium, transgenic plant cells can be placed directly onto the medium or can be placed onto a filter that is then placed in contact with the medium. When using liquid medium, transgenic plant cells can be placed onto a flotation device, e.g., a porous membrane that contacts the liquid medium.

When transiently transformed plant cells are used, a reporter sequence encoding a reporter polypeptide having a reporter activity can be included in the transformation procedure and an assay for reporter activity or expression can be performed at a suitable time after transformation. A suitable time for conducting the assay typically is about 1-21 days after transformation, e.g., about 1-14 days, about 1-7 days, or about 1-3 days. The use of transient assays is particularly convenient for rapid analysis in different species, or to confirm expression of a heterologous polypeptide whose expression has not previously been confirmed in particular recipient cells.

Techniques for introducing nucleic acids into monocotyledonous and dicotyledonous plants are known in the art, and include, without limitation, *Agrobacterium*-mediated transformation, viral vector-mediated transformation, electroporation and particle gun transformation, U.S. Pat. Nos. 5,538,880; 5,204,253; 6,329,571; and 6,013,863. If a cell or cultured tissue is used as the recipient tissue for transformation, plants can be regenerated from transformed cultures if desired, by techniques known to those skilled in the art.

A population of transgenic plants can be screened and/or selected for those members of the population that have a trait or phenotype conferred by expression of the transgene. For example, a population of progeny of a single transformation event can be screened for those plants having a desired level of expression of a steviol or steviol glycoside biosynthesis polypeptide or nucleic acid. Physical and biochemical methods can be used to identify expression levels. These include Southern analysis or PCR amplification for detection of a

polynucleotide; Northern blots, Si RNase protection, primer-extension, or RT-PCR amplification for detecting RNA transcripts; enzymatic assays for detecting enzyme or ribozyme activity of polypeptides and polynucleotides; and protein gel electrophoresis, Western blots, immunoprecipitation, and enzyme-linked immunoassays to detect polypeptides. Other techniques such as in situ hybridization, enzyme staining, and immunostaining also can be used to detect the presence or expression of polypeptides and/or nucleic acids. Methods for performing all of the referenced techniques are known. As an alternative, a population of plants comprising independent transformation events can be screened for those plants having a desired trait, such as production of a steviol glycoside or modulated biosynthesis of a steviol glycoside. Selection and/or screening can be carried out over one or more generations, and/or in more than one geographic location. In some cases, transgenic plants can be grown and selected under conditions which induce a desired phenotype or are otherwise necessary to produce a desired phenotype in a transgenic plant. In addition, selection and/or screening can be applied during a particular developmental stage in which the phenotype is expected to be exhibited by the plant. Selection and/or screening can be carried out to choose those transgenic plants having a statistically significant difference in a steviol glycoside level relative to a control plant that lacks the transgene.

The nucleic acids, recombinant genes, and constructs described herein can be used to transform a number of monocotyledonous and dicotyledonous plants and plant cell systems. Non-limiting examples of suitable monocots include, for example, cereal crops such as rice, rye, sorghum, millet, wheat, maize, and barley. The plant may be a non-cereal monocot such as asparagus, banana, or onion. The plant also may be a dicot such as stevia (*Stevia rebaudiana*), soybean, cotton, sunflower, pea, geranium, spinach, or tobacco. In some cases, the plant may contain the precursor pathways for phenyl phosphate production such as the mevalonate pathway, typically found in the cytoplasm and mitochondria. The non-mevalonate pathway is more often found in plant plastids [Dubey, et al., 2003 *J. Biosci.* 28 637-646]. One with skill in the art may target expression of steviol glycoside biosynthesis polypeptides to the appropriate organelle through the use of leader sequences, such that steviol glycoside biosynthesis occurs in the desired location of the plant cell. One with skill in the art will use appropriate promoters to direct synthesis, e.g., to the leaf of a plant, if so desired. Expression may also occur in tissue cultures such as callus culture or hairy root culture, if so desired.

In one embodiment, one or more nucleic acid or polypeptides described herein are introduced into *Stevia* (e.g., *Stevia rebaudiana*) such that overall steviol glycoside biosynthesis is increased or that the overall steviol glycoside composition is selectively enriched for one or more specific steviol glycosides (e.g., rebaudioside D). For example, one or more recombinant genes can be introduced into *Stevia* such that a EUGT11 enzyme (e.g., SEQ ID NO: 152 or a functional homolog thereof) is expressed alone or in combination with one or more of: a UGT91D enzyme such as UGT91D2e (e.g., SEQ ID NO:5 or a functional homolog thereof), UGT91D2m (e.g., SEQ ID NO:10); a UGT85C enzyme such as a variant described in the "Functional Homolog" section, a UGT76G1 enzyme such as a variant described in the "Functional Homolog" section, or a UGT74G1 enzyme. Nucleic acid constructs typically include a suitable promoter (e.g., 35S, e35S, or ssRUBISCO promoters) operably linked to a nucleic acid encoding the

UGT polypeptide. Nucleic acids can be introduced into *Stevia* by *Agrobacterium*-mediated transformation; electroporation-mediated gene transfer to protoplasts; or by particle bombardment. See, e.g., Singh, et al., *Compendium of Transgenic Crop Plants: Transgenic Sugar, Tuber and Fiber*, Edited by Chittaranjan Kole and Timothy C. Hall, Blackwell Publishing Ltd. (2008), pp. 97-115. For particle bombardment of *stevia* leaf derived callus, the parameters can be as follows: 6 cm distance, 1100 psi He pressure, gold particles, and one bombardment.

Stevia plants can be regenerated by somatic embryogenesis as described by Singh et al., 2008, *supra*. In particular, leaf segments (approximately 1-2 cm long) can be removed from 5 to 6-week-old in vitro raised plants and incubated (adaxial side down) on MS medium supplemented with B5 vitamins, 30 g sucrose and 3 g Gelrite. 2,4-dichlorophenoxyacetic acid (2,4-D) can be used in combination with 6-benzyl adenine (BA), kinetin (KN), or zeatin. Proembryonic masses appear after 8 weeks of subculture. Within 2-3 weeks of subcultures, somatic embryos will appear on the surface of cultures. Embryos can be matured in medium containing BA in combination with 2,4-D, *a*-naphthaleneacetic acid (NAA), or indolbutyric acid (IBA). Mature somatic embryos that germinate and form plantlets can be excised from calli. After plantlets reach 3-4 weeks, the plantlets can be transferred to pots with vermiculite and grown for 6-8 weeks in growth chambers for acclimatization and transferred to greenhouses.

In one embodiment, steviol glycosides are produced in rice. Rice and maize are readily transformable using techniques such as *Agrobacterium*-mediated transformation. Binary vector systems are commonly utilized for *Agrobacterium* exogenous gene introduction to monocots. See, for example, U.S. Pat. Nos. 6,215,051 and 6,329,571. In a binary vector system, one vector contains the T-DNA region, which includes a gene of interest (e.g., a UGT described herein) and the other vector is a disarmed Ti plasmid containing the *vir* region. Co-integrated vectors and mobilizable vectors also can be used. The types and pretreatment of tissues to be transformed, the strain of *Agrobacterium* used, the duration of the inoculation, the prevention of overgrowth and necrosis by the *Agrobacterium*, can be readily adjusted by one of skill in the art. Immature embryo cells of rice can be prepared for transformation with *Agrobacterium* using binary vectors. The culture medium used is supplemented with phenolic compounds. Alternatively, the transformation can be done in planta using vacuum infiltration. See, for example, WO 2000037663, WO 2000063400, and WO 2001012828.

IV. METHODS OF PRODUCING STEVIOL GLYCOSIDES

Recombinant hosts described herein can be used in methods to produce steviol or steviol glycosides. For example, if the recombinant host is a microorganism, the method can include growing the recombinant microorganism in a culture medium under conditions in which steviol and/or steviol glycoside biosynthesis genes are expressed. The recombinant microorganism may be grown in a fed batch or continuous process. Typically, the recombinant microorganism is grown in a fermentor at a defined temperature(s) for a desired period of time. Depending on the particular microorganism used in the method, other recombinant genes such as isopentenyl biosynthesis genes and terpene synthase and cyclase genes may also be present and expressed. Levels of substrates and intermediates, e.g., isopentenyl diphosphate,

dimethylallyl diphosphate, geranylgeranyl diphosphate, kaurene and kaurenoic acid, can be determined by extracting samples from culture media for analysis according to published methods.

After the recombinant microorganism has been grown in culture for the desired period of time, steviol and/or one or more steviol glycosides can then be recovered from the culture using various techniques known in the art. In some embodiments, a permeabilizing agent can be added to aid the feedstock entering into the host and product getting out. If the recombinant host is a plant or plant cells, steviol or steviol glycosides can be extracted from the plant tissue using various techniques known in the art. For example, a crude lysate of the cultured microorganism or plant tissue can be centrifuged to obtain a supernatant. The resulting supernatant can then be applied to a chromatography column, e.g., a C-18 column, and washed with water to remove hydrophilic compounds, followed by elution of the compound(s) of interest with a solvent such as methanol. The compound(s) can then be further purified by preparative HPLC. See also WO 2009/140394.

The amount of steviol glycoside (e.g., rebaudioside D) produced can be from about 1 mg/L to about 1500 mg/L, e.g., about 1 to about 10 mg/L, about 3 to about 10 mg/L, about 5 to about 20 mg/L, about 10 to about 50 mg/L, about 10 to about 100 mg/L, about 25 to about 500 mg/L, about 100 to about 1,500 mg/L, or about 200 to about 1,000 mg/L. In general, longer culture times will lead to greater amounts of product. Thus, the recombinant microorganism can be cultured for from 1 day to 7 days, from 1 day to 5 days, from 3 days to 5 days, about 3 days, about 4 days, or about 5 days.

It will be appreciated that the various genes and modules discussed herein can be present in two or more recombinant microorganisms rather than a single microorganism. When a plurality of recombinant microorganisms is used, they can be grown in a mixed culture to produce steviol and/or steviol glycosides. For example, a first microorganism can comprise one or more biosynthesis genes for producing steviol while a second microorganism comprises steviol glycoside biosynthesis genes. Alternatively, the two or more microorganisms each can be grown in a separate culture medium and the product of the first culture medium, e.g., steviol, can be introduced into second culture medium to be converted into a subsequent intermediate, or into an end product such as rebaudioside A. The product produced by the second, or final microorganism is then recovered. It will also be appreciated that in some embodiments, a recombinant microorganism is grown using nutrient sources other than a culture medium and utilizing a system other than a fermentor.

Steviol glycosides do not necessarily have equivalent performance in different food systems. It is therefore desirable to have the ability to direct the synthesis to steviol glycoside compositions of choice. Recombinant hosts described herein can produce compositions that are selectively enriched for specific steviol glycosides (e.g., rebaudioside D) and have a consistent taste profile. Thus, the recombinant microorganisms, plants, and plant cells described herein can facilitate the production of compositions that are tailored to meet the sweetening profile desired for a given food product and that have a proportion of each steviol glycoside that is consistent from batch to batch. Microorganisms described herein do not produce the undesired plant byproducts found in *Stevia* extracts. Thus, steviol glycoside compositions produced by the recombinant micro-

organisms described herein are distinguishable from compositions derived from Stevia plants.

V. FOOD PRODUCTS

The steviol glycosides obtained by the methods disclosed herein can be used to make food products, dietary supplements and sweetener compositions. For example, substantially pure steviol or steviol glycoside such as rebaudioside A or rebaudioside D can be included in food products such as ice cream, carbonated beverages, fruit juices, yogurts, baked goods, chewing gums, hard and soft candies, and sauces. Substantially pure steviol or steviol glycoside can also be included in non-food products such as pharmaceutical products, medicinal products, dietary supplements and nutritional supplements. Substantially pure steviol or steviol glycosides may also be included in animal feed products for both the agriculture industry and the companion animal industry. Alternatively, a mixture of steviol and/or steviol glycosides can be made by culturing recombinant microorganisms separately or growing different plants/plant cells, each producing a specific steviol or steviol glycoside, recovering the steviol or steviol glycoside in substantially pure form from each microorganism or plant/plant cells and then combining the compounds to obtain a mixture containing each compound in the desired proportion. The recombinant microorganisms, plants, and plant cells described herein permit more precise and consistent mixtures to be obtained compared to current Stevia products. In another alternative, a substantially pure steviol or steviol glycoside can be incorporated into a food product along with other sweeteners, e.g., saccharin, dextrose, sucrose, fructose, erythritol, aspartame, sucralose, monatin, or acesulfame potassium. The weight ratio of steviol or steviol glycoside relative to other sweeteners can be varied as desired to achieve a satisfactory taste in the final food product. See, e.g., U.S. Patent Publication No. 2007/0128311. In some embodiments, the steviol or steviol glycoside may be provided with a flavor (e.g., citrus) as a flavor modulator. For example, Rebaudioside C can be used as a sweetness enhancer or sweetness modulator, in particular for carbohydrate based sweeteners, such that the amount of sugar can be reduced in the food product.

Compositions produced by a recombinant microorganism, plant, or plant cell described herein can be incorporated into food products. For example, a steviol glycoside composition produced by a recombinant microorganism, plant, or plant cell can be incorporated into a food product in an amount ranging from about 20 mg steviol glycoside/kg food product to about 1800 mg steviol glycoside/kg food product on a dry weight basis, depending on the type of steviol glycoside and food product. For example, a steviol glycoside composition produced by a recombinant microorganism, plant, or plant cell can be incorporated into a dessert, cold confectionary (e.g., ice cream), dairy product (e.g., yogurt), or beverage (e.g., a carbonated beverage) such that the food product has a maximum of 500 mg steviol glycoside/kg food on a dry weight basis. A steviol glycoside composition produced by a recombinant microorganism, plant, or plant cell can be incorporated into a baked good (e.g., a biscuit) such that the food product has a maximum of 300 mg steviol glycoside/kg food on a dry weight basis. A steviol glycoside composition produced by a recombinant microorganism, plant, or plant cell can be incorporated into a sauce (e.g., chocolate syrup) or vegetable product (e.g., pickles) such that the food product has a maximum of 1000 mg steviol glycoside/kg food on a dry weight basis. A steviol glycoside composition

produced by a recombinant microorganism, plant, or plant cell can be incorporated into a bread such that the food product has a maximum of 160 mg steviol glycoside/kg food on a dry weight basis. A steviol glycoside composition produced by a recombinant microorganism, plant, or plant cell can be incorporated into a hard or soft candy such that the food product has a maximum of 1600 mg steviol glycoside/kg food on a dry weight basis. A steviol glycoside composition produced by a recombinant microorganism, plant, or plant cell can be incorporated into a processed fruit product (e.g., fruit juices, fruit filling, jams, and jellies) such that the food product has a maximum of 1000 mg steviol glycoside/kg food on a dry weight basis.

For example, such a steviol glycoside composition can have from 90-99% rebaudioside A and an undetectable amount of stevia plant-derived contaminants, and be incorporated into a food product at from 25-1600 mg/kg, e.g., 100-500 mg/kg, 25-100 mg/kg, 250-1000 mg/kg, 50-500 mg/kg or 500-1000 mg/kg on a dry weight basis.

Such a steviol glycoside composition can be a rebaudioside B-enriched composition having greater than 3% rebaudioside B and be incorporated into the food product such that the amount of rebaudioside B in the product is from 25-1600 mg/kg, e.g., 100-500 mg/kg, 25-100 mg/kg, 250-1000 mg/kg, 50-500 mg/kg or 500-1000 mg/kg on a dry weight basis. Typically, the rebaudioside B-enriched composition has an undetectable amount of stevia plant-derived contaminants.

Such a steviol glycoside composition can be a rebaudioside C-enriched composition having greater than 15% rebaudioside C and be incorporated into the food product such that the amount of rebaudioside C in the product is from 20-600 mg/kg, e.g., 100-600 mg/kg, 20-100 mg/kg, 20-95 mg/kg, 20-250 mg/kg, 50-75 mg/kg or 50-95 mg/kg on a dry weight basis. Typically, the rebaudioside C-enriched composition has an undetectable amount of stevia plant-derived contaminants.

Such a steviol glycoside composition can be a rebaudioside D-enriched composition having greater than 3% rebaudioside D and be incorporated into the food product such that the amount of rebaudioside D in the product is from 25-1600 mg/kg, e.g., 100-500 mg/kg, 25-100 mg/kg, 250-1000 mg/kg, 50-500 mg/kg or 500-1000 mg/kg on a dry weight basis. Typically, the rebaudioside D-enriched composition has an undetectable amount of stevia plant-derived contaminants.

Such a steviol glycoside composition can be a rebaudioside E-enriched composition having greater than 3% rebaudioside E and be incorporated into the food product such that the amount of rebaudioside E in the product is from 25-1600 mg/kg, e.g., 100-500 mg/kg, 25-100 mg/kg, 250-1000 mg/kg, 50-500 mg/kg or 500-1000 mg/kg on a dry weight basis. Typically, the rebaudioside E-enriched composition has an undetectable amount of stevia plant-derived contaminants.

Such a steviol glycoside composition can be a rebaudioside F-enriched composition having greater than 4% rebaudioside F and be incorporated into the food product such that the amount of rebaudioside F in the product is from 25-1000 mg/kg, e.g., 100-600 mg/kg, 25-100 mg/kg, 25-95 mg/kg, 50-75 mg/kg or 50-95 mg/kg on a dry weight basis. Typically, the rebaudioside F-enriched composition has an undetectable amount of stevia plant-derived contaminants.

Such a steviol glycoside composition can be a dulcoside A-enriched composition having greater than 4% dulcoside A and be incorporated into the food product such that the amount of dulcoside A in the product is from 25-1000

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mg/kg, e.g., 100-600 mg/kg, 25-100 mg/kg, 25-95 mg/kg, 50-75 mg/kg or 50-95 mg/kg on a dry weight basis. Typically, the dulcoside A-enriched composition has an undetectable amount of stevia plant-derived contaminants.

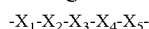
Such a steviol glycoside composition can be a composition enriched for rubusoside xylosylated on either of the two positions—the 13-O-glucose or the 19-O-glucose. Such a composition can have greater than 4% of the xylosylated rubusoside compound, and can be incorporated into the food product such that the amount of xylosylated rubusoside compound in the product is from 25-1000 mg/kg, e.g., 100-600 mg/kg, 25-100 mg/kg, 25-95 mg/kg, 50-75 mg/kg or 50-95 mg/kg on a dry weight basis. Typically, the xylosylated rubusoside enriched composition has an undetectable amount of stevia plant-derived contaminants.

Such a steviol glycoside composition can be a composition enriched for compounds rhamnosylated on either of the two positions—the 13-O-glucose or the 19-O-glucose, or compounds containing one rhamnose and multiple glucoses (e.g., steviol 13-O-1,3-diglycoside-1,2-rhamnoside). Such a composition can have greater than 4% of the rhamnosylated compound, and can be incorporated into the food product such that the amount of rhamnosylated compound in the product is from 25-1000 mg/kg, e.g., 100-600 mg/kg, 25-100 mg/kg, 25-95 mg/kg, 50-75 mg/kg or 50-95 mg/kg on a dry weight basis. Typically, the composition enriched for rhamnosylated compounds has as an undetectable amount of stevia plant-derived contaminants.

In some embodiments, a substantially pure steviol or steviol glycoside is incorporated into a tabletop sweetener or “cup-for-cup” product. Such products typically are diluted to the appropriate sweetness level with one or more bulking agents, e.g., maltodextrins, known to those skilled in the art. Steviol glycoside compositions enriched for rebaudioside A, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F, dulcoside A, or rhamnosylated or xylosylated compounds, can be packaged in a sachet, for example, at from 10,000 to 30,000 mg steviol glycoside/kg product on a dry weight basis, for tabletop use.

In some embodiments, this disclosure relates to the following items:

1. A recombinant host cell comprising a nucleic acid sequence, said nucleic acid comprising a heterologous insert sequence operably linked to an open reading frame, wherein the heterologous insert sequence has the general formula (I):



wherein X_2 comprises at least 4 consecutive nucleotides complementary to at least 4 consecutive nucleotides of X_4 ,

wherein X_3 comprises zero nucleotides, or one or more nucleotides forming a hairpin loop,

wherein X_1 and X_5 each individually consists of zero nucleotides, or one or more nucleotides, wherein the open reading frame encodes a squalene synthase (EC 2.5.1.21).

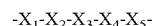
2. The recombinant cell according to item 1, said nucleic acid comprises in 5' to 3' order, a promoter sequence operably linked to a heterologous insert sequence operably linked to an open reading frame, wherein the heterologous insert sequence and the open reading frame are as defined in item 1.

3. A cell comprising a nucleic acid sequence, said nucleic acid comprising

- i) a promoter sequence operably linked to
- ii) a heterologous insert sequence operably linked to
- iii) an open reading frame operably linked to
- iv) a transcription termination signal,

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wherein the heterologous insert sequence has the general formula (I):



wherein X_2 comprises at least 4 consecutive nucleic acids being complementary to, and forms a hairpin secondary structure element with at least 4 consecutive nucleic acids of X_4 , and

wherein X_3 comprises unpaired nucleic acids thus forming a hairpin loop between X_2 and X_4 , and

wherein X_1 and X_5 individually and optionally comprises one or more nucleic acids, and

wherein the open reading frame upon expression encodes a polypeptide sequence having at least 70% identity to a squalene synthase (EC 2.5.1.21) or a biologically active fragment thereof, said fragment having at least 70% sequence identity to said squalene synthase in a range of overlap of at least 100 amino acids.

4. The cell according to any one of items 1 to 3, wherein the heterologous insert sequence comprises 10-50 nucleotides, preferably 10-30 nucleotides, more preferably 15-25 nucleotides, more preferably 17-22 nucleotides, more preferably 18-21 nucleotides, more preferably 18-20 nucleotides, more preferably 19 nucleotides.

5. The cell according to any one of items 1 to 4, wherein X_2 and X_4 consists of the same number of nucleotides.

6. The cell according to any one of items 1 to 5, wherein all X_2 consists of in the range of 4 to 25, such as in the range of 4 to 20, for example of in the range of 4 to 15, such as in the range of 6 to 12, for example in the range of 8 to 12, such as in the range of 9 to 11 nucleotides.

7. The cell according to any one of items 1 to 6, wherein all X_4 consists of in the range of 4 to 25, such as in the range of 4 to 20, for example of in the range of 4 to 15, such as in the range of 6 to 12, for example in the range of 8 to 12, such as in the range of 9 to 11 nucleotides.

8. The cell according to any one of items 1 to 7, wherein X_2 consists of a nucleotide sequence, which is complementary to the nucleotide sequence of X_4 .

9. The cell according to any one of items 1 to 8, wherein X_4 consists of a nucleotide sequence, which is complementary to the nucleotide sequence of X_2 .

10. The cell according to any one of items 1 to 9, wherein X_3 is absent, i.e. X_3 consists of zero nucleotides.

11. The cell according to any one of items 1 to 9, wherein X_3 consists of in the range of 1 to 5, such as in the range of 1 to 3 nucleotides.

12. The cell according to any one of items 1 to 11, wherein X_1 is absent, i.e. X_1 consists of zero nucleotides.

13. The cell according to any one of items 1 to 11, wherein X_1 consists of in the range of 1 to 25, such as in the range of 1 to 20, for example in the range of 1 to 15, such as in the range of 1 to 10, for example in the range of 1 to 5, such as in the range of 1 to 3 nucleotides.

14. The cell according to any one of items 1 to 13, wherein X_5 is absent, i.e. X_5 consists of zero nucleotides.

15. The cell according to any one of items 1 to 11, wherein X_5 consists of in the range 1 to 5, such as in the range of 1 to 3 nucleotides.

16. The cell according to any one of items 1 to 15, wherein the heterologous insert sequence comprises a sequence selected from the group consisting of SEQ ID NO: 181, SEQ ID NO: 182, SEQ ID NO: 183 and SEQ ID NO: 184.

17. The cell according to any one of items 1 to 16, wherein the heterologous insert sequence is selected from the group

consisting of SEQ ID NO: 181, SEQ ID NO: 182, SEQ ID NO: 183 and SEQ ID NO: 184.

18. The cell according to any one of items 1 to 17, wherein the squalene synthase is at least 75%, such as at least 80%, such as at least 85%, such as at least 87%, such as at least 90%, such as at least 91%, such as at least 92%, such as at least 93%, such as at least 94%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99%, such as 100% identical to a squalene synthase selected from the group consisting of SEQ ID NO: 192, SEQ ID NO:193, SEQ ID NO:194, SEQ ID NO:194, SEQ ID NO:195, SEQ ID NO:196, SEQ ID NO:197, SEQ ID NO:198, SEQ ID NO:199, SEQ ID NO:200, SEQ ID NO:201, and SEQ ID NO:202.

19. The cell according to any one of items 1 to 18, wherein said promoter is a constitutive or inducible promoter.

20. The cell according to any one of items 1 to 19, wherein said promoter is selected from the group consisting of an endogenous promoter, GPD1, PGK1, ADH1, ADH2, PYK1, TPI1, PDC1, TEF1, TEF2, FBA1, GAL1-10, CUP1, MET2, MET14, MET25, CYC1, GAL1-S, GAL1-L, TEF1, ADH1, CAG, CMV, human UbiC, RSV, EF-1alpha, SV40, Mt1, Tet-On, Tet-Off, Mo-MLV-LTR, Mx1, progesterone, RU486 and Rapamycin-inducible promoter.

21. The cell according to any one of items 1 to 20, wherein the nucleic acid sequence further comprises a polyadenyl sequence.

22. The cell according to item 21, wherein the 5' end of said polyadenyl sequence is operably linked to the 3' end of the nucleic acid of item 1.

23. The cell according to any one of items 1 to 22, wherein the nucleic acid sequence further comprises a post-transcriptional regulatory element.

24. The cell of item 23, wherein said post-transcriptional regulatory element is a Woodchuck hepatitis virus post-transcriptional regulatory element (WPRE).

25. The cell of any of items 1 to 24, wherein the nucleic acid comprises a 5' terminal repeat and a 3' terminal repeat.

26. The cell of item 25, wherein the 5' and 3' terminal repeats are selected from Inverted Terminal Repeats [ITR] and Long Terminal Repeats [LTR].

27. The cell according to any one of items 1 to 26, wherein the nucleic acid sequence is integrated in a vector.

28. The cell of item 27, wherein the vector is an expression vector.

29. The cell of item 27, wherein the vector is selected from the group consisting of plasmid vectors, cosmids, artificial chromosomes and viral vectors.

30. The cell of item 29 wherein the plasmid vector can be maintained and replicated in bacteria, fungi and yeast.

31. The cell of item 29, wherein the viral vector is selected from the group consisting of vectors derived from the Retroviridae family including lentivirus, HIV, SIV, FIV, EAIIV, CIV.

32. The cell of item 31, wherein the viral vector is selected from the group consisting of alphavirus, adenovirus, adeno associated virus, baculovirus, HSV, coronavirus, Bovine papilloma virus, Mo-MLV and adeno associated virus.

33. The cell of any of items 27 to 32, wherein said vector is functional in mammalian cells.

34. The cell of any of the preceding items wherein the cell is transformed or transduced with the vector of any of the items 27 to 33.

35. The cell of any of items 1 to 34, wherein said cell is a eukaryotic cell.

36. The cell of any of items 1 to 34, wherein said cell is a prokaryotic cell.

37. The cell of item 35, wherein said cell is selected from the group consisting of fungi cells such as yeast and *aspergillus*; microalgae such as *Chlorella* and *Prototheca*; plant cells; and mammalian cells, such as human, feline, porcine, simian, canine, murine, rat, mouse and rabbit cells.

38. The cell of item 37, wherein the yeast is selected from the group consisting of *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Yarrowia lipolytica*, *Candida glabrata*, *Ashbya gossypii*, *Cyberlindnera jadinii*, and *Candida albicans*.

39. The cell of item 37, wherein the cell is selected from the group consisting of CHO, CHO-K1, HEI193T, HEK293, COS, PC12, HiB5, RN33b, BHK cells.

40. The cell of item 36, wherein said cell is *E. coli*, *Corynebacterium*, *Bacillus*, *Pseudomonas* or *Streptomyces*.

41. The cell of any of items 35 to 40, wherein said prokaryotic cell, or said fungi cell, has been genetically modified to express at least a portion of the enzymes of the mevalonate independent pathway.

42. The cell according to any one of items 1 to 41, wherein the cell furthermore comprises a heterologous nucleic acid encoding GGPPS operably linked to a nucleic acid sequence directing expression of GGPPS in said cell.

43. The cell according to item 42, wherein said GGPPS is selected from the groups consisting of SEQ ID NO: 126, SEQ ID NO:123, SEQ ID NO:203, SEQ ID NO:167, and functional homologues thereof sharing at least 75% sequence identity with any of the aforementioned.

43. A method for producing a terpenoid compound synthesized through the squalene pathway, in a cell culture, said method comprising the steps of

(a) providing the cell of any one of items 1 to 42,

(b) culturing the cell of (a).

(c) recovering the terpenoid product compound.

44. A method for producing a terpenoid derived from a terpenoid precursor selected from the group consisting of Farnesyl-pyrophosphate (FPP), Isopentenyl-pyrophosphate (IPP), Dimethylallyl-pyrophosphate (DMAPP), Geranyl-pyrophosphate (GPP) and/or Geranylgeranyl-pyrophosphate (GGPP), said method comprising:

(a) contacting said precursor with an enzyme of the squalene synthase pathway,

(b) recovering the terpenoid product.

45. The method of any of items 44 and 45, wherein the terpenoid product is selected from the group consisting of hemiterpenoids, monoterpenes, sesquiterpenoids, diterpenoids, sesterpenes, triterpenoids, tetraterpenoids and polyterpenoids.

46. The method of item 44, wherein the terpenoid is selected from the group consisting of farnesyl phosphate, farnesol, geranylgeranyl, geranylgeraniol, isoprene, prenol, isovaleric acid, geranyl pyrophosphate, eucalyptol, limonene, pinene, farnesyl pyrophosphate, artemisinin, bisabolol, geranylgeranyl pyrophosphate, retinol, retinal, phytol, taxol, forskolin, aphidicolin, lanosterol, lycopene and carotene.

47. The method of item 46, wherein said method further comprises dephosphorylating the farnesyl phosphate to produce farnesol.

48. The method of item 44, wherein the enzyme of the squalene synthase pathway is selected from the group consisting of Dimethylallyltransferase (EC 2.5.1.1), Isoprene synthase (EC 4.2.3.27) and Geranyltransferase (EC 2.5.1.10).

49. A method for reducing the translation rate of a functional squalene synthase (EC 2.5.1.21) said method comprising:

(a) providing the cell of any one of items 1 to 42,

(b) culturing the cell of (a).

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50. A method for decreasing turnover of farnesyl-pp to squalene, said method comprising:

(d) providing the cell of any one of items 1 to 42,

(e) culturing the cell of (a).

51. A method for enhancing accumulation of a compound selected from the group consisting of Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate and Geranylgeranyl-pyrophosphate, said method comprising the steps of:

(a) providing the cell of any one of items 1 to 42, and

(b) culturing the cell of (a).

52. The method of item 51, further comprising recovering the Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate or Geranylgeranyl-pyrophosphate compound.

53. The method of any one of items 51 and 52, further comprising recovering a compound synthesized through the squalene pathway, said compound being derived from said Farnesyl-pyrophosphate, Isopentenyl-pyrophosphate, Dimethylallyl-pyrophosphate, Geranyl-pyrophosphate and/or Geranylgeranyl-pyrophosphate.

54. The method of any of items 43 to 53, wherein the step of culturing the cell is performed in the presence of a squalene synthase inhibitor.

55. The method of any of items 43 to 54, wherein the cell additionally is genetically modified to enhance activity of and/or overexpress one or more enzymes selected from the group consisting of Phosphomevalonate kinase (EC 2.7.4.2), Diphosphomevalonate decarboxylase (EC 4.1.1.33), 4-hydroxy-3-methylbut-2-en-1-yl diphosphate synthase (EC 1.17.7.1), 4-hydroxy-3-methylbut-2-enyl diphosphate reductase (EC 1.17.1.2), Isopentenyl-diphosphate Delta-isomerase 1 (EC 5.3.3.2), Short-chain Z-isoprenyl diphosphate synthase (EC 2.5.1.68), Dimethylallyltransferase (EC 2.5.1.1), Geranyltranstransferase (EC 2.5.1.10) and Geranylgeranyl pyrophosphate synthetase (EC 2.5.1.29).

56. The method of any of items 43 to 55, wherein the cell additionally is genetically modified to enhance activity of and/or overexpress one or more enzymes selected from the group consisting of acetoacetyl CoA thiolase, HMG-CoA reductase or the catalytic domain thereof, HMG-CoA synthase, mevalonate kinase, phosphomevalonate kinase, phosphomevalonate decarboxylase, isopentenyl pyrophosphate isomerase, farnesyl pyrophosphate synthase, D-1-deoxyxylulose 5-phosphate synthase, and 1-deoxy-D-xylulose 5-phosphate reductoisomerase and farnesyl pyrophosphate synthase.

57. The method of any of items 43 to 56, wherein the cell comprises a mutation in the ERG9 open reading frame.

58. The method of any of items 43 to 57, wherein the cell comprises an ERG9[Delta]::HIS3 deletion/insertion allele.

59. The method of any of items 43 to 58, wherein the step of recovering the compound comprises purification of said compound from the cell culture media.

VI. EXAMPLES

The invention will be further described in the following examples, which do not limit the scope of the invention described in the claims. In the examples described herein, the following LC-MS methodology was used to analyze steviol glycosides and steviol pathway intermediates unless otherwise indicated.

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1) Analyses of Steviol Glycosides

LC-MS analyses were performed using an Agilent 1200 Series HPLC system (Agilent Technologies, Wilmington, DE, USA) fitted with a Phenomenex® kinetex C18 column (150×2.1 mm, 2.6 µm particles, 100 Å pore size) connected to a TSQ Quantum Access (ThermoFisher Scientific) triple quadrupole mass spectrometer with a heated electrospray ion (HESI) source. Elution was carried out using a mobile phase of eluent B (MeCN with 0.1% Formic acid) and eluent A (water with 0.1% Formic acid) by increasing the gradient from 10->40% B from min 0.0 to 1.0, increasing 40->50% B in min 1.0 to 6.5, 50->100% B from min 6.5 to 7.0 and finally washing and re-equilibration. The flow rate was 0.4 ml/min and the column temperature 30° C. The steviol glycosides were detected using SIM (Single Ion Monitoring) in positive mode with the following m/z-traces.

Description	Exact Mass	m/z trace	compound (typical t_R in min)
Steviol + 1 Glucose	[M + H] ⁺ 481.2796 [M + Na] ⁺ 503.2615	481.2 ± 0.5 503.1 ± 0.5	19-SMG (6.1), 13-SMG (6.4)
Steviol + 2 Glucose	[M + Na] ⁺ 665.3149	665 ± 0.5	Rubusoside (4.7) Steviol-1,2-bioside (5.2) Steviol-1,3-bioside (5.8)
Steviol + 3 Glucose	[M + Na] ⁺ 827.3677	827.4 ± 0.5	1,2-Stevioside (4.0) 1,3-Stevioside (4.4) Rebaudioside B (5.0)
Steviol + 4 Glucose	[M + Na] ⁺ 989.4200	989.4 ± 0.5	Rebaudioside A (3.9)
Steviol + 5 Glucose	[M + Na] ⁺ 1151.4728	1151.4 ± 0.5	Rebaudioside D (3.3)

The level of steviol glycosides was quantified by comparing with calibration curves obtained with authentic standards from LGC Standards. For example, standard solutions of 0.5 to 100 µM Rebaudioside A were typically utilized to construct a calibration curve.

2) Analyses of Steviol and Ent-Kaurenoic Acid

LC-MS analyses of steviol and ent-kaurenoic acid were performed on the system described above. For the separation, a Thermo Science Hypersil Gold (C-18, 3 µm, 100×2.1 mm) column was used and a 20 mM ammonium acetate aqueous solution was used as eluent A and acetonitrile as eluent B. The gradient conditions were: 20->55% B in min 0.0 to 1.0, 55->100 in min 1.0-7.0 and finally washing and re-equilibration. The flow rate was 0.5 mL/min and the column temperature 30° C. Steviol and ent-kaurenoic acid were detected using SIM (Single Ion Monitoring) in negative mode with the following m/z-traces.

Description	Exact Mass	m/z trace	typical t_R in min
Steviol	[M - H] ⁻ 317.2122	317.4 ± 0.5	3.3
Ent-kaurenoic acid	[M - H] ⁻ 301.2173	301.4 ± 0.5	5.5

3) HPLC Quantification of UDP-Glucose

For the quantification of UDP-glucose, an Agilent 1200 Series HPLC system was used, with a Waters XBridge BEH amide (2.5 µm, 3.0×50 mm) column. Eluent A was a 10 mM ammonium acetate aqueous solution (pH 9.0) and Eluent B acetonitrile. The gradient conditions were: 95% B holding from min 0.0-0.5, decreasing from 95-50% B in min 0.5-4.5,

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holding 50% B from min 4.5-6.8 and finally re-equilibrating to 95% B. The flow rate was 0.9 mL/min and the column temperature 20° C. UDP-glucose was detected by UV_{262nm} absorbance.

The amount of UDP-glucose was quantified by comparing with a calibration curve obtained with a commercially available standard (e.g., from Sigma Aldrich).

Example 1—Identification of EUGT11

Fifteen genes were tested for RebA 1,2-glycosylation activity. See Table 10.

TABLE 10

Name	Source	GenBank Accession No.
EUGT2	<i>Oryza sativa</i> UGT91 homolog	AP003270
EUGT3	<i>Oryza sativa</i> UGT91 homolog	AP005171
EUGT4	<i>Oryza sativa</i> UGT91 homolog	AP005643
EUGT6	<i>Oryza sativa</i> UGT91 homolog	AP005259
EUGT7	<i>Oryza sativa</i> UGT91 homolog	AP005171
EUGT8	<i>Oryza sativa</i> UGT91 homolog	XM_470006
EUGT9	<i>Oryza sativa</i> UGT91 homolog	AP005643
EUGT10	<i>Oryza sativa</i> UGT91 homolog	AC133334
EUGT11	<i>Oryza sativa</i> UGT91 homolog	AC133334
EUGT12	<i>Oryza sativa</i> UGT91 homolog	AC133334
EUGT15	<i>Petunia</i> × hybrid UGT79 homolog	Z25802
EUGT16	<i>Arabidopsis thaliana</i> UGT79 homolog	AC004786
EUGT17	<i>Dianthus caryophyllus</i> UGT79 homolog	AB294391
EUGT18	<i>Ipomoea nil</i> UGT79 homolog	AB192314
EUGT19	<i>Oryza sativa</i> UGT79 homolog	NM_001074394

In vitro transcription and translation of these genes was performed, and the resulting UGTs incubated with RebA and UDP-glucose. Following incubation, the reactions were analyzed by LC-MS. The reaction mixture containing EUGT11 (Rice, AC133334, SEQ ID NO:152) was shown to convert significant quantities of RebA to RebD. See LC-MS chromatograms in FIG. 4. As shown in the left panel of FIG. 4, UGT91D2e produced a trace amount of RebD when RebA was used as the feedstock. As shown in the right panel of FIG. 4, EUGT11 produced a significant amount of RebD when RebA was used as the feedstock. Preliminary quantification of the amount of RebD that was produced indicated that EUGT11 was approximately 30 times more efficient than UGT91D2e at converting RebA to RebD.

To further characterize EUGT11 and for quantitative comparison to UGT91D2e, the nucleotide sequence encoding EUGT11 (SEQ ID NO: 153, non-codon optimized, FIG. 7) was cloned into two *E. coli* expression vectors, one containing an N-terminal HIS-tag and one containing an N-terminal GST-tag. EUGT11 was expressed using both systems and purified. When the purified enzymes were incubated with UDP-glucose and RebA, RebD was produced.

Example 2—Identification of EUGT11 Reactions

EUGT11 was produced by in vitro transcription and translation, and incubated with various substrates in the RebD pathway. Similar experiments were carried out using in vitro transcribed and translated UGT91D2e. FIG. 3 shows a schematic overview of 19-O-1,2-diglycosylation reactions performed by EUGT11 and UGT91D2e. Compounds 1-3 were identified solely by mass and expected retention time. The numbers shown in FIG. 3 are the average peak height of the indicated steviol glycoside obtained from a LC-MS chromatogram, and, although not quantitative, can be used

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to compare the activity of the two enzymes. EUGT11 and UGT91D2e were not able to use steviol as a substrate. Both enzymes were able to convert steviol 19-O-monoglucoside (SMG) to compound 1, with EUGT11 being about ten times more efficient than UGT91D2e at converting 19-SMG to compound 1.

Both enzymes were able to convert rubusoside to steviolside with comparable activity but only EUGT11 was able to convert rubusoside to compound 2 and compound 3 (RebE). See FIG. 5. The left panel of FIG. 5 contains LC-MS chromatograms of the conversion of rubusoside to steviolside. The right panel of FIG. 5 contains chromatograms of the conversion of rubusoside to steviolside, to compound 2, and to compound 3 (RebE). Conversion of rubusoside to compound 3 requires two consecutive 1, 2-O-glycosylations at the 19- and 13-positions of steviol. UGT91D2e was able to produce a trace amount of compound 3 (RebE) in one experiment whereas EUGT11 produced a significant amount of compound 3.

Both enzymes were able to convert RebA to RebD. However, EUGT11 was approximately 30 times better at converting RebA to RebD. Overall, it appears that EUGT11 produces more product than UGT91D2e in all reactions (with similar time, concentrations, temperature, and purity of enzyme) except the conversion of rubusoside to steviolside.

Example 3—Expression of EUGT11 in Yeast

The nucleotide sequence encoding EUGT11 was codon-optimized (SEQ ID NO:154) and transformed into yeast along with nucleic acids encoding all four UGTs (UGT91D2e, UGT74G1, UGT76G1, and UGT85C2). The resulting yeast strain was grown in medium containing steviol and steviol glycosides that accumulated were analyzed by LC-MS. EUGT11 was required for the production of RebD. In other experiments, RebD production has been observed with UGT91D2e, UGT74G1, UGT76G1, and UGT85C2.

Example 4—UGT Activity on 19-O-1,2-Diglycosylated Steviol Glycosides

The 19-O-1,2-diglycosylated steviol glycosides produced by EUGT11 need further glycosylation to be converted to RebD. The following experiments were performed to determine if other UGTs could use these intermediates as substrates.

In one experiment, compound 1 was produced in vitro from 19-SMG by either EUGT11 or UGT91D2e in the presence of UDP-glucose. After boiling the sample, UGT85C2 and UDP-glucose were added. The sample was analyzed by LC-MS and compound 2 was detected. This experiment indicated that UGT85C2 can use compound 1 as a substrate.

In another experiment, compound 2 was incubated with UGT91D2e and UDP-glucose. The reaction was analyzed by LC-MS. UGT91D2e was not able to convert compound 2 to compound 3 (RebE). Incubation of compound 2 with EUGT11 and UDP-glucose results in the production of compound 3. UGT76G1 was able to use RebE as a substrate to produce RebD.

This shows that the 19-O-1,2-diglycosylation of the steviol glycosides is able to take place at any time during production of RebD as the downstream enzymes are able to metabolize the 19-O-1,2-diglycosylated intermediates.

Example 5—Comparison of EUGT11 and UGT91D2e Sequence

The amino acid sequence of EUGT11 (SEQ ID NO:152, FIG. 7) and the amino acid sequence of UGT91D2e (SEQ ID NO:5) were aligned using the FASTA algorithm (Pearson and Lipman, Proc. Natl. Acad. Sci., 85:2444-2448 (1998)). See FIG. 6. EUGT11 and UGT91D2e are 42.7% identical over 457 amino acids.

Example 6—Modification of 19-1,2-Diglycosylating Activity of UGT91D2e

Crystal structures are available for a number of UGTs. Generally, the N-terminal half of a UGT is primarily involved with substrate binding whereas the C-terminal half is involved in binding the UDP-sugar donor.

Modeling the secondary structure of UGT91D2e onto the secondary structure of the UGTs that have been crystalized revealed a conserved pattern of secondary structure, despite a highly diverged primary sequence as shown in FIG. 8. The crystal structures of UGT71G1 and UGT85H2 (see, for example H. Shao et al, *The Plant Cell* November 2005 vol. 17 no 11 3141-3154 and L. Li et al., *J Mol Biol.* 2007 370(5):951-63) have been reported. Known loops, alpha-helices and beta-sheets are indicated on UGT91D2e in FIG. 8. Although the homology at the primary structure level of these UGTs is fairly low, the secondary structure appears to be conserved, allowing predictions regarding the locations of amino acids involved in substrate binding on UGT91D2e based on the location of such amino acids in UGT85H2 and UGT71G1.

Regions commonly involved in substrate binding were superimposed on UGT91D2e and largely shown to coincide with the 22 amino acid differences from UGT91D1 (GenBank Accession No. Protein Accession number AAR06918, GI:37993665). UGT91D1 is highly expressed in *Stevia* and thought to be a functional UGT. However, its substrate is not a steviol glycoside. This suggests that UGT91D1 has a different substrate, which may be defined by the 22 amino acids with which it differs from UGT91D2e. FIG. 9 is an alignment of the amino acid sequences of UGT91D1 and UGT91D2e. The boxes represent areas that are reported to be involved in substrate binding. The amino acids highlighted in dark grey show the 22 amino acid differences between UGT91D1 and UGT91D2e. Stars denote amino acids that have been shown to be involved in substrate binding in UGTs that have had their crystal structure resolved (more stars under one particular amino acids means substrate binding has been shown with more than one structure-resolved UGTs). There is a strong correlation between the 22 amino acid differences between the two UGT91s, the regions known to be involved in substrate binding, and the actual amino acids involved in substrate binding in the crystal structure-resolved UGTs. This suggests that the 22 amino acid differences between the two UGT91s are involved in substrate binding.

All 22 altered 91D2es were expressed in a XJb Autolysis *E. coli* strain from a pGEX-4T1 vector. In order to assess the activity of the enzymes, two substrate feeding experiments were performed—in vivo and in vitro. Most mutants had lower activity than wild type, however, 5 mutants showed increased activity. This was reproduced by in vitro transcription and translation (IVT) and showed that C583A, C631A and T857C have approximately 3-fold higher stevioside-forming activity than the wild-type UGT91D2e, whereas C662t and A1313C had approximately twice the stevioside-

forming activity (nucleotide numbering). These changes result in amino acid mutations corresponding to L195M, L211M, V286A; and S221F and E438A, respectively. The increased activity differed depending on substrate, with C583A and C631A showing almost a 10-fold increase using 13-SMG as substrate and about a 3-fold increase using rubusoside as substrate, whereas T857C showed a 3-fold increase when using either 13-SMG or rubusoside as substrate.

To investigate if these mutations were additive, a range of double mutants were made and analyzed for activity (FIG. 10). In this particular experiment, a higher wild type level of activity was observed than the previous four experiments; however, the relative activities of the mutations remain the same. As rubusoside accumulates in many of the *S. cerevisiae* strains expressing the 4 UGTs (UGT74G1, UGT85C2, UGT76G1, and UGT91D2e), the stevioside-forming activity may be more important for increasing steviol glycoside production. As such, the double mutant C631A/T857C (nucleotide numbering) may be useful. This mutant has been named UGT91D2e-b, which contains the amino acid modifications L211M and V286A. The experiments have been reproduced in vitro using *S. cerevisiae*-expressed UGT91D2e-mutants.

To improve 19-1,2-diglycosylating activity of UGT91D2e, a directed saturated mutagenic screen of UGT91D2e of the 22 amino acid differences between UGT91D2e and UGT91D1 was performed. GeneArt's® (Life Technologies, Carlsbad, CA) site-saturation mutagenesis was used to obtain a library containing each of the mutations. The library was cloned into the BamHI and NotI sites of pGEX4T1 bacterial expression plasmid expressing the mutated versions of 91D2e as GST fusion proteins, resulting in a new library (Lib #116). Lib #116 was transformed into XJbAutolysis *E. coli* strain (ZymoResearch, Orange, CA) to produce approximately 1600 clones containing the 418 expected mutations (i.e., 22 positions with 19 different amino acids at each position). Other plasmids expressing GST-tagged versions of 91D2e (EPSC1314), 91D2e-b (EPSC1888) or EUGT11 (EPSC1744) as well as the empty pGEX4T1 (PSB12) were transformed as well. Screening by LC-MS

To analyze the approximately 1600 mutant clones of UGT91D2e, the *E. coli* transformants were grown overnight at 30° C. in 1 ml of NZCYM containing ampicillin (100 mg/l) and chloramphenicol (33 mg/l), in 96-well format. The next day, 150 µl of each culture was inoculated into 3 ml NZCYM containing ampicillin (100 mg/l), chloramphenicol (33 mg/l), arabinose 3 mM, IPTG 0.1 mM and ethanol 2% v/v, in 24-well format, and incubated at 20° C. and 200 rpm for ~20 h. The following day, cells were spun down and pellets were resuspended in 100 µl of lysis buffer containing 10 mM Tris-HCl pH 8, 5 mM MgCl₂, 1 mM CaCl₂ and complete mini protease inhibitor EDTA-free (3 tablets/100 ml) (Hoffmann-La Roche, Basel, Switzerland) and frozen -80° C. for at least 15 minutes to promote cell lysis. Pellets were thawed at room temperature and 50 µl of DNase mix (1 µl of 1.4 mg/ml DNase in H₂O (~80000 u/ml), 1.2 µl of MgCl₂ 500 mM and 47.8 µl of 4×PBS buffer solution) was added to each well. Plates were shaken at 500 rpm for 5 min at room temperature to allow degradation of genomic DNA. Plates were spun down at 4000 rpm for 30 min at 4° C. and six µl of the lysates were used in UGT in vitro reactions as described for GST-91D2e-b, using rubusoside or rebaudioside A as substrates. In each case, the resulting compounds, stevioside or rebaudioside D (rebD), were measured by LC-MS. Results were analyzed in com-

parison with the stevioside or rebD produced by the lysates expressing the corresponding controls (91D2e, 91D2e-b, EUGT11 and the empty plasmid). Clones showing activity similar to or higher than the ones expressing 91D2e-b were selected as primary hits.

Half of the 1600 clones and the corresponding controls were assayed for their capacity to glycosylate rubusoside and rebaudiosideA. Stevioside and RebD were quantified by LC-MS. Under the conditions used, lysates from clones expressing the native UGT91D2e show activity just around background with both substrates (approximately 0.5 μ M stevioside and 1 μ M RebD), while clones expressing UGT91D2e-b show consistently improved product formation (>10 μ M Stevioside; >1.5 μ M RebD). Clones expressing EUGT11 consistently display a higher level of activity, especially using RebA as substrate. Cutoff for considering clones as primary hits in the screening was generally set at 1.5 μ M for both products, but in some cases was adjusted for each independent assay.

Example 7—EUGT11 Homologs

A Blastp search of the NCBI nr database using the EUGT11 protein sequence revealed approximately 79 potential UGT homologs from 14 plant species (one of which is the *Stevia* UGT91D1, approximately 67% identical to EUGT11 in conserved UGT regions but less than 45% overall). Homologs with greater than 90% identity in conserved regions were identified from corn, soybean, *Arabidopsis*, grape, and Sorghum. The overall homology of the full-length EUGT11 homologs, at the amino acid level, was only 28-68%.

RNA was extracted from plant material by the method described by Iandolino et al. (Iandolino et al., *Plant Mol Biol Reporter* 22, 269-278, 2004), the RNeasy Plant mini Kit (Qiagen) according to the manufacturer's instructions, or using the Fast RNA Pro Green Kit (MP Biomedicals) according to the manufacturer's instructions. cDNA was produced by AffinityScript QPCR cDNA Synthesis Kit (Agilent) according to the manufacturer's instructions. Genomic DNA was extracted using the FastDNA kit (MP Biomedicals) according to the manufacturer's instructions. PCR was performed on cDNA using either the Dream Taq polymerase (Fermentas) or the Phusion polymerase (New England Biolabs) and a series of primers designed to amplify the homologs.

PCR-reactions were analyzed by electrophoresis in SyberSafe-containing agarose-TAE gels. DNA was visualized by UV-irradiation in a trans illuminator. Bands of the correct size were cut out, purified through spin columns according to the manufacturer's specifications, and cloned into TOPO-Zero blunt (for Phusion polymerase-generated products) or TOPO-TA (for Dream Taq-generated products). The TOPO-vectors containing the PCR-products were transformed into *E. coli* DH5B α and plated on LB-agar plates containing the appropriate selective antibiotics. DNA was extracted from surviving colonies and sequenced. The genes with the correct sequence were cut out by restriction digest with SbfI and Ascl, cloned into similarly digested IVT8 vector and transformed into *E. coli*. PCRs were performed on all cloned genes to amplify the gene and flanking regions required for in vitro transcription and translation. Proteins were produced from the PCR products by in vitro transcription and translation using the Promega L5540, TNT T7 Quick for PCR DNA Kit according to the manufacturer's instructions. Production of protein was evaluated by incorporation of ³⁵S-

methionine followed by separation by SDS-PAGE and visualization on a Typhoon phosphor-imager.

Activity assays were set up totaling 20% (by volume) of each in vitro reaction, 0.1 mM rubusoside or RebA, 5% DMSO, 100 mM Tris-HCl pH 7.0, 0.01 units Fast alkaline phosphatase (Fermentas), and 0.3 mM UDP-glucose (final concentrations). Following incubation at 30° C. for one hour, the samples were analyzed by LC-MS for production of stevioside and RebD as described above. The UGT91D2e and UGT91D2e-b (double mutant described in Example 6) were used as positive controls, along with EUGT11. Under the initial assay conditions, clone P 64B (see Table 11) produced a trace amount of product using rubusoside and RebA. Table 11 lists the percent identity at the amino acid level compared to EUGT11 for the whole length of the UGTs, which ranges from 28-58%. High amounts of homology (96-100%) were observed over shorter stretches of sequences, which may indicate highly conserved domains of plant UGTs.

TABLE 11

List of cloned EUGT11 homologs and their amino acid percent identity to EUGT11.

UGT	Accession	% identity to EUGT11
P44G	XP_002297733.1	32.16
P54A	XP_002532392.1	34.20
P51H	XP_002325254.1	32.53
P55D	XP_002533517.1	31.90
P5F	AAM12787.1	31.73
P48G	XP_002318358.1	33.20
P52F	XP_002334090.1	32.80
P48F	XP_002318358.1	33.00
T4B-b	NP_565540.4	31.19
P56C	XP_002533518.1	32.60
T67H	XP_002270294.1	34.06
T65E	CAN80742.1	34.98
T74G	XP_002270331.1	35.48
T65D	CAN80742.1	34.98
T69F1	XP_003635103.1	34.69
P6B	Q66PF2.1	33.20
P6D variant	Q66PF2.1	33.60
P64B	ACE87855.1	34.64
T3F	AT5G65550	34.94
P53H	XP_002527371.1	33.40
P53F	XP_002527371.1	33.40
P46H	XP_002303861.1	32.40
2-b	NP_199780.1	35.79
T70F	XP_002275802.1	36.67
T72A	XP_002275850.1	36.42
T71G	XP_002275824.1	37.25
P49G	XP_002320817.1	35.15
P57H	XP_002511902.1	36.23
45 Pop	XP_002302598.1	34.21
P50G	XP_002323718.1	32.86
P50H	XP_002323718.1	32.66
T73G	XP_002281094.1	32.05
63	XP_002458816.1	37.25
P78B	NP_001147674.1	35.33
62	XP_002458815.1	34.06
P9F	BAJ84800.1	37.92
T7H	NP_001240857.1	31.30
16-1	BAJ93155.1	58.03
T16H	BAJ93155.1	58.03
31TA	BAD35324.1	51.81
P41G	NP_001174664.1	35.40
P37G	NP_001051010.1	56.71
P60aH	XP_002466606.1	57.35
12	BAJ89368.1	44.35
P12A	BAJ89368.1	44.35
P12H	BAJ89368.1	44.35
P10B	BAJ86656.1	45.16
P58aF	XP_002463702.1	43.71

TABLE 11-continued

List of cloned EUGT11 homologs and their amino acid percent identity to EUGT11.		
UGT	Accession	% identity to EUGT11
P59aG	XP_002463705.1	43.51
P76H	NP_001140711.1	28.81

Example 8—Cell-Free Biocatalytic Production of Reb-D

The cell-free approach is an in vitro system where RebA, stevioside or a steviol glycoside mixture is enzymatically converted to RebD. The system requires stoichiometric amounts of UDP-glucose and therefore UDP-glucose regeneration from UDP and sucrose using sucrose synthase can be used. Additionally, sucrose synthase removes UDP produced during the reaction, which improves conversion to glycosylated products by alleviated product inhibition observed for glycosylation reactions. See, WO 2011/153378.

Enzyme Expression and Purification

UGT91D2e-b (described in Example 6) and EUGT11 are key enzymes that catalyze the glycosylation of RebA yielding RebD. These UGTs were expressed in bacteria (*E. coli*) but one of ordinary skill in the art will appreciate that such proteins also can be prepared using different methods and hosts (e.g., other bacteria such as *Bacillus* sp., yeast such as *Pichia* sp. or *Saccharomyces* sp., other fungi (e.g., *Aspergillus*), or other organisms). For example, the proteins can be produced by in vitro transcription and translation or by protein synthesis.

The UGT91D2e-b and EUGT11 genes were cloned in pET30a or pGEX4T1 plasmids. Resulting vectors were transformed into an XJb (DE3) Autolysis *E. coli* strain (ZymoResearch, Orange, CA). Initially, *E. coli* transformants were grown overnight at 30° C. in NZCYM medium, followed by induction with 3 mM arabinose and 0.1 mM IPTG, and further incubation overnight at 30° C. The corresponding fusion proteins were purified by affinity chromatography using included 6HIS- or GST-tags and standard methods. One skilled in the art will appreciate that other protein purification methods such as gel filtration or other chromatography techniques also can be used, along with precipitation/crystallization or fractionation with e.g., ammonium sulfate. While EUGT11 was expressed well using the initial conditions, UGT91D2e-b required several modifications to the base protocol to increase protein solubility, including lowering the temperature of the overnight expression from 30° C. to 20° C. and adding 2% ethanol to the expression medium. Generally 2-4 mg/L of soluble GST-EUGT11 and 400-800 µg/L of GST-UGT91D2e-b were purified with this method.

Stability of EUGT11

Reactions were conducted to explore the stability of EUGT11 under various RebA to RebD reaction conditions. Omitting the substrate from the reaction mixture, EUGT11 was pre-incubated for various periods of time before substrate was added. Following a pre-incubation of the enzyme in 100 mM Tris-HCl buffer, substrate (100 µM RebA) and other reaction components (300 µM UDP-glucose, and 10 U/mL Alkaline Phosphatase (Fermentas/Thermo Fisher, Waltham, MA)) were added (0, 1, 4 or 24 hours after the incubation was started). The reaction was then allowed to proceed for 20 h, after which the reactions were stopped and

RebD product-formation measured. Experiments were repeated at different temperatures: 30° C., 32.7° C., 35.8° C. and 37° C.

The activity of EUGT11 was reduced rapidly when the enzyme was pre-incubated at 37° C., reaching approximately half activity after 1 hour, and having almost no activity after 4 hours. At 30° C., the activity was not significantly reduced after 4 hours and after 24 hours, approximately one-third of the activity remained. This suggests that EUGT11 is heat-labile.

To assess the thermal stability of EUGT11 and to compare it with the other UGTs in the steviol glycosylation pathway, denaturation temperatures of the proteins were determined using differential scanning calorimetry (DSC). Use of DSC thermograms to estimate denaturation temperatures, TD, is described, for example, by E. Freire in *Methods in Molecular Biology* 1995, Vol. 40 191-218. DSC was performed (using 6HIS-purified EUGT11, yielding an apparent TD of 39° C.; while when GST-purified 91D2e-b was used, the measured TD was 79° C. For reference, the measured TD when using 6HIS-purified UGT74G1, UGT76G1 and UGT85C2 was 86° C. in all cases. One of skill in the art will recognize that enzyme immobilization or addition of thermal protectants can be added to the reactions to improve stability of the protein. Non-limiting examples of thermal protectants include trehalose, glycerol, ammonium sulphate, betaine, trimethylamine oxide, and proteins.

Enzyme Kinetics

A series of experiments were performed to determine kinetic parameters of EUGT11 and 91D2e-b. For both enzymes, 100 µM RebA, 300 µM UDP-glucose, and 10 U/mL Alkaline Phosphatase (Fermentas/Thermo Fisher, Waltham, MA) were used in the reactions. For EUGT11, the reactions were performed at 37° C. using 100 mM Tris-HCl, pH 7, and 2% enzyme. For 91D2e-b, the reactions were performed at 30° C. using 20 mM Hepes-NaOH, pH 7.6, 20% (by volume) enzyme. The initial velocities (V_0) were calculated in the linear range of a product versus time plot.

To first investigate the linearity intervals, initial time-courses were done for each enzyme. EUGT11 was assayed at 37° C. for 48 h at initial concentrations of 100 µM RebA and 300 µM UDP-glucose. UGT91D2e-b was assayed at 37° C. for 24 h at initial concentrations of 200 µM RebA and 600 µM UDP-glucose. Based on these range-finding studies, it was determined that the initial 10 minutes in the case of EUGT11, and the initial 20 minutes in UGT91D2e-b would be in the linear range with respect to product formation, and therefore initial velocities of each reaction were calculated in those intervals. In the case of EUGT11, RebA concentrations assayed were 30 µM, 50 µM, 100 µM, 200 µM, 300 µM and 500 µM. Concentration of UDP-glucose was always three times the concentration of RebA and incubation was performed at 37° C. By plotting the calculated V_0 as a function of the substrate concentrations, Michaelis-Menten curves were generated. By plotting the reciprocal of V_0 and the reciprocal of $[S]$, a Lineweaver-Burk graphic was obtained, with $y=339.85x+1.8644$; $R^2=0.9759$.

V_{max} and K_M parameters were determined from the curve-fit Lineweaver-Burk data, calculated from the x- and y intercepts ($x=0$, $y=1/V_{max}$) and ($y=0$, $x=-1/K_M$). Additionally, the same parameters also were calculated by a non-linear least squares regression, using the SOLVER function in Excel. The results obtained with both methods for EUGT11 and RebA are presented in Table 12, along with all the kinetic parameters of this example. Results from both Nonlinear Least Square Fit method and Lineweaver-Burk

plot are presented in Table 12. K_{cat} is calculated based on V_{max} divided by the approximate amount of protein in the assay.

TABLE 12

Comparison of kinetic parameters for EUGT11 and UGT 91D2e-b, with RebA or UDP-glucose as substrate.								
	Nonlinear Least Square Fit				Lineweaver-Burk plot			
	Reb A		UDP-glucose		Reb A		UDP-glucose	
	EUGT11	91D2e-b	EUGT11	91D2e-b	EUGT11	91D2e-b	EUGT11	91D2e-b
V_{max} ($\mu\text{M} \cdot \text{min}^{-1}$)	0.52	0.34	0.79	0.19	0.54	0.44	0.78	0.18
K_{cat} (min^{-1})	8.11	0.32	12.32	0.2	8.42	0.41	12.1	0.19
K_M (μM)	162.5	1150	130	45.1	182.3	1580	118	41.9
K_{cat}/K_M ($\text{min}^{-1} \cdot \mu\text{M}^{-1}$)	0.05	0.000275	0.095	0.00454	0.046	0.000258	0.102	0.00463

In order to investigate the influence of UDP-glucose concentration in the glycosylation reaction, as well as the affinity of EUGT11 for UDP-glucose, similar kinetics analysis were performed. EUGT11 was incubated with increasing amounts of UDP-glucose (20 μM , 50 μM , 100 μM , and 200 μM), maintaining an excess of RebA (500 μM). The kinetic parameters were calculated as described above, and shown in Table 12.

In the case of UGT91D2e-b, RebA concentrations assayed were 50 μM , 100 μM , 200 μM , 300 μM , 400 μM and 500 μM . Concentration of UDP-glucose was always three times the concentration of RebA and incubation was performed at 30° C., in the reaction conditions described above for UGT91D2e-b. The kinetic parameters were calculated as previously described; and the resulting kinetic parameters are shown in Table 12. Additionally, kinetic parameters of UGT91D2e-b towards UDP-glucose were determined. UGT91D2e-b was incubated with increasing amounts of UDP-glucose (30 μM , 50 μM , 100 μM , and 200 μM), maintaining an excess of RebA (1500 μM). incubation was performed at 30° C., in optimal conditions for UGT91D2e-b. The kinetic parameters were calculated as previously described and results are presented in Table 12.

By comparison of the kinetics parameters for EUGT11 and 91D2e-b, it was concluded that 91D2e-b has a lower K and has lower affinity for RebA (higher K_M) although the K_M for UDP-glucose of 91D2e-b is lower than EUGT11. UGT91D2e-b has a lower K_{cat}/K_M which is a measure of catalytic efficiency, combining information on rate of catalysis with a particular substrate (K_{cat}) and the strength of enzyme-substrate binding (K_M).

Determining the Limiting Factor in Reactions

Under the conditions described above for EUGT11, approximately 25% of the RebA administered was converted to RebD. The limiting factor in these conditions could be either the enzyme, UDP-glucose or RebA. Experiments were set up to distinguish between these possibilities. A standard assay was allowed to run its course during 4 hours. This was followed by addition of either extra RebA substrate, extra enzyme, extra UDP-glucose or extra enzyme and UDP-glucose. Addition of extra enzyme resulted in a relative increase of the conversion of around 50%, adding extra RebA or UDP-glucose alone did not increase the conversion significantly, but the simultaneous addition of enzyme and UDP-glucose increased the conversion approximately 2-fold.

Experiments were conducted to examine the limit to this benefit of adding bolus amounts of UDP-glucose and fresh enzyme in the conversion of RebA to RebD reaction. Addi-

tional enzyme or enzyme and UDP-glucose were added after 1, 6, 24 and 28 hours. In the case of the addition of both extra EUGT11 and UDP-glucose, a conversion of more than 70%

was achieved. No other components had a significant effect on the conversion. This indicates that EUGT11 is a primary limiting factor for the reaction but UDP-glucose also is limiting. As UDP-glucose is present at 3-fold higher concentration than RebA, this indicates that UDP-glucose may be somewhat unstable in the reaction mixture, at least in the presence of EUGT11. Alternatively, as explained below, EUGT11 may be metabolizing the UDP-glucose.

Inhibition Studies

Experiments were conducted to determine if factors such as sucrose, fructose, UDP, product (RebD) and impurities in the less pure Stevia extracts raw materials inhibited the extent of the conversion of steviol glycoside substrates to RebD. In a standard reaction mixture, excess of the potential inhibitors (sucrose, fructose, UDP, RebD, or a commercial blend of steviol glycosides (*Steviva*, Steviva Brands, Inc., Portland, OR)) were added. Following incubation, RebD-production was quantified. Addition of 500 $\mu\text{g}/\text{ml}$ of the commercial Steviva mix (approximately 60% 1,2-stevioside, 30% RebA, 5% Rubusoside, 2% 1,2-bioside, less than 1% of RebD, RebC and others, as evaluated by LC-MS) was not found to be inhibitory, but rather increased the overall RebD production (to around 60 μM from around 30 μM without any addition) well beyond the RebD originally added with the blend (around 5 μM). From the molecules tested, only UDP was shown to have an inhibitory effect on RebD-production at the concentration used (500 μM), as measured by LC-MS. The RebD that was produced was less than 7 μM . This inhibition can be alleviated in the in vivo or in vitro reactions for RebD production, by including an UDP recycling system to UDP-glucose, either by yeast or by an added SUS (sucrose synthase enzyme) in conjunction with sucrose. Moreover, when working with lower amounts of UDP-glucose (300 μM), the addition of alkaline phosphatase to remove UDP-G does not increase the amount of RebD produced in the in vitro glycosylations substantially, suggesting that the UDP produced may not be inhibitory at these concentrations.

RebA Vs Crude Steviol Glycoside Mix

In some experiments, a crude steviol glycoside mix was used as a source of RebA instead of purified RebA. As such a crude steviol glycoside mix contains a high percentage of stevioside along with RebA, UGT76G1 was included in the reactions. In vitro reactions were performed as described above using 0.5 g/l of the *Steviva*® mix as substrate and enzyme (UGT76G1 and/or EUGT11) and incubated at 30° C. The presence of steviol glycosides was analyzed by LC-MS.

When only UGT76G1 was added to the reactions, stevioside was converted to RebA quite efficiently. An unknown penta-glycoside (with a retention time peak at 4.02 min) also was detected. When only EUGT11 was added to the reaction, large amounts of RebE, RebA, RebD and an unknown steviol-pentaglycoside (with a retention time peak at 3.15 min) were found. When both EUGT11 and UGT76G1 were added to the reactions, the stevioside peak was reduced, and almost entirely converted to RebA and RebD. There were trace amounts of the unknown steviol-pentaglycoside (peak at 4.02 min). No RebE was detected nor was the second unknown steviol-pentaglycoside (peak at 3.15 min). This result indicated that the use of stevia extracts as a substrate to produce RebD in vitro is possible when EUGT11 and UGT76G1 are used in combination.

Non-Specific UDP-Glucose Metabolism

To determine if EUGT11 can metabolize UDP-glucose independently of the conversion of RebA to RebD, GST-purified EUGT11 was incubated in the presence or absence of RebA substrate, and UDP-glucose usage was measured as UDP-release, using the TR-FRET Transcreener® kit (Bell-Brook Labs). The Transcreener® kit is based on a tracer molecule bound to an antibody. The tracer molecule is displaced by UDP or ADP in a highly sensitive and quantitative manner. The FP kit includes an Alexa633 tracer bound to an antibody. The tracer is displaced by UDP/ADP. The displaced tracer freely rotates leading to a decrease in fluorescence polarization. Therefore, UDP production is proportional to a decrease in polarization. The FI kit includes a quenched Alexa594 Tracer bound to an antibody, which is conjugated to an IRDye® QC-1 quencher. The tracer is displaced by UDP/ADP, whereby the displaced tracer is un-quenched, leading to a positive increase in fluorescence intensity. Therefore, UDP production is proportional to an increase in fluorescence. A TR-FRET kit includes a HiLyte647 Tracer bound to an Antibody-Tb conjugate. Excitation of the terbium complex in the UV range (ca. 330 nm) results in energy transfer to the tracer and emission at a higher wavelength (665 nm) after a time delay. The tracer is displaced by UDP/ADP causing a decrease in TR-FRET.

It was observed that UDP-glucose measured was the same independent of the presence of RebA substrate. UDP release was not detectable in the absence of enzyme. This indicates a non-specific degradation of UDP-glucose by EUGT11. Nevertheless, RebD was still produced when RebA was added, suggesting that EUGT11 would preferentially catalyze RebA glycosylation over the non-specific UDP-glucose degradation.

Experiments were set up to find out the destiny of the glucose molecule in the absence of RebA or other obvious glycosylation substrates. One common factor in all previous reactions was the presence of Tris buffer and/or trace amounts of glutathione, which both contain potential glycosylation sites. The effect of these molecules on the non-specific UDP-glucose consumption was assayed using GST-purified EUGT11 (with glutathione) and HIS-purified enzyme (without glutathione) in in vitro reactions, in the presence or absence of RebA. UDP-glucose usage was measured as UDP-release, using the TR-FRET Transcreener® kit. UDP release occurred in all cases and was independent of the presence of RebA. UDP release was slower when the HIS-purified enzyme was used, but the overall catalytic activity of the enzyme in conversion of RebA to RebD was also lower, suggesting a lower amount of active soluble enzyme present in the assay. Therefore, it appears that the UDP-glucose metabolism by EUGT11 is

independent of the presence of substrate and independent of the presence of glutathione in the reaction, under the conditions tested.

To test the effect of Tris on the metabolism of UDP-glucose by EUGT11, GST-EUGT11 was purified using a Tris- or a PBS-based buffer for the elution, obtaining similar amounts of protein in both cases. Tris- and PBS-purified enzymes were used in in vitro reactions using Tris and HEPES as buffers respectively, in the presence or absence of RebA in a similar manner as above. In both conditions, the UDP release was the same in the reactions whether RebA was added or not, indicating that the metabolism of UDP-glucose by EUGT11 is independent of both the presence of RebA and Tris in the reaction. This suggests that the UDP-release detected may somehow be an artifact caused by a property of EUGT11 or, alternatively, EUGT11 may be hydrolyzing UDP-glucose. EUGT11 is still efficient at converting RebA to RebD preferentially and the loss of UDP-G can be compensated by addition of the sucrose synthase recycling system described below.

RebA Solubility

The solubility of RebA determines the concentration that can be used both for the whole-cell approach and for the cell-free approach. Several different solutions of RebA in water were made and left at room temperature for several days. After 24 hours of storage, RebA precipitated at concentrations of 50 mM or higher. Twenty-five mM RebA started to precipitate after 4-5 days, while 10 mM or lower concentrations remained in solution, even when stored at 4° C.

RebD Solubility

The solubility of RebD was assessed by making several different solutions of RebD in water were made and incubated at 30° C. for 72 hours. RebD was found to be soluble in water initially in concentrations of 1 mM or lower while concentrations of 0.5 mM or less were found to be stable for longer periods of time. One with skill in the art will recognize that the solubility can be influenced by any number of conditions such as pH, temperature, or different matrices.

Sucrose Synthase

Sucrose synthase (SUS) has been used to regenerate UDP-glucose from UDP and sucrose (FIG. 11) for other small molecule glycosylations (Masada Sayaka et al. *FEBS Letters* 581 (2007) 2562-2566). Three SUS1 genes from *A. thaliana*, *S. rebaudiana* and coffee (*Coffea arabica*) were cloned into pGEX4T1 *E. coli* expression vectors (see FIG. 17 for the sequences). Using methods similar to those described for EUGT11, around 0.8 mg/l of GST-AtSUS1 (*A. thaliana* SUS1) was purified. Initial expression of CaSUS1 (*Coffea arabica* SUS1) and SrSUS1 (*S. rebaudiana* SUS1) followed by GST-purification did not produce significant amounts of protein although, when analyzed by western blot, the presence of GST-SrSUS1 was verified. When GST-SrSUS1 was expressed at 20° C. in the presence of 2% ethanol, approximately 50 µg/l of enzyme was produced.

Experiments were performed to evaluate the UDP-glucose regenerating activity of the purified GST-AtSUS1 and GST-SrSUS1. In vitro assays were conducted in 100 mM Tris-HCl pH=7.5 and 1 mM UDP (final concentration). Either ~2.4 µg of purified GST-AtSUS1, ~0.15 µg of GST-SrSUS1, or ~1.5 µg commercial BSA (New England Biolabs, Ipswich, MA) were also added. Reactions were done in presence or absence of ~200 mM sucrose and incubated at 37° C. for 24 h. Product UDP-glucose was measured by HPLC as described in the analytical section. AtSUS1 produced ~0.8 mM UDP-glucose when sucrose was present. No

UDP-glucose was observed when SrSUS1 or the negative control (BSA) was used. The lack of activity observed for SrSUS1 could be explained by the poor quality and concentration of the purified enzyme. UDP-glucose production by AtSUS1 was sucrose dependent and, therefore, it was concluded that AtSUS1 can be used in a coupled reaction to regenerate the UDP-glucose used by EUGT1 or other UGTs for small molecule glycosylation (FIG. 11, above).

SUS catalyzes the formation of UDP-glucose and fructose from sucrose and from UDP as depicted in FIG. 11. This UDP-glucose then can be used by EUGT11 for glycosylation of RebA to produce RebD. In vitro assays as described above were performed, adding ~200 mM sucrose, 1 mM UDP, 100 μ M RebA, ~1.6 μ g purified GST-AtSUS1 and ~0.8 μ g GST-EUGT11. Formation of product, RebD, was evaluated by LC-MS. When AtSUS, EUGT11, sucrose and UDP were mixed with RebA, 81 ± 5 μ M of RebD was formed. The reaction was dependent on the presence of AtSUS, EUGT11 and sucrose. The conversion rate was similar to what has been observed previously using UDP-glucose provided extraneously. This shows that AtSUS can be used to regenerate UDP-glucose for RebD-formation by EUGT11.

Example 9: Whole-Cell Biocatalytic Production of RebD

In this example, several parameters were studied that are factors for using whole cell biocatalytic systems in the production of RebD from RebA or other steviol glycosides. The ability of raw materials to cross the cell membrane and availability of UDP-glucose are two such factors. Permeabilizing agents were studied as well as different cell types to ascertain which systems may be the most beneficial for RebD production.

Permeabilizing Agents

Several different permeabilization agents have previously been shown to allow intracellular enzymatic conversion of various compounds that are normally not able to cross a cell membrane (Chow and Palecek, *Biotechnol. Prog.* 2004 March-April; 20(2):449-56). In several cases, the approaches resemble a partial lysis of the cells and, in yeast, often rely on the removal of the cell membrane by a detergent and the encapsulation of the enzymes inside of the remaining cell wall, which is permeable to smaller molecules. Common to these methods is the exposure to the permeabilizing agent followed by a centrifugation step to pellet cells before the addition of the substrate. See, for example, Flores et al., *Enzyme Microb. Technol.*, 16, pp. 340-346 (1994); Presecki & Vasic-Racki, *Biotechnology Letters*, 27, pp. 1835-1839 (2005); Yu et al., *J Ind Microbiol Biotechnol*, 34, 151-156 (2007); Chow and Palecek, *Cells. Biotechnol. Prog.*, 20, pp. 449-456 (2004); Fernandez et al., *Journal of Bacteriology*, 152, pp. 1255-1264 (1982); Kondo et al., *Enzyme and microbial technology*, 27, pp. 806-811 (2000); Abraham and Bhat, *J Ind Microbiol Biotechnol*, 35, pp. 799-804 (2008); Liu et al., *Journal of bioscience and bioengineering*, 89, pp. 554-558 (2000); and Gietz and Schiestl, *Nature Protocols*, 2, pp. 31-34 (2007) regarding permeabilization of yeast. See, Naglak and Wang, *Biotechnology and Bioengineering*, 39, pp. 732-740 (1991); Alakomi et al., *Applied and environmental Microbiology*, 66, pp. 2001-2005 (2000); and Fowler and Zabin, *Journal of bacteriology*, 92, pp. 353-357 (1966) regarding permeabilization of bacteria. As described in this example, it was determined if cells could remain viable and therefore could retain de novo UDP-glucose biosynthesis.

Experiments were done to establish conditions for permeabilization in *E. coli* and in yeast. Growing cells (*S. cerevisiae* or *E. coli*) were treated with different concentrations/combinations of permeabilization agents: toluene, chloroform and ethanol for permeabilization of *S. cerevisiae*, and guanidine, lactic acid, DMSO and/or Triton X-100 for permeabilization of *E. coli*. Tolerance of both model organisms to high concentrations of RebA and other potential substrates also was evaluated. The permeabilization was measured by the amount of RebD produced from a EUGT11-expressing organism after incubation in a RebA containing medium (feeding experiment). Enzyme activity was monitored before and after exposure to the permeabilizing agents by lysing the cells and analyzing the activity of the released UGTs in an in vitro assay.

In yeast, none of the permeabilization conditions tested resulted in an increase on RebD above the detected background (i.e., contaminating RebD levels present in the RebA stock used for feeding). This indicates that, under the tested conditions, yeast cells remain impermeable to RebA and/or the reduced cell viability caused by the solvents results in a decrease of EUGT11 activity as well.

In *E. coli*, none of the conditions tested resulted in permeabilization of the cells and subsequent production of RebD above background levels. Detectable levels of RebD were measured when lysates from strains expressing EUGT11 were used in the in vitro reactions (data not shown), indicating that EUGT11 enzyme is present and active even after all permeabilization treatments (though the level of activity varies). The permeabilization treatments had little or no effect on cell viability, except treating cultures with 0.2 M guanidine and 0.5% TritonX-100, which severely decreased viability.

S. cerevisiae also was subjected to permeabilization assays not allowing further growth of the cells using Triton X-100, N-lauryl sarcosine (LS), or Lithium acetate+polyethylene glycol (LiAc+PEG). That is, under these conditions, permeabilization renders the cells unviable by removing the cell membrane altogether while retaining the cell wall as a barrier to keep enzymes and gDNA inside. In such methods, UDP-glucose can be supplemental or recycled as described above. The advantage of permeabilization versus the purely in vitro approach is that individual enzymes do not need to be separately produced and isolated.

N-Lauryl sarcosine treatment resulted in inactivation of EUGT11 and only a minor increase in RebD was detected when LiAc/PEG was applied (data not shown). Treatment with Triton X-100 0.3% or 0.5%, however, increased the amount of RebD above background levels (see FIG. 18) while sustaining the activity of EUGT11. For Triton X-100 assays, overnight cultures were washed three times in PBS buffer. Cells corresponding to 6 OD₆₀₀ units were resuspended in PBS containing 0.3% or 0.5% Triton X-100 respectively. Treated cells were vortexed and incubated 30 minutes at 30° C. After treatment, cells were washed in PBS buffer. Cells corresponding to 5 OD₆₀₀ units were used in an in vitro assay, as described for GST-EUGT11 and 0.6 OD₆₀₀ units were resuspended in reaction buffer and incubated overnight at 30° C. as described for the LS treated samples. Untreated samples were used as controls.

Lysates from transformants expressing EUGT11 were able to convert some RebA into RebD (8 to 50 μ M were measured in the reactions) when cells were untreated or after treatment with LiAc/PEG or Triton X100. However, no RebD was measured in lysates of cell pellets treated with LS. Permeabilized but non-lysed cells were able to produce some RebD (1.4 to 1.5 μ M measured) when treated with

0.3% or 0.5% Triton X100 (FIG. 18) while no RebD was found on the samples treated with LS or LiAC/PEG. These results show that RebD can be produced from RebA biocatalytically using whole cells and using Triton X100 as the permeabilizing agent.

Example 10—Assessment of Codon Optimized UGT Sequences

Optimal coding sequences for UGT 91d2e, 74G1, 76G1, and 85C2 were designed and synthesized for yeast expression using two methodologies, supplied by GeneArt (Regensburg, Germany) (SEQ ID NOs: 6, 2, 8, and 4, respectively) or DNA 2.0 (Menlo Park, CA) (SEQ ID NOs: 84, 83, 85, and 82, respectively). The amino acid sequences of UGT 91d2e, 74G1, 76G1, and 85C2 (SEQ ID NOs: 5, 1, 7, and 3, respectively) were not changed.

The wild-type, DNA 2.0, and GeneArt sequences were assayed for in vitro activity to compare reactivity on substrates in the steviol glycosides pathway. UGTs were inserted in high copy (2p) vectors and expressed from a strong constitutive promoter (GPD1) (vectors P423-GPD, P424-GPD, P425-GPD, and P426-GPD). The plasmids were transformed individually into the universal Watchmaker strain, EFSC301 (described in Example 3 of WO2011/153378) and assays were carried out using cell lysates prepared from equal amount of cells (8 OD units). For the enzymatic reactions, 6 μ L of each cell lysate were incubated in a 30 μ L reaction with 0.25 mM steviol (final concentration) to test UGT74G1 and UGT85C2 clones, and with 0.25 mM 13-SMG (13SMG) (final concentration) to test 76G1 and 91D2e UGTs. Assays were carried out for 24 hours at 30° C. Prior to LC-MS analysis, one volume of 100% DMSO was added to each reaction, samples were centrifuged at 16000 g, and the supernatants analysed.

The lysates expressing the GeneArt-optimized genes provided higher levels of UGT activity under the conditions tested. Expressed as a percentage of the wild-type enzyme, the GeneArt lysates showed equivalent activity to the wild-type for UGT74G1, 170% activity for UGT76G1, 340% activity for UGT85C2 and 130% activity for UGT91D2e. Using UGT85C2 may improve the overall flux and productivity of cells for production of Reb-A and Reb-D when expressed in *S. cerevisiae*.

Further experiments were conducted to determine if the codon-optimized UGT85C2 could reduce 19-SMG accumulation and increase rubusoside and higher glycosylated steviol glycosides production. The production of 19-SMG and rubusoside were analysed in a steviol-feeding experiment of *S. cerevisiae* strain BY4741 expressing the wild type UGT74G1 as well as the codon-optimized UGT85C2 from high copy (2p) vectors under strong constitutive promoter (GPD1) (vectors P426-GPD and P423-GPD, respectively). Whole culture samples (without cell removal) were taken and boiled in an equal volume of DMSO for total glycosides levels. Intracellular concentrations reported were obtained by pelleting cells, and resuspending in 50% DMSO to the volume of the original culture sample taken, followed by boiling. The “total” glycosides level and the normalized intracellular level then were measured using LC-MS. Using wild type UGT74G1 and wild type UGT85C2, approximately 13.5 μ M rubusoside was produced in total with a maximum normalized intracellular concentration of about 7 μ M. In contrast, when wild type UGT74G1 and codon-optimized UGT85C2 were used, a maximum of 26 μ M rubusoside was produced, or approximately double of what was produced using the wild type UGT85C2. Additionally,

the maximum normalized intracellular concentration of rubusoside was 13 μ M, again an approximate doubling of what was produced using wild type UGT85C2. Intracellular concentration of 19-SMG was significantly reduced from a maximum of 35 μ M using the wild type UGT85C2 to 19 μ M using the codon-optimized UGT85C2. Consequently, about 10 μ M less total 19-SMG was measured for the codon-optimized UGT85C2. This shows that more 19-SMG is converted into rubusoside and confirms that the wild type UGT85C2 is a bottleneck.

During diversity screening, another homolog of UGT85C2 was discovered during *Stevia rebaudiana* cDNA cloning. The homolog has the following combination of conserved amino acid polymorphisms (with respect to the amino acid numbering of the wild-type *S. rebaudiana* UGT85C coding sequence set forth in Accession No. AY345978.1): A65S, E71Q, T270M, Q289H, and A389V. This clone, termed UGT85C2 D37, was expressed through coupled in vitro transcription-translation of PCR products (TNT[®]T7 Quick for PCR DNA kit, Promega). The expression product was assayed for glycosylation activity using steviol (0.5 mM) as the sugar acceptor, as described in WO/2011/153378 with the exception that assays were allowed to incubate for 24 hours. As compared to the wildtype UGT85C2 control assay, the D37 enzyme appears to have approximately 30% higher glycosylation activity.

Example 11—Identification of a Novel & *Rebaudiana* KAH

A partial sequence (GenBank Accession No. BG521726) was identified in the *Stevia rebaudiana* EST data base that had some homology to a *Stevia* KAH. The partial sequence was blasted against raw *Stevia rebaudiana* pyrosequencing reads using CLC main workbench software. Reads that partially overlapped with the ends of the partial sequence were identified and used to increase the length of the partial sequence. This was done several times until the sequence encompassed both the start- and the stop codons. The complete sequence was analyzed for frameshift mutations and nucleotide substitutions that may have been introduced by blasting the complete sequence against the raw pyrosequencing reads. The resulting sequence was designated SrKAHe1. See FIG. 12.

Activity of the KAH encoded by SrKAHe1 was assessed in vivo in *S. cerevisiae* background strain CEN.PK 111-61A, which expresses genes encoding enzymes constituting the entire biosynthetic pathway from the yeast secondary metabolites isopentenyl pyrophosphate (IPP) and farnesyl pyrophosphate (FPP) to steviol-19-O-monoside, except the steviol synthase enzyme that converts ent-kaurenoic acid to steviol.

Briefly, the *S. cerevisiae* strain CEN.PK 111-61A was modified to express an *Aspergillus nidulans* GGPPS, a 150 nt truncated *Zea mays* CDPS (with a new start codon, see below), a *S. rebaudiana* KS, a *S. rebaudiana* KO and the *S. rebaudiana* UGT74G1 from chromosomally integrated gene copies, with TPI1 and GPD1 yeast promoters driving transcription. The CEN.PK 111-61A yeast strain that expresses all of these genes was designated EFSC2386. Thus, strain EFSC2386 contained the following integrated genes: *Aspergillus nidulans* Geranyl geranyl pyrophosphate synthase (GGPPS); *Zea mays* ent-Copalyl diphosphate synthase (CDPS); *Stevia rebaudiana* ent-Kaurene synthase (KS); *Stevia rebaudiana* ent-kaurene oxidase (KO); and *Stevia*

rebaudiana UGT74G1; in combination with the pathway from IPP and FPP to steviol-19-O-monoside, without a steviol synthase (KAH).

Expression of different steviol synthases (from episomal expression plasmids) was tested in strain EFSC2386 in combination with the expression of various CPRs (from episomal expression plasmids), and production of steviol-19-O-monoside was detected by LC-MS analysis of culture sample extracts. The nucleic acids encoding the CPRs were inserted in the multi cloning site of the p426 GPD basic plasmid while the nucleic acids encoding the steviol synthases were inserted in the multi cloning site the p415 TEF basic plasmid (p4XX basic plasmid series by Mumberg et al., *Gene* 156 (1995), 119-122). Production of steviol-19-O-monoside occurs when a functional steviol synthase enzyme is present.

The KAHs that were expressed from episomal expression plasmids in strain EFSC2386 were “indKAH” (Kumar et al., Accession no. DQ398871; Reeja et al., Accession No. EU722415); “KAH1” (*S. rebaudiana* steviol synthase from Brandle et al., U.S. Patent Publication No. 2008/0064063 A1); “KAH3” (*A. thaliana* steviol synthase from Yamaguchi et al., U.S. Patent Publication No. 2008/0271205 A1); “SrKAHe1” (*S. rebaudiana* steviol synthase cloned from *S. rebaudiana* cDNA as described above); and “DNA2.0.SrKAHe1” (codon optimized sequence (DNA2.0) encoding *S. rebaudiana* steviol synthase, see FIG. 12B).

The CPRs that were expressed from episomal expression plasmids in strain EFSC2386 were “CPR1” (*S. rebaudiana* NADPH dependent cytochrome P450 reductase (Kumar et al., Accession no. DQ269454); “ATR1” (*A. thaliana* CPR, Accession No. CAA23011, see also FIG. 13); “ATR2” (*A. thaliana* CPR, Accession No. CAA46815, see also FIG. 13); “CPR7” (*S. rebaudiana* CPR, see FIG. 13, CPR7 is similar to “CPR1”); “CPR8” (*S. rebaudiana* CPR, similar to *Artemisia annua* CPR, see FIG. 13); and “CPR4” (*S. cerevisiae* NCP1 (Accession No. YHR042W, see also FIG. 13).

Table 13 provides the levels of steviol-19-O-monoside (μM) in strain EFSC2386 with the various combination of steviol synthases and CPRs.

TABLE 13

19-SMG Production	
Strain	19-SMG production (μM)
“indKAH” CPR1	0.000
“indKAH” ATR1	0.000
“indKAH” ATR2	0.000
“indKAH” CPR7	0.000
“indKAH” CPR8	0.000
“indKAH” CPR4	0.000
“KAH1” CPR1	0.000
“KAH1” ATR1	0.000
“KAH1” ATR2	0.000
“KAH1” CPR7	0.000
“KAH1” CPR8	0.000
“KAH1” CPR4	0.000
“KAH3” CPR1	5.300
“KAH3” ATR1	5.921
“KAH3” ATR2	0.000
“KAH3” CPR7	5.693
“KAH3” CPR8	0.000
“KAH3” CPR4	0.000
“SrKAHe1” CPR1	20.129
“SrKAHe1” ATR1	15.613
“SrKAHe1” ATR2	40.407
“SrKAHe1” CPR7	33.724
“SrKAHe1” CPR8	41.695
“SrKAHe1” CPR4	28.949

TABLE 13-continued

19-SMG Production	
Strain	19-SMG production (μM)
“DNA2.0.SrKAHe1” CPR1	26.065
“DNA2.0.SrKAHe1” ATR1	26.974
“DNA2.0.SrKAHe1” ATR2	54.354
“DNA2.0.SrKAHe1” CPR7	30.797
“DNA2.0.SrKAHe1” CPR8	50.956
“DNA2.0.SrKAHe1” CPR4	30.368

Only KAH3 and the steviol synthase encoded by SrKAHe1 had activity when expressed in *S. cerevisiae*. The DNA 2.0. codon optimized SrKAHe1 sequence encoding steviol synthase resulted in a level of steviol-19-O-monoside accumulation that was approximately one order of magnitude higher as compared with a codon optimized KAH3 when each were co-expressed with optimal CPRs. In the experiments presented in this example, the combination of KAH1 and ATR2 CPR did not result in the production of steviol-19-O-monoside.

Example 12—Pairings of CPRs and KO

The CEN.PK *S. Cerevisiae* EFSC2386 strain and the CPRs referred to in this Example are described in the Example 11 (“Identification of *S. rebaudiana* KAH”). EFSC2386 contained the following integrated genes: *Aspergillus nidulans* Geranyl geranyl pyrophosphate synthase (GGPPS); *Zea mays* ent-copalyl diphosphate synthase (CDPS); *Stevia rebaudiana* ent-kaurene synthase (KS); and *Stevia rebaudiana* ent-kaurene oxidase (KO). This strain produces ent-kaurenoic acid that was detected by LC-MS analysis.

A collection of cytochrome P450 reductases (CPRs) were expressed and tested in strain EFSC2386; “CPR1” (*S. rebaudiana* NADPH dependent cytochrome P450 reductase, Kumar et al., Accession no. DQ269454); “ATR1” (*A. thaliana* CPR, Accession No. CAA23011); “ATR2” (*A. thaliana* CPR, Accession No. CAA46815); “CPR7” (*S. rebaudiana* CPR, CPR7 is similar to “CPR1”); “CPR8” (*S. rebaudiana* CPR, similar to *Artemisia annua* CPR; and “CPR4” (*S. cerevisiae* NCP1, Accession No. YHR042W).

Overexpression of the *S. cerevisiae* endogenous native CPR (referred to as CPR4 in Table 14), and especially overexpression of one of the *A. thaliana* CPRs namely ATR2, gives good activation of the *Stevia rebaudiana* kaurene oxidase (the latter called KO1 in Table 14) and results in increased accumulation of ent-kaurenoic acid. See Table 14, which presents the area under curve (AUC) of the ent-kaurenoic acid peak in the LC-MS chromatograms. KO1 is an ent-kaurenoic acid producing yeast control strain without additional overexpression of CPRs.

TABLE 14

Effect of Different Cytochrome P450 Reductase Enzymes with KO-1	
Cytochrome P450 Reductase	Ent-Kaurenoic Acid (AUC)
CPR-1	14113
ATR-1	13558
ATR-2	29412
CPR-7	18918
CPR-8	12590

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TABLE 14-continued

Effect of Different Cytochrome P450 Reductase Enzymes with KO-1	
Cytochrome P450 Reductase	Ent-Kaurenoic Acid (AUC)
CPR-4	25103
Control	16593

Example 13—Evaluating KS-5 and KS-1 in Steviol Pathways

The yeast strain EFSC1972 is a CEN.PK 111-61A *S. cerevisiae* strain that has the biosynthetic pathway from IPP/FPP to rubusoside expressed by integrated gene copies encoding the *Aspergillus nidulans* GGPPS (internal name GGPPS-10), the *Stevia rebaudiana* KS (KS1, SEQ ID NO:133), the *Arabidopsis thaliana* KAH (KAH-3, SEQ ID NO:144), the *Stevia rebaudiana* KO (KO1, SEQ ID NO:138), the *Stevia rebaudiana* CPR (CPR-1, SEQ ID NO:147), the full length *Zea mays* CDPS (CDPS-5, SEQ ID NO:158), the *Stevia rebaudiana* UGT74G1 (SEQ ID NO:1) and *Stevia rebaudiana* UGT85C2 (SEQ ID NO:3). Furthermore EFSC1972 has down regulation of the ERG9 gene expression by displacement of the endogenous promoter with the copper inducible promoter CUP1.

When EFSC1972 is transformed with a CEN/ARS-based plasmid that expresses the *Stevia rebaudiana* SrKAHe1 from a TEF1 promoter, and simultaneously transformed with 2 μ -based plasmids that express the *Synechococcus* sp GGPPS (GGPPS-7) and a truncated version of the *Zea mays* CDPS (truncated CDPS-5) from a GPD promoter, the result is growth-impaired *S. cerevisiae* producer of rubusoside (and 19-SMG). This strain is referred to as the “enhanced EFSC1972” in the following text. To determine whether the slow growth rate is caused by accumulation of the toxic pathway intermediate ent-copalyl diphosphate, a collection of kaurene synthase (KS) genes was expressed in the “enhanced EFSC1972” strain then growth and steviol glycoside production was assessed.

Expression of the *A. thaliana* KS (KS5) results in improved growth and steviol glycoside production of the “enhanced EFSC1972” strain. See FIG. 16. The same positive effect on growth cannot be achieved by further overexpression of the *Stevia rebaudiana* kaurene synthase (KS-1) in the enhanced EFSC1972 (data not shown).

Example 14—Yeast Strain EFSC1859

Saccharomyces cerevisiae strain EFSC1859 contains GGPPS-10, CDPS-5, KS-1, KO-1, KAH-3, CPR-1 and UGT74G1 coding sequences integrated into the genome and expressed from the strong constitutive GPD1 and TPI promoters. See Table 15. In addition, the endogenous promoter for the yeast ERG9 gene was replaced with the copper inducible promoter CUP1 for downregulation of the ERG9 squalene synthase. In standard yeast growth medium, the ERG9 gene is transcribed at very low levels, since the concentration of copper in such medium is low. The decrease in ergosterol production in this strain results in increased amounts of isoprene units available for isoprenoid biosynthesis. In addition, strain EFSC1859 also expresses UGT85C2 from a 2 micron multicopy vector using a GPD1 promoter. EFSC1859 produces rubusoside and steviol 19-O-glycoside.

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Zea mays CDPS DNA, with and without the chloroplast signal peptide, was expressed from a 2 micron multicopy plasmid using the GPD promoter. The nucleotide sequence and amino acid sequence of the *Zea mays* CDPS are forth in FIG. 14. The chloroplast signal peptide is encoded by nucleotides 1-150 and corresponds to residues 1 to 50 of the amino acid sequence.

TABLE 15

Gene Source	Enzyme	Designation	gi Number	Accession No.
<i>Aspergillus nidulans</i>	GGPP synthase	GGPPS-10 (C301)	29468175	AF479566
<i>Zea mays</i>	CDP synthase	CDPS-5 (EV65)	50082774	AY562490
<i>Stevia rebaudiana</i>	Kaurene synthase	KS-1	4959241	AAD34295
<i>Stevia rebaudiana</i>	KO	KO-1	76446107	ABA42921
<i>Arabidopsis thaliana</i>	KAH	KAH-3	15238644	NP_197872
<i>Stevia rebaudiana</i>	UGT74G1			
<i>Stevia rebaudiana</i>	UGT85C2			
<i>Stevia rebaudiana</i>	CPR	CPR-1	93211213	ABB88839

EFSC1859+maize full-length CDPS plasmid, and EFSC+maize truncated CDPS plasmid were grown in selective yeast medium with 4% glucose. Rubusoside and 19-SMG production were measured by LC-MS to estimate the production level. The removal of the plastid leader sequence did not appear to increase steviol glycoside production as compared to the wild-type sequence, and demonstrates that the CDPS transit peptide can be removed without causing a loss of steviol glycoside biosynthesis.

Example 15—Yeast Strain EFSC1923

Saccharomyces cerevisiae strain CEN.PK 111-61A was modified to produce steviol glycosides by introduction of steviol glycoside pathway enzymes from various organisms. The modified strain was designated EFSC1923.

Strain EFSC1923 contains an *Aspergillus nidulans* GGPP synthase gene expression cassette in the *S. cerevisiae* PRP5-YBR238C intergenic region, a *Zea mays* full-length CDPS and *Stevia rebaudiana* CPR gene expression cassette in the MPT5-YGL176C intergenic region, a *Stevia rebaudiana* kaurene synthase and CDPS-1 gene expression cassette in the ECM3-YOR093C intergenic region, an *Arabidopsis thaliana* KAH and *Stevia rebaudiana* KO gene expression cassette in the KIN1-INO2 intergenic region, a *Stevia rebaudiana* UGT74G1 gene expression cassette in the MGA1-YGR250C intergenic region and a *Stevia rebaudiana* UGT85C2 gene expression cassette integrated by displacing the TRP1 gene ORF. See Table 15. In addition, the endogenous promoter for the yeast ERG9 gene was replaced with the copper inducible promoter CUP1.

Strain EFSC1923 produced approximately 5 μ M of the steviol glycoside, steviol 19-O-monoside, on selective yeast medium with 4% glucose.

Example 16—Expression of a Truncated Maize CDPS in Yeast Strain EFSC1923

The 150 nucleotides at the 5' end of the *Zea mays* CDP synthase coding sequence in Table 15 (SEQ ID NO:157, see

FIG. 14) was deleted, the remainder of the coding sequence was provided with a new translation start ATG, and the truncated sequence was operably linked to the GPD1 promoter in the multicopy plasmid p423GPD in *Saccharomyces cerevisiae* EFSC1923. Plasmid p423GPD is described in Mumberg, D et al, *Gene*, 156: 119-122(1995). EFSC1923 and EFSC1923 plus p423GPD-Z.m.tCDPS were grown in for 96 hours in selective yeast medium containing 4% glucose. The amount of steviol 19-O-monoside produced by EFSC1923+p423GPD-Z.m.tCDPS (the truncated *Zea mays* CDPS) under these conditions was approximately 2.5 fold more than that produced by EFSC1923 without the plasmid.

The *Arabidopsis thaliana* KAH coding sequence from Table 15 was inserted in a multicopy plasmid designated p426GPD, under the control of the GPD1 promoter. Plasmid p426GPD is described in Mumberg, D et al, *Gene*, 156: 119-122(1995). No significant difference was observed between the amount of steviol 19-O-monoside produced by EFSC1923+p426GPD-A.t.KAH, and EFSC1923 lacking the plasmid.

EFSC1923 was transformed with both p423GPD-Z.m.tCDPS and p426 p426GPD-A.t.KAH. Surprisingly, the amount of steviol 19-O-monoside produced under these conditions by EFSC1923 harboring both plasmids (i.e., the truncated *Zea mays* CDPS and *Arabidopsis* KAH) was more than 6 fold greater than the amount produced by EFSC1923 alone.

A bifunctional CDPS-KS from *Gibberella fujikuroi* (NCBI Accession no: Q9UVY5.1, FIG. 15) was cloned and compared to the truncated CDPS-5. The bifunctional *Gibberella* CDPS-KS was cloned into a 2p plasmid with a GPD promoter and transformed with a plasmid expressing the *Arabidopsis thaliana* KAH-3 from a 2μ based-plasmid from a GPD promoter into EFSC1923. In shake flask studies, this bifunctional CDPS-KS was about 5.8 times more active in producing steviol 19-O-monoside than strain EFSC1923 with the KAH-3 alone. However, it was found to be less optimal than the KAH-3 and truncated CDPS combination under the conditions tested. Therefore, further strains were constructed with KS-5 and truncated CDPS.

Example 17—Toxicity of Intermediates

The effect on *S. cerevisiae* vitality of geranyl geranyl pyrophosphate (GGPP), ent-copalyl diphosphate (CDP) or ent-kaurene production was investigated by expression of *Synechococcus* sp GGPPS alone (GGPP production), the GGPPS and the 50 amino acid N-terminally truncated *Zea mays* CDPS (see Example 16) together (CDP production), or the GGPPS, truncated CDPS and the *Arabidopsis thaliana* kaurene synthase (KS5) together (ent kaurene production) in the laboratory *S. cerevisiae* strain CEN.PK background. Genes were expressed from 2μ plasmids with GPD promoters driving transcription of truncated CDPS and KS5, while transcription of the GGPPS was driven by the ADH1 promoter. The growth of *S. cerevisiae* CEN.PK transformed with various combinations of these plasmids (GGPPS alone; GGPPS+truncated CDPS; or GGPPS+truncated CDPS+KS5) or plasmids without gene insertions was observed. GGPP production, and especially CDP production, was toxic to *S. cerevisiae* when produced as end products. Interestingly, ent-kaurene appeared to not be toxic to yeast in the amounts produced in this experiment.

Example 18—Disruption of Endogenous Phosphatase Activity

The yeast genes DPP1 and LPP1 encode phosphatases that can degrade FPP and GGPP to farnesol and gera-

nylgeraniol, respectively. The gene-encoding DPP1 was deleted in strain EFSC1923 (described in Example 15) to determine if there was an effect on steviol glycoside production. When this dpp1 mutant strain was further transformed with a plasmid expressing the *Z. mays* CDPS lacking the chloroplast transit sequence (Example 16), both small and large transformants emerged. Strains of the “large colony” type produced ~40% more 19-SMG as compared to “small colony” type and the non-DPP1 deleted strain, under the conditions tested. These results indicate that deletion of DPP1 can have a positive effect on steviol glycoside production and that the degradation of prenol pyrophosphates in yeast therefore could influence steviol glycoside production negatively.

Example 19—Construction of a Genetically Stable Yeast Reporter Strain Producing Vanillin Glucoside from Glucose with a Disrupted SUC2 Gene

A yeast strain producing vanillin glucoside from glucose was created basically as described in Brochado et al. ((2010) *Microbial Cell Factories* 9:84-98) (strain VG4), but with additional integration into the ECM3 inter-locus region in the yeast genome of an expression cassette with *E. coli* EntD PPTase controlled by the yeast TPI1 promoter (as described in Hansen et al. (2009) *Appl. Environ. Microbiol.* 75(9): 2765-2774), disruption of SUC2 by replacing coding sequence with a MET15 expression cassette, and disruption of LEU2 by replacing coding sequence with a Tn5ble expression cassette conferring resistance to phleomycin. The resulting yeast strain was called V28. This strain also encodes a recombinant *A. thaliana* UDP-glycosyltransferase (UGT72E2, GenBank Accession No. Q9LVR1) having the amino acid sequence set forth in FIG. 19 (SEQ ID NO:178).

Example 20—Expression of Sucrose Transporter and Sucrose Synthase in Yeast Already Biosynthesizing Vanillin Glucoside

A sucrose transporter SUC1 from *Arabidopsis thaliana* was isolated by PCR amplification from cDNA prepared from *A. thaliana*, using proof-reading PCR polymerase. The resulting PCR fragment was transferred by restriction digestion with SpeI and EcoRI and inserted into the corresponding in the low copy number yeast expression vector p416-TEF (a CEN-ARS based vector), from which the gene can be expressed from the strong TEF promoter. The resulting plasmid was named pVAN192. The sequence of the encoded sucrose transporter is set forth in FIG. 19B (GenBank Accession No. AEE35247, SEQ ID NO:179).

A sucrose synthase SUS1 from *Coffea arabica* (Accession No. CAJ32596) from was isolated by PCR amplification from cDNA prepared from *C. arabica*, using proof-reading PCR polymerase. The PCR fragment was transferred by restriction digestion with SpeI and SaI and inserted into the corresponding position in the high copy number yeast expression vector p425-GPD (a 2 μm based vector), from which the gene can be expressed from the strong GPD promoter. The resulting plasmid was named pMUS55. The sequence of the encoded sucrose synthase is set forth in FIG. 19C (GenBank Accession No. CAJ32596; SEQ ID NO:180).

pVAN192 and pMUS55 were introduced into the yeast strain V28 by genetic transformation, using a lithium acetate transformation protocol, creating the yeast strain V28::

pVAN192::pMUS55. A control strain was made by transforming V28 with the empty plasmids P146-TEF and P425-GPD.

These two yeast strains were grown in 200 ml cultures in 500 ml Erlenmeyer shake flasks using SC (synthetic complete) growth medium without aromatic amino acids supplemented with 2% glucose and 2% sucrose and adjusted to pH 5.0. Cultures were incubated at moderate revolution (150 rpm), at 30° C. for 72 hours. Samples were taken at 72 hours, and the content of vanillin glucoside determined. As can be seen from the table below, VG production in the control strain (containing empty plasmids p416-TEF and p425-GPD) was 330 mg/L VG, while the yeast strain V28::pVAN192::pMUS55 expressing sucrose synthase and sucrose transporter produced 445 mg/l VG, corresponding to a 34.8% increase in VG production.

Strain	Vanillin glucoside (g/L after 72 h)
V28 (p416-TEF + P425-GPD)	330
V28::pVAN192::pMUS55	445

This indicates that co-expression of a sucrose synthase and a sucrose transporter together with a glucosyltransferase increased the ability to glycosylate a small molecule aglycon, and concentration of the glycosylated aglycon was significantly increased. In this case, a significant improvement in vanillin glucosylation was achieved, resulting in a significant increase in titer of the end product, vanillin-O-β-glucoside.

Example 21—Improved Steviol Glycoside Producing Strains

Strain Construction of *Saccharomyces cerevisiae* EFSC2763

EFSC2763 yeast strain is derived from a wild type *Saccharomyces cerevisiae* strain containing three auxotrophic modifications, namely the deletions of URA3, LEU2 and HIS3. The genetics of the strain have been stabilized and can be used as a regular diploid or haploid yeast strain. EFSC2763 has been converted to a steviol glycoside producing yeast by genomic-integration of four DNA constructs. Each construct contains multiple genes that were introduced into the yeast genome by homologous recombination. Furthermore, construct one and two were assembled by homologous recombination.

The first construct contains eight genes and is inserted in the DPP1 locus and disrupts and partially deletes DPP1 (see Example 18). The DNA inserted contains: the *A. gossypii* TEF promoter expressing the NatMX gene (selectable marker) followed by the TEF terminator from *A. gossypii*; Gene Art codon optimized *S. rebaudiana* UGT85C2 (see Example 10) expressed from the native yeast GPD1 promoter and followed by the native yeast CYC1 terminator; *S. rebaudiana* CPR-8 (see FIG. 13) expressed using the TPI1 promoter followed by the native yeast TDH1 terminator; *A. thaliana* Kaurene synthase (KS-5, see Example 13, SEQ ID NO:156) expressed from the PDC1 promoter and followed by the native yeast FBA1 terminator; *Synechococcus* sp. GGPPS (GGPPS-7) expressed using the TEF2 promoter and followed by the native yeast PFI1 terminator; DNA2.0 codon-optimized *S. rebaudiana* KAHe1 (see Example 11, SEQ ID NO:165), expressed from the TEF1 promoter and followed by the ENO2 terminator; *S. rebaudiana* KO-1 expressed using the FBA1 promoter and followed by the

native yeast TDH2 terminator; and *Zea mays* truncated CDPS (see Example 14) expressed using the PGK1 promoter and followed by the native yeast ADH2 terminator.

The second construct was inserted at the YPRCA15 locus and contains the native yeast TEF promoter from *A. gossypii* in front expressing the KanMX gene (selectable marker) followed by the TEF terminator from *A. gossypii*, the Gene Art codon optimized *A. thaliana* ATR2 (see FIG. 13B) expressed from the PGK1 promoter followed by the yeast ADH2 terminator; *S. rebaudiana* UGT74G1 expressed from the TPI1 promoter followed by the yeast TDH1 terminator; Gene Art codon-optimized *S. rebaudiana* UGT76G1 expressed from the TEF1 promoter followed by the yeast ENO2 terminator, and GeneArt codon-optimized *S. rebaudiana* UGT91D2e-b (see Example 6) expressed from the GPD1 promoter and followed by the yeast CYC1 terminator.

The first and the second construct were combined in the same spore clone by mating and dissection. This yeast strain was subsequently transformed with construct three and four in two successive events.

Construct three was integrated between genes PRP5 and YBR238C and contained the TEF promoter from *A. gossypii* in expressing the *K. lactis* LEU2 gene followed by the TEF terminator from *A. gossypii*, the GPD1 promoter expressing the DNA2.0-optimized *S. rebaudiana* KAHe1 followed by the CYC1 terminator, and the TPI1 promoter expressing the *Zea mays* truncated CDPS. Construct four was integrated in the genome between genes ECM3 and YOR093C with an expression cassette containing the TEF promoter from *A. gossypii* expressing the *K. pneumoniae* hph gene followed by the TEF terminator from *A. gossypii*, *Synechococcus* sp. GGPPS expressed from the GPD1 promoter followed by the CYC1 terminator, and the TPI1 promoter expressing the *A. thaliana* Kaurene synthase. The four utilized genetic markers were subsequently removed.

As analyzed by LC-MS following the DMSO-extraction of total steviol glycosides from cells and broth, EFSC2772 produces between 40-50 μM or 2-3 μM/OD600 Rebaudioside A, after growth for four days in 3 ml SC (Synthetic Complete) media at 30° C. with 320 RPM shaking in deep-well plates.

Strain Construction of *Saccharomyces cerevisiae* EFSC2772

EFSC2772 is very similar to strain 2763 with the exception that the genetic markers were not removed, and the strain was made prototrophic by introduction of the two plasmids p413TEF (public domain CEN/ARS shuttle plasmid with HIS3 marker) and p416-TEF (public domain CEN/ARS shuttle plasmid with URA3 marker) by transformation, and designated EFSC2772.

As analyzed by LC-MS following the DMSO-extraction of total steviol glycosides from cells and broth, EFSC2772 produces similar levels of Rebaudioside A as 2763, after growth in deep-well plates. Higher optical densities and higher titers were obtained through aerobic fed-batch growth in 2 L (working volume) fermentors which included a ~16 hour growth phase in the base medium (Synthetic Complete media) followed by ~100 hours of feeding with glucose utilized as the carbon and energy source combined with trace metals, vitamins, salts, and Yeast Nitrogen Base (YNB) and/or amino acid supplementation. The pH was kept near pH 5 and the temperature setpoint was 30° C. As evidenced by LC-MS, combined cellular and extracellular product concentrations were between 920-1660 mg/L of Reb-A and approximately 300-320 mg/L of Reb-D in the two different experiments, approximately 700 mg/L of Reb-A was detected in the broth when the higher titer results were obtained. Additionally a large peak was seen for Reb-B, and

one skilled in the art will recognize that additional copies of UGT74G1 or upregulation of UGT74G1 will further increase the conversion of RebB to RebA.

Strain EFSC2743 was made in a similar manner as above, but without the two plasmids conferring prototrophy and with the addition of a p416 (CEN/ARS)-based plasmid expressing EUGT11 from the TEF promoter. This strain was grown in a fed-batch fermentation as above. This strain produced a total amount of RebD of 920 mg/L and furthermore approximately a 9:1 ratio of RebD to RebA was seen. Approximately 360 mg/L of RebD was found in the broth.

Example 22—UDP-Glucose Capacity

In Example 21, it was shown that yeast can fully glycosylate over 1 mM steviol e.g., to RebD, RebB, and RebA. Similarly, *Saccharomyces* strains are able to glycosylate as much as 60 mM of other small molecule products (data not shown). However, the glycosylation limit of the yeast native UDP-glucose regenerating system is unknown, or the rate at which it replenishes the UDP-glucose pool needed for cell wall synthesis. Therefore, experiments were designed to investigate if an increase in UDP-glucose production would increase the glycosylation rate in yeast. A *suc2* deletion mutant was transformed with plasmids harboring the *A. thaliana* *suc1* gene encoding a sucrose transporter, UGT74G1 and *A. thaliana* SUS. UGT74G1 can rapidly glycosylate steviol to steviol 19-O-monoglucoside (19-SMG). Transformants were pre-grown overnight in 13-ml culture tubes containing 4 ml of SC medium lacking leucine, histidine and uracil. The next day, cells corresponding to 2 OD₆₀₀ units were spun down and resuspended in 2 ml of fresh media was containing 2% sucrose and or 100 μM steviol. Cultures were shaken at 30° C. for 3 days in culture tubes. After 1 h, 3 h, 6 h, 21 h and 46 h, aliquots were taken. Aliquots of 100 μl of culture were spun down and an equal volume of DMSO was added. Samples were vortexed, heated at 80° C. for 15 minutes, centrifuged, and the 19-SMG content analyzed by LC-MS. No difference in the rate of glycosylation of steviol was observed between wild-type and SUS1-augmented strains at the time points tested. This suggests that glycosylation of steviol by UGT74G1 proceeds at a slower rate than UDP-glucose is regenerated by the yeast and that extra UDP-glucose may not be needed to achieve high titers of small molecule glycosylation in vivo. Nevertheless, the use of a SUS to recycle UDP-glucose in vitro is shown in Example 8 and therefore its use in an in vivo system is expected to increase the rate of production of steviol glycosides, if UDP-glucose should become limiting.

Example 23—Reb-C and Reb-F Production In Vivo from Glucose

Production of RebC from Steviol

Previous experiments (Publication No. WO/2011/153378) have shown that recombinantly expressed *Arabidopsis thaliana* RHM2 (rhamnose synthetase, locus tag AT1G53500) is able to convert UDP-glucose to UDP-rhamnose. This UDP-rhamnose can be used to produce steviol-13-O-glucopyranosyl-1,2-rhamnoside, when incubated with UGT91D2e and steviol-13-O-monoglucoside in vitro.

Further experiments were conducted to confirm production of RebC from steviol by expressing all 4 UGTs and the RHM2 in yeast in vivo, followed by steviol feeding. EFSC301 strain (MAT alpha, lys2ADE8 his3ura3leu2trp1) was transformed with the following plasmids expressing

wild type gene sequences: p424GPD expressing wild type UGT74G1 (Accession no: AY345982); p423GPD expressing wild type 85C2 (Accession no.: AY345978.1); and a p426GPD derived-plasmid expressing wildtype UGT76G1 (Accession no: AY345974) and UGT91D2e under GPD promoters. Plasmid p425GPD expressing either RHM2 or an empty p425GPD control plasmid was cotransformed with the UGTs. Transformants were pre-grown overnight in 13 mL culture tubes containing 2-3 ml of SC medium lacking leucine, histidine, tryptophan and uracil. The next day, after growth had reached 0.4 OD₆₀₀ units, cells were spun down, resuspended in fresh medium containing 25 μM steviol and shaken at 30° C. for 3 days in culture tubes. An aliquot of 100 μL of culture was spun down. An equal volume of DMSO was added to the supernatant of this sample while 200 μL of 50% DMSO was added to the pellet. Samples were vortexed, heated at 80° C. for 15 minutes, centrifuged, and the steviol glycoside content analyzed by LC-MS. RebC was detected in growth media and cellular extracts only when the RHM2 gene was coexpressed with the UGTs. Quantification showed that approximately equal amounts of RebA and RebC were produced. This shows that RHM2 is able to produce significant quantities of UDP-rhamnose in vivo and that UGT91D2e is capable of efficient rhamnosylation in vivo. Two other compounds were observed via LC-MS with retention times of 5.64 and 5.33 minutes and m/z ratios corresponding to steviol with 1 glucose- and 1 rhamnose (steviol-1,2 rhamnobioside), and 2 glucoses- and 1 rhamnose (Dulcoside A), respectively. This suggests that the remaining UGTs in the steviol glycoside pathway are capable of accepting rhamnosylated intermediates, i.e, the rhamnosylation step does not need to occur last.

In addition, a series of sequential in vitro experiments were conducted to determine whether any dead-end reactions occur in the rebaudioside C pathway. See FIG. 2B. For example, the rhamnosylation activity of UGT91D2e on rubusoside and subsequent conversion of the product to RebC by UGT76G1 was demonstrated using in vitro reactions. In this experiment, UGT91D2e and RHM2 recombinantly expressed in *E. coli* and purified were incubated overnight with rubusoside, NADPH, NAD⁺ and UDP-glucose. The reaction mixture was subsequently boiled to denature the enzymes. An aliquot of the reaction was added to an enzyme preparation of UGT76G1 with UDP-glucose. The rubusoside was converted in the presence of UGT91D2e and RHM2 to a compound with m/z corresponding to steviol with 2-glucoses and 1-rhamnose. Subsequently, this compound was converted in the presence of UGT76G1 to RebC, which indicates that the intermediate is Dulcoside A. This experiment therefore demonstrates that UGT91D2e is able to rhamnosylate rubusoside and that UGT76G1 is able to convert the product to RebC.

Similarly, it was shown through in vitro reactions that rhamnosylation of 13-SMG by UGT91D2e (forming a steviol compound with one glucose and one rhamnose) and subsequent formation of a compound with 2 glucoses and 1 rhamnose by UGT76G1. This compound has a unique retention time (4.56 min) and is thought to be steviol 13-O-1,3-diglycoside-1,2-rhamnoside. This compound also was observed when steviol was fed to yeast expressing the four UGTs and RHM2.

From the current data, it is shown that UGT91D2e is able to rhamnosylate 13-SMG and rubusoside. It is also shown that UGT74G1 and UGT76G1 are able to metabolize the rhamnosylated compound produced by UGT91D2e from 13-SMG. When these compounds are incubated with the remaining UGT (UGT74G1 or UGT76G1 depending on

which UGT was used for the previous step), RebC is formed. This indicates that the order of glycosylation is of little importance as UGT74G1 and UGT76G1 are able to glycosylate rhamnosylated substrates.

Production of RebC from Glucose

Plasmids expressing RHM2 and UGTs 76G1 and 91D2e were transformed into a stable rubusoside producer, the EFSC1923 strain (see Example 15). This yeast is a *Saccharomyces cerevisiae* CEN.PK 111-61A derivative with the UGTs 85C2 (Accession no.: AY345978.1) and 74G1 (Accession no.: AY345982) integrated into the genome as well as auxotrophic modifications. In strain EFSC1923 (see Example 15), expression of squalene synthase, which is encoded by ERG9, was downregulated by displacement of the endogenous promoter with the CUP1 copper-inducible promoter. Strain EFSC1923 also contains an *Aspergillus nidulans* GGPP synthase (GGPPS-10) expression cassette in the *S. cerevisiae* PRP5-YBR238C intergenic region, a *Zea mays* full-length CDPS (CDPS-5) and *Stevia rebaudiana* CPR (CPR-1) gene expression cassette in the MPT5-YGL176C intergenic region, a *Stevia rebaudiana* Kaurene synthase and CDPS (KS-1/CDPS-1) gene expression cassette in the ECM3-YOR093C intergenic region, an *Arabidopsis thaliana* KAH (KAH-3) and *Stevia rebaudiana* KO (KO-1) gene expression cassette in the KIN1-INO2 intergenic region, a *Stevia rebaudiana* UGT74G1 gene expression cassette in the MGA1-YGR250C intergenic region and a *Stevia rebaudiana* UGT85C2 gene expression cassette integrated by displacing the TRP1 gene ORF20. Inserted steviol pathway genes are described in Table 11 of published PCT WO/2011/153378.

EFSC1923 strain was transformed with a p423GPD-derived plasmid expressing wildtype UGT74G1 and UGT85C2 sequences using GPD promoters and a p426GPD-derived plasmid expressing wildtype UGT76G1 (Accession no.: AY345974) and UGT91D2e (see SEQ ID NO:5) under the control of GPD promoters. Plasmid p425GPD expressing *Arabidopsis thaliana* RHM2 (enzyme locus tag AT1G53500) or an empty p425GPD control plasmid was co-transformed. Transformants were pre-grown overnight in 13-ml culture tubes containing 2-3 ml of SC medium lacking leucine, histidine and uracil. The next day, when the culture reached an OD₆₀₀ of 0.4 units it was centrifuged, resuspended in fresh medium, and shaken at 30° C. for 3 days in culture tubes. One hundred μ L of culture were spun down; to this an equivalent volume of DMSO was added to the supernatant while 200 μ L of 50% DMSO was added to the pellet. Samples were vortexed, heated at 80° C. for 15 minutes, spun down and the steviol glycoside content analyzed by LC-MS.

Analyses of the medium and normalized intracellular content of this strain showed production of RebC. Approximately 8 μ M RebC and 4 μ M RebA was produced as determined by LC-MS. Furthermore, the intermediates produced following steviol feeding were not detected in this experiment. Accumulation of RebC was strictly dependent on expression of RHM2. This example demonstrates de novo biosynthesis of RebC from glucose.

Production of Additional Steviol Glycosides from Steviol and Glucose

Using the same GPD-based plasmids described above, the stable steviol-producing strain EFSC1923 containing UGT74G1 and UGT85C2 was transformed with the UGTs required to produce RebB (UGT76G1 and UGT91D2e/EUGT11), RebE (UGT91D2e/EUGT11) and dulcoside A (RHM2, UGT91D2e/EUGT11). Wildtype EUGT11 (NCBI: NP_001051007), which was found to have higher diglyco-

sylation activity, was cloned into p424GPD for this experiment. Transformants were pre-grown overnight in 13-ml culture tubes containing 2-3 ml of SC medium lacking leucine, histidine, tryptophan and uracil. The next day, after growth had reached 0.4 OD₆₀₀ units, cells were spun down, resuspended in fresh medium containing 25 μ M steviol (except for glucose experiments) and shaken at 30° C. for 3 days in culture tubes. An aliquot of 100 μ L of culture was spun down. An equal volume of DMSO was added to the supernatant of this sample while 200 μ L of 50% DMSO was added to the pellet. Samples were vortexed, heated at 80° C. for 15 minutes, centrifuged, and the steviol glycoside content analyzed by LC-MS. LC-MS analyses confirmed in vivo production of RebB, RebE, and Dulcoside A in *S. cerevisiae* from glucose or steviol. See, e.g., FIGS. 2A and 2B. A higher concentration of steviol-glycosides was observed following steviol-feeding (as judged by chromatograms).

Characterization of RebF Pathway Intermediates Using EUGT11.

The xylosylating properties of UGT91D2e and EUGT11 were compared in vitro. By using UDP-xylose as the sugar-donor, UGT91D2e was previously shown to xylosylate steviol-13-O-monoglucoside forming a key intermediate in RebF biosynthesis (Publication No. WO/2011/153378). Similar in vitro experiments using EUGT11 and UGT91D2e have shown that these UGTs are capable of xylosylating rubusoside. When UGT91D2e is used, the LC-MS analysis shows a new peak with an m/z ratio corresponding to steviol with 2 glucose molecules and 1 xylose. See, FIG. 26. Because of the shift in the retention time this peak is thought to correspond to rubusoside xylosylated on the 13-O-glucose. When EUGT11 is used, the LC-MS analysis shows two new, similar sized peaks at retention time 3.99 and 4.39 minutes with m/z ratios corresponding to steviol with 2 glucoses and 1 xylose. These products most likely correspond to rubusoside xylosylated on either of the two positions—the 13-O-glucose or the 19-O-glucose.

Production of RebF from Glucose

In vivo production of RebF requires cloning of UGD1 (UDP-glucose dehydrogenase) and USX3 (UDP-glucuronic acid decarboxylase) from *Arabidopsis* for production of UDP-xylose. UGD1 and UXS3 were inserted in a high copy (2 μ) vector, derived from P425-GPD, containing two expression cassettes, and expressed from strong constitutive promoters (TPI1 and GPD1, respectively). The plasmid was transformed into the RebA producer strain EFSC2763 (described in Example 21) and cultivated during 3 days in selection medium (SC-leu). The LC-MS results clearly show the appearance of a new peak at retention time 4.13 minutes with m/z ratios corresponding to steviol with 3 glucoses and 1 xylose and identified as RebF (based on a commercial RebF standard), as well as other new peaks with m/z ratios corresponding to steviol with 2 glucoses and 1 xylose (as above), indicating that UGT91D2e was capable of carrying out xylosylation in vivo. These peaks were not seen in the negative controls.

Example 24—Effect of Squalene Synthase (ERG9) Down Regulation Using a Heterologous Insert

In yeast such as *Saccharomyces cerevisiae*, the mevalonate pathway produces a number of isoprenoid phosphate intermediates in the biosynthetic pathway to squalene (See FIG. 20). The squalene synthase in yeast is ERG9. See GenBank Accession No. P29704.2 for the *Saccharomyces cerevisiae* squalene synthase; P36596 for the *Schizosaccha-*

romyces pombe squalene synthase; Q9Y753 for the *Yarrowia lipolytica* squalene synthase; Q9HGZ6 for the *Candida glabrata* squalene synthase; Q752X9 for the *Ashbya gossypii* squalene synthase; 074165 for the *Cyberlindnera jadinii* squalene synthase; P78589 for the *Candida albicans* squalene synthase; P38604 for the *Saccharomyces cerevisiae* lanosterol synthase; P37268 for the *Homo sapiens* squalene synthase; P53798 for the *Mus musculus* squalene synthase; and Q02769 for the *Rattus norvegicus* squalene synthase. See FIG. 25 (SEQ ID NOs:192-202).

Introduction of Stemloop Structure in 5'UTR of ERG9 Gene

The wild-type ERG9 promoter region was replaced with the CYC1 promoter sequence and a 5'UTR sequence by homologous recombination. The 5'UTR region contains a sequence that can form a stemloop structure. See SEQ ID NOs. 181-183. SEQ ID NO:184 is another sequence that also can be used.

(heterologous insert 1):
SEQ ID NO: 181
TGAATTCGTTACGAATTC

(heterologous insert 2):
SEQ ID NO: 182
TGAATTCGTTACGAATTC

(heterologous insert 3):
SEQ ID NO: 183
TGAATTCGTTACGAAGTC

(heterologous insert 4):
SEQ ID NO: 184
TGAATTCGTTACGAAATT

Without being bound to a particular mechanism, the stemloop may partially block the 5'-3' directed ribosomal scanning for the AUG and reduce the translation of the transcript. Stemloops with different degree of basepairing were tested to find stemloops that reduced the ERG9 transcript translation sufficiently to boost FPP levels without affecting the growth of the yeast strain.

DNA fragments encompassing an ERG9 promoter upstream sequence (for homologous recombination), an expression cassette for the gene (NatR) that confers resistance to Nourseothricin, a CYC1 promoter (SEQ ID NO: 185, FIG. 21), a 5' UTR sequence with a stemloop structure, and an ERG9 ORF sequence (for homologous recombination) were generated by PCR. DNA fragments that contained either the CYC1 promoter or the KEX2 promoter (SEQ ID NO: 186) but no stemloops were also generated as controls. The flanking ERG9 sequences for recombination as well as the stemloop structure were introduced via the PCR oligos. An overview of the construct for homologous recombination is shown in FIG. 22. The DNA fragments were transformed into an *S. cerevisiae* host strain that subsequently was selected on nourseothricin containing growth plates. Clones with successful exchange of the native ERG9 promoter with the CYC1 promoter and stemloop-containing 5'UTR sequence were identified. Overview and sequence of the stem-loop region is provided in FIG. 23. The sequence identified as 5% corresponds with the heterologous insert having SEQ ID NO:181; the sequence identified as 20% corresponds with the heterologous insert having SEQ ID NO:182; and the sequence identified as 50% corresponds with the heterologous insert having SEQ ID NO:183. Assessment of FPP Accumulation (Boosting Effect)

The *Amorpha-4,11*-diene Synthase (ADS) gene catalyzes the chemical reaction that turns one FPP molecule into *Amorpha-4,11*-diene in the plant *Artemisia annua*. The gene

is functional and efficient in *S. cerevisiae* and can be used to indirectly assess the accumulation of FPP in the strains with the stemloop structure introduced in the heterologous 5'UTR of the ERG9 gene. An *S. cerevisiae* codon optimized nucleic acid encoding ADS (GenBank Accession No. AAF61439) was cloned on a multicopy plasmid (2 μ) under the control of the PGK1 promoter and transformed in the wild type and engineered *S. cerevisiae* strains. *Amorpha-4,11*-diene production was measured and compared to the standard compound caryophyllene, as described by (Ro et al. 2006. *Nature* 440(7086):940-943; Paradise et al. *Biotechnol Bioeng.* 2008 Jun. 1; 100(2):371-8; Newman et al. *Biotechnol Bioeng* 95(4):684-691).

Chemicals

Dodecane and caryophyllene were purchased from Sigma-Aldrich (St. Louis, MO). Complete Supplement Mixtures for formulation of Synthetic Complete (SC) media were purchased from Formedium (UK). All others chemical were purchased from Sigma-Aldrich.

Yeast Cultivation

Engineered yeast strains were grown in SC 2% glucose with uracil dropped out. Cultures were grown at 30° C. overnight and then used to inoculate main cultures in 250 mL shake flasks containing 25 mL SC medium, and grown to an optical density of 0.1 at 600 nm. The main cultures were grown for 72 h at 30° C. Because amorphaadiene at very low concentrations is volatile from aqueous cultures, 2.5 mL dodecane was added to each culture flask in order to trap and retain the amorphaadiene produced. 10 μ L of the dodecane layer was sampled and diluted 100 fold in ethyl acetate for quantification by GC-MS

GC-MS Analysis of Amorphaadiene

GC-MS was used to measure amorphaadiene production from yeast cultures. Samples were analysed using the method as follow: The GC oven temperature program used 80° C. for 2 min, followed by a ramping of 30° C./min to 160° C., then 3° C./min up to 170° C., and finally 30° C./min up to 300° C. with a 2 min final hold. Injector and MS quadrupole detector temperatures were 250° C. and 150° C., respectively. 1 μ L was injected in split less mode. The MS was operated in full scan mode. Amorphaadiene concentration was calculated in (-)-tran-caryophyllene equivalents using a caryophyllene standard curve using the total ions.

The analysis of the different strains, including the different promoter constructs, showed an increased production of amorphaadiene ((2.5 $\times$) when using the heterologous insert having the nucleotide sequence set forth in SEQ ID NO: 181 compared to either no insert or the inserts having the nucleotide sequences set forth in SEQ ID NO:182 and 183. See FIG. 24. The heterologous insert set forth in SEQ ID NO:181 has the most stable secondary structure. For comparison the wild type yeast, with unmodified ERG9, was also analyzed (FIG. 24: CTRL-ADS) and this strain showed even lower production of amorphaadiene. Conversely, the construct that comprised the very weak promoter ScKex2 showed an even higher level of amorphaadiene (6 $\times$).

Example 25—Analysis of the Effect of Squalene Synthase (ERG9) Down Regulation and GGPPS Overexpression on GGPP Production

Assessment of GGPP Accumulation

S. cerevisiae contains a GGPPS (BTS1). In addition to BTS1 there are several heterologous GGPPS enzymes that are functional and efficient in *S. cerevisiae*. When a functional GGPPS is overexpressed in *S. cerevisiae*, it leads to accumulation of GGPP, which may be converted to gera-

nylgeraniol (GGOH) by the *S. cerevisiae* enzymes DPP1 and LPP1. The GGOH is partly exported to the yeast culture medium. GGOH can be measured by GC-MS and its accumulation can indirectly be used to assess the potential pool of GGPP that is available for enzymes that use GGPP as substrate.

Four different GGPPSs (GGPPS-1 (*S. acidocaldarius*, see Table 7), GGPPS-2 (*A. nidulans*, FIG. 25, SEQ ID NO:203), GGPPS-3 (*S. cerevisiae*, BTS1, FIG. 25, SEQ ID NO:167), and GGPPS-4 (*M. musculus*, see Table 7)) were assessed. The nucleotide sequences encoding GGPPS-1, GGPPS-2, and GGPPS-4 were *S. cerevisiae* codon optimized. All nucleic acids encoding the GGPPS polypeptides were cloned on a multicopy plasmid (2p) under the control of the PGK1 promoter and transformed in two different ERG9 down regulated strains: KEX2-ERG9 and CYC1(5%)-ERG9 (see Example 24).

Engineered yeast strains were grown in SC 2% glucose with uracil dropped out. Complete Supplement Mixtures for formulation of Synthetic Complete (SC) media were purchased from Formedium (UK). All others chemical were purchased from Sigma-Aldrich (St. Louis, MO). All optical density measurements were done at OD 600 nm. Cultures were grown at 30° C. overnight and then used to inoculate 250 ml unbaffled culture flasks containing 25 ml SC medium at an OD600 of 0.1. The main cultures were grown for 72 h at 30° C.

To measure GGOH accumulation, yeast cells (pellet) and yeast culture medium (supernatant) were extracted sepa-

rately and then combined before analysis by GC-MS. The supernatant was extracted with Hexane in a 1:1 ratio. The pellet was first subjected to a saponification in solution containing 20% KOH and 50% Ethanol and the lysed cells were finally extracted with Hexane in a 1:1 ratio. The GC oven temperature program used was 80° C. for 2 min, followed by a ramp to 160° C. at 30° C./min, then to 170° C. at 3° C./min and finally to 300° C. at 30° C./min with a 2 min hold. Injector and MS quadrupole detector temperatures were 250° C. and 150° C., respectively. 2 ul was injected in split less mode. The MS was operated in full scan mode.

When the GGPPS were overexpressed in the CYC1(5%)-ERG9 strain or KEX2-ERG9 strain, there was a significant increase in GGOH (GGPP) production observed with all four GGPPS polypeptides compared to the control where no GGPPS was expressed. Notably, the CYC1(5%)-ERG9 strain showed a 2-4 fold higher GGOH (GGPP) accumulation than the KEX2-ERG9 strain. The results are shown in FIG. 26.

OTHER EMBODIMENTS

It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

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tgtttgacg gagccccagg gtttagattc gaaacaattc ctgacgggtg ttcacattcc 240
ccagaggcct ccatacccaat aagagagagt ttactgaggt caatagaaac caactttttg 300
gatcgtttca ttgacttggt cacaaaaact ccagaccac caacttgcac aatctctgat 360
ggctttctgt cagtgtttac tatcgacgct gccaaaaagt tgggtatccc agttatgatg 420
tactggactc ttgctgcatg cggtttcatg ggtttctatc acatccattc tcttatcgaa 480
aagggttttg ctccactgaa agatgcatca tacttaacca acggctacct ggatactgtt 540
attgactggt taccaggatg ggaaggtata agacttaaaag attttccttt ggattgggtc 600
acagacctta atgataaaat attgatgttt actacagaag ctccacaaag atctcataag 660
gtttcacatc atatctttca cacctttgat gaattggaac catcaatcat caaaaccttg 720
tctctaagat acaatcatat ctacactatt ggtccattac aattacttct agatcaaat 780
cctgaagaga aaaagcaaac tggatttaca tccttacacg gctactcttt agtgaaaagag 840
gaaccagaat gttttcaatg gctacaaaag aaagagccta attctgtggt ctacgtcaac 900
tcggaagta caacagtcac gtccttgcaa gatatgactg aatttggttg gggccttgct 960
aattcaaatc attcaatttc attgatttat aggtccaatt tggtaaatag ggaaaacgcc 1020
gtattacctc cagaattgga ggaacacatc aaaaagagag gtttcattgc ttctgggtgt 1080
tctcaggaat aggtattgaa acatccttct gttggtggtt tccttactca ttgcggttgg 1140
ggctctacaa tcgaatcact aagtgcagga gttccaatga tttgttggtc atattcatgg 1200
gaccaactta caaattgtag gtatatctgt aaagagtggg aagttggatt agaaatggga 1260
acaaagggtt aacgtgatga agtgaaaaga ttggttcagg agttgatggg ggaaggtggc 1320
cacaagatga gaaacaagcg caaagatttg aaggaaaaag ccagaattgc tattgtcctc 1380
aacgggtcat cctctctaaa cattgataag atggtcaaaag agattacagt cttagccaga 1440
aactaa 1446

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SEQ ID NO: 5          moltype = AA  length = 473
FEATURE              Location/Qualifiers
source               1..473
                    mol_type = protein
                    organism = Stevia rebaudiana

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SEQUENCE: 5
MATSDSIVDD RKQLHVATFP WLAFGHILPY LQLSKLIAEK GHKVSFLSTT RNIQRLSSHI 60
SPLINVVQLT LPRVQELPED AEATTDVHPE DIPYLKASD GLQPEVTRFL EQHSPDWIY 120
DYTHYWLPSI AASLGISRAH FSVTTPWAlA YMGPSADAMI NGSDGRTVE DLTTPPKWFP 180
FPTKVCWRKH DLARLVPIKA PGISDGYRMG LVLKGSDDLK SKCYHEFGTQ WLPLLETLHQ 240
VPVVPVGLLP PEIPGDEKDE TWSIHKWLD GKQKGSVVYV ALGSEVLVSQ TEVVELALGL 300
ELSGLPFVWA YRKPKGPAKS DSVLPDGFV ERTRDRGLVW TSWAPQLRIL SHESVCGFLT 360
HCGSGSIVEG LMFHGLPLIML PIFGDQLPNA RLLEDKQVGI EIPRNEEDGC LTKESVARSL 420
RSVVVEKEGE IYKANARELS KIYNDTKVEK EYVSQFVDYL EKNARAVAlD HES 473

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SEQ ID NO: 6          moltype = DNA  length = 1422
FEATURE              Location/Qualifiers
source               1..1422
                    mol_type = other DNA
                    organism = Stevia rebaudiana

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SEQUENCE: 6
atggctacat ctgattctat tgttgatgac aggaagcagt tgcattgtggc tactttccct 60
tggttgctct tcggtcatat actgccttac ctacaactat caaaactgat agctgaaaaa 120
ggacataaag tgtcattcct ttcaacaact agaaacattc aaagattatc ttccacata 180

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tcaccattga ttaacgtcgt tcaattgaca cttccaagag tacaggaatt accagaagat 240
gctgaagcta caacagatgt gcatcctgaa gataccctt acttgaaaaa ggcatccgat 300
ggattacagc ctgagggtcac tagattcctt gagcaacaca gtccagattg gatcatatac 360
gactacactc actattgggt gcttcaatt gcagcatcac taggcatttc tagggcacat 420
ttcagtgtaa ccacaccttg ggccattgct tacatgggtc catccgctga tgctatgatt 480
aacggcagtg atggtagaac taccgttgaa gatttgacaa cccacacaaa gtgggtttcca 540
ttccaacta aagtctgttg gagaaaacac gacttagcaa gactgggttc atacaaggca 600
ccaggaatct cagacggcta tagaatgggt ttagtcttta aagggtctga ctgcctattg 660
tctaagtgtt accatgagtt tgggacacaa tgggtaccac ttttggaaac attacaccaa 720
gttctgtcgc taccagttgg tctattacct ccagaaatcc ctggtgatga gaaggacgag 780
acttgggttt caatcaaaaa gtgggttagac gggaagcaaa aaggctcagt ggtatatgtg 840
gcactgggtt ccgaagtttt agtatctcaa acagaagttg tggaaactgc cttaggtttg 900
gaactatctg gattgccatt tgtctgggcc tacagaaaac caaaaggccc tgcaaagtcc 960
gattcagttg aattgccaga cggctttgtc gagagaaacta gagatagagg gttggtatgg 1020
acttcatggg ctccacaatt gagaatcctg agtcacgaat ctgtgtgcgg tttcctaaca 1080
cattgtggtt ctggttctat agtgaagga ctgatgtttg gtcatccact tatcatgttg 1140
ccaatctttg gtgaccgaca tttgaatgca cgtctgttag aagataaaca agttggaatt 1200
gaaatccccc gtaatgagga agatggatgt ttaaccaagg agtctgtggc cagatcatta 1260
cgttccgttg tcgttgaaaa ggaaggcgaa atctacaagg ccaatgcccg tgaactttca 1320
aagatctaca atgacacaaa agtagagaag gaatatgttt ctcaatttgt agattaccta 1380
gagaaaaacg ctagagccgt agctattgat catgaatcct aa 1422

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SEQ ID NO: 7          moltype = AA  length = 458
FEATURE              Location/Qualifiers
source               1..458
                    mol_type = protein
                    organism = Stevia rebaudiana

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SEQUENCE: 7
MENKTETTIVR RRRRIILFPV PFQGHINPIL QLANVLYSKG PSITIFHTNF NKPKTSNYPH 60
FTFRFILDNM PQDERISNLP THGPLAGMRI PIINEHGADE LRRELELMLL ASEDEEVSC 120
LITDALWYFA QSVADSLNLR RLVMLTSSLF NFHAHVSLPQ FDELGYLDPD DKTRLEEQAS 180
GFPMLKVVDI KSAYSNWQIL KEILGKMIQK TKASSGVIWN SFKELEESSEL ETVIREIPAP 240
SFLIPLPKHL TASSSSLLDH DRTVFQWLDQ QPPSSVLYVS FGSTSEVDEK DFLEIARGLV 300
DSKQSFLLWV RPGEVFGSTW VEPLPDGFLG ERGRIVKWVP QQEVLAHGAI GAFWTHSGWN 360
STLESVCEGV PMIFSDFGLD QPLNARYMSD VLKVGYYLEN GWERGEIANA IRRVMVDEEG 420
EYIRQNAVRL KQKADVSLMK GGSYSLESLS LVSYISLL 458

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SEQ ID NO: 8          moltype = DNA  length = 1377
FEATURE              Location/Qualifiers
source               1..1377
                    mol_type = other DNA
                    organism = Stevia rebaudiana

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SEQUENCE: 8
atggaaaaca agaccgaaac aacagttaga cgtaggcgta gaatcattct gtttccagta 60
ccttttcaag ggcacatcaa tccaatacta caactagcca acgttttgta ctctaagggt 120
ttttctatta caatctttca caccaatttc aacaacacaa aaacatccaa ttaccacat 180
ttcacattca gattcatact tgataatgat ccacaagatg aacgtatttc aaacttacct 240
acccacggtc ctttagctgg aatgagaatt ccaatcatca atgaacatgg tgcgatgag 300
cttagaagag aattagagtt acttatgttg gcatccgaag aggacgagga agtctcttgt 360
ctgattactg acgctctatg gtactttgcc caatctgttg ctgatagttt gaatttgagg 420
agattggtag taatgacatc cagtctgttt aactttcacg ctcatgttag ttaccacaa 480
tttgacgaat tgggataact ggaccctgat gacaagacta ggtagagga acaggcctct 540
ggttttccta tggtgaaagt caaagatata aagtctgcct attctaattg gcaaatcttg 600
aaagagatct taggaaagat gatcaaacag acaaaaggctt catctggagt gatttgggaa 660
agtttcaaa agttagaaga gtctgaattg gagactgtta tcagagaaat tccagcacct 720
tcattctcga taccattacc aaaacatttg actgcttcct ctctctcttt gttggatcat 780
gacagaacag tttttcaatg gttggaccaa caaccaccta gttctgtttt gtacgtgtca 840
tttggtagta cttctgaagt cgatgaaaag gacttccttg aaatcgcaag aggccttagt 900
gatagtaagc agtcattcct ttgggtcgtg cgtccagggt tcgtgaaagg ctcaacatgg 960
gtcgaaccac ttccagatgg ttttctaggg gaaagaggta gaatagtaa atgggttcct 1020
caacaggaag ttttagctga tggcgctatt ggggcattct ggactcattc cggatgggaa 1080
tcaactttag aatcagtatg cgaaggggta cctatgatct tttcagattt tgggtcttgat 1140
caaccactga acgcaagata catgtctgat gttttgaaag tgggtgtata tctagaaaat 1200
ggctgggaaa ggggtgaaat agctaagca ataagacgtg ttatgggtga tgaagagggg 1260
gagtatatca gacaaaacgc aagagtgtcg aagcaaaagg ccgacgtttc tctaatagag 1320
ggaggtctct catacgaatc cttagaatct cttgtttcct acatttcac 1377

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SEQ ID NO: 9          moltype = DNA  length = 1422
FEATURE              Location/Qualifiers
source               1..1422
                    mol_type = other DNA
                    organism = Stevia rebaudiana

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SEQUENCE: 9
atggctacca gtgactccat agttgacgac cgtaagcagc ttcatgttgc gacgttccca 60
tgggttgctt tgggtccatc cctcccttac cttcagcttt cgaaattgat agctgaaaag 120
ggtcacaaag tctcgtttct tttaccacc agaaacattc aacgtctctc ttctcatatc 180
tcgccactca taaatgttgt tcaactcaca cttccacgtg tccaagagct gccggaggat 240
gcagaggcga ccaactgacg ccaccctgaa gatattccat atctcaagaa ggcttctgat 300
ggtcttcaac cggaggtcac ccggtttcta gaacaacact ctccggactg gattatttat 360

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gattatactc actactgggt gccatccatc gcggctagcc tcggtatctc acgagccac 420
ttctccgtca ccactccatg ggcattgct tatatgggac cctcagctga cgccatgata 480
aatgggtcag atgggtcgaa cacgggtgag gatctcacga caccgcccac gtgggttccc 540
tttccgacca aagtatgctg gcggaagcat gatcttgccc gactgggtgcc ttacaaagct 600
ccgggggatat ctgatggata ccgtatgggg ctgggttctta agggatctga ttgtttgctt 660
tccaaatggt accatgagtt tggaaactca tggctacctc ttttgagac actacaccaa 720
gtaccggtgg ttccggtggg attactgcca ccggaatac ccggagacga gaaagatgaa 780
acatgggtgt caatcaagaa atggctcgat ggtaaacaaa aaggcagtggt ggtgtacgtt 840
gcattaggaa gcgaggtttt ggtgagccaa accgaggttg ttgagttagc attgggtctc 900
gagctttctg ggttgccatt tgtttgggct tatagaaaac caaaaggtcc cgcaagtca 960
gactcgggtg agttgccaga cgggttcgtg gaacgaactc gtgaccgtgg gttggtctgg 1020
acgagttggg cacctcagtt acgaatactg agccatgagt cggtttgggg tttcttgact 1080
cattgtggtt ctggatcaat tgtggaaggg ctaatgtttg gtcaccctct aatcatgcta 1140
ccgatttttg gggaccaacc tctgaatgct cgattactgg aggacaaaca ggtgggaatc 1200
gagataccaa gaaatgagga agatggttgc ttgaccaagg agtcggttgc tagatcactg 1260
aggtccgttg ttgtgaaaaa agaaggggag atctacaagg cgaacgcgag ggagctgagt 1320
aaaaatctata acgacactaa ggttgaaaaa gaatatgtaa gccaatcggt agactatttg 1380
gaaaagaatg cgcgtgcggt tgccatcgat catgagagtt aa 1422

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SEQ ID NO: 10      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Stevia rebaudiana

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SEQUENCE: 10
MATSDSIVDD RKQLHVATFP WLAFGHILPY LQLSKLIAEK GHKVSFLSTT RNIQRLSSHI 60
SPLINNVQLT LPRVQELPED AEATTDVHPE DIPYLKKASD GLQPEVTRFL EQHSPDWIYY 120
DYTHYWLPSI AASLGISRAH FSVTPWAIYA YMGPSADAMI NGSDGRTTVE DLTTPPKWFP 180
FPTKVCWRKH DLARLVPIKA PGISDGYRMG LVLKGSDDLK SKCYHEFGTQ WLPLETLHQ 240
VPVVPVGLLP PEVPGDEKDE TWVSIKKWLD GKQKGSVVYV ALGSEVLVSQ TEVVELALGL 300
ELSGLPFWVA YRKPKGPAKS DSVLPDGFV ERTRDRGLVW TSWAPQLRIL SHESVCGFLT 360
HCGSGSIVEG LMFHGLPLML PIFGDQPLNA RLLEDKQVGI EIPRNEEDGC LTKESVARSL 420
RSVVVEKEGE IYKANARELS KIYNDTKVEK EYVSQFVDYL EKNTRAVID HES 473

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SEQ ID NO: 11      moltype = DNA length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 11
atggctacca gtgactccat agttgacgac cgtaagcagc ttcatgttgc gacgttccca 60
tgggttgctt tcggtcacat cctcccttac cttcagcttt cgaaattgat agctgaaaag 120
gggtcacaaa gttcgtttct ttctaccacc agaaacattc aacgtctctc ttctcatatc 180
tcgccactca taaatgttgt tcaactcaca cttccacgtg tccaagagct gccggaggat 240
gcagaggcga ccactgacgt ccaccctgaa gatattccat atctcaagaa ggcttctgat 300
ggctcttcaac cggaggtcac cgggtttcta gaacaacact ctcgggactg gattatttat 360
gattatactc actactgggt gccatccatc gcggctagcc tcggtatctc acgagccac 420
ttctccgtca ccactccatg ggcattgct tatatgggac cctcagctga cgccatgata 480
aatgggtcag atgggtcgaa caccggtgag gatctcacga caccgcccac gtgggttccc 540
tttccgacca aagtatgctg gcggaagcat gatcttgccc gactgggtgcc ttacaaagct 600
ccgggggatat ctgatggata ccgtatgggg ctgggttctta agggatctga ttgtttgctt 660
tccaaatggt accatgagtt tggaaactca tggctacctc ttttgagac actacaccaa 720
gtaccggtgg ttccggtggg attactgcca ccggaagtac ccggagacga gaaagatgaa 780
acatgggtgt caatcaagaa atggctcgat ggtaaacaaa aaggcagtggt ggtgtacgtt 840
gcattaggaa gcgaggtttt ggtgagccaa accgaggttg ttgagttagc attgggtctc 900
gagctttctg ggttgccatt tgtttgggct tatagaaaac caaaaggtcc cgcaagtca 960
gactcgggtg agttgccaga cgggttcgtg gaacgaactc gtgaccgtgg gttggtctgg 1020
acgagttggg cacctcagtt acgaatactg agccatgagt cggtttgggg tttcttgact 1080
cattgtggtt ctggatcaat tgtggaaggg ctaatgtttg gtcaccctct aatcatgcta 1140
ccgatttttg gggaccaacc tctgaatgct cgattactgg aggacaaaca ggtgggaatc 1200
gagataccaa gaaatgagga agatggttgc ttgaccaagg agtcggttgc tagatcactg 1260
aggtccgttg ttgtgaaaaa agaaggggag atctacaagg cgaacgcgag ggagctgagt 1320
aaaaatctata acgacactaa ggttgaaaaa gaatatgtaa gccaatcggt agactatttg 1380
gaaaagaata cgcgtgcggt tgccatcgat catgagagtt aa 1422

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SEQ ID NO: 12      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Stevia rebaudiana

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SEQUENCE: 12
MATSDSIVDD RKQLHVATFP WLAFGHILPF LQLSKLIAEK GHKVSFLSTT RNIQRLSSHI 60
SPLINNVQLT LPRVQELPED AEATTDVHPE DIQYLKKAVD GLQPEVTRFL EQHSPDWIYY 120
DFTHYWLPSI AASLGISRAH FSVITPWTIA YLAPSSDAMI NDSGRTTVE DLTTPPKWFP 180
FPTKVCWRKH DLARMEPYEA PGISDGYRMG MVFKGSDCLL FKCYHEFGTQ WLPLETLHQ 240
VPVVPVGLLP PEIPGDEKDE TWVSIKKWLD GKQKGSVVYV ALGSEALVSQ TEVVELALGL 300
ELSGLPFWVA YRKPKGPAKS DSVLPDGFV ERTRDRGLVW TSWAPQLRIL SHESVCGFLT 360
HCGSGSIVEG LMFHGLPLML PLFGDQPLNA RLLEDKQVGI EIPRNEEDGC LTKESVARSL 420
RSVVVNEGE IYKANARELS KIYNDTKVEK EYVSQFVDYL EKNARAVID HES 473

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SEQ ID NO: 13      moltype = DNA length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = other DNA
                  organism = Stevia rebaudiana

SEQUENCE: 13
atggctacca gtgactccat agttagcgac cgttaagcagc ttcattgtgc gacgttccca 60
tggtttgctt tcgggtccat cctccctttc cttcagcttt cgaaattgat agctgaaaag 120
ggtcacaaaag tctcgtttct tcttaccacc agaaacattc aacgtctctc ttctcatatc 180
tcgccactca taaatgttgt tcaactcaca cttccacgtg tccaagagct gccggaggat 240
gcagaggcga ccactgacgt ccaccctgaa gatattcaat atctcaagaa ggctgttgat 300
ggctttcaac cggagggtcac ccggtttcta gaacaacact ctccggactg gattatttat 360
gattttactc actactgggtt gccatccatc gcggctagcc tcggtatctc acgagcctac 420
ttctgcgtca tcactccatg gaccattgct tatttggcac cctcatctga cgccatgata 480
aatgattcag atggttgaac cacggttgag gatctcacga caccgcccac gtggtttccc 540
tttcgacca aagtatgtgt gcggaagcat gatcttgccc gaatggagcc ttacgaagct 600
ccagggatat ctgatggata ccgtatgggg atggttttta agggatctga ttggttgctt 660
ttcaaatggt accatgagtt tggaaactca tggctacctc ttttgagac actacaccaa 720
gtaccgggtg ttccgggtgg attactgccg ccggaataac ccggagacga gaaagatgaa 780
acatgggtgt caatcaagaa atggctcgat ggtaaacaaa aaggcagtggt ggtgtacgtt 840
gcattaggaa gcgagggtct ggtgagccaa accgaggttg ttgagttagc attgggtctc 900
gagctttctg ggttgccatt tgtttgggct tatagaaaac caaaaggctc ccggaagtca 960
gactcgggtg agttgccaga cgggttcgtg gaacgaactc gtgaccgttg gttggtctgg 1020
acgagttggg cactcagttt acgaatactg agccatgagt cggtttgttg tttcttgact 1080
cattgtggtt ctggtatcaat tctggaaggg ctaattgttg gtcacctctc aatcatgcta 1140
ccgctttttg gggaccaacc tctgaatgct cgattactgg aggcacaaac ggtgggaatc 1200
gagataccaa gaaatgagga agatggttgc ttgaccaagg agtcggttgc tagatcactg 1260
aggtccgttg ttgtggaaaa cgaaggggag atctacaagg cgaacgcgag ggagctgagt 1320
aaaatctata acgacactaa ggtggaaaaa gaatatgtaa gccaatctgt agactatttg 1380
gaaaagaatg cgcgtgcggt tgccatcgat catgagagtt aa 1422

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SEQ ID NO: 14      moltype = AA length = 485
FEATURE           Location/Qualifiers
source            1..485
                  mol_type = protein
                  organism = Stevia rebaudiana

SEQUENCE: 14
MYNVTYHQS KAMATSDSIV DDRKQLHVAT FPWLAFGHIL PFLQLSKLIA EKGHKVSFLS 60
TTRNIQRLSS HISPLINVVQ LTLPRVQELP EDAAETTDVH PEDIQYLKKA VDGLQPEVTR 120
FLEQHSPDWI IYDFTHYWLP SIAASLGISR AYFCVITPWT IAYLAPSSDA MINDSDGRRT 180
VEDLTTPPKW PFPPTKVCWR KHDLMARPEY EAPGISDGYR MGMVFKGSDC LLFKCYHEFG 240
TQWLPLLETL HQVPVVPVGL LPPEIPGDEK DETWVSIKWK LDGKQKGSVV YVALGSEALV 300
SQTEVVLELAL GLELSGLPFV WAYRKPKGPA KSDSVLPDGL FVERTDRDGL VWTSWAPQLR 360
ILSHESVCGF LTHCGSSGIV EGLMFGHPLI MLPIFCDQPL NARLLEDKQV GIEIPRNEED 420
GCLTKESVAR SLRSVVVENE GEIYKANARA LSKIYNDTKV EKEYVSQFVD YLEKNARAVA 480
IDHES 485

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SEQ ID NO: 15      moltype = DNA length = 1458
FEATURE           Location/Qualifiers
source            1..1458
                  mol_type = other DNA
                  organism = Stevia rebaudiana

SEQUENCE: 15
atgtacaacg ttacttatca tcaaaattca aaagcaatgg ctaccagtga ctccatagtt 60
gacgaccgta agcagcttca tgttgcgacg ttcccatggc ttgctttcgg tcacatcctc 120
cctttccttc agctttcgaa attgatagct gaaaagggtc acaaagctct gtttctttct 180
accaccagaa acattcaagc tctctcttct catatctcgc cactcataaa tgttgttcaa 240
ctcacacttc cacgtgtcca agagctgccg gaggatgcag aggcgaccac tgacgtccac 300
cctgaagata ttcaatatct caagaaggct gttgatggct ttcaaccgga ggtcaccctg 360
tttctagaac aacactcttc ggactggatt atttatgatt ttactcata ctgggttgcca 420
tccatcgccg ctagctctcg tatctcacga gctacttct gcgtcatcac tccatggacc 480
attgcttatt tggcaccctc atctgacgcc atgataaatg attcagatgg tcgaaccacg 540
gttgaggatc tcacgacacc gcccaagtgg ttccctttc cgaccaaaag atgctggcgg 600
aagcatgata ttgcccgaat ggagccttac gaagctccgg ggatatctga tggataccgt 660
atggggatgg tttttaaggg atctgattgt ttgcttttca aatgttacca tgagtttgga 720
actcaatggc tactcttttt ggagacacta caccaagtac cgggtggttc ggtgggatta 780
ctgcgcgcgg aaataccctg agacgagaaa gatgaaacat ggggtgtcaat caagaatagg 840
ctcgatggta aacaaaaagg cagtgtggtg tacgttgcat taggaagcga ggcctttggtg 900
agccaaaccg aggttgttga gttagcattg ggtctcgagc tttctgggtt gccatttgtt 960
tggtcttata gaaaaccaaa aggtcccgcg aagtcagact cgggtggagt gccagacggg 1020
ttcgtggaac gaactcgtga ccgtgggttg gttcggacga gttgggcacc tcagttacga 1080
atactgagcc acgagtcagt ttgtggtttc ttgactcatt gtggttctgg atcaatttgt 1140
gaagggtcaa tgtttggtca cctctaate atgctaccga ttttttgtga ccaacctctg 1200
aatgctcgat tactggagga caaacagggt ggaatcgaga taccagaaa tgaggaagat 1260
ggttgcttga ccaaggagtc ggttgctaga tcaactgaggt ccgttgttgt gaaaaacgaa 1320
ggggagatct acaaggcgaa cgcgagggcg ctgagtaaaa tctataacga cactaagggt 1380
gaaaaagaat atgtaagcca attcgtagac tatttggaaa agaatgcgcg tcggttggtc 1440
atcgatcatg agagttaa 1458

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SEQ ID NO: 16          moltype = AA  length = 473
FEATURE               Location/Qualifiers
source                1..473
                     mol_type = protein
                     organism = Stevia rebaudiana

SEQUENCE: 16
MATSDSIVDD RKQLHVATFP WLAFGHILPF LQLSKLIAEK GHKVSFLSTT RNIQRLSSHI 60
SPLINVVQLT LPRVQELPED AEATTDVHPE DIQYLKKAVD GLQPEVTRFL EQHSPDWIYY 120
DFTHYWLPSI AASLGISRAY FCVITPWTIA YLAPSSDAMI NDSGRTTVE DLTPPKWFP 180
FPTKVCWRKH DLARMEPYEA PGISDGYRMG MVFKGSDCLL FKCYHEFGTQ WLPLLETLHQ 240
VPVVPVGLLP PEIPGDEKDE TWVSIKKWLD GKQKGSVVYV ALGSEALVSQ TEVVELALGL 300
ELSGLPFWWA YRKPKGPAKS DSVELPDGFV ERTRDRGLVW TSWAPQLRIL SHESVCGFLT 360
HCGSGSIVEG LMFHGHLIML PIFCDQPLNA RLLEDKQVGI EIPRNEEDGC LTKESVARSL 420
RSVVVENEGE IYKANARALS KIYNDTKVEK EYVSQFVDYL EKNARAVAIH HES 473

SEQ ID NO: 17          moltype = DNA  length = 1422
FEATURE               Location/Qualifiers
source                1..1422
                     mol_type = other DNA
                     organism = Stevia rebaudiana

SEQUENCE: 17
atggctacca gtgactccat agttgacgac cgtaagcagc ttcattgtgc gacgttccca 60
tggcttgctt tcggtcacat cctccctttc cttcagcttt cgaaattgat agctgaaaag 120
ggtcacaaaag tctcgtttct ttctaccacc agaaacattc aacgtctctc ttctcatatc 180
tcgccactca taaatgttgt tcaactcaca cttccacgtg tccaagagct gccggaggat 240
gcagaggcga ccactgacgt ccaccctgaa gatattcaat atctcaagaa ggctgttgat 300
ggctcttaac cggagggtcac ccggtttcta gaacaacact ctccggactg gattatttat 360
gattttactc actactgggt gccatccatc gcggctagcc tcggtatctc acgagcctac 420
ttctgcgtca tcactccatg gaccattgct tatttggcac cctcatctga cgccatgata 480
aatgattcag atggtcgaac cacggttgag gatctcacga caccgcccac gtggtttccc 540
tttcgcacca aagtatgctg gcggaagcat gatcttgccc gaattggagc ttacgaagct 600
ccgggggatc ctgatggata ccgtatgggg atggttttta agggatctga ttgtttgctt 660
ttcaaatggt accatgagtt tggaaactca tggctacctc ttttgagac actacaccaa 720
gtaccgggtg ttccgggtgg attaactgcc cggaataac ccggagacga gaaagatgaa 780
acatgggtgt caatcaagaa atggctcgat ggtaaacaaa aaggcagtggt ggtgtacgtt 840
gcattaggaa gcgaggcttt ggtgagccaa accgaggttg ttgagtagc attgggtctc 900
gagctttctg ggttgccatt tgtttgggct tatagaaaac caaaaggctc ccggaagtca 960
gactcggttg agttgccaga cgggttcgtg gaacgaactc gtgaccgtgg gttggtctgg 1020
acgagttggg cacctcagtt cgaactactg agccacgagt cagtttgttg tttcttgact 1080
cattgtggtt ctggatcaat tgtggaaggg ctaatgtttg gtcacctctc aatcatgcta 1140
ccgatttttt gtgaccaacc tctgaatgct cgattactgg aggcacaaac ggtgggaatc 1200
gagataccaa gaaatgagga agatggttgc ttgaccaagg agtcggttgc tagatcactg 1260
agggtccgtg ttgtggaaaa cgaaggggag atctacaagg cgaacgcgag ggcgctgagt 1320
aaaaatctata acgacactaa ggtggaaaaa gaatatgtaa gccaatcgat agactatttg 1380
gaaaagaatg cgcgtgcggt tgccatcgat catgagagtt aa 1422

SEQ ID NO: 18          moltype = DNA  length = 1086
FEATURE               Location/Qualifiers
misc_feature          1..1086
                     note = Description of Artificial Sequence: Synthetic
                           polynucleotide
source                1..1086
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 18
atggcttttg taaacccaac cgctcttttc tatggtacct ctatcagaac aagacctaca 60
aaacttactaa atccaactca aaagctaaga ccagtttcat catcttcctt accttctttc 120
tcatcagtta gtgcgattct tactgaaaaa catcaatcta atccttctga gaacaacaat 180
ttgcaaaactc atctagaaac tcctttcaac tttgatagtt atatgttgga aaaagtcaac 240
atggttaacg aggcgcttga tgcattctgc ccactaaaag acccaatcaa aatccatgaa 300
tccatgagat actctttatt ggcaggcggg aagagaatca gaccaatgat gtgtattgca 360
gcctgcgaaa tagtcggagg taatatcctt aacgccatgc cagccgcatg tgcggtggaa 420
atgattcata ctatgtcttt ggtgcatgac gatcttccat gtatggataa tgatgacttc 480
agaagaggta aacctatttc acacaaggtc tacggggagg aaatggcagt attgaccggc 540
gatgctttac taagtttatc ttccgaacat atagctactg ctacaaaggg tgtatcaaag 600
gatagaatcg tcagagctat aggggagttg gcccgttcag ttggctccga aggttttagt 660
gctggacaag ttgtagatat cttgtcagag ggtgctgatg ttggattaga tcacctagaa 720
tacattcaca tccacaaaac agcaatggtt cttgagtcct cagtagttat tggcgctatc 780
atgggaggag gatctgatca gcagatcgaa aagttgagaa aattcgctag atctatttgt 840
ctactattcc aagtttgga tgacattttg gatgttacaa aatctaccga agagttgggg 900
aaaacagctg gtaaggattt gttgacagat aagacaactt acccaaagtt gttaggtata 960
gaaaagtcca gagaatttgc cgaaaaactt aacaagggaag cacaagagca attaatgtgc 1020
tttgatagac gtaaggcagc tcctttgatc gcgttagcca actacaatgc gtaccgtcaa 1080
aattga 1086

SEQ ID NO: 19          moltype = DNA  length = 1029
FEATURE               Location/Qualifiers
misc_feature          1..1029

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note = Description of Artificial Sequence: Synthetic
      polynucleotide
source 1..1029
       mol_type = other DNA
       organism = synthetic construct

SEQUENCE: 19
atggctgagc aacaaatata taacttgctg tctatgtttg atgcttcaca tgctagtcag 60
aaattagaaa ttactgtcca aatgatggac acataccatt acagagaaac gcttcagat 120
tcctcatctt ctgaaggcgg ttcatgtgtc agatacgacg agagaagagt ctctttgcct 180
ctcagtcata atgctgcctc tccagatatt gtatcacaaac tatgtttttc cactgcaatg 240
tcttcagagt tgaatcacag atggaaatct caaagattaa aggtggcoga ttctccttac 300
aactatatcc taacattacc atcaaaagga attagagggtg cctttatcga ttccctgaac 360
gtatggttgg aggttccaga ggaatgaaca tcagtcatac aggaagtatt tggatgtctc 420
cacaaactctt cattaatcat tgatgacttc caagataatt ctccacttag aagaggaaag 480
ccatctaccc atacagtctt cggccctgcc caggctatca atactgtac ttacgttata 540
gttaaagcaa tcgaaaagat acaagacata gtgggacacg atgcattggc agatgttacg 600
ggtagctatta caactatttt ccaaggtcag gccatggact tgggtgggac agcaaatgca 660
atcggtccat caatacagga atacttactt atggtaaacg ataaaccggg tgctctcttt 720
agactgagtt tggagttgtt agctctgaat tccgaagcca gtatttctga ctctgcttta 780
gaaagtttat tagtgctgtt ttctctgcta ggtcaatact tccaaatcag agacgactat 840
atgaacttga tcgataacaa gtatacagat cagaagggtc tctgcgaaga tcttgatgaa 900
ggcaagtact cactaacact tattcatgcc ctccaaactg attcatccga tctactgacc 960
aacatccttt caatgagaag agtgcaagga aagttaacgg cacaaaagag atgttggttc 1020
tggaatatga                                     1029

SEQ ID NO: 20      moltype = DNA length = 903
FEATURE            Location/Qualifiers
misc_feature       1..903
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..903
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 20
atggaaaaga ctaaggagaa agcagaacgt atcttgctgg agccatacag atacttatta 60
caactaccag gaaagcaagt ccgttctaaa ctatcacaaag cgttcaatca ctgggtaaaa 120
gttcctgaag ataagttaca aatcattatt gaagtcacag aaatgctaca caatgcttct 180
ttactgatcg atgatataga ggattcttcc aaactgagaa gaggtttttc tgcgctcat 240
tccatatacg gggtaaccaag tgtaatcaac tcagctaatt acgtctactt ctggggattg 300
gaaaaagtat tgacattaga tcatccagac gctgtaaagc tattaccag acaacttctt 360
gaattgcatc aaggtcaagg tttggatata tattggagag acacttatac ttgccaaca 420
gaagaggagt acaagcaat ggttctacaa aagactggcg gtttggtcgg acttgccgtt 480
ggcttgatgc aacttttctc tgattacaag gaggacttaa agcctctgtt ggataccttg 540
ggcttggttt tccagattag agatgactac gctaaactac attcaagga atattcagaa 600
aacaatcat tctgtgaaga tttgactgaa gggaaagtta gttttccaac aatccacgcc 660
atttgggtcaa gaccagaatc tactcaagtg caaaacattc tgcgtcagag aacagagaat 720
attgacatca aaaagtattg tgttcagtag ttggaagatg ttggttcttt tgcctacaca 780
agacatacac ttagagaatt agaggcaaaa gcatacaagc aaatagaagc ctgtggaggc 840
aatccttctc tagtggcatt ggttaaacat ttgtccaaaa tgttcaccga ggaaaacaag 900
taa                                     903

SEQ ID NO: 21      moltype = DNA length = 1020
FEATURE            Location/Qualifiers
misc_feature       1..1020
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..1020
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 21
atggcaagat tctattttct taacgcacta ttgatggtta tctcattaca atcaactaca 60
gccttcactc cagctaaact tgcctatcca acaacaacaa cagctctaaa tgcgcctcc 120
gccgaaactt ctttcagtct agatgaatac ttggcctcta agataggacc tatagagtct 180
gccttggaag catcagtcac atccagaatt ccacagaccg ataagatctg cgaatctatg 240
gcctactctt tgatggcagg aggcaagaga attagaccag tgttgtgtat cgctgcatgt 300
gagatgttgc gtggatccca agatgtcgct atgcctactg ctgtggcatt agaaatgata 360
cacacaatgt ctttgattca tgatgatttg ccatccatgg ataacgatga cttgagaaga 420
ggtaaaccaa caaacatgt cgttttcggc gaagatgtag ctattcttgc aggtgactct 480
ttattgtcaa cttccttcga gcacgtcgct agagaaacaa aaggagtgtc agcagaaaag 540
atcggtgatg ttatcgctag attaggcaaa tctgttggtg ccgagggcct tgctggcggg 600
caagttatgg acttagaatg tgaagctaaa ccaggtacca cattagacga cttgaaatgg 660
attcatatcc ataaaaccgc tacattgtta caagttgctg tagcttctgg tgcagttcta 720
ggtgtgtcaa ctctgaaga ggttgctgca tgcaggtgtt ttgctatgaa tataggtctt 780
gcctttcaag ttgccgacga tatccttgat gtaaccgctt catcagaaga tttgggtaaa 840
actgcaggca aagatgaagc tactgataag acaacttacc caaagttatt aggattagaa 900
gagagtaagg catacgaag acaactaatc gatgaagcca aggaaagtgt ggctcctttt 960
ggagatagag ctgccccttt attggccatt gcagatttca ttattgatag aaagaattga 1020

SEQ ID NO: 22      moltype = DNA length = 1068

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FEATURE          Location/Qualifiers
misc_feature      1..1068
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1068
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 22
atgcacttag caccacgtag agtccctaga ggtagaagat caccacctga cagagttcct 60
gaaagacaag gtgccttggg tagaagacgt ggagctggct ctactggctg tgcccgtgct 120
gctgctggtg ttcaocgtag aagaggagga ggcgaggctg atccatcagc tgctgtgcat 180
agaggctggc aagcgggtgg tggcaccggt ttgcctgatg aggtgggtgc taccgcagcc 240
gccttagaaa tgtttcatgc ttttgcctta atccatgatg atatcatgga tgatagtga 300
actagaagag gctccccaac tgttcacaga gccctagctg atcgtttagg cgtgctctg 360
gaccagatc aggcgggtca actaggagtt tctactgcta tcttggttgg agatctggct 420
ttgacatggt ccgatgaatt gttatacgtc ccattgactc cacatagact ggcagcagta 480
ctaccattgg taacagctat gagagctgaa accgttcctg gccaatatct tgatataact 540
agtgtctaga gacctgggac cgatacttct cttgcattga gaatagccag atataagaca 600
gcagcttaca caatggaacg tccactgcac attggtgcag ccctggctgg ggcaagacca 660
gaactattag cagggtcttc agcatacggc ttgccagctg gagaagcctt ccaattggca 720
gatgacctgc taggcgtctt cgggtgatcca agacgtacag ggaaacctga cctagatgat 780
cttagagggtg gaaagcctac attattgggt acacacaggtc ttgatagaca aggcgcttca 900
cagagacaca cactggatgc attattgggt acacacaggtc ttgatagaca aggcgcttca 900
agactaagat gcgtattggt agcaactggt gcaagagccg aagccgaaag acttattaca 960
gagagaagag atcaagcatt aactgcattg aacgcattaa cactgccacc tcctttagct 1020
gaggcattag caagattgac attagggtct acagctcacc ctgcttaa 1068

SEQ ID NO: 23      moltype = DNA length = 993
FEATURE            Location/Qualifiers
misc_feature        1..993
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..993
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 23
atgtcatatt tcgataacta cttcaatgag atagttaatt ccgtgaacga catcattaag 60
tcttacatct ctggcgacgt accaaaacta tacgaagcct cctaccattt gtttacatca 120
ggaggaaaga gactaagacc attgatcctt acaatttctt ctgatctttt cggtggacag 180
agagaaagag cactactatgc tggcgacgca atcgaagttt tgcacacatt cactttggtt 240
cacgatgata tcattggatca agataacatt cgtagaggctc ttctactgtg acatgtcaag 300
tatggcctac ctttggccat tttagctggt gacttattgc atgcaaaagc ctttcaattg 360
ttgactcagg cacttagagg tctaccatct gaaactatca tcaaggcgtt tgatatcttt 420
acaagatcta tcattatcat atcagaaggt caagctgtcg atatggaatt cgaagataga 480
attgatatac aggaacaaga gtatttggat atgatattct gtaaaaccgc tgccttattc 540
tcagcttctt cttccattgg ggcgttgata gctggagcta atgataacga tgtgagatta 600
atgtccgatt tcggtacaaa tcttggggtc gcatttcaaa ttgtagatga tatacttgg 660
ttaacagctg atgaaaaaga gctaggaaaa cctgttttca gtgatattcag agaaggtaaa 720
aagaccatag tagtcattaa gacttttaga ttgtgtaagg aagacgagaa aaagattgtg 780
ttaaaagcgc taggcaacaa gtcagcatca aaggaagagt tgatgagttc tgcgtgacat 840
atcaaaaagt actcattgga ttacgcctac aacttagctg agaaatacta caaaaacgcc 900
atcgattctc taaatcaagt ttcaagttaa agtgatattc cagggaagggc attgaaatat 960
cttctgtaat tcaccatcag aagacgtaag taa 993

SEQ ID NO: 24      moltype = DNA length = 894
FEATURE            Location/Qualifiers
misc_feature        1..894
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..894
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 24
atggctgcac aaactttcaa cctggatacc tacttatccc aaagacaaca acaagttgaa 60
gaggccctaa gtgctgctct tgtgccagct tatcctgaga gaataacga agctatgaga 120
tactccctcc tggcaggtgg caaaagatta agacctatct tatgtttagc tgcttgcaaa 180
ttggcagggtg gttctgttga acaagccatg ccaactgcgt gtgcacttga aatgatccat 240
acaatgtcac taattcatga tgacctggca gccatggata acgatgattt cagaagagga 300
aagccaacta atcacaaggt gttcggggaa gatatagcca tcttagcggg tgatgcgctt 360
ttagcttacg cttttgaaca tattgttctt caaacaagag gagtaccacc tcaattgggt 420
ctacaagtta ttgctagaat cggacacgcc gttgctgcaa caggcctcgt tggaggccaa 480
gtcgtagacc ttgaatctga aggtaaagct atttccttag aaacattgga gtatattcac 540
tcacataaga ctggagcctt gctggaagca tcagttgtct caggcggtat tctcgcagg 600
gcagatgaag agcttttggc cagattgtct cattacgcta gagatatagg cttggctttt 660
caaatcgtcg atgatattct ggaatttact gctacatctg aacagttggg gaaaaaccgt 720
ggtaaaagcc aggcagccgc aaaggcaact tatccaagtc tattgggttt agaagcctct 780
agacagaagc cggaagaggt gattcaatct gctaaggaag ccttaagacc ttacggttca 840
caagcagagc cactcctagc gctggcagac ttcacacac gtcgtcagca ttaa 894

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SEQ ID NO: 25      moltype = DNA length = 1116
FEATURE            Location/Qualifiers
misc_feature       1..1116
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..1116
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 25
atggcctctg ttactttggg ttcctggatc gtcgtccacc accataacca tcaccatcca 60
tcactctatc taactaaatc tcgttcaaga tccgtgccta ttacactaac caaaccaatc 120
tcttttcgtt caaagagaac agtttctctc agtagttcta tcgtgtcctc tagtgtcgtc 180
actaaggaag acaatctgag acagctcgaa ccttcttctc ttgatttcat gtcataatc 240
attactaagg cagaactagt gaataaggct cttgattcag cagttccatt aagagagcca 300
ttgaaaatcc atgaagcaat gagatactct cttctagctg gcgggaagag agtcagacct 360
gtactctgca tagcagcgtg cgaattagtt ggtggcgagg aatcaaccgc tatgctgcc 420
gcttgtgctg tagaaatgat tcatacaatg tcactgatac acgatgattt gccatgtatg 480
gataacgatg atctgagaag gggtaagcca actaaccata aggttttcgg cgaagatggt 540
gccgtcttag ctggtgatgc tttgttatct ttcgcgttcg aacatttggc atccgcaaca 600
tcaagtgtag ttgtgtcacc agtaagagta gttagagcag ttggagaact ggctaaagct 660
attggaactg aggggttagt tgcagggtcaa gtcgtcgata tctcttcgga aggtcttgat 720
ttgaatgatg taggtcttag acatctcgaa ttcattccac ttcacaagac agctgcactt 780
ttagaagcca gtgcgggttc cggcgcaatt gttggcggag ggagtgtgga cgaaattgag 840
agattgagga agtttgctag atgtatagga ttactgttcc aagtagtaga cgataacta 900
gatgtgacaa agtcttccaa agagttggga aaaacagctg gtaaagattt gattgcccag 960
aaattgacct accctaagat tatgggggcta gaaaaatcaa gagaatttgc cgagaaactc 1020
aatagagagg cgcgtgatca actgttgggt ttcgattctg ataaagtgc accactctta 1080
gccttagcca actacatcgc ttacagacaa aactaa 1116

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SEQ ID NO: 26      moltype = DNA length = 1086
FEATURE            Location/Qualifiers
source             1..1086
                   mol_type = other DNA
                   organism = Stevia rebaudiana

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SEQUENCE: 26
atggctcttg taaatcccac agctttgttc tatggaacct ccataagaac cagaccacca 60
aacttgctca acccgaccga aaaacttcga cccgtttcct cgtcttcttt gccttccttc 120
tcttcagttt ctgcaatctt gacggaaaaa caccaatcaa acccatcaga aaacaataac 180
ttgcaaaccc atctcgaaac accattcaat ttcgactctt acatgctgga gaaagtaaac 240
atggtgaatg aagctctgga cgctcgggtt ccaactcaaa acccgataaa gatccatgaa 300
tccatgcggt actccctctc agctggcggg aaacgcaccc gaccgatgat gtgcacgcc 360
gcttgcgaaa tagtcggagg caacatatta aacgccatgc cagctgcagc cgcggtcgag 420
atgattcaca ccatgtcaact agttcatgac gaccttccat gcatggataa cgacgacttc 480
cgacgtgga aaccaataag ccacaagggt tacggtgaag aaatggcggt tctaaccggg 540
gacgctgtac tctcattatc ctctgaacat atcgcgaccg cgacaaaagg cgtatccaaa 600
gacaggatcg tccgagccat tggtagaact gcaagggtccg ttggtcggga gggtttgggtc 660
gccggtcagg tgggtgatat tttatccgaa gggggtgatg ttgggttaga ccacttgagg 720
tatattcata tacacaagag tccaatgttg cttgagagct cggctcgtgat cggcgcgagc 780
atgggcgggt ggtctgacca acagatcgaa aagttgcgaa agtttgcgag atcgattggt 840
ttgttggttc aggttggtaga tgatattctt gatgtcacia agtcgactga ggaattgggg 900
aaaacggcgg gaaaagattt gctgacggac aagacaacgt atccgaagtt gttgggggac 960
gaaaaatcga gagaatttgc ggagaaatta aacaagggaag cgcaagaaca attgtcgggg 1020
tttgatcgcc gcaaggcggc tccgttaatt gcccttgcta attacaatgc ttataggcaa 1080
aactga 1086

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SEQ ID NO: 27      moltype = DNA length = 1029
FEATURE            Location/Qualifiers
source             1..1029
                   mol_type = other DNA
                   organism = Gibberella fujikuroi

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SEQUENCE: 27
atggctgaac aacagatctc caaccttctt tcaatgtttg atgtttctca cgcaagccag 60
aagttggaga ttacgggttc gatgatggat acctaccatt acagagaaac tctccagac 120
tcttctctc cagaaggcgg ttcttatctc cgtatgatg agcgacgggt ctcccttcgg 180
ctctctcaca atgcagcctc cccagacata gtctcccagt tatgcttctc aacagctatg 240
agctcggagc tcaatcacag gtggaagtca cagcgcctca aggttgctga ctctccctac 300
aactacatcc tgactcttcc atctaaagggt attcgtgggg ctttcattga ctactgaat 360
gtctgggtcg aggtcccgga agacgagacc tcgggtgatca aagaggtgat tggcatgctc 420
cacaactcgt ctctcataat cgatgacttc caagacaact ccccaactcg gcggggcaag 480
ccatctacac atactgtctt cgttcagca caagcaatca acacagcaac atatgtcatc 540
gtcaaggcca tcgagaaaat acaggatatt gtcggtcacg atgcattggc agatgtaact 600
ggcactataa ccacaatctt ccagggtcag gcaatggatc tgtgggtggac tgctaagtc 660
attgttcctg ctatccaaga atatctcctg atggccaatg acaagactgg tgcctctgtc 720
aggttatcgc ttgaactact ggcgctgaac tctgaagcat ccatcagtga cagcgcgctt 780
gaatctctca gcagcgtgt ctcaactgctc gggcagattt tccagataag agatgattac 840
atgaatctca ttgacaacaa gtatactgat cagaaaggat tttgcgagga tctggacgag 900
gggaaatact cgttgactct aatccatgct ctgcagaccg actccagcga cttctcacc 960
aacatcttat cgatgagaag agtccaagga aaacttacgg cgcaagaaag atgctgggtt 1020
tggaagtga 1029

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SEQ ID NO: 28          moltype = DNA length = 903
FEATURE               Location/Qualifiers
source                1..903
                     mol_type = other DNA
                     organism = Mus musculus

SEQUENCE: 28
atggagaaaa ctaagagaa agctgagagg attcttctag agccctatag gtacttactt 60
cagttaccag gtaaacagg gagaagcaaa ctttcacagg catttaatca ctggctgaaa 120
gttccagaag acaagctaca gattatcatt gaagtgactg aaatgttgca taatgccagt 180
ttactcattg atgatattga agacagttca aagctccgac gtggtttccc agtggctcac 240
agcatctatg gtgtcccatc tgtcattaat tctgccaat cgtctactt ccttggactg 300
gaaaaagtct taacccttga tcacccggtt gcggtgaagc ttttcacacg ccagcttctg 360
gaacttcacg agggacaagg cctcgatatt tactggaggg acacctacac ttgtccaact 420
gaagaagaat ataaagccat ggtgttgtag aagacaggtg gtttgtttgg attagcagta 480
ggtcttatgc agctgttctc tgattacaaa gaagatctaa agccactgct tgacacactt 540
gggtcttttt tccagattag agatgattat gccaatctac actccaaaga atacagttaa 600
aacaagaagt tctgtgaaga cttgcacagaa gggaagttct cattcccac tatccatgcc 660
atttgggtcaa ggcagagaa caccaggtta cagaacatcc tgcgccagag aacagagaat 720
atagatatta aaaagtattg tgtgcagtag ctggaggatg taggttcttt tgcatacact 780
cgacacactc tttagagagct tgaagctaaa gcctacaac aaattgaggg ctgtggtggg 840
aacccttcac tagtggcttt agtcaagcac ttaagtaaga tgttcacaga agaaaataaa 900
taa                                                903

SEQ ID NO: 29          moltype = DNA length = 1020
FEATURE               Location/Qualifiers
source                1..1020
                     mol_type = other DNA
                     organism = Thalassiosira pseudonana

SEQUENCE: 29
atggctcgtt tctacttctt gaacgctctc ctcatgggtg tttctttaca aagcaccacg 60
gcattcaccc cggcaaaact cgctacacca acaaccacca ctgcattaaa cgttgccctt 120
gccgaaacat catttagcct cgatgaatac ctagcctcca aaatcggacc cattgaatca 180
gctctcgagg catctgtcaa atctcgcat cctcaaaactg acaagatag cgagtctatg 240
gcatactcac tcattggctg aggaaagcgt atccgtcccg ttttgtgcat tgcgtcttgt 300
gaaatgtttg ggggaagtca agatgtggct atgcccagcg ctgtggcttt ggagatgatt 360
catactatga gtcttattca tgacgatttg cttcaatgg acaacgatga tctccgacga 420
ggaaagccaa ctaatcatgt tgtctttgga gaggatgttg ctattcttgc tggggattct 480
cttctcagta cgtcttttga acatgttgcc cgtgaaacca aaggagtgtc agctgaaaag 540
attgtagatg ttatcgctcg cctcgggaag tctgtgggtg cagagggtct tgcgtgtgga 600
caggttatgg atcttgatg tgaggcgaa cagggaacta ccctcgacga tctcaagtgg 660
attcacatcc acaaaactgc cactcttctt caagtggcag tggcatcagg tgcgttctt 720
ggagggggcca caccagagga ggttgctgct tgtgaactgt tcgcaatgaa tattggactt 780
gcctccagg tcgctgatga tattttggac gtgacggcat cgagtggaga tcttggcaaa 840
actgtcgtaa aggatgaagc cacagataag acaacttacc ctaagctttt gggattggag 900
gagagtaagg catcagctcg acaactcata gacgaagcaa aggaatcttt ggtcctttc 960
ggtgatcgta ctgctccatt gttggcaatt gccgacttta tcattgatcg aaagaactag 1020

SEQ ID NO: 30          moltype = DNA length = 1068
FEATURE               Location/Qualifiers
source                1..1068
                     mol_type = other DNA
                     organism = Streptomyces clavuligerus

SEQUENCE: 30
atgcacctgg ctccccgcg agtaccgcgc ggccgctgaa gccacactga ccgcgttctt 60
gaacgccaa gaggcgtcgg ccgcccgcgg ggggcccgtt ccacaggatg tgcccgcgct 120
gctgcgggag ttcacggcg ccgggggggg ggggaagcgg atccgtccgc tgcgtgcat 180
cgcgctggc aggcggcgcg ggaaacagga ctgccggacg aggtggtgtc cacagcggcg 240
gcgctggaga tgttccacgc gttcgcgctg atccacgacg acatcatgga tgactccgcg 300
accaggcgcg gcagcccgac ggtgcaccgg gcactcgccg accggctcgg cgcgcgtctc 360
gaccccgacc aagccggaca actggggggtg agcacggcga tctcgtcgg ggaacctcgc 420
ctgacctggt cggacgaact gctgtacgct cccctgaccc cccaccggct ggccgcggtg 480
ctgccccctg tcacggccat gcgcgcggaa acggtccacg gccagtaacct ggacatcacc 540
tccgcccgcg ggcgcggcac ggaacacctc ctggcgctgc gaatcgcgcg ctacaaaacc 600
gctgcttaca ccatggaacg cccctgcac atcggagcag cgctcgccgg cgcacgacgg 660
gaactcctgg cagggtctag cgcctacgcg ctgccggcgg gcgaggcatt ccagctcgcc 720
gacgacctcc tgggagtgtt cggcgatcca cggagaaccg gcaaaaccga cctcgacgac 780
ctccgcgcg gcaagcacac cgtcctcgty gccctcgccc gggaacacgc cacacctgaa 840
cagcggcaca cctggacac cctgctcggc acaccaggcc tcgaccggca ggccgcgtcc 900
cggctgcgct gcgtcctcgt cgcacccggg gcccgggcgg aagccgaacg cctgatcacc 960
gaacggcgcg accaggccct caccgcgctc aacgccctga cactgcccc accgctcgcc 1020
gaggcactcg cccgcctcac cctcgggagt accgcacacc cggcctga 1068

SEQ ID NO: 31          moltype = DNA length = 993
FEATURE               Location/Qualifiers
source                1..993
                     mol_type = other DNA
                     organism = Sulfolobus acididarius

SEQUENCE: 31

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-continued

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atgagttact ttgacaacta ttttaatgag attgttaatt ctgtaaacga cattattaag 60
agctatatat ctggagatgt tcttaacta tatgaagcct catatcatgt gtttacatct 120
ggaggtaaga gggttaagacc attaatctta actatatcat cagatttatt cggaggacag 180
agagaaagag cttattatgc aggtgcagct attgaagttc ttcatacttt tacgcttggt 240
catgatgata ttatggatca agataatcct agaagagggt taccacaggt ccacgtgaaa 300
tacggcttac ccttagcaat attagctggg gatttactac atgcaaaaggc ttttcagctc 360
ttaaccaggc cttcttagagg tttgccaagt gaaaccataa ttaaggcttt cgatattttc 420
actcgttcaa taataattat atccgaagga caggcagtag atatggaatt tgaggacaga 480
attgatataa aggagcagga atacccttgac atgatctcac gtaagacagc tgcattattc 540
tcggcatcct caagtatagg cgcacttatt gctgggtgcta atgataatga tgtaagactg 600
atgtctgatt tcggtacgaa tctaggtatt gcatttcaga ttgttgacga tatcttaggt 660
ctaacagcag acgaaaagga acttggaag cctgttttta gtgatattag ggagggtaaa 720
aagactatac ttgtaataaa aacactggag ctttgtaaaagg aggacgagaa gaagattgtc 780
ctaaaggcgt tagtataaa gtacgctca aaagaagaat taatgagctc agcagatata 840
attaagaaat actctttaga ttatgcatac aatttagcag agaaatatta taaaaatgct 900
atagactctt taaatcaagt ctctctaag agtgatatac ctggaaggc tttaaaaat 960
ctagctgaat ttacgataag aaggagaaaa taa 993

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SEQ ID NO: 32      moltype = DNA length = 894
FEATURE           Location/Qualifiers
source            1..894
                  mol_type = other DNA
                  organism = Synechococcus sp.

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SEQUENCE: 32
ttggttgccc aaaccttcaa cctggacacc tacttgagcc aacgccagca acagggtgaa 60
gaggcgcttt ctgcggcatt ggttcccgcc tatccggagc gcatttacga ggcatgagc 120
tacagcctgc tggcgggggg gaaacgcctg aggcgatcc tctgtctggc ggctgtgag 180
ttggccggcg gctctgtgga gcaggccatg cccacgcct gcgcctgga gatgatccac 240
accatgtcgc tgatccacga cgaatcgccg cgaatggaca acgacgattt tcgcccgccg 300
aagccccacca atcacaaggt attcggcgag gatatcgcca ttttgccagg agatgccctg 360
ttggcctatg cctttgagca tatcgccagc caaacgcggg gggtgcccgc gcagtgtgtg 420
ctgcaagtca ttgcccgcgt tggccatgct gtggcgccaa ccgctctggt agggggccag 480
gtggtggatc tggagtcgga aggcaaaaggc atttccctag aaactttgga gtacatccac 540
agtcacaaga cgggtgctct gctggaggcc tcggtggttt cgggagggat cctggcaggg 600
gccgatgagg agctgctggc gcggtgagc cactacgctc gggacatcgg cctggctttt 660
cagatcgtgg acgacatttt ggaatgtact gccaccagcg agcaactggg caaaacggca 720
ggcaaggatc aagctgcgcg caaagccacc taccacagct tgttgggctc agaggcttcc 780
cggcagaaaag ctgaggaact gatccaatcg gcccaaggag cgttgccgcc ctacggatcc 840
caggccgagc ccctgttggtc tctggccgat ttcacacccc gccgccagca ttga 894

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SEQ ID NO: 33      moltype = DNA length = 1116
FEATURE           Location/Qualifiers
source            1..1116
                  mol_type = other DNA
                  organism = Arabidopsis thaliana

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SEQUENCE: 33
atggcttcag tgactctagg ttcattggatt gttgttcacc accacaatca tcatcatcca 60
tcttcaatcc ttacaaaatc cagatccaga tcttgtccta taactcttac taaaccatc 120
tcctttcgat caaaacgcac cgtttcatca tcttcttcaa tcgtttcttc ttccgttggt 180
acaaaagaag acaatctacg ccaatctgaa ccatcctctt tcgatttcat gtcgtacac 240
atcaccaaaag ccgaattagt caacaaagct ttagattcag ctgttctctc ccgtgagcca 300
ctcaagatcc acgaagcgat gcgttactct cttctcgccg gtggcaaaag agttagacca 360
gtttctctga tcgctgcttg tgaactcgtc ggaggtgaag aatcaaccgc tatgccagca 420
gcttgccgctc tcgagatgat tcacacatg tcgttgatcc acgacgatct ccctgtgatg 480
gataacgcag atctccgcgc tggaaaaaccg accaaccaca aagtgtttgg tgaagacgtc 540
gctgtttttg ccggagagcg gcttctctct ttcgctttcg agcatttagc ttccggcgagc 600
agttctgatg ttgtttctcc ggtgagagtg gttcgagccg ttggagaatt ggctaaagcg 660
ataggaaacag aagggttagt ggcgggtcaa gtcgtggata ttagtagtga agggtagat 720
ttaaacgacg tcggtttaga gcatttgagg tttatccatt tgcataaaac ggccggcgtg 780
cttgaagctt ctgctgtttt gggagctatt gttggtggag gaagttagta tgagattgag 840
agggttaagaa agtttgcgag atgtattggt ttgtgttttc aggtggttga tgatatcttg 900
gatgtgacga aatcgtcgaa agagttaggg aaaactgctg ggaagatttt gattgctgat 960
aagttgacgt atcctaagat tatgggtttg gagaatcga gagagtttgc tgagaaattg 1020
aatagagagg ctcgtagtca gcttttaggg ttgattctg ataaggttgc tcctttgttg 1080
gctttggcta attacattgc ctatagacag aactga 1116

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SEQ ID NO: 34      moltype = DNA length = 2364
FEATURE           Location/Qualifiers
misc_feature      1..2364
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..2364
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 34
atgaaaaccg gggttatctc accagcaaca gtatttcac acagaatctc accagcgacc 60
actttcagac atcacttacc acctgctact acaaaactca caggcattgt cgccttaaga 120
gacatcaact tcagatgtaa agcagtttct aaagagtagt ctgatctggt gcagaaagat 180
gaggcttctt tcacaaaaatg ggacgatgac aagggtgaaag atcatcttga taccacaaaa 240

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aacttatacc caaatgatga gattaaggaa tttgttgaat cagtaaaggc tatgttcggg 300
agtatgaatg acggggagat aaacgtctct gcatacgata ctgcatgggt tgctttgggt 360
caagatgtcg atggatcagg tagtcctcag ttcccttctt ctttagaatg gattgccaac 420
aatcaattgt cagatggatc atggggagat catttgcgtg tctcagctca cgatagaatc 480
atcaacacat tagcatgcgt tattgcactt acaagttgga atgttcatcc ttctaagtgt 540
gaaaagggtt tgaattttct gagagaaaac atttgcaaat tagaagatga aaacgcagaa 600
catatgccaa ttggttttga agtaacattc ccatacacta ttgatatcgc gaaaaagttg 660
aacattgaag tacctgagga tactccagca cttaaagaga tctacgcacg tagagatc 720
aagttaacta agatcccaat ggaagtctct cacaagggtac ctactacttt gttacattct 780
ttggaaggaa tgctgattt ggagtgaggaa aaactgttaa agctacaatg taaagatgg 840
agtttcttgt ttccccatc tagtacgcga ttgcgcctaa tgcaaacaaa agatgagaaa 900
tgcttacagt atctaacaaa tatcgtcact aagttcaacg gtggcgtgcc taatgtgtac 960
ccagtcgatt tgtttgaaca tatgtgggtt gtgatagac tgcagagatt ggggattgcc 1020
agatacttca aatcagagat aaaagattgt gttagagata tcaataagta ctggacccaa 1080
aatggaattt gttgggctag aaatactcac gttcaagata tcgatgatac agccatggga 1140
ttcagagtgt tgagagcgca cggttatgac gtcactccag atgtttttag acaatttgaa 1200
aaagatggta aattcgtttg ctttgcaggg caatcaacac aagccgtgac aggaatggtt 1260
aacgtttaca gagcctctca aatgttggtt ccaggggaga gaattttgga agatgccaaa 1320
aagttctctt acaattactt aaaggaaaag caaagtacca acgaattgct ggataaatgg 1380
ataatcgcta aagatctacc tgggtgaagt ggttatgctc tggatatccc atggtatgct 1440
tccttaccaa gattggaac tcgttattac cttgaacaat acggcgggtg agatgatgct 1500
tggataggca agacattata cagaatgggt tacgtgtcca ataacacata tctagaaatg 1560
gcaaaagctg attacaataa ctatgttgca gtccttcaat tagaatggta cacaatacaa 1620
caatggtagc tcgatattgg tatagagaag ttgcaatctg acaacatcaa gtcagtcctg 1680
gtttcttact acttggtctg ggccttcaata ttgcaacctg agagatctaa ggagagaatc 1740
gcttgggcaa agacaacaat cttagtcgat aagatcacat caattttcga ttccctctag 1800
tcaagtaagg aagatattac tgcctttatt gacaagtttc gtaacaagtc ctccctctaa 1860
aagcactcta tcaacggtga accatggcat gaagtattgg tagctttgaa aaagacotta 1920
cacggctttg ctctggatgc tcttatgact cattctcaag atatacatcc acagttacat 1980
caagcctggg aaatgtgggt gactaaacta caagacggcg tagatgttac tgctgagcta 2040
atggtccaaa tgatcaacat gactgctggc agatgggtat caaaggaatt acttactcat 2100
ccacaatc acaaagtgtc tactgtgaca aattctgtgt gtcacgatat taccaaactt 2160
cacaatttca aggagaattc caccacagtg gattcaagg ttcaggaact agtccagttg 2220
gttttttagtg acacaccaga tgatttggat caagatatga aacaaacatt cctgacagtg 2280
atgaagacat tctactacaa ggcgtgggtg gatccaaaca ctataaacga tcatatatct 2340
aaagttttcg aaatcgtaat ttga 2364

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SEQ ID NO: 35      moltype = DNA length = 1584
FEATURE            Location/Qualifiers
misc_feature       1..1584
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..1584
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 35
atgctgatg cacaogatgc tccacctcca caaataagac agagaacact agtagatgag 60
gtaccccaac tgctaactga gtccgcagaa gatgcatggg gtgaagtcag tgtgtcagaa 120
tacgaaacag caaggctagt tgcccctgct acatgggttag gtggacacgc cacaagagtg 180
gccttctctc tggagagaca acacgaagac ggggtcatgg gtccaccagg tggatatagg 240
tagtcctcta cattatctgc tgttcacgca ttattgacat gtcttgcttc tctgtctcag 300
gatcatggcg ttccacatga tagactttta agagctgttg acgcaggctt gactgccttg 360
agaagattgg ggacatctga ctccccacct gatactatag cagttgagct ggttatccca 420
tctttgctag agggcattca acacttactg gacctgtctc atcctcatag tagaccagcc 480
ttctctcaac atagaggctc tcttgtttgt cctggtggac tagatgggag aactctagga 540
gctttgagat cacacgcgcg agcaggtaca ccagttaccg gaaaagtcg gcacgcttcc 600
gagacttttg gcttgagtac cgaagctgct tctcacttgc aaccagccca aggtataatc 660
ggttgctctg ctgctgccac agcaacatgg ctaaccaggg ttgcaccatc tcaacagtca 720
gattctgcca gaagatacct tgaggaatta caacacagat actctggccc agttccttcc 780
attaccccta tcacatactt cgaagagaca tgggtattga acaattttgc agcagccggt 840
gttctctgtg aggtctcagc tgctttgttg gattccttag aagcagcact tacaccacaa 900
ggtgctctcg ctggagcagg attgctctca gatgctgatg atacagccgc tgtgttgctt 960
gcattggcaa cacatgggag aggtagaaga ccagaagtac tgatggatta caggactgac 1020
gggtatttcc aatgctttat ttgggaaaagg actccatcaa ttccaacaaa cgctcacgta 1080
ttggaacat tagggcatca tgtggcccaa catccacaag atagagccag atacggatca 1140
gcatgggata ccgcatcagc ttggctgctg gcagctcaaa agcaagatgg ctcttggtta 1200
gataaatggc atgcctcacc atactacgct actgtttgtt gcacacaagc cctagccgct 1260
catgcaagtc ctgcaactgc accagctaga cagagagctg tcagatgggt ttagccaca 1320
caaaagatccg atggcggttg ggttctatgg cattcaactg ttgaagagac tgcttatgcc 1380
ttacagatct tggccccacc ttctgtgtgt ggcaatatcc cagtccaaca agcacttact 1440
agaggcagag caagatgtgt tggagccttg ccactgactc ctttatggca tgataagga 1500
ttgtatactc cagtaagagt agtcagagct gccagagctg ctgctctgta cactaccaga 1560
gatctattgt taccaccatt gtaa 1584

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SEQ ID NO: 36      moltype = DNA length = 1551
FEATURE            Location/Qualifiers
misc_feature       1..1551
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide

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source          1..1551
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 36
atgaacgccc      tatcogaaca  cattttgtct    gaattgagaa    gattattgtc    tgaatgagt     60
gatggcgggat     ctgttggtcc    atctgtgtat    gatacggccc    aggcocctaag   attccacggt    120
aacgtaacag      gtagacaaga    tgcataatgct   tgggtgatcg    cccagcaaca    agcagatgga    180
ggttggggct     ctgcogactt    tccactcttt    agacatgctc    caacatgggc    tgcacttctc    240
gcattacaaa      gagctgatcc    acttctctggc   cgacgagacg    cagttcagac    cgcaacaaga    300
ttcttgcaaa      gacaaccaga    tccatacgcct   catgcccgttc   ctgaggatgc    ccctatttgt    360
gctgaactga      tcttgctcca    gttttgtgga    gaggctgctt    ggttgttggg    aggtgtggcc    420
ttccctagac      acccagccct    attaccatta    agacaggctt    gtttagtcaa    actgggtgca    480
gtcgccatgt      tgccttcagg    acaccattg     ctccactcct    gggaggcatg    gggtaacttct   540
ccaacaacag      cctgtccaca    cgatgatggt    tctataggtta  tctcaccagc    agctacagcc    600
gcctggagag      cccagggtgt    gaccagaggc    tcaactcctc    aagtgggcag    agctgacgca    660
tacttacaaa      tggttcaag     agcaacgaga    tcaggcatag    aaggagtctt    ccctaattgt    720
tggcctataa      acgctatcga    accatgctgg    tcactgtaca    ctctccatct    tgcgggtctg    780
ttcgcccatc      cagcactggc    tgaggctgta    agagtatatc    ttgctcaact    tgaagcaaga    840
ttgggagtg      atggcctcgg    accagcttta    cattttgctg    ccgacgctga    tgatactgca    900
gttgccctat      gcgttctcgca  tttggctggc    agagatcctg    cagttgacgc    attgagacat    960
tttgaattg       gtgagctctt    tgttacattc    ccaggagaga    gaaatgctag    tgtctctacg    1020
aacattcacg      cctctcatgc    tttgagattg    ttaggtaaac    cagctgccgg    agcaagtgca    1080
tacgtcgaag      ccaatagaaa    tccacatggt    ttgtgggaca    acgaaaaatg    gcacgtttca    1140
tggctttatc      caactgcaca    cgccgttgca    gctctagctc    aaggcaagcc    tcaatggaga    1200
gatgaaagag      cactagccgc    tctactacaa    gctcaaagag    atgatggtgg    ttggggagct    1260
ggtagaggat      ccactttcga    ggaaccggcc    tacgctcttt    tcgctttaca    cgttatggac    1320
ggatctgag       aagccacagg    cagaagaaga    atcgctcaag    tcgtcgcaag    agccttagaa    1380
tggatgctag      ctgacatgc     cgcacatgga    ttaccacaaa    caccactctg    gattggttaag   1440
gaatgtact       tctctactag    agtcgtaaga    gtagctgagc    tagctggcct    gtgggttagca   1500
ttaagatggg      gtagaagagt    attagctgaa    ggtgctggtg    ctgcacctta    a              1551

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SEQ ID NO: 37      moltype = DNA length = 2364
FEATURE           Location/Qualifiers
source            1..2364
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 37
atgaagaccg      gcttcacatc    tcccgccacc    gtcttccacc    accgtatttc    tccggcaacc     60
accttccgcc      accacctttc    tccggcgacc    accaactcca    ctggaattgt    agctcttaga    120
gacatcaact      acggtgttaa    agcggtatcc    aaagagtact    ctgatttact    acaaaaagat    180
gaggettcac      ttaccaagtg    ggaacgatgac    aaagtgaagg    accatttgga    cacaataaag    240
aatttgatc       caaacgatga    gatcaaggag    tttgttgaga    gcgtgaaagc    aatgttttgt    300
tctatgaatg      acggagaaat    aaatgtgtca    gcgtatgata    cggcttgggt    tgcactcgtg    360
caagatgttg      atggaagtgg    ttccctcaa     tttccatcaa    gtttggagtg    gatcgcgaaac   420
aatcaactct      cagatggggt    ttggggcgat    catttggtat    tttcggtcca    tgataggatc    480
attaacacgt      tggcatgtgt    tatagcgcct    acttcttgga    acgtccatcc    aagtaaatgt    540
gaaaaaggac      tgaattttct    tagagaaaac    atatgtaaac    tcgaagacga    gaacgcggaa    600
catatgccaa      ttggttttga    agtcacggtt    ccgtcgctaa    tagatatcgc    aaagaagcta    660
aatattgaag      ttctgagga    tactcctgcc    ttaaaagaaa    tttatgcaag    aagagacata    720
aaactcacaa      agataccaat    ggaagtattg    cacaaagtc    ccacaacttt    acttcatagt    780
ttggaaggaa      tgccagattt    ggaatgggaa    aaactctga    aattgcaatg    caaagatgga    840
tcatttctgt      tttctcaatc    atctactgct    tttgactca    tgcaaacaaa    agatgaaaag    900
tgtcttcagt      atttgacaaa    tattgttacc    aaattcaatg    gtggagtctc    gaatgtgtac    960
ccggtggatc      tatcogaaca    tatttgggta    gttgatcgac    ttcaacgact    tgggattgtct   1020
cgttatttca      aatcagagat    caaagattgc    gttgaatata    ttaacaagta    ttggacaaaag   1080
aatgggattt      gttgggcaag    aaacacgcac    gtacaagata    ttgatgatac    cgcaatggga    1140
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aatgtgtata      gagcgtcaca    aatgctcttt    cccggagaaa    gaattcttga    agatgcaaaag   1320
aaattttcat      ataattattt    gaaagaaaaa    caatcgacaa    atgagcttct    tgataaatgg    1380
atcatcgcca      aagactttac    tggagagggt    ggatatgcgc    tagacatacc    atggtatgca    1440
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gcgtgggcta      aaaccaccat    attagttgac    aagatcacct    caatttttga    ttcacacaaa    1800
tctcaaaaag      aggacataac    agcctttata    gacaaaatta    ggaacaaatc    gtcttctaag    1860
aagcattcaa      taaatggaga    accatggcac    gaggtgatgg    ttgcaactgaa  aaagacccta    1920
cagcgcttcg      ctttggatgc    actcatgact    catagtcaag    acatccaccc    gcaactccat    1980
caagcttggg      agatgtgggt    gacgaaattg    caagatggag    tagatgtgac    agcggaaatta   2040
atggtacaaa      tgataaatat    gacagctggg    cgttgggtat    ccaaagaact    ttaactcat     2100
cctcaatacc      aacgctctc    aaccgtcaca    aatagtgtgt    gtcacgatat    aactaagctc    2160
cataacttca      aggagaattc    cagcacggta    gactcgaaag    tcaagaact    agtgcaactt    2220
gtgttttagcg     acacgcccga    tgatcttgat    caggatatga    aacagacgtt    tctaaccgtc    2280
atgaaaacct      tctactacaa    ggcgtgggtg    gatccgaaca    cgataaatga    ccatactctc    2340
aagggtgttcg     agattgtaat    atga              2364

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SEQ ID NO: 38      moltype = DNA length = 1584

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FEATURE
source

Location/Qualifiers
1..1584
mol_type = other DNA
organism = *Streptomyces clavuligerus*

SEQUENCE: 38

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gcgacgcaac	tctctcagga	gtcggccgag	gacgcctggg	gtgaggtgtc	cgtgtccgag	120
tacgaaacgg	cgcggctggg	ggcccacgcc	acctggctcg	gcggtcacgc	cacacgggtg	180
gccttctctg	tggagcggca	gcatgaggac	ggctcgtggg	gcccgcgccg	cgggtaccgt	240
ctcgtaccga	cgtctagtgc	cgtacacgcc	ctgtcacct	gtctggcgct	tcccgcgcag	300
gaccacggag	tgcctcatga	cgggtcctg	cgcgcagtgt	acgcgggcct	gacggcactg	360
cgtcgtcttg	ggacgagcga	cagcccgcgc	gacaccattg	cgttcgaact	ggtcataccc	420
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ggtggctccg	cgccgcggac	agcaacatgg	ctcaccaggg	tcgccccgtc	gcaacagagc	720
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gcgcttgcca	cgcacggccg	cgggcgcctg	cccgaggtcc	tcattgacta	ccgcacggac	1020
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caacgctcgg	acggcggtgt	gggcctgtgt	cactccacgg	tcgaggagac	cgcctacgcc	1380
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ctgtacacgc	cggtagctgt	gtcccgccgc	gcccgtgccg	ccgccttgta	caccaccgct	1560
gacctgcttc	tgccgcccct	gtga				1584

SEQ ID NO: 39

FEATURE
source

Location/Qualifiers
1..1551
mol_type = other DNA
organism = *Bradyrhizobium japonicum*

SEQUENCE: 39

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gatggcgcca	gcgtcgttcc	gtccgtctac	gacacggcgc	aggcgctgcg	cttccacggc	120
aacgtccacc	gtcggcagga	cgcatacgcg	tggtcctatc	cgcagcaaca	ggccgacggc	180
ggatggggaa	gcgcggactt	cccgtgttcc	cgccatgcgc	ccacgtgggc	ggcggtactg	240
gcattgcagc	gtgcgcgacc	tcttcccgga	gctgcggagc	cagtcacagc	tgcaacgagg	300
ttctccacgc	gccagcccga	tcctacgcga	catgcgggtc	cagaagacgc	gcccgatcgc	360
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gtggcgatgt	tgccgagcgg	ccatccgttg	ctacactcct	gggaagcctg	ggggacgtcg	540
ccgaccaccg	catgcgccga	tgacgacggc	agcatcgcca	tcagtcgggc	ggccaccgcc	600
gcgtggcggt	cccaggccgt	gacacggggg	agcacgcgcg	aggtcggggc	cggcgatcgc	660
tatctgcaga	tggcatcgcg	ggcgacgcgc	agcggcatcg	aaggtgtctt	tcccaacgtc	720
tggccgatca	atgtgttcga	gcatgtctgg	tcgctgtaca	ccctgcatct	ggccgggctt	780
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gttgcggtgt	gcgtcctgca	ccttgacagg	cgtgaaccgc	cggtcgatgc	ggtgcgcat	960
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tacgtcgagg	ccaatcgcaa	cccgcacggc	ctatgggaca	acgaaaaatg	gcacgtttcg	1140
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ggtcgcgggt	ccacgtttcga	ggaaaaccgc	tatgcgctgt	ttgcgttgca	cgtgatggat	1320
gggagcgaa	aggcgacagg	gcgcggcggc	atcgcgcagg	tggtggcgcg	tgcgctggag	1380
tggtatgctg	cccgccatgc	ggcgcatgga	ttgccgcaga	cgcgcgtgtg	gatcggaag	1440
gaactgtatt	gcaccaactc	ggtcgtgcgc	gtggccgaac	tcgcggggtt	gtggctggcg	1500
cttcgttggg	ggcgcgcgct	cctggccgag	ggggcaggag	cggcgccatg	a	1551

SEQ ID NO: 40

FEATURE
misc_feature

Location/Qualifiers
1..2355
note = Description of Artificial Sequence: Synthetic polynucleotide

source

Location/Qualifiers
1..2355
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 40

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taaatcgata	cgaccaaggga	gagaatacaa	aaacaattca	aaaatgttga	aatttcagtt	180
tcttcttatg	atactgcgtg	ggttgccatg	gttccatcac	ctaattctcc	aaagtctcca	240

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tgtttccag aatgtttgaa ttggttgatt aacaaccagt tgaatgatgg atcttgggg 300
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ttggcttgca tcgtggccct aaagagatgg aacgtaggtg aggtacagat taacaagggg 420
cttagtttca ttgaatctaa cttggcttcc gcgactgaaa aatctcaacc atctccaata 480
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ctgtctaagc aaactgattt ctactaatg ttacacaaga gagaattaga acaaaagaga 600
tgtcattcaa acgaaatgga tggttaccta gcttatatct ctgaaggtct tggtaatctt 660
tacgattgga atatggtgaa aaagtaccag atgaaaaatg gctcagtttt caattccctt 720
tctgcaactg cggcagcatt cattaacccat caaaatccag gatgcctgaa ctatttgaat 780
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tactttgtag gacaaaagct atccgaggaa atcgtcgaaat catcagataa ccataaactg 1980
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atgaactcaa tcttcgaaga gaacggttca attgttctca gagcatgtaa ggatgcattt 2220
tggaacatgt gtcattgtgt aaactttttc tacgcaaacg acgatggttt tactgggaac 2280
acaatactag atacagtaaa agacatcata tacaaccctt tggctcttagt aaacgaaaaa 2340
gaggagcaaa gataa 2355

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SEQ ID NO: 41      moltype = DNA length = 2355
FEATURE            Location/Qualifiers
misc_feature       1..2355
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..2355
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 41
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ataatcgata ctactaagga gagaatccaa aagctattca aaaatgttga aatctcagta 180
tcattcttatg acaccgcagtg ggttgcaatg gtgccatcac ctaattcccc aaaaagtcca 240
tgttttccag agtgcttgaa ttgggttaatc aataatcagt taaacgatgg ttcttggggg 300
ttagtcaacc acactcataa ccacaatcat ccattattga aggactcttt atcatcaaca 360
ttagcctgta ttgttgcatg gaaaagatgg aatgtaggtg aagatcaaat caacaagggg 420
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aaattgatac acaagggaagt agaaaaacga ctaaagtttc ctattaacac tggtttagag 1260
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gttgctgcta ccctttcatc ccacagaattg tctgatgcca gaataagttg ggccaaaaat 1560
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gaacatgtga gaatactttt cctggctcta aaagatgcaa tatgttggat tggcgacgag 1740
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gaatacatgg aaaaacgcta cgtctcattt gcttgggtc ctattgttaa gccagccata 1920
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ttcaagttaa tgtcaacaca aggcagactt ctgaacgata tccactcctt caaaagagaa 2040
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gtggaagagg aagtcgttga ggaatgatg atgatgatca aaaacaagag aaagggaattg 2160
atgaaattga ttttcgagga aaatggttca atcgtaccta gagcttgtaa agatgctttt 2220
tggaatatgt gccatgttct taacttcctt tacgctaattg atgatggctt cactggaaat 2280
acaatattgg atacagttaa agatatcatc tacaaccacac ttgttttggg caatgagaac 2340
gaggaacaaa gataa 2355

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SEQ ID NO: 42      moltype = DNA length = 1773
FEATURE
misc_feature      Location/Qualifiers
                  1..1773
                  note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source            1..1773
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 42
atggctatgc cagtgaagct aacacctgctg tcattatcct taaaagctgt gtgctgcaga 60
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cctacccaaa gatctacttc ttctctact actagaccag ctgccgaagt gtcacacagt 180
aagagtaaac aacatgatca ggaagctagt gaagcgacta tcagacaaca attacaacct 240
gtggatgtcc tggagaatat ggaatatatc agacattttg ctgcagagat aaagtgcata 300
ctagacagaa cttacagatc ttgggttaca agacacgagg aaatcatgct ggacactatg 360
acatgtgcta tggcttttag aatcctaaga ttgaacggat acaacgtttc atcagatgaa 420
ctataccacg ttgtagaggc atctggctctg cataattcct tgggtgggta tcttaacgat 480
accgaacac tacttgaatt acacaaggct tcaacagtta gtatctctga ggatgaatct 540
atcttagatt caattggctc tagatccaga acattgctta gagaacaatt ggagtctggt 600
ggcgactga gaaagccttc ttattcaaa gaggttgaac atgcactgga tggactttt 660
tacaccacac ttgatagact tcatcatagg tggaaatttg aaaacttcaa cattattgag 720
caacacatgt tggagactcc atacttatct aaccagcata catcaaggga tatcctagca 780
ttgtcaatta gagatttttc ctctccacaa ttcacttata aacaagagct acagcatctg 840
gagagttggg ttaaggaatg tagattagat caactacagt tcgcaagaca gaaattagcg 900
tacttttacc tatcagccgc aggcaccatg tttctctctg agctttctga tgcgagaaca 960
ttatgggcca aaaacggggg gttgacaact attgttgatg atttcttga tgttgccggg 1020
tctaagaggg aattggaaaa cttagtcatg ctggctgaaa tgtgggatga acatcacaaa 1080
gttgaaattc attctgagca ggtcgaaatc atctctctct ccatctacga ttctgtcaac 1140
caattgggtg agaaggcctc ttgggttcaa gacagatcaa ttacaaaaca ccttggtgaa 1200
atatgggtag acttggttaa gtccatgatg acggaagtgt aatggagact gtcaaaatac 1260
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gttttaccag ctttgtattt cgttggtcca aagatttcag aaagtatagt aaaggaccca 1380
gaatatgatg aattgttcaa actaatgtca acatgttgga gattgttgaa tgacgtgcaa 1440
acgttcgaaa gagaatacaa tagaggtaaa ctgaattctg tcagtctatt ggttctctac 1500
ggaggcccaa tgtctatttc agacgcaaa aggaatttac aaaagcctat tgatacgtgt 1560
agaagagatc ttctttcttt ggctcctaga gaagagtctg tagtaccag accatgtaag 1620
gaactattct ggaaaatgtg taaagtgtgc tatttctttt actcaacaac tgatgggttt 1680
tctagtcaag tcgaaagagc aagaagagga gacgctgtca taaatgagcc actgaagttg 1740
caaggttctc atacactggt atctgatgtt taa 1773

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SEQ ID NO: 43      moltype = DNA length = 2232
FEATURE
misc_feature      Location/Qualifiers
                  1..2232
                  note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source            1..2232
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 43
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acaccttggt ttccagaatg tactaaatgg atcctagaaa atcagttggg tgatggtagt 180
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ttcatagaac tcaactctgc tagtgttaac gataacgaac aacacaaacc aattggattt 360
gacattatct ttcaggatg gattgaatac gctatagact tagacctgaa tctaccacta 420
aaaccaactg acattaactc catgttgcat cgtagagccc ttgaattgac atcaggtgga 480
ggcaaaaatc tagaaggtag aagagcttac ttggcctacg tctctgaagg aatcggttaag 540
ctgcaagatt gggaaatggc tatgaaatac caacgtaaaa acggatctct gttcaatagt 600
ccatcaacaa ctgcagctgc attcatccat atacaagatg ctgaatgctc ccactatatt 660
cgttctcttc tccagaaatt tggaaacgca gtccctacaa tataccctct cgatatctat 720
gccagacttt caatggtaga tgccctggaa cgtcttggtg ttgatagaca tttcagaaag 780
gagagaaagt tcgttctgga tgaacaatac agattttggg tgcaaggaga agaggagatt 840
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gtctctcttg aagatcactt ctctaaactc ctgggctgggt acttaaggga ctgaggagca 960
gctttagaac tgtacagagc cttccaattg tcttaccacg acgagtccct cctggaaaag 1020
caaaattcta gaactttcta cttcttaaaa caaggtttat ccaatgtctc cctctgtggt 1080
gacagattgc gtaaaaacat aattggagag gtgcatgatg ctttaaaact ttccgaccac 1140
gctaacttac aaagattagc tattcgtaga aggattaagc attacgctac tgacgatata 1200
aggattctaa aaacttcta cagatgctca accaagattt tctaaaactt 1260
gcagtggaag atttcaatat ctgtcaatca atacaagag aggaattcaa gcatattgaa 1320

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agatgggtcg ttgaagacg tctagacaag ttaaagtctg ctagacaaaa agaggcctat 1380
tgctatttct cagocgcagc aacattgttt gccctgaat tgtctgatgc tagaatgtct 1440
tgggccaaaa atggtgtatt gacaactgtg gttgatgatt tottcogagt cggaggtctc 1500
gaagaggaat tagttaactt gatagaattg atcgagcggt gggatgtgaa tggcagtgc 1560
gatttttgta gtgaggaagt tgagattatc tattctgcta tccactcaac tatctctgaa 1620
ataggtgata agtcatttgg ctggcaaggt agagatgtaa agtctcaagt tatcaagatc 1680
tggctggact tattgaaatc aatgttaact gaagctcaat ggtcttcaaa caagtctgtt 1740
cctaccctag atgagtatat gacaaccgcc catgtttcat tgcacttgg tccaattgta 1800
cttcagcctt tatacttcgt tggcccaaaag ttgtcagaag aggttgcagg tcatcctgaa 1860
ctactaaacc tctacaaagt cacatctact tgtggcagac tactgaatga ttggagaagt 1920
ttaaagagag aatccgagctc aggttaagctc aacgctatta gtttatacat gatccactcc 1980
ggtggtgctt ctacagaaga ggaacaatc gaacatttca aaggtttgat tgattctcag 2040
agaaggcaac tgttacaatt ggtgttgcaa gagaaggata gtatcatacc tagaccatgt 2100
aaagatctat ttgggaatat gattaagtta ttacacactt tctacatgaa agatgatggc 2160
ttcacctcaa atgagatgag gaatgtagtt aaggcaatca ttaacgaacc aatctcactg 2220
gatgaattat ga 2232

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SEQ ID NO: 44      moltype = DNA length = 2355
FEATURE           Location/Qualifiers
source            1..2355
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 44
atgaatcttt cactatgcat cgcgtcccct ttgttaacca aatcaaatcg acccgcggtc 60
ctgtcagcta ttcatacagc atcaacttca catggtggac aaactaatcc cactaatctg 120
atcattgata caaccaaaga acggatccaa aaacagttta aaaatgtaga aatttctgtt 180
tcttcataatg acacagcatg ggtagccatg gtcccttctc caaactcacc caaatcgctc 240
tgtttccctg agtgtctcaa ttgggttaatt aataatcagc ttaatgatgg ttcattgggt 300
ctgtttaatc acactcataa tcataatcac ccgttgctta aagattctct atcttcaaca 360
tagcatgta ttgttgcat aaaaagatgg aatgttgggg aagatcaaat aaataaagg 420
ctaaagtttta ttgagtcaaa tcttgcttca gctactgaaa aaagtcaacc atctccatt 480
ggttttgaca tcatatttcc tggtttgctt gagtatgcga aaaacttggg cataaacctc 540
ctttcaaaac aaacagattt tagtttgatg ctacataaga gggaaattgga gcaaaaaaga 600
tgccattcaa atgagatgga tggatacttg gcgtatatct ctgaaggact cggtaattta 660
tatgattgga atagtgtgaa gaaatatcag atgaaaaatg gttctgtttt caactcacca 720
tcagcaacag ctgctgcttt cattaatcat caaaatcctg gttgtcttaa ttatttaaat 780
tcacttttgg acaagtttgg taatgcagtc ccaacagttt atcctcatga tttatttatc 840
cgactttcta tgggtgacac aattgaaaga ttaggaattt cacaccattt cagagtggaa 900
attaaaaatg ttttagatga aacatacaga tgttgggtgg aacgagatga gcaaatattc 960
atggatgttg taacatgtgc tttagccttt cggttattaa ggaatcaatg gtaagaagt 1020
tcccagatc cattggctga aattactaat gaattagctt tgaagacga atatgcagct 1080
cttgaacat atcatgcgtc acatatatta taccagagg atttatcttc tggaaaaaaa 1140
atcttgaagt cagctgattt cctcaaaag ataatatcca ctgattcaaa caggctttct 1200
aaattaattc acaagaggtt ggaatgctt cttaagtcc ctatcaatc cggtttagaa 1260
cgcataaaca ctagacgaaa tatacagctt tacaatgtag acaatacaag aattctgaaa 1320
actacatatc actcatcaaa tattagtaac actgattacc taagggtggc tgttgaagat 1380
ttctacacct gccaatctat ttatcgtgaa gaattaaaag gtcttgaaag gtgggtggta 1440
gagaataagt tggaccagct caagtttgct agggcaaaaga ccgcctactg ttatttctct 1500
gttgcctgaa cactttcgtc tcccgaatta tcagatgcgc gtatttcatg ggccaaaaat 1560
ggcatattaa ctacagatg ttgatgactt tttgatctg gtggtacaat cgatgaattg 1620
accaacctga ttcaatgtgt tgaaaaatgg aatgtagatg tcgacaagga ttgttgctca 1680
gagcatgttc ggattttatt tttagcatta aaagatgcaa tctgttggat tggagatgaa 1740
gcttttaaat ggcaagcgcg cgatgtaact agccatgtta ttcaaaactg gttggaacta 1800
atgaatagta tgttgagaga agctatatgg acaagagatg cttatgtgcc aacattaaat 1860
gaatatatgg aaaaacgcta cgtgtcattt gcattaggcc cgattgtcaa gccggctatt 1920
tactttgtgg ggcccaaat atcagaggag attgttgaaa gctctgaata tcataatcta 1980
tttaagctaa tgagcacgca gggtcgactt ctaaaacgata tccatagctt caagagggaa 2040
tttaaggaa gcaaatataa ccggttagca ttcgatttga gtaacggaga aagtgggaaa 2100
gtggaagaag aggttgtgga ggagatgatg atgatgatta aaaacaagag gaaagaatta 2160
atgaaattaa tttttgaaga aaatgtagc attgttctca gagcttgtaa agatgcattt 2220
tggaaacatgt gtcacgtggt gaattttttt tacgcaaacg atgacgggtt tactggaaac 2280
acgattcttg atactgtgaa ggacatcatt tacaacccgt tgggtgctgt gaatgaaaat 2340
gaagaacaaa ggtaa 2355

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SEQ ID NO: 45      moltype = DNA length = 2355
FEATURE           Location/Qualifiers
source            1..2355
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 45
atgaatcttt cactatgcat tgcgtcccct ttgttaacca aatcaagtctg acccacggct 60
ctgtcagcta ttcatacagc atcaacttca catggtggac aaactaatcc cactaatctg 120
atcattgata caaccaaaga acggatccaa aaactgttta aaaatgtaga aatttctgtt 180
tcttcataatg acacagcatg ggtagccatg gtcccttctc caaactcacc caaatcgctc 240
tgtttccctg agtgtctcaa ttgggttaatt aataatcagc ttaatgatgg ttcattgggt 300
ctgtttaatc acactcataa tcataatcac ccgttgctta aagattctct atcttcaaca 360
tagcatgta ttgttgcat aaaaagatgg aatgttgggg aagatcaaat aaataaagg 420
ctaaagtttta ttgagtcaaa tcttgcttca gcaactgaca aaagtcaacc atctccatt 480
ggttttgata tcatatttcc tggtttgctt gagtatgcga aaaacttggg cataaacctc 540

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ctttcaaaac aaacagattt tagttttagt ctacataaga ggggaattgga gcaaaaaaga 600
tgccatttcaa atgagatttga tggataacttg gcgtatatct ctgaaggact cggtaattta 660
tatgatttga atattggtgaa gaatatcag atgaaaaatg gttctgtttt caactcacca 720
tcagcaacag cagctgcttt cattaatcat caaaatcccg gttgtcttaa ttatttaaat 780
tcacttttgg acaagtttgg taatgcagtc ccaacagttt atcctcttga tttatatatc 840
cggttttcta tgggtgacac aattgaaaga tttaggaattt cacaccattt cagagtggaa 900
attaaaaatg ttttagatga aacatacaga tgttgggtgg aacgagatga gcaaatattc 960
atggatgttg taacatgtgc tttagccttt cggttattaa ggatccacgg gtataaagtc 1020
tccccagatc aattgggtga aattactaat gaattagctt tcaaagacga atacgcagct 1080
cttgaacat atcatgcatc acagatatta taccaagagg atttatcttc tggaaaacaa 1140
atcttgaagt cagctgattt cctcaaaggg atattatcca ctgattcaaa caggctttct 1200
aaattaattc acaagaggt ggaatatgct cttaagtctc ctatcaatac cggtttagaa 1260
cgcataaaca ctagacgaaa tatacagctt tacaatgtag acaatacaag aattctgaaa 1320
actacatcct actcatcaaa tattagtaac acttattacc taagggtggc tgttgaagat 1380
ttctacacct gccaatctat ttatcgtgaa gaattaaaag gtcttgaag gtgggtggt 1440
cagaataagt tggaccagct caagtttgc agggcaaaaga ccgcctactg ttatttctct 1500
gttgcgtgca cactttgtgc tccgaatta tcagatgcgc gtatttctat ggccaaaaat 1560
ggcatattaa ctacagtagt tgatgacttt tttgatctgc gtgtgacaat cgtatgaattg 1620
accaacctga tccaatgtgt tgaaaaatgg aatgtagatg tcgacaagga ttgttgttca 1680
gagcatgttc ggtatttatt tttagcatta aaagatgcaa tctgttggat tggagatgaa 1740
gcttttaaat ggcaagcgcg cgtatgaact agccatgtta ttcaaacctg gttggaacta 1800
atgaatagta tgttgagaga agctatatgg acaagagatg cttatgtgcc aacattaaat 1860
gaatatatgg aaaacgctta cgtgtcattt gcattaggcc cgattgtcaa gccggctatt 1920
tactttgtgg ggcccaaat atcagaggag attgttgaaa gctctgaata tcataatcta 1980
tttaagctaa tggagcagca ggttcgactt ctaaaagata tccatagctt caagagggaa 2040
tttaaggaa gcaaatataa cgcggttagc ttgcatttga gtaacggaga aagtgggaaa 2100
gtggaagaag aggttgttga ggagatgatg atgatgatta aaaacaagag gaaagaatta 2160
atgaaattaa tttttgaaga aaatggtagc attgttccta gagcttgtta agatgcattt 2220
tggaaacatg gtcacgtggt gaattttttt tacgcaaacg atgacgggtt tactggaaac 2280
acgattcttg atactgtgaa ggacatcatt tacaacccgt tgggtgctgt gaatgaaat 2340
gaagaacaaa ggtaa 2355

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SEQ ID NO: 46      moltype = DNA length = 1773
FEATURE
source             Location/Qualifiers
                   1..1773
                   mol_type = other DNA
                   organism = Zea mays

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SEQUENCE: 46
atggccatgc cagtgaagct gactcctgcc tccctctcgc tgaaggcggg ctgctgccgc 60
ttcagctcgc gagggcagtc gctgcgcttc ggctcgtcgc taccgtgctg gaggaggacg 120
ccgacgcacac ggaacacgctc gctgctctac acgcgccctg cggctgaggt tagctctggc 180
aaaagcaagc agcacgatca agaagcatcg gaggctacga taagacagca gctccagcta 240
gtcagatgtgc ttgagaacat ggggatttct cggcattttg ctgctgaaat caaatgcctc 300
cttgacagga catacagaag ttgggttacag agacatgagg aaattatgct ggacacaatg 360
acctgtgcga tggcatttgc tattctaagg ttgaatggat acaatgtctc ttctgatgag 420
ttgtatcatg ttgttgaaag ttccggactc cataaattcac ttggaggata tctcaatgat 480
acaagaacct tgttagaatt acacaaggcc tcgacagtta gtatctctga agatgagctc 540
atcctggata gcataggctc aaggctcacgt acctactga gggaaacaact agagtctggt 600
ggtgctctac gaaaaccttc actctttaa gagggtggaac atgctctgga cggtcctctc 660
tacaccacat tggaccgtct acaccatagg tggaaacatc aaaatttcaa tattatagag 720
cagcacatgc tagagacacc atacttgtca aatcaacata ccagtagaga tattctagcg 780
ttgagtatta gagacttcag ttctctcag agcaagaact tcaacatctt 840
gaaagctggg tgaagagtg cagggttagac cagctacaat ttgcgcgaca gaagtgggca 900
tacttctact tgtctgtctc tggcaccatg ttctctcctg agctgtctga tgcctgaact 960
ttgtgggcca aaaatggtgt gctcacaact attgttgacg acttcttga tgttgcggga 1020
tcaaaaagag aacttgaaaa ccttgtcatg ttgggttga tgtgggacga gcatcacaaa 1080
gttgagttct actcagaaca agtagagatt atattttctt caatttatga ctcagttaac 1140
caacttggtg aaaagcgctc ttgtgtacaa gaccgcagta ttaccaaaac cctagtagaa 1200
atatgggttg atttgctaaa gtctatgatg acagaggtag agtggcggtt gagcaaatat 1260
gtgccaacag agaaggaata catgataaat gcacttttaa tatttggaat aggccccatt 1320
gtattgcccag cattatatt ttgttgggcca aagatctcag agtctattgt taaagatcca 1380
gaatatgatg aattgttcaa actgatgagc acatgtgtgc gcctcttgaa tgatgttctc 1440
acttttgaga gggagtacaa cgagggcaag ttgaatagtg tttctctcct cgttcttctc 1500
gggtggcccca ttcccatatc agacgccaaa aggaatttac agaagcccat agacacatgc 1560
agaagagacc tcctaagttt agttcttctg taagaaagtg ttgttcttag gccctgcaag 1620
gaattatttt ggaaaaatgt caaggtgtgc tacttctct actcgacgac ggaatgggtt 1680
agctcacaag tggagagggc taaagaagtg gatgcgggtga tcaatgagcc actaaagcta 1740
caaggaagtc atacgctggt gtcctgatgtg tga 1773

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SEQ ID NO: 47      moltype = DNA length = 2232
FEATURE
source             Location/Qualifiers
                   1..2232
                   mol_type = other DNA
                   organism = Populus trichocarpa

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SEQUENCE: 47
atgcagaact ttcattgaac taaggaaagg atcaagaaga tgtttgataa gattgaattg 60
tcagtgtctt catatgacac tgcttgggtg gcaatggctc catctccaga ttgtccggaa 120
actccttggt ttccagagtg cacaaaaatg attttggaaa atcaacttgg tgatggctcc 180
tggagtcttc ctcatggcaa tccattatta gttaaggatg ctctttctac tacattagcg 240

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tgcaccccttg cattgaagcg atgggggtatc ggtgaagaac aaataaataa aggccttcga 300
tttatttgagt tgaattccgc ttccagttacg gataacgagc aacataaacc aattggattt 360
gatataaatat ttccctggcat gattgaaatg gccatagatt tggatttgaa cctccctttg 420
aagccgcagag atataaatcc catgctccac aggagggcctt tggagcttac aagtggcgg 480
ggcaagaact tggagggaag aagagcctac ttagcatatg tttcggaagg aattggaaaa 540
ttacaggattt gggaaatggc catgaaatat caaagaaaga atggatcact gttcaattca 600
ccatccacca cagcagctgc ctttattcat attcaagatg ctgagtgtct ccattatatt 660
cgttcactct tacagaagtt tgggaatgca gttccaaacca tttatccttt ggatatatat 720
gctcgtcttt ctatgggtga tgctcttgaa aggttgggaa tcgatcgga ttttaggaag 780
gaaagaaaat ttgttttggg cgaacatac cgattttggt tgcaggggga ggaagagata 840
ttttctgata atgccaactt tgctttggca tttaggatat tacgtttgaa cggatatgat 900
gtctctctag aagatcattt ctctaattca ctgggaggat atttgaagga ttccggagct 960
gccttagagt tgtacagagc ttgacagcta agttatccag atgaatcact tctggaaaaa 1020
caaaattctc ggacaagcta tttcctgaaa cagggattat ccaacgtttc actttgtgga 1080
gataggcttc gtaaaaatat tatcggagag gtgcatgatg ctctcaattt ttctgacct 1140
gcaaatttgc aacgcttagc taccagaaga agaattaaac attatgctac agatgatacg 1200
aggattttga aaacttcgta tcgtttgttcg actattggta accaggattt tctcaaatg 1260
gctgtagaa actcaatat ctgtcaatca atacagcgtg aagaatttaa acatcgcag 1320
aggtgggttg tagagaggag actggacaag ctaaagtgtg ctaggcagaa ggaggcctac 1380
tgttactctc ctgctgcagc aactctcttc gctccagaac tatctgatgc acgcatgtca 1440
tgggcaaaaa atggtgtgct tactactgtt gttgatgact tctttgatgt tgggtgttct 1500
gaagaagaac tggtaaacct tattgaattg attgagaggt gggatgtcaa tggcagtgtc 1560
gatttttgtt ctgaggaagt tgagatcata tattcggcaa ttcacagcac tataagttag 1620
ataggagaca aatctttcgg atggcaagga cgcgatgtga aaagtacagt tatcaagatt 1680
tggttgattt tgctcaaatc actgttgaca gaagcacaaat ggtcaagtaa caaatcagt 1740
ccgacccctg atgaatatat gacaacttgc catgtatcgt tcgctctagg gctattgtt 1800
cttcagctc tgtattttgt ggggcctaag ctttcagagg aggttgcctg acatcctgaa 1860
ttgcttaatc tatacaaggt tacagcact tgcgggcgtc tgctcaatga ctggagaagc 1920
tttaagagag aatctgaaga agggaaattg aatgccatct cattgtacat gattccagc 1980
ggtgtgtgct caactgaaga agagaccatc gaacatttta aaggattgat cgacagccag 2040
agaagacaat tgcctcaatt agttttgcag gaaaaggata gtataattcc tagacctgc 2100
aaggatttgt ttgtgaacat gataaaaata ttgcacacgt tctacatgaa ggatgatgga 2160
ttcacttcaa acgagatgag aaatgtgttc aaggcaataa taaatgaacc catctctcta 2220
gatgaattat aa 2232

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SEQ ID NO: 48      moltype = DNA length = 2952
FEATURE            Location/Qualifiers
misc_feature       1..2952
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..2952
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 48
atggaatttg atgaaccatt ggttgacgaa gcaagatctt tagtgcagcg tactttacaa 60
gattatgatg acagatacgg cttcgggtact atgtcatgtg ctgcttatga tacagcctgg 120
gtgtctttag ttacaaaaaac agtcgatggg agaaaaacat ggcttttccc agagtgtttt 180
gaatttctac tagaaacaca atctgatgcc ggaggatggg aaatcgggaa ttcagacca 240
atcgacggta tattgaatac agctgcatcc ttacttgctc taaaacgtca cgttcaaaact 300
gagcaaatca tccaaactca acatgacctt aaggatctag caggtagagc tgaacgtgcc 360
gctgcatctt tgagagcaca attggctgca ttggatgtgt ctacaactga acacgtcgg 420
tttgagataa ttgttctctg aatgctagac ccattagaag ccgaagatcc atctctagt 480
ttcgattttc cagctaggaa acccttgatg aagattcatg atgctaagat gtagtagattc 540
aggccagaat acttgtatgg caaacaacca atgaccgcct tacattcatt agaggctttc 600
ataggcaaaa tcgacttcga taaggtaaga caccaccgta cccatgggtc tatgatgggt 660
tctccttcac ctaccgcagc ctacttaatg cacgcttcac aatgggtagg tgactcagag 720
gcttacctta gacacgtgat taaacacgca gcaggggcagg gaactggtgc tgtaccatct 780
gctttcccat caacacattt tgaagtcatc ttgattctta ccacattgtt tagagctgga 840
ttttcagctt ctcatcttgc ctgtgatgag ttgaacaagt tggctcagat acttgagggc 900
tcattcgaga aggaaggtgg ggcaatcggg tacgctccag ggtttcaagc agatgttgat 960
gatactgcta aaacaataag tacattagca gtccttgga gagatgctac accaagacaa 1020
atgatcaagg tatttgaagc taatacacat tttagaacat accctggtga aagagatcct 1080
tctttgacag ctaattgtta tgctctatca gccttactac accaaccaga tgcagcaatg 1140
tatggatctc aaattcaaaa gattacaaa tttgtctgtg actattgggt gaagtctgat 1200
ggtaagatta aagataagtg gaacacttgc tacttgtacc catctgtctt attagttgag 1260
gttttggttg atcttgttag ttattggag cagggtaaat tgctgatgtt ttggatcaa 1320
gagcttcaat acagatgcgc catcacattg ttccaagcat gtttaaggcc attactagac 1380
caagatgccg aaggatcatg gaacaagtct atcgaaagcca cagcctacgg catccttatc 1440
ctaactgaag ctaggagagt ttgtttcttc gacagattgt ctgagccatt gaatgaggca 1500
atccgtagag gtatcgcttt gcgcgactct atgtctggaa ctgaagctca gttgaactac 1560
atttggatcg aaaaggttag ttacgcacct gcatattga ctaaatccta tttgttagca 1620
gcaagatggg ctgctaagtc tcttttaggc gcttccgtag gctcttcttt gtggactcca 1680
ccaagagaag gatggataa gcatgtcaga ttattccatc aagctgatgt attcagatcc 1740
cttcagaaat gggcaattag agctccatg attgaagcag ctttgttcc accacttcta 1800
agagcacata gactagacgt ttccctaga caagatgtag gtgaagacaa atatcttgat 1860
gtagttccat tcttttgagc tgccgctaac aacagagata gaacttacgc ttccactcta 1920
ttcctttacg atatgtgtt tatcgcaatg ttaaaactcc agttagacga attcatggag 1980
gccacagcgg gtatcttatt cagagatcat atggatgatt tgaggcaatt gattcatgat 2040
cttttggcag agaaaacttc cccaaagagt tctggtagaa gtagtcaggc cacaaaagat 2100

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gctgactcag gtatagagga agacgtgtca atgtccgatt cagcttcaga ttcccaggat 2160
agaagtccag aatocagatt ggtttttcagt gcattgagta cctttacaaa acatgtcttg 2220
caacacccat ctatacaaa tgctctgtga tgggatagaa aactacttgc tagagagatg 2280
aaggcttact tacttgctca tatccaacaa gcagaagatt caactccatt gtctgaattg 2340
aaagatgtgc ctcaaaagac tgatgtaaca agagtttcta catctactac taccttcttt 2400
aactgggtta gaacaacttc cgcagacccat atatcctgcc catactcctt ccactttgt 2460
gcatgccatc taggcgcagc attgtcacct aaagggtcta acggtgattg ctatccttca 2520
gctggtgaga agttcttggc agctgcagtc tgcagacatt tggccaccat gtgtagaatg 2580
tacaacgato ttggatcagc tgaacgtgat tctgatgaag gtaatttgaa ctcttggac 2640
ttccctgaat tcgcccattc cgcaggaaac ggagggatag aaattcagaa ggccgctcta 2700
ttaagggttag ctgagtttga gagagattca tacttagagg ccttcgcgtc tttacaagat 2760
gaatccaata gagttcacgg tcacgcccgt ggtgatgaag ccagattgtc cagaaggaga 2820
atggcaatcc ttgaattctt cgcacagcag gtagatttgt acggtcaagt atcgtcatt 2880
agggatattt ccgtcgtat tcctaaaaac gaggttgaga aaaagagaaa attggatgat 2940
gctttcaatt ga 2952

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SEQ ID NO: 49      moltype = DNA length = 2646
FEATURE           Location/Qualifiers
misc_feature       1..2646
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..2646
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 49
atggcttcta gtacacttat ccaaaacaga tcatgtggcg tcacatcatt tatgtcaagt 60
tttcaaatct tcagagggtca accactaaga ttctctggca ctagaacccc agctgcagtt 120
caatgcttga aaaagaggag atgccttagg ccaaccgaat cgtactaga atcatctcct 180
ggctctggtt caataagaat agtaactggc ccttctggaa ttaaccctag ttctaaccgg 240
cacttgcaag agggttctct gactcacagg ttaccaatca caatggaaaa atctatcgat 300
aacttccaat ctactctata ttgttcagat atttggtctg aaacactaca gagaactgaa 360
tgttgtctac aagtaactga aaacgtccag atgaatgagt ggattgagga aattagaatg 420
tactttagaa atatgacttt aggtgaaatt tccatgtccc cttacgacac tgcctgggtg 480
gctagagttc cagcgttgga cggttctcat gggcctcaat tccacagatc tttgcaatgg 540
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gatagagttt gtaatacttt agcctgtgtg attgctgtga aaacatgggg tgttggggca 660
caaaacgttg aaagaggaaat tcagttccta caatctaaca tatacaagat ggagggaagt 720
gacgctaata atatgccaat aggatccgaa atcgtattcc ctgctatgat ggaagatgcc 780
aaagcattag ttgttgattt gccatacgat gctactattt tgcaacagat ttcagccgaa 840
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ttgatcaacg gtttgccaga gcaagctaaa atcttggtta tgggcttata caaaacagtt 2040
aacacaattg cagagggaag atctatggca cagaaaagag acgtccatca tcatttgaaa 2100
cactattggg acaagttgat aacaagtggc ctaaaaggag ccgaatgggc agagtccaggt 2160
tacgtcccaa catttgatga atacatggaa gtagetgaaa tttctgttgc tctagaacca 2220
attgtctgta gtaccttgtt ctttgcgggt catagactag atgaggatgt tctagatagt 2280
tacgattacc atctagtatt gcatttggtg aacagagtcg gtagaatctt gaatgatata 2340
caaggcatga agagggaggg ttccacaagg atagatctcat cagttcaaat ctacatggag 2400
gaacatccat ctgttccatc tgaggccatg cgcagtcgct atcttcaaga gttagtgtat 2460
aattcaatgc agcaattgac atacgaagtt cttaggttca ctgcccgttc aaaaagtgtg 2520
aagagaatcc acttgaatat ggctaaaatc atgcatgcct tctacaagga tactgatgga 2580
ttctcatccc ttactgcaat gacaggattc gtcaaaaagg ttcttttcga acctgtgcct 2640
gagtaa 2646

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SEQ ID NO: 50      moltype = DNA length = 2952
FEATURE           Location/Qualifiers
source            1..2952
                  mol_type = other DNA
                  organism = Phomopsis amygdali

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SEQUENCE: 50
atggagttcg atgaaccact tgtggacgag gcgaggtcct tggtoaaaag aacctgcaa 60
gattatgacg accgctatgg ctttggcact atgagctgtg cggcctatga cacagcatgg 120

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gtatcgctgg tgactaaaac agtcgatggg cgtaaacaat ggttggtccc tgagtgttcc 180
gaattttccc tagaaacgca gtcgatgctt ggcggctggg aaatcgccaa cagcgacccc 240
atcgatggga tccctaacac tgctgcttca ctgctggcat tgaagcgcca cgtccaaaca 300
gagcagatta ttcagccgca acacgacccat aaagacctgg ccgggctgtc ggaaagagcg 360
gcggcgctct tgcgagcaca gttggcggtc ctggatgtgt cgacaacgga gcatgtgggc 420
ttcgaaatca tcgtcccggc catgctcgac cctctcgagg ccgaagaccc gtctttggtg 480
ttcgacttcc cagcacgcaa accactgatg aagatccacg acgctaagat gtcgcgatcc 540
cgaccagagt acctctacgg taaacagccg atgacggcat tgcattcgct cgaggccttt 600
atcgggaaaa tagacttcga caaagtacgg catcacagga cacacgggtc gatgatgggg 660
tcgccctcgt cgcagggctg atacctgatg catgctttcc agtgggacgg cgactctgag 720
gcctatctac gccatgtcat caagcacgca gctggccagg gcacggagc tgttccgagt 780
gcatttcctt cgacgcattt cgagtcttct tggattttga caacattggt tcgagctggg 840
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tcatttgaga aagaaggggg agccatcggt tatgctcctg ggtttcaagg agatgtggat 960
gataccgcaa agaccatctc cactttgggt gtgcttggga gagatgccac tccccggcaa 1020
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tacggcagcc agatccagaa gatcacaaag tttgtttgtg actactgggt gaaaagtgc 1200
ggcaaaatca aggacaagtg gaatacctgc tacttgatc catcggtcct cctcgtcgag 1260
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gagctgcaat acagggctgc cattacgtta ttccaggcct gcttgcgacc gctacttgat 1380
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cttacggagg cgcggcgagt atgctttttt gaccgtctga gtgagcctct gaatgaggct 1500
attcgacgcg ggatttgcgt tcgagattcg atgagcggtg ctgaagctca gctgaattat 1560
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gctcgggtgg cggcaaaagtc cccgcttggc gcttccgttg gatccagcct ttggacgcct 1680
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cgtgcgcata ggtcggtagt atttccacgc caagacgtcg gcgaggacaa gtacctggac 1860
gttgtgccgt tcttctggac ggcgcgcaat aaccgcgcat gcacgtacgc atccactctg 1920
tttctgtatg acatgtgctt tatcgccatg cttaacttcc agctggatga gttcatggag 1980
gtacacagcg gatacctctt cccgggacct atggatgatt tgcgccaaat catccacgac 2040
ctgcttgcgc aaaagacgag ccccaagtca tcgggcagaa gttagccaaag aaccaagac 2100
gcggaactcg gcactgaaga agacgtttct atgagcgact cagcgtcaga ctccaggac 2160
cgcagccctg aatacgacct ggtcttctct gcgctctcta ccttcaccaa acatgtcctg 2220
cagcaccctt caatccagtc agccagtgtc tgggatagga aactactcgc tcgcgagatg 2280
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gcgtgtcacc tcggcgccgc gctgagcccc aagggcagca acggcgactg ttaccgctca 2520
gcccgtgaaa agtctctcgc ggcgcgcgta tggccccatt tggccacgat gtgcccgatg 2580
tacaatgact tgggactcgc ggagcgcgac agtgacgagg gaaatttgaa ttcaactcag 2640
tttcccgagt tcgcgcgact agcggggaat ggtgggattg agatccagaa agctgccttg 2700
ctcaggctgg ccgagttcga acgcgactcg tatctcgagg ctttccggcg acttcaggat 2760
gaaagcaacc gcgttcacgg accggtgtgt ggggatgaag ccagactcag caggcggcgc 2820
atggccatcc ttgagttctt tgcccagcag gtggaacttg atggccaggt ctacgttatt 2880
cgcgatatct cggccaggat tccaaagaac gaggttgaga agaaaaggaa actagatgat 2940
gctttcaatt ag 2952

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SEQ ID NO: 51      moltype = DNA length = 2646
FEATURE            Location/Qualifiers
source              1..2646
                    mol_type = other DNA
                    organism = Physcomitrella patens

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SEQUENCE: 51
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tttcagattt ttccaggggca acctctacgt tttccaggca ctagaactcc tgctgcagtt 120
caatgcctaa agaagcgtcg atgtttgcca cctactgaat cagtcctcga gagctctcct 180
ggtagcggat cttacaggat tgtaactgga cctccggca tcaatccttc ttcaaacggc 240
cacttgcaag aggggtccct tactcacaga cttccgatac ccatggaaaa atccattgat 300
aaacttcagt ctactttgta cgtatcagac atatggtcag aaaccttgca aagaacggaa 360
tgtttgttgc aggtgactga gaatgtacag atgaacgagt ggattgagga aatcagaatg 420
tacttccgaa atatgacact gggggaaata tccatgtctc catcacgac agcttgggta 480
gcgcgagtcg cagcgctgga tggtccacat ggccctcagt tccatcggtc tttgcagtgg 540
attattgata atcagctccc ggtggcgcat tggggtgaa cgtctctttt ccttggatac 600
gatcgcgttt gcaacactct cgcctgtgta attgccctga aaacttgggg tgttggggct 660
cagaacgtag agcgtggaat ccagtttctg caatctaaca tctacaaaaa ggaggaaagt 720
gacgccaatc atatgccgat tggatttgag attgtcttcc cagcgatgat ggaagatgcc 780
aaggcactgg gactggattt accatacagat gccactatct tgcaacaaat ctcggtgaa 840
agagagaaga aaatgaaaaa gattcctatg gcgatgggtg acaagtaccc cactactttg 900
ctgcattctc tgggaaggcct gcaccgggaa gtggactgga acaagctcct ccagctacag 960
tccgagaatg gctcctttct gtattcaccg gcactccact catgcgcat tatgtacaca 1020
aaagatgtga agtgcctcga ctacttgaa cagctcctca tcaagttcga ccacgcttgt 1080
ccaaacgtgt accccgttga tctcttcgag cgtttgtgga tggtagaccg cctacaaagg 1140
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aaaaaggcca gaaccttttc cagaaaacttt ttgagaacca agcatgaaaa caatgaatgc 1500
ttcgacaagt ggataatcac gaaggatcta gcgggcgagg tggaatacaa tctcacattc 1560
ccttggtatg ctageccttc tctgtctttag catcgcacct acttggaacca atatgggatt 1620
gatgatattc ggattggcaa gtcgtctctac aaaatgccgg ccgtcaccaa cgaagtgttt 1680
ctcaaattgg ccaaagccga cttcaacatg tgccaagctc ttcacaagaa ggaactcgag 1740
cagggtcatca aatggaatgc cagctgcctaa tttagagacc tcgagtttgc tagacagaaa 1800
tccgtggagt gctacttcgc aggcgtgca accatgtttg agcccgaaat ggtgcaggcg 1860
aggctcgttt gggcacgctg ttgcgtgctc accacgcttc tagacgatta cttcgatcac 1920
ggtacacctg tggaagagct tccgggtttt gtgcaggccg taaggacttg gaatcccgag 1980
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cattactggg acaaatgat cacttcagct ttgaaagaag ccgaatgggc agagtccggc 2160
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gagcatccaa gtgtgccttc agaggccatg gccatcgctc atctgcagga attggtcgac 2460
aactccatgc aacagctgac atacgaagtg ctgcgcttca ctgcagtcgc gaagtccgtg 2520
aagagaatcc atttaacat ggcgaagatc atgcacgctt tctacaagga cactgatggg 2580
ttttcgtcac tgacagccat gacagggttt gtgaagaagg tgctcttcga gccagtacct 2640
gaatag

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SEQ ID NO: 52      moltype = DNA length = 1542
FEATURE            Location/Qualifiers
misc_feature        1..1542
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1542
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 52
atggatgctg tgacgggttt gttaaactgtc ccagcaaccg ctataactat tgggtggaact 60
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agatcccaat caaatcatct tccaagagtg cctgaagtcc cagggtgttc attgttagga 180
aatctgttac aattgaagga gaaaaagcca tacatgactt ttacgagatg ggcagcgaca 240
tatggacctc tctatagtat caaaactggg gctacaagta tgggttgtgt atcatctaat 300
gagatagcca aggaggcatt ggtgaccaga ttccaatcca tatctacaag gaacttatct 360
aaagccctga aagtacttac agcagataag acaatggctc caatgtcaga ttatgatgat 420
tatcataaaa cagttaagag acacatactg accgcgctct tgggtcctaa tgcacagaaa 480
aagcatagaa ttcacagaga tatcatgatg gataacatat ctactcaact tcatgaattc 540
tgaaaaaaca acccagaaca ggaagaggtg gaccttagaa aaatctttca atctgagtta 600
tcgggcttag ctatgagaca agccttagga aaggatgttg aaagtttgta cgttgaagac 660
ctgaaaatca aattgcaaga agacgaaatc ttcaagtcc ttgttgttga tccaatgatg 720
ggagcaatcg atgttgattg gagagacttc ttccataacc taaagtgggt cccaaacaaa 780
aagttcgaaa ataactattc acaaatgtac atcagaagag aagctgttat gaaatcttta 840
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cttttatctg aagctcaaac tttaaccgat cagcaactat tgatgtcctt gtgggaacca 960
atcattgaaat cttcagatac aacaatggct acaacagaat gggcaatgta cgaattagct 1020
aaaaacccta aattgcaaga taggtgtgac agagacatta agtccgtctg tggatctgaa 1080
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ctaggcggtc accatgttcc tgctggcaca gaacttgccg ttaacatcta cggttgcaac 1260
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ggttccttgc aagccctttt aactgcatct attgggattg ggagaatggg tcaagagttc 1440
gaatggaaac tgaaggatat gactcaagag gaagtgaaca cgataggcct aactacacaa 1500
atgttaagac catgagagcg tattatcaaaa ctaggatct aa 1542

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SEQ ID NO: 53      moltype = DNA length = 1530
FEATURE            Location/Qualifiers
misc_feature        1..1530
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1530
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 53
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atcttttttc tcaaaaagtt acttagtttt agtaggaaaa acatgtcaga agtttctact 120
ttgccaagtg ttccagtagt gcctggtttt ccagttattg ggaatttgtt gcaactaaaag 180
gagaaaaagc ctcataaaac ttccactaga tggtcagaga tatatggacc tatctactct 240
ataaagatgg gttcttcacg tcttattgta ttgaacagta cagaaactgc taaggaaagca 300
atgggtcacta gattttcatc aatatctacc agaaaaattg caaacgcctt aacagttcta 360
acctgcgata agtctatggg cgccacttct gattatgatg acttcacaaa attagttaag 420
agatgtttgc taaatggact tcttgggtgt aatgctcaaa agagaaaaag acactacaga 480
gatgctttga ttgaaaatgt gagttccaag ctacatgcac acgctagaga tcatccacaa 540
gagccagtta acttttagagc aattttcgaa cacgaattgt ttggtgtagc attaaagcaa 600
gccttcggta aagacgtaga atccatatac gtcaaggagt taggcgtaac attatcaaaa 660
gatgaaatct ttaaggtgct tgtacatgat atgatggagg gtgcaattga tgtagattgg 720

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agagatttct tcccatattt gaaatggatc cctaataagt cttttgaagc taggatacaa 780
caaaagcaca agagaagact agctgttatg aacgcactta tacaggacag attgaagcaa 840
aatgggtctg aatcagatga tgattgttac cttaacttct taatgtctga ggctaaaaca 900
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accttagtca caactgaatg ggccatatac gagctagcca aacatccatc tgtgcaagat 1020
aggttgtgta aggagatcca gaacgtgtgt ggtggagaga aattcaagga agagcagttg 1080
tcacaagttc cttaccttaa cggcgttttc catgaaacct tgagaaaata ctcacctgca 1140
ccattagttc ctattagata cgcccacgaa gatacacaaa tcggtggcta ccatgttcca 1200
gctgggtccg aaattgtctat aaacatctac gggtgcaaca tggcaaaaa gagatgggaa 1260
agaccagaag attggtggcc agaagattc ttagatgatg gcaaataatga aacatctgat 1320
ttgcataaaa caatggcttt cggagctggc aaaagagtgt gtgccgtgct tctacaagcc 1380
tcocctaatg ctggtatcgc tattgtgtaga ttggtccaag agttcgaatg gaaacttaga 1440
gatggtgaag aggaaaatgt cgatacttat gggttaacat ctcaaaagtt ataccacta 1500
atggcaatca tcaatcctaa aagatcctaa 1530

SEQ ID NO: 54      moltype = DNA length = 1578
FEATURE            Location/Qualifiers
misc_feature       1..1578
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..1578
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 54
atgagtaagt ctaatagtat gaattctaca tcacacgaaa ccctttttca acaattggct 60
tgggtttgg accgatggc attgatggat gttcactggt tgatctacgt tgccttcggc 120
gcatggttat gttcttatgt gatacatggt ttatcatctt cctctacagt aaaagtggca 180
gttggtggat acaggtctgt attcgaaact acatggttgc ttagacttag attcgtctgg 240
gaaggtggct ctactcatag tcaaggggtac aataaagtta aagactctat tttccaagtt 300
aggaaattgg gaactgatat tgcattata ccaacctaac atatttgatga agtgagaaaa 360
ttgtcacagg ctgatattga atcagttgaa cctttcatta atgattttgc aggtcaatac 420
acaagaggca tgggtttctt gcaatctgac ttacaaaaacc gtgttataca acaaagacta 480
actccaaaat tggtttccct gaccaaggtc atgaagggaag agttggatta tgccttaaca 540
aaagagatgc ctgatattga aaatgacgaa tgggtagaag tagatatcag tagtataatg 600
gtgagattga tttccaggat ctccgccaga gtctttctag ggctgaaca ctgtcgtaac 660
caggaatggt tgactactac agcagaatat tcagaatcac ttttcattac aggggtttatc 720
ttaagagtgt taccctcatat cttaaagcca ttcctcgccc ctctattacc ttcatacagg 780
actctactta gaaacgtttc aagtgttaga agagtcacgt gtgacatcat aagatctcag 840
caaggggagt gtaacgaaga tatactttcc tggatgagag atgctgccac aggagaggaa 900
aagcaaatcg ataactctgc tcagagaatg ttaattcttt ctttagcatc aatccacact 960
actgcgatga ccatgacaca tgcctatgtg gatctatgtg cttgccctga gtacattgaa 1020
ccattaagag atgaagttaa atctgttgtt ggggcttctg gctgggacaa gacagcgtaa 1080
aacagatttc ataagttgga tccttctcta aaagagtcac aaagattcaa cccgatttc 1140
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ccatctggaa cactgattgc tgttccatca cagcgaatgt tgcaagattc tgcacatgtc 1260
ccaggtccaa cccacatcac tgaatttgat ggattcagat atagtaagat acgttctgat 1320
agtaactacg cacaaaagta cctattctcc atgaccgatt cttcaaacat ggctttcgga 1380
tacggcaagt atgcttgtcc aggtagattt tacgcgtcta atgagatgaa actaacatta 1440
gccattttgt tgctacaatt tgagttcaaa ctaccagatg gtaaaaggtcgc tcctagaaat 1500
atcactatcg attctgatat gattccagac ccaagagcta gactttgcgt cagaaaaaga 1560
tcacttagag atgaatga 1578

SEQ ID NO: 55      moltype = DNA length = 1500
FEATURE            Location/Qualifiers
misc_feature       1..1500
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..1500
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 55
atggaagatc ctactgtctt atatgcttgt cttgccattg cagttgcaac tttcgttgtt 60
agatggtaca gagatccatt gagatccatc ccaacagttg gtggttcoga tttgcctatt 120
ctatcttaca tcggcgcaact aagatggaca agacgtggca gagagatact tcaagaggga 180
tatgatggct acagaggatc tacattcaaa atcgcgatgt tagaccgttg gatcgtgatc 240
gcaaatggtc ctaactagc tgatgaagtc agacgtagac cagatgaaga gttaaacttt 300
atggacggat taggagcatt cgtccaaact aagtacacct taggtgaagc tattcataac 360
gatccatacc atgtcogatat cataaagaaa aaactaacia gaggccttcc agcgcgtgct 420
cctgatgtca ttgaagagtt gacacttgcg gttagacagt acattccaac agaaggtgat 480
gaatgggtgt ccgtaaaactg ttcaaaaggcc gcaagagata ttgttgctag agcttctaatt 540
agagtctttg taggtttgccc tgcttgccaga aaccaaggtt acttagattt ggcaatagac 600
tttacattgt ctgttgtcaa ggatagagcc atcatcaata tgtttccaga attgttgaag 660
ccaatagtgt gcagagttgt aggtaacgcc accagaaatg ttcgttagagc tgttcccttt 720
gttgctccat tgggtggagga aagacgtaga cttatggaag agtacgggtg agactgggtc 780
gaaaaaccta atgatattgt acagtggtga atggatgaag ctgcatccag agatagttca 840
gtgaaggcaa tcgcagagag attgttaatt gtgaacttcg cggctattca tactcatca 900
aacactatca ctcatgcttt gtaccacctt gccgaaatgc ctgaaacttt gcaaccactt 960
agagaagaga tcgaaccatt agtcaaaagag gagggctgga ccaaggctgc tatgggaaaa 1020
atgtggtggt tagattcatt tctaagagaa tctcaaagat acaatggcat taacatcgta 1080

```

tctttaaacta	gaatgggga	caagaatatt	acattgaagt	atggcacatt	tttgcacaaa	1140
ggctactctag	tggccgcttc	agcgtattct	actcatagat	atgatcgctg	ctacgctgat	1200
gccttagtat	tcgactcttc	cagatctcca	cgtatgagag	cagagaaagg	tgaaggtgat	1260
aagccacaggt	tcgttaatac	ttcaagtctg	tcagctccat	ttggtcacgg	aaagcatgct	1320
ttgcagagac	gattcttcgc	cgcaaacgaa	tgaagaccaa	tgttggtcta	catgtgtcta	1380
aactatgatg	taaaagtTgcc	tggtagcggg	aaacgctccat	tgaacatgta	ttggggctca	1440
acaggtttgc	ctgcaccagc	aggccaagta	ttggtcagaa	agagacaagt	tagtctataa	1500

```

SEQ ID NO: 56          moltype = DNA length = 1542
FEATURE                Location/Qualifiers
source                 1..1542
                        mol_type = other DNA
                        organism = Stevia rebaudiana

```

SEQUENCE: 56				
atggatgcgcg	tcaaccggttt	gctgacagatt	cgcggcaaccg	caataaacat cggcgctgcag 60
gcgcgtgcacg	tcgcgcgtcgc	tctgatattc	tggtacctca	aaagctacac atctgcacgc 120
aggagccaat	caaaccatct	ccctcgggtt	cccgagggtac	ctgggtgtgcc attattgggg 180
aattttatgc	agttgaagga	gaagaaacct	tacatgactt	ttaacaagat ggcgcgcaact 240
tatggtccga	ttttatctc	taaaaccgga	gcgaactcta	tgggtgtcgt cagttccaat 300
gaaattgcaa	aggaggcatt	ggttaccaga	tttcaatcta	tctcaaccag aaacctatca 360
aaagccattca	aggtttatcc	acgcagataaa	accatgggtg	cgatgagbta tttatgatgat 420
tatcataa	ctgtcaaacg	catatactgc	accgcgtgtt	tgggcaacaa tgcctagaag 480
aaacacccga	tccatagggga	catcatgatg	gataatatat	caacccaact tcatgaatt 540
tttgaataat	atctctgaaca	agaggaagtg	gatctaagga	aaatatctca atccgaact 600
tttggaattg	ctatgagaca	agcatggga	aaaggtgtgg	agagcttata tgttgaggat 660
cttaaaatca	ccatgaaccg	agacgagata	tttcagggtat	tggttgttga cccgatgatg 720
gaagtgcattg	acgtgcagca	gagagatttc	ttcccgattc	taaagttgggt ccgcgaataaa 780
aaggtttgaaa	acacgactca	acaaattgat	atccggagag	aagctgtgat gaagtctct 840
attaaagaac	ataaaaaaacg	tattgcattc	ggagagaaat	taaacagcta cattgattac 900
tgtcatctcg	aagcacaaac	gtttaacgat	caacaactac	tattgtctct attggaacct 960
attattgaat	catcacagac	catcattggt	acaactgaat	gggctatgta tgaacttgc 1020
aaaaacccca	aacttcagga	tcggtttgat	cgggatatca	aaagttgttg cgggtcagag 1080
aagattacag	aagaacactc	gtctcaactg	ccatacataa	ctgcctattt tcatgaacct 1140
ttgagaagcg	atagctcagt	tcctataatt	ccattaaagc	atgctgcattg agacacagc 1200
ttaggagggt	accattgtgc	agctggaacc	gagctagcgg	taaacaatta tggatgtaac 1260
atggatga	attgtgtgga	gaatcctgaa	gagtggaact	gcagagagat catgaaggaa 1320
actgaagaag	tagatttcca	gaaaacaatg	cggtttgcag	tgtgaaagcg cgtatgtcgt 1380
ggttcgcttc	aagcattggt	gactgcttcc	attggaattg	gaagaatggt gcaagagatt 1440
gagtggaagaa	tgaagayatt	gacccaagaa	gaagtttaata	cgattgggct taccgaccag 1500
atgcttcctc	cactgcgqqc	cataataaag	cccaagatqa	

```
SEQ ID NO: 57      moltype = DNA length = 1530
FEATURE            Location/Qualifiers
source             1..1530
                  mol_type = other DNA
                  organism = Arabidopsis thaliana
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SEQUENCE: 57						
atggcgctctt	cttccatgat	ctccattctc	cttggtcttg	tatatctctc	cttcattctc	60
atctctctct	tcaagaacct	ttctctcttc	tcagagaaga	acatgctcga	agtcctcact	120
ctccccctctg	ttccagtggt	accagggttt	cctgttatatg	ggaacttgct	gcaactaaaa	180
gagaagaanaa	ctccacaagc	tttctactaga	tggtcagaga	tttatggctc	tatttactct	240
ataaagatgg	gttctctctc	ttctattgtg	ctcaattcta	ctgagactgc	caagaaggcc	300
atggtgacgc	ggtttctcgtc	tatctccaacg	aggaagttgt	caaatgcgtt	gacgtctctt	360
actcttgaca	aatctctatgtt	tgctactagt	gatttatgatg	atttccacaa	gttggtgaaa	420
cgggtgtctct	tgaacggtct	tttgggtgct	aatgcacaga	acgaagaaa	acattacaga	480
gatgcactca	ttgaaaatgt	gtcttccaag	ttgcattgcc	atgctaggga	ccatccacaa	540
gaacctgttaa	acttcagagc	tatatcttgag	ctgagcctt	tcggtgtgac	atttgaagcaa	600
gcttttggga	aagatgttga	atcatcttat	gttaagaagc	tcggtgtgac	tttgcgaaa	660
gacgagatct	tccaagtttt	agtcacatgac	atgatggaa	gtgcgaattga	tgttgattgg	720
agagactctt	ctccatactt	gaattggatt	ccaaataaaa	gttttgagc	agaataccag	780
caaaagcata	aacgtagact	cgcggtgatg	aatgctctga	ttcaagatgc	actgaagcag	840
aatggttctag	aatcggatga	tgtattgctat	ctcaactctt	tgatgtcggga	agcgaanaac	900
cttaaccaag	agcaaatatgc	tatcttgggt	tgggagacga	tatcagagac	agctgacact	960
actttgggtta	acaactgaatg	ggcactctat	gagctcgcta	agcatccaga	tgctccaagt	1020
cgctctgtgta	aagaatactca	aagattctcgc	ggaggagaaa	agttccaaga	agagcaattg	1080
ttccaagtct	cttatctcaa	tgaggtattc	ctgaanaacg	ttaggaagata	cagtcctgct	1140
ctctatgctc	ccattcgcta	tgcccacgaa	gatacgcaaa	tcgggaagta	tcattgctct	1200
gcaggaaagt	agattgcaat	aaacatctat	ggatgcaaca	tggatagaaa	gcgttgggag	1260
agaccacagc	actgggtggcc	ggagcgggtt	cttgatgatg	gcaaaatcga	aacgtcggat	1320
cttccacaaga	caatgcggct	tggaagcggga	aagagggttt	gtgctgggtg	ttcttcaaga	1380
ttctctcatgg	caggcattgc	cattggggagg	ttagtgcgaag	aattcagagt	gaagctttaga	1440
gcagggtgaag	aagagaatgt	ggatacatat	ggcttgacct	ctcagaagct	ttatctctct	1500
atggctatta	tcaatccaag	cgctctctaa				1530

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SEQ ID NO: 58      moltype = DNA  length = 1578
FEATURE            Location/Qualifiers
source             1..1578
                  mol_type = other DNA
                  organism = Gibberella fujikuroi

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-continued

SEQUENCE: 58

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atgagtaagt ccaacagcat gaacagtacc agccatgaaa cgttattcca gcagctcgtc 60
ttaggtcttg acagaatgcc gctaattggac gttaactggc tgatctacgt ggctttggc 120
gcttggttat gctcttatgt catccatgtc ctatcgctct cttctacagt caaagtggcc 180
gtcgtaggct accgcagcgt ctttgagcct acatggcttc tccgtttgcg ctttgtttg 240
gaaggggcat ctatcatcgg ccaaggctac aacaaattta aagactctat cttccaggtg 300
cgaaagcttg gtaccgatat cgtcatcctc ccgccaaact acatcgatga ggtcagaaaag 360
ctgtcccaag acaagactcg ctccgtcgag ccttccatca atgactttgc gggacagtat 420
acacggggca tggctcttct gcaaaagtgt ttgcagaacc gtgtgatcca gcagcggtg 480
acgccaaaac tcgtatcggt gacaaaggta atgaaggagg agcttgacta tgccttgacc 540
aaagagatgc ctgaacatgaa gaatgatgaa tgggttgaa tgcacatttc ttcctcatg 600
gtcaggctca tatcacgcat ctacagccaga gtgtttctcg gtccagagca ctgccgcaac 660
caagaatggt tgacgaccac tcagagtagc agcgagagcc tgttcataac tggctttatt 720
ctccgcttg ttcccatat tctaagacca ttcatagccc cgctgtacc ctctacaga 780
acactacttc gcaacgcttc gtccaggtcg agagtattg gagacatcat tcgctcccag 840
caagtgatg gcaacgagga catcctgtca tggatgagg atgctgcgac aggggaagaa 900
aagcaaatg acaaacatgc ccagcggtg cttatcctga gtctcgctc tattcacact 960
acggcaatga cgatgacgca tgcctatgat gacttatgtg cttgccctga gtacatagag 1020
cctcttagag atgaggtaaa aagtgtcggt ggcgctagt gtgggacaa gacggcggtg 1080
aatcgattcc acaaacatgc cagctttctc aaagagtcac aacgcttcaa ccccggttc 1140
ctcttaacgt tcaatcgcat ttatcaccaa tccatgacac tctcagatgg caccacatc 1200
ccatcaggca ctgcactcgc ggttccctct cagcgatgc ttcaggactc agcgcatgct 1260
ccaggcccg cgcaccaaac cgagtttgat ggatttagat actcaaatg tcgctcagac 1320
tcaaacatg cacagaata tctcttctcc atgactgatt ctagtaacat ggcgtttggg 1380
tatgggaaat acgctcgccc agggcggttc tatgcatcta atgagatgaa gctgactttg 1440
gcgatactcc ttttacaaat tgagttcaag ttgccagatg ggaagggaag accacgaaat 1500
atcactattg atagtgcac gatacctgat ccgagagcta ggctgtgcgt taggaagcga 1560
tcactgagag atgaatga 1578

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SEQ ID NO: 59

moltype = DNA length = 1500

FEATURE

Location/Qualifiers

source

1..1500

mol_type = other DNA

organism = Trametes versicolor

SEQUENCE: 59

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atggaggatc ccacgcgtact ctacgcttgc ctgccatcgt ctgctcgctac ttctggtgct 60
agatggtaca gagaccgcgt tcggtccatt cctacgggtg ggggctctga ccttcccatc 120
ctctcataca gtcggggcgt cagggtggacc cgcgcggaa gagagatact gcaagaaggt 180
tatgatgggt atcgcggatc caggttcaag atcgcgatgc tcgaccgggt gatcgctcctc 240
gccaacggcc caaagctcgc cagcaggtg aggaggcgct ctgacgaaga gctaaaacttc 300
atggacggac tgggagcggt cgtgcagacg aagtataccc ttggggaagc aatccacaat 360
gaccgcgtac acgtggacat tattcgtagg aagctgacgc gaggcctccc ggcagtcctg 420
ccggacgtca tcgaggaact cacgctagcc gttcgccagt acatcccgcg ggaaggagat 480
gaatgggtca gcgtgaactg ctccaaagca gcgcgggaca tcgtcgcccg ggcaagcaac 540
cgctctttg tcgggttgcc cgcttgccgc aaccagggtt atctcgacct cgccattgac 600
ttcaccctga gcgttgctca agacagggag atcatcaata tgttcccgga gttgctgaaa 660
cctatcgctg gacgcgtggt tggaaatgcc actaggaacg tgcgccgcgc ggtcccattc 720
gtagcgccgt tgggtgagga acgtcgccgc ctcatggagg agtaccgtga ggattggctg 780
gagaaaacga acgacatgct ccagtggtat atggacgagg cagcctcgcg ggactcctcc 840
gtcaaaagca tcgctgagcg tcttctcatg gtcaactttg ccgcaattca cagctcgctg 900
aacaccatca cccacgctct ttaccacctc gccagatgc cggagaccct acagccgctg 960
cgggaagaga tcagaccgct cgtcaaggaa gaaggctgga cgaaggccgc catgggcaag 1020
atgtgggtgc tcgacagctt cctgcgggag tcacagcgt acaatggcat caacatcgct 1080
tccctgacgc gcatggccga caaggacata acgctcagcg acggcacggt cctcccgaag 1140
ggcacgctcg tcgcggtccc cgctactcg acgcaccgcg acgacgggt gtacgcggac 1200
gcgctggctc tcgaccggt ccgcttctcc cgcctgcgcg cccgcgaggg cgagggcacg 1260
aagcaccagt tcgtcaaacac ctccgtggag tacgtgccct tcggccacgg gaagcacgcc 1320
tgccccgggc ggttctctcg ggccaacgag ctgaaggcga tgcctcgcta catcgctc 1380
aactacgacg ttaagctgcc cggcgatggc aagcgcccc tgaacatgta ctggggcccc 1440
acggtcttgc ctgctccgca tgggcagggt ctcttccgca agaggcaggt gtcgctgtag 1500

```

SEQ ID NO: 60

moltype = DNA length = 1578

FEATURE

Location/Qualifiers

misc_feature

1..1578

note = Description of Artificial Sequence: Synthetic polynucleotide

source

1..1578

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 60

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atgggtttgt tccattaga ggattcctac gcgctggtct ttgaaggact agcaataaca 60
ctggctttgt actatctact gtctttcatc tacaaaacat ctaaaaagac atgtacacct 120
cctaaagcat cttgtgaaat cattccaatt acaggaatca tattgaatct gctatctggc 180
tcaagtggct tactattat cttagcaact gcctctttag cagacagatg tggctctatt 240
ttcaccatta ggctgggtat taggagagt ctagttagt caaattggga aatcgctaag 300
gagattttca ctaccacga ttgatagtt tctaatagac caaataactt agccgctaag 360
attcttgggt tcaattatgt ttcatctct ttcgctccat acggcccata ttgggtcgga 420
atcagaaaaga ttattgctac tcttcttcca gacttcagaa gttgcaatt 480
gtaagagttt ttgaactaga aaactctatg aaactctatc gagaatcatg gaaggagaaa 540

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-continued

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aaggatgaag agggaaaggt attagttgag atgaaaaagt gggtctgga actgaatatg 600
aacatagtg taaggacagt tgctggtaaa caatacactg gtacagtga tgatgcgat 660
gcaaagcgta tctccgagtt attcagagaa tgggttcact acactggcag atttgcgtt 720
ggagacgctt tctcttttct aggttggttg gacctgggag gatacaaaaa gacaatggaa 780
ttagttgcta gtgattgga ctcaatggtc agtaaatggt tagatgagca tcgtaaaaag 840
caagctaacc atgacaaaaa ggaggatatg gatttcattg atatcatgat ctccatgaca 900
gaagcaaat caccacttga aggatacggc actgatacta ttatcaagac cacatgtatg 960
actttgattg tttcaggagt tgatacaacc tcaatcgta ttacttgggc cttatcactt 1020
ttgttaaaaca acagagatgac tttgaaaaag gcacaagagg aattagatat gtgcgtaggt 1080
aaaggaagac aagtcaacga gtcgtatctt gtttaactga tatacttggg agcagtgtct 1140
aaagaggctt taagacttta ccacgacgag ttcttaggag gaccaagagc attcttgaa 1200
gattgtactg ttgctgtgta tagaattcca aagggcacct gcttgttgat taacatgtg 1260
aaactgcata gagatccaaa ccttggaggt gatccttgag aattcaagcc agaaagattt 1320
ttgacaccta atcaaaagga tgttgatgag atcggatgag atttgcgaat gataccattt 1380
gggtccggca gaagatattg tccaggtaact agattggctt tacagatggt gcatacgta 1440
ttagcgacat tgctgcaaaa cttcgaaatg tcaacaccaa acgatgcgc agtcgatatg 1500
actgcttctg ttgcatgac aaatgccaaa gcatacact tagaagtctt gctatcacct 1560
cgtgttaaat ggtcctaa 1578

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SEQ ID NO: 61      moltype = DNA length = 1431
FEATURE            Location/Qualifiers
misc_feature       1..1431
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..1431
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 61
atgatacaag ttttaactcc aattctactc ttctctcatc tcttcgtttt ctggaaagtc 60
tacaacatc aaaagactaa aatcaatcta ccaccagggt ccttcggctg gccatttttg 120
gggtgaaact tagccttact tagagcaggc tgggattctg agccagaaag attcgtaaga 180
gagcgtatca aaagcatggt atctccactt gttttcaaga catcactatt tggagacaga 240
tccgtgttcc tttgcggtcc agcttgtaat aagtttttgt tctgcaacga aaacaaatta 300
gtggcatctt ggtggcaggt cctgtaagg aagttgttcg gtaaaagtgt actcacaata 360
agaggagatg aagcaaaatg gatgagaaaa atgctattgt cttacttggg tccagatgca 420
tttgccacac attatgccgt tactatggat gttgtaaac gtagacatat tgatgtccat 480
tggaggggca agaggaagt taatgtattt caaacagtta agttgtacgc attcgaatta 540
gctttagatg tattcatgaa cctagatgac ccaaaaccac tcgcgaaact cggtagtctt 600
tcaacatttt tctccaaagg gatcatcgag cttcctatag acgttctcgg aactagattt 660
tactccagta aaaaggccgc agctgccatt agaattgaat tgaaaagct cattaaagct 720
agaaaactcg aattgaagga gggtaaggcg tcttcttcac aggacttgct ttctcatcta 780
tcaacatcac ctgatgagaa tgggatgttc ttgacagaag aggaatagtg cgataacatt 840
ctacttttgt tattcgtcgg tccagatacc tctgcactat caataacact tttgatgaaa 900
accttaggtg aacacagtga tgtgtacgac aaggttttga aggaacaatt agaaatttcc 960
aaaacaaagg aggttggtg atcactaaag tgggaagata tccagaagat gaagtactca 1020
tgggtcagta tctgtgaaat catgagattg aatcctcctg tcatagggac atacagagag 1080
gctgtggttg atatcgacta tgcgtgttac actatcccaa aaggatggaa gttgcatttg 1140
tcagctgttt ctactcaaag agacgaagcc aattcogaag atgtaactag attcgatcca 1200
tccagatttg aagggcgagg cctacttcca ttcacatttg tgcctttcgg tggaggtcct 1260
agaatgtgtt taggcaaaag gtttgccagg ttagaagtgt tagcatttct ccacaacatt 1320
gttccaactt ttaagtggga tcttctaact cctgatgaga agatcgaata tgatccaatt 1380
gctactccag ctaagggtctt gccaattaga cttcatccac accaagtcta a 1431

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SEQ ID NO: 62      moltype = DNA length = 1578
FEATURE            Location/Qualifiers
misc_feature       1..1578
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..1578
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 62
atggagtctt tagtggttca tacagtaaag gctatctggt gtattgtaat cgtcgggatt 60
ttctcagttg gttatcacgt ttacggtaga gctgtggtcg aacaatggag aatgagaaga 120
tactgaagc tacaaggtgt taaaggccca ccaccatcca tcttcaatgg taacgtctca 180
gaaatgcaac gtatccaatc cgaagctaaa cactgctctg gcgataacat tatctcacat 240
gattattctt cttcattatt ccacacttc gatcactgga gaaaacagta cggcagaatc 300
tacacatact ctactggatt aaagcaacac ttgtacatca atcatccaga aatgggtgaag 360
gagctatctc agactaacac attgaacttg ggtagaatca cccatataac caaaagattg 420
aatctatct taggtaacgg aatcataacc tctaattggt ctcattgggc ccacagcgt 480
agaattatc cctacgagtt tactcatgat aagatcaagg gtatggttgg tttgatggtt 540
gagctctgta tgcctatgtt gaataagtgg gaggagatgg taaagagagg cggagaaatg 600
ggatgcgaca taagagtgtg tagaggactg aaagatgttt cagcagatgt gattgcaaaa 660
gctgtgttcg gatcctcatt ttctaaaggt aaggctattt tctctatgat aagagatttg 720
ttacagctga tcacaaagag aagtgttcta ttcagattca acggattcac tgatattggtc 780
tttgggagta aaaagcatgg tgacgttgat atagacgctt tagaaatgga attggaatca 840
tccatttggg aaactgtcaa ggaacgtgaa atagaatgta aagatactca caaaaaggat 900
ctgatgcaat tgattttgga aggggcaatg cgttcatgtg acggtaacct ttgggataaa 960
tcagcatata gaagatttgt ttagataaat tgtaaatcta tctacttcgc agggcatgat 1020

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-continued

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agtcacagctg tctcagtgtc atgggtgtttg atgttactgg ccctaaaccc atcatggcaa 1080
gttaagatcc gtgatgaaat tctgtcttct tgcaaaaatg gtattccaga tgccgaaagt 1140
atcccaaaccc ttaaaacagt gactatgggtt attcaagaga caatgagatt ataccctcca 1200
gcaccaatcg tcggggagaga agcctctaaa gatatcagat tgggcgatct agttgttctt 1260
aaaggcgctc gtatatggac actaatacca gctttacaca gagatcctga gatttgggga 1320
ccagatgcaa acgatttcaa accagaaaga ttttctgaag gaatttcaaa ggcttgtaag 1380
tatcctcaaa gttacattcc atttgggtctg ggtcctagaa catgcgttgg taaaaacttt 1440
ggcatgatgg aagtaaaagg tcttgtttcc ctgattgtct ccaagttctc ttctactcta 1500
tctctacct accaacatag tcctagtcac aaacttttag tagaaccaca acatggggtg 1560
gttaattagag tgggttaa 1578

```

```

SEQ ID NO: 63      moltype = DNA length = 1590
FEATURE
misc_feature      Location/Qualifiers
                  1..1590
                  note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source            1..1590
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 63
atgtacttcc tactacaata cctcaacatc acaaccgttg gtgtctttgc cacattgttt 60
ctctcttatt gtttacttct ctggagaagt agagcgggta acaaaaagat tgcccagaa 120
gtgcccgtcg catggcctat tatcgccac ctccacttac ttgcaggtgg atcccatcaa 180
tcaccacata ttacattggg taacatggca gataagtacg gtccctgtatt cacaatcaga 240
ataggcttgc atagagctgt agttgtctca tcttgggaaa tggcaaaagga atgttcaaca 300
gctaagtatc aagtgtcttc ttcaagacct gaactattag cttctaagtt gttgggttat 360
aactacgcca tgtttgtttt ttaccatac ggttcatact ggagagaaat gagaaagatc 420
atctctctcg aattactatc taattccaga ttggaactat tgaaagatgt tagagcctca 480
gaagttgtca catctattaa ggaactatac aaattgtggg cggaaaagaa gaatgagtca 540
ggattggttt ctgtcgagat gaacaatgg ttcggagatt tgactttaaa cgtgatcttg 600
agaatggtgg ctggtaaaag atacttctcc gcgagtgcag cttcagaaaa caaacaggcc 660
cagcgttgta gaagagctct cagagaattc ccggcttgtt tgtggttgct 720
gatgctatac ctttcttgg atggctcgat tggggaagac acgagaagac cttgaaaaag 780
accgccatag aaatggattc catcgcccag gagtggcttg aggaacatag acgtagaaaa 840
gattctggag atgataatc tacccaagat ttcattggag ttatgcaatc tgtgctagat 900
ggcaaaaatc taggcggata cgtatctgat acgattaaca aggtacatag ctttaactctt 960
atatcagggtg gcagtataac tactgtagtt tctttgacat gggctcttag tcttgtgtta 1020
aacaatagag atactttgaa aaaggcacag gaagagttag acatccaagt cggtaaggaa 1080
agattggtta accagcaaga catcagtaag ttagtttact tgcaagcaat agtaaaagag 1140
acactcagac tttatccacc aggtcctttg ggtggtttga gacaattcac tgaagattgt 1200
acactaggtg gctatcacgt tccaaaagga actagattaa tcatgaactt atccaagatt 1260
caaaaagatc cagctatttg gtctgaccc actgaattcc aaccagagag attccttacg 1320
actcataaag atgtcgatcc acgtggtaaa cactttgaat tcattccatt cgggtgcagga 1380
agacgtgcat gtcttggtat ccatttcgga ttacaagtac tacatctaac attggcatct 1440
ttcttgcatg cgtttgaatt ttcaacacca tcaaatgagc aggttaacat gagagaatca 1500
ttaggtctta cgaatatgaa atctacccca ttagaagttt tgatttctcc aagactatcc 1560
cttaattgct tcaaccttat gaaaatttga 1590

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SEQ ID NO: 64      moltype = DNA length = 1440
FEATURE
misc_feature      Location/Qualifiers
                  1..1440
                  note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source            1..1440
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 64
atggaaccta acttttactt gtcattacta ttgtgttctg tgaccttcatt ttctttaagt 60
ctgtttttca tcttttacaa acaaaagtcc ccattgaatt tgccaccagg gaaaatgggt 120
taccctatca taggtgaaag tttagaattc ctatccacag gctggaaggg acatccctgaa 180
aagttcatat ttgatagaat gcgtaagtac agtagtgagt tattcaagac ttctattgta 240
ggcgaatcca cagttgtttg ctgtggggca gctagtaaca aattcctatt ctctaacgaa 300
aacaactggg taactgcctg gtggccagat tctgttaaca aaatcttccc acaacttca 360
ctggattctc atttgaagga ggaatctata aagatgagaa agttgctgcc acagttcttc 420
aaaccagaag cacttcaaaag atacgtcggc gttatggatg taatcgaca aagacathtt 480
gtcactcact gggacaacaa aaatgagatc acagtttacc cacttgctaa aagatacact 540
ttcttgcttg cgtgtagact gttcatgtct gttgaggatg aaaaatcatgt ggcgaaatc 600
tcagaccatc tccaactaat cgtcgaggc atcatttcac ttctatcgta tcttctggt 660
actccattca acaaggccat aaaggcttca aatttcatta gaaaagagct gataaagatt 720
atcaaacaaa gacgtgttga tctggcagag ggtacagcat ctccaaccca ggatatcttg 780
tcacatatgc tattaacatc tgatgaaaac ggtaaaatcta tgaacgagtt gaacattgcc 840
gacaagatcc ttggactatt gataggaggc cagcagacag cttcagtagc ttgcacattt 900
ctagtgaagt acttaggaga attaccacat atctacgata aagttcacc aagacaaatg 960
gaaattggca agtccaaaac tgctggggaa ttgttgaatt gggatgactt gaaaaagatg 1020
aagtattcat ggaatgtggc atgtgaggta atgagattgt caccaccttt acaaggtggg 1080
tttagagagg ctataactga ctttatgttt aacggtttct ctattccaaa aggggtggaag 1140
ttatactggt ccgccaactc tacacacaaa aatgcagaat gtttcccaat gctgagaaa 1200
tcgatcccta ccagatttga aggtaatggc ccagcgcctt atacatttgt accattcggt 1260
ggaggcccta gaatgtgtcc tggaaaggaa tacgctagat tagaaatctt ggttttcatg 1320

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-continued

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cataatctggt tcaaacgttt taagtgggaa aagggtattc cagacgaaaa gattattgtc 1380
gatccattcc caatoccagc taaagatctt ccaatocggt tgtatcctca caaagcttaa 1440

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SEQ ID NO: 65      moltype = DNA length = 1572
FEATURE           Location/Qualifiers
source            1..1572
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 65
atgggtctct tccctttgga agatagttac gcactcgtct ttgaagggtt agcaataact 60
actctagctc tctactactt attatccttc atctataaaa cctctaaaaa gacttgtact 120
ccacctaaag caagcgggtga gcacccctata acaggccact taaacctctc tagtggttca 180
tccggtcttc cccatctagc cttagcatct ttggctgacc gatgtgggac catattcacc 240
atccgacttg gcatacgtag agttttgggt gttagtaatt gggaaattgc taaggagatc 300
ttcactaccc atgatttgat tgtttcaaac cgtcccaaat acctcgtgac aaagattttg 360
ggattcaact atgtgtcctt ttcgtttgct ccatatggcc cctattgggt tggatccgt 420
aagatcactg ccacaaaact gatgtcaagt agcagggtcc agaagcttca gtttgtccga 480
gttttcgaac tagaaaaact catgaaaagc atacgcgagt cttggaaaga gaaaaaagac 540
gaagaaggta aagtgttggt ggagatgaaa aaatgggttt gggaattgaa tatgaatata 600
gttcttagaa ctgttgctgg taaacagtac actggaaactg ttgatgatgc ggatgcgaag 660
aggatttagt aattgtttag agaatgggtt cattacacag gaaggtttgt tgtgggagat 720
gcttttcctt tctttgggtg gttggatttg ggtggatata agaagaccat ggaactagt 780
gcttccagac tagattccat ggtctcaaaa tgggttagac agcatcgcaa aaagcaggct 840
aacgcagaca aaaaagagga catggatttc atggacatca tgatatcgat gactgaagcc 900
aattcccttc tggagggtta tggtaacgat acaataatta aaaccacttg catgactctt 960
attgtcagtg gtgtagatag aacctccatc gtgctaactt gggcactctc gttactactg 1020
aacaaccgtg acactcttaa gaaagctcaa gaagagctag acatgtgtgt gggaaaaggt 1080
cgacaagtaa acgaatcaga tctagtaaac ctaatctacc ttgaagccgt attaaaagaa 1140
gcattgacac tataccagc agcattcctt ggaggtccta gagccttttt agaagactgc 1200
accgtggcag ggtaccgtat cccaaaaggc acatgtctac ttattaacat gtggaaactt 1260
catcgtgatc caaacatgat gtcagaccca tgtgagttta aaccagagag gttcttaacc 1320
ccaaacccaa aggcagtaga tgttatttga atggattttg agttaatccc atttggtgag 1380
ggaaagaggt attgtccagg gacacgtttg gcattacaaa tgttacacat agttctggcc 1440
actctactac aaaactttga gatgtcaact ccaaatgatg caccggttga tatgacgcgc 1500
agtgttgtaa tgacaaatgc gaaggcaagt ccacttgaag ttctactttc gccacgtgtt 1560
aagtggctat ag                                     1572

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SEQ ID NO: 66      moltype = DNA length = 1431
FEATURE           Location/Qualifiers
source            1..1431
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 66
atgattcaag ttctaaccac gatccttctc ttcctcattt tcttcgtttt ctggaagggt 60
tacaagcacc agaaaaccac aatcaatctt ccaccgggaa gcttcggatg gccattttctg 120
ggcgaaactc tggcactcct acgtgcaggt tgggaactcag agccggagag atttggtcgt 180
gaacggatca agaaacacgg aagtccctta gtgtttaaga cgtcgttgtt tggcgaccgt 240
tttgcggtgt tgtgtggacc tgcgggaaac aagttcctgt tctgcaacga gaacaagctg 300
gtggcgctgt ggtggccggt tccggtgagg aagcttttct gcaagtctct gctcacgatt 360
cgtggtgatg aagctaaagt gatgaggaag atgtgtttat cgtatctcgg tcttgatgct 420
ttcgcaactc attatgcctt caccatggac gtctgcaccc gtccgcatat cgacgttcat 480
tggcgaggga aggaagcgtt caacgtattc caaacgggta agttatatgc ctttgagctt 540
gcatgtcgtt tatcatgaa cctagacgac ccaaaccaca ttgcaaaact cgttctcttg 600
ttcaacattt tcttgaagg catcattgag cttccaatcg acgtccagg gacacgattt 660
tatagctcca aaaaagcagc agcagctatc aggattgaac taaaaaattt gattaaagca 720
agaaaactgg aactgaaaga aggggaaggc tcatcttcac aagacctctt atcacattt 780
cttacatctc cagatgaaaa tggatatgtt ctaaccgaag aagagattgt agacaacatc 840
ttgttactac tctttgctgg tcatgatacc tccgctcttt caatcacttt gctcatgaag 900
actcttggcg aacattctga tgtttatgac aagggtgtta aagagcaact agagatatcg 960
aagacgaaag aagcatggga gtccctgaaa tgggaggaca taaaaagat gaaatactcc 1020
tggagtgtta tatgtgaagt catgagacta aatccacctg ttataggaa ctagagagag 1080
gcccttgtgg atattgatg tgcgggttat accatcccca aaggatggaa gctgcactgg 1140
agtgtctgat cgacacaaag ggacgaggct aactttgaag acgtaacacg ttttgaccca 1200
tcacggtttg aaggcgagg accgactcca ttcaccttgg ttcggtttgg aggggggct 1260
agaatgtgtt tagggaaaga atttgcctga ttggaagtac ttgcgtttct tcacaatatt 1320
gtcaccatgt tcaaatggga cctgttgata cctgatgaga aaatagaata tgatcccatg 1380
gctacccagc caaaggggct tccaattcgt cttcatcccc atcaagtttg a 1431

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SEQ ID NO: 67      moltype = DNA length = 1578
FEATURE           Location/Qualifiers
source            1..1578
                  mol_type = other DNA
                  organism = Arabidopsis thaliana

```

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SEQUENCE: 67
atggagaggt ttggtgttca tacggtaaat gcaatttggg gcatagttat tgtcggaatc 60
ttcagcgtag gttatcatgt gtatggaaga gcggtgggtg agcagtgagg gatgcggagg 120
agtttaaaat tgcaggcggt gaagggtcct ccaccgtcga tctttaacgg caatgtgtcg 180
gagatgcaac ggattcagtc ggaggctaaa cactgttccg gcgataacat catttctcat 240
gactattctt cttctctatt tctctatttc gatcactggc gaaaacaata cggaaggatt 300

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tacacatact caacgggggtt aaagcagcac ctttacataa accacccgga aatggtgaag 360
gagcttagcc aaaccaacac acttaacctt ggtagaatca ctcacatcac caaacgcctt 420
aaccoccatc tcggcaatgg catcatcacc tctaattgggc ctcatctggc ccatcaacgt 480
cgtatcattg cctatgagtt taccacgac aaaatcaagg gaatggttgg tttaatggtg 540
gaatctgcca tgccaatggt gaacaaatgg gaagagatgg tgaaaagagg aggagaaatg 600
ggttggtgaca taagagtggg cgaagacctt aaggatgtct cagctgatgt catcgctaag 660
gcttgctttg ggagctcttt tcaaaaaggc aaagcaatat tctctatgat tagggatctt 720
ttaaccggca ttaactaaac aagcgctctc ttcagattca atggcttcac tgatatgggtg 780
tttggaagta agaagcatgg tgatgtggat attgatgcgc ttgagatgga attagaatct 840
tctatatggg aaacgggtta ggagagggaa attgaatgta aggatactca caagaaggat 900
ctaattgcagt tgatactcga gggagcgaat cgaagctgcg atggtaactt gtgggacaag 960
tcagcctata gacgggttgt ggtggacaat tgcaagagca tctatttcgc cggacatgat 1020
tcaaccgcag tctcagtgtc ttggtgcctt atgctcctcg ctctcaatcc tagttggcag 1080
gttaaaattc gcgatgaaat cttgagttct tgcaagaatg gcattcccgga cgcgaatca 1140
attcctaatt tcaaaacggg gacaatggta atacaagaaa caatgagact ataccacca 1200
gcaccaatcg tgggaagaga agcatccaaa gacataagac ttggagacct tgtggtgcca 1260
aaaggagtgt gcatttggac ctccattcct gccttacacc gagacccgga gatctgggga 1320
ccagacgcaa acgacttcaa gccagagagg tttagtgagg gaatctctaa ggcttgcaaa 1380
taccctcagt catacatccc atttggcctt ggaccaagaa catgcgtagg caaaaacttt 1440
ggtatgatgg aagtgaagaa ctttgtttca cttattgtct caaagttcag ttttactctt 1500
tcccgcagct atcagcactc tccaagccat aaactccttg tagagcctca acatggtggt 1560
gtcattaggg ttggttga                                     1578

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SEQ ID NO: 68      moltype = DNA length = 1590
FEATURE           Location/Qualifiers
source            1..1590
                  mol_type = other DNA
                  organism = Vitis vinifera

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SEQUENCE: 68
atgtatttcc ttctccaata cctaaacatc accacggctc gagtctttgc cacacttttc 60
ctttcctact gtctattatt attgaggtct agagctggta acaaaaaaat agcacctgaa 120
gctctgctg ctgagcccat aatcggtcac ctacacctgt tagctgggtg ttctcatcag 180
cttccccaca taaccttggg aaacatggcc gacaaatatg gaccggtctt cacaattcgg 240
attgggttgc atcgagctgt ggtggtaagt tcttgggaga tggctaagaa atgctcgacc 300
gccaatgacc aggttttcac ctccgctccc gaacttttag cctcaaaact tttgggctac 360
aactacgcca tgtttgtgtt ctctccatac ggttcttact ggctgaaat ggcgaagata 420
atcagcctag agctactctc taacagccgc ttagagctgc tgaaggacct ccgagcttca 480
gaagtgttga catccataaa agagctatac aagctctggg cagagaaaaa aaatgaatcg 540
ggccttgtct cggtggagat gaagcagtggt tttggagact tgactctgaa cgtaatctct 600
aggatggtgg cagggaagcg ttatttcagt gcttcagatg caagtgaaaa taaacaggcg 660
cagaggtgcc ggagagtggt caggggaattc tttcatttgt cagggctctt tgtggtggcg 720
gacgctattc catttcttgg atggctcgac tggggggagac atgagaaaaa cctaaagaag 780
acagcaatag aaatggacag tattgtctca gaatggttag aggagcaccg tcggaggaaa 840
gactccggtg atgataatag tacgcaagac ttcattggatg tgatgcagtc agttcttgat 900
ggcaaaaaac ttggtgtgta cgcagctgat accatcaata aagccacatg cctgactcta 960
atctccggag gtacgcacac aactgttgtc tctctaaccat gggcactctc tcttgtaact 1020
aacaaccgtg acaccttaaa aaaagctcaa gaagaattag acatccaagt tggtaaggaa 1080
agattagtga atgaacaaga tataagtaag ttggtctatc tccaagccat tgttaagag 1140
acattacggt tatatccacc aggaacctt ggaggactac gccaatttac cgaggattgc 1200
accttgggtg gataccatgt ctctaaaggg acccgtttaa taatgaacct ttccaagatc 1260
caaaaggatc caagaatttg gtcagatccg acagaattcc aaccagagag gtttctcacc 1320
accataaag atgttgatcc tcggggaaaa cattttgagt ttataccatt tggagctggt 1380
cgaagagcat gtccaggaat aacttttgggt cttcaagtat tacatttaac attggctagt 1440
ttcttacatg cgtttgtaatt ttcaactcca tcaaatgaac aggtcaatat ggcgcagagc 1500
cttggactta caaatatgaa atctacccca cttgaagttc tcatttctcc acgcttatca 1560
ttgaattggt ttaacctaat gaagatataa                                     1590

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SEQ ID NO: 69      moltype = DNA length = 1440
FEATURE           Location/Qualifiers
source            1..1440
                  mol_type = other DNA
                  organism = Medicago trunculata

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SEQUENCE: 69
atggagccta attttctatc ctcccttctc cttctctttg tcactttcat atctctctct 60
ctttttttca tattctacaa acagaaatct ccattaaaat tgccacctgg taaaatgggt 120
taccacatca taggtgaag ccttgagttc ttatcaacag gatggaaagg acatcctgaa 180
aaattcattt cgcacgctat gcgtaaatat tcctcagaac tctttaaac atcaatcgta 240
ggagaaatca cggtggtttg ttgcggagca gcaagtaaca agtttttgggt ttcaaacgag 300
aataaaactg tgactgcatg gtggccagat agtgtaaaaa aaatcttccc tactacttct 360
cttgactcta acttgaagga agaatccatc aagatgagaa aattgcttcc acaattcttt 420
aaaccggaag cctcacacg ttatgttgtt gtcattggatg ttattgctca aagacatttt 480
gttactcatt gggataataa aaatgaaatc accgtctacc ccttggccaa gaggtacacc 540
ttttgttag cttgtcgggt gttcatgagc gttgaagacg agaatcatgt agcaaaattt 600
agtgatccat ttcagttaat tgcggccgga atcatatctc taccaattga tttgccagga 660
acaccattca acaagctat aaaggcctca aactttataa gaaaggaggt gattaagatc 720
ataaagcaaa ggagggtaga ttggcgagaa gggacagcat caccaacaca agatatattg 780
tctcacatgt tgttgacaag tgatgaaaat ggaaagagta tgaatgaact taatattgct 840
gataagatc ttggcctttt gatcggagga catgacactg ctagcgtcgc atgcactttc 900
cttgtcaaat atctcggcga gttacctcac atttatgata aagtctatca agagcaaatg 960

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gaaattgcaa aatcgaaacc agcaggagaa ttgttgaatt gggatgacct gaagaaaatg 1020
aaatactctt ggaacgtagc ttgtgaagta atgagacttt cccctccact ccaaggaggt 1080
ttcagggaag ccatacactga ctttatgttc aatggattct caattccctaa gggatggaag 1140
ctttattgga gtgcaaatc aacacataag aacgcagaat gttttcccat gccagagaaa 1200
tttgacccaa caagatttga aggaaatgga ccagctcctt atacttttgt tcccttttgt 1260
ggaggaccaa ggaagtgtcc tggaaaagag tatgcaagat tagaataact tgttttcagt 1320
cacaatttgg tgaaaagggt taagtgggaa aaggtgattc cagatgagaa gattattgtt 1380
gatccattcc ccatacctgc aaaggatctt ccaattcgcc ttatccaca caaagcttaa 1440

SEQ ID NO: 70      moltype = DNA length = 2133
FEATURE           Location/Qualifiers
misc_feature       1..2133
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..2133
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 70
atgcaatcag attcagtcac agtctctcca ttgtatttgg tttccgctgc tatgaatggc 60
aaggcaatgg aaaagtgtga cgctagttaa tctgaagatc caacaacatt gcctgcacta 120
aagatgctag ttgaaaatag agaattgttg acactgttca caacttcctt cgagttctt 180
attgggtgtc ttgtatttct aatgtggaga cgttcatcct ctaaaaagct ggtacaagat 240
ccagttccac aagttatcgt tgtaaagaag aaagagaagg agtcagaggt tgatgacggg 300
aaaaagaaag tttctatttt ctacggcaca caaacaggaa ctgccgaagg ttttgctaaa 360
gcattagtgc aggaagcaaa agtgagatat gaaaagacct cttcaagggt tatcgatcta 420
gatgactacg ctgcagatga tgatgaatat gaggaataac tgaaaaagga atccttagcc 480
ttcttcttct tggccacata cggtagtggt gaacctactg ataagtgtgc taactctac 540
aagtgggtta cagaaggcca gataaaagg gtatgggtga aaaagttaca atacggagta 600
tttggtttag gtaacagaca atatgaacat ttcaacaaga tcgctattgt agttgatgat 660
aaacttactg aaatgggagc caaaagatta gtaccagtag gattagggga tgatgatcag 720
tgtatagaag atgacttcac gccttggaag gaattggtat ggcagaaatt ggatcaactt 780
ttaagggagc aagatgatac ttcttgtagt acccctatac ctgcagccgt attggagtag 840
agagtgtgtt accatgataa accagcagac tcatatgtcg aagatcaaac ccatacaaac 900
ggctatgttg ttcatgatgc acagcatcct tcaagatcta atgtggcttt caaaaaggaa 960
ctacacacct ctcaatcaga taggtcttgt actcacttag aattcgatat ttctcacaca 1020
ggactgtctt acgaaactgg cgatcacggt ggcgtttatt ccgagaactt gtcggaagt 1080
gtcgatgaag cactaaaact gttagggtta tcaccagaca catacttctc agtccatgct 1140
gataaggagg atgggacacc tatcgggtgt gcttcactac caccaccttt tctccttgc 1200
acattgagag acgctctaac cagatacgca gatgtcttat cctcacctaa aaaggtagct 1260
ttgctggcat ttgctgtcta tgcagtgtat cctagtgaag ccgataggtt aaagtctctg 1320
gcttcaccag ccggaagaaga tgaatatgca caatggatcg tcgccaacca acgttctttg 1380
ctagaagtga tgcaaaagtt tccatctgcc aagcctccat tagtgtgtgt cttcgagaga 1440
gtagctccac gtttacaacc aagatactac tctatcagtt catctcctaa gatgtctcct 1500
aacagaatac atgtttacatg tgcctttgtg tacgagacta ctccagcagg cagaattcac 1560
agaggattgt gttcaacctg gatgaaaaat gctgtccctt taacagagtc acctgattgc 1620
tctcaagcat ccatttttgt tagaacatca aatttcagac ttccagtgga tccaaaagt 1680
ccagtcatta tgataggacc aggcactggt cttgccccat tcaggggctt tcttcaagag 1740
agattggcct tgaaggaaac ttgtacagaa ttgggttctt ctatcttttt ctttggttgc 1800
cgtaatatga aagttgactt tatctacgag gacgagctta acaattttgt tgagacagga 1860
gcattgtcag aattgatcgt cgcattttca agagaaggga ctgccaaga gtacgttcag 1920
cacaagatga gtcaaaaagc ctccgatata tggaaacttc taagtgaagg tgcctatctt 1980
tatgtctgtg gcgatgcaaa gggcatggcc aaggatgtcc atagaactct gcatacaatt 2040
gttcaggaac aaggaggtct ggattcttcc aagggtgaat tgtacgtcaa aaacttacag 2100
atgtctgga gatacttaag agatgtttgg taa 2133

SEQ ID NO: 71      moltype = DNA length = 2079
FEATURE           Location/Qualifiers
misc_feature       1..2079
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..2079
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 71
atgacttctg cactttatgc ctccgatctt ttcaacaact tgaaaagtat catgggaacg 60
gattctttgt ccgatgatgt tgtattagtt attgtacaaa cttctctggc actggttgc 120
ggtttcgttg tcttattgtg gaaaaagacc acggcagatc gttccggcga gctaaagcca 180
ctaagtatcc ctaagtctct gatggcgaaa gatgaggtg atgacttaga tctagttct 240
ggaaaaacga gagtctctat cttcttcggc acacaaaccg gaacagccga aggattcgct 300
aaagcacttt cagaagagat caaagcaaga tacgaaaagg cggctgtaaa agtaatcgat 360
ttggatgatt acgctgcgga tgatgaccaa tatgaggaag agttgaaaaa ggaacattg 420
gctttctttt gtgtagccac gtatgtgtgt ggtgaaccaa ccgataacgc cgcaagattc 480
tacaagtgtt ttactgaaga gaacgaaaag gatatacaag tgcagcaact tgcctacggc 540
gtttttgcct tagttaacag acaatacagc cactttaaca agataggtag tgtcttagat 600
gaagagttaa gcaaaaaggg tgcgaagaga ttgattgaag tcggtttagg agatgatgat 660
caatctatcg aggatgactt taatgcatgg aaggaaactt tgtggtctga attagataag 720
ttacttaagg acgaagatga taaatccgtt gccactccat acacagccgt cattccagaa 780
tatagagtga ttactcatga tccaagattc acaacacaga aatcaatgga aagtaatgtg 840
gctaattgta atactacat cgatattcat catccatgta gtagtagcgt tgcagttcaa 900

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aaggaattgc acactcatga atcagacaga tcttgcatat atcttgaatt tgatatatca 960
cgtactggta tcaacttacga aacaggtgat cacgtgggtg tctacgctga aaaccatgtt 1020
gaaattgttag aggaagctgg aaagtgtgtg ggccatagtt tagatcttgt tttctcaatt 1080
catgccgata aagaggatgg ctcaccacta gaaagtgcag tgctccacc atttccagga 1140
ccatgcaccc taggtacogg tttagctcgt tacgcggatc tgttaaatcc tccacgtaaa 1200
tcagctctag tggccttggc tgcgtacgcc acagaaacct ctgaggcaga aaaactgaaa 1260
catctaactt caccagatgg taaggatgaa tactcacaat ggatagtagc tagtcaacgt 1320
tctttactag aagttatggc tgctttccca tccgctaaac ctcttttggg tgttttcttc 1380
gccgcaatag cgcttagact gcaaccaaga tactattcaa tttcatcctc acctagactg 1440
gcaccatcaa gagttcatgt cacatccgct ttaggtgacg gtccaaactcc tactggtaga 1500
atccataaag gcgtttgttc aacatggatg aaaaacgcgg ttccagcaga gaagtctcac 1560
gaatgttctg gtgtctcaat ctttatcaga gccccaact tcaaaactgcc ttccaatcct 1620
tctactccta ttgtcatggg cggctcctgt acaggtcttg ctccattcag aggtttctta 1680
caagagagaa tggccttggg ggaggatgg gatcttcttt gttgtttttc 1740
ggctgtagaa acagacaat ggatttcac tacgaagatg aactgaataa cttttagat 1800
caaggagtta tttcagagtt gataatggct ttttctagag aagtgctca gaaggagtac 1860
gtccaacaca aaatgatgga aaaggccgca caagtttggg acttaataca agagggaaggc 1920
tatctatatg tctgtgtgga tgcaagggt atggcaagag atgttcacag aacacttcat 1980
actatagtc aggaacagga aggcgttagt tcttctgaag cggaagcaat tgtgaaaaag 2040
ttacaaacag agggaagata cttgagagat gtgtggtaa 2079

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SEQ ID NO: 72      moltype = DNA length = 2142
FEATURE           Location/Qualifiers
misc_feature      1..2142
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..2142
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 72
atggcagaat tagatacact tgatatagta gtattaggtg ttatcttttt gggtagctgtg 60
gcatacttta ctaagggtaa attgtgggtt gttaccaagg atccatacgc taacggattc 120
gctgcaggtg gtgcttccaa gccctgcaga actagaacaa tcgtcgaagc tatggaggaa 180
tcaggtaaaa actgtgttgt ttctacggc agtcaaacag gtacagcggg ggattacgca 240
tcaagacttg caaaggaagg aaagtcgaga ttccgtttga acactatgat cggcgatcta 300
gaagattatg acttcgataa cttagacact gttccatctg ataacatcgt tatgtttgta 360
ttggctactt acggtgaaag cgaaccaaca gataacgcgc tggatttcta tgagttcatt 420
actggcgaag atgctctttt caatgagggc aacgacctc cactaggtaa cttgaattac 480
gttgcgttgc gtctgggcaa caatacctac gaacactaca actcaatggt caggaaactt 540
aacaaggctc tagaaagttt agagagctcat agaattggag aagcaggtga gggtagacac 600
ggagctggaa ctatgggaag ggaactttta gcttggaaag atccaatgtg ggaagccttg 660
gctaaaaaga tgggcttgga ggaagagaaa gctgtatatg aacctatttt cgctatcaat 720
gagagagatg atttgacccc tgaagcgaat gaggtatact tgggagaacc taataagcta 780
cacttgggaag gtacacgcaa aggtccattc aactccaca acccatatat cgcaccaatt 840
gcagaatcat acgaactttt ctacgctaag gatagaaatt gtctgcatac ggaattgat 900
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tacgatgcta tattgagata ccatctggaa atatcgctc cagtttctag acagtttgtc 1140
tcaactttag cagcattcgc ccctaattgat gatatacaag ctgagatgaa ccgtttggga 1200
tcagacaaag attacttcca cgaagagaca ggaccacatt actacaatat cgctagattt 1260
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aaaaagatta gtattactgc tgttgtcgaa tctcagcaaa ttccaggtag agatgaccca 1440
ttcagaggtg tgcgactgaa ctactgttc gctttgaagc agaacaacaa cggtgatcca 1500
aatccagctc cttttggcca atcatacgag ttgacaggac caaggaataa gtatgatggg 1560
atcatgttc cagtcocatg aagacattct aactttaagc taccatctga tccaggcaaa 1620
cctattatca tgaatcggtc aggtaccggt gttgccctt ttagaggctt cgtccaagag 1680
agggcaaaac aagccagaga tgggtgtaga gttggtaaaa cactgctgtt ctttggatgt 1740
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caacacagac tgaaggaaag atcaaaaggaa gtttctgac ttctatccca aaaagcatac 1920
ttctacgttt gccgagacgc cgcacatatg gcacgtgaag tgaacactgt gtttagcacag 1980
atcatagcag aagccctgtg tgtatcagaa gccaaagggtg aggaattgtt caaaaacatg 2040
agatcagcaa atcaatacca agtgtgttct gatttcgtaa ctttacactg taaagagaca 2100
acatacgcca attcagaatt gcaagaggat gtctggagtt aa 2142

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SEQ ID NO: 73      moltype = DNA length = 2133
FEATURE           Location/Qualifiers
source            1..2133
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 73
atgcaatcag attccgtaaa agtgtcgccg ttcatctcgt tatctgcagc tatgaacgga 60
aaagcaatgg agaaattgaa cgcacggaag tcggaagatc cgacgacgct accggcggtg 120
aagatgctgg tggagaatcg cgagctgctg acactgttta cgacgtcgtt tgctgtattg 180
atcgggatgtc tctgtttttt gatgtggcgg agatcgtcct cgaagaaact gggtcaggat 240
ccggtgcccgc aggttaactgt tgttaagaag aaagagaagg agtctgaggt tgatgatggc 300
aagaagaagg tttcgtatatt ctacgggaact caaacaggaa ccgctgaagg ttttgccaa 360

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gcactttag aggaagctaa agttcgatat gaaaagacat cctttaaagt tattgatctg 420
gatgattatg ctgctgatga cgatgagtat gaggagaagc ttaagaaaga atctttggcg 480
tttttctttt tggcaacgta tggagatggg gaaccaacag ataatgcagc caatttttac 540
aaatggttta cagagggaga tgacaaaggc gaatggctga agaaacttca atatggcgtg 600
tttggcctcg gtaacagaca atatgagcat ttcaataaga ttgcaatagt ggttgatgac 660
aaactcacag aaatgggcgc aaaacgcctt gttcctgttg gtcttggaga tgacgatcaa 720
tgtatagaag atgactttac agcatggaaa gagttagtgt ggcccgagtt ggatcaattg 780
ttgcgtgatg aggatgacac gagtgttacg actccttaca ctgctgcggg tttggaatac 840
cgagttgtat atcatgataa acctgcagac tctgatgcag aagatcaaac tcatacaaat 900
ggtcattgtt ttcattgatgc tcaacatcca tctagatcca atgtggcatt taaaaaggaa 960
ttgcacacct ctcaatctga ccggtcttgc actcatttgg aatttgatat ctctcacacc 1020
gggctatcat acgagacggg ggatcatggt ggtgtctaca gtgagaatct aagtgaagtt 1080
gtagatgaag ctttaaaatt actcgggttg tcacccgaca cttatttctc agtccatgct 1140
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cttgaagtta tgcagtcatt tccgtcagct aaaccgccac ttggtgtttt cttcgcagct 1440
gtcgcccccac gtttcaaac tgcatttacc tcgatttctt cttctccaaa gatgtcacca 1500
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ccggtcatca tgatcggttc cgggactggg ttgcccctt tcagaggttt tcttcaagaa 1740
cgggttagctt tgaaggaaat tggaaaccgaa ctccgggtcat ctattttctt tttcggatgc 1800
agaaaccgca aagtggaatt tatatacgaa gacgaactaa acaacttgtt ggagaccggt 1860
gctttatcgg agcttattgt tgcattctcc cgtgaaggaa ccgcaaaagga gtatgtgcaa 1920
cataaaatga gccagaaggc ttcagatata tggaaagtgc tttcagaggg agcatattta 1980
tatgtatgtg gtgatgctaa aggcattggc aaagatgtac acagaacctc tcacacaatt 2040
gtacaagaac aggatctctc agattcttcc aaggcagaat tgtatgtaaa gaacctacaa 2100
atgtcgggaa gatattcttc tgaatttttg taa 2133

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SEQ ID NO: 74      moltype = DNA length = 2079
FEATURE            Location/Qualifiers
source              1..2079
                    mol_type = other DNA
                    organism = Arabidopsis thaliana

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SEQUENCE: 74
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ggatttttgg tgttgttatg gaagaaaacg acggcgggac ggagcgggga gctgaagcct 180
ttgatgatcc ctaagtctct tatggctaag gacgaggtat atgatttgga tttgggatcc 240
gggaagacta gagtctctat cttcttcggg acgcagactg gaacagctga gggattttgt 300
aaggcattat ccgaagaaat caagcgaga tatgaaaaag cagcagtcac agtcattgac 360
ttggatgact atgctgcoga tgaatgacag tatgaagaga aattgaagaa ggaaactttg 420
gcatttttct gtgttgctac ttatggagat ggagagccta ctgacaatgc tgcagatttt 480
tacaaatggt ttacggagga aatgaacgg gatataaagc ttcaacaact agcatatggt 540
gtgtttgtct ttggttaatc ccaatatgaa cattttaata agatcgggat agttcttgat 600
gaagagttat gtaagaaagg tgcaaaagcg cttatttgaag tcggtctagg agatgatgat 660
cagagcattg aggatgatt ttatgctctg aaagaatcac tatggtctga gctagacaag 720
ctctcctcaag acgaggtatg taaaagtgtg gcaactcctt atacagctgt tattcttgaa 780
taccgggttg tgactcatga tctcgggttt acaactcaaa aatcaatgga atcaaatgtg 840
gccaatggaa atactactat tgacattcat catcctgca gagttgatgt tgcgtgacag 900
aaggagcttc acacacatga atctgatcgg tcttgcatct atctcgagtt cgacatatcc 960
aggacgggta ttacatatga aacaggtgac catgtagggt tatatgtgta aaatcatggt 1020
gaaatagttg aagaagctgg aaaaatgctt ggccactctt tagatttagt attttccata 1080
catgctgaca aggaagatgg ctccccattg gaaagcgcag tgcgcctcc tttccctggg 1140
ccatgcacac ttgggactgg tttggcaaga tacgcagacc ttttgaacc tctcgaaaag 1200
tctgcgttag ttgccttggc ggcctatgcc actgaaccaa gtgaagccga gaaacttaag 1260
cacctgacat cacctgatgg aaaggatgag tactcacaat ggattgttgc aagtcagaga 1320
agtcttttag aggtgatggc tgcctttcca tctgcaaaac cccactagg tgtattttt 1380
gctgcaatag ctctcgtctt acaacctcgt tactactcca tctcatctc gccaaagatt 1440
ggccaagta gagttcatgt tacatccgca ctagtatatg gtccaactcc tactggtaga 1500
atccacaagg gtgtgtgttc tacgtggatg aagaatgcag ttctgcgga gaaaagtcac 1560
gaatgtatgt gagcccaat ctttattcga gcatctaatt tcaagttacc atccaacct 1620
tcaactccaa tcttatggtt gggacctggg actgggctgg cactttttag aggtttttct 1680
caggaaagga tggcactaaa agaagatgga gaagaactag gttcatcttt gctctcttt 1740
gggtgtagaa atcgacagat ggactttata tacgaggtat agtcaataa tttgttgat 1800
caaggctgaa tatctgagct catcatggca ttctccgtg aaggagctca gaaggagtat 1860
gttcaacata agatgatgga gaagcgagca caagtttggg atctaataa ggaagaagga 1920
tatctctatg tatcggtgta tgcctaaggc atggcgaggg acgtccaccg aactctacac 1980
accattgttc aggagcagga aggtgtgagt tgcgcagag cagaggctat agttaagaaa 2040
cttcaaacgg aaggaagata cctcagagat gtctggtgta 2079

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SEQ ID NO: 75      moltype = DNA length = 2142
FEATURE            Location/Qualifiers
source              1..2142
                    mol_type = other DNA
                    organism = Giberella fujikuroi

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SEQUENCE: 75

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atggctgaac  tegacactct  ggacatcgtc  gtctctggcg  ttatcttctt  cggaacgggt  60
gcatacttta  caaagggcaa  gctatgggt  gttaccaagg  atccctacgc  gaatggcttc  120
gctgcccggc  ggcgttctaa  gccgggtcgc  acgaggaaca  tgcctgaggg  aatggaagaa  180
tccggcaaga  actgtgttgt  cttctatggt  tctcagaccg  gtactgctga  agattatgct  240
tctcgctcgc  ccaaggaggg  taagagtcga  ttcggactaa  acaccatgat  tgccgatctt  300
gaggactacg  atttcgacaa  cctggatacc  gttcccagtg  acaacattgt  catgttcgtt  360
ctcgcaactt  atggtgaagg  tgagcctacc  gataacggcg  tcgacttcta  tgaattcatt  420
acggcgaggg  atgccagctt  caatgagggc  aatgactctc  cgctgggcaa  cctcaactac  480
gttgcttttc  gtctcggaaa  caacacgtac  gagcactaca  actctatggt  ccgcaatggt  540
aacaaggctc  tcgagaagct  tggcgctcac  cgcctcggtg  aagctgggtg  gggatgatgat  600
gggtgctgga  ccatggaaga  ggacttcttg  gcctggaagg  atcccatgtg  ggaagccctc  660
gctaagaaaa  tgggactgga  agagcgtgaa  cgactctacg  agcctatgtt  tgccattaac  720
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catctcgaag  ggcacgccaa  gggaccattc  aactctcaca  acccctacat  tgcccctatc  840
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atcagcgggt  ctaacctcaa  gtacgaaact  ggagaccata  ttgctatctg  gcctaccaac  960
cctgggtgag  aggtcaacaa  attcctggat  attctcgacc  tctctggaaa  gcagcacagc  1020
gttgctactg  tcaaggctct  cgagcctacc  gccaaagggt  ctttcccaaa  ccctacaacc  1080
tacgatgcca  tctgcgata  ccacctcgag  atctgcgctc  ctgtttcagc  tcaattcgtc  1140
tctactctgc  ccgcatattg  tcccaacgat  gatataaagg  ctgagatgaa  ccgcccctgg  1200
agcgataaag  attatttcca  cgagaagact  ggcccgcatt  actacaacat  tgcccgtttc  1260
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aagaaaaatc  cgatcactgc  cgtcgttgaa  tcccagcaga  ttctcggcgg  ggatgatcct  1440
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aaacctgcac  cttttgtgca  gagctacgag  cttacagggc  cccgcaataa  gtatgatggc  1560
atccacgttc  ctgtccatgt  tcgtcactcc  aactccaagc  tcccctcgga  ccccggtaa  1620
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cgtgctaagc  aagcccggtg  tgggtgttgag  gttggaaaag  cactcttggt  ctttggttgc  1740
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ggcgataagt  ttaaaatgat  caccgccttt  tctcgagagg  gctccaagaa  ggtttatggt  1860
cagcaccgac  ttaaggagcg  atctaaggag  gtcacggatc  tgctctccca  gaaggcttat  1920
ttctatgtct  gcggtgatgc  agcccacatg  gcccgcgagg  tcaataccgt  cttggcacia  1980
atcattgccg  agggacgtgg  ggtgtctgag  gccaaaggcg  aggagatcgt  gaagaacatg  2040
agatcagcga  accaatacca  ggtatgtagt  gactttgtta  ctcttcactg  caaagaaacc  2100
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SEQ ID NO: 76      moltype = AA  length = 459
FEATURE            Location/Qualifiers
source              1..459
                    mol_type = protein
                    organism = Ipomoea purpurea

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SEQUENCE: 76
MGSQATTYHM  AMYPWFGVGH  LTGFFRLANK  LAGKGHRISF  LIPKNTQSKL  ESFNLHPHLI  60
SFVPIVVPPI  PGLPPGAEIT  SDVPPFSTHL  LMEAMDKTQN  DIEILKDLK  VDVVFYDFTH  120
WLPPLARKIG  IKSIFYSTIS  PLMHGYALSP  ERRVVGKQLT  EADMMKAPAS  FPDPSIKLHA  180
HEARGFTART  VMKFGGDITF  FDRIFTAVSE  SDGLAYSTCR  EIEGQFCDYI  ETQFQKPVLL  240
AGPALPVPSK  STMQKQSDW  LGKFKEGSKI  YCAFGSECTL  RKDKFQELLW  GLELTGMPFF  300
AALKPPFETE  SVEAIIPEEL  KEKIQGRGIV  HGEWVQQQLF  LQHPVSGCFV  SHCGWASLSE  360
ALVNDQCIVL  LPQVGDQIIN  ARIMSVSLKV  GVEVEKGEED  GVFSRESVCK  AVKAVMDEKS  420
ETGREVVRNH  DKLRGFLMNA  DLDKYMDSF  NQKLQDLLG  459

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SEQ ID NO: 77      moltype = DNA  length = 1380
FEATURE            Location/Qualifiers
source              1..1380
                    mol_type = other DNA
                    organism = Ipomoea purpurea

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SEQUENCE: 77
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ctaattccta  aaaacactca  atctaagtta  gaatctttca  acctcatcc  acacttaatc  180
tcttttctgc  ctatcgttgt  cccaagtata  ccaggcctgc  cactcgttgc  agagactaca  240
tcagatgttc  ctttcccaag  tacacatttg  ctaatggaa  caatggacaa  gactcaaaac  300
gatatagaga  ttatccctga  ggaacttaaa  gtagatgttg  tttctatga  tttactcac  360
tgggtgcctt  ctctggccag  aaagattggc  attaagagtg  tcttttactc  caccatttct  420
cctttaatgc  atggatatgc  tttatacca  gaaagacgtg  tagttggtaa  gcaattgaca  480
gaggcagata  tggatgaagg  cccagcttct  tcccagacc  catccattaa  gctacatgca  540
catgaagcta  ggggttttac  agccagaacc  gttatgaaat  tcggtgggtg  catcaccttt  600
ttcgatagaa  tattcacagc  agtttccgaa  agtgatggcc  tggcctactc  tactttgtaga  660
gagatcgagg  gacaattctg  tgattacatt  gaaacacaa  tccagaagcc  agtcttggtg  720
gcggtccag  ctttgcaggt  cccatccaaa  tccactatgg  aacaaaagtg  gtcagattgg  780
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gcccttgtaa  tgattgttca  aatcgtgtta  cttccacaag  ttggcgatca  gattatcaac  1140
gccagaataa  tgtcagtatc  acttaagtg  ggcgtggaag  ttgaaaagg  tgaggaggac  1200
ggtgtctttt  caagagaatc  tgtgtgcaag  gctgttaaag  cagtaatgga  tgaaaaatct  1260

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gaaatcggta gagaagtcag aggtaatcat gataaaactga ggggttttctt gatgaatgca 1320
gaacttagatt caaagtagcat ggattcattc aatcaaaaagc tacaagattt gctaggttaa 1380

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SEQ ID NO: 78      moltype = AA  length = 438
FEATURE           Location/Qualifiers
source            1..438
                  mol_type = protein
                  organism = Bellis perennis

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SEQUENCE: 78
MDSKIDSKTF RVVMLPWLAY SHISSFLVFA KRLTNHNFHI YICSSQTNMQ YLKNLTSQY 60
SKSIQLIELN LPSSSELPLQ YHTHGLPPH LTKTSLDDYQ KSGPDFETIL IKLNPFLVIY 120
DFNQLWAPEV ASTLHPSIQ LLSGCVALYA LDAHLYTKPL DENLAKFFFP EIYPKNRDIP 180
KGGSKYIERF VDCMRRSCEI ILVRSTMELE GKYIDYLSKT LGKKVLPVGP LVQEASLLQD 240
DHIWIMKWL D KKEESSVVFV CFGSEYILSD NEIEDIAYGL ELSQVSFVWA IRAKTSALNG 300
FIDRVGDKGL VIDKWVQAN ILSHSSTGGF ISHCGWSSTM ESIRYGVPII AMPMQFDQPY 360
NARLMETVGA GIEVGRDGEG RLKREEIAAV VRKVVEDSG ESIREKAKEL GEIMKKNMEA 420
EVDGIVIEINL VKLCEMNN 438

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SEQ ID NO: 79      moltype = DNA  length = 1317
FEATURE           Location/Qualifiers
source            1..1317
                  mol_type = other DNA
                  organism = Bellis perennis

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SEQUENCE: 79
atggattcta aaatcgattc aaagacattc agagtcgta tgttgcttg gcttgcatatc 60
tcacacattt catcattcct agtgtttgcc aagagactaa caaatcataa cttccacatc 120
tacatttggt cctctcaaac aaatatgcaa tacctgaaaa acaacttgac gtctcagtat 180
tcaaaatcta tacaactgat tgagttgaat cttccatcta gttccgaatt gcctctgcag 240
tatcatacta ctcacggact accaccacac cttacgaaaa cattgtctga tgattatcaa 300
aagtccggac ctgactttga aaccattttg atcaaatga acccacatct ggtaatctac 360
gactttaatc aactttgggc tccagaggtt gctagtagac ttcatatcc atccatcacg 420
ttactgtctg gttgcgtcgc cttatatgcc ttagacgccc atctgtacac aaagccacta 480
gacgaaaact tggctaagtt tcctttccca gaaatctatc ctaaaaacag agatattcct 540
aaggggaggt gtaaatatcat gaaaagggtc gtagactgta tgagaagatc ttgtgaaatc 600
atattagtcg gaagtacat ggaactgaa ggaaaataca ttgattactt gtctaagaca 660
taggggaaaa aggtgttgcc agtagggcct ctggtgcaag aggtctcttt gttgcaagat 720
gatcatatat ggattatgaa ttggttagac aaaaaggagg agtcatccgt cgtgtttggt 780
tgttttggtt ctgagtacat cttatcagac aacgaaatag aagatattgc ttatggccta 840
gagttgtccc aagtaagttt cgtttgggca ataagagcta agacttctgc cttaaatggc 900
ttcattgata gagtgggtga taaaggctta gtcacgata aatgggttcc acaggctaac 960
atcttatctc actctctcac tggtaggattc attagtcatt gcggttggtc atcaacaatg 1020
gaatctatta gatatggggt tcctattatc gccatgccaa tgcaattcga tcaaccttac 1080
aatgctaggt tgatggaaac gtttggtgca ggtatcgaa tggcagaga tggcgaaggt 1140
agattgaaaa gagaagagat tegtgcgtg gttagaaaag tcgtgttgta agattctggg 1200
gaatccataa gggagaaggc aaaggaattg ggagaaatca tgaaaaaaa catggaggcc 1260
gaagtagatg gtatagtagt tgaatatcta gttaagctat gtgagatgaa caattaa 1317

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SEQ ID NO: 80      moltype = DNA  length = 2490
FEATURE           Location/Qualifiers
source            1..2490
                  mol_type = other DNA
                  organism = Zea mays

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SEQUENCE: 80
atggttttgt cttctctctt tactacagta ccacacttat cttcattagc tgtcgtgcaa 60
cttggtcctt ggagcagtag gattaaaaag aaaaccgata ctgttgagc accagcgcgt 120
gcagggaagt ggagaagggc cttggctaga gcacagcaca catcagaatc cgcagctgtc 180
gcaaggggca gcagtttgac ccttatagtg agaactgacg ctgagtcagg gagaacaaga 240
tggccaacgc atgacgatga cgccgaacct ttagtggatg agatcagggc aatgcttact 300
tccatgtctg atggtgacat ttcctgagc gcatacgata cagcctgggt cggattgggt 360
ccaagattag acggcgttga aggtcctcaa tttccagcag ctgtgagatg gataaagaat 420
aaccagttgc ctgacgggaa ttggggcgat gccgcattat tctctgccta tgacaggcct 480
atcaataccc ttgcctgcgt tgttaacttg acaagggtgt ccctagaacc agagatgaga 540
ggtagaggac tatctttttt gggtaggaac atgtggaaat tagcaactga agatgaagag 600
tcaatgccta ttggtctcga attagcattt ccatctttga tagagcttgc taagagccta 660
ggtgtccatg acttccctta tgatcaccag gccctacaag gaatctactc ttcaagagag 720
atcaaaatga agaggattcc aaaagaagtg atgcataccg ttccaacatc aatattgcac 780
agtttgaggg ttagtcctgg cctagattgg gctaaactac ttaaaactca gagcagcgac 840
ggaggttttt gttctcacc agctgccact gcatatgctt taatgaatac cggagatgac 900
agggtgttta gctacatcga tagaacagta aagaaattca acggcggcgt ccctaattgt 960
tatccagtgg atctatttga acatatattg gccgttgata gacttgaaag attaggaatc 1020
tccaggtact tccaaaagga gatcgaacaa tgcattggatt atgtaaacag gcattggact 1080
gaggacggta tttgttgggc aaggaactct gatgtcaaag aggtggacga cacagctatg 1140
gcctttagac ttcttaggtt gcacggctac agcgtcagtc ctgatgtgtt taaaaacttc 1200
gaaaaggacg gtgaattttt cgactttgtc ggacagtcta atcaagctgt taccggtatg 1260
tacaacttaa acagagcaag ccagatatcc ttcccaggcg aggatgtgct tcatagagct 1320
ggtgccttct catatgagtt cttgaggaga aaagaagcag agggagcttt gagggacaag 1380
tggatcattt ctaaaagatc acctggtgaa gttgtgtata ctttgattt tccatggtag 1440
ggcaacttac ctagatcga gccacagagc tacctagagc aatacggagg tggtagtagc 1500
gtttggattg gcaagacatt gtatagtagt ccacttgtaa acaatgatgt atatttggaa 1560

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ttggcagaa tggatttcaa cactgcccag gctttgcac agttagagtg gcaaggacta 1620
aaaagatggt atactgaaaa taggttgatg gactttgggt tcgcccaga agatgcocct 1680
agagcttatt ttcttgagc cgcactgtgt tacgagcctt gtagagctgc cgagaggctt 1740
gcatgggcta gagccgcaat actagctaac gccgtgagca cccacttaag aaatagccca 1800
tcattcagag aaagggttaga gcattctctt aggtgtagac ctagtgaaga gacagatggc 1860
tcctggttta actcctcaag tggctctgat gcagttttag taaaggctgt cttaagactt 1920
actgattcat tagccaggga agcacagcca atccatggag gtgaccaga agatattata 1980
cacaagtgt taagatctgc ttgggcccag tgggttaggg aaaaggcaga cgtgcccag 2040
agcgtgtgca atggtagtgc tgcagtagaa caagagggat caagaatggt ccatgataaa 2100
cagacctgtc tattattggc tagaatgac gaaatttctg ccggtagggc agctggtgaa 2160
gcagccagtg aggcggcga tagaagaata attcaattaa caggctccat ctgacagct 2220
cttaagcaaa aaatgctagt ttacacaggc cctgaaaaaa atgaagagat gatgtctcac 2280
gtggatgacg aattgaagtt gaggattaga gagttcgttc aatatttgc tagactaggt 2340
gaaaaaaaga ctggatctag cgaaccagg caaacatttt taagtatagt gaaatcatgt 2400
tactatgctg ctcatgtccc acctcatgtc gttgatagac acatttagtag agtgattttc 2460
gagccagtaa gtgccgcaaa gtaaccgagg 2490

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SEQ ID NO: 81      moltype = AA length = 827
FEATURE           Location/Qualifiers
source            1..827
                  mol_type = protein
                  organism = Zea mays

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SEQUENCE: 81
MVLSSSCTTV PHLSSLAVVQ LGPWSSRIKK KTDTVAVPAA AGRWRRALAR AQHTSESAAV 60
AKGSLTPIV RTDAESRRTR WPTDDDDAEP LVDEIRAMLT SMSDGDISVS AYDTAWVGLV 120
PRLDGGEGPQ FPAAVRWIRN NQLPDGSGWD AALFSAYDRL INTLACVVTL TRWSLEPEMR 180
GRGLSFLGRN MWKLATEDEE SMPIGFELAF PSILIELAKSL GVHDFPYDHQ ALQGIYSSRE 240
IKMKRIPKEV MHTVPTSILH SLEGMPLDWD AKLLKLQSSD GSFLFSPAAT AYALMNTGDD 300
RCFSYIDRTV KKFNGGVPNV YPVDLFEHIW AVDRLERLGI SRYFQKEIEQ CMDYVNRHWT 360
EDGICWARSN DVKEVDDTAM AFRLLRLHGY SVSPDVFKNF EKDGEFFAFV GQSNQAVTGM 420
YNLNRASQIS FPGEDVLHRA GAFSYEFLRR KEAEGALRDK WIISKDLPE VVYTLDFPWY 480
GNLPRVEARD YLEQYGGGDD VWIGKTLYRM PLVNNVDVYLE LARMDFNHCQ ALHQLEWQGL 540
KRWYTENRML DFGVAQEDAL RAYFLAAASV YEPCAAERL AWARAAILAN AVSTHLRNSP 600
SPRERLEHSL RCRPSEETDG SWFNSSSGSD AVLVKAVLRL TDSLAREAPQ IHGGDPEDII 660
HKLLRSAAWE WVREKADAAD SVCNGSSAVE QEGSRMVHDK QTCLLLARMI EISAGRAAGE 720
AASEDGDRII IQLTGSICDS LKQKMLVSQD PEKNEEMMSH VDDELKLRIR EFVQYLLRLG 780
EKKTGSSETR QTFLSIVKSC YYAAHCPPHV VDRHISRVIF EPVSAAK 827

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SEQ ID NO: 82      moltype = DNA length = 1461
FEATURE           Location/Qualifiers
misc_feature       1..1461
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1461
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 82
actagtaaaa tggatgcaat ggcaactact gagaaaaagc ctcatgtgat cttcattcca 60
tttctctcac aatctcacat aaaggcaatg ctaaagttag cacaactatt acaccataag 120
ggattacaga taactttcgt gaataccgac ttcatccata atcaatttct ggaatctagt 180
ggccctcatt ttgtggacgg agccccagggt ttagatttcg aaacaattcc tgacggtgtt 240
tcacattccc cagagggcttc catcccaata agagagaggtt tactgagggtc aatagaaacc 300
aacttttttg atcgtttcat tgacttggtc acaaaaacttc cagaccacc aacttcata 360
atctctgatg gctttctgtc agtgtttact atcgacgctg ccaaaaagt gggtatocca 420
gttatgatgt actggactct tgcgtcatgc ggtttcatgg gtttctatca catccattct 480
cttatcgaaa agggttttgc tccactgaaa gatgcatcat acttaacca cggctacctg 540
gatactgtta ttgactgggt accaggtatg gaaggtataa gacttaaaaga ttttctttg 600
gattgggtcta cagaccttaa tgataaagta ttgatgttta ctacagaagc tccacaaaga 660
tctcataaagg tttcacatca tatctttcac acctttgatg aattggaacc atcaatcatc 720
aaaaccttgt ctctaagata caatcatatc tacactattg gtccattaca attacttcta 780
gatcaaatc ctgaagagaa aaagcaaaact ggtattacat ccttacacgg ctactcttta 840
gtgaaagagg aaccagaatg ttttcaatgg ctacaaaagta aagagcctaa ttctgtggtc 900
tacgtcaact tcggaagtac aacagtcagt tccttggaag atatgactga atttggttgg 960
ggccttgcta attcaaatca ttactttcta tggattatca ggtccaattt ggtaaatagg 1020
gaaaacgccc tattacctcc agaattggag gaacacatca aaaagagagg ttctattgct 1080
tcctggtgtt ctacagaaaa ggtattgaaa catccttctg ttggtggtt ccttactcat 1140
tgcggttggg gctctacaat cgaatcacta agtgaggag ttccaatgat ttgttggcca 1200
tattcatggg accaacttac aaattgtagg tatatctgta aagagtggga agttggatta 1260
gaaatgggaa caaagggtta acgtgatgaa gtgaaaagat tggttcagga gttgatgggg 1320
gaaggtggcc acaagatgag aaacaaggcc aaagattgga aggaaaaagc cagaatttgt 1380
attgctccta acgggtcatc ctctctaacc attgataaga tggtaaaaga gattacagtc 1440
ttagccagaa actaagtcga c 1461

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SEQ ID NO: 83      moltype = DNA length = 1398
FEATURE           Location/Qualifiers
misc_feature       1..1398
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1398

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 83
actagtaaaa tggcagagca acaaaagatc aaaaagtcac ctacagtcctt acttattcca 60
tttcctctgc aaggacatat caaccatttc atacaatttg ggaaaagatt gattagtaag 120
gggtgtaaaa caacactggg aaccactatc cacactttga attctactct gaaccactca 180
aataactacta ctacaagtat agaaattcaa gctatatcag acggatgcga tgagggtggc 240
tttatgtctg ccggtgaatc ttacttggaa acattcaagc aagtgggac caagtctctg 300
gccgatctaa tcaaaaagtt acagagtgaa ggcaccacaa ttgacgccat aatctacgat 360
tctatgacag agtgggtttt agacgttgct atcgaatttg gtattgatgg aggttccttt 420
ttcacacaag catgtgttgt gaattctcta tactaccatg tgcataaagg gttaatctct 480
ttaccattgg gtgaaactgt ttacgttcca gggtttccag tgttacaacg ttgggaaacc 540
ccattgatct tacaatatca tgaacaaata caatcacctt ggtcccagat gttgttttgt 600
caattcgcta acatcgatca agcaagatgg gtctttacta attcattcta taagttagag 660
gaagaggtta ttgaatggac taggaagatc tgggaattga aagtcattgg tccaacattg 720
ccatcaatgt atttggacaa aagacttgat gatgataaag ataatggttt caatttgtac 780
aaggctaatic atcacgatga tatgaatttg ctggatgaca aaccaaagga atcagttgta 840
tatgttgctt tcggtctctt tgttaaacat ggtccagaac aagttaggga gattacaaga 900
gcacttatag actctgacgt aaactttttg tgggtcatta agcacaaga ggaggggaaa 960
ctgccagaaa accctttctga agtgataaag accgaaaag gtctaactgt tgcttggtgt 1020
aaacaattgg atgttttagc tcatgaatct gttagctgtt ttgtaacaca ttgcggattc 1080
aactctacac tagaagcctt ttcttaggc gtacctgtcg ttgcaatgcc tcagttctcc 1140
gatcagacaa ccaacgctaa acttttggac gaaatactag gggtaggtgt cagagttaaa 1200
gcagacgaga atggtatcgt cagaagaggg aacctagctt catgtatcaa aatgatcatg 1260
gaagagggaaa gagaggttat ctaaggaaa aacgcagtta agtggaagga tcttgcaaa 1320
gttgccgtcc atgaaggcgg ctcttcagat aatgatattg ttgaatttgt gtccgaacta 1380
atcaaacgct aagtcgac                                     1398

SEQ ID NO: 84      moltype = DNA length = 1437
FEATURE            Location/Qualifiers
misc_feature        1..1437
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1437
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 84
actagtaaaa tggctacatc tgattctatt gttgatgaca ggaagcagtt gcatgtggct 60
actttccctt ggcttgtttt ccgtcatata ctgocctacc tacaactatc aaaactgata 120
gctgaaaaag gacataaagt gtcattcctt tcaacaacta gaaacattca aagattatct 180
tcccacatat caccattgat taacgtcgtt caattgacac ttccaagagt acaggaatta 240
ccagaagatg ctgaagctac aacagatgtg catcctgaag atatccctta cttgaaaaag 300
gcaccogatg gattacagcc tagagtcact agattccttg agcaacacag tccagatttg 360
atcatatacg actacactca ctattgggtg ccttcaattg cagcatcact aggcatttct 420
agggcacatt tcagtgtaac cacaccttgg gccattgctt acatgggttc atccgctgat 480
gctatgatta acggcagtg tggtagaact accgttgaag atttgacaac cccaccaaag 540
tgggttccat ttccaactaa agtctgttgg agaaaacacg acttagcaag actggttcca 600
tacaaggcac caggaatctc agacggctat agaattgggt tagtctctaa agggctctgac 660
tgcttattgt ctaagtgtta ccatgagttt gggacacaaat ggctaccact tttggaaca 720
ttacaccaag ttctgtcgt accagtttgt ctattacctc cagaaatccc tggatgatgag 780
aaggacgaga cttgggtttc aatcaaaaag tgggttagac ggaagcaaaa aggcctcagt 840
gtatatgtgg cctcgggttc cgaagtttta gtatctcaaa cagaagttgt ggaacttgcc 900
ttaggtttgg aactatctgg attgccattt gtctgggctt acagaaaacc aaaagggcct 960
gcaaagtcgg attcagttga attgccagac ggctttgtcg agagaactag agatagaggg 1020
ttggtatgga cttcagtggt tccacaattg agaactctga gtcacgaatc tgtgtgctgt 1080
ttcctaaccac attgtgtgtc tgggtctata gttgaaggac tgatgttttg tcatccactt 1140
atcatgttgc caatctttgg tgaccagcct ttgaatgcac gtctgttaga agataaacia 1200
gttgaatttg aaatcccacg taatgaggaa gatggatgtt taaccaagga gtctgtggcc 1260
agatcattac gttccgttgt cgttgaaaaa gaaggcgaaa tctacaaggc caatgccctg 1320
gaactttcaa agatctacaa tgacacaaaa gttagagaag aatatgttct tcaatttgta 1380
gattacctag agaaaaacgc tagagccgta gctattgatc atgaatccta agtcgac 1437

SEQ ID NO: 85      moltype = DNA length = 1392
FEATURE            Location/Qualifiers
misc_feature        1..1392
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1392
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 85
actagtaaaa tggaaaacaa gaccgaaaca acagtttagc gtaggcgtag aatcattctg 60
tttcagtagc cttttcaagg gcacatcaat ccaatactac aactagccaa cgttttgtac 120
tctaaagggt ttctatttac aatctttcac accaatttca acaaaccaaa aacatccaat 180
taccacattt tcacattcag attcatactt gataatgatc cacaagatga acgtatttca 240
aacttaccta cccacggtcc tttagctgga atgagaattc caatcatcaa tgaacatggt 300
gccgatgagc ttagaagaga attagagtta cttatgttgg catccgaaga ggacgaggaa 360
gtctcttgct tgattactga cgtctatagg tactttgccc aatctgtggc tgatagtttg 420
aatttgagga gattggtact aatgacatcc agtctgttta actttcacgc tcatgttagt 480

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ttaccacaat ttgacgaatt gggatacttg gaccctgatg acaagactag gttagaggaa 540
caggcctctg gttttcctat gttgaaagtc aaagatatca agtctgccta ttctaattgg 600
caaatcttga aagagatctt aggaaagatg atcaaacaga caaaggcttc atctggagtg 660
atttggaaca gtttcaaaga gttagaagag tctgaattgg agactgtaat cagagaaatt 720
ccagcacctt cattoctgat accattacca aaacatttga ctgcttcctc ttctctttg 780
ttggatcatg acagaacagt ttttcaatgg ttggaccaac aaccacctag ttctgtttg 840
tacgtgtcat ttggtagtac ttctgaagtc gatgaaaagg acttccttga aatcgcaaga 900
ggcttagtcg atagtaagca gtcattcctt tgggtcgtgc gtccaggttt cgtgaaaggg 960
tcaacatggg tcgaaccact tccagatggg tttctaggcg aaagaggtag aatagtcaaa 1020
tgggttcctc aacaggaagt tttagctcat ggcgctattg gggcattctg gactcattcc 1080
ggatggaatt caactttaga atcagtatgc gaaggggtac ctatgatctt ttcagatttt 1140
ggtcttgatc aaccactgaa cgcaagatac atgtctgatg ttttgaaagt ggggtgtatat 1200
ctagaaaaatg gctgggaaag ggggtgaaata gctaatagcaa taagacgtgt tatgggtgat 1260
gaagaggggg agtatatcag acaaaacgca agagtgtgta agcaaaaggg cgacgtttct 1320
ctaatagaag gaggtctctc atacgaatcc ttagaatctc ttgtttccta catttcatca 1380
ctgtaagtgc ac 1392

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SEQ ID NO: 86      moltype = AA length = 12
FEATURE           Location/Qualifiers
REGION            1..12
                  note = Description of Artificial Sequence: Synthetic peptide
source            1..12
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 86
TSFAEYWNLL SP 12

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SEQ ID NO: 87      moltype = DNA length = 1602
FEATURE           Location/Qualifiers
misc_feature       1..1602
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1602
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 87
atgaccagct ttgccgagta ttggaatctg ttaagtccca cttcttttgc agaatatgg 60
aaccttctat caccgacgag ttctgcggag tactggaatt tgttttctcc aacatcgctc 120
gctgaatact ggaacttact cagccctgct agtaaaatgg atgcaatggc aactactgag 180
aaaaagcctc atgtgatctt cattccattt cctgcacaat ctacataaaa ggcaatgcta 240
aagtagcac aactattaca ccataaggga ttacagataa ctttcgtgaa taccgacttc 300
atccataatc aatttctgga atctagtggc cctcattgtt tggacggagc cccagggttt 360
agattcgaaa caattcctga cgggtgttca cattccccag aggcctccat cccaataaga 420
gagagtttac tgagggtcaat agaaaccaac tttttggatc gtttcattga cttggtcaca 480
aaacttccag accccaacac ttgcataatc tctgatggct ttctgtcagt gtttactatc 540
gacgtcgcca aaaagtggg tatcccagtt atgatgtact ggactcttgc tgcattcggt 600
ttcatgggtt tctatcacat ccattctctt atcgaaaagg gttttgctcc actgaaagat 660
gcatcactac taaccaacgg ctacctggat actgttattg actgggtacc aggtatggaa 720
ggataaagac ttaagatttt tcctttggat tgggtctacag acctaatga taaagtattg 780
atgtttacta cagaagctcc acaaaagatc cataagggtt caccatcatat ctttcacacc 840
tttgatgaat tggaaacctc aatcatcaaa acctgtgctc taagatacaa tcatatctac 900
actatttggtc cattctagat caaattcctg aagagaaaaa gcaaaactggt 960
attacatcct tacacggcta ctcttttagt aaagaggaaac cagaatgttt tcaatggcta 1020
caaagtaaa agcctaattc ttggtgtctc gtcaacttcg gaagtacaac agtcatgtcc 1080
ttggaagata tgactgaaat cttgtggggc cttgctaatt caaatcatta ctttctattg 1140
attatcaggt ccaatttggg aataggggaa aacgcgctat tacctccaga attggaggaa 1200
cacatcaaaa agagaggttt cattgcttcc tgggtgtctc aggaaaaggt attgaaacat 1260
ccttctgttg gtggtttctc tactcattgc ggttggggct ctacaatcga atcactaagt 1320
gcaggagttc caatgatttg ttggccatat tcatggggacc aacttacaaa ttgtagggat 1380
atctgtaaa agtgggaagt tggattagaa atgggaacaa aggttaaacg tgatgaagt 1440
aaaaagattg ttcaggaggt gatgggggaa ggtggccaca agatgagaaa caaggccaaa 1500
gattggaagg aaaaagccag aattgtctat gctcctaacy ggctcatctc tctaaacatt 1560
gataagatgg tcaaagagat tacagtctta gccagaaact aa 1602

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SEQ ID NO: 88      moltype = AA length = 533
FEATURE           Location/Qualifiers
REGION            1..533
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..533
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 88
MTSFAEYWNL LSPTSFAEYW NLLSPTSFAE YWNLFSPSTF AEYWNLLSPA SKMDAMATTE 60
KKPHVIFIPF PAQSHIKAML KLAQLLHHKG LQITFVNTDF IHNQFLESSG PHCLDGAPGF 120
RFETIPDGVF HSPFASPIR ESLLRSIETN FLDRFIDLVT KLPDPPTCII SDGLSVFTI 180
DAAKKGIPV MMYWTLAAGC FMGFYHIHSL IEKGFAPLKD ASYLTNGYLD TVIDWVPOME 240
GRLKDFPLD WSTDLNDKVL MFTTEAPQRS HKVSHHIFHT FDELEPSIIK TSLRYNHIY 300
TIGPLQLLLD QIPEKKQTG ITS LHGYSLV KEEPECFQWL QSKEPNSVVY VNFGSTTVMS 360

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LEDMTFEGWG LANSNHYFLW IIRSNLVIGE NAVLPPELEE HIKKRGFIAS WCSQEKVLKH 420
 PSVGGFLTHC GWGSTIESLS AGVPMICWPY SWDQLTNCRY ICKEWEVGLE MGTKVKRDEV 480
 KRLVQELMGE GGHKMRNKAK DWEEKARIAI APNGSSSLNI DKMVKEITVL ARN 533

SEQ ID NO: 89 moltype = DNA length = 1893
 FEATURE Location/Qualifiers
 misc_feature 1..1893
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1893
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 89

atgtgcaata	ccaacatgtc	tgtacctact	gatggtgctg	taaccacctc	acagattcca	60
gcttcggaac	aagagaccct	ggtagacca	aagccattgc	ttttgaagtt	attaaagtct	120
gttggtgcac	aaaaagacac	ttatactatg	aaagagggtc	tttttatctt	tggccagtat	180
attatgacta	aacgattata	tgatgagaag	caacaacata	ttgtatatgt	ttcaaatgat	240
cttctaggag	atttgtttgg	cgtgccaaag	ttctctgtga	aagagcacag	gaaaatatat	300
accatgatct	acaggaacct	ggtagtagtc	aatcagcagg	aatcatcgga	ctcaggtaca	360
tctgtgagtg	agaacagggt	tcaccttgaa	ggtgggagtg	atcaaaagga	ccttgtaaca	420
gagcttcagg	aagagaaacc	ttcatcttca	catttggttt	ctagaccatc	taccgggtgt	480
agcggatcct	ctggaggcag	tgctagtata	atggcagagc	aacaaaagat	caaaaagtca	540
cctcacgtct	tacttatttc	atttctcttg	caaggacata	tcaacccatt	catacaattt	600
gggaaaaagt	tgatttagta	gggtgtaaag	acaacactgg	taaccactat	ccacactttg	660
aattctactc	tgaaccactc	aaatactact	actacaagta	tagaaattca	agctatatca	720
gacggatgag	atgaggggtg	ctttatgtct	gccggtgaat	cttacttgga	aacattcaag	780
caagtgggat	ccaagtctct	ggccgatcta	atcaaaaagt	tacagagtga	aggcaccaca	840
attgacgcca	taatctacga	ttctatgaca	gagtgggttt	tagacgttgc	tatcgaattt	900
ggtattgatg	gaggttcott	tttcacacaa	gcatgtgttg	tgaattctct	atactaccat	960
gtgcataaag	gggttaatctc	tttaccattg	ggtgaaactg	tttcagttcc	aggtttttcca	1020
gtgttacaac	ggtgggaaac	cccattgatc	ttacaaaatc	atgaacaaat	acaatcacct	1080
tgggtcccaga	tgtgtgtttg	tcaattcgct	aacatcgatc	aagcaagatg	ggtctttact	1140
aattcattct	ataagttaga	ggaagaggta	attgaatgga	ctaggaagat	ctggaatttg	1200
aaagtctatt	gtccaaacat	gccatcaatg	tatttggaac	aaagacttga	tgatgataaa	1260
gataatggtt	tcaatttgta	caaggctaag	catcacgaat	gtatgaattg	gctggatgac	1320
aaaccaaaag	aatcagttgt	atatgttgct	ttcggctctc	ttgttaaaca	tggtccagaa	1380
caagttgagg	agattacaag	agcacttata	gactctgacg	taaacttttt	gtgggtcatt	1440
aagcacaaaag	aggaggggaa	actgccagaa	aacctttctg	aagtataaaa	gaccggaaaa	1500
ggtctaatcg	ttgcttggtg	taaaccaattg	gatgttttag	ctcatgaatc	tgtaggttgt	1560
tttgtaacac	attgcggatt	caactctaca	ctagaagcca	tttccttagg	cgtacctgtc	1620
gttgcaatgc	ctcagttctc	cgatcagaca	accaacgcta	aacttttgga	cgaataacta	1680
ggggtgggtg	tcagagttaa	agcagacgag	aatggtatcg	tcagaagagg	gaacctagct	1740
tcattgtatc	aaatgatcat	ggaagaggaa	agaggagtta	tcataaggaa	aaacgcagtt	1800
aagtgggaag	atcttgcaaa	gggtgcccgc	catgaaggcg	gctcttcaga	taatgatatt	1860
gttgaatttg	tgtccgaact	aatcaaaagg	taa			1893

SEQ ID NO: 90 moltype = AA length = 630
 FEATURE Location/Qualifiers
 REGION 1..630
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..630
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 90

MCNTNMSVPT	DGAVTTSQIP	ASEQETLVRP	KPLLLKLLKS	VGAQKDTYTM	KEVLFPYLGQY	60
IMTKRLYDEK	QQHIVYCSND	LLGDLFGVPS	FSVKEHRKIY	TMIYRNLVVV	NQQESSDSGT	120
SVSENRCHE	GGSDQKDLVQ	ELQEEKPSSS	HLVSRPSTGG	SGSSGGSASK	MAEQQKIKKS	180
PHVLLIPFPL	QGHINPFIQF	GKRLISKGVK	TTLVTTIHTL	NSTLNHSNTT	TTSIEIQAIS	240
DGCDEGGFMS	AGESYLETFK	QVGSKSLADL	IKKLQSEGTT	IDAIYDSMT	EWVLDVAIEF	300
GIDGGSFFTQ	ACVVNSLYYH	VHKGLISLPL	GETVSVPGFP	VLQRWETPLI	LQNHEQIQSP	360
WSQMLFGQFA	NIDQARWVFT	NSFYKLEEEV	IEWTRKIWNL	KVIGPTLPSP	YLDKRLDDDK	420
DNGFNLYKAN	HHECMNWLDD	KPKESVYVVA	FGSLVKHGPE	QVEEITRALI	DSDVNFLWVI	480
KHKEEGKLPE	NLSVVIKTGK	GLIVAWCKQL	DVLAHESVGC	FVTHCGFNST	LEAISLGVVP	540
VAMPQFSQDT	TNAKLLDEIL	GVGVRVKADE	NGIVRRGNLA	SCIKMIMEEE	RGVIIRKNVAV	600
KWKDLAKVAV	HEGGSSNDNI	VEFVSELIKA				630

SEQ ID NO: 91 moltype = DNA length = 1932
 FEATURE Location/Qualifiers
 misc_feature 1..1932
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1932
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 91

atgtgcaata	ccaacatgtc	tgtacctact	gatggtgctg	taaccacctc	acagattcca	60
gcttcggaac	aagagaccct	ggtagacca	aagccattgc	ttttgaagtt	attaaagtct	120
gttggtgcac	aaaaagacac	ttatactatg	aaagagggtc	tttttatctt	tggccagtat	180

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attatgacta aacgattata tgatgagaag caacaacata ttgtatatgg ttcaaatgat 240
cttctaggag atttggttgg cgtgccaaagc ttctctgtga aagagcacag gaaaatata 300
accatgatct acaggaactt ggtagtagtc aatcagcagg aatcatcgga ctccaggtaca 360
tctgtgagtg agaacagggt tcaccttgaa ggtgggagtg atcaaaagga cctgtgtaca 420
gagcttcagg aagagaaacc ttcatcttca catttggttt ctgaccatc taccggtggg 480
agcggatcct ctggaggcag tgcagtaaaa atggctacat ctgattctat tgttgatgac 540
aggaagcagt tgcattgtgc tactttccct tggcttgctt tgggtcatat actgccttac 600
ctacaactat caaaactgat agctgaaaaa ggacataaag tgtcattcct ttcaacaact 660
agaaacatct aaagattatc ttcccacata tcaccattga ttaacgtcgt tcaattgaca 720
cttccaagag tacaggaatt accagaagat gctgaagcta caacagatgt gcatcctgaa 780
gatatccctt acttgaaaaa ggcatccgat ggattacagc ctgaggtcac tagattcctt 840
gagcaacaca gtccagattg gatcatatac gactacactc actattggtt gccttcaatt 900
gcagcatcac taggcatttc tagggcacat ttcagtgtaa ccacaccttg ggccattgct 960
tacctgggtc catccgtgta tgctatgatt aacggcagtg atggtagaac taccgttgaa 1020
gatttgacaa cccaccacaa gtgggtttcca ttccaacta aagtctgttg gagaaaacac 1080
gacttagcaa gactggttcc atacaaggca ccaggaatct cagacggcta tagaatgggt 1140
ttagtcctta aagggtctga ctgcctattg tctaagtggt accatgagtt tgggacacaa 1200
tgggtaccac ttttggaac attacaccaa gtctctgtcg taccagttgg tctattacct 1260
ccagaatcc ctggtgatga gaaggacgag acttgggttt caatcaaaaa gtggttagac 1320
gggaagcaaa aaggctcagt ggtatatgtg gcactgggtt ccgaagtttt agtatctcaa 1380
acagaagtgt tggaaactgc cttaggtttg gaactatctg gattgccatt tgtctgggcc 1440
tacagaaaac caaaagccc tgcaagtcg gattcagttg aattgccaga cggtttgtc 1500
gagagaacta gagatagagg gttggtatgg acttcatggg ctccacaatt gagaatcctg 1560
agtccagaa ctgtgtgcgg ttctcaaca cattgtggtt ctggttctat agttgaagga 1620
ctgatgtttg gtcattccact tatcatgttg ccaatctttg gtgaccagcc ttggaatgca 1680
cgtctgttag aagataaaca agttggaatt gaaatccacc gtaatgagga agatggatgt 1740
ttaaccaagg agtctgtggc cagatcatta cgttccgttg tcgttgaaaa ggaaggcgaa 1800
atttacaagg ccaatgcccg tgaactttca aagatctaca atgacacaaa agtagagaag 1860
gaatatgttt ctcaatttgt agattaccta gagaaaaacg ctgagacggt agctattgat 1920
catgaatcct aa 1932

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SEQ ID NO: 92      moltype = AA length = 643
FEATURE           Location/Qualifiers
REGION            1..643
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..643
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 92
MCNTNMSVPT DGAVTTSQIP ASEQETLVRP KPLLLKLLKS VQAQKDYTM KEVLFYLGQY 60
IMTKRLYDEK QQHIVYCSND LLGDLFGVPS FSVKEHRKIY TMIYRNLVVV NQQESSDSGT 120
SVSENKCHLE GGSQKDLVQ ELQEEKPSSS HLVSRPSTGG SGSSGGSASK MATSDSIVDD 180
RKQLHVATFP WLAFGHILPY LQLSKLIAEK GHKVSFLSTT RNIQRLSSHI SPLINVVQLT 240
LPRVQELPED AEATTDVHPE DIPYLKASD GLQPEVTRFL EQHSPDWIIY DYTHYWLPSI 300
AASLGI SRAH FSVTPWAIA YMGPSADAMI NGSDGRTTVE DLTTPPKWFP FPTKVCWRKH 360
DLARLVPIKA PGISDGYRMG LVLKGSDDL LSKYHEFGTQ WLPLETLHQ VPVVPVGLLP 420
PEIPGDEKDE TWVSIKKWLD GKQKGSVVYV ALGSEVLVSQ TEVVELALGL ELSGLPFVWA 480
YRKPKGPAKS DSVLELPGFV ERTDRGLVW TSWAPQLRIL SHESVCGFLT HCGSGSIVEG 540
LMFGHPLIML PIFGDQPLNA RLLEDKQVGI EIPRNEEDGC LTKESVARSL RSVVVEKEGE 600
IYKANARELS KIYNDTKVEK EYVSQFVDYL EKNARAVAIH HES 643

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SEQ ID NO: 93      moltype = DNA length = 1887
FEATURE           Location/Qualifiers
misc_feature       1..1887
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1887
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 93
atgtgcaata ccaacatgtc tgtacctact gatggtgctg taaccacctc acagattcca 60
gcttcggaac aagagaccct ggttagacca aagccattgc ttttgaagtt attaaagtct 120
gttggtgcac aaaaagacac ttatactatg aaagagggtc ttttttatct tggccagtat 180
attatgacta aacgattata tgatgagaag caacaacata ttgtatatgg ttcaaatgat 240
cttctaggag atttggttgg cgtgccaaagc ttctctgtga aagagcacag gaaaatata 300
accatgatct acaggaactt ggtagtagtc aatcagcagg aatcatcgga ctccaggtaca 360
tctgtgagtg agaacagggt tcaccttgaa ggtgggagtg atcaaaagga cctgtgtaca 420
gagcttcagg aagagaaacc ttcatcttca catttggttt ctgaccatc taccggtggg 480
agcggatcct ctggaggcag tgcagtaaaa atggaaaaa agaccgaaac aacaggttaga 540
cgtaggcgta gaatcattct gtttccagta ccttttcaag ggacatcaa tccaatacta 600
caactagcca acgttttcta ctctaaaggt ttttctatta caatctttca caccaatttc 660
aacaaccaa aaacatccaa ttaccacat ttcacattca gattcatact tgataatgat 720
ccacaagatg aacgtatttc aaacttacct acccaggtc ctttagctgg aatgagaatt 780
ccaatcatca atgaacatgg tgccgatgag cttagaagag aattagagtt acttatgttg 840
gcatccgaag aggacgagga agtctcttgt ctgattactg acgctctatg gtactttgcc 900
caatctgtgg ctgatagttt gaatttgagg agattggtag taatgacatc cagtctgttt 960
aactttcacg ctcatgttag ttaccacaa tttgacgaat tgggatactt ggacctgat 1020
gacaagacta ggtagagga acaggcctct ggttttctta tgttgaaagt caaagatatc 1080

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aagtctgcct attctaattg gcaaatcttg aaagagatct taggaaagat gatcaaacag 1140
acaaaggctt catctggagt gatttggaa acgttcaaa agttagaaga gtctgaattg 1200
gagactgtaa tcagagaaat tccagcacct tcattcctga taccattacc aaaacatttg 1260
actgcttctt cttcctcttt gttggatcat gacagaacag ttttcaatg gttggacca 1320
caaccaccta gttctgtttt gtacgtgtca tttgtagta cttctgaagt cgtgaaaag 1380
gacttccttg aaatcgcaag aggccttagtc gatagtaagc agtcattcct ttgggtcgtg 1440
cgtccagggt tcgtgaaaag ctcaacatgg gtccaaccac ttcagatgg tttcttaggc 1500
gaaagaggta gaatagtc aaatgggttc caacaggaag ttttagctca tggcgctatt 1560
ggggcattct ggactcattc cggatggaat tcaactttag aatcagtatg cgaaggggta 1620
cctatgatct tttcagattt tggctctgat caaccactga acgcaagata catgtctgat 1680
gttttgaaa ggggtgtata tctagaaaa ggctgggaaa ggggtgaaat agctaatagca 1740
ataagacgtg ttatgggtga tgaagagggg gagtatatca gacaaaacgc aagagtgtgt 1800
aagcaaaagg ccgacgtttc tctaataga ggaggtcttt catacgaatc cttagaatct 1860
cttgtttctt acatttcac tctgtaa 1887

SEQ ID NO: 94      moltype = AA length = 628
FEATURE           Location/Qualifiers
REGION            1..628
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..628
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 94
MCNTNMSVPT DGAVTTSQIP ASEQETLVRP KPLLLKLLKS VGAQKDTYTM KEVLFYLGQY 60
IMTKRLYDEK QQHIVYCSND LLGDLFGVPS FSVKEHRKIY TMIYRNLVVV NQQESSDSGT 120
SVSENRCHLE GGSQDKDLVQ ELQEEKPSSS HLVSRPSTGG SGSSGGGASK MENKTETTVR 180
RRRRRIILFPV PQGHINPIL QLANVLYSKG FSITIFHTNF NKPKTSNYPH FTRFILDND 240
PQDERISNLP THGPLAGMRI PIINEHGADE LRRELELLML ASEDEEVSC LITDALWYFA 300
QSVADSLNLR RLVLMTSSLF NFHAHVSLPQ FDELGYLDPD DKTRLEEQAS GFPMLKVKI 360
KSAYSNWQIL KEILGKMIKQ TKASSGVIWN SPKELEESL ETVIREIPAP SFLIPLPKHL 420
TASSSSLLDH DRTVFQWLDQ QPPSSVLYVS FGSTSEVDEK DFLEIARGLV DSKQSFLWV 480
RPGFVKGSTW VEPLPDGFLG ERGRIVKWVP QQEVLAHGAI GAFWTHSGWN STLESVCEGV 540
PMIFSDFGLD QPLNARYMSD VLKGVVYLEN GWERGEIANA IRRVMVDEEG EYIRQNARVL 600
KQKADVSLMK GGSSYESLES LVSYISSL 628

SEQ ID NO: 95      moltype = AA length = 473
FEATURE           Location/Qualifiers
source            1..473
                  mol_type = protein
                  organism = Stevia rebaudiana

SEQUENCE: 95
MATSDSIVDD RKQLHVATFP WLAFGHILPY LQLSKLIAEK GHKVSFLSTT RNIQRLSSHI 60
SPLINVVQLT LPRVQELPED AEATTDVHPE DIPYLKKASD GLQPEVTRFL EQHSPDWIY 120
DYTHYWLPSI AASLGISRAH FSVTTPWAI A YMGPSADAMI NGSDGRTTVE DLTTPPKWFP 180
FPTKVCWRKH DLARLVYPKA PGISDRCRMG LVLKGSDDL SKCYHEFGTQ WLPLETLHQ 240
VPVVPVGLLP PEIPGDEKDE TWSIKKWLD GKQKGSVVYV ALGSEVLVSQ TEVVELALGL 300
ELSGLPFWVA YRKPKGPAKS DSVLPDGFV ERTRDRGLVW TSRAPQLRIL SHESVCGFLT 360
HCGSGSIVEG LMFHGLPML PIFGDQPLNA RLLEDKQVGI EIPRNEEDG LTKESVARSL 420
RSVVVEKEGE IYKANARELS KIYNDTKVEK EYVSQFVDYL EKNARAVAI HES 473

SEQ ID NO: 96      moltype = DNA length = 1422
FEATURE           Location/Qualifiers
source            1..1422
                  mol_type = other DNA
                  organism = Stevia rebaudiana

SEQUENCE: 96
atggctacca gtgactccat agttgacgac cgtaagcagc ttcattgtgc gacgttccca 60
tgggttgctt tgggtcacat cctcccttac cttcagcttt cgaaattgat agctgaaaag 120
ggtcacaaag tctcgtttct tctaccacc agaaacattc aacgtctctc ttctcatatc 180
tcgccactca taatgttgt tcaactcaca cttccacgtg tccaagagct gccggaggat 240
gcagaggcga ccactgacgt ccaccctgaa gatattccat atctcaagaa ggtctctgat 300
ggtcttcaac cggaggtcac ccggtttcta gaacaacact ctcggactgt gattatttat 360
gattatactc actactggtt gccatccatc gcggctagcc tcggtatctc acgagccac 420
ttctccgtca ccactccatg ggccattgct tatatgggac cctcagctga cgccatgata 480
aatggttcag atggtcgaa caccggttag gatctcacga caccgcccaa gtggtttccc 540
ttccgacca aagtatgctg gcggaagcat gatcttgccc gactggtgac ttacaaagct 600
ccggggatat ctgatcgtg ccgtatgggg ctggttctta agggatctga ttgtttgctc 660
tccaaatggt accatgagtt tggaaactca tggctacctc ttttgagagc actacaccaa 720
gtaccggtgg ttccggtggg attactgcca ccggaaatac ccggagacga gaaagatgaa 780
acatgggtgt caatcaagaa atggctcgat ggtaaacaaa aaggcagtg ggtgtacgtt 840
gcattaggaa gcgaggtttt ggtgagccaa accgaggttg ttgagtttag attgggtctc 900
gagctttctg ggttgccatt tgtttgggct tatagaaaa caaaaggctc ccggaagtca 960
gactcggtag agttgcaga cgggttcgtg gaacgaactc gtgaccgtgg gttggtctgg 1020
acgagtcggg cacctcagtt agccatagct cgggtttgtg gttcttgacg 1080
cattgtggtt ctggatcaat tctggaaggg ctaatgtttg gtcacctctc aatcatgcta 1140
ccgatttttg gggaccaacc tctgaatgct cgattactgg aggacaaaca ggtgggaatc 1200
gagataccaa gaaatgagga agatggttgc ttgaccaagg agtcggttgc tagatactg 1260
aggtccgttg ttgtgaaaaa agaaggggag atctacaagg cgaacgcgag ggagctgagt 1320

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aaaaatctata acgacactaa ggttgaaaaa gaatatgtaa gccaatctgt agactatttg 1380
gaaaagaatg cgcgtgcggt tgccatcgat catgagagtt aa 1422

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SEQ ID NO: 97      moltype = DNA length = 1380
FEATURE           Location/Qualifiers
source            1..1380
                  mol_type = other DNA
                  organism = Stevia rebaudiana

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SEQUENCE: 97
atggaaaata aaacggagac caccgttcgc cggcgccgga gaataatatt attcccggta 60
ccatttcaag gccacattaa cccaattctt cagctagcca atgtgttgta ctctaaagga 120
ttcagtatca ccattctttca caccaacttc aacaaaccca aaacatctaa ttaccctcac 180
ttcactttca gattcatcct cgacaacgac ccacaagacg aacgcatttc caatctaccg 240
actcatggtc cgctcgctgg tatgcggatt ccgattatca acgaacacgg agctgacgaa 300
ttacgacgcg aactggaact gttgatgtta gcttctgaag aagatgaaga ggtatcgtgt 360
ttaatcacgg atgctctttg gtacttcgcg caatctgttg ctgacagtct taacctccga 420
cggcttggtt tgatgacaag cagcttggtt aattttcatg cacatgtttc acttccctcag 480
tttgatgagc ttggttaacct cgatcctgat gacaaaaccc gtttgggaaga acaagcgagt 540
gggtttccta tgctaaaagt gaaagacatc aagtctgcgt attcgaactg gcaaaactc 600
aaagagatat tagggaagat gataaaacaa acaaaagcat cttcaggagt catctggaac 660
tcatttaagg aactcgaaga gtctgagctc gaaactgtta tccgtgagat cccggctcca 720
agtttcttga taccactccc caagcatttg acagcctctt ccagcagctt actagaccac 780
gatcgaaccc gttttccatg gttagaccaa caaccgtcac gttcggtagt gtatgttagt 840
tttggttagt gtactgaagt actggatgag aaagatttct tggaaatagc tcgtgggttg 900
gttgatagca agcagtcggt tttatgggtg gttcgacctg ggtttgtcaa gggttcgacg 960
tgggtcgaaac cgttgccaga tgggttcttg ggtgaaagag gacgtattgt gaaatgggtt 1020
ccacagcaag aagtgtctagc tcatggagca ataggcgcat tctggactca tagcggatgg 1080
aactctacgt tggaaagcgt ttgtgaaggt gttcctatga tttctcggga ttttgggtc 1140
gatcaaccgt tgaatgctag atacatgagt gatgttttga aggtaggggg gtatttggaa 1200
aatgggtggg aaagaggaga gatagcaaat gcaataagaa gagttatggt ggatgaagaa 1260
ggagaataca ttgacagaa tgcaagaggt ttgaaacaaa aggcagatgt tctttgatg 1320
aagggtggtt cgtcttacga atcattagag tctctagttt cttacatttc atcgttgtaa 1380

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SEQ ID NO: 98      moltype = AA length = 459
FEATURE           Location/Qualifiers
source            1..459
                  mol_type = protein
                  organism = Stevia rebaudiana

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SEQUENCE: 98
MENKTETTVR RRRRIILFPV PFQGHINPIL QLANVLYSKG FSITIFHTNF NKPKTSNYPH 60
FTFRFILDND PQDERISNLP THGPLAGMRI PIINEHGADE LRRELELLML ASEDEEVSC 120
LITDALWYFA QSVADSLNLR RLVLMTSSLF NFHAHVSLPQ FDELGYLDPD DKTRLEEQAS 180
GFPMLKVKDI KSAYSNWQIL KEILGKMIKQ TKASSGVIWN SFKELEESL ETVIREIPAP 240
SFLIPLPKHL TASSSSLLDH DRTVFPWLDQ QPSRSVLVYS FSGTEVLDE KDFLEIARGL 300
VDSKQSFLLW VRPGFVKGST WVEPLPDGFL GERGRIVKWV PQQEVLAHGA IGAFTWTHSGW 360
NSTLESVCEG VPMIFSDFGL DQPLNARYMS DVLKVGIVLE NGWERGEIAN AIRRMVWDEE 420
GEYIRQNARV LKQKADVSLM KGGSSYESLE SLVSYISL 459

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SEQ ID NO: 99      moltype = DNA length = 1380
FEATURE           Location/Qualifiers
modified_base     861..863
                  mod_base = OTHER
                  note = a, c, t, g, unknown or other
misc_feature      861..863
                  note = n is a, c, g, or t
source            1..1380
                  mol_type = other DNA
                  organism = Stevia rebaudiana

```

```

SEQUENCE: 99
atggaaaata aaacggagac caccgttcgc cggcgccgga gaataatatt attcccggta 60
ccatttcaag gccacattaa cccaattctt cagctagcca atgtgttgta ctctaaagga 120
ttcagtatca ccattctttca caccaacttc aacaaaccca aaacatctaa ttaccctcac 180
ttcactttca gattcatcct cgacaacgac ccacaagacg aacgcatttc caatctaccg 240
actcatggtc cgctcgctgg tatgcggatt ccgattatca acgaacacgg agctgacgaa 300
ttacgacgcg aactggaact gttgatgtta gcttctgaag aagatgaaga ggtatcgtgt 360
ttaatcacgg atgctctttg gtacttcgcg caatctgttg ctgacagtct taacctccga 420
cggcttggtt tgatgacaag cagcttggtt aattttcatg cacatgtttc acttccctcag 480
tttgatgagc ttggttaacct cgatcctgat gacaaaaccc gtttgggaaga acaagcgagt 540
gggtttccta tgctaaaagt gaaagacatc aagtctgcgt attcgaactg gcaaaactc 600
aaagagatat tagggaagat gataaaacaa acaaaagcat cttcaggagt catctggaac 660
tcatttaagg aactcgaaga gtctgagctc gaaactgtta tccgtgagat cccggctcca 720
agtttcttga taccactccc caagcatttg acagcctctt ccagcagctt actagaccac 780
gatcgaaccc gttttcaatg gttagaccaa caaccgcaa gttcggtagt gtatgttagt 840
tttggtagta ctagtgaagt nnnnggatgag aaagatttct tggaaatagc tcgtgggttg 900
gttgatagca agcagtcggt tttatgggtg gttcgacctg ggtttgtcaa gggttcgacg 960
tgggtcgaaac cgttgccaga tgggttcgtg gccgaaagag ggcgtattgt gaaatgggtt 1020
ccgcaacagg aagtgatagc tcatggagca atcgggtgat tctggactca tagcggatgg 1080
aactctacat tggaaagcgt ttgtgaaggt gttcctatga tttctcggga ttttgggtc 1140
gatcaaccgt tgaatgctag atacatgagt gatgttttga aggtaggggg gtatttggaa 1200

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-continued

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aatgggtggg aaagaggaga gatagcaaat gcaatacgaa gagttatggt ggatgaagaa 1260
ggagaataca ttagacagaa tgcaagagtt ttgaaacaaa aggcagatgt ttctttgatg 1320
aagggtggtt catcttacga atcattagag tctctagttt cttacatttc atcgttgtaa 1380

```

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SEQ ID NO: 100      moltype = AA  length = 459
FEATURE            Location/Qualifiers
MOD_RES            288
                    note = Any amino acid
source             1..459
                    mol_type = protein
                    organism = Stevia rebaudiana

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```

SEQUENCE: 100
MENKTETTVR RRRRIILFPV PFQGHINPIL QLANVLYSKG FSITIFHTNF NKPKTSNYPH 60
FTFRFILDND PQDERISNLP THGPLAGMRI PIINEHGADE LRRELELLML ASEDEEVSC 120
LITDALWYFA QSVADSLNLR RLVMTSSLF NFHAHVSLPQ FDELGYLDPD DKTRLEEQAS 180
GFPMLKVKDI KSAYSNWQIL KEILGKMIKQ TKASSGVIWN SFKELEESSEL ETVIREIPAP 240
SFLIPLPKHL TASSSSLLDH DRTVFQWLDQ QPPSSVLYVS FGSTSEVXDE KDFLEIARGL 300
VDSKQSFLWV VRPGFVKGST WVEPLPDGFV AERGRIVKWV PQQEVIAHGA IGAFWTHSGW 360
NSTLESVCEG VPMIFSDGFL DQPLNARYMS DVLKVGIVLE NGWERGEIAN AIRRMVDEE 420
GEYIRQNARV LKQKADVSLM KGGSSYESLE SLVSYISSL 459

```

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SEQ ID NO: 101      moltype = DNA  length = 1380
FEATURE            Location/Qualifiers
modified_base      861..863
                    mod_base = OTHER
                    note = a, c, t, g, unknown or other
misc_feature       861..863
                    note = n is a, c, g, or t
source             1..1380
                    mol_type = other DNA
                    organism = Stevia rebaudiana

```

```

SEQUENCE: 101
atggaaaata aaacggagac caccgttcgc cggcgccgga gaataatatt attcccggtg 60
ccatttcaag gccacattaa cccaattctt cagctagcca atgtgttgta ctctaaagga 120
ttcagtatca ccacttttca caccaacttc aacaaaccca aaacatctaa ttaccctcac 180
ttcactttca gattcatcct cgacaacgac ccacaagacg aacgcatttc caatctaccg 240
actcatggtc cgctcgctgg tatgctggatt cgcgattatca acgaacacgg agctgacgaa 300
ttacgacgcg aactggaact gttgatggtta gcttctgaag aagatgaaga ggtatcgtgt 360
ttaatcacgg atgctctttg gtacttcgcg caatctgttg ctgacagtct taacctccga 420
cggcttggtt tgatgacaag cagcttggtt aattttcatg cacatgttcc acttcctcag 480
tttgatgagc ttggttacct cgatcctgat gacaaaaccc gtttggaaga acaagcgagt 540
gggtttccta tgctaaaagt gaaagacatc aagctgcgtt attcgaactg gcaaataactc 600
aaagagatat tagggaagat gataaaacaa acaaaagcat cttcaggagt catctggaac 660
tcatttaagg aactcgaaga gctctgagctc gaaactgtta tccgtgagat cccgggtcca 720
agtttcttga taccactccc caagcatttg acagcctctt ccagcagctt actagaccac 780
gatcgaaccg tttttcaatg tttagaccaa caaccgccaa gttcggtagt gtatgttagt 840
tttggtagta ctagtgaagt nnnnggatgag aaagatttct tggaaatagc tctgtgggtg 900
gttgatagca agcagtcggt tttatgggtg gttcgacctg ggtttgtcaa gggttcgacg 960
tgggtcgaac cgttgccaga tgggttcttg ggtgaaagag gacgtattgt gaaatgggtt 1020
ccacagcaag aagtgcctagc tcatggagca ataggcgcat tctggactca tagcggatgg 1080
aactctacgt tggaaagcgt ttgtgaaggt gttcctatga tttctcggga ttttgggctc 1140
gatcaaccgt tgaatgctag atacatgagt gatgttttga aggtagggggt gtatttgtaa 1200
aatgggtggg aaagaggaga gatagcaaat gcaataagaa gagttatggt ggatgaagaa 1260
ggagaataca ttagacagaa tgcaagagtt ttgaaacaaa aggcagatgt ttctttgatg 1320
aagggtggtt cgtcttacga atcattagag tctctagttt cttacatttc atcgttgtaa 1380

```

```

SEQ ID NO: 102      moltype = AA  length = 459
FEATURE            Location/Qualifiers
MOD_RES            288
                    note = Any amino acid
source             1..459
                    mol_type = protein
                    organism = Stevia rebaudiana

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```

SEQUENCE: 102
MENKTETTVR RRRRIILFPV PFQGHINPIL QLANVLYSKG FSITIFHTNF NKPKTSNYPH 60
FTFRFILDND PQDERISNLP THGPLAGMRI PIINEHGADE LRRELELLML ASEDEEVSC 120
LITDALWYFA QSVADSLNLR RLVMTSSLF NFHAHVSLPQ FDELGYLDPD DKTRLEEQAS 180
GFPMLKVKDI KSAYSNWQIL KEILGKMIKQ TKASSGVIWN SFKELEESSEL ETVIREIPAP 240
SFLIPLPKHL TASSSSLLDH DRTVFQWLDQ QPPSSVLYVS FGSTSEVXDE KDFLEIARGL 300
VDSKQSFLWV VRPGFVKGST WVEPLPDGFL GERGRIVKWV PQQEVLAHGA IGAFWTHSGW 360
NSTLESVCEG VPMIFSDGFL DQPLNARYMS DVLKVGIVLE NGWERGEIAN AIRRMVDEE 420
GEYIRQNARV LKQKADVSLM KGGSSYESLE SLVSYISSL 459

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SEQ ID NO: 103      moltype = DNA  length = 1323
FEATURE            Location/Qualifiers
misc_feature       1..1323
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..1323

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-continued

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 103
atgcattcta ccagacatat cttaaagacaa agggccgtcc tagttacagg cgctagaaca 60
ccattcgtag aatcatttgg ggctcttatg aaagcagata ccttgggaatt ggcacagca 120
tcagtcgctg ggttgctgaa caagacctca ctggacccta gagatatcga tcatatcggt 180
tggggtaatg ttgtacttca aggatcagct cataactgcy ccagagaaat agttatcgac 240
cttaacatgc ctaaaaagat catcggtaat ttgacatcta tggcctgtgc ttcaggctta 300
tcttctttgt cacaagcctg tatgctaata gagggtggtc atgcccagtg cgtcattgct 360
ggcggttctg attcagtcct caaactgaa gtgcctttgc caagatccgt cacttaagggt 420
ctaagtatgg cccaaaggaa ggggtgttat ggcttcttta aggaagcagg atacaacca 480
ttcaaatggt ttccaggcgg tattgtctta accgaacgta gtacaggaaa aactatgggt 540
tggcatggag acttaattgc tgagttaaact tctatatcta gagatgacca ggaagccctg 600
gctgtggctt ctcatgcgaa tgcgtctaga gcagaaaaag ctgggtactt taaggaggaa 660
attgtacctg tgacaatcga caaaaagggc aaaaagactg aagtaacatg tgatgatgtt 720
atgcaagagag atacagaaaa gatgaaggcc aagatgccat cattgaagcc tgttttcaga 780
aaagaggggag gtacaataac agcagccact tccagtactc tgactgatgg tggctctgca 840
atgttgggta tgcagagga aaaggccaaa aagttgggtt atccaactga tgtctgctg 900
aagtcctggt attcagtggt tatcgatcct taccacaac tttgttagc accagttcta 960
ggttggggct cagctttgaa aaaggccgga ttaaccoccta aagatcaga tttgtacgaa 1020
attcacgaag catttgctgc acaagtctta gccacaatta agtgtttgaa gtctcaggaa 1080
ttcttcgata ggtacgctaa cgttgcaaaag ccagtattaa ctgaggatat tgatctttct 1140
aaactaaatg ttaatggcgg ttctctagca cttggccacc cattcgccgc tacaggagggt 1200
agaatcgtaa tctctctagc aaatgagttg agaagatccg gaaagagaca cgggctgggtc 1260
agtatttggt cagctggagg gttaggcgga gtagctatag ttgagcatag agcaagtaag 1320
taa 1323

SEQ ID NO: 104      moltype = AA length = 440
FEATURE             Location/Qualifiers
source              1..440
                    mol_type = protein
                    organism = Leishmania infantum

SEQUENCE: 104
MHSTRHILRQ RAVLVTGART PFVKSEFGALM KADTLELASA SVAGLLNKTS LDPRDIDHIV 60
WGNVVLQGS AHCAREIVID LNMPPKIIGN LTSMACASGL SLSQACMLI EGGHADVVIA 120
GGSDSVSNT E VPLPRSVTYG LMMAPRKQVM GFFKEAGYNP FKWPPGGIAL TERSTGKTMG 180
WHGDLIAELN SISRDDQREAL AVASHANAAR AEKAGYFKEE IVPVTIDKKG KKTEVTCCDV 240
MQRDTEKMK A KMPSLKPVFR KEGGTITAAT SSTLTDGSGA MLVMSEKAK KLGYPDVVCV 300
KSWYFSGIDP YPQLLLAPVL GWGPALKKAG LTPKDIDLYE IHEFAFAQVL ATIKCKLSQE 360
FFDRYANGAK PVLTEIDLS KLVNNGSLA LGHPFAATGG RIVISLANEL RRSGRKHLV 420
SICAAGGLG VAILEHTASK 440

SEQ ID NO: 105      moltype = DNA length = 1584
FEATURE             Location/Qualifiers
misc_feature        1..1584
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1584
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 105
atggcagctg accaattggt gaaaaactgaa gtcaccaaga agtcttttac tgctcctgta 60
caaaaggctt ctacaccagt ttaaccaat aaaacagtc tttctggatc gaaagtcaaa 120
agtttatcat ctgcgcaatc gagctcatca ggaccttcat catctagtga ggaagatgat 180
tcccgcgata ttgaaagctt ggataagaaa atacgtcctt tagaagaatt agaagcatta 240
ttaagtatgt gaaatacaaa acaattgaag aacaagaggg tcgctgcctt ggttattcac 300
ggtaagttac cttgtacgc tttggagaaa aaattagggt atactacgag agcggttgcy 360
gtacgtagga aggtcttttc aattttggca gaagctcctg tattagcatc tgatcgttta 420
ccatataaaa attatgacta cgaccgcgta tttggcgctt gttgtgaaaa tgttatagg 480
tacatgcctt tgcccgttgg tgttataggc cctctgggta tcgatggtag atcttatcat 540
ataccaatgg caactacaga gggttgtttg gtagcttctg ccatgcgtgg ctgtaaggca 600
atcaatgctg gcggtggtgc aacaactgtt ttaactaagg atggtatgac aagaggccca 660
gtagtcgctt tcccaacttt gaaaagatct ggtgcctgta agataggtt agactcagaa 720
gagggacaaa acgcaattaa aaaagctttt aactctacat caagatttgc acgtctgcaa 780
catattcaaa cttgtctagc aggagattta ctcttcata gatttagaac aactactggt 840
gacgcaatgg gtatgaatat gatttctaaa ggtgtcgaat actcattaaa gcaaatggta 900
gaagagtatg gctgggaaga tatggagggt gtctccggtt ctggttaacta ctgtaccgac 960
aaaaaaccag ctgcccacta ctggatcgaa ggtcgtggtg agagtgtcgt cgcagaagct 1020
actattcctg gtgatgttgt cagaaaagtg ttaaaaagtg atgtttccgc attggttgag 1080
ttgaacattg ctaagaattt ggttggtatc gcaatggcgt ggtcgtgttg tggatttaac 1140
gcacatgcag ctaatttagt gcagactgtt ttcttggcat taggacaaga tctgcacaa 1200
aatgttgaaa gttccaactg tataacattg atgaaagaag tggacgggtg tttgagaatt 1260
tccgtatcca tgccatccat cgaagtaggt accatcggtg gtggtactgt tctagaacca 1320
caaggtgcc aattggactt attaggtgta agaggccgc atgctaccgc tctgggtacc 1380
aacgcagctc ttttagcaag aatagttgcc tgtgcgctct tggcagggtg attatcctta 1440
tgtgctgccc tagcagccgg ccatttggtt caaagtcata tgaccacaa caggaaacct 1500
gctgaaccaa caaacctaa caatttggtg gccactgata taaatcggtt gaaagatggg 1560
tccgtcacct gcattaaatc ctaa 1584

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-continued

SEQ ID NO: 106 moltype = AA length = 527
 FEATURE Location/Qualifiers
 source 1..527
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 106

MAADQLVKTE	VTKKSFTAPV	QKASTPVLTN	KTVISGSKVK	SLSSAQSSSS	GPSSSSSEEDD	60
SRDIESLDKK	IRPLEELEAL	LSSGNTKQLK	NKEVAALVIH	GKLPYALEK	KLGDTRAVA	120
VRRKALSILA	EAPVLASDRL	PYKNYDYDRV	FGACCENVIG	YMLPVGIVIG	PLVIDGTSYH	180
IPMATTEGCL	VASAMRGCKA	INAGGGATTV	LTkdGMTRGP	VVRPPTLKRS	GACKIWLDS	240
EGQNAIKKAF	NSTSRFARLQ	HIQTCLAGDL	LFMRFRTTTG	DAMGMNMISK	GVEYSKQMV	300
EEYGWEDMEV	VSVSGNYCTD	KKPAAINWIE	GRGKSVVAEA	TIPGDVVRKV	LKSDVSALVE	360
LNIANKNLVGS	AMAGSVGGFN	AHAANLVTAV	FLALGQDPAQ	NVESSNCITL	MKEVDGDLRI	420
SVSMPSIEVG	TIGGGTVLEP	QGAMLDLLGV	RGPHTAPGT	NARQLARIVA	CAVLAGELSL	480
CAALAAGHLV	QSHMTHNRKP	AEPTKPNMLD	ATDINRLKDG	SVTCIKS		527

SEQ ID NO: 107 moltype = DNA length = 3681
 FEATURE Location/Qualifiers
 misc_feature 1..3681
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..3681
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 107

atgagagctg	tccttagatt	gttatcaaca	catactgttt	tctctcctat	tgaacaatt	60
gtatctgttt	tcgtgttagc	tacattagct	tacttcacac	tctgtgccgg	aatcaagcac	120
tcaagtttct	ttgcattctc	tcacctcctc	gctatcacag	ctgcttttgc	acatctgacc	180
aacggggaat	gggttgccgt	ctcccaacat	gattggactg	aagcatggaa	gcacccctggc	240
ggttcacttg	atgcattaga	acttcaacaa	gtagttttca	cttttagatga	caagactcaa	300
ccatctgctg	tgctagatgc	atccgcaatt	agtcagcact	tagtttccaa	tgttccctgca	360
ttatctggaa	aagcctactc	ttcattgtgc	caccatccaa	atgtatcagg	cacctcctgt	420
tttacatcag	tttctgttcc	aggagcttca	ccaatcttga	cactgagttt	taagcctgga	480
actagagacg	actggtagtg	atcatcaagg	aaggagaaaa	ctatcacact	agatgggggt	540
aagtacgacg	ttggagccgg	aaaaagacaa	gagtcacatc	gcgatatgga	atcatctaag	600
tgggttgctt	atgcattatc	agctttggta	cttagatttt	gggaattaac	aaaggcagat	660
tccttagata	tactagtggg	tttaactggg	tacatcctaa	tgacgcgaac	attcatgaga	720
ttgttcttgg	catccagagc	acttggcagt	aaacttttgg	tatcagctgg	catattctcc	780
tcgcacaaca	tttcttctct	attcacttta	ccaatgtgta	gatctatgga	tattccactt	840
gatccaattg	ccttgacaga	agccctgcca	ttcttgggtg	gtacgcgtag	ttttgacaaa	900
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ggtaggtaga	aagctgccgg	agatgtcatt	cttgaggcac	tggacagagt	tggttaacatg	1020
atattgagag	attacgcttt	agagatcgca	gttctattcg	ttggcgttaa	ctccagagtt	1080
ggcgggtctt	aggaattttg	tgtgttagct	gcagcattac	ttgctatgga	cagattaatg	1140
acattcacac	tttatacagc	agtgttaacc	atcagtggtg	aggtaaggcg	tatcaaaaag	1200
gtcagagata	tgtataaggc	agttcttcta	ttaccgccgt	tacagccaac		1260
ggcacccgca	taagaggcgt	tttgagtaga	aaactctcaa	aacaatctgt	gacagaacca	1320
gagacaacta	aaaaccttaag	acaaagagcc	actgattcag	ccatcggtgt	taagggttca	1380
ttgctgaaag	attgaggcag	attgcaggaa	gccgaggaga	atccaatggc	aagattaaag	1440
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tcagccacag	ctaacgcaag	acatcaaaag	catcctttta	gaaccgttca	agaggtagta	1560
ccaatttcta	gagttgacat	tactacccca	gccatagcca	atatcttgtc	tcactatgct	1620
gtggctcagg	aacctatgtt	cactgttgtt	ggcagtgaa	ctatcgaa	cttgtttaa	1680
gtcgtgtctc	cagtcctacg	ccatgtctta	ccattggccc	ctgtcttaag	agcttcaaac	1740
actaataact	gagaagctat	tgaaaacttt	atgagttcat	ggtctagtct	ggtaggtgac	1800
ccagttgtta	gtaagtggat	cgtagcattg	ctagctgtct	ctgttgca	gaatggatac	1860
ttgttaaagg	gtatagccgc	aggttccggg	ttggctgcca	tgagagctgt	tagatctcaa	1920
gggttctcgt	tcagatctag	agctagaagt	atcgtaaaag	tatctgatga	acctgagcca	1980
gagccagaac	actctatcga	cccagcacca	gtagtgttct	tcgcttccgc	agcaccagct	2040
gtagaggccc	ctgctccagg	tctgcacct	gaaccagaac	caccagtcac	cagaccacca	2100
ccattgacta	ttttctcaag	accactgaac	ttagaaaacag	tggacaaaaa	gttacaagat	2160
gctctgccaa	taagatcccc	accacctgtt	gaaccaatca	ctccagaatc	tagagaagtg	2220
gaaccaaccc	aagtagaagt	aagatctcta	gctgaatgtg	tggatgtgtt	cgagaatggg	2280
ccaagaccag	tctcagtggt	tttaagact	ctgaatgatg	aggaagttaa	cctgctttgc	2340
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gtacgtgtca	gaagagcact	tatttagtaga	gcttcacgta	caaaaacttt	agaaaactca	2460
ctggttccta	tgaagatta	tgattacgcc	agagtcattg	gtgcctgttg	tgaaaacgtt	2520
atcgatata	tgccattacc	actagggatt	gcaggtccat	tgaagattga	tggtctgtag	2580
tatcctatac	caatggcaac	cgcagaaggt	accttggttg	catctacttc	taggggctgt	2640
aaggccttaa	atgctgtgtg	aggggtcaca	actgtcttga	cagcagatgg	catgacaaga	2700
gggcccagcta	tagactttcc	ttccatcgtc	agagctgcag	aggctaaggc	cttcattgaa	2760
tcagaagatg	gatacgttac	aatcagggag	gctttcgagt	ctacttctag	atttgccaag	2820
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acaggagatg	ccatgggtat	gaacatgatt	tctaaggcta	ccgaaaaggc	acttgatgtc	2940
ctgagtcacg	agttccctga	aatggctcgt	cttgctttgt	ctggtaacta	ctgcacagac	3000
aaaaagcctg	cagctatttc	atggatcgaa	ggtaggggaa	aatctatgt	agcagaagca	3060
gttattctctg	tgaaggtcgt	taagtcagtc	ctgaaaacaa	cagtcagatc	tctttgcaat	3120
gtcaacacta	agaaaaacct	gattgtttca	gccatggcag	gttctgtgtg	tggtttcaac	3180
gctcatgccg	ccaacatcct	aacagctgtg	ttcctagcca	caggtcagga	tcctgtctca	3240

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aatgtcgaat cttctaattg catgacttta atggaaccaa caaacggcgg tgaggatttg 3300
ctaattgacaa tttcaatgcc atgtatagag gtaggaaccg ttggtggagg gacaattctg 3360
gaaccacaag gtgcagtttt ggatttggtt ggcgttagag gggctcacc cactaatcct 3420
ggtaaaaacg ctcaacagtt agccagaatt atcgcatcag ctgtaatggc aggcgaattg 3480
tctttgataa gtgcotttag cgcaggtcat ttggttagag ctcactctgc ccacaatcgt 3540
tctcaattga atacaccaat gccatccaga ccacatactc ctggccctga ggatgtctca 3600
catgtgcagc agctacctac accatctgca tctgatgata aaggtgttac agctcaagggt 3660
tacgttgtcg aagcaaaata a 3681

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SEQ ID NO: 108      moltype = AA length = 1226
FEATURE             Location/Qualifiers
source              1..1226
                   mol_type = protein
                   organism = Ganoderma lucidum

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SEQUENCE: 108
MRAVLRLLLST HTVFSPiETI VSVFVLATLA YPHILSGIKH SSFFASSHPP AIRPAFAHLT 60
NGEWVAVSQH DWTEAWKHGP GSLDALELQQ VVFTLDDKTQ PSAVLDASAI SQHLVSNVPA 120
LSGKAYSSLC HHPNVSGTSC FTSVSGPGAS PILTSLFKPG TRDDWLGLSLR KEKITLDGV 180
KYDVGAGKRQ ESIGDMESSK WYAYALSALV LRFWELTKAD SLDILVVLTG YILMHVTFMR 240
LFLASRALGS NFWLSAGIFS SATISFLFTL PMCRSMDIPL DPIALTEALP FLVCTVGFDK 300
PLRLARAVMA HPNLIKPODD GRMKAAGDVI LEALDRVGNM ILRDYALEIA VLFVGVNSRV 360
GGLKEFCAVA AALLAMDRML TFTLYTAVLT IMVEVRIKK VRDMTKARSR SSSITAVTAN 420
GTAIRGVLSR KSSKQSVTEP ETTKNLRQRA TDSAIGVKGS LLKDGGRQLQE AEENPMARLK 480
LLLIASFLLT HILNFCTTLT SATANARHQR HPFRTVQEVV PIPRVDITTP AIANILSHLA 540
VAQEPMTTVV GSEPIELLVK VAAPVVYHAL PLAPALRASN TNTGEAIENF MSSWSSLVGD 600
PVVSKWIVAL LAVSVALNGY LLKGIAAGSG LAAMRAVRSQ GVRFRSRARS IVKISDEPEP 660
EPEHSIDPAP VVFPASAAPA VEAPAPAPAP EPEPPVNRPP PLTIFSRPLN LETVDKKLQD 720
ALPIRSPPPV EPITPESREV EPTQVEVRSI AECVDVFENG PRPVSVALKT LNDEEVILLC 780
QTGKIAPYAL VKMLADFDRV VRVRRALISR ASRRTKLENS LVPMDYDYA RVMGACCENV 840
IGYMLPLPGI AGPLKIDGLM YPIPMATAEG TLVASTSRGC KALNAGGGVT TVLTADGMTR 900
GPAIDFPSIV RAAEAKAFIE SEDGYATIRE AFESTSRFAK LQKIKCALAG RTLFVRFATR 960
TGDAMGMNMI SKATEKALDV LSHEFPPEMVV LALSGNYCTD KKPAAISWIE GRGKSIVAEA 1020
VIPGVVKVSV LKTIVESLCN VNTKKNLIGS AMAGSVGGFN AHAANILTAV FLATGQDPAQ 1080
NVESNMCMTL MEPTNGGEDL LMTISMPCIE VGTVGGGTIL EPQGAVLDDL GVRGAHPTNP 1140
GQNAQQLARI IASAVMAGEL SLISALAAGH LVRAHLAHRN SQLNTPMPFSR PHTPGPEDVS 1200
HVQQLPTPSA SDDKGVTAQG YVVEAK 1226

```

```

SEQ ID NO: 109      moltype = DNA length = 2667
FEATURE             Location/Qualifiers
misc_feature         1..2667
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source              1..2667
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 109
atgttatcaa gattgttcag aatgcatggt ctatttggtt cttctcacc ttgggaagta 60
atagttggta ctgtaacatt aacgatctgt atgatgtcta tgaacatggt taccggaaac 120
aacaagattt gtggttgga aatatagatgt cctaagctgg aagaggatgt gttgagtcca 180
gacatcatca tacttactat aacaagatgc attgcaatat tgtatatcta cttccaattt 240
caaaacctta gacaattggc tagtaaatat atcctaggca tcgccggatt gttcactatt 300
ttctctagtt ttgttttttc aaccgtcggt attcactttt tggacaaaga gtttaactggt 360
ttgaacgaag ctctaccatt cttcttgctg ctggtagatt tgtccagagc ttccgcttta 420
gctaaattcg ctctgtcttc taattctcaa gatgaagtta gagagaatat agcaagggga 480
atggccatct ttggacatct ttccacactt gatgcccttg tcgaatggtt ggttattggg 540
gttggcacia tgctccggct tagacagtta gaaatcatgt gttgttttgg ctgtatgagt 600
gtcttggtta actactttgt ctttatgaca ttctttccag cttgcgtttc ttgggtattg 660
gagctgtcaa gagaatcaag agaaggcaga ccaatatggc aactatcaca ttccgccaga 720
gtgttagaag aggaggaaaa caaacctaat cctgtcacac agagagtga aatgatcatg 780
tctttgggtt tagtcctagt gcatgctcat tctagatgga tcgcagatcc atcccctcag 840
aattctacag ctgataactc taaagttagt ttaggtttag atgaaaatgt aagtaagagg 900
attgaacctt ccgtgtcttt gtggcaattc tacttatcaa aaatgatttc catggatatt 960
gaacaagtga taacgttgtc ttggcttta ttgttagccg ttaagtacat tttctttgag 1020
caagccgaaa ccgaatctac attatcactg aaaaacccaa ttacatcccc agtcgttacc 1080
cagaaaaaga taactgatga ttgctgtaga agagatccag tgttggtcag gaatgatcaa 1140
aagttccacg ccatggagga ggaactagg aaaaacagag aaaggaaagt tgaagtattc 1200
aagcctctat tagcagaaaa tgacacttca catagggcca ctttcgttgt cggcaattca 1260
gtctttttag gtacgtcatt ggagctggaa acacaggaac cagaaatgga actaccagtt 1320
gaaccaagac caaatgagga atgtttgcaa atactagaga acgctgaaaa gggagccaag 1380
ttcctatctg atgcgcagat tatccagctg gtcaatgcca agcaccattc tgcctacaag 1440
ttgaaacccc ttatggagac acatgagaga ggtgtgtcta ttaggagaca attactatct 1500
aaaaagttac ctgaaccaag ttccctacaa tacctgcctt atagagatta caattactcc 1560
ttggtaattg gagcttgttg tgaatatgtc attgggtaca tgccaattcc agtgggtgtc 1620
gccgtccac tatgtttgga cggttaaggaa ttcaagtagc ctatggcaac gactgaaggc 1680
tgcttagttg catctacaaa cagaggttgt agagccattg gattagggtg cggtgcttct 1740
tcaagagctt tggctgacgg tatgactaga ggtcctgttg tgagatttcc tagggcctgt 1800
gactctcgag aagttaaggc ttggttgga actccagaag gtttcaccgt aatcaaaag 1860
gcctttgatt ccacatcaag ggtggccaga ttacaaaaac tacacatgtc tgcctgctgg 1920
agaaatctgt atatcagatt tcaatccaga tccggcgacg caatgggtat gaatatgatt 1980

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tcaaaaggga	cagaaaaggc	ttgttcaaa	ctgcaggagt	atttccaga	gatgcaaatc	2040
ttggcgctat	ctggccaata	ttgcacgac	aaaaagctcg	ccgcgatcaa	ctggagttaa	2100
ggaagagga	aactctgtgt	ttgtgaagct	gtaattccg	cogaattctt	tagaagaagt	2160
ttaaagacca	caacacgaagc	tatgattgaa	gtaaacataa	acaaaaacct	agtaggggtc	2220
gcatctgtgc	gttcaattgt	aggatacaac	gctcatgtcg	ccaatatgtg	aacccgtatc	2280
tacatcgcat	gttgacaaga	tgctgcctaa	aatctcggtt	ctccaattgt	catcacattg	2340
atggaagcat	ctggccctac	aaacgaggat	ttgtatatca	gttgacaat	gccatctata	2400
gaaataggga	ctgtggggag	aggacaatac	ttaactccac	acgaagctcg	cttacaattg	2460
ctgggttgta	aaggagcctg	tagagataat	cacggggaga	acgctagaca	actgtccaga	2520
attgtttgtg	ggacagttat	ggctggtgaa	cttagtctaa	tggcagcttt	ggctgctggg	2580
caacctggta	gatctcatat	gattcataat	agaagttaaga	ttaaccttca	agatttgcaa	2640
gqttactgtta	cqaaaaaqqc	gtctcaaa				2676

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SEQ ID NO: 110      moltype = AA  length = 888
FEATURE             Location/Qualifiers
source              1..888
                   mol_type = protein
                   organism = Bos taurus
```

SEQUENCE: 110

MLSRLFRMHG	LFVASHPEVE	IVGTVTLTIC	MMSMMNFTGN	NKICGWNIEC	PKLEEDVLSS	60
DIILITITRC	LAILYLYYQF	QNLRLQSGSY	ILTGALGTFI	FSSVFSTVTV	IHFLDKELTG	120
LNEALPFPLL	LDVLSRASAL	AKFALSSNQX	DEVRENTART	MALIGTFTFL	DALVECLVIG	180
VGTMSGVRQL	EIMCCFGCMS	VLANYVFVMT	FFPACVSLVL	ELSRESREGR	PIWQLSHFAR	240
LVEEENKPN	PVTKQVKMID	SLGLVLVLAH	SRWDADSPQ	NSTADNSKVS	LGLDENVYSR	300
IEPSVLSQWF	LXKMSIMDI	EQVITLSHAL	LLAVKYIFFE	QATETSLSL	KNPITSPVTT	360
QKKITDDCCR	RDPVLVRNDQ	KFHAMEEETR	KNRERKEVEI	KPLLAENDTS	HRATFVVGNS	420
SLGLTSELE	TQPEMRLVL	EPRNEECQL	ILYNAEKAGK	FLSDABEILV	VNAKHIPAYK	480
LETLMTSEB	GVSEIRQLLS	KKLPESSQL	TLPRVDYNYL	LWMDAGCENV	IGYMPIPVGV	540
AGPLCLDGKE	FQVPMATTEG	CLVASTNRGC	RAIGLGGGAS	SRVLADGMTR	GPVVRFPRAC	600
DASEVKAWE	TPEGFTVIKE	APDSTSRVAR	LQKLHMSVAG	RNLRYFQFSR	SCDAMGMNMI	660
SKSKEETEL	LQEVFPEMKI	LAVSGNYCTD	KKPAANWIE	GRKGSVCSBA	VIPAKVVRVZ	720
LKTTTEAMIE	VNINKNLVGS	AMAGSIGGYN	AHAANIVTAI	YIACGQDAAQ	NVGSSNCITL	780
MEASGPTNED	LYISCTMPSI	EIGTVGGGNT	LRPQACQLQM	LGTVGQACRDN	PGENARALTR	840
IEVCTVMAGE	LSLMAALAA	HLVRSHMIHI	LLKNLODLO	TKCTKKAA		888

SEQ ID NO: 111 moltype = DNA length = 1704

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SEQ ID NO: 111      motifpe = DNA 100g
FEATURE             Location/Qualifiers
```

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misc feature      1..1704
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note = Description of Artificial Sequence: Synthetic polynucleotide

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source 1..1704
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mol type = other DNA
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organism = synthetic construct

SEQUENCE: 111

atggaatttga	gaagggaatt	accacctaa	ctcccatctt	caacaacaac	aaaacagcca	60
agtcataagtg	cccattctcc	tacgccaatt	ccaaaggctt	catagtcagtg	gctctttcca	120
ttgtacctga	ccaatacgtg	tttttctaact	ctttttcttt	ctgcgtacata	ttacctgttg	180
cataggttga	gagacaagat	tagatccgga	acacctttac	acgtttgtgac	actgactgaa	240
ctatcccgca	tttgtactgt	gattgtctcc	ttcatctatc	tttttaggtt	tttcggtatt	300
gatttttgat	aactcttcc	atcaagagaa	atgagcaac	taaacacaag	tgatcacaa	360
gtcgtgtcaa	caacaatgt	tttatctgat	agaagggtag	tttacgacata	tggatttcgat	420
gtgcaggagg	acaacagata	cgataatgat	gacgagtga	ttgtgaaaag	tgtcgtttct	480
gggggaattg	attcttatag	tttgaggctg	tcctaggag	attgttacag	agccgc aaag	540
attagaaga	gagccgtcga	gagaattgtc	gggagagaag	tattaggctt	gggtttccag	600
ggatttgatt	atgaatctat	ctcggggcaa	tgttgtgaaa	tgccatctcg	gtacgtccaa	660
gtgccaggat	gtgtcgtcgg	acctttatgt	ttaattgttg	gggaattcat	gggtccaatg	720
gtcacaactg	aaggctgtct	tgtactctcc	actaataag	gtttgaaacc	catatgtcta	780
tgactggtg	ccactgccat	attgtcaaaa	gatggtatga	caagagcccc	agtagtgaga	840
ttcgccacag	ctgagagagc	ttcacaaacta	aagttttact	tggaaagatg	tgtcaatttc	900
gatacattct	ctgtgtttct	taacaaaagt	tcaagatttg	ccagattgca	aaacattcaa	960
tgtcctaattg	ccggtataaaa	cttgtacatt	aggttttact	gtccacagcc	cgacgcctgt	1020
ggatgaacca	tggtttcaaa	aggagtcaaa	aatgtattag	actttttaca	aaatgatttt	1080
ctgtgattgg	actgaatttg	gatctctctg	aagttctctg	ctgcaaaaaa	gccaacagctt	1140
gtcaactcga	tgtaggcgac	agggaaagtc	gtcgttttcc	agggcgtaat	taccaaaaag	1200
gtggttagaa	agttctgcact	gaacctccaa	acttgcaact	gtagaacttt	gacctgttta	1260
agaccattat	tgtttctcgt	actcttgggt	tttgtagtgg	acttaaatgca	tatgcttcat	1320
atcgtgtctg	ccgtgttcat	cgctaccggg	caagatccag	ctcagaatat	cgaatctagt	1380
cactgtatca	ctatgatgga	ggctgtcaac	aatggtaaag	atttgacgtt	taatgttacg	1440
atgccatctc	tagaaggctg	cacggtgtgga	ggtggcaact	agctagcctc	tcaatcagcc	1500
ttgttgaaat	tgcttgtgtg	aaagggtggc	tgatagaact	cccaggagtc	aaacgccacg	1560
ttgttagctc	gaactcgttc	tggttgtgtt	ctggcaggcg	aattaagttt	gatgtcagct	1620
ataaagtctg	ggcaactagt	taaatctcat	atgaaataca	ataggtcttg	tagagatatg	1680
tcagcaaatg	cttctaaggt	ctaa				1740

SEQ ID NO: 112 moltype = AA length = 567

FEATURE	Location/Qualifiers
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source      1..567
```

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mol type = protein
```

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organism = Artemisia annua
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SEQUENCE: 112
MDLRRLPKP PPSSTTTKQP SHRSHTPTPI PKASDALPLP LYLTNTFFFT LFFSVAYYLL 60
HRWRDKIRSG TPLHVVLTE LSAIVLLIAS FIYLLGFFGI DFVQSFTSRE NEQLNDDHN 120
VVSTNNVLSR RRLVVDYGFV VTGDNDNDND DDVIVKSVVS GEVNSYSLEA SLGDCYRAAK 180
IRKRAVERIV GREVLGLGFE GFDYESILGQ CCEMPIGYVQ VPVGAVAGPLL LNGGEFVPM 240
ATTEGCLVAS TNRGCKAICL SGGATAILLK DGMTRAPVVR FATAERASQL KFYLEDGVNF 300
DTLSVVFNKS SRFARLQNIQ CSIAGKNLYI RFTCTGDAM GMNMVSKGVQ NVLDFLQDNF 360
PDMDVIGISW KFCSDKKPTA VNWIEGRGKS VVFQAVITKK VVRKSALNPQ TCTCRTLTCL 420
RPLLVLALLV LLVLDLHMLH IVSAVFIATG QDPAQNISS HCITMMEAVN NGKDLHVNVT 480
MPSIEVGTVG GGTQLASQSA CLNLLGVKGA CIESPGSNAQ LLARIVAGSV LAGELSLMSA 540
ISAGQLVKSH MKYNRSSRDM SAIASKV 567

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SEQ ID NO: 113      moltype = DNA length = 1308
FEATURE            Location/Qualifiers
misc_feature        1..1308
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1308
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 113
atgtttagaa gagctatact gttaggatgc tctgctgcca agacaccatg gtctgagtgt 60
tctaaccgtc aattagattga tgcagttaag tctagaaaga tctcattcta cggtcttgaa 120
caagccttgg aaccagatta tagaagggtc atcgaagtaa ggagagaggt tgtctctgaa 180
atcgccctac aacagccaga agcaaaaaag aagcaatccg cattgcacac aataccattt 240
gagaattatg attggaataa ggtcggtggc caaaactgtg aaaacattat tggatacgct 300
ccaataccac tgggcgttgc tggccctatt ttgattgatg gtaaagagta cccaatacca 360
atggctacaa cagaaggcgc ttgtgtcgct agtactcata gaggtgctag agctattaca 420
agatccggag gttgtaagac attgttatta ggtgaaggta tgacaagagc accagtggtt 480
gaattgcctt cattagagga agctgggcgt ttgcacaagt actgtaatga gaacttctta 540
tctttaaagg aagcatttga atcaactacc caatatggaa aacttaattc tttaaagtgc 600
gtactagctg gtagaaaaag ataccctaga ttcagagcca ctacaggcga tgcattgggc 660
atgaacatga taacaaaggg ttagacaaaa gcaatgtctg ttctacagca acatttcctt 720
tcaatggaaa tctagccct aagtggtaat tactgtaccg acaaaaagcc atctgctgta 780
aattggattg atggcagagg taaatcagtg gttgcagaag ccactttatt ggctgatgtt 840
gtcgaagata ctctgaaatg tacagtcgat tctttggtat ccttgaatat cgacaaaaac 900
ctgtttgggt cagctatggc tggttctggt ggaggtttta acgcccaggc tgcaaacgct 960
gtggcagcca ttttcattgc aaccggtcaa gatcctgctc aagtggtaga aagttcaatg 1020
tgtatcacta caatgtccaa ggtaggtaac gatctattga tctctgtgac catgccttct 1080
atcgaggtcg gggctgctgg aggagggact ggtcttgctg cccaaagagg atgcttagag 1140
ttaaataggt gcggaggccc atctaaggag tctcctggtg ctaatgcccc acttctaagt 1200
agagttgttg cagctggcgt tttatcagcc gaactttcct tgatgtccgg actggcagca 1260
ggtcatctat tgtcagcaca tatgagattg aacagaaaga agaaataa 1308

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SEQ ID NO: 114      moltype = AA length = 435
FEATURE            Location/Qualifiers
source              1..435
                    mol_type = protein
                    organism = Trypanosoma cruzi

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```

SEQUENCE: 114
MFRRAILLGC SAAKTPWSEC SNAQLVDAVK SRKISFYGLE QALEPDYRRA IEVRREVVS 60
IASQQPEAKK KQSALHTIPF ENYDWNKVVQ QNCENIIGYV PIPLGVAGPI LIDGKEYPIP 120
MATTEGALVA STHRGARAIT RSGGCKTLLL GEGMTRAPVV ELPSLEEAGR LHKYCENFL 180
SLKEAFESTT QYGLNLSLKC VLAGRKAYLR FRATTGDAMG NMNITKGVDK ALSVLQQHFP 240
SMEILALSGN YCTDKKPSAV NWIDGRGKSV VAEATLLADV VEDTLKCTVD SLVSLNIDKN 300
LVGSAMAGSV GGFNAQAANA VAAIFIATGQ DPAQVVESSM CITMMSKVG N DLLISVTMPS 360
IEVGVVGGGT GLAQRGCLLE LIGCGGPSKE SPGTNAQLLS RRVVAGVLSA ELSLMSGLAA 420
GHLLSAHMRL NRKKK 435

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SEQ ID NO: 115      moltype = DNA length = 1281
FEATURE            Location/Qualifiers
misc_feature        1..1281
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1281
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 115
atgcaatccc tggacaaaaa ctttagacac ttatcaagac aacagaagtt acaacagcta 60
gttgataaac aatggctatc agaggaacaa ttcaatattc tacttaacca cccacttatt 120
gatgaagagg tagcaaaactc attgatagaa aatgtcatcg cacaggggcg actgcctggt 180
ggtttactac caaatatcat cggtgatgac aaagcatacg tcgtgcctat gatggtggaa 240
gagccatctg ttgttgccgc tgcttcatac ggcgctaaat tggtgaaacca aacaggtggt 300
ttcaaaacgg tgtctcaga acgtatcatg ataggtcaaa tagtatttga tggagtcgat 360
gataccgaga aactgtctgc agatatcaag gctcttgaaa aacaaatcca tcagattgca 420
gatgaggctt acccttctat taaggccaga ggtggaggct atcaaaaggat cgccatcgat 480
acattcccag aacaacagtt gctttcattg aaggttttct ttgatactaa ggatgctatg 540
ggcgtaataa tgttaaacac aatcctagaa gcaatcacag cctttttgaa aaacgaattc 600
ccacaatctg atatcttgat gtctatcctt tccaaccacg caacagccag tgttgtcaag 660

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gtccaggggtg aaatagacgt taaggatttg gcaagaggag aacgtactgg agaagaggtc 720
gctaagagaa tggaaagagc atctgtgtta gctcaagtgg acattcatag agcagcaaca 780
cacaataaag gtgttatgaa tggcattcat gctgtagtct tggctacagg taatgatact 840
agaggtgcag aagcctctgc tcacgcttac gcttccaaag acggtcaata tagagggata 900
gctacatgga gatacgatca agagagacaa aggttaatag gaactataga agttccaatg 960
actctggcca ttgttggtgg cggtagcaag gtactgccta ttgctaaggc ctctttagaa 1020
ctgttaaacy tagaaagtgc ccaagagttg ggacatgttg tcgctgccgt tggactagct 1080
caaaaacttcg ctgcatgtag agctttgggt tccgaaggta ttcaacaagg gcataatgtct 1140
ttgcaataca agtctttagc cctcgtagtc ggggctaagg gcgatgaaat tgctcaggta 1200
gccgaagcac taaagcaaga gcccaagagca aacactcaag ttgcagagag aattttgcaa 1260
gatttgagaa gtcaacaata a 1281

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```

SEQ ID NO: 116      moltype = AA length = 426
FEATURE            Location/Qualifiers
source              1..426
                    mol_type = protein
                    organism = Staphylococcus aureus

```

```

SEQUENCE: 116
MQSLDKNFRH LSRQKQLQQL VDKQWLSEEQ FNILLNHPLI DEEVANSIE NVIAQGALPV 60
GLLPNIIVDD KAYVVPMMVE EPSVVAAS Y GAKLVNQTGG FKTVSSERIM IGQIVFDGVD 120
DTEKLSADIK ALEKQIHQIA DEAYPSIKAR GGGYQRIAD TFPEQQQLSL KVFVDTKDAM 180
GANMLNTILE AITAFLEKNEF PQSDILMSIL SNHATASVVK VQGEIDVKDL ARGERTGEEV 240
AKRMERASVL AQVDIHRAT HNKGVMMNGIH AVVLATGNDT RGAEASAHAY ASKDGQYRGI 300
ATWRVDQERQ RLIGTIEVPM TLAIVGGGK VLPIAKASLE LLNVESAQEL GHVVAAVGLA 360
QNFAACRALV SEGIQGHMS LQYKSLAIVV GAKGDEIAQV AEALKQEPRA NTQVAERILQ 420
DLRSQQ 426

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SEQ ID NO: 117      moltype = DNA length = 1311
FEATURE            Location/Qualifiers
misc_feature        1..1311
                    note = Description of Artificial Sequence: Synthetic
                        polynucleotide
source              1..1311
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 117
atgcaggtct taagattgga taggagacat tacaaaagtg gcaagattag aagagcaatg 60
agttctagaa ttcttggttt ctacaaattg tcagtcgagg aaagactgaa aaaggttgct 120
gaatttgcag gggtatctga tgaggaagtg aaagctgttt tgtcacaagg ttacaccttg 180
gacgtagctg atagaatgat cgaaaatgtg atcggtacat ttgaattacc acttggtata 240
gcaaccaatt tccttatgta tggcaaggat tatctaattc ctatggctat agaggaacca 300
tcagtagttg cagctgcttc taacgcagct agaattggcca gagagctcgg cgggtttaca 360
actgattaca caggggtccct gatgattggt caaattcaag tcacaaaact gttgaatcca 420
aatgcagcta agttcgaagt tctacgtcaa aaagacgaaa tcatagaaag agcaaatgag 480
tgtgatccaa tgttggtgaa ttggggcggg ggatgtaaa atatagaagc aaggggtgatc 540
gatacaatca tgggtaagat gctaattggt catctgatcg ttgatgttaa agacgctatg 600
ggtgcaaatg tgtcacaacac tatgtgtgaa aaagttgctc ctttcacga acgtattact 660
gggggaaagg tctatcttag aatcatttcc aacttggctg catatagact tgctagagca 720
aaggccgttt ttgacaaaaga cgttattggc ggagaggagg ttgtagaagg gatcatgctt 780
gcatacgctc tcgctgccgc tgaccattt cgttgcgcca cccacaataa ggggtatcatg 840
aatgcatatc cagccttaat gatcgctaca ggaaacgact ttagagccat tgaagcagga 900
gtcattctct atgctgcaat aggttgatgc aaaccactaa ctacctacga agttgataga 960
aaagtaatc tagtaggcac aattgaaata cctatggcag taggcgtgat tgggtggtgca 1020
accaaagtc accactagc caagatctct cttaaagata taggagtga cactgctgaa 1080
gagttagcca gagtcgcagc cgctctaggt ttggctcaaa actttgctgc cttaaagagcc 1140
ttggccacag aaggtatcca aagaggtcac atggaattac atgccaggaa cttagcaatc 1200
atggctggag ctactggaga tgaggttgac agagttgtag agattatggt gagagatggc 1260
aaaatcagat tggactacgc taaggaaagta ttggagagac tgcgttccta a 1311

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SEQ ID NO: 118      moltype = AA length = 436
FEATURE            Location/Qualifiers
source              1..436
                    mol_type = protein
                    organism = Archaeoglobus fulgidus

```

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SEQUENCE: 118
MQVLRDLRRH YKSGKIRRAM SSRIPGFYKL SVEERLKKVA EFAGLSDEEV KAVLSQGLPL 60
DVADRMIENV IGTFFELPLGI ATNFLIDGKD YLIPMAIEEP SVVAAASNAA RMARESGGFT 120
TDYTGSLMIG QIQVTKLLNP NAAKFEVLRQ KDEIIERANE CDPMLVNLGG GCKDIEARVI 180
DTIMGKMLIV HLIVDVKDAM GANAVNTMCE KVAPPIERIT GGKVVLRIS NLAAAYRLARA 240
KAVFDKDVIG GEEVVEGIML AYAFAAADPF RCATHNKGIM NGISALMIAT GNDFRAIEAG 300
AHSYAAIGGY KPLTTYEVDR KGNLVGTIEI PMAVGVI GGA TKVNPLAKIS LKILGVNTAE 360
ELARVAAALG LAQNFAALRA LATEGIQRGH MELHARNLAI MAGATGDEVD RVVEIMVRDG 420
KIRLDYAKEV LERLRS 436

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SEQ ID NO: 119      moltype = DNA length = 1287
FEATURE            Location/Qualifiers
misc_feature        1..1287
                    note = Description of Artificial Sequence: Synthetic
                        polynucleotide

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source          1..1287
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 119
atgtccttag attcaagact gccagcttgc agaaatctgt ctccagctgc aagactagat 60
cacattggcc aacttttggg actaagtcac gacgacgttt ccccttttagc aaacgccggt 120
gctttacca tggatategc taatggtatg attgaaaatg taatcgggac ctttgaactg 180
ccatatgcag tggccagtaa ctttcagatc aatggccgtg acgtcttagt accattagtt 240
gtggaggaa ctagtatcgt tgctgcagcc tcttacatgg caaagttagc tagagccaat 300
ggtagggttc ctacatcttc atctgctcca ctaatgcatg cacaagtaca aattgtcggc 360
attcaggatc cactaaacgc aagattgtct ttactgcgta gaaaggatga gatcatagaa 420
ttagccaata ggaaggacca acttctgaat tcatgggctg gtggttcgag agacatagag 480
gtgcatacat ttgccgatac tccaagagga ccaatgcttg tagcacacct tattgtcgat 540
gtgcgtgatg ccatgggagc taatactgtt aacactatgg ctgaagcagt agcacctctg 600
atggaagcca taacaggtgg ccaggtaaga ttgagaatcc tttccaattt ggctgatctt 660
agattggcca gagcccaagt gagaatcact cctcagcaat tggaaaatgc cgaattctca 720
ggtagggcag taattggagg tatcttggac gcatatgctt ttgccgctgt ggacccttac 780
agagccgcta cccacaacaa aggcataatg aacggtatcg atcctttgat cgtcgctaca 840
ggaaatgatt tgagagctgt tgaggcagga gctcatgcat acgctttag atccggacat 900
tacggttcat taacaacatg ggaataagat aacaatggac acttggctcg gacattggaa 960
atgcctatgc cagttggttt agttgggggt gctacaaaaa cccatcctct tgctcaattg 1020
tctttgagga taattggtgt caaaactgct caagcactag cgaatattgc cgttgctgtt 1080
ggtttggcac aaaacttggg tgcaatgctg gctttagcta cagaaggcag ccaaagagga 1140
catatggctc tacacgctag aaacattgca gttgttcgag gagccagagg tgatgaggtt 1200
gattgggtgg ctagacaact tgcgaatat catgatgtca gacgagacag ggctgtggca 1260
ttactgaaac agaagagagg tcaataa 1287

SEQ ID NO: 120      moltype = AA length = 428
FEATURE            Location/Qualifiers
source             1..428
                  mol_type = protein
                  organism = Pseudomonas mevalonii

SEQUENCE: 120
MSLDSRLPAF RNLSPAARLD HIGQLLGLSH DDVSLLANAG ALPMDIANGM IENVIGTFEL 60
PYAVASNFQI NGRDVLVPLV VEEPSIVAAA SYMAKLARAN GGFTTSSSAP LMHAQVQIVG 120
IQDPLNARLS LLRRKDEIIE LANRKDQLLN SLGGGCRDIE VHTFADTPRG PMLVAHLIVD 180
VRDAMGANTV NTMAEAVAPL MEAITGGQVR LRILSNLADL RLARAQVRIT PQQLETAEFS 240
GEAVIEGILD AYAFAAVDPY RAATHNKGIM NGIDPLIVAT GNDWRAVEAG AHAYACRS GH 300
YGSLTWEKID NNGHLVGTLE MPMPVGLVGG ATKTHPLAQL SLRILGVKTA QALAEIAVAV 360
GLAQLNLGAMR ALATEGIQRG HMALHARNIA VVAGARGDEV DWVARQLVEY HDVRADRAVA 420
LLKQKRGQ 428

SEQ ID NO: 121      moltype = AA length = 361
FEATURE            Location/Qualifiers
source             1..361
                  mol_type = protein
                  organism = Stevia rebaudiana

SEQUENCE: 121
MALVNPALF YGTSIRTRPT NLLNPTQKLR PVSSSLPSF SSVSAILTEK HQSNPSENNN 60
LQTHLETFFN FDSYMLEKVN MVNEALDASV PLKDKPIKHE SMRYSLLAGG KRIRPMMCIA 120
ACEIVGGNIL NAMPAACAVE MIHTMSLVHD DLPCMDNDDF RRGKPISHKV YGEEMAVLTG 180
DALLSLSPFH IATAKGVSK DRIVRAIGEL ARSVGSEGLV AGQVVDILSE GADVGLDHLE 240
YIHIKHTAML LESSVVIGAI MGGGSDQQIE KLRKFARSIG LLFQVVDIL DVTKSTEELG 300
KTAGKDLLTD KTTYPKLGI EKSREFAEKL NKEAQEQLSG FDRRKAAPLI ALANYNAYRQ 360
N 361

SEQ ID NO: 122      moltype = AA length = 342
FEATURE            Location/Qualifiers
source             1..342
                  mol_type = protein
                  organism = Gibberella fujikuroi

SEQUENCE: 122
MAEQQISNLL SMFDASHASQ KLEITVQMMD TYHYRETPPD SSSSEGGSL SRYDERRVSLP 60
LSHNAASPD I VSQLCFSTAM SSELNHRWKS QRLKVADSPY NYILTLP SKG IRGAFIDSLN 120
VWLEVPEDET SVIKEVIGML HNSSLIIDDF QDNSPLRRGK PSTHTVFGPA QAINATYVI 180
VKAIEKIQDI VGHADALAVT GTTITIFQGG AMDLWWTANA IVPSIQEYLL MVNDKTGALF 240
RLSLELLALN SEASIDSAL ESLSSAVSLL GQYFQIRDDY MNLIDNKYTD QKGFCEDLDE 300
GKYSLTLIHA LQTDSSDLLT NILSMRRVQG KLTAQKRCWF WK 342

SEQ ID NO: 123      moltype = AA length = 300
FEATURE            Location/Qualifiers
source             1..300
                  mol_type = protein
                  organism = Mus musculus

SEQUENCE: 123
MEKTKEKAER ILLEPYRYLL QLPQKQVRSK LSQAFNHWLK VPEDKLQIII EVTEMLHNAS 60
LLIDDIEDSS KLRGFPVAH SIYGVPSVIN SANYVYFLGL EKVLTLDHPD AVKLFTRQLL 120
ELHQGGGLDI YWRDTYTCPT EEEYKAMVLQ KTGGLFGLAV GLMQLFSDYK EDLKPPLD TL 180
GLFFQIRDDY ANLHKEYSE NKSFCEDLTE GKPSFPTIHA IWSRPESTQV QNLRQR TEN 240

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IDIKKYCVQY LEDVGSFAYT RHTLRELEAK AYKQIEACGG NPSLVALVKH LSKMFTTEENK 300

SEQ ID NO: 124 moltype = AA length = 339
FEATURE Location/Qualifiers
source 1..339
 mol_type = protein
 organism = *Thalassiosira pseudonana*

SEQUENCE: 124
MARFYFLNAL LMVISLQSTT AFTPAKLAYP TTTTALNVAS AETSFSLDEY LASKIGPIES 60
ALEASVKRSI PQTDKICESM AYSLMAGGKR IRPVLCIAAC EMFGGSQDVA MPTAVALEMI 120
HTMSLIHDDL PSMDNDDLRR GKPTNHVVFG EDVAILAGDS LLSTSFEHVA RETKGVSAAEK 180
IVDVIARLKG SVGAEGLAGG QVMDLECEAK PGTTLDDLKW IHIKKTATLL QVAVASGAVL 240
GGATPEEVAA CELFAMNIGL AFQVADDILD VTASSEDLGK TAGKDEATDK TTPKLLGLE 300
ESKAYARQLI DEAKESLAPF GDRAAPLLAI ADFIIDRKN 339

SEQ ID NO: 125 moltype = AA length = 355
FEATURE Location/Qualifiers
source 1..355
 mol_type = protein
 organism = *Streptomyces clavuligerus*

SEQUENCE: 125
MHLAPRRVPR GRRSPDPVPR ERQALGRRR GAGSTGCARA AAGVHRRRG GADPSAAVH 60
RGWQAGGGTG LPDEVSTAA ALEMFHAFAL IHDDIMDDSA TRRGSPVTHR ALADRLGAAL 120
DPDQAGQLGV STAILVGDLA LTWSDELLYA PLTPHRLAAV LPLVTAMRAE TVHGQYLDIT 180
SARRPGTDT S LALRIARYKT AAYTMERPLH IGAAALAGARP ELLAGLSAYA LPAGEAFQLA 240
DDLLGVFGDP RRTGKPDLD LRGGKHTVLV ALAREHATPE QRHTLDTLLG TPGLDRQGAS 300
RLRCVLVATG ARAEAERLIT ERDQALTAL NALTLPPLA EALARLTLSG TAHPA 355

SEQ ID NO: 126 moltype = AA length = 330
FEATURE Location/Qualifiers
source 1..330
 mol_type = protein
 organism = *Sulfolobus acidicaldarius*

SEQUENCE: 126
MSYFDNYPNE IVNSVNDIHK SYISGDVPKL YEASYHLFTS GGKRLRPLIL TISSDLFGGQ 60
RERAYAGAA IEVHFTFTLV HDDIMDQDNI RRGLPTVHVK YGLPLAILAG DLLHAKAFQL 120
LTQALRGLPS ETIIKAFDIF TRSIIIISEG QAVDMEFEDR IDIKEQEYLD MISRKTAALF 180
SASSSIGALI AGANDNDVRL MSDFGTNLGI AFQIVDDILG LTADEKELGK PVFSDIREGK 240
KTILVIKTLE LCKEDEKKIV LKALGNKSAS KEELMSSADI IKKYSLDYAY NLAEKYYKNA 300
IDSLNQVSSK SDIPGKALKY LAEFTIRRRK 330

SEQ ID NO: 127 moltype = AA length = 297
FEATURE Location/Qualifiers
source 1..297
 mol_type = protein
 organism = *Synechococcus* sp.

SEQUENCE: 127
MVAQTFNLDY YLSQRQQQVE EALSAAALVA YPERIYEAMR YSLLAGGKRL RPILCLAACE 60
LAGGSVEQAM PTACALEMIH TMSLIHDDL AMDNDDFRRG KPTNHKVFE DIALAGDAL 120
LAYAFEHIAS QTRGVPPQLV LQVIARIGHA VAATGLVGGQ VVDLESEGKA ISLETLEYIH 180
SHKTGALLEA SVVSGGILAG ADEELLARLS HYARDIGLAF QIVDDILDVT ATSEQLGKTA 240
GKDQAAKAT YPSLLGLEAS RQKAELIQS AKEALRPYGS QAEPLALAD FITRRQH 297

SEQ ID NO: 128 moltype = AA length = 371
FEATURE Location/Qualifiers
source 1..371
 mol_type = protein
 organism = *Arabidopsis thaliana*

SEQUENCE: 128
MASVTLGSI VVHHNHHP SSILTKSRSR SCPITLTKPI SFRSKRTVSS SSSIVSSSVV 60
TKEDNLRQSE PSSFDMSYI ITKAELVNKA LDSAVPLREP LKIHEAMRYS LLAGGKRVPR 120
VLCIAACELV GGEESTAMPA ACAVEMIHTM SLIHDDLPCM DNDDLRRGKP TNHKVFGEDV 180
AVLAGDALLS FAFEHLASAT SSDVSPVRV VRVAGELAKA IGTEGLVAGQ VVDISSEGLD 240
LNDVGLHLE FIHLHKTAL LEASAVLGAI VGGGSDEIE RLRKFARCIG LLFQVVDIL 300
DVTKSSKELG KTAGKDLIAD KLTPKIMGL EKSREFAEKL NREARDQLLG FDSKVAPLL 360
ALANYIAYRQ N 371

SEQ ID NO: 129 moltype = AA length = 787
FEATURE Location/Qualifiers
source 1..787
 mol_type = protein
 organism = *Stevia rebaudiana*

SEQUENCE: 129
MKTGFISPAT VFHHRISPAT TFRHHLSPAT TNSTGIVALR DINFRCKAVS KEYSDDLQKD 60
EASFTKWDDD KVKDHLDTNK NLYPNDEIKE FVESVKAMFG SMNDGEINVS AYDTAWVALV 120
QVDGSGSPQ FSSLEWIAN NQLSDGSGWD HLLFSADRI INTLACVIAL TSWNVHPSKC 180
EKGLNFLREN ICKLEDENAE HMPIGFEVTF PSLIDIAKKL NIEVPEDTPA LKEIYARRDI 240
KLTKIPMEVL HKVPTLLHS LEGMPDLEWE KLLKLQCKDG SFLFSPSTA FALMQTKDEK 300
CLQYLTNIIVT KFGGVPNVY PVDLFEHIWV VDRLQRLGIA RYFKSEIKDC VEYINKYWTK 360

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NGICWARNTN VQDIDDTAMG FRVLRAHGYD VTPDVFRQFE KDGKFCVCFAG QSTQAVTGMF 420
NVYRASQMLF PGERILEDK KFSYNYLKEK QSTNELLDKW IIAKDLPGEV GYALDIPWYA 480
SLPRLETRY LEQYGGEDDV WIGKTLYRMG YVSNNTYLEM AKLDYNNYVA VLQLEWYTIQ 540
QWYVDIGIEK PESDNISVL VSYLAAASI FEPERSKERI AWAKTTILVD KITSIFDSSQ 600
SSKEDITAFI DKFRNKSSSK KHSINGEPWH EVMVALKKTLL HGFALDALMT HSQDIHPQLH 660
QAWEMWLTKL QDGVDTVTEL MVQMINMTAG RWVSKELLTH PQYQRLSTVT NSVCHDITKL 720
HNFKENSTTV DSKVQELVLQ VFSPTPDDLD QDMKQTFLLTV MKTFYKAWC DPNTINDHIS 780
KVFEIVI 787

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SEQ ID NO: 130      moltype = AA length = 527
FEATURE            Location/Qualifiers
source             1..527
                  mol_type = protein
                  organism = Streptomyces clavuligerus

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SEQUENCE: 130
MPDAHDAAPP QIRQRTLVDE ATQLLTESAE DAWGEVSVSE YETARLVAHA TWLGGHATRV 60
AFLERQHEH GSWGPPGGYR LVPTLSAVHA LLTCLASPAQ DHGVPHDRLL RAVDAGLTAL 120
RRLGTSDSP DTIAVELVIP SLLEGIQHLL DPAHPHSRPA FSQHRGSLVC PGGLDGRTL 180
ALRSAAAGT PVPKGVHVA ETGLSTEEA SHLQPAQGII GGSAAATATW LTRVAPSQS 240
DSARRYLEEL QHRYSGPVPS ITPITYFERA WLLNNFAAAG VPCEAPAALL DSLEAALTPQ 300
GAPAGAGLPP DADDTAAVLL ALATHGRGRR PEVLMDYRTD GYFQCFIGER TPSISTNAHV 360
LETGLGHVQA HPQDRARYGS AMDTASAWLL AAQKQDGSWL DKWHASPYA TVCCTQALAA 420
HASPATAPAR QRAVRVWLAT QRSDDGGWGLW HSTVEETAYA LQILAPPSGG GNIPVQQALT 480
RGRARLCGAL PLTPLWHDKD LYTPVRVVR ARAAALYTTR DLLLLPPL 527

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SEQ ID NO: 131      moltype = AA length = 516
FEATURE            Location/Qualifiers
source             1..516
                  mol_type = protein
                  organism = Bradyrhizobium japonicum

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SEQUENCE: 131
MNALSEHILS ELRRLLEMS DGGSVGPSVY DTAQALRPHG NVTGRQDAYA WLIAQQQADG 60
GWGSADFPFL RHAPTWAALL ALQRADPLPG AADAVQTATR FLQRQPDPIA HAVPEDAPIG 120
AELILPQFCG EAAWLLGGVA FPRHPALLPL RQACLVKLGA VAMLP SGHPL LHSWEAWGTS 180
PTTACPDGDD SIGISPAATA AWAQAVTRG STPQVGRADA YLQMASRATR SGIEGVFPNV 240
WPINVFEPWC SLYTLHLAAG FAHPALAEAV RVIVAQLEAR LGVHGLGPAL HFAADADDTA 300
VALCVLHLAG RDPADVLRH FEIGELFVTF PGERNASVST NIHALHALRL LGKPAAGASA 360
YVEANRNPBG LWDNEKWHVS WLYPTAHAVA ALAQGKQWR DERALAALLQ AQRDDGGWGA 420
GRGSTFEETA YALFALHVMG GSEEAATGRRR IAQVVARALE WMLARHAAHG LPQTPLWIGK 480
ELYCPTVRVR VAEALGLWLA LRWGRRVLAE GAGAAP 516

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SEQ ID NO: 132      moltype = AA length = 784
FEATURE            Location/Qualifiers
source             1..784
                  mol_type = protein
                  organism = Stevia rebaudiana

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SEQUENCE: 132
MNLSLCIASP LLTKSNRPAA LSAIHTASTS HGGQTNPTNL IIDTTKERIQ KQFKNVEISV 60
SSYDTAWVAM VSPNSPKSP CFPECLNWLII NNQLNDGSWG LVNHTHNHNL PLLKDSLSST 120
LACIVALKRW NVGEDQINKG LSFIESNLAS ATEKSQSPSI GFDIIFPGLL EYAKNLDINL 180
LSKQTDPSLM LHKRELEQKR CHSNEMDGYL AYISEGLGNL YDWNMVKKYQ MKNGSVFNPS 240
SATAAAFINH QNPGCLNYLN SLDDKFGNAV PTVPYPHDLFI RLSMVDITIER LGISHHFRVE 300
IKNVLDETYR CWVERDEQIF MDVVTALAF RLLRINGYEV SPDPLAETN ELALKDEYAA 360
LETYHASHIL YQEDLSSGKQ ILKSADFLKE IISTDSNRLS KLIHKEVENA LKFPINTGLE 420
RINTRRNIQL YNVNTRILK TTYHSSNIN TDYLRALAVD FYTCQSIYRE ELKGLERWV 480
ENKLDQLKFA RQKTAYCYFS VAATLSSPEL SDARISWAKN GILTTVVDDF FDIGGTIDEL 540
TNLIQCVEKW NVDVDKCCS EHVRIILFLAL KDAICWIGDE AFKWQARDVT SHVIQTWLEL 600
MNSMLREAIW TRDAYVPTLN EYMENAYVSF ALGPIVKPAI YFVGPKLSEE IVESSEYHNL 660
FKLMSTQGR LNDIHSFKRE FKEGKLNAVA LHLSNGESGK VEEVVEEMM MMIKNKRKEL 720
MKLIFEENG IVPACKDAF WNMCHVLNFF YANDDGFTGN TILDTVKDII YNPLVLVNEN 780
EEQR 784

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SEQ ID NO: 133      moltype = AA length = 784
FEATURE            Location/Qualifiers
source             1..784
                  mol_type = protein
                  organism = Stevia rebaudiana

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```

SEQUENCE: 133
MNLSLCIASP LLTKSSRPTA LSAIHTASTS HGGQTNPTNL IIDTTKERIQ KLFKNVEISV 60
SSYDTAWVAM VSPNSPKSP CFPECLNWLII NNQLNDGSWG LVNHTHNHNL PLLKDSLSST 120
LACIVALKRW NVGEDQINKG LSFIESNLAS ATDKSQSPSI GFDIIFPGLL EYAKNLDINL 180
LSKQTDPSLM LHKRELEQKR CHSNEIDGYL AYISEGLGNL YDWNMVKKYQ MKNGSVFNPS 240
SATAAAFINH QNPGCLNYLN SLDDKFGNAV PTVPYPLDYI RLSMVDITIER LGISHHFRVE 300
IKNVLDETYR CWVERDEQIF MDVVTALAF RLLRIHGYKV SPDQLAETN ELAFKDEYAA 360
LETYHASQIL YQEDLSSGKQ ILKSADFLKG ILTDSNRLS KLIHKEVENA LKFPINTGLE 420
RINTRRNIQL YNVNTRILK TTYHSSNIN TYLRLAVED FYTCQSIYRE ELKGLERWV 480
QNKLDQLKFA RQKTAYCYFS VAATLSSPEL SDARISWAKN GILTTVVDDF FDIGGTIDEL 540
TNLIQCVEKW NVDVDKCCS EHVRIILFLAL KDAICWIGDE AFKWQARDVT SHVIQTWLEL 600
MNSMLREAIW TRDAYVPTLN EYMENAYVSF ALGPIVKPAI YFVGPKLSEE IVESSEYHNL 660

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FKLMSTQGR	LNDIHSFKRE	FKEGKLNVA	LHLSNGESGK	VEEEVVEEM	MMIKNKRKEL	720
MKLIFEENG	IVPRACKDAF	WNMCHVLNPF	YANDDGFTGN	TILDTVKDII	YNPLVLVNEN	780
EEQR						784

SEQ ID NO: 134 moltype = AA length = 590
 FEATURE Location/Qualifiers
 source 1..590
 mol_type = protein
 organism = Zea mays

SEQUENCE: 134

MAMPVKLTPA	SLSLKAVCCR	FSSGGHALRF	GSSLPCWRR	PTQRSTSSST	TRPAAEVSSG	60
KSKQHDQEAS	EATIRQQLQL	VDVLENMGIS	RHFAAEIKCI	LDRTYRSWLQ	RHEEIMLDTM	120
TCAMAFRILR	LNGYNVSSDE	LYHVVEASGL	HNSLGGYLND	TRTLELHKA	STVSISEDES	180
ILDSIGSRSR	TLLREQLESG	GALRKPSLKF	EVEHALDGPF	YTTLDRLHHR	WNIENFNIE	240
QHMLETPYLS	NQHTSRDILA	LSIRDFSSSQ	FTYQQELQHL	ESWVKECRLD	QLQFARQKLA	300
YFYLSAAGTM	FSPELSDART	LWAKNGVLT	IVDDFFDVAG	SKEELENLVM	LVEMWDEHHK	360
VEFYSEQVEI	IFSSIIDSVN	QLGEKASLVQ	DRSITKHLVE	IWLDDLKSM	TEVEWRLSKY	420
VPTEKEYMIN	ASLIFGLGPI	VLPALYFVGP	KISESIVKDP	EYDELFLKMS	TCGRLLNDVQ	480
TFEREYNEKG	LNSVSLVLH	GGPMSISDAK	RKLQKPIDTC	RRDLLSLVLR	EESVVPKCK	540
ELFWKMCKVC	YFFYSTTDGF	SSQVERAKEV	DAVINEPLKL	QGSHTLVSDV		590

SEQ ID NO: 135 moltype = AA length = 743
 FEATURE Location/Qualifiers
 source 1..743
 mol_type = protein
 organism = Populus trichocarpa

SEQUENCE: 135

MCNPHGKTER	IKKMFDKIEL	SVSSYDTAWV	AMVPSDPCPE	TPCFPECTKW	ILENLQGDGS	60
WSLPHGNPLL	VKDALSSTLA	CILALKRWGI	GEEQINKGLR	FIELNSASVT	DNEQHKPIGF	120
DIIFPGMIEY	AIDLNLNPL	KPTDINSMLH	RRAELTSGG	GKNLEGRRAY	LAYVSEGIGK	180
LQDWMAMKY	QRKNGSLFNS	PSTTAAAFIH	IQDAECLHYI	RSLLQKFGNA	VPTIYPLDIY	240
ARLSMVDAL	RLGIDRHRFK	ERKFVLDETY	RFWLQGEETI	FSDNATCALA	FRILRLNGYD	300
VSLEDHFSNS	LGGYLKDSGA	ALELYRALQL	SYDESLELEK	QNSRTSYFLK	QGLSNVSLCG	360
DRLRKNIIGE	VHDALNFSDH	ANLQRLAIR	RIKHYATDDT	RILKTSYRCS	TIGNQDFLKL	420
AVEDFNICQS	IQREEFKHIE	RWVVERRLDK	LKFARQKEAY	CYFSAAATLF	APELSDARMS	480
WAKNGVLTIV	VDDFFDVGG	EEELVNLIEL	IERWDVNGSA	DFCEEEVEII	YSAIHSTISE	540
IGDKSFGWQ	RDVKSQVIKI	WLDLLKSM	EAQWSSNKSV	PTLDEYMTTA	HVSFALGPV	600
LPALYFVGPK	LSEEVAGHPE	LLNLKYVTST	CGRLLNDWRS	FKRESEEGKL	NAISLYMIHS	660
GGASTEETET	EHFKGLIDSQ	RRQLQLVLQ	EKDSIIPRPC	KDLFWNMIKL	LHTFYMKDDG	720
FTSNEMRNVV	KAIINEPISL	DEL				743

SEQ ID NO: 136 moltype = AA length = 983
 FEATURE Location/Qualifiers
 source 1..983
 mol_type = protein
 organism = Phomopsis amygdali

SEQUENCE: 136

MEFDEPLVDE	ARSLVQRTLQ	DYDDRYGFGT	MSCAAYDTAW	VSLVTKTVDG	RKQWLFPECF	60
EFLLLETQSDA	GGWEIGNSAP	IDGILNTAAS	LLALKRHRVQT	EQIIQPQHDH	KDLAGRAERA	120
AASLRAQLAA	LDVSTTEHVG	FEIIVPAMLD	PLEAEDPSLV	FDFPARKPLM	KIHDAKMSRF	180
RPEYLYGKQP	MTALHSLEAF	IGKIDFDKVR	HRHTGSMGM	SPSSTAAYLM	HASQWDGDS	240
AYLRHVIKHA	AGQGTGAVPS	AFPSTHFESS	WILTTLFRAG	FSASHLACDE	LNKLVEILEG	300
SFEKEGGAIG	YAPGFQADVD	DTAKTISTLA	VLGRDATPRQ	MIKVFEANTH	FRTYPGERDP	360
SLTANCNALS	ALLHQPDAA	YGSQIQKITK	FVCDYWWKSD	GKIKDKWNTC	YLPSVLLVE	420
VLVDLVSLLE	QGKLPDVLQ	ELQYRVAITL	FQACLRPLLD	QDAEGSWNKS	IEATAYGILI	480
LTEARRVCFP	DRLSEPLNEA	IRRGIAFADS	MSGTEAQLNY	IWIEKVSYP	ALLTKSYLLA	540
ARWAAKSPGL	ASVGSSTWTP	PREGLDKHVR	LFHQAELEFRS	LPEWELRASM	IEAALPTPL	600
RAHRLDVFP	QDVGEDKYLD	VVPFFWTAAN	NRDRTYASTL	FLYDMCFIAM	LNQQLDEFME	660
ATAGILFRDH	MDDLRLQIHD	LLAEKTSPPK	SGRSSQGTKD	ADSGIEEDVS	MSDSASDSQD	720
RSPEYDLVFS	ALSTFTKHVL	QHPISQASV	WDRKLLAREM	KAYLLAHIQ	AEDSTPLSEL	780
KDVPQKTDVT	RVSTSTTTF	NWVRTTSADH	ISCPYSFHEV	ACHLGAALSP	KGSNGDCYPS	840
AGEKFLAAAV	CRHLATMCRM	YNDLGSARD	SDEGNLNSLD	PPEFADSAGN	GGIEIQKAAL	900
LRLAEFERDS	YLEAFRRLLQ	ESNRVHGPA	GDEARLSRRR	MAILEFFAQ	VDLYGQVYVI	960
RDISARIPKN	EVEKKRKLDD	AFN				983

SEQ ID NO: 137 moltype = AA length = 881
 FEATURE Location/Qualifiers
 source 1..881
 mol_type = protein
 organism = Physcomitrella patens

SEQUENCE: 137

MASSTLIQNR	SCGVTSSMSS	FQIFRGQPLR	FPGRTRPAAV	QCLKRRCLR	PTESVLESSP	60
GGSGYRIVTG	PSGINPSSNG	HLQEGSLTHR	LPIPMKESID	NFQSTLYVSD	IWSETLQRT	120
CLLQVTENVQ	MSEWIEEIRP	YFRNMTLGEI	SMSPYDTAWV	ARVPALDGSH	GPQFHRSLQW	180
IIDNQLPDGD	WGEPFLFLGY	DRVNTLACV	IALKTWGVGA	QNVERGIQFL	QSNIIYKMEED	240
DANHMPIGFE	IVFPAMMEDA	KALGLDLPYD	ATILQQISAE	REKKMKKIPM	AMVYKPTTL	300
LHSLEGLHRE	VDWNKLLQLQ	SENGSFLYSP	ASTACALMYT	KDVKCFDYLN	QLLIKFDHAC	360
PNVYPVDLFE	RLWMVDRLQ	LGISRYFERE	IRDCLQYVYR	YWKDCGIGWA	SNSSVQDVDD	420
TAMAFRLRLT	HGFDVKEDCF	RQPFKDGFEF	CFAGQSSQAV	TGMPNLSRAS	QTLFPGESLL	480

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KKARTFSRNF LRTKHENNEC FDKWIITKDL AGEVEYNLTF PWYASLPRLE HRTYLDQYGI 540
DDIWIGKSLY KMPAVTNEVF LKLAKADFNM CQALHKKLE QVIKWNASCQ FRDLEFARQK 600
SVECYFAGAA TMFEPVMVQA RLVARCCVL TTVLDDYFDH GTPVEELRVF VQAVRTWNPE 660
LINGLPEQAK ILFMGLYKTV NTIAEEAFMA QKRDVHHHLK HYWDKLITSA LKEAEWAESG 720
YVPTFDEYME VAEISVALEP IVCSTLFFAG HRLDEDVLDS YDYHLVMHLV NRVGRILNDI 780
QGMKREASQG KISSVQIYME EHPSVPSEAM AIAHLQELVD NSMQQLTYEV LRFTAVPKSC 840
KRIHLNMAKI MHAIFYKDTDG FSSLTAMTGF VKKVLFEVPV E 881

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SEQ ID NO: 138      moltype = AA length = 513
FEATURE            Location/Qualifiers
source              1..513
                    mol_type = protein
                    organism = Stevia rebaudiana

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SEQUENCE: 138
MDAVTGLLTV PATAITIGGT AVALAVALIF WYLKSYTSAR RSQSNHLPRV PEVPGVPLLG 60
NLLQLKEKKP YMTPTRWAAAT YGPIYSIKTG ATSMVVVSSN EIAKEALVTR FQSISTRNLS 120
KALKVLTADK TMVAMSDYDD YHKTVKRHIL TAVLGPNAQK KHRIHRDIMM DNISTQLHEF 180
VKNNPEQEEV DLKRFQSEL FGLAMRQALG KDVESLYVED LKTMNRDEI FQVLVVDPM 240
GAIDVDWRDF FPYLKWVPNK KFCNTIQQMY IRREAVMKSL IKEHKRIAS GEKLSYIDY 300
LLSEAQTLTD QQLLMSLWEP IIESSDTTMV TTEWAMYELA KNPQLQDRLY RDIKSVCGSE 360
KITEEHLSQL PYITAI FHET LRRHSPVPII PLRHVHEDTV LGGYHVPAGT ELAVNIYGCN 420
MDKNVWENPE EWNPERFMKE NETIDFQKTM AFGGKRVCA GSLQALLTAS IGIGRMVQEF 480
EWKLKDMTQE EVNTIGLTTQ MLRPLRAIIK PRI 513

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SEQ ID NO: 139      moltype = AA length = 509
FEATURE            Location/Qualifiers
source              1..509
                    mol_type = protein
                    organism = Arabidopsis thaliana

```

```

SEQUENCE: 139
MAFFSMISIL LGFVISSFIF IFFFKLLSF SRKNMSEVST LPSVPVVPGF PVIGNLLQLK 60
EKKPHKTFTR WSEIYGPIYS IKMGSSSLIV LNSTETAKEA MVTRFSSIST RKLSNALTVL 120
TCDKSMVATS DYDDFHLVK RCLNLGLLGA NAQKRKRHYR DALIENVSSK LHAHARDHPQ 180
EPVNFRAIFE HELFGVALKQ AFGKDVESYI VKELGVTLSE DEIFKVLVHD MMEGAIDVDW 240
RDFFPYLKWI PNKSFEARIQ QKHKKRLAVM NALIQDRLKQ NGSESDDDCY LNFLMSEAKT 300
LTKEQIALLV WETIETADT TLVTTEWAIY ELAKHPSVQD RLCKEIQNVC GGEKFKEEQL 360
SQVPYLVGVF HETLRKYSPA PLVPIRYAHE DTQIGGYHVP AGSEIAINIY GCNMDKKRWE 420
RPEDWWPERF LDDGKYETSD LHKTMAGFAG KRVCAALQA SLMAGIAIGR LVQEFEWKLR 480
DGEENVVDY GLTSQKLYPL MAIINPRRS 509

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SEQ ID NO: 140      moltype = AA length = 525
FEATURE            Location/Qualifiers
source              1..525
                    mol_type = protein
                    organism = Gibberella fujikoro

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SEQUENCE: 140
MSKSNMNST SHETLFQQLV LGLDRMPLMD VHWLIYVAFG AWLCSYVIHV LSSSSTVKVP 60
VVGYSRVFEP TWLLRLRFVW EGGSIIGQGY NKFKDSIFQV RKLGTDIVII PPNYIDEVRK 120
LSQDKTRSVF FINDFAGQY TRGMVFLQSD LQNRVIQQLR TPKLVSLTKV MKEELDYALT 180
KEMPMKNDK WVEVDISSIM VRLISRISAR VFLGPEHCRN QEWLTTTAEY SESLFITGFI 240
LRVVPILRP FIAPLLPSYR TLLRNVSSEGR RVIGDIIRSQ QGDGNEDILS WMRDAATGEE 300
KQIDNIAQRM LILSLASHT TAMTMTTHAMY DLCACEYIE PLRDEVKSVV GASGWDKTAL 360
NRFHKLDSPL KESQRFNPVF LLTFNRIYHQ SMTLSDGTNI PSGTIRIAPVS HAMLQDSAHV 420
PGTTPPTEFD GFRYSKIRSD SNYAQKYLFS MTDSSNMAFG YGKYACPGRF YASNEMKLT 480
AILLQQFEFK LPDGKGRPRN ITIDSDMIPD PRARLCVRKR SLRDE 525

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SEQ ID NO: 141      moltype = AA length = 499
FEATURE            Location/Qualifiers
source              1..499
                    mol_type = protein
                    organism = Trametes versicolor

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SEQUENCE: 141
MEDPTVLYAC LAIAVATFVW RWYRDLPLRSI PTVGGSDLPI LSYIGALRWT RRGREILQEG 60
YDGYRGSTFK IAMLDRWIVI ANGPKLADEV RRRPDEELNF MDGLGAFVQT KYTLGEAIHN 120
DPYHVDIIRE KLTRGLPAVL PDVIEELTLA VRQYIPTEGD EWVSVNCSKA ARDIVARASN 180
RVFVGLPACR NQGYLDLAD FTLSSVKDRA IINMFPPELLK PIVGRVVGNA TRNVRRAVPF 240
VAPLVEERRR LMEYGEDWS EKPNDMLQWI MDEAASRDSS VKAIAERLLM VNFAAHTSS 300
NTITHALYHL AEMPETIQPL REEIEPLVKE EGWTKAAMGK MWLDSPLRE SQRYNGINIV 360
SLTRMADKDI TLDGTFLPK GTLVAVPAYS THRDDAVYAD ALVFDPPFRS RMRAREGEGT 420
KHQFVNTSVE YVPPGHGKHA CPGRFFAANE LKAMLAYIVL NYDVKLPGDG KRPLNMYWGP 480
TVLPAPAGQV LFRKRQVSL 499

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SEQ ID NO: 142      moltype = AA length = 525
FEATURE            Location/Qualifiers
source              1..525
                    mol_type = protein
                    organism = Stevia rebaudiana

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```

SEQUENCE: 142
MGLFPLEDSY ALVFEGLAIT LALYYLLSFI YKTSKKTCTP PKASGEIPII TGIILNLLSG 60

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SSGLPIILAL	ASLADRCGPI	FTIRLGIRRV	LUVSNWEIAK	EIFTTHDLIV	SNRPKYLAAC	120
ILGFNYVSFS	FAPYGPYVWG	IRKIIATKLM	SSSRLQKLQF	VRVFELENSM	KSIRESWKEK	180
KDEEGKVLVE	MKKWFELNM	NIVLRTVAGK	QYTGTVDDAD	AKRISELFRE	WFHYTGRFVV	240
GDAPFPGLWL	DLGGYKKTME	LVASRLDSMV	SKWLDEHRKK	QANDDKKEDM	DFMDIMISMT	300
EANSPLLEGY	TDIIKTTCM	TLIVSGVDTT	SIVLTWALS	LLNNRDTLKK	AQEELDMCVG	360
KGRQVNESDL	VNLIYLEAVL	KEALRLYPAA	FLGGPRAPLE	DCTVAGYRIP	KGTCCLLINMW	420
KLHRDPNIWS	DPCEFKPERF	LTPNQKQDV	IGMDFELIPF	GAGRRYCPGT	RLALQMLHIV	480
LATLLQNPEM	STPNDAVDM	TASVGMTNAK	ASPLEVLLSP	RVKWS		525

SEQ ID NO: 143 moltype = AA length = 476
 FEATURE Location/Qualifiers
 source 1..476
 mol_type = protein
 organism = Stevia rebaudiana

SEQUENCE: 143

MIQVLTPIILL	PLIFFVFWKV	YKHQKTKINL	PPGSPGWPF	GETLALLRAG	WDSEPERFVR	60
ERIKKHGSPL	VFKTSLFGDR	FAVLCGPAGN	KFLFCNENKL	VASWWVPVVR	KLFGKSLTTI	120
RGDEAKWMRK	MLLSYLGPD	FATHYAVTMD	VVTRRHIDVH	WRGKEEVNVF	QTVKLYAFEL	180
ACRLPMNLDD	PNHIAKLGS	FNIFLKGII	LPIDVPGTRF	YSSKAAAAI	RIELKKLIKA	240
RKLELKEGKA	SSSQDLLSHL	LTSPEDEMGF	LTEEEIVDNI	LLLLFAGHDT	SALSITLLMK	300
TLGEHSDVVD	KVLKEQLAIS	KTKEAWESLK	WEDIQMKKYS	WSVICEVMRL	NPPVIGTYRE	360
ALVDDIDYAGY	TIPKGWKLHW	SAVSTQRDEA	NFEDVTRFDP	SRFEGAGTPT	FTFVPPGGGP	420
RMCLGKEFAR	LEVLAFLHNI	VTNFKWDLII	PDEKIEYDPM	ATPAKGLPIR	LHPHQV	476

SEQ ID NO: 144 moltype = AA length = 525
 FEATURE Location/Qualifiers
 source 1..525
 mol_type = protein
 organism = Arabidopsis thaliana

SEQUENCE: 144

MESLVVHTVN	AIWCIVIVGI	FSVGYHVG	AVVEQWRMR	SLKLQGVKGP	PPSIFNGNVS	60
EMQRIQSEAK	HCSGDNIIIS	DYSSSLFPFH	DHWRKQYGR	YTYSTGLKQH	LYINHPMEMV	120
ELSQTNTLNL	GRITHITKRL	NPILGNGIIT	SNGPHWAHQ	RIIAYEPTH	KIKGMVGLMV	180
ESAMPMLNWK	EEMVKRGEM	GCDIRVDEDL	KDVSADVIK	ACFGSSFSKG	KAIFSMIRDL	240
LTAITKRSVL	PRFNGFTDMV	FGSKKHGDVD	IDALEMELES	SIWETVKERE	IECKDTHKKD	300
LMQLILEGAM	RSCDGNLWDK	SAYRRFVVDN	CKSIYPAGHD	STAVSVSWCL	MLLALNPSWQ	360
VKIRDEILSD	CKNGIPDAES	IPNLKTVTMV	IQETMRLYPP	APIVGREASK	DIRLGDLVVP	420
KGVCIWTLIP	ALHRDPBIWG	PDANDFKPER	FSEGISKACK	YPQSYIPFGL	GPRTCVGKNP	480
GMMEVKVLVS	LIVSKFSFTL	SPTYQHSPPH	KLLVEPQHG	VIRVV		525

SEQ ID NO: 145 moltype = AA length = 529
 FEATURE Location/Qualifiers
 source 1..529
 mol_type = protein
 organism = Vitis vinifera

SEQUENCE: 145

MYFLLQYLN	ITTVGVFATLF	LSYCLLLWRS	RAGNKKIAPE	AAAAPPIIGH	LHLLAGGSHQ	60
LPHITLGNMA	DKYGPVFTIR	IGLHRAVVVS	SWEMAKECST	ANDQVSSSRP	ELLASKLLGY	120
NYAMFGFSFY	GSYVREMRKI	ISLELLSNSR	LELLKDVRAS	EVVTSIKELY	KLWAEKKNES	180
GLVSVEMKQW	FGDLTLNVIL	RMVAGKRYFS	ASDASENKQA	QRCRRVPREF	FHLGSLFVVA	240
DAIPFLGLWD	WGRHEKTLKK	TATEMSIAQ	EWLEEHRRRK	DSGDDNSTQD	FMDVMQSVLD	300
GKNLGGYDAD	TINKATCLTL	ISGSDTTTV	SLTWALS	LLNNRDTLKK	AEELDIQVGKE	360
RLVNEQDISK	LVLQAIIVKE	TLRLYPPGGL	GGLRQFTEDC	TLGGYHVS	SKGTRLIMNLSKI	420
QKDPRIWSDP	TEFQPERFLT	THKDVPDRGK	HFEFIPFGAG	RRACPGITFG	LQVLHLTLAS	480
FLHAFEFSTP	SNEQVNMRES	LGLTNMKSTP	LEVLI	SPRLSLNCFNL	LMKI	529

SEQ ID NO: 146 moltype = AA length = 479
 FEATURE Location/Qualifiers
 source 1..479
 mol_type = protein
 organism = Medicago trunculata

SEQUENCE: 146

MEPNFYLSLL	LLFVTFISLS	LFFIFYKQKS	PLNLPPGKMG	YPIIGESLEF	LSTGWKGHP	60
KPIFDRMRKY	SSELFKTSIV	GESTVVCCGA	ASNKFLFSNE	NKLVTAWWD	SVNKFPTTS	120
LDSNLKEESI	KMRKLLPQFF	KPEALQRYVG	VMDVIAQRHF	VTHWDNKNEI	TVYPLAKRYT	180
FLACRLFMS	VEDENHAKF	SDPFQLIAAG	IISLPIDLP	TPFNKAIKAS	NFIRKELIKI	240
IKQRRVDLAE	GTASPTQDIL	SHMLLTSDEN	GKSMNELNIA	DKILGLLIGG	HDASVACTF	300
LVKYLGLLPH	LYDKVYQEQM	EIAKSKPAGE	LLNWDDLKHM	KYSWNVACEV	MRLSPPLQGG	360
FREAITDFMF	NGFSIPKQWK	LYWSANSTHK	NAECFPMPEK	FDPTRFEGNG	PAPYTFVPFG	420
GGPRMCPGKE	YARLEILVFM	HNLVVRPKWE	KVIPDEKIIV	DPFPIPAKDL	PIRLYPHKA	479

SEQ ID NO: 147 moltype = AA length = 710
 FEATURE Location/Qualifiers
 source 1..710
 mol_type = protein
 organism = Stevia rebaudiana

SEQUENCE: 147

MQSDSVKVP	FDLVSAAMNG	KAMEKLNASE	SEDPTLPAL	KMLVENRELL	TLFTTSFAVL	60
IGCLVFLMWR	RSSSKLVQD	PVPQVIVVKK	KEKESEVDDG	KKKVSIFYGT	QTGTAEFGAK	120

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ALVEEAKVRY	EKTSFKVIDL	DDYAADDDEY	EEKLKESLA	FFFLATYGDG	EPTDNAANFY	180
KWFTGGDDKG	EWLKKLQYGV	FGLGNRQYEH	FNKIAIVDD	KLTEMGAKRL	VPVGLGDDQ	240
CIEDDFTAWK	ELVWPELDQL	LRDEDDTSVT	TPYTAAVLEY	RVVYHDKPAD	SYAEDQHTN	300
GHVVHDAQHP	SRSNVAFKKE	LHTSQSDRSC	THLEFDISHT	GLSYETGDHV	GVYSENLSEV	360
VDEALKLLGL	SPDITYFSVHA	DKEDGTPIGG	ASLPPPPPPC	TLRDALTRYA	DVLSSPKKVA	420
LLALAAHASD	PSEADRLKFL	ASPAGKDEYA	QWIVANQRSL	LEVMSQSFSA	KPPLGVFFAA	480
VAPRLQPRY	SISSSPKMSP	NRIHVTCLV	YETTPAGRIH	RGLCSTWMKN	AVPLTESPDC	540
SQASIFVVRTS	NFRLPVPDPKV	PVIMIGPGTG	LAPFRGFLQE	RLALKESGTE	LGSSIFFFGC	600
RNRKVDIFYE	DELNNFVETG	ALSELIVAFS	REGTAKEYVQ	HKMSQKASDI	WKLLSEGAYL	660
YVCGDAKGMA	KDVHRTLHTI	VQEQGSLDSS	KAELYVKNLQ	MSGRYLRDVG		710

SEQ ID NO: 148 moltype = AA length = 692
 FEATURE Location/Qualifiers
 source 1..692
 mol_type = protein
 organism = Arabidopsis thaliana

SEQUENCE: 148

MTSALYASDL	FKQLKSIMGT	DSLSDDVVLV	IATTSALVA	GFVVLLWKKT	TADRSSELKP	60
LMIPKSLMAK	DEDDDLDLGS	GKTRVSIFFG	TQTGTAEFGA	KALSEEIKAR	YEKAAVKVID	120
LDDYAADDQ	YEEKLKKETL	AFPCVATYGD	GEPTDNAARF	YKWFTEENER	DIKLQQLAYG	180
VPALGNRQYE	HFNKIGIVLD	EELCKKGAKR	LIEVGLGDD	QSIEDDFNAW	KESLWSELDK	240
LLKDEDDKSV	ATPYTAUIPE	YRVVTHDPRF	TTQKSMESNV	ANGNTTIDIH	HPCRVDVAVQ	300
KELHTHESDR	SCIHLEFDS	RTGITYETGD	HVGUYAENHV	EIVVEAGKLL	GHSLLDLVFSI	360
HADKEDGSP	ESAVPPPPFG	PCTLGTGLAR	YADLLNPPRK	SALVALAAYA	TEPSEAEKLG	420
HLTSPDGKDE	YSQWIVASQR	SLEVMMAAFP	SAKPLGVVFF	AAIAPRLQPR	YYSISSSPRL	480
APSRVHVTS	LVYGTPTTGR	IHKGVCSTWM	KNAVPAEKSH	ECSGAPIFIR	ASNFKLPSNP	540
STPIVMVGPG	TGLAPFRGFL	QERMALKEDG	EELGSSLLFF	GCRNRQMDFI	YEDELNNFVD	600
QGVISELIMA	FSREGAQKEY	VQHKMEKAA	QVWDLIKEEG	YLYVCGDAKG	MARDVHRTLH	660
TIVQEQEGVS	SSEAEIVKK	LQTEGRYLRD	VW			692

SEQ ID NO: 149 moltype = AA length = 713
 FEATURE Location/Qualifiers
 source 1..713
 mol_type = protein
 organism = Giberella fujikuroi

SEQUENCE: 149

MAELDTLDIV	VLGVIFLGTV	AYFTKGKLWG	VTKDPYANGF	AAGGASKPGR	TRNIVEAMEE	60
SGKNCVVFFYG	SQTGAEDYA	SRLAKEGKSR	FGLNTMIADL	EDYDFDNLDT	VPSDNIVMFV	120
LATYGEGETP	DNAVDFYEFI	TGEDASFNEG	NDPPLGNLNY	VAFGLGNNTY	EHYNSMVRNV	180
NKALEKLGAH	RIGEGEGDD	GAGTMEEDFL	AWKDPMEAL	AKKMGLEERE	AVYEPIFAIN	240
ERDDLTPKAN	EVYLGEPNKL	HLEGTAKGPF	NSHNPYIAPI	AESYELFSAK	DRNCLHMEID	300
ISGSNLKYET	GDHIAIWPTN	PGEEVNKFLD	ILDLSGKQHS	VVTVKALEPT	AKVPPPNPTT	360
YDAILRYHLE	ICAPVSRQFV	STLAAPAPND	DIKAEMNRLG	SDKDYFHEKT	GPHYNYIARF	420
LASVSKGEKW	TKIPFSAPIE	GLTKLQPRYY	SISSSSLVQP	KKISITAVVE	SQQIPGRDDP	480
FRGVATNYLF	ALKQKQNGDP	NPAPFGQSYE	LTGPRNKYDG	THVPVHVRHS	NFKLPSDPGK	540
PIIMIGPGTG	VAPFRGFVQE	RAKQARDGVE	VGKTLLEFGC	RKSTEDFMYQ	KEWQEYKEAL	600
GDKFEMITAF	SREGSKKVVY	QHRLEKERSKE	VSDLLSQKAY	FYVCGDAAHM	AREVNTVLAQ	660
IIAEGRGVSE	AKGEEIVKNM	RSANQYQVCS	DFVTLHCKET	TYANSELQED	VWS	713

SEQ ID NO: 150 moltype = AA length = 453
 FEATURE Location/Qualifiers
 source 1..453
 mol_type = protein
 organism = Arabidopsis thaliana

SEQUENCE: 150

MGGLKFHVLM	YPWFATGHMT	PFLFLANKLA	EKGHTVTFL	PKKSLKQLEH	FNLFPHNIVF	60
RSVTVPVVDG	LPVGTETASE	IPVTSTDLLM	SAMDLTRDQV	EAVVRAVEPD	LIFDFFAHWI	120
PEVARDFGLK	TVKYVVVSAS	TIASMLVPGG	ELGVPPPGYP	SSKVLRLKQD	AYTMKKLEPT	180
NTIDVGNPLL	ERVTTSLMNS	DVIAIRTARE	IEGNFCDYIE	KHCRKKVLLT	GPVFPEPDKT	240
RELEERWVKW	LSGYEPDSVV	FCALGSQVIL	EKDQFQELCL	GMELTGSFPL	VAVKPPRGSS	300
TIQEALPEGF	EERVKGKGLV	WGGWVQQPLI	LSHPSVGCFF	SHCGFGSMWE	SLLSDCQIVL	360
VPQLGDQVLN	TRLLSDELKV	SVEVAREETG	WFSKESLCDA	VNSVMKRDSE	LGNLVRKNHT	420
KWRETVASPG	LMTGYVDAFV	ESLQDLVSGT	THD			453

SEQ ID NO: 151 moltype = DNA length = 1362
 FEATURE Location/Qualifiers
 source 1..1362
 mol_type = other DNA
 organism = Arabidopsis thaliana

SEQUENCE: 151

atgggtggtt	tgaagtttca	tgtacttatg	tatccatggt	tcgcaacagg	ccatattgacc	60
ccgttccttt	ttcttgccaa	caaattggct	gagaaaggtc	atacggtcac	tttcttgctt	120
cccaagaat	ctctgaaaca	gttggaacat	ttcaatctgt	ttccacacaa	cattgtcttt	180
cgctctgtca	cgctccctca	tgtggatggt	ctcccggtg	gcacagagac	agcctctgag	240
atccctgtga	catcaactga	tctcttgatg	tctgctatgg	atctcacacg	tgatcaagtt	300
gaagctgtgg	tccgagccgt	tgaaccggac	ctgatcttct	ttgactttgc	tcattggatt	360
ccagaagtag	ctagggactt	cggccttaag	actgtaaagt	acgtcgtggt	gtctgcatcg	420
actatagcta	gtatgcttgg	ccaggtggtg	gagttaggtg	ttcctccacc	gggatattcca	480
tcatacaagg	tgctgcttcg	taaacaagat	gcttacacta	tgaagaacct	ggagcctaca	540

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aatacaatcg atgtcggacc aaacctcttg gaacgagtca ctacaagtct tatgaactct 600
gatgtcattg cgataaggac agccagagaa atcgaaaggaa acttttgcga ctatatagaa 660
aaacattgca ggaataaagggt tctcttgaca ggtccgggtg tccctgagcc agacaagact 720
agagagctag aggaacgatg ggttaagtgg ctaagtgggt atgaaccaga ctcaagtggg 780
ttttgtgcac tgggtccaca agtcatttta gagaaagatc aattccaaga actctgotta 840
ggaatggagc taacaggttc accgtttctt gtacggtta agccccctag aggctcatca 900
acgattcaag aagcacttcc tgaaggattc gaagagcggg ttaaaggaag aggccctgtt 960
tggggaggat ggggtcaaca accattgata ttgtctcatc catcagtcgg gtgctttgtg 1020
agccattgtg ggtttggatc aatgtgggag tctttgctga gtgattgtca gatagtctta 1080
gtaccacagt tgggtgatca agtcctgaac acaagattgc tgagtgcga actcaagggt 1140
tcggttgaag tggcaagaga ggaacaggga tgggtctcga aagagagctt gtgcgatgct 1200
gtcaatagtg tgatgaaaag ggacagcgag ctccgggaacc tggtagaggaa gaatcacacc 1260
aagtggaggg agacagtagc tagtctcgga ctaatgactg gttatgtcga tgctttcgta 1320
gagtcattgc aggatctgtg ctctgggacc acccatgact ga 1362

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SEQ ID NO: 152      moltype = AA length = 462
FEATURE
source              Location/Qualifiers
                    1..462
                    mol_type = protein
                    organism = Oryza sativa

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SEQUENCE: 152
MDSGYSSSYA AAAGMHVVIC PWLAFGHLPL CLDLAQLRLS RGHRSVSVST PRNISRLPPV 60
RPALAPLVAF VALPLPRVEG LPDGAESTND VPHDRPDMVE LHRRAFDGLA APFSEFLGTA 120
CADWVIVDVF HHWAAAAALE HKVPCAMMLL GSAHMIASIA DRRLEAETE SPAAAGQGRP 180
AAPTFEVAR MKLIRTKGSS GMSLAERFSL TLRSSSLVVG RSCVEFEPET VPLLSTLRGK 240
PITFLGLMPP LHEGRREDGE DATVRWLDAQ PAKSVVYVAL GSEVPLGVEK VHELALGLEL 300
AGTRFLWALR KPTGVSDADL LPAGFEERTR GRGVVATRVV PQMSILAHAA VGAFLETHCGW 360
NSTIEGLMFG HPLIMLPFG DQGNARLIE AKNAGLQVAR NDGGSFDRE GVAAAIRAVA 420
VEEESKVFQ AKAKKLQEV ADMACHERYI DGFIIQLRSY KD 462

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SEQ ID NO: 153      moltype = DNA length = 1389
FEATURE
source              Location/Qualifiers
                    1..1389
                    mol_type = other DNA
                    organism = Oryza sativa

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SEQUENCE: 153
atggactcgc gctactcttc ctctacgcc gccgcgcgcg gcatgcacgt cgtgatctgc 60
ccgtggctcg ccttcggcca cctgctcccg tgcctcgacc tcgcccagcg cctcgcgctg 120
cggggccacc gcgtgctggt cgtctccacc cgcgggaaca tatcccgctt cccgcgcgtg 180
cgccccgcgc tcgcgcgcgt cgtcgccctc gtggcgctgc cgtcccgcg cgtcgagggg 240
ctccccgacg gcgcgcgac caccacacgc gtccccacgc acaggccgga catggtcgag 300
ctccaccgga gggccttcga cgggctcgcc gcgcctctct cggagttctt gggcacccgc 360
tgccgcgact gggctcctgt cgactcttc caccactggg ccgcagccgc cgtctcgag 420
cacaagggtc catgtgcaat gatgtgtgtt ggctctgcac atatgatcgc ttccatagca 480
gacagacggc tcagagcgcg ggagacagag tcgctcgccg ctgcccggca gggacgcccc 540
gcggcggcgc caacgttcga ggtggcgagg atgaagttga tacgaaccaa aggcctatcg 600
ggaatgtccc tcgcgcgag cttctccttg acgctctcga ggagcagcct cgtcgctcgg 660
cggagctcgc tggagttcga gccggagacc gtcccgcctc tgcgacgct cccgcgtaag 720
cctattacct tccttggcct ttgcccgcgc gcccgcgga ggaaggcgag 780
gatgccaccg tccgctgggt cgacgcgcag ccggccaagt ccgtcgtgta cgtcgcgcta 840
ggcagcgagg tgccactggg agtgagagaag gtccacgagc tcgcgctcgg gctggagctc 900
gccgggagcg gcttctctgt ggtctctagg aagcccaact gcgtctccga cgcgacctc 960
ctccccgcgc gcttcgagga gcgcacgcgc ggccgcggcg tcgtggcgac gagatgggtt 1020
cctcagatga gcatactggc gcacgcgcgc gtgggcgcgt tcttgaccca ctgcggtcgt 1080
aactcgacca tcgagggggt catgttcggc caccgcgcta tcatgctgcc gatcttcg 1140
gaccagggac cgaacgcgcg gctaactcag gcgaagaacg ccggattgca ggtggcaaga 1200
aacgacggcg atggatcgt cgaccgagaa ggcgtcgccg cggcgatcgc tcagtcgcgc 1260
gtggaggaag aaagcagcaa agtggttcaa gccaaagcca agaagctgca ggagatcg 1320
gcggaatagg cctgccatga gaggtacatc gacggattca ttcagcaatt gagatcttac 1380
aaggattga 1389

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SEQ ID NO: 154      moltype = DNA length = 1389
FEATURE
misc_feature        Location/Qualifiers
                    1..1389
                    note = Oryza sativa sequence that has been codon optimized
                        for yeast
source              1..1389
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 154
atggatagtg gctactcttc atcttatgct gctgccgctg gtatgcacgt tgtgatctgc 60
ccttggttgg cctttgtgta cctgttacca tgtctggatt tagcccaaag actggcctca 120
agaggccata gagtatcatt tgtgtctact cctagaaata tctctcgttt accaccagtc 180
agactcgctc tagtctctct agttgcattc ttgtctcttc cacttccaag agtagaagga 240
ttgccagacg gcgctgaatc tactaatgac gtaccacatg atagacctga catggtcgaa 300
ttgcatagaa gagcctttga tggattggca gctccatttt ctgagttcct gggcacagca 360
tgtgcagact ggggttatagt cgatgtatct catcactggg ctgctgcagc cgcattggaa 420
cataaggctc cttgtgctat gatgttgtaa gggtcagcac acatgatcgc atccatagct 480
gatagaagat tggaaagagc tgaacagaaa tccccagccg cagcaggaca aggtaggcca 540

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-continued

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gctgccgccc caacctttga agtggctaga atgaaattga ttcgtactaa aggtagtcca 600
gggatgagtc ttgctgaaag gttttctctg acattatcta gatcatcatt agttgtagg 660
agatcctcgcg tcgagttcga acctgaaaca gtacctttac tatctacttt gagaggcaaa 720
cctattactt tecttggctt aatgectcca ttacatgaag gaaggagaga agatggtgaa 780
gatgctactg ttaggtgggt agatgcccac cctgctaagt ctgttggtta cgttgcaattg 840
ggttctgagg taccactagg ggtggaaaag gtgcatgaat tagcattagg acttgagctg 900
gccggaacaa gattcctttg ggctttgaga aaaccaaccg gtgtttctga cgccgacttg 960
ctaccagctg gggtogaaga gagaacaaga ggccgtgggtg tcgttgctac tagatgggtc 1020
ccacaaatga gtattctagc tcatgcagct gtaggggcct ttctaaccac ttgcggttgg 1080
aactcaacaa tagaaggact gatgtttggt catccactta ttatgttacc aatctttggc 1140
gatcagggac ctaacgcaag attgattgag gcaaagaacg caggtctgca ggttgcaagt 1200
aatgatggtg atggttccct tgatagagaa ggcggttgca ctgccatcag agcagtcgcc 1260
gttgaggaag agtcatctaa agttttccaa gctaaggcca aaaaattaca agagattgtg 1320
gctgacatgg cttgtcacga aagatacatc gatggtttca tccaacaatt gagaagttat 1380
aaagactaa 1389

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SEQ ID NO: 155      moltype = DNA length = 2506
FEATURE            Location/Qualifiers
source              1..2506
                    mol_type = other DNA
                    organism = Arabidopsis thaliana

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SEQUENCE: 155
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ggaacgagga ttggactcag aagtacagac aagagctaac aatgtgagct ttgagcaaac 180
aaaggagaag attaggaaga tgttgagaa agtggagctt tctgtttcgg cctacgatac 240
tagttgggta gcaatggttc catcaccgag ctcccacaaat gctccacttt tcccacagt 300
tgtgaaatgg ttattggata atcaacatga agatggatct tggggacttg ataaccatga 360
ccatcaatct cttagaagg atgtgttatc atctacactg gctagtatcc tcgcgttaaa 420
gaagtgggga attggtgaaa gacaaataaa caagggtctc cagtttattg agctgaattc 480
tgcattagtc tatctggcct atgttttaga ggggacaaga aacctaaaag attgggattt 540
gatgattaaa tatgctagag atttgaatct gacgattcca ttgggctcag aagtgggtga 600
tgacatgata cgaaaaagag attctggatct taaatgtgat agtgaagaat ttccaaaggg 660
aagagaagca tatctggcct atgttttaga ggggacaaga aacctaaaag attgggattt 720
gatagtcaaa tatcaaaagga aaaaatgggtc actgtttgat tctccagcca caacagcagc 780
tgcttttact cagtttggga atgatgggtg tctccgttat ctctgttctc tccctcagaa 840
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tgtcactctt gaaagcttag gaattgatag agatttcaaa accgaaatca aaagcatatt 960
ggatgaaacc tatagattat ggcttcgtgg ggatgaagaa atatgtttgg acttgccac 1020
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gttttctgtg ttagaattat ttaaggctgc tcaaagttat ccacatgaat cagctttgaa 1200
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cgaaacagga gacaaagcat tcacctatca aggacgcaat gtgacacacc acatttgtga 1860
aatttggttg gatctgtcca agtctatgtt gagagaagcc gagtggtcca gtgacaagt 1920
aacaccaagc ttggaggatt acatggaaaa tgcgtacata tcatttgcat taggaccaat 1980
tgtctctcca gctacattatc tgatcgagcc tccacttcca gagaagacag tcgatagcca 2040
ccaatataat cagctctaca agctcgtgag cactatgggt cgtcttctaa atgacataca 2100
agggttttaag agagaaagcg cggaaggga gctgaatgcg gtttcattgc acatgaaaca 2160
cgagagagac aatcgagca aagaagtgt catagaatcg atgaaagggt tagcagagag 2220
aaagagggaa gaattgcata agctagtgtt ggaggagaaa ggaagtgtgg ttccaaggga 2280
atgcaaaagaa gcgttcttga aaatgagcaa agtgttgaac ttattttaca ggaaggacga 2340
tggtattaca tcaaatgatc tgatgagctt tgttaaatca gtgatctacg agcctgttag 2400
cttacagaaa gaatctttaa cttgatccaa gttgatctgg caggtaaact cagtaaatga 2460
aaataagact ttggtcttct tctttgttgc ttcagaacaa gaagag 2506

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SEQ ID NO: 156      moltype = AA length = 785
FEATURE            Location/Qualifiers
source              1..785
                    mol_type = protein
                    organism = Arabidopsis thaliana

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SEQUENCE: 156
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VAMVPSPSQ NAPLFPQCVK WLLDNQHEHG SWGLDNHDHQ SLKKDVLSSST LASILALKKW 120
GIGERQINKG LQFIELNSAL VDTETIQKPT GFDIIFPGMI KYARDLNLTI PLGSEVDDMM 180
IRKRDLCLK DSEKFSKGRE AYLAYVLEGT RNLKDWDLIV KYQRKNGSLF DSPATTAAPF 240
TQFGNDGCLR YLCSLLQKFE AAVPSVYPFD QYARLSIIVT LESLGIDRDF KTEIKSILDE 300
TYRYWLRGDE EICLDLATCA LAFRLLLAHG YDVSYPDKP FAESGFSST LEGYVKNTFS 360
VLELFKAQS YPHESALKKQ CWTQKYLEM ELSSWVKTSV RDKYLKKEVE DALAFPSYAS 420
LERSDHRRIK LNSAVENTR VTKTSYRLHN ICTSDILKLA VDDFNFCQSI HREEMERLDR 480

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WIVENRLQEL	KFARQKLAYC	YFSGAATLFS	PELSDARISW	AKGGVLTTVV	DDFFDVGGSK	540
EELNLIHLV	EKWDLNGVPE	YSSEHVEIIF	SVLRDTILET	GDKAFTYQGR	NVTHHIVKIW	600
LDLLKSMLRE	AEWSSDKSTP	SLEDYMENAY	ISFALGPVIL	PATYLIGPPL	PEKTVDSHQY	660
NQLYKLVSTM	GRLNDIQGF	KRESAEGKLN	AVSLHMKHER	DNRSKEVII	SMKGLAERKR	720
EELHKLVLVEE	KGSVVPRECK	EAFLKMSKVL	NLFYRKDDGF	TSNDLMSLVK	SVIYEPVSLQ	780
KESLT						785

SEQ ID NO: 157 moltype = DNA length = 2490
 FEATURE Location/Qualifiers
 source 1..2490
 mol_type = other DNA
 organism = Zea mays

SEQUENCE: 157

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cttggtcctt	ggagcagtag	gattaaaaag	aaaaccgata	ctgttgccagt	accagccgct	120
gcaggagagt	ggagaagggc	cttggtctaga	gcacagcaca	catcagaatc	cgcagctgtc	180
gcaaaagggc	gcagtttgac	ccctatagtg	agaactgacg	ctgagtcaag	gagaacaaga	240
tggccaaccg	atgacgatga	cgcgaacct	ttagtggtatg	agatcagggc	aatgcttact	300
tccatgtctg	atggtgacat	ttccgtgagc	gcatacgata	cagcctgggt	cggattggtt	360
ccaagattag	acggcgggtga	aggtcctcaa	ttccagcag	ctgtgagatg	gataagaaat	420
aaccagttgc	ctgacggaag	ttggggcgat	gccgcattat	tctctgccta	tgacaggctt	480
atcaataccc	ttgcctcgct	tgtaaacttg	acaagtggtg	ccctagaacc	agagatgaga	540
ggtagaggac	tatctttttt	gggttaggaac	atgtggaaat	tagcaactga	agatgaagag	600
tcaatgccta	ttggcttcga	attagcattt	ccatctttga	tagagcttgc	taagagccta	660
gggtgccatg	acttccctta	tgatccaccg	gccctacaag	gaatctactc	ttcaagagag	720
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tatccagtgg	atctatttga	acatatttgg	gccgttgata	gacttgaaga	attaggaatc	1020
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gaaaaggacg	gtgaatttga	cccatgtgtc	ggacagtcta	atcaagctgt	taccggtatg	1260
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gggtgccttt	catatgagtt	cttgaggaga	aaagaagcag	agggagcttt	gagggacaag	1380
tggatcattt	ctaaagatct	acctggtgaa	gttgtgtata	ctttggattt	tccatggtac	1440
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gtttggattg	gcaagacatt	gtataggatg	ccacttgtaa	acaatgatgt	atatttgtaa	1560
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gcattggcga	gagccgcaat	actagctaac	gccgtgagca	cccacttaag	aaatagccca	1800
tcattcagag	aaagggttaga	gcattctctt	aggtgtagac	ctagtgaaga	gacagatggc	1860
tcctgggtta	actcctcaag	tggctctgat	gcagttttag	taaaggctgt	cttaagactt	1920
actgattcat	ctaaaggaga	agccacagca	atccatggag	gtgaccocaga	agatattata	1980
cacaagtgtg	taagatctgc	ttgggcccag	tgggttaggg	aaaaggcaga	cgtgcccgat	2040
agcgtgtgca	atggtagttc	tgcagtagaa	caagagggat	caagaatggt	ccatgataaa	2100
cagacctgtc	tattattggc	tagaatgatc	gaaatttctg	cggtagggc	agctgggtgaa	2160
gcagccagtg	aggacggcga	tagaagaata	attcaattaa	caggctccat	ctgcgcagct	2220
cttaagcaaa	aaatgctagt	ttcacaggac	cctgaaaaaa	atgaagagat	gatgtctcac	2280
gtggatgacg	aattgaagtt	gaggattaga	gagttcgttc	aatatttctg	tagactaggt	2340
gaaaaaaaaga	ctggatctag	cgaaccaggg	caaacatttt	taagtatagt	gaaatcatgt	2400
tactatgctg	ctcattgccc	acctcatgtc	gttgatagac	acattagtag	agtgattttc	2460
gagccagtaa	gtgccgcaaa	gtaaccgcgg				2490

SEQ ID NO: 158 moltype = AA length = 827
 FEATURE Location/Qualifiers
 source 1..827
 mol_type = protein
 organism = Zea mays

SEQUENCE: 158

MVLSSTCTTV	PHLSSLAVVQ	LGPWSSRIKK	KTDTVAVPAA	AGRWRRLAR	AQHTSESAV	60
AKGSSLTPIV	RTDAESRRTR	WPTDDDDAEP	LVDEIRAMLT	SMSDGDISVS	AYDTAWVGLV	120
PRLDGGEGPQ	PPAAVRWIRN	NQLPDGSGWD	AALFSAYDRL	INTLACVVTL	TRWSLEPEMR	180
GRGLSFLGRN	MWKLATDEEE	SMPIGFELAF	PSLIELAKSL	GVHDFPYDHQ	ALQGIYSSRE	240
IKMKRIPKEV	MHTVPTSILH	SLEGMPLDWD	AKLLKLQSSD	GSFLFSPAAT	AYALMNTGDD	300
RCFSYIDRTV	KHFNQGVPNV	YPVDLFEHIW	AVDRLERLGI	SRYPQKEIEQ	CMDYVNRHWT	360
EDGICWARRN	DVKEVDDTAM	AFRLRLRHGY	SVSPDVFKNF	EKDGEFFAFV	GQSNQAVTGM	420
YNLNRASQIS	PPGEDVLHRA	GAFSYEFLRR	KEAEGALRDK	WIISKDLFGE	VVYTLDFPWY	480
GNLPRVEARD	YLEQYGGGDD	VWIGKTYLYRM	PLVNNDVYLE	LARMDFNHCQ	ALHQLEDWQGL	540
KRWYTENRLM	DFGVAQEDAL	RAYFLAAASV	YEPCAAERL	AWARAAILAN	AVSTHLRNSP	600
SFRERLEHSL	RCRPSSETDG	SWFNSSSGSD	AVLVKAVLRL	TDLSAREAQP	IHGDPEDII	660
HKLLRSAAWE	WVRKADAAD	SVCNGSSAVE	QEGSRMVHDK	QTCLLLARMI	EISAGRAAGE	720
AASEDGDRRI	IQLTGSICDS	LKQKMLVSQD	PEKNEEMMSH	VDELKLRIR	EFVQYLLRLG	780
EKKTGSSETR	QTFLSIVKSC	YYAAHCPPHV	VDRHISRVIF	EPVSAAK		827

SEQ ID NO: 159 moltype = DNA length = 2570
 FEATURE Location/Qualifiers

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source          1..2570
                mol_type = other DNA
                organism = Arabidopsis thaliana

SEQUENCE: 159
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cttcttcttt ccttaccatc tcaggatctc ctctcaatgt cgctagagac aaatccagaa 180
gcggttccat acattgttca aagcttcgaa ctcaagaata cattaattct caagagggtc 240
aacatgattt gcctctaata catgagtggtc aacagcttca aggagaagat gctcctcaga 300
ttagtggttg aagtaatagt aatgcattca aagaagcagt gaagagtgtg aaaacgatct 360
tgagaaacct aacggacggg gaaattacga tatcggctta cgatacagct tgggttgcat 420
tgatcgatgc cggagataaa actccggcgt ttccctccgc cgtgaaatgg atcgccgaga 480
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acaaaggaat cacgtttttc cgggaaaaata ttgggaagct agaagacgaa aatgatgagc 660
atatgccaat cggattcgaa gtagcattcc catcgttgct tgagatagct cgaggaataa 720
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ctttcctctt cttctcttcc tctaccgctt ttgcatctat gcagaccoga gacagtaact 960
gcctcgagta tttgcgaaat gccgtcaaac gtttcaatgg aggagttccc aatgtctttc 1020
ccgtggatct tttcgagcgc atatggatag tggatcggtt acaacgttta gggatatcga 1080
gatactttga agaagagatt aaagagtgtc ccacagatat tggacgcgaa 1140
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tatgtggcga tcactctcaa actcacatct ccaaagctct gtttcaaaaa gtctagtaac 2460
ctcatcatca tcactgatcc attaacaatc agtggatcga tgtatccata gatcggtgaa 2520
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SEQ ID NO: 160      moltype = AA length = 802
FEATURE            Location/Qualifiers
source             1..802
                  mol_type = protein
                  organism = Arabidopsis thaliana

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SEQUENCE: 160
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SQEVQHDLP L IHEWQQLQGE DAPQISVGSN SNAPKEAVKS VKTILRNLTG GEITISAYDT 120
AWVALIDAGD KTPAPPSAVK WIAENQLSDG SWGDAYLFSY HDRLINTLAC VVALRSWNLF 180
PHQCNKGITF FRENIGKLED ENDEHMPIGF EVAFPSLLEI ARGINIDVPY DSPVLKDIYA 240
KKELKLTRIP KEIMHKIPTT LLHSLEGMRD LDWEKLLKLQ SQDGSFLFSP SSTAFAFMQT 300
RDSNCLEYLR NAVKRFNGGV PNVFPVDLFE HIWIVDRLQR LGISRYFEEE IKECLDYVHR 360
YWTDNGICWA RCSVQDIDD TAMAFRLLRQ HGYQVSADVF KNFEKEGEFF CFVGQSNQAV 420
TGMFNLRYAS QLAPFREIL KNAKEFSYNY LLEKREREEL IDKWIIMKDL PGEIGFALEI 480
PWYASLPRVE TRFYIDQYGG ENDVWIGKTL YRMPYVNNNG YLELAKQDYN NCQAQHLEW 540
DIFQKWYEEN RLSEWGVRRS ELLECYLLAA ATIFESERSH ERMVWAKSSV LVKAISSSF 600
ESSDSRRSFS DQFHEYIANA RRDHDFNDR NMRLDRPGSV QASRLAGVLI GTLNQMSFDL 660
FMSHGRDVNN LLYLSWDGWM EKWKLYGDEG EGELMVKMII LMKNNDLTNF FTHTHFVRLA 720
EIINRICLPR QYLKARNRDE KEKTIKSMEK EMGMVLELAL SESDTFRDVS ITPLDVAKAF 780
YYFALCGDHL QTHISKVLFO KV 802

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SEQ ID NO: 161      moltype = DNA length = 2859
FEATURE            Location/Qualifiers
source             1..2859
                  mol_type = other DNA
                  organism = Gibberella fujikuroi

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SEQUENCE: 161
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tgctcaactt catgtccaagt ttatgataca gcttggggtg caatgattcc aaaaacaaga 180
gataatgtaa aacagtgggtt gtttcagaaa tgtttccatt acctcttaaa aacacaagcc 240

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gcagatggct catgggggttc attgectaca acacagacag cgggtatcct agatacagcc 300
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ccagatgaaa tgggggtgag aatagaacac ggtgtcacat ccttgaaacg tcaattagca 420
gtttggaatg atgtggagga caccaaccat attggcgctg agtttatcat accagcctta 480
ctttccatgc tagaaaagga attagatggt ccatcttttg aatttccatg taggtccatc 540
ttagagagaa tgcacgggga gaaattaggt catttcgacc tggacaacgt ttacggcaag 600
ccaagctcat tgttgactc attggaagca ttctcggta agctagattt tgatcgacta 660
tcacatcacc tataccacgg cagtatgatg gcatctccat cttcaacggc tgcttatctt 720
attggggcta caaaatggga tgacgaagcc gaagattacc taagacatgt aatgcgtaat 780
gggtgcaggac atgggaatgg aggtatttct ggtacatttc caactactca tttcgaatgt 840
agctggatta tagcaacggt gttaaagggt ggctttactt tgaagcaaat tgacggcgat 900
ggcttaagag gtttatcaac catcttactt gaggcgcttc gtgatgagaa tgggtgcaat 960
ggctttgccc ctagaacagc agatgtagat gacacagcca agctctatt ggccttgta 1020
ttggtaaac ccagcagtgct acctgatatc atgattaagg tctttgaggg caaagacct 1080
tttaccactt ttggttcaga aagagatcca tcattgactt ccaacctgca cgtcctttta 1140
tctttactta aacaactctaa ctgtctcaa taccatctc aaatcctcaa aacaacatta 1200
ttcaacttga gatgggtggg ggggtccgat cattgtgtca aagacaaatg gaatttgagt 1260
cacctatata caactatggt gttgggtgaa gccttcactg aagtgtccca tctcattgac 1320
gggtggaat tgtctagtct gtttgatgaa tcctttaagt gtaagattgg tcttagcatc 1380
tttcaagcgg tacttagaat aatcctcacc caagacaacg acggtctctt gagaggatag 1440
agagaacaga cgtgttacgc aatattggct ttagttcaag cgagacatgt atgctttttc 1500
actcacatgg ttgacagact gcaatcatgt gttgatcgag gttctcatg gttgaaatct 1560
tgctcttttc attctcaaga cctgacttgg acctctaaaa cagcttatga agtgggtttc 1620
gtagctgaag catataaac agctgcttta caatctgctt ccctggaggt tctgctgcc 1680
accattggac attctgtcac gtcctgccgtt ccatcaagtg atctgaaaa atacatgaga 1740
ttgggtgagaa aaactgcggt attctctcca ctgggatgag ggggtctaat ggcttctatc 1800
atcgaaatct catttttctg accattactg caggcacaaa gagttgaaat ataccctaga 1860
gataatatca aggtggacga agataagta tttcttatta tccattcac atgggtcgga 1920
tgcaataata ggtctagaac tttcgcaagt aacagatggc tatacgatat gatgtacctt 1980
tcattactcg gctatcaaac cgacgagtag atggaagctg tagctgggcc agtggttgag 2040
gatgtttcct tgttacatca aacaattgat aaggtgattg ataatacaat gggtaacctt 2100
gcgagagcca atggaacagt acacagtggg aatggacatc agcacgaatc tctaatata 2160
ggccaagtgc aggcacctt gactcgtttc acaaatcag tcttgatca caaagacgtc 2220
cttaactcta gctcatctga tcaagatact ttgagaagag agtttagaac attcatgcac 2280
gctcatataa cacaaatcga agataactca cgattcagta agcaagcctc atccgatgag 2340
ttttcctctc ctgaacaatc ttactttcaa tgggtgaact caactgggtg ctacatgctc 2400
gcttgccgct attcatttgc cttctctaatt tgccctatgt ctgcaaatat gttgcagggt 2460
aaagacgcat ttccaagcgg aacgcaaaag tacttaatct cctctgttat gagacatgcc 2520
acaaacatgt gtagaatgta taacgacttt ggtctctatt ccagagacaa cgctgagaga 2580
aatgttaata gtattctatt tcttgagttt actctctgta acggaacttc tcaaaacctc 2640
gatgaaagga aggaagact tctgaaaatc gcaacttacg aacaagggtg tttggataga 2700
gcactagagg ccttggaag acagagtaga gatgatgccg gagacagagc tggatctaaa 2760
gatatgagaa agttgaaat cgtttaagta ttctgtgatg ttacggactt atacgatcag 2820
ctctacgta tcaagattt gtcactctct atgaagtaa 2859

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SEQ ID NO: 162      moltype = AA length = 952
FEATURE             Location/Qualifiers
source              1..952
                    mol_type = protein
                    organism = Gibberella fujikuroi

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SEQUENCE: 162
MPGKIENGTP KDLKTGNDFV SAAKSLLDRA FKSHHSYYGL CSTSCQVYDT AWMAMIPKTR 60
DNVKKQLFPE CFHYLLKTAQ ADGSWGS LPT TQTAGILDTA SAVLALLCHA QEPLQILDVS 120
PDEMGLRIEH GVTSLKKQLA VMNDVEDTNH IGVFIIIPAL LSMLEKELDV PSFEPPCRSI 180
LERMHGKELG HFDLEQVYVK PSSLLHSLEA FLGKLDLDFRL SHHLYHGSMM ASPSSTAAYL 240
IGATKWDEEA EDYLRHVMRN GAGHGNGGIS GTFPTTHFEC SWIIATLLKV GFTLKQIDGD 300
GLRGLSTILL EALRDENSVI GFAPRTADV DTAKALLALS LVNQPVSPDI MIKVFEKDH 360
FTTFGSGERL SLTSNLHVLL SLLKQSNLSQ YHPQILKTTL FPCRWWGSD HCVKDKWNLS 420
HLYPTMLLVE AFTEVLHLID GGELSSLFDE SFKCKIGLSI FQAVLRILT QDNDGSGRGY 480
REQTCTYALA LVQARHVCFF THMVDRLQSC VDRGFSWLKS CSFHSQDLTW TSKTAYEVGF 540
VAEAYKLAAL QSASLEVPAA TIGHSVTSVA PSSDLEKYMR LVRKTALFSP LDEWGLMASI 600
IESSFFVPLL QAQRVEIYPR DNIKVDKDY LSIIPFTWVG CNNSRTFAS NRWLYDMMYL 660
SLLGVTQDEY MEAVAGPVFG DVSLHLQITD KVIDNTMGNL ARANGTVHSG NGHQHESPNI 720
GQVEDTLTRF TNSVLNKHVD LNSSSSDQDT LRREFRTFMH AHITQIEDNS RFSKQASSDA 780
FSSPEQSYFQ WVNSTGGSHV ACAYSFAFSN CLMSANLLQG KDAFPSTGQK YLISSVMRHA 840
TNMCRMYNDF GSIARDNAER NVNSIHPEF TLCNGTSQNL DERKERLLKI ATYEQGYLDR 900
ALEALERQSR DDAGDRAGSK DMRKLKIVKL FCDVTDLYDQ LYVIKDLSSS MK 952

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SEQ ID NO: 163      moltype = DNA length = 1503
FEATURE             Location/Qualifiers
source              1..1503
                    mol_type = other DNA
                    organism = Stevia rebaudiana

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SEQUENCE: 163
atggaggctt catatctata catttccatt cttctgcttc tagcttcgta cctcttcacc 60
acccaacttc gtcgtaaatc cgccaatcta ccgccgacgg tgttccatc catccccaata 120
atcgggtcac tctacctctc caaaaaacca ctctatagaa cactagccaa aattgccgcc 180
aaatacggcc ctatcctcca actccaaacta gggtagccgc gtgtcctcgt aatctcctcc 240
ccttcgccgc ccgaagaatg cttcaccaac aacgacgta tcttcgccaa cgtccgaag 300

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-continued

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acgctattcg gaaaaattgt aggggggtacc agcctcgggt cgctgtcgtta cggcgaccag 360
tggcgcaacc tccgcccggt tgcattccatc gagattctat cagtcacccg gctcaacgag 420
tttcacgaca tacgtgttga cgaaaaccgg cttctgatcc gtaaaactaa atccagttct 480
tctccggtga cttctgataac ggtgttttac gcactaacgt taaatgtgat tatgagaatg 540
atctccggaa agagggtattt cgactcgggt gatcgggaat tggaggagga agggaagcga 600
ttccgggaga tactcgatga gacacttttg ctccggggtg cttctaattg tggggattac 660
ttgccgattc tgaattgggt ggggggtgaag agcttggaag agaagcta atcgattgcag 720
aaaaagagag atgatttctt tcagggactg atcgagcaag ttcgggaatc tagaggggct 780
aaagtgggaa aagggaaggaa gacgatgatc gagttgtgtg tatccctaca agaactctga 840
cctgagtatt acactgacgc catgatccga tcatttgcgc tgggtttatt agcagcaggg 900
agtgatacat cggctggaac tatggaatgg gcgatgtctc ttttggtaaa cccccgcac 960
gtattaaaaa aggcacaggc tgaattgatc cgagtcacgc gcaacaaccg tctaattgat 1020
gagtcgacga tagggaatat accttacatt ggttgcacga taaacgagac gctcagattg 1080
tacctgtcgg gcccggtgct atttcccatc gagtcacag cggaactgtg tatcagcggg 1140
tacaacatcc ctctgtggac gatgcttatt gtcaaccaat gggcgatata tcatgaccca 1200
aagggtgtggg acgacctga gacattcaaa ccggaagat tccaagggct tgaagggaca 1260
cgagacgggt ttaagcttgg gcccttggg tccggaagga ggggttgtcc aggggagggg 1320
ttggcgattc gtttgccttg gatgactctc gggtcgggta tccaatgctt tgattgggaa 1380
cgagtcgggt acgagatggt tgatatgacc gaaggtcttg gggtcacgtt gcctaaagct 1440
gtaccattag tcgcaaatg caagcccgct tccgaaatga cgaatctact ctctgagcta 1500
tga 1503

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SEQ ID NO: 164      moltype = AA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = protein
                    organism = Stevia rebaudiana

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SEQUENCE: 164
MEASYLYISI LLLASYLFT TQLRRKSANL PPTVFPSIPI IGHLYLLKKP LYRTLAKIAA 60
KYGPILQLQL GYRRVLVISS PSAAECEFTN NDVIFANRPK TLFGKIVGGT SLGSLSYGDQ 120
WRNLRRVASI EILSVHRLNE FHDIRVDENR LLIRKLRSSE SPVTLITVVFY ALTLNVIMRM 180
ISGKRYFDSG DRELEEEGKR FREILDETL LAGASNVGDY LPILNWLGVK SLEKKLIALQ 240
KKRDDFFQGL IEQVRKSRGA KVGKGRKTM ELLLSLQESE PEYTYDAMIR SFVLGLLAAG 300
SDTSAGTMEW AMSLLVNHPH VLKKAQAEID RVIGNNRLID ESDIGNIPIYI GCIINETLRL 360
YPAGPLLPFH ESSADCVISG YNIPRGTM LI VNQWAIHHPD KVVDDPETFK PERFQGLEGT 420
RDGFKLMPFG SGRRCGPGE LAILRLGMTL GSVIQCDFWE RVGDEMVDMT EGLGVTLPKA 480
VPLVAKCKPR SEMTNLSEL 500

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SEQ ID NO: 165      moltype = DNA length = 1503
FEATURE            Location/Qualifiers
source              1..1503
                    mol_type = other DNA
                    organism = Stevia rebaudiana

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SEQUENCE: 165
atggaagcct cttacctata catttctatt ttgcttttac tggcatcata cctgttcacc 60
actcaactta gaaggaagag cgctaatac caccacaaccg tgtttccatc aataccaatc 120
attggacact tatacttact caaaaagcct ctttatagaa ctttagcaaa aattgccgct 180
aagtacggac caatactgca attacaactc ggctacagac gtgttctggt gatttcctca 240
ccatcagcag cagaagagtg ctttaccat aacgatgtaa tcttcgcaaa tagacctaag 300
acattgtttg gcaaaatagt ggggtggaac tcccttgcca gtttatccta cggcgatcaa 360
tggcgtaatc taaggagagt agcttctatc gaaatccat cagttcatag gttgaacgaa 420
tttcatgata tcagagtggg tgagaacaga ttgttaatta gaaaacttag aagttcatct 480
tctcctgtta ctcttataac agtcttttat gctctaactc tgaacgtcat tatgagaatg 540
atctctggca aaagatatatt cgacagtggt gatagagaat tggaggagga aggtaagaga 600
tttcgaaaaa tcttagacga aacgttgcct ctacgggtg cttctaattg tggcgactac 660
ttaccaatat tgaactggtt gggaggttaag tctcttgaaa agaaattgat cgttttcgag 720
aaaaagagag atgacttttt ccagggtttg attgaacagg ttagaaaaatc tcgttggtgct 780
aaagtaggca aaggtagaaa aacgatgatc gaactcttat tatctttgca agagtcagaa 840
cctgagtact atacagatgc tatgataaga tcttttgcct taggtctgct ggctgcaggt 900
agtgatactt cagcgggcac tatggaatgg gccatgagct tactggtcaa tcaccacat 960
gtattgaaga aagctcaagc tgaatctgat agagttatcg gtaataacag attgattgac 1020
gagtcagaca ttggaaatat cccttacatc ggggtgtatta tcaatgaaac tctaagactc 1080
tatccagcag ggccattgtt gttccacatc gaaagttctg ccgactcgct tatttccggt 1140
tacaatatat ctataggtac aatgttaatc gtaaaccaat gggcgattca tcacgatcct 1200
aaagtctggg atgatcctga aacctttaa cctgaaagat ttcaaggatt agaaggaact 1260
agagatggtt tcaaaacttat gccattcggt tctgggagaa gaggatgtcc aggtgaaggt 1320
ttggcaataa ggctgttagg gatgacacta ggctcagtga tccaatgttt tgattgggag 1380
agagtaggag atgagatggt tgacatgaca gaaggttttg gtgtcacact tcctaaggcc 1440
gttccattag ttgccaatg taagccacgt tccgaaatga ctaatctcct atccgaactt 1500
taa 1503

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SEQ ID NO: 166      moltype = AA length = 691
FEATURE            Location/Qualifiers
source              1..691
                    mol_type = protein
                    organism = Saccharomyces cerevisiae

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SEQUENCE: 166
MPFGIDNTDF TVLAGLVAV LLYVKRNSIK ELLMSDDGDI TAVSSGNRDI AQVVTENNKN 60
YLVLYASQTG TAEDYAKKFS KELVAKFNLN VMCADVENYD FESLNDVPVI VSIFISTYGE 120

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-continued

GDFPDGAVNF	EDFICNAEAG	ALSNLRYNMF	GLGNSTYEFF	NGAAKKAKEH	LSAAGAIRLG	180
KLGEADDGAG	TTDEDYMAWK	DSILEVLKDE	LHLDEQEAKE	TSQFQYTVLN	EITDSMSLGE	240
PSAHYLPESHQ	LNRNADGIQL	GPFDLSQPYI	APIVKSRELF	SSNDRNCIHS	EFDLSGSNIK	300
YSTGDHLAVW	PSNPLEKVEQ	FLSIFNLDPE	TIFDLKPLDP	TVKVFPPTPT	TIGAAIKHYL	360
EITGPVSRQL	FSSLIQFAPN	ADVKEKLTLL	SKDKDQFAVE	ITSKYFNIAI	ALKYLSDGAK	420
WDTVPMQPLV	ESVPQMTFRY	YSISSSSLSE	KQTVHVTISV	ENFPNPBLPD	APPVVGVTN	480
LLRNIQLAQN	NVNIAETNLP	VHYDLNGPRK	LFANYKLPVH	VRRSNFRLPS	NPSTPVIMIG	540
PGTGVAFFRG	FIRERVAFLE	SQKKGNNVS	LGKHILFYGS	RNTDDFLYQD	EWPEYAKKLD	600
GSFEMVVAHS	RLPNTKKVYV	QDKLKDYEDQ	VFEMINNGAF	IYVCGDAKGM	AKGVSTALVG	660
ILSRGKSITT	DEATELIKML	KTSGRYQEDV	W			691

SEQ ID NO: 167 moltype = AA length = 335
 FEATURE Location/Qualifiers
 source 1..335
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 167

MEAKIDELIN	NDPVWSSQNE	SLISKYPNHI	LLKPGKNFRL	NLIVQINRVM	NLPKDQLAIV	60
SQIVELLHNS	SLLLDDIEDN	APLRRGQTTS	HLIFGVPSTI	NTANYMYFRA	MQLVSQLTTK	120
EPLYHNHITI	FNEELINLHR	GQGLDIYWRD	FLPEIIPTEQ	MYLNMVMNKT	GGLFRLTLRL	180
MEALSPSSH	GHSVPPFINL	LGIIVQIRDD	YLNKDFQMS	SEKGFPAEDT	EGKLSFPIVH	240
ALNFTKTKGQ	TEQHNEILRI	LLLRSTDKDI	KLKLIQILEF	DTNSLAYTKN	FINQLVNMKI	300
NDNENKYLDP	LASHSDTATN	LHDELLYIID	HLSEL			335

SEQ ID NO: 168 moltype = AA length = 712
 FEATURE Location/Qualifiers
 source 1..712
 mol_type = protein
 organism = *Arabidopsis thaliana*

SEQUENCE: 168

MSSSSSSSTS	MIDLMAAIK	GEPVIVSDPA	NASAYESVAA	ELSSMLIENR	QFAMIVTTSI	60
AVLIGICVML	VWRRSGSGNS	KRVEPLKPLV	IKPREEEIDD	GRKKVTIFFG	TQTGTAEFGA	120
KALGEEAKAR	YEKTRFKIVD	LDDYAADDDE	YEEKLKKEDV	AFFFLATYGD	GEPTDNAARF	180
YKWFTEGNDR	GEWLKLNLYG	VFGLGNRQYE	HFNKVAKVVD	DILVEQGAQR	LVQVGLGDDD	240
QCIEDDFTAW	REALWPELDT	ILREEGDTAV	ATPYTAAVLE	YRVSIHSDSE	AKFNDITLAN	300
GNGYTVFDAQ	HPYKANVAVK	RELHTPESDR	SCIHLEFDIA	GSGLTMKLG	HVGVLCNLS	360
ETVDEALRL	DMSPDYFSL	HAEKEDGTPI	SSSLPPPPFP	CNLRALTTRY	ACLLSSPKKS	420
ALVALAAHAS	DPTAERLKH	LASPAGKDEY	SKWVVESQRS	LLEVMAEFPS	AKPPLGVFFA	480
GVAPRLQPRF	YSISSSPKIA	ETRIHVTCAL	VYEKMPTGRI	HKGVCSTWMK	NAVPEKSEK	540
LFLGRPIFVS	QSNPKLPSDS	KVPIIMIGPG	TGLAPFRGFL	QERLALVESG	VELGPSVLFF	600
GCRNRRMDFI	YEEELQRVE	SGALAELSVA	FSREGPTKEY	VQHKMMDKAS	DIWNMISQGA	660
YLYVCGDAKG	MARDVHRS	LH	TIAQEQGSMD	STKAEGFVK	LQTSGRYLDR	712

SEQ ID NO: 169 moltype = AA length = 709
 FEATURE Location/Qualifiers
 source 1..709
 mol_type = protein
 organism = *Stevia rebaudiana*

SEQUENCE: 169

MQSESVEAST	IDLMTAVLKD	TVIDTANASD	NGDSKMPPAL	AMMFEIRDLL	LILTTSVAVL	60
VGCFVVLVWK	RSSGKKSQKE	LEPPKIVVPK	RRLEQEVDDG	KKKVTIFFGT	QTGTAEFGAK	120
ALFEEAKARY	EKAAPKVIDL	DDYAADLDEY	AEKLLKETYA	FFFLATYGDG	EPTDNAAKFY	180
KWFTEGDEKG	VWLQKLQYGV	FGLGNRQYEH	FNKIGIVVDD	GLTEQGAARI	VPVGLGDDQ	240
STEDDFSAWK	ELVWPELDDL	LRDEDDKAAA	TPYTAAIPEY	RVVFHDKPDA	FSDDTQTNG	300
HAVHDAQHPC	RSNVAVKKE	HTPESDRSCT	HLEFDISHTG	LSYETGDHVG	VYCENLIEVV	360
EEAGKLLGLS	TDYFSLHID	NEDGSPLGGP	SLQPPPPPCT	LRKALTNYAD	LLSSPKKSTL	420
LALAAHASDP	TEADRLRFLA	SREGKDEYAE	WVVANQRSL	EVMEAFPSAR	PPLGVFFAAV	480
APRLQPRYYS	ISSSPKMEPN	RIHVTALVY	EKTPAGRIKH	GICSTWMKNA	VPLTESQDCS	540
WAPIFVRTSN	FRLPIDPKVP	VIMIGPGTGL	APFRGFLQER	LALKESGTEL	GSSILFFGCR	600
NRKVDYIYEN	ELNNFVENGA	LSELDVAFSR	DGPTKEYVQH	KMTQKASEIW	NMLSEGAYLY	660
VCGDAKGMK	DVHRTLHTIV	QEQSLDSSK	AELYVKNLQM	SGRYLRDVW		709

SEQ ID NO: 170 moltype = AA length = 707
 FEATURE Location/Qualifiers
 source 1..707
 mol_type = protein
 organism = *Stevia rebaudiana*

SEQUENCE: 170

MQSNSVKISP	LDLVTALFSG	KVLDTSNASE	SGESAMLPTI	AMIMENRELL	MILTTSVAVL	60
IGCVVVLVWR	RSSTKKSAL	PPVIVVPKRV	QEEEVDDGKK	KVTVFPGTQT	GTAEGFAKAL	120
VVEAKARYEK	AVFKVIDLDD	YAADDDEYEE	KLKKESLAFF	FLATYGDGEP	TDNAARFYKW	180
FTEGDAKGEW	LNLQYGVFG	LGNRQYEHFN	KIAKVVDGL	VEQGAARLVP	VGLGDDQCI	240
EDDFTAWKEL	VWPELDQLLR	DEDDTTVATP	YTAARVYRV	VFHEKPDALS	EDYSYTNHGA	300
VHDAQHPCRS	NVAVKKEHLS	PESDRSCTHL	EFDISNTGLS	YETGDHVG	CENLSEVVD	360
AERLVGLPPD	TYSSIHTDSE	DGSPLGGASL	PPPPPPCTLR	KALTCYADVL	SSPKKSALLA	420
LAAHATDPSE	ADRLKFLASP	AGKDEYSQWI	VASQRSLEEV	MEAFPSAKPS	LGVFASVAP	480
RLQPRYYSIS	SSPKMAPDRI	HVTALVYEEK	TPAGRIHKG	CSTWMKNVAV	MTESQDCSWA	540
PIYVRTSNFR	LPSDPKVPVI	MIGPGTGLAP	FRGFLQERLA	LKEAGTDLGL	SILFFGCRNR	600
KVDIFIYENL	NNFVETGALS	ELIVAFSREG	PTKEYVQHKM	SEKASDIWNL	LSEGAYLYVC	660

-continued

GDAKGMADKV HRTLHTIVQE QGSLDSSKAE LYVKNLQMSG RYLRDVW 707

SEQ ID NO: 171 moltype = DNA length = 2079
 FEATURE Location/Qualifiers
 misc_feature 1..2079
 note = Arabidopsis thaliana sequence that has been codon
 optimized for yeast
 source 1..2079
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 171

atgacttctg	cactttatgc	ctccgatctt	ttcaaacat	tgaaaagtat	catgggaacg	60
gattctttgt	cgatgatgt	tgtattagtt	attgctacaa	cttctctggc	actgggtgct	120
ggtttcgttg	tcttattgtg	gaaaaagacc	acggcagatc	gttccggcga	gctaaagcca	180
ctaagtatcc	ctaagtctct	gatggcgaaa	gatgaggatg	atgacttaga	tctaggttct	240
ggaaaaacga	gagttcttat	cttctctggc	acacaaaccg	gaacagccga	aggattcgct	300
aaagcacttt	cagaagagat	caaagcaaga	tacgaaaagg	cggctgtaaa	agtaatcgat	360
ttggatgatt	acgctgccga	tgatgaccaa	tatgaggaaa	agttgaaaaa	ggaacattg	420
gctttctttt	gtgtagccac	gtatggtgat	ggtgaaccaa	cgataacgc	cgcaagattc	480
tacaagtgtt	ttactgaaga	gaacgaaaga	gatatcaagt	tgcagcaact	tgcttacggc	540
gtttttgctt	taggtaacag	acaatacag	cactttaaca	agataggtag	tgcttagat	600
gaagagtatt	gcaaaaagg	tgcaagaga	ttgattgaag	tcggtttagg	agatgatgat	660
caatctatcg	aggatgactt	taatgcattg	aagggaatctt	tggtggtctga	attagataag	720
ttacttaagg	acgaagatga	taaatccgtt	gccactccat	acacagccgt	cattccagaa	780
tatagagtag	ttactcatga	tccaagattc	acaacacaga	aatcaatgga	aagtaatgtg	840
gctaattgta	atactacat	cgataattcat	catccatgta	gagtagacgt	tgcaagttca	900
aagggaattgc	acactcatga	atcagacaga	tcttgcatat	atcttgaatt	tgatatatca	960
cgtactggta	tacttaacga	aacaggtgat	cacgtgggtg	tctacgctga	aaacctggtt	1020
gaaattgtag	aggaagctgg	aaagtgtgtg	ggccatagtt	tagatcttgt	tttctcaatt	1080
catgccgata	aaggagatgg	ctcaccacta	gaaagtgcag	tgccctccac	atttccagga	1140
ccatgcaccc	taggtaccgg	tttagctcgt	tacgcggatc	tggtaaatcc	tccacgtaaa	1200
tcagctctag	tggtcttgcc	tgcgtagccc	acagaacctt	ctgaggcaga	aaaactgaaa	1260
catctaaact	caccagatgg	taaggatgaa	tactcacaat	ggatagtagc	tagtcaacgt	1320
tctttactag	aagttatggc	tgctttccca	tccgctaacc	ctcctttggg	tggtttcttc	1380
gccgcaatag	cgcttagact	gcaaccaaga	tactattcaa	tttcatcttc	acctagactg	1440
gcaccatcaa	gagttcatgt	cacatccgct	ttagtgtacg	gtccaactcc	tactggtaga	1500
atccataagg	gcgtttgttc	aacatggatg	aaaaacgcgg	ttccagcaga	gaagtctcac	1560
gaatgttctg	gtgtctcaat	ctttatcaga	gcctccaact	tcaaaactgc	ttccaatcct	1620
tctactccta	ttgtcatggt	cggtcctggt	acaggtcttg	ctccattcag	aggttttcta	1680
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gtccaacaca	aaatgatgga	aaaggccgca	caagtttggg	acttaatcaa	agagggaaggc	1920
tatctatatg	tctgtggtga	tgcaaaagggt	atggcaagag	atgttcacag	aacacttcat	1980
actatagtag	aggaacagga	aggcgttagt	tctctggaag	cggaagcaat	tgtagaaaag	2040
ttacaacacg	agggaagata	cttgagagat	gtgtggttaa			2079

SEQ ID NO: 172 moltype = DNA length = 2139
 FEATURE Location/Qualifiers
 misc_feature 1..2139
 note = Arabidopsis thaliana sequence that has been codon
 optimized for yeast
 source 1..2139
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 172

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gaattgtctt	caatgttgat	cgaaaacaga	caattcgcca	tgatcgtaac	tacatcaatc	180
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ggtagaaaaga	aagttacaat	atttttcggg	acccaaactg	gtacagctga	aggttttgca	360
aaagccttag	gtgaagaagc	taaggcaaga	tacgaaaaga	ctagattcaa	gatagtcgat	420
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gatattttgg	tcgaacaagg	tgctcaaa	ttagtccaag	taggtttggg	tgacgatgac	720
caatgtatag	aagatgactt	tactgcctgg	agagaagcgt	tgtggcctga	attagacaca	780
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tacagagttt	ccatccatga	tagtgaagac	gcaaaagtta	atgatatac	tttgcccaat	900
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agagaattac	atacaccaga	atccgacaga	agttgtatac	acttggaatt	tgatattcgt	1020
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cacgctgaaa	aagaagatgg	tacaccaatt	tccagttctt	taccacctcc	attccctcca	1200
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gaaactagaa	ttcatgttac	atgtgcatta	gtctacgaaa	agatgccaac	cggtagaatt	1560
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gttcaacata	agatgatgga	taaggcatcc	gacatatgga	acatgatcag	tcaaggtgct	1980
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acaattgctc	aagaacaagg	ttccatggat	agtaccaaag	ctgaagggtt	cgtaaaagac	2100
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SEQ ID NO: 173 moltype = DNA length = 2130
 FEATURE Location/Qualifiers
 source 1..2130
 mol_type = other DNA
 organism = Stevia rebaudiana

SEQUENCE: 173

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gcatgatgtg	tcgaaattcg	tgaactgttg	ctgattttga	ctacgtcagt	tgctgttttg	180
gtcggatgtt	tcgttgtttt	ggtgtggaag	agatcgctccg	ggaagaagtc	cggaaggaa	240
ttggagccgc	cgaagatcgt	tgtgccgaag	aggcggctgg	agcaggaggt	tgatgatggt	300
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cgcgtcgtat	ttcatgacaa	accgatgctg	ttttctgatg	atcatactca	aaccaatggt	900
catgctgttc	atgatgctca	acatccatgc	agatccaatg	tggctgttaa	aaaagagctt	960
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gaagtcattg	aagcttttcc	gtcagctaga	cgcgcaactg	gtgttttctt	tgacgaggtt	1440
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caagaacagg	gaagtttggg	ctcgtctaaa	gcggagttgt	atgtgaagaa	tctacaaatg	2100
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SEQ ID NO: 174 moltype = DNA length = 2124
 FEATURE Location/Qualifiers
 source 1..2124
 mol_type = other DNA
 organism = Stevia rebaudiana

SEQUENCE: 174

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gcatgatgta	tggagaatcg	tgaactgttg	atgatactca	caacgtcggt	tgctgtattg	180
atcggatgcg	ttgtcgtttt	ggtgtggcgg	agatcgtcta	cgaagaagtc	ggcgttggag	240
ccaccggtga	ttgtggttcc	gaagagagtg	caagaggagg	aagttgatga	tgtaagaaga	300
aaagttacgg	ttttcttcgg	cacccaaact	ggaacagctg	aaggcttcgc	taaggcaact	360
gttgaggaa	ctaagctcg	atatgaaaag	gctgtcttta	aagtaattga	tttggatgat	420
tatgctgctg	atgacgatga	gtatgaggag	aaactaaaga	aagaattctt	ggcctttttc	480
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ttgggttaaca	gacaatatga	acattttaac	aagatcgcaa	aagtgggtga	tgatggtctt	660
gtgaacacgg	gtgcaaaagg	tcttgttctc	gttggacttg	gagatgatga	tcaatgtatt	720
gaagatgact	tcaccgatg	gaagagtgta	gtatggccgg	agttggatca	attacttcgt	780
gatgaggatg	acacaactgt	tgctactcca	tacacagctg	ctgttgacga	atatcgcgtt	840
gtttttcatg	aaaaaccaga	cgcgctttct	gaagattata	gttatacaaa	tgcccatgct	900
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ggtgatgcca aaggcactgg caaagatgta catcgacccc tccacacaat tgtgcaagaa 2040
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agatacctcc gtgacgtttg gtaaa 2124

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SEQ ID NO: 175      moltype = DNA length = 2427
FEATURE             Location/Qualifiers
source              1..2427
                    mol_type = other DNA
                    organism = Arabidopsis thaliana

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SEQUENCE: 175
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aaaggatttt tacacaaaaa ccagatcatt gctgaattcg aagctttgcc tgaacaaacc 180
cggaagaaac ctgaaggttg tcttctctt acccttctca aatccactca ggaagcaatt 240
gtgttgccac ctgggttgcc tctagctgtg agggcaaggc ctggtgtttg ggaatactta 300
cgagtcattc tccatgctct tctcgttgaa gaactccaac ctgctgagtt tcttcatctc 360
aaggaaagaa cgtttgatgc agttaagaat ggtaatttca ctcttgagct tgatttcgag 420
ccattcaatg cgtctatccc tegtccaaca ctccacaat acattggaaa tgggtgttgac 480
tcccttaacc gtcattttat ggctaagctc tcccatgaca aggagagttt gcttccattg 540
cttaagttcc ttcgtcttca cagccaccag ggcaagaacc tgatgttgag cgagaagatt 600
cagaacctca acactctgca acacaccttg aggaaagcag aagagtatct agcagagctt 660
aagtcgcaaa cactgtatga agagtttgag gccaaagttg aggagattgg tcttgagagg 720
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gaggcgcttg atccttgcac tcttgagact tttcttgcaa gagtaccat ggtgttcaac 840
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aggctcttgc cattgactgg tgtgatgga ttctggaagc atgtctcgaa ccttgaccgt 2340
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SEQ ID NO: 176      moltype = DNA length = 2421
FEATURE             Location/Qualifiers
source              1..2421
                    mol_type = other DNA
                    organism = Stevia rebaudiana

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SEQUENCE: 176
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atatgaaagc cctatcaagt tatgactgaa tttgaagcta tctgcaagaa agatcagagc 180
aaactctctg atggtgcttt ttatgaagtt cttaaatgca cacaggaagc aatagtgcaa 240

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ctcccatggg ttgcactcgc gatccgtctt cgaccgggtg tttgggaata tggtagagtc 300
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gaattgggtta atgggaacatc gaattggcaac ttcgtgttgg aactggattt tgaacctttt 420
accgcacgtt ttctctcgacc aactttaacc aagtctattg gtaatgggtg tgagtttcta 480
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ctcaatgctc tacaatcggt gttgcgaaag gcgtcagagt acttatcaac actcgacgcg 660
gcaacaccgt actctcagtt tgaacataag tttcaagaaa tcgggttgga gagaggttgg 720
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ttgacgcttg ccggagttta tggattctgg aagcatgtgt cgaagcttga caggctcgag 2340
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ccattggctg ttgatgagtg a
2421

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SEQ ID NO: 177      moltype = DNA length = 2421
FEATURE             Location/Qualifiers
source               1..2421
                    mol_type = other DNA
                    organism = Coffea arabica

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SEQUENCE: 177
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atcctgaaac cccaccaact tttggctgag tttagaagaa ttaacaagga tggtaaacaa 180
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aatgtccatg cactcgttgt tgaggagtta accgtgccag agtacctgca tttcaaggaa 360
gaactcgttg atggaagcaa aaatgggaat tttgttttgg aactggactt cgaacctatt 420
acagcctctt ttcccagcgc aactctaact aagtacatag gcgacggagt tgagttcttc 480
aacaggcacc tctctgccaa aatgttccat gacaaggaga gcatggcccc tctccttgat 540
tttctccgtg ttccaacata caagggcaag acaatgatgc ttaacgacag gatcaaggac 600
cttaacactc tccaagcagt tctgagggaag gcagaggagt acctaacaac actctctgca 660
gatacaccat actctgaatt cgagcacaaa ttccaagaaa ttggactgga gagaggttgg 720
ggtgatactg ctgagcgtgt cttggaaatg atctgcatgc ttctggatct tctgggggct 780
cctgatactg gcacactaga gaaattccta gggagaatcc ctatggtatt caatgttgtt 840
attctttccc cccatggata ctttggcccag gaaaacgtat tgggttatcc tgataccggt 900
ggccaggttg tttacatatt ggaatcaagt cctgccttgg agcgtgagat gctgaagagg 960
ataaaggaa aaggacttga tgtcaagcca cgcattctaa ttataactag gctgcttctt 1020
gatgcccctg gaaccacttg ttgtcaacgg cttgagaaag tatatggatc agagtactcc 1080
catatactca gagtcccctt cagaactgag aagggagttg ttgcgaaatg gatctctcgc 1140
tttgaagttt ggcctcatct ggaacatttt actgaggatg ttgcaaaaga agtcaactgc 1200
gaattacagg caaagccaga tttggttatt ggtaactaca gtgagggtta ccttgttgcc 1260
tccttgcttg ctcacaagtt aggtgtaaca cagtgtacca ttgctcatgc tttggaaaaa 1320
accaagtatc ctgattctga tatttatttg agcaaatatt atgaaaagta ccacttctca 1380
tgtcagttca ctgcggatct tatcgcaatg aaccatacag atttcattat cactagcact 1440
ttccaagaaa tagctggaag caaggacact gttgggcaat atgaaagcca tatggccttc 1500
acaatgccag gattatacag agttgtacat ggcattgatg tttttgatcc aaaattcaac 1560
attgtctcac ctggagctga tacaacctc tacttccac acacagagaa ggaaaagaga 1620
ttgacatctt tccatcctga aattgaggag ttgcttttca gcgatgtgga gaatgaggaa 1680
cacctatgtg tgctaaaaga caaaaagaag cctatcttat tcaccatggc aagactggat 1740
cgcgtaaaaga attgacagg gcttgttgaa ttgtatgcta agaaccctaa actaagggaa 1800
ttggtcaatc ttgtcgttgt ttgtggagac cgaagggaag aatccaaaga tttggaagaa 1860
caagctgaga tgaagaaat gtattcattg atagaaactt acaacttgaa cggccaattc 1920
agatggattt cttctcagat gaacagggtt agaaatgggt aactctatcg gtacattgct 1980
gacaccaggg gagcatctcg gcaacctgca ttttatgagg catttgggtt gactgtggtc 2040
gaggccatga catgtgtgtt gccaacattt gcaaccaacc atggtgtgtc tgctgagatc 2100

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attattcatg ggaaatctgg ttccacatt gatccatacc atggtgagca ggteagcgag 2160
ctccttgcca atttctttga aaggtgcaaag aaagagcctt cttactggga caccatttca 2220
gocggtggct tgaagcggtat ccaggaaaag tacacctggc aaatttactc agatcggttg 2280
ctgacgctgg ctggagttta tggattctgg aaatgtgttt ccaagcttga tcgccaggag 2340
atccgccggt atctggaaat gttttatgct ctcaagtatc gcaagttggc tgaagctgtt 2400
ccattggctg ttgatcagta a 2421

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SEQ ID NO: 178      moltype = AA  length = 481
FEATURE            Location/Qualifiers
source              1..481
                    mol_type = protein
                    organism = Arabidopsis thaliana

```

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SEQUENCE: 178
MHITKPHAAM FSSPGMGHVI PVIELGKRLS ANNGFHVTVF VLETDAASQA SKFLNSTGVD 60
IVKLPSPIY GLVDPDDHV TKIGVIMRAA VPALRSKIAA MHQKPTALIV DLFGTDALCL 120
AKEFNMLSV FIPNARFLG VSIYYPNLDK DIKEHTVQR NPLAIPGCEP VRFEDTLDAY 180
LVPDEPVYRD FVRHGLAYPK ADGILVNTWE EMEPKSLKSL LNPKLGRVA RVPVYPIGPI 240
CRPIQSSETD HPVLDWLNEQ PNESVLYISF GSGGCLSAKQ LTELAWGLEQ SQQRFVWVVR 300
PPVDGSCCSE YVSSANGGTE DNTPEYLPEG FVSRTSDRGF VVPSWAPQAE ILSHRAVGGF 360
LTHCGWSSTL ESVVGGVPMI AWPLFAEQNM NAALLSDELG IAVRLDDPKE DISRWKIEAL 420
VRKVMTEKEG EAMRRKVKKL RDSAEMSLSI DGGGLAHESL CRVTKECQRF LERVVDLSRG 480
A 481

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SEQ ID NO: 179      moltype = AA  length = 513
FEATURE            Location/Qualifiers
source              1..513
                    mol_type = protein
                    organism = Arabidopsis thaliana

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SEQUENCE: 179
MGAYETEKPT KDAAALETQS PEDFDQPSPL RKIISVASIA AGVQFGWALQ LSLTTPYVQL 60
LGIPHKWSSL IWLCPVSGM IVQPIVGFHS DRCRSKFGRR RPFATGAAL VAVAVFLIGY 120
AADFGYKMGD KLEEKVKVRA IGIFALGFWI LDVANNTLQG PCRAFLADLA AGDAKRTRVA 180
NAFFSFFMAV GNVLGYAAGS YTNLHKMFPP TMTKACDIYC ANLTKCFPLS ITLLLVTVT 240
SLWYVNDKQW SPPPRNADDD EKTSSVPLEG EIFGAFKVMK RPMWMLLIVT ALNWIWFFP 300
LLFDTDWMGR EVFGGSDSGN ERSKKLYSLG VQSGAMGLMF NSIVLGFMSL GVEWIGRKL 360
GAKRLWGIWN FILAAGLAMT VLVTKFAEDH RKTAGDLAP SASVKAGALS LFAVLGIPLA 420
ITFSTPFALA SIFSSCSGAG QGLSLGVLNL AIVIPQMIVS LGGGPPDALF GGNLPAFIV 480
AAIAAAISGV LALTVLPSP PDAPKATTMG GFH 513

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SEQ ID NO: 180      moltype = AA  length = 806
FEATURE            Location/Qualifiers
source              1..806
                    mol_type = protein
                    organism = Coffea arabica

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SEQUENCE: 180
MAERVLTRVH SLRERLDATL AAHRNDVLLF MSRLETHGKG ILKPHQLLAE FERINKDGKQ 60
KIHDAHAFEEV LKSTQEAIVL PPWVALAIRL RPGVWEYVRV NVHALVVEEL TVPEYLHFKE 120
ELVDGSKNGN FVLELDFEPP TASFPKPTLT KYIGDGVEFL NRHLSAKMFH DKESMAPLLD 180
FLRVHQYKKG TMMLNDRIDK LNTLQAVLRK AEEYLTLSA DTPYSEFEHK FQEIGLERGW 240
GDTAERVLEM ICMLLDLLGA PDSCTLEKFL GRIPMVFNVV ILSPHGYFAQ ENVLGYPTDG 300
GQVYILDQV PALEREMLRK IKEQGLDVKP RILIIITRLP DAPGTTTCQR LEKVYGYSEYS 360
HILRVFPRTE KGVVRKWISR FEVWPYMETF TEDVAKEVTA ELQAKPDLVI GNYSEGNLVA 420
SLLAHKLGV TQCTIAHALEK TKYPDSDIYL SKFDEKYHFS CQFTADLIAM NHTDFIITST 480
PQEIAGSKDT VGQYESHMAF TMPGLYRVVH GIDVDPKFN IVSPGADTNL YFPHTEKEKR 540
LTSFPEIEE LLFSDVNEE HLCVLKDKKK PILFTMARLD RVKNLTGLVE LYAKNPKLRE 600
LVNLVVVGDD RRKESKDL EEQAEMKKMYSL IETYNLNGQF RWISSQMN RVNGLERYRIA 660
DTRGAFVQPA FYEAFGLTV EAMTCGLPTF ATNHGGPAEI IHHKSGFHI DPYHGEQVSE 720
LLANFFERCK KEPSYWDTIS AGGLKRIQEK YTWQIYSDRL LTLAGVYGFW KCVSKLDRQE 780
IRRYLEMFFA LKYRKLAEEV PLAVDQ 806

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SEQ ID NO: 181      moltype = DNA  length = 19
FEATURE            Location/Qualifiers
misc_feature        1..19
                    note = Hairpin loop forming sequence
source              1..19
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 181
tgaattcggtt aacgaattc 19

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SEQ ID NO: 182      moltype = DNA  length = 19
FEATURE            Location/Qualifiers
misc_feature        1..19
                    note = Hairpin loop forming sequence
source              1..19
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 182
tgaattcggtt aacgaactc 19

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SEQ ID NO: 183      moltype = DNA  length = 19
FEATURE            Location/Qualifiers
misc_feature        1..19
                    note = Hairpin loop forming sequence
source              1..19
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 183
tgaattcgtt aacgaagtc                                     19

SEQ ID NO: 184      moltype = DNA  length = 19
FEATURE            Location/Qualifiers
misc_feature        1..19
                    note = Hairpin loop forming sequence
source              1..19
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 184
tgaattcgtt aacgaaatt                                     19

SEQ ID NO: 185      moltype = DNA  length = 287
FEATURE            Location/Qualifiers
source              1..287
                    mol_type = other DNA
                    organism = Saccharomyces cerevisiae

SEQUENCE: 185
gcgttggttg gtggatcaag cccacgcgta ggcaatcctc gagcagatcc gccaggcgtg   60
tatatatagc gtggatggcc aggcaacttt agtgctgaca catacaggca tatatatatg   120
tgtgcgacga cacatgatca tatggcatgc atgtgctctg tatgtatata aaactcttgt   180
tttcttcttt tctctaaata ttcttctcct atacattagg acctttgcag cataaattac   240
tatacttcta tagacacaca aacacaaata cacacactaa attaata                 287

SEQ ID NO: 186      moltype = DNA  length = 491
FEATURE            Location/Qualifiers
source              1..491
                    mol_type = other DNA
                    organism = Saccharomyces cerevisiae

SEQUENCE: 186
ggtcagcagc tctgatgtag atacacgtat ctcgacatgt tttattttta ctatacatat   60
ataaaagaaa taaaaaatga taacgtgtat attattattc atataatcaa tgagggtcat   120
tttctgaaac gcaaaaaaac gtaaatggaa aaaaaataaa gatagaaaaa gaaaacaaac   180
aaaggaaagg ttagcatatt aaataactga gctgatactt caacagcatc gctgaagaga   240
acagtattga aaccgaaaca ttttctaaag gcaaacaagg tactccatat ttgctggacg   300
tggtctttct ctcgtttcat atgcataatt ctgtcataag cctgttcttt ttctgggctt   360
aaacatcccg ttttgtaaaa gagaaatcta ttccacatat ttcattcatt cggctacat   420
actaaggata aactaatccc gttgtttttt ggccctcgta cataattata aactactaac   480
cattatcag a                                                    491

SEQ ID NO: 187      moltype = DNA  length = 28
FEATURE            Location/Qualifiers
misc_feature        1..28
                    note = Hairpin loop forming sequence
source              1..28
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 187
cacaaataca cacactaaat taataatg                             28

SEQ ID NO: 188      moltype = DNA  length = 47
FEATURE            Location/Qualifiers
misc_feature        1..47
                    note = Hairpin loop forming sequence
source              1..47
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 188
cacaaataca cacactaaat taatatgaat tcgttaacga attcatg       47

SEQ ID NO: 189      moltype = DNA  length = 47
FEATURE            Location/Qualifiers
misc_feature        1..47
                    note = Hairpin loop forming sequence
source              1..47
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 189
cacaaataca cacactaaat taatatgaat tcgttaacga actcatg       47

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SEQ ID NO: 190 moltype = DNA length = 47
 FEATURE Location/Qualifiers
 misc_feature 1..47
 note = Hairpin loop forming sequence
 source 1..47
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 190
 cacaaatata cacactaaat taatatgaat tcgttaacga agtcatg 47

SEQ ID NO: 191 moltype = DNA length = 47
 FEATURE Location/Qualifiers
 misc_feature 1..47
 note = Hairpin loop forming sequence
 source 1..47
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 191
 cacaaatata cacactaaat taatatgaat tcgttaacga aattatg 47

SEQ ID NO: 192 moltype = AA length = 444
 FEATURE Location/Qualifiers
 source 1..444
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 192
 MGKLLQLALH PVEMKAALKL KFCRTPLFSI YDQSTSPYLL HCFELLNLTS RSFAAVIREL 60
 HPRLNRCVTL FYLILRALDT IEDDMSIEHD LKIDLLRHFH EKLLLTKWSF DGNAPDVKDR 120
 AVLTDFESIL IEPHKLKPEY QEVIKEITEK MGNGMADYIL DENYNLNLQ TVHDYDVYCH 180
 YVAGLVGDGL TRLIVIAKFA NESLYSNEQL YESMGLFLQK TNIRDYNEED LVDGRSFWPK 240
 EIWSQYAPQL KDFMKPENEQ LGLDCINHLV LNALSHVIDV LTYLAGIHEQ STQFCAIPQ 300
 VMAIATLALV FNNREVLHGN VKIRKGTTCY LILKSRTLGR CVEIFDYLYR DIKSKLAVQD 360
 PNFLKLNLIQI SKIEQFMEEM YQDKLPPNVK PNETPIFLKV KERSRYDDEL VPTQQEEYK 420
 FNMVLSIILS VLLGFYYIYT LHRA 444

SEQ ID NO: 193 moltype = AA length = 460
 FEATURE Location/Qualifiers
 source 1..460
 mol_type = protein
 organism = *Schizosaccharomyces pombe*

SEQUENCE: 193
 MSLANRIEII RCLCQYKLWN DLPSYGEDEN VPQNIRRCYQ LLDMTSRSAF VVIKELPNGI 60
 REAVMIFYLV LRGLDTVEDD MTLPLDKKLP ILRDFYKTIE VEGWTFNESG PNEKDRQLLV 120
 EPDVVIKEYL NLSEGYRNV I SNITKEMGDG MAYYASLAEK NDGFSVETIE DFNKYCHYVA 180
 GLVGIQLSRL FAQSKLEDPD LAHSQAISNS LGLFLQKVNI IRDYREDFDD NRHFWPREIW 240
 SKYTSSFGLD CLPDNSEKAL ECLSDMTANA LTHATDALVY LSQKLTQEIF NFCAIPQVMA 300
 IATLAAVFRN PDVFQTNVKI RKGQAVQIIL HSVNLKQVCD LFLRYTRDIH YKNTPKDPNF 360
 LKISIECGKI EQVSESLFPR RFREMYEKAY VSKLSEQKKG NGTQKAILND EQKELYRKDL 420
 QKLGISILFV FFILVLCLAV IFVFNIRIH WSDPKELNLF 460

SEQ ID NO: 194 moltype = AA length = 445
 FEATURE Location/Qualifiers
 source 1..445
 mol_type = protein
 organism = *Yarrowia lipolytica*

SEQUENCE: 194
 MGKLIELLLH PSELASAIHY KLWRQPLHPR DLSKESTELR RCYELLNVCS RSFAAVIREL 60
 HPEVRDAVML FYLILRALDT IEDDMTSLRD IKIPILRDFT KCMKTPGWKF TSDPNERNDR 120
 VVLQEFPPVM TEFNKLKPKY QEVIYDITDR MGNGMADYVI DDDFNNNGVD TIAAYDLYCH 180
 HVAGIVGEGE TRITILAGFG TDVLHENPRL QESMGLFLQK VNIIRDYRED IDVNRAPWPR 240
 EIWHKYAEEM RDFKDPKYSK KALHCTSDLV ANALGHATDC LDYLDNVTDI STFTFCAIPQ 300
 VMAIATLDLV YRNPVDFQKN VKLRKGTTVS LILEASNVSG VCDIFTRYAR KVKKSDPNP 360
 PNYFRVSVLC GKIEQHAALI KRQRGPPAKT IAQLEGERKE MALSLIVCLA VIFMSGLMA 420
 YIAYVSGFRW SPREIFDSKM FPLRD 445

SEQ ID NO: 195 moltype = AA length = 443
 FEATURE Location/Qualifiers
 source 1..443
 mol_type = protein
 organism = *Candida glabrata*

SEQUENCE: 195
 MGKVLDLALH PLELRAALKL KFIQPLFST NDTRATPQLE RCYELLNLTS RSFAAVIMEL 60
 HPRLNRCVTL FYLILRALDT VEDDMTIDPQ LKVVLREFD SKLDTTDSWF DGNLKEKDR 120
 VVLTEFPCLL GEYHKLKPEY QKVIKRTIGL MGNGMADYIL DENFNLNGVQ TVKDYDKYCH 180
 YVAGLVGDGL TELIVLAGFG SDDLHYHGKNS FQLYESMGLF LQKTNIIIRDY AEDLDDGRSF 240
 WPKEIWSYEA TKLTDFRDPK NTQKGVDCIN HVLNLNALTHV IDVLTYLSSI HEQSSQFCA 300
 IPQVMAIATL AKVFNNPEVL RKNVKIRKGT TCDLILNRSR LKGCVDIFQY YLRDMKQRLP 360
 VEDPNYLKFN IQVAKIEQFI EEMFQDNLPA GVEPRETMII LKVQERLKWQ TVQIPRVQEE 420
 DYKFNMAISV VFCVLLSFYF FTK 443

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SEQ ID NO: 196 moltype = AA length = 441
 FEATURE Location/Qualifiers
 source 1..441
 mol_type = protein
 organism = *Ashbya gossypii*

SEQUENCE: 196

MGKVVQLFTH	PLELKAALKL	KFLREPLYPA	DDTQGSSELK	RCYQLLQRTS	RSFAAVIMEL	60
HPELRNAVML	FYLILRALDT	VEDDMTISPK	VKVPLLRREFD	QKLKLDTWSF	DGNAKTEKDR	120
DVLVEFSTIL	AEFHKLKPEY	QQVIADITHK	MGNMGADYIL	DEKFNLSGLE	TIQDYDRYCH	180
YVAGLVGDGL	THLIMLAKFS	SPGLYYDSPD	LYESMGLFLQ	KTNIIRDYAE	DLADGRSFWP	240
KEIWSHYADD	LASFSSKPENA	TAGVYCINHL	VLNALGHVQH	VLTYLASLRE	QSSFQFCAIP	300
QVMAIATLAL	VFGNERVLQT	SVKIRKGTTC	YLILKSRTFQ	GCVEIFEHYL	RDIRKRLTVA	360
DPNYLKLNIE	IAKLDKFIEE	MYQDKLPVGA	KPQETEIYKK	VRERSAYDLE	VLPREQEEEF	420
KFNVLLSILF	TVFGALYWA	K				441

SEQ ID NO: 197 moltype = AA length = 443
 FEATURE Location/Qualifiers
 source 1..443
 mol_type = protein
 organism = *Cyberlindnera jadinii*

SEQUENCE: 197

MGKLLQLALH	PDELASIVQF	KLFRKNENAR	NPATESAELI	RCYELLNLTS	RSFAAVIEEL	60
HPELRNVIMV	FYLVLRALDT	VEVDMISIENS	VKLPLVRQFPH	EKLDTKDWF	DGNSPNEKDR	120
CVLVEFDRIL	GQYHELKPQY	QKVIKEITEK	MGNMGADYIE	NENFNNSGLL	TIEDYDLYCY	180
YVAGLVGDGL	TQLIVLAKFG	NSELSVNKQL	FKSMGLFLQK	TNIIRDYEED	QVDGRAFWPK	240
ETWGYANEL	SDFMKPEQNS	QGLWCISELV	CNALDHVIDV	LQYLALVEEQ	TSFNFCAIPQ	300
VMAIATLELV	PQNPOVLQTH	SVKIRKGTTS	LILESRTLEG	CARIFRRYLR	KIHHKSHPSD	360
PNYLRLGITI	GKIEQFLDGM	YPHYVPKGIT	PQTTSIRTQV	VKRLQLDEPM	KRDIDEIEILK	420
TRILLLSLGV	AVFGVVGTV	RII				443

SEQ ID NO: 198 moltype = AA length = 448
 FEATURE Location/Qualifiers
 source 1..448
 mol_type = protein
 organism = *Candida albicans*

SEQUENCE: 198

MGKFLQLLSH	PTELKAVIQL	FGFRQPLHPG	KRDVNDKELG	RCYELLNLTS	RSFAAVIEEL	60
HPELRDAMI	FYLVLRALDT	IEDDMTIKSS	IKIPLLRREFD	TKLNTKNWTF	DGYGPNEKDR	120
TVLVEFDKIL	NVYHRLKPQY	QDIKISITFK	MGNMGADYIL	DEEFNVYGVA	TVEDYNLYCH	180
YVAGLVGEGE	TNLFVLANFG	DKTLTENNFA	KADSMGLFLQ	KTNIIRDYHE	DLQDGRSFWP	240
REIWSKYTEN	LQDFHKVKT	AKEFAGVSCI	NELVLNALGH	VTDCLDYLSL	VKDPSSFSFC	300
AIPQVMAVAT	LAEPVNNKFK	LHGTVVKIRK	TTCLRLILESR	TLPGVVKIFK	EYIQVINHKS	360
SVRDPNYLKI	GKICGEIEQY	CEMIYPNKQA	LPPSMKSLPE	NKFTKIVASR	ESIDLSVQRR	420
IEPGNFNCNV	VLFGIGALIL	SLIYFVLY				448

SEQ ID NO: 199 moltype = AA length = 731
 FEATURE Location/Qualifiers
 source 1..731
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 199

MTEFYSDTIG	LPKTDPLRWR	LRTDELGRES	WEYLTPQQAA	NDPPSTFTQW	LLQDPKFPQP	60
HPERNKHSPI	FSAPDACHNG	ASFFKLLQEP	DSGIFPCQYK	GPMTMTIGYV	AVNYIAGIEI	120
PEHERIELIR	YIVNTAHPVD	GGWGLHSVDK	STVFGTVLNY	VILRLGLPK	DHPVCAKARS	180
TLLRLGGAIG	SPHWGKIWLS	ALNLYKWEV	NPAPPETWLL	PYSLPMHPGR	WWVHTRGVYI	240
PVSYSLSLVK	SCPMPTLLEE	LRNEIYTKPF	DKINFSKNRN	TVCGVDLYYP	HSTTLNIANS	300
LVVFEYKYL	NRFIYSLSKK	KVYDLIKTEL	QNTDSLCAIP	VNQAFCALVT	LIEEGVDSEA	360
FORLQYRFKD	ALFHGPQGMT	IMGTNGVQW	DCAFAIQYFF	VAGLAERPEF	YNTIVSAYKF	420
LCHAQFDTEC	VPGSYRDKRK	GAWGFSKTKQ	GTVADCTAE	AIKAIIMVK	SPVSEVHHM	480
ISSERLFEGI	DVLLNLQNI	SFEYGSFATY	EKIKAPLAME	TLNPAEVFNG	IMVEYPYVEC	540
TDSSVLGLTY	PHKYFDYRKE	EIRTRIRIAI	EFIKKSQLPD	GSWYGSWGIC	FTYAGMFALE	600
ALHTVGETYE	NSSTVRKQCD	FLVSKQMKDG	GWGESMKSE	LHSYVDSEKS	LVVQTAWALI	660
ALLFAEYPNK	EVDRGIDL	KNRQESGEW	KFESVEGVFN	HSCAIEYPSY	RFLFPIKALG	720
MYSRAYETH	L					731

SEQ ID NO: 200 moltype = AA length = 417
 FEATURE Location/Qualifiers
 source 1..417
 mol_type = protein
 organism = *Homo sapiens*

SEQUENCE: 200

MEFVKCLGHP	EEFYNLVFR	IGGKRKVMKP	MDQDSLSSSL	KTCYKYNQ	SRSFAAVIQA	60
LDGEMRNAVC	IFYLVLRALD	TLEDDMTISV	EKKVPLLNHF	HSFLYQPDWR	FMESKEKDRQ	120
VLEDFFTISL	EFNRNLAKEY	TVIADICRRM	GIGMAEFLDK	HVTSEQEWK	YCHYVAGLVG	180
IGLSRLFSAS	EFEDPLVGED	TERANSMLGF	LQKTNIIRDY	LEDQGGREF	WPQEVWSRYV	240
KKLGDFAKPE	NIDLAVQCLN	ELITNALHHI	PDVITYLSRL	RNQSVFNFCA	IPQVMAIATL	300
AACYNNQQVF	KGAVKIRKGQ	AVTLMMDATN	MPAVKAIYQ	YMEIYHRI	DSDPSSSKTR	360
QIISTIRTQN	LPNCQLISRS	HYSPIYLSFV	MLLAALSWQY	LTTLSQVTED	YVQTGEH	417

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SEQ ID NO: 201 moltype = AA length = 416
 FEATURE Location/Qualifiers
 source 1..416
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 201

MEFVKCLGHP	EEFYNLLRFR	MGGRRNFIPK	MDQDSLSSSL	KTCYKYLNT	SRSFAAVIQA	60
LDGDIRHAIC	VFYLVLRALD	TVEDDMSISV	EKKIPLLCNF	HTFLYDPEWR	FTESKEKDRQ	120
VLEDPPTISL	EFRLAKEYQ	TVIDDICHRM	GCGMAEFVDK	DVTSKQDWDK	YCHYVAGLVG	180
IGLSRLFSAS	EFEDPIVGED	IECANSMLGF	LQKTNIIIRDY	LEDQQEGRKF	WPQEVWGRYI	240
KKLEDFAKPE	NVDVAVQCLN	ELITNTLQHI	PDVLTYSRL	RNQSVFNFCA	IPQVMAIATL	300
AACYNNQQVF	KGVKIRKQG	AVTLMMDATN	MPAVKAIYQ	YIEETIYHRIP	NSDPSSSKTK	360
QVISKIRTQN	LPNCQLISRS	HYSPIYLSFI	MLLAALSWQY	LSTLSQVTED	YVQREH	416

SEQ ID NO: 202 moltype = AA length = 416
 FEATURE Location/Qualifiers
 source 1..416
 mol_type = protein
 organism = Rattus norvegicus

SEQUENCE: 202

MEFVKCLGHP	EEFYNLLRFR	MGGRRNFIPK	MDRNSLSNSL	KTCYKYLDQT	SRSFAAVIQA	60
LDGDIRHAVC	VFYLILRAMD	TVEDDMAISV	EKKIPLLRNF	HTFLYEPEDWR	FTESKEKHRV	120
VLEDPPTISL	EFRLAKEYQ	TVIADICHRM	GCGMAEFLNK	DVTSKQDWDK	YCHYVAGLVG	180
IGLSRLFSAS	EFEDPIVGED	TECANSMLGF	LQKTNIIIRDY	LEDQQEGRQF	WPQEVWGRYI	240
KKLEDFVKPE	NVDVAVKCLN	ELITNALQHI	PDVITYLSRL	RNQSVFNFCA	IPQVMAIATL	300
AACYNNHQVF	KGVKIRKQG	AVTLMMDATN	MPAVKAIYQ	YIEETIYHRVP	NSDPSSSKAK	360
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SEQ ID NO: 203 moltype = AA length = 396
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SEQUENCE: 203

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AAIDIIYQEL	LNLHRGQGMD	LFWRDTLTCP	SEDEYLEMVG	NKTGGLFRLA	VKLMQAESST	240
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LINILRQRTK	DEEVKRYALQ	YMESTGSFKH	TQDVVRQLRA	RALQLIEEIE	NSENGEQPEE	360
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SEQ ID NO: 204 moltype = AA length = 485
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 mol_type = protein
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SEQUENCE: 204

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VEDLTPPKW	PPFPTKVCWR	KHDLARMEPY	EAPGISDGYR	MGMVFKGSDC	LLFKCYHEFG	240
TQWLPLLETL	HQVPVVPVGL	LPPEIPGDEK	DETWSIKKW	LDGKQKGSV	YVALGSEALV	300
SQTEVVELAL	GLELSGLPFV	WAYRKPKGPA	KSDSVLEPDG	FVERTDRDRI	VWTSWAPQLR	360
ILSHESVCGF	LTHCGSGSIV	EGLMFGHPLI	MLPIFCQPL	NARLLEDKQV	GIEIPRNEED	420
GCLTKESVAR	SLRSVVVENE	GEIYKANARA	LSKIYNDTKV	EKEYVSQFVD	YLEKNARAVA	480
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SEQ ID NO: 205 moltype = DNA length = 1673
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 source 1..1673
 mol_type = other DNA
 organism = Stevia rebaudiana

SEQUENCE: 205

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aaggctattt	aaagccaaat	tatgtgatga	ataaagcaat	tgattccagc	cgt	1673

What is claimed is:

1. A method for producing a target steviol glycoside or a target steviol glycoside composition, comprising contacting a starting composition comprising a precursor steviol glycoside having a 13-O-glucose, a 19-O-glucose, or both a 13-O-glucose and a 19-O-glucose, and/or a mixture thereof with a first recombinant uridine 5'-diphospho (UDP) glycosyl transferase polypeptide capable of beta 1,2 glycosylation of a C2' of the 13-O-glucose, the 19-O-glucose, or both the 13-O-glucose and the 19-O-glucose of the precursor steviol glycoside and having at least 80% identity to the amino acid sequence set forth in SEQ ID NO: 152 and one or more sugar moiety donor, under suitable reaction conditions to transfer one or more sugar moieties from the one or more sugar moiety donor to the precursor steviol glycoside, thereby producing the target steviol glycoside or the target steviol glycoside composition;

wherein the precursor steviol glycoside is steviol-13-O-glucoside, rubusoside, stevioside or rebaudioside A;

wherein the target steviol glycoside is or the target steviol glycoside composition comprises stevioside, rebaudioside E, rebaudioside D, steviol-1,2 bioside, steviol-1,2-xylobioside, steviol-1,2-rhamnobioside, or an isomer thereof;

wherein the first recombinant 5'-UDP glycosyl transferase polypeptide is capable of converting Rebaudioside A (RebA) to Rebaudioside D (RebD) at a rate that is at least 20 times faster than the rate at which a UDP glycosyl transferase polypeptide having the amino acid sequence set forth in SEQ ID NO:5 is capable of converting RebA to RebD under corresponding reaction conditions; and/or

wherein the first recombinant 5'-UDP glycosyl transferase polypeptide is capable of converting higher amounts of RebA to RebD compared to the UDP glycosyl transferase polypeptide having the amino acid sequence set forth in SEQ ID NO:5 under corresponding reaction conditions.

2. The method of claim 1, wherein:

(a) the precursor steviol glycoside is rubusoside, wherein the additional sugar moiety is glucose, and stevioside is produced upon transfer of the additional glucose moiety;

(b) the precursor steviol glycoside is stevioside, the additional sugar moiety is glucose, and rebaudioside E is produced upon transfer of the additional glucose moiety;

(c) the precursor steviol glycoside is stevioside, the additional sugar moiety is glucose, the stevioside is contacted with the polypeptide capable of beta 1,2 glycosylation of the C2' of the 13-O-glucose, 19-O-glucose,

or both 13-O-glucose and 19-O-glucose of a steviol glycoside and a polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose of a steviol glycoside, and rebaudioside D is produced upon transfer of the additional glucose moiety;

(d) the precursor steviol glycoside is steviol-13-O-glucoside, the additional sugar moiety is glucose, and steviol-1,2 bioside is produced upon transfer of the additional glucose moiety;

(e) the precursor steviol glycoside is steviol-13-O-glucoside, the additional sugar moiety is xylose, and steviol-1,2-xylobioside is produced upon transfer of the additional sugar moiety;

(f) the precursor steviol glycoside is steviol-13-O-glucoside, the additional sugar moiety is rhamnose, and steviol-1,2-rhamnobioside is produced upon transfer of the additional sugar moiety;

(g) the precursor steviol glycoside is rebaudioside A, the additional sugar moiety is glucose, and rebaudioside D is produced upon transfer of the additional glucose moiety; or

(h) the precursor steviol glycoside is rubusoside, the additional sugar moiety is xylose, and 1,2-stevioxyloside is produced upon transfer of the sugar moiety.

3. The method of claim 1, wherein the starting composition is further contacted with:

(a) a second 5'-UDP glycosyl transferase polypeptide capable of beta 1,2 glycosylation of the C2' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside;

(b) a 5'-UDP glycosyl transferase polypeptide capable of glycosylating steviol or the precursor steviol glycoside at its C-19 carboxyl group;

(c) a 5'-UDP glycosyl transferase polypeptide capable of glycosylating steviol or the precursor steviol glycoside at its C-13 hydroxyl group; and/or

(d) a 5'-UDP glycosyl transferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

4. The method of claim 3, wherein at least one of the polypeptides is expressed in a recombinant host comprising one or more genes encoding the one or more polypeptides.

5. The method of claim 4, wherein the sugar moiety donor comprises ADP-sugar, CDP-sugar, GDP-sugar, and/or UDP-sugar.

6. The method of claim 3, wherein:

(a) the second 5'-UDP glycosyl transferase polypeptide comprises a polypeptide having at least 80% sequence

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identity to the amino acid sequence set forth in SEQ ID NO:5 or the amino acid sequence set forth in SEQ ID NO:76 or 78;

- (b) the 5'-UDP glycosyl transferase polypeptide capable of glycosylating the steviol or the precursor steviol glycoside at its C-19 carboxyl group comprises a polypeptide having at least 90% sequence identity to the amino acid sequence set forth in SEQ ID NO: 1;
- (c) the 5'-UDP glycosyl transferase polypeptide capable of glycosylating the steviol or the precursor steviol glycoside at its C-13 hydroxyl group comprises a polypeptide having at least 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:3; and/or
- (d) the 5'-UDP glycosyl transferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside comprises a polypeptide having at least 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:7.

7. The method of claim 1, wherein the first recombinant 5'-UDP glycosyl transferase polypeptide is expressed by a recombinant host comprising a recombinant gene encoding the first recombinant 5'-UDP glycosyl transferase polypeptide.

8. The method of claim 1, wherein the method is an in vitro method, further comprising supplying the sugar moiety donor or a cell-free system for regeneration of the sugar moiety donor.

9. The method of claim 8, wherein the target steviol glycoside is or the target steviol glycoside composition comprises RebD, the starting composition comprises RebA as the precursor steviol glycoside, wherein the starting composition is contacted with the first recombinant 5'-UDP glycosyl transferase polypeptide in stoichiometric or excess amount relative to RebD.

10. The method of claim 8, wherein the target steviol glycoside is or the target steviol glycoside composition comprises RebD, the starting composition comprises a *stevia* extract having at least one of RebA and stevioside as the precursor steviol glycoside,

wherein the starting composition is contacted with the first recombinant 5'-UDP glycosyl transferase polypeptide, a 5'-UDP glycosyl transferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside and UDP-glucose in stoichiometric or excess amount relative to the steviol glycoside.

11. The method of claim 8, wherein the target steviol glycoside is RebA, RebD, rebaudioside B (RebB), steviol-1,2-bioside, stevioside, rebaudioside E (RebE), dulcoside A, rebaudioside C (RebC), rebaudioside F (RebF), or a mixture of two or more of these compounds.

12. The method of claim 8, wherein the first recombinant 5'-UDP glycosyl transferase polypeptide is provided in soluble form or in immobilized form.

13. The method of claim 8, wherein the in vitro method is a whole cell in vitro method, wherein the whole cells are fed raw materials comprising the steviol and/or the precursor steviol glycoside.

14. The method of claim 13, wherein the whole cells are fed raw materials comprising the steviol and/or the precursor steviol glycoside is derived from plant extract.

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15. The method of claim 13, wherein the whole cell used in the whole cell in vitro method is:

- (a) in suspension or immobilized;
- (b) entrapped in a calcium or sodium alginate bead;
- (c) linked to a hollow fiber tube reactor system;
- (d) concentrated and entrapped within a membrane reactor system; or (d)
- (e) in fermentation broth or in a reaction buffer.

16. The method of claim 13, further comprising permeabilizing the whole cell by using a permeabilizing agent, wherein the permeabilizing agent is a solvent, a detergent, or a surfactant, by a mechanical shock, an electroporation, or an osmotic shock.

17. The method of claim 13, wherein the whole cell is microorganism being a prokaryote or a eukaryote.

18. The method of claim 13, wherein the whole cell is an *Escherichia coli* cell, a *Saccharomyces cerevisiae* cell, or a *Yarrowia lipolytica* cell.

19. The method of claim 13, wherein the starting composition further comprises steviol that is converted to RebA, RebD and/or RebE and the whole cell is a recombinant cell expressing:

- (a) the first recombinant 5'-UDP glycosyl transferase polypeptide;
- (b) a 5'-UDP glycosyl transferase polypeptide capable of glycosylating steviol or the precursor steviol glycoside at its C-19 carboxyl group;
- (c) a 5'-UDP glycosyl transferase polypeptide capable of glycosylating steviol or the precursor steviol glycoside at its C-13 hydroxyl group; and
- (d) a 5'-UDP glycosyl transferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

20. The method of claim 19, wherein the recombinant cell further expresses a second 5'-UDP glycosyl transferase polypeptide capable of beta 1,2 glycosylation of the C2' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

21. The method of claim 13, wherein RebA as the precursor steviol glycoside is converted to RebD as the target steviol glycoside and the whole cell is the recombinant cell expressing the first recombinant 5'-UDP glycosyl transferase polypeptide.

22. The method of claim 13, wherein rubusoside or stevioside RebA as the precursor steviol glycoside is converted to RebD as the target steviol glycoside and the whole cell is a recombinant cell expressing:

- (a) the first recombinant 5'-UDP glycosyl transferase polypeptide; and
- (b) a 5'-UDP glycosyl transferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

23. The method of claim 22, wherein the recombinant cell further expresses a second 5'-UDP glycosyl transferase polypeptide capable of beta 1,2 glycosylation of the C2' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

24. The method of claim 13, wherein steviol-13-O-glucoside (13-SMG) is converted to RebD and the whole cell is a recombinant cell expressing:

- (a) the first recombinant 5'-UDP glycosyl transferase polypeptide;
- (b) a 5'-UDP glycosyl transferase polypeptide capable of glycosylating steviol or the precursor steviol glycoside at its C-19 carboxyl group; and
- (c) a 5'-UDP glycosyl transferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose,

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19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

25. The method of claim 24, wherein the recombinant cell further expresses a second 5'-UDP glycosyl transferase polypeptide capable of beta 1,2 glycosylation of the C2' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

26. The method of claim 13, wherein steviol-19-O-glucoside (19-SMG) is converted to RebD and the whole cell is a recombinant cell expressing:

- (a) the first recombinant 5'-UDP glycosyl transferase polypeptide;
- (b) a 5'-UDP glycosyl transferase polypeptide capable of glycosylating steviol or the precursor steviol glycoside at its C-13 hydroxyl group; and
- (c) a 5'-UDP glycosyl transferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

27. The method of claim 26, wherein the recombinant cell further expresses a second 5'-UDP glycosyl transferase polypeptide capable of beta 1,2 glycosylation of the C2' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside.

28. A method of transferring an additional sugar moiety to a C2' of the 13-O-glucose, the 19-O-glucose, or both the 13-O-glucose and the 19-O-glucose of a precursor steviol glycoside having a 13-O-glucose, a 19-O-glucose, or both a 13-O-glucose and a 19-O-glucose, comprising contacting the precursor steviol glycoside with a first recombinant uridine 5'-diphospho (UDP) glycosyl transferase polypeptide capable of beta 1,2 glycosylation of a C2' of the 13-O-glucose, the 19-O-glucose, or both the 13-O-glucose and the 19-O-glucose of the precursor steviol glycoside and having at least 80% identity to the amino acid sequence set forth in SEQ ID NO:152 and one or more sugar moiety donor, under suitable reaction conditions to transfer of the additional sugar moiety to the precursor steviol glycoside;

wherein the precursor steviol glycoside is steviol-13-O-glucoside, rubusoside, stevioside or rebaudioside A; and

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wherein stevioside, rebaudioside E, rebaudioside D, steviol-1,2 bioside, steviol-1,2-xylobioside, steviol-1,2-rhamnobioside, an isomer thereof, and/or a steviol glycoside composition thereof is produced upon transfer of the additional sugar moiety.

29. The method of claim 28, wherein:

- (a) the precursor steviol glycoside is rubusoside, the second additional sugar moiety is glucose, and stevioside is produced upon transfer of the additional second glucose moiety;
- (b) the precursor steviol glycoside is stevioside, the second additional sugar moiety is glucose, and RebE is produced upon transfer of the additional second glucose moiety; and/or
- (c) the steviol glycoside is RebA, the additional second sugar moiety is glucose and RebD is produced upon transfer of the additional second glucose moiety.

30. The method of claim 28, wherein the starting composition comprising the precursor steviol glycoside is further contacted with:

- (a) a second 5'-UDP glycosyltransferase polypeptide having at least 80% sequence identity to the amino acid sequence set forth in SEQ ID NO:5 or the amino acid sequence set forth in SEQ ID NO:76 or 78;
- (b) a 5'-UDP glycosyltransferase polypeptide capable of glycosylating the steviol or the precursor steviol glycoside at its C-19 carboxyl group has the amino acid sequence having at least 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:1;
- (c) a 5'-UDP glycosyltransferase polypeptide capable of glycosylating the steviol or the precursor steviol glycoside at its C-13 hydroxyl group has the amino acid sequence having at least 90% sequence identity to the amino acid sequence set forth in SEQ ID NO:3; and/or
- (d) a 5'-UDP glycosyltransferase polypeptide capable of beta 1,3 glycosylation of the C3' of the 13-O-glucose, 19-O-glucose, or both 13-O-glucose and 19-O-glucose of the precursor steviol glycoside has the amino acid sequence having at least 90% sequence identity to the amino acid sequence set forth in SEQ ID NO: 7.

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